

BT505 – MIDTERM MERGED PPT (LECTURE 01 TO 102) Biosensors

Prepared By:

All About Biotech

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Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

Biosensors

Characteristics of Biosensor



Characteristics of Biosensor

Characteristics of Biosensor.

1. Sensitivity
 2. Selectivity
 3. Stability
 4. Reproducibility
 5. Response
 6. Biocompatibility
 7. Low background
- 
- The background of the slide features a faint, stylized illustration. On the right side, there is a depiction of a microchip or circuit board with various components and lines. Overlaid on and around this is a molecular structure, represented by blue spheres of different sizes connected by thin lines, suggesting a chemical or biological process. The overall aesthetic is technical and scientific.

Characteristics of Biosensor

Sensitivity

1. Response to per unit
2. Change in Analyte
3. Detect an in a mixture
4. Main point in designing.
5. Binding characteristics
6. $\Delta S / \Delta c$ analyte (S = signal)

Characteristics of Biosensor

Selectivity

1. Respond to Analyte
2. Most important feature
3. Changes in analyte conc
4. Example: In Glucometer
 - I. Glucose
 - II. Vitamin C

In a drop of blood

Characteristics of Biosensor

Stability

Is the degree of susceptibility to ambient disturbances

1. In and
2. Around the Biosensor.

Characteristics of Biosensor

Long term Stability

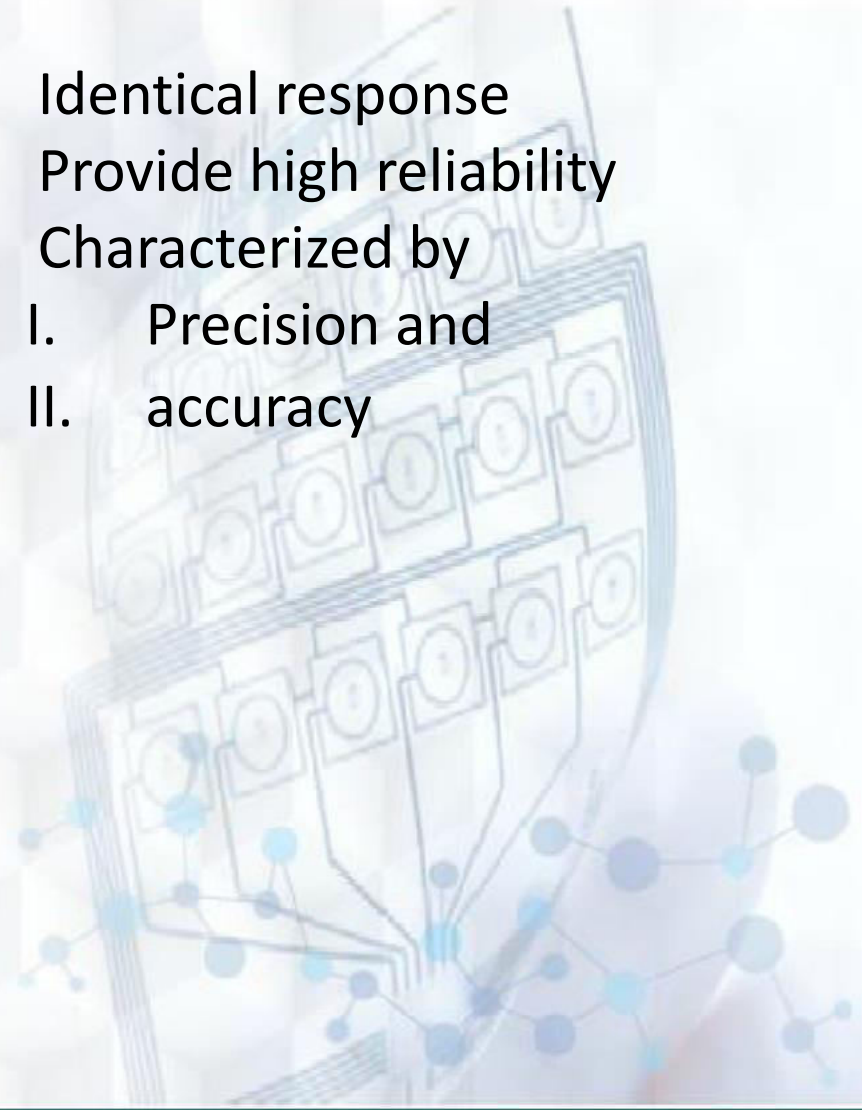
Not subjected to:

1. Fouling
2. Poisoning
3. Oxide formation
4. Moisture content
5. Interference with signal

Characteristics of Biosensor

Reproducibility

1. Identical response
2. Provide high reliability
3. Characterized by
 - I. Precision and
 - II. accuracy



Characteristics of Biosensor

Response

For example:

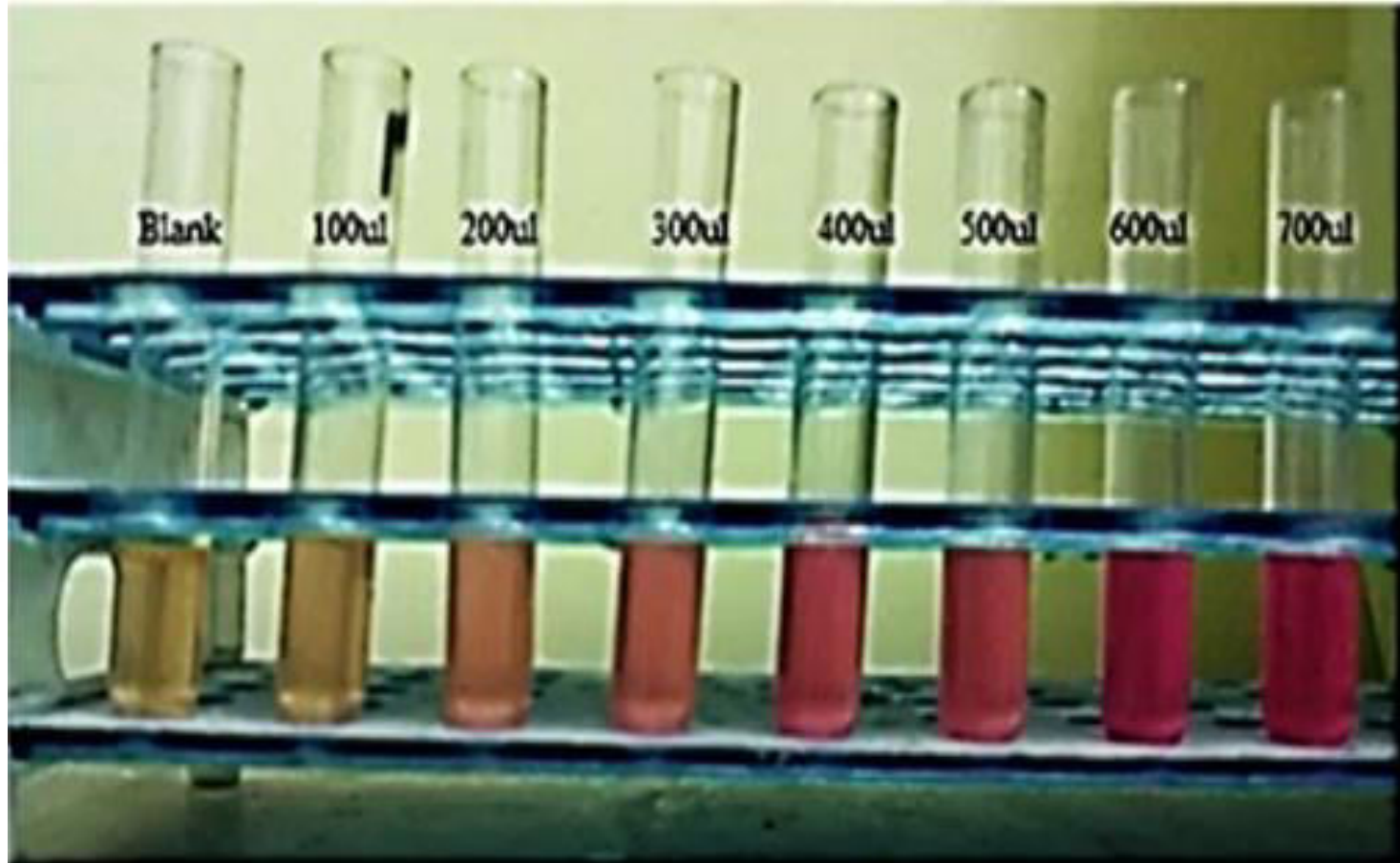
In case of Prostate specific antigen (PSA) concentration of 4ng/ml in blood, is associated with prostate cancer

Characteristics of Biosensor

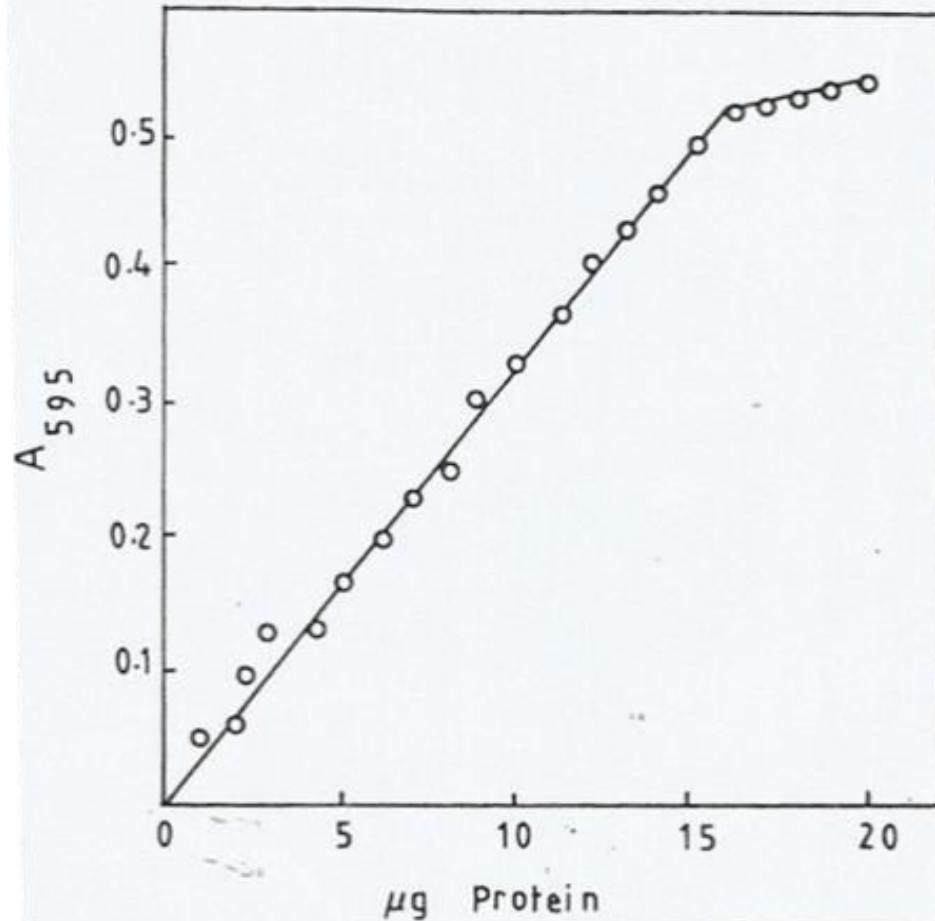
Linear Response

1. $\Delta S / \Delta c$ Analyte constant
2. Range of Analyte
3. Accuracy of the response
4. $Y = mc$
 - C is concentration
 - M is sensitivity
 - Y is output signal

Characteristics of Biosensor



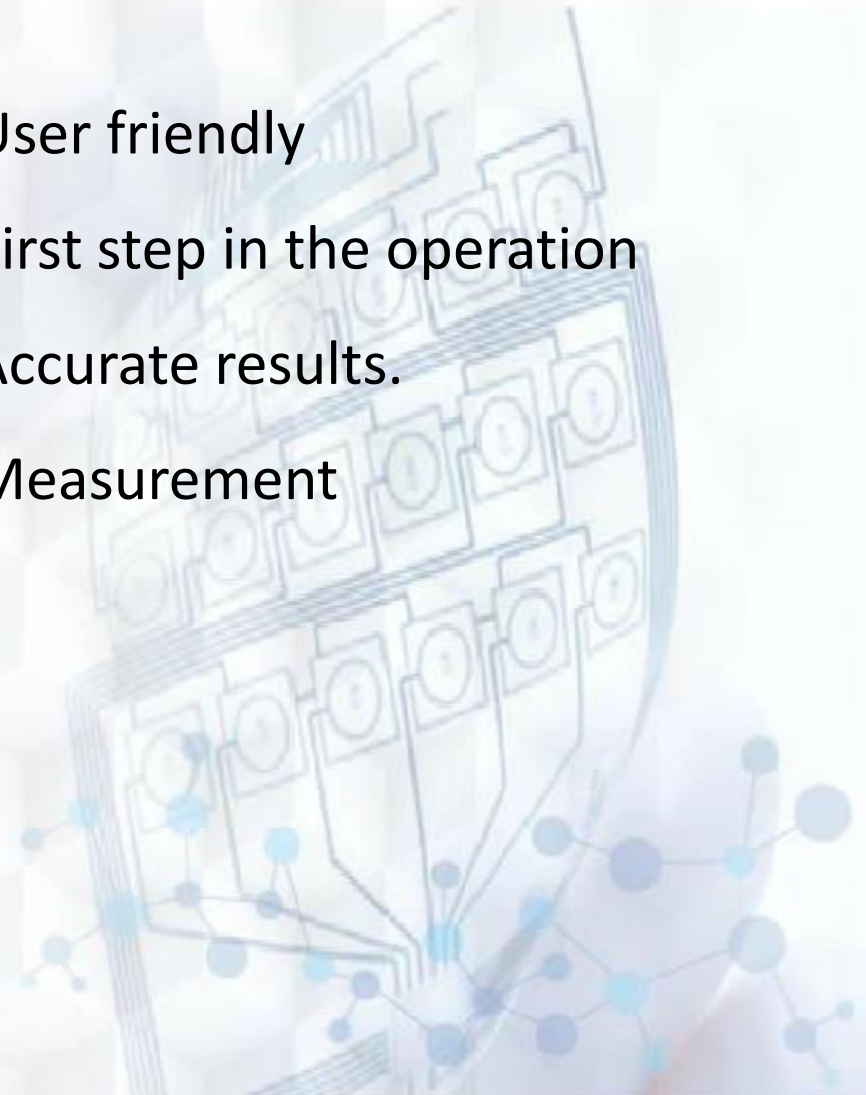
Characteristics of Biosensor



Characteristics of Biosensor

Simple calibration

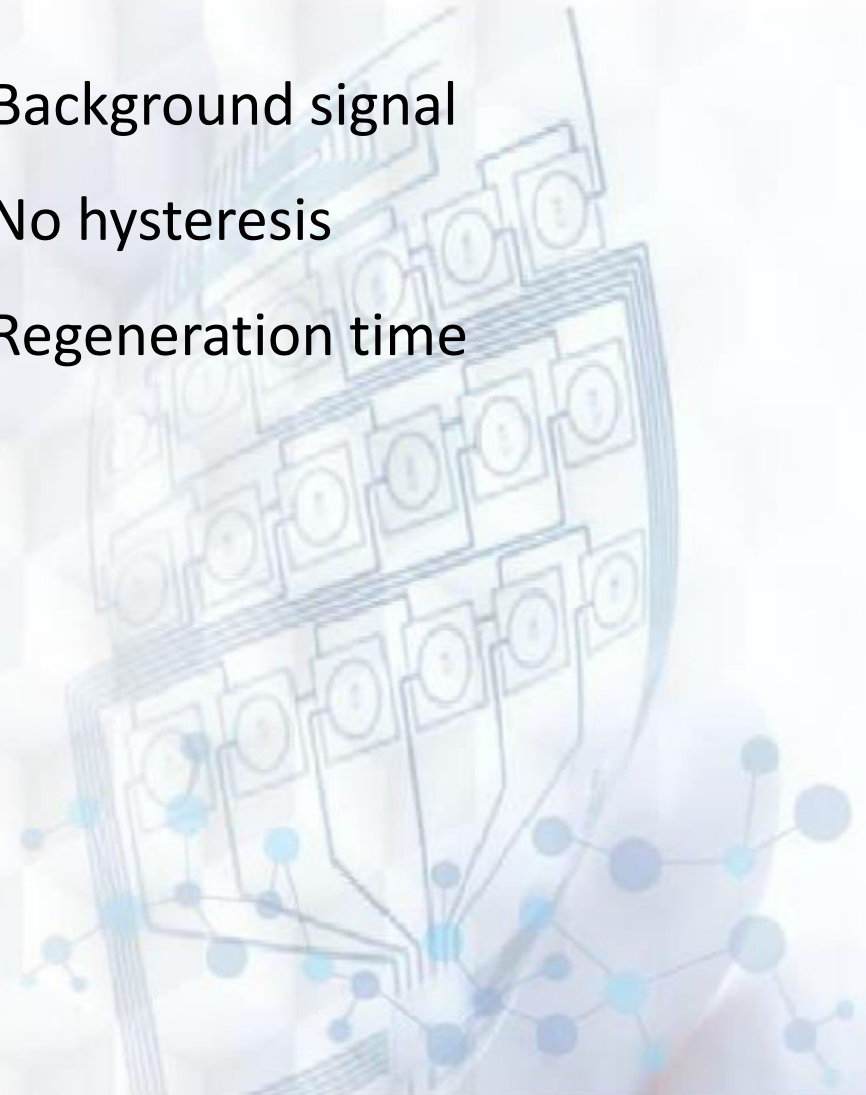
1. User friendly
2. First step in the operation
3. Accurate results.
4. Measurement



Characteristics of Biosensor

Low Noise

1. Background signal
2. No hysteresis
3. Regeneration time



Characteristics of Biosensor

Low Noise

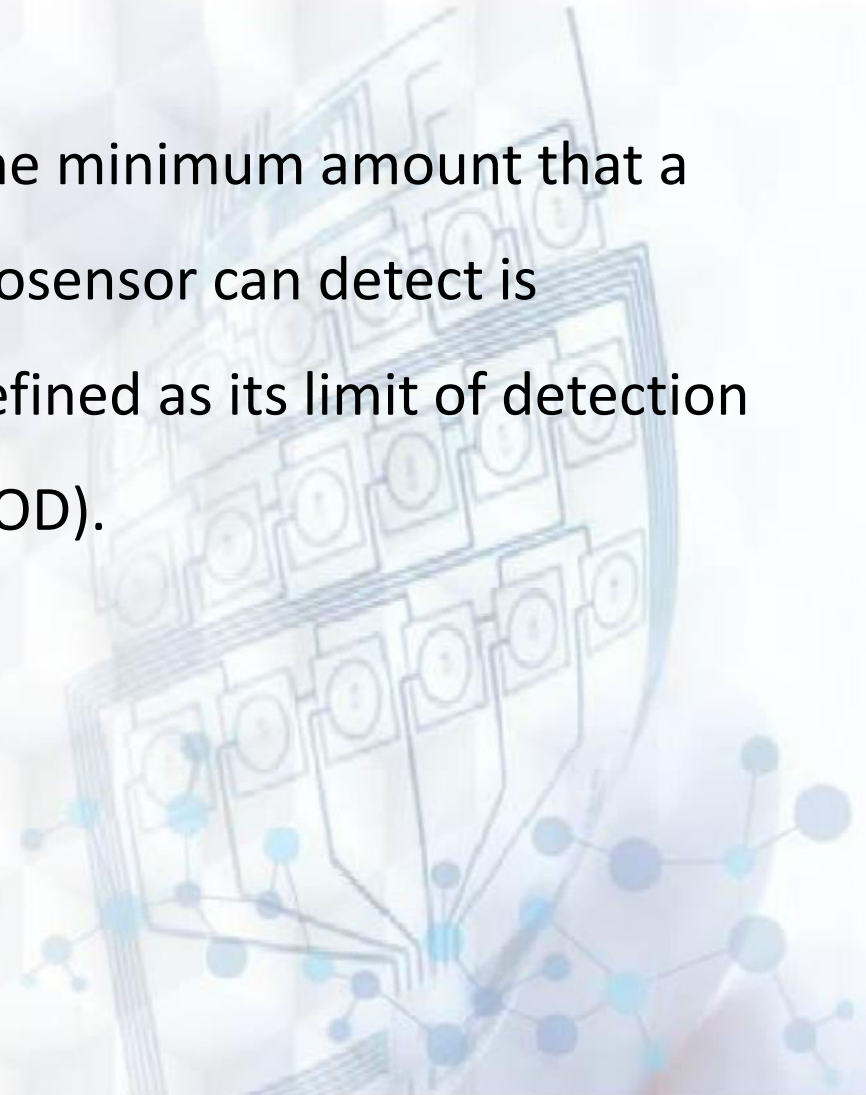
1. Accuracy is mean value
2. Precision alike results



Characteristics of Biosensor

Detection Limit

The minimum amount that a biosensor can detect is defined as its limit of detection (LOD).



Characteristics of Biosensor

Example Glucometer

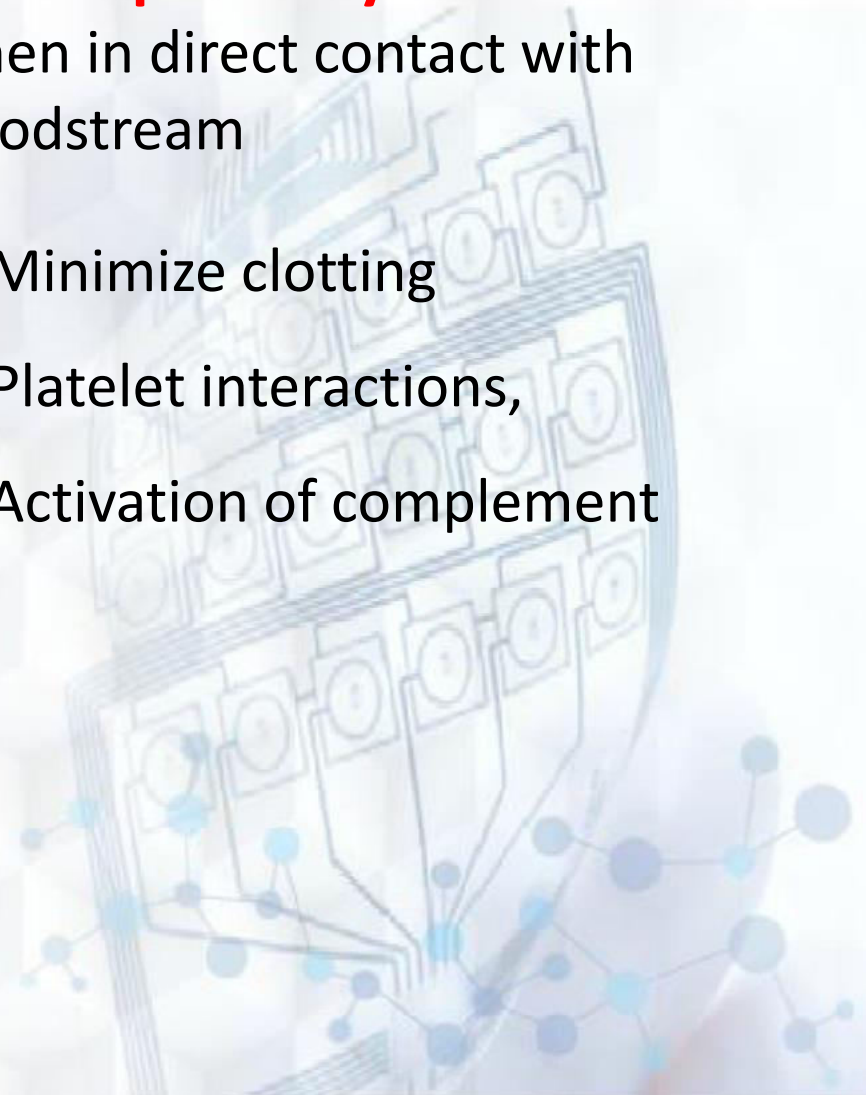
1. can measure 200-400 mg/dl.
2. Normal range 70-140 mg/dl.

Characteristics of Biosensor

Biocompatibility

When in direct contact with bloodstream

1. Minimize clotting
2. Platelet interactions,
3. Activation of complement



Characteristics of Biosensor

Lifetime

Life Time is the time period over which biosensor can be used

1. Without deterioration
2. In performance

Biosensors

Background



Background

Analyte

1. Biosensors are designed for the detection of Target element at an analytical level.
2. Analyte is a target element
3. Analyte could be a
 - i. Molecule
 - ii. Toxin
 - iii. Pathogen or others.

Background

Analyte

1. In a system, an Analyte binds with its Receptor.
2. It generates a recognizable signal.

Example.

Glucose molecule is an Analyte for a Glucometer .

Background

Examples

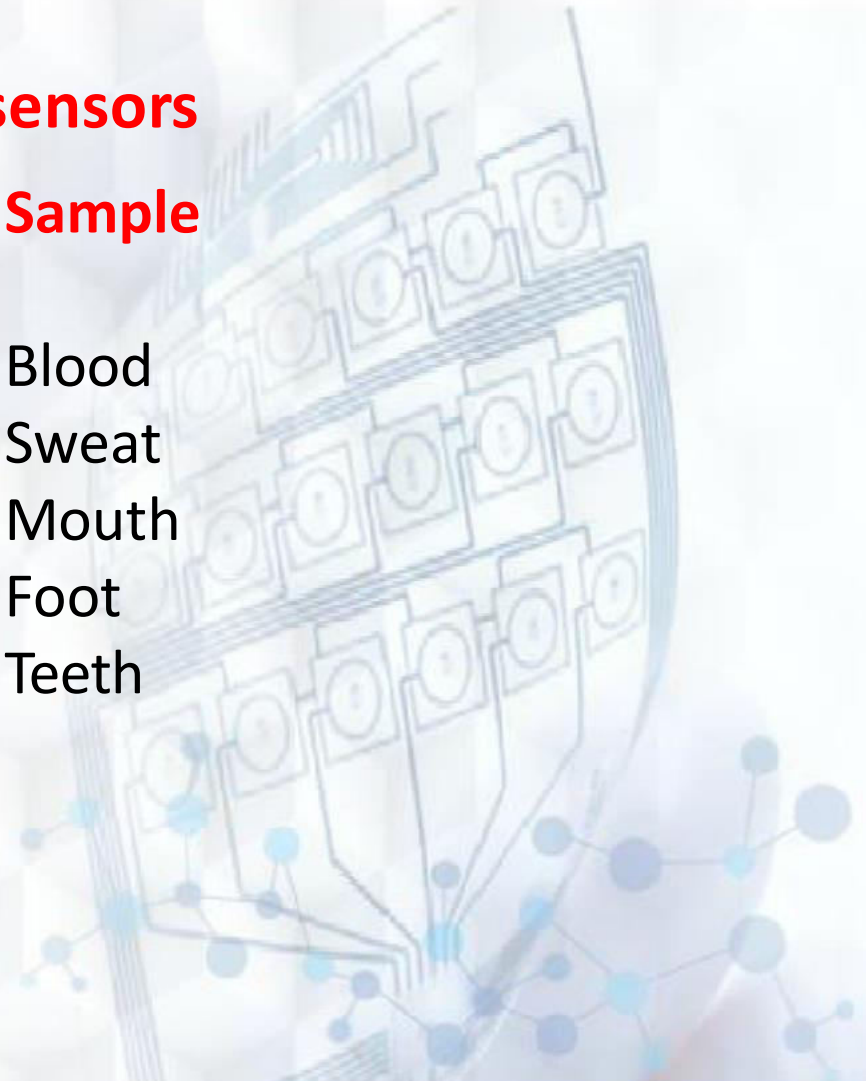
Wearable Biosensors

Analyte

Glucose
Lactic Acid
Uric Acid
Cortisol
Bacteria

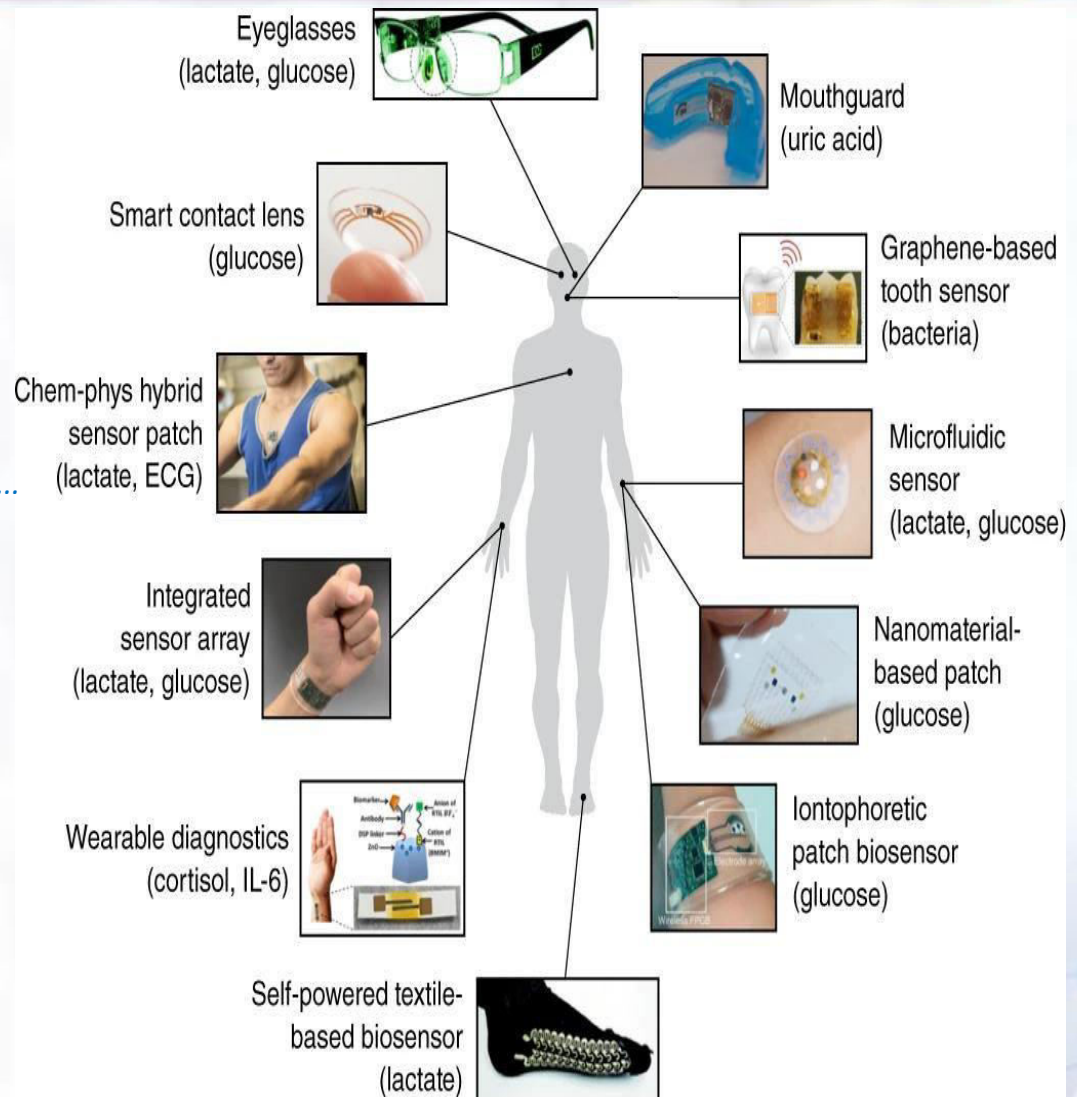
Sample

Blood
Sweat
Mouth
Foot
Teeth



Background

*Wearable biosensors for healthcare ...
nature.com*



Background

Designing of Biosensor

Biosensors are first designed in laboratory involving the fundamental and applied subjects:

1. Chemistry
2. Biochemistry
3. Physics
4. Microelectronics and Miniaturization

Effort By: AARISH ALI

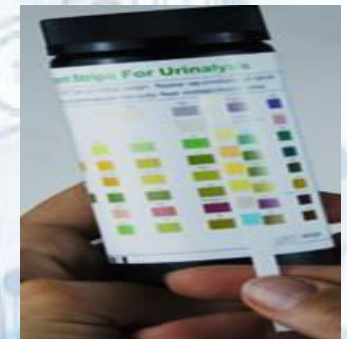
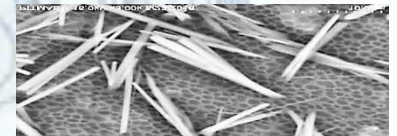
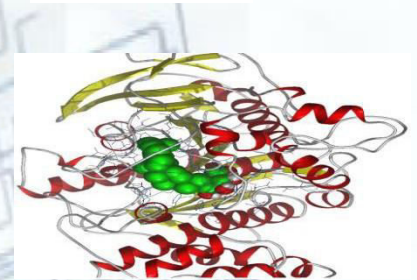
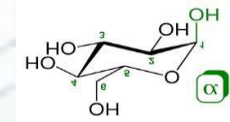
Background

Chemistry and Biochemistry

1. Characteristics of
 - a. Analyte
 - b. Bioreceptor Element
 - c. Support material
2. Clinical Chemistry

Example.

Rapid Urine test for glucose by dipstick method uses a thin plastic strip treated with chemicals.



Background

Physics

Physical components are the basic units of the Biosensor.

1. Detector
2. Signal converter
3. Amplifier
4. Measuring of signals
5. Display unit

Background

Microelectronics and Miniaturization

This is the manufacturing of smaller electronic devices with full/ improved functionality.

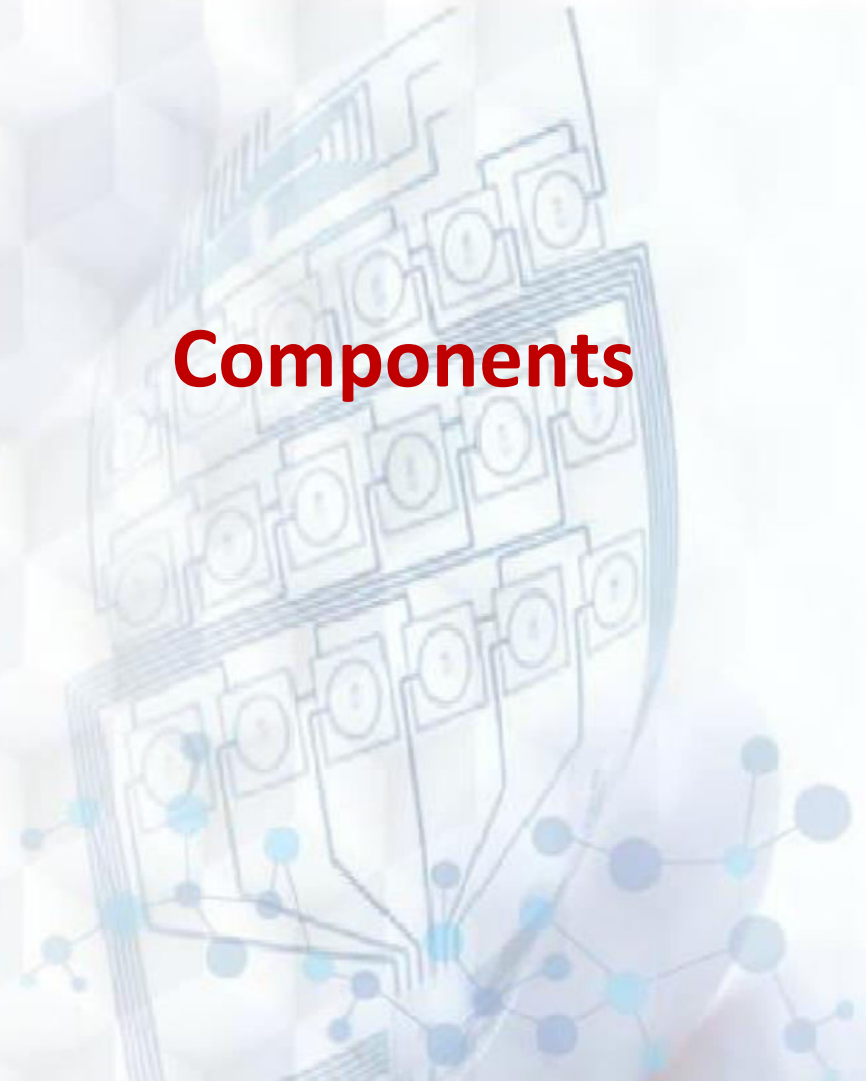
Example



END

Biosensors

Components

A stylized illustration of a biosensor chip. The chip is depicted as a rectangular grid of various components, including circular elements and rectangular blocks, connected by lines. Below the chip, there are several molecular structures represented by blue spheres and connecting lines, suggesting the interaction between the chip and biological molecules. The background features a faint, repeating pattern of hexagons.

Components of Biosensors

Two types of components

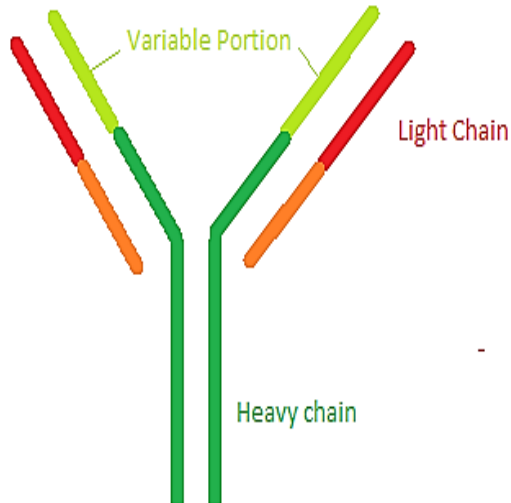
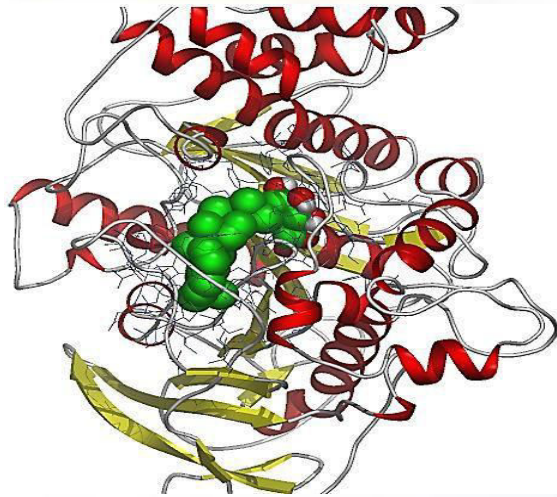
1. Biological component that is a Biological Recognition Element (BRE)
2. Physical components
 - i. Input
 - ii. Output devices.

Components of Biosensors

1. Biological Recognition Element (BRE)

- i. Tissue
- ii. Microbial cell
- iii. A derivative
- iv. Protein
- v. Nucleic acids

Components of Biosensors



Example of Proteins

1. Enzymes are biological catalysts produced by living cells specific binding sites specific substrate
2. Antibody (blood Proteins)
Produced by immune system counteracting an antigen specific binding sites for the antigen molecule

Components of Biosensors



**Plant
Tissues**

Microbial cell



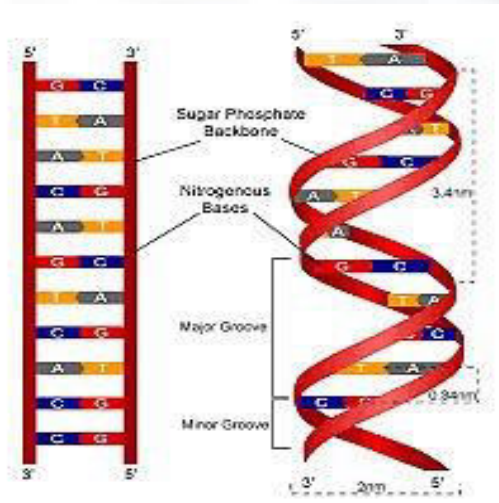
Microbial cell, Tissues and Virus as BRE

- i. Contains all the enzymes
- ii. No purification required
- iii. Stable environment
- iv. Contains co factors
- v. Cells can be regenerated

Nucleic Acids

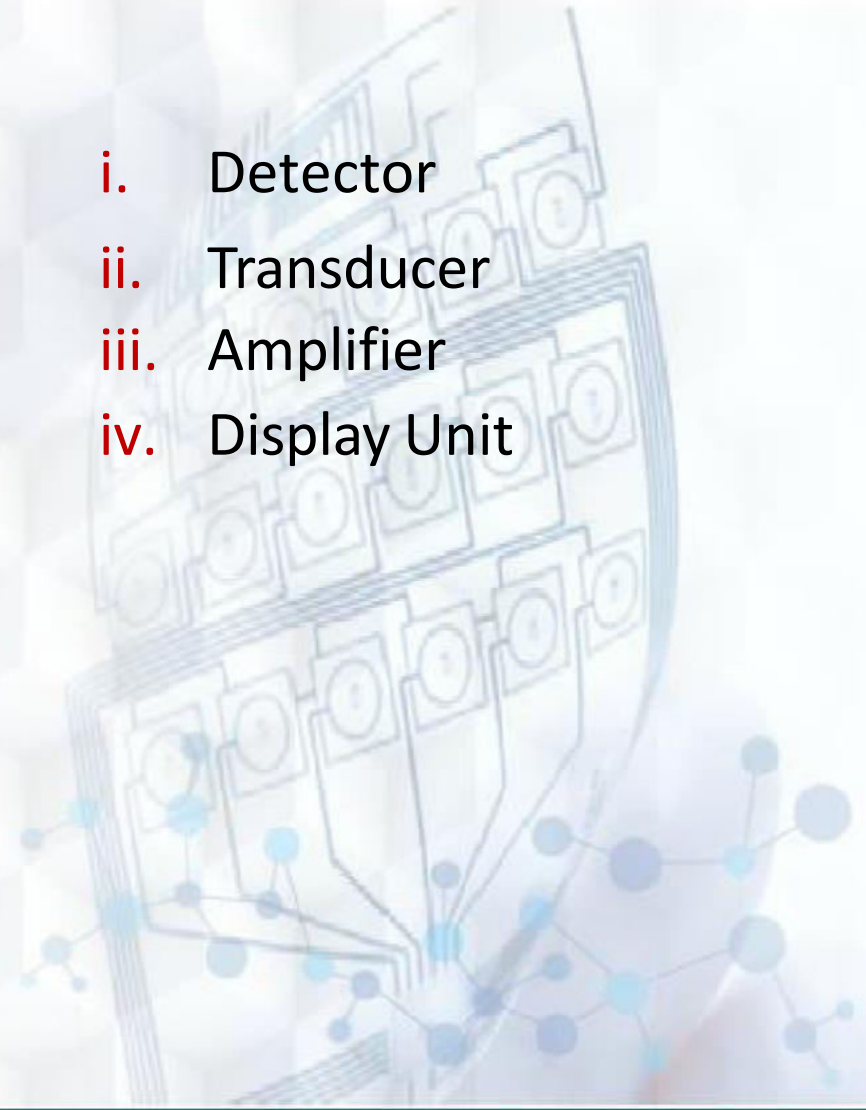
- i. DNA binds specifically with its complementary sequence.
- ii. **A** binds with **T**
- iii. **G** binds with **C**

DNA



2. Physical Components

- i. Detector
- ii. Transducer
- iii. Amplifier
- iv. Display Unit



Components of Biosensors

Physical Components

1. **Detector** detects the signal, produced by Biorecognition process.

1. Signal could be a

i. Chemical

ii. Electrical

iii. Magnetic others

2. **Transducer**

Converts signals of various physical forms to electronic signals, and vice versa.

Components of Biosensors

3. Display Unit

The display unit consists of a combination of

- i. Hardware
- ii. Software

Function is to generate the results in a user friendly manner.

The display unit exhibit output signals.

END

Biosensors

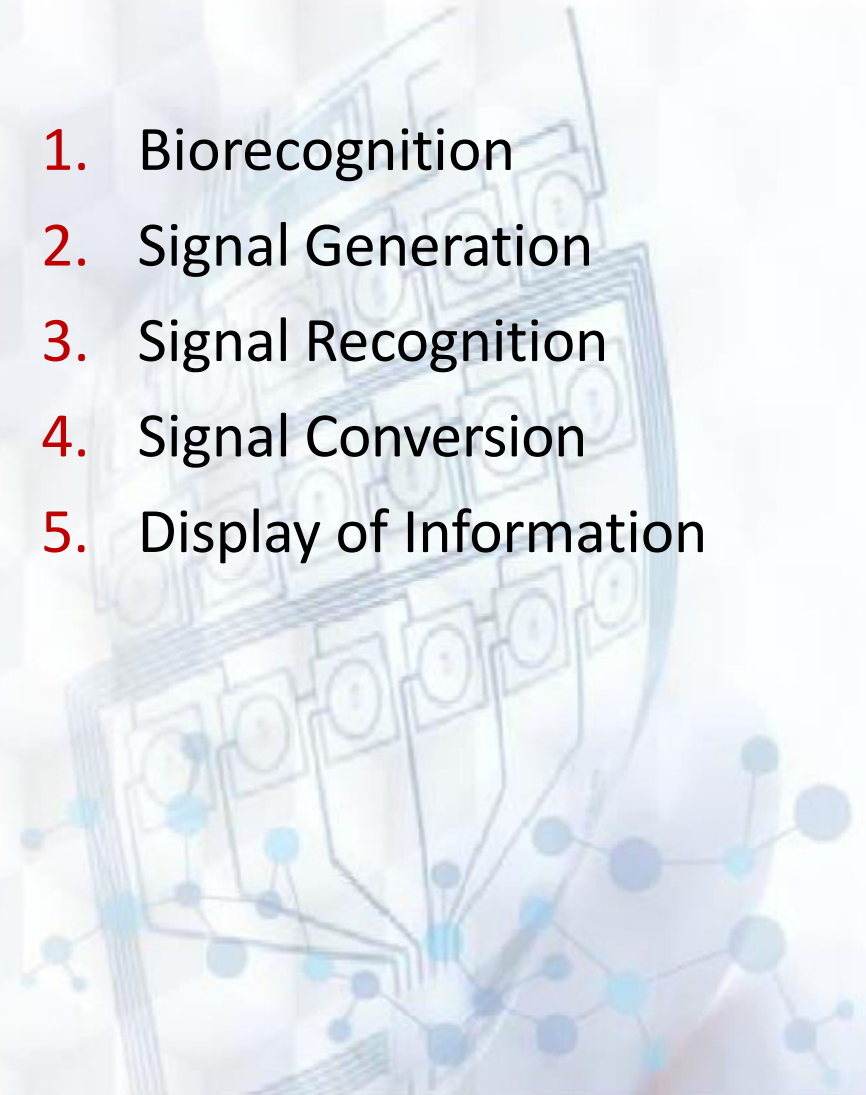
Working of Biosensor



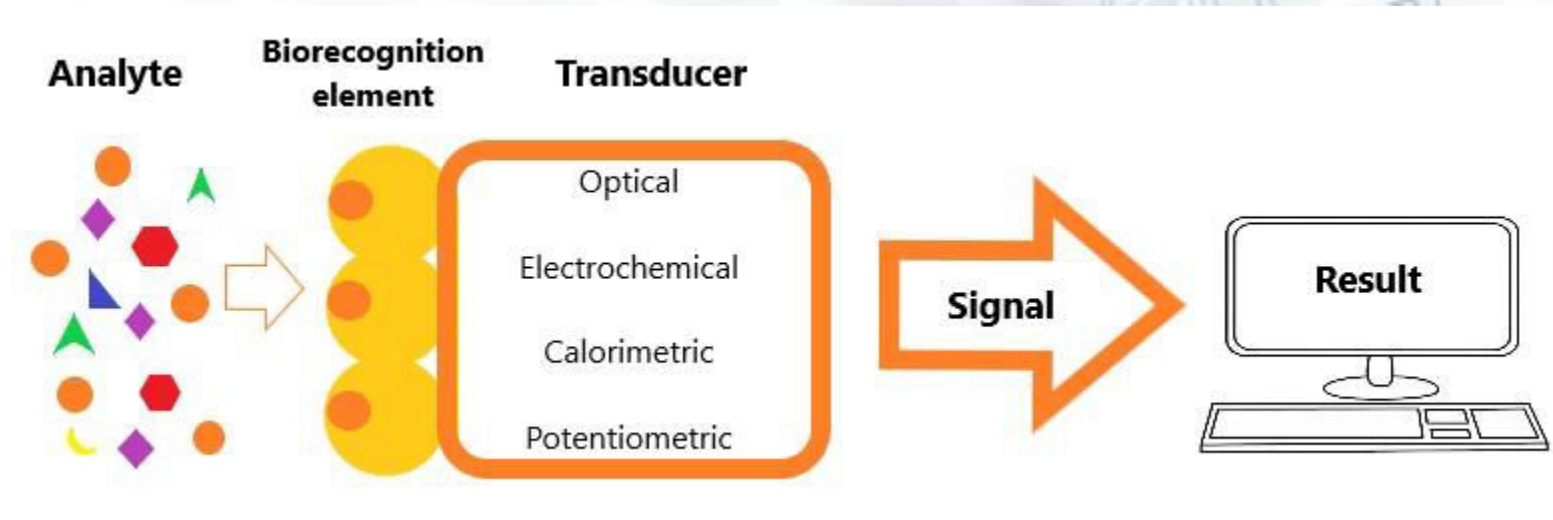
Working of Biosensors

There are five main steps

1. Biorecognition
2. Signal Generation
3. Signal Recognition
4. Signal Conversion
5. Display of Information



Working of Biosensors



Working of Biosensors

Example

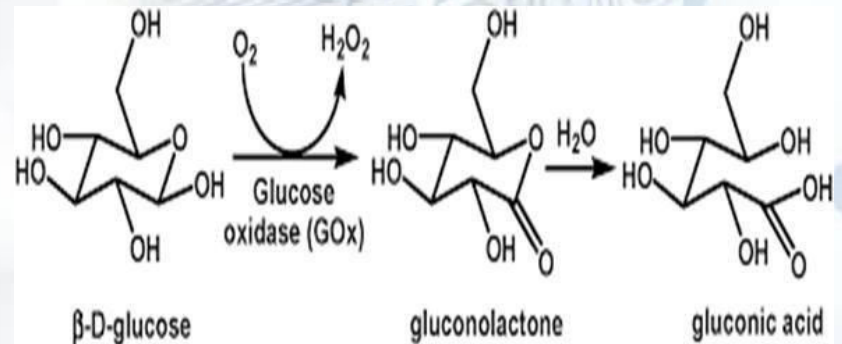
Glucometer

1. Blood sample.
2. Analyte is a glucose molecule.
3. B.R. E is an Enzyme.
4. Biorecognition signal is the Generation of electrons
5. Detector is an Electrode
6. Physical units composed of software and hardware with a powerful processor.

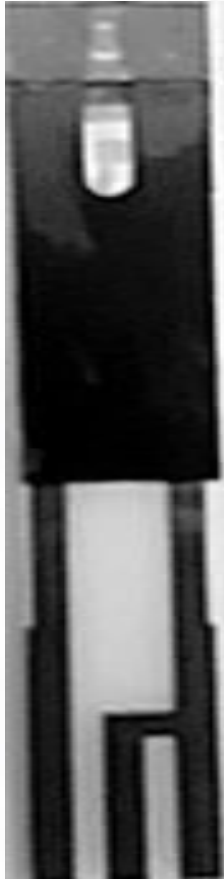
Working of Biosensors

Glucose oxidase Enzyme

1. Glucose oxidase (Enzyme Commission No. 1.1.3.4)
2. It is obtained from the fungus, *Aspergillus niger*.
3. Molecular weight of about 160kD. It is a dimer
4. Enzyme binds with glucose.



Working of Biosensors



1. Wet Platform

- i. The disposable Test strip is a wet platform.
- ii. Blood travels through Microfluidic channel.
- iii. Glucose oxidase is immobilized on the test strip.
- iv. Electrodes are fixed on strip and in contact with the enzyme.

Working of Biosensors



2.Dry Platform

The Dry platform consists of

- i. A software
- ii. A hardware.

It has a powerful processing unit with a

- Voltage converter
- Amplifier
- Filter
- Display unit

Working of Biosensors

Working of Glucometer

1. When blood is placed on the test strip, a chemical reaction takes place on working electrode.
2. An electrical current is generated.
3. Current is directly proportional to the concentration of glucose.
4. Information is displayed in mg/dl mmol/l.

END

Biosensors

Characteristics



Characteristics

Parameters

1. Sensitivity
 2. Range
 3. Linearity
 4. Selectivity
 5. Reproducibility
 6. Response time
 7. Stability
- 
- The background of the slide features a faint, artistic illustration. On the right side, there is a detailed drawing of a circuit board with various components like resistors and capacitors. Overlaid on and around the circuit board is a network of blue dots connected by thin lines, resembling a molecular structure or a data network. The overall color scheme is light blue and white, with a dark green header.

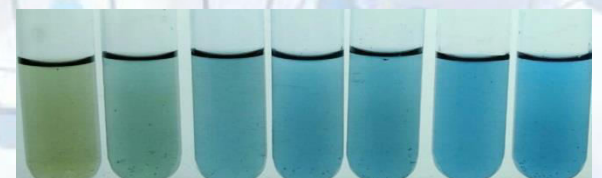
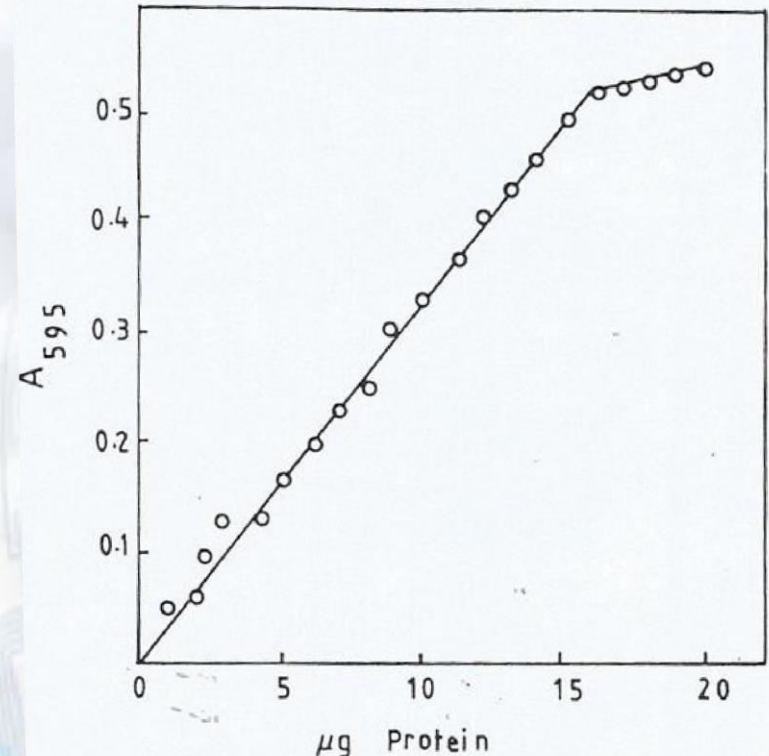
Characteristics

1. Sensitivity

The minimum input of an Analyte that will create a detectable output change.

2. Linearity

The linearity of an analytical method is its ability to deduce test results for Analyte in a sample, within a given range.



Characteristics

3. Range

Range of the sensor is the maximum and minimum values of applied parameter that can be measured.

Example

A Glucometer can measure 20-400 mg/dl. Normal physiological range of blood glucose level is 70-140 mg/dl.

Characteristics

4. Selectivity

The ability of the sensor to respond only to the target molecule.

Example

In a glucometer only Glucose molecule is detected and measured.

Selectivity of the biosensor is mainly dependent on the selection of Biological recognition element.

Characteristics

5. Reproducibility

It is the ability of a biosensor to obtain identical response for duplicate experimental setup.

Reproducible signals provide high reliability.

6. Response times

The times a Biosensor takes to produce a result.

A glucometer takes 3 to 60 seconds to display the results for different models.

Characteristics

7. Stability

Factors effecting stability

- i. Fouling is accumulation of living or non living materials.
- ii. Temperature
- iii. Humidity
- iv. pH
- v. Altitude, others .

END

Biosensors

**Economics and
Impact**

The background of the slide features a light gray hexagonal grid pattern. Overlaid on this is a faint, stylized illustration of a circuit board with various components and lines. In the bottom right corner, there is a molecular structure diagram with blue and gray spheres connected by lines.

Economics and Impact



Biosensors Market

- Biosensor market is mainly a health care market. As 90% of market share belongs to health care.
- The aim of healthcare market is to benefit the consumers that is public, current or future patients.
- The objective is to improve the population's health by encouraging the value of a person's health.

Economics and Impact

Biosensors Market:

1. Half of all biosensors produced worldwide are glucose monitors.
2. First commercial biosensor was introduced in 1987, was not accepted initially by patient and physicians.
3. The global self monitoring blood glucose system market is currently experiencing a boost.

Economy and Impact

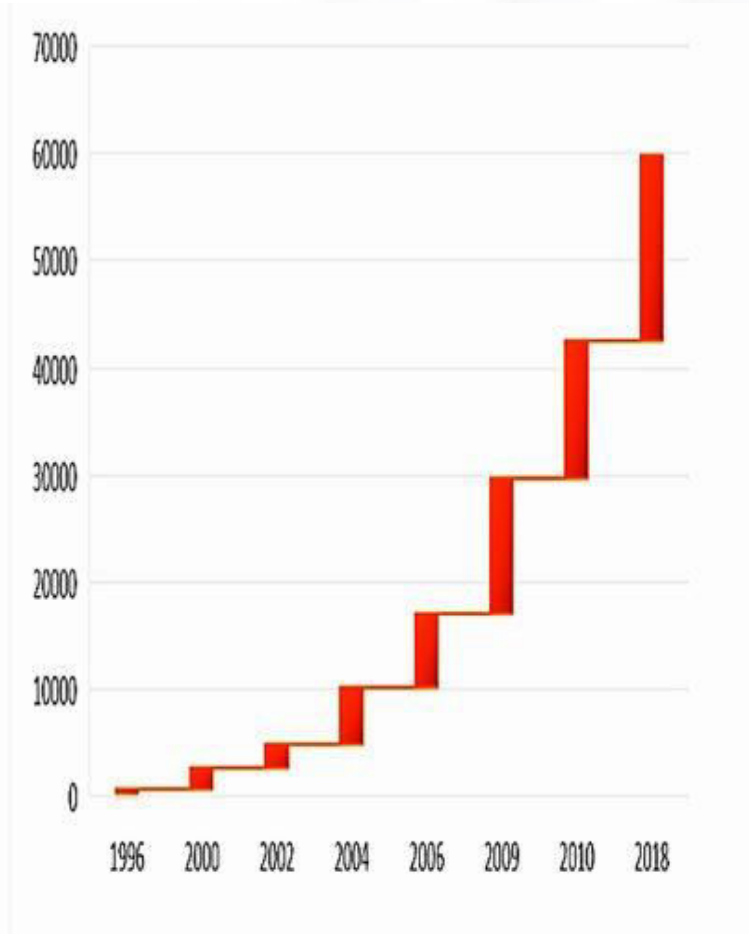
Biosensor Market

Is categorized as a growth market. The major growth areas are

1. Point-of-care (POC) testing
2. Diagnostics

They make real time remote health monitoring.

For example a diabetic patient has to test the blood three times.



Economics and Impact

POC testing

1. Point-of-care testing is undertaken close to the patient.
2. The specimen is obtained, the test performed, and the result reviewed where care is provided.
3. It involves testing by non-laboratory personnel using specialized technologies adapted for near-patient beside applications.

Economics and Impact

Technology Push

1. deals with the induction of technology without realizing the demand of users.
2. These products are developed with the desire to create a market demand.
3. Generally less successful and profitable until the product develops a need for its own distinct market

Economics and Impact

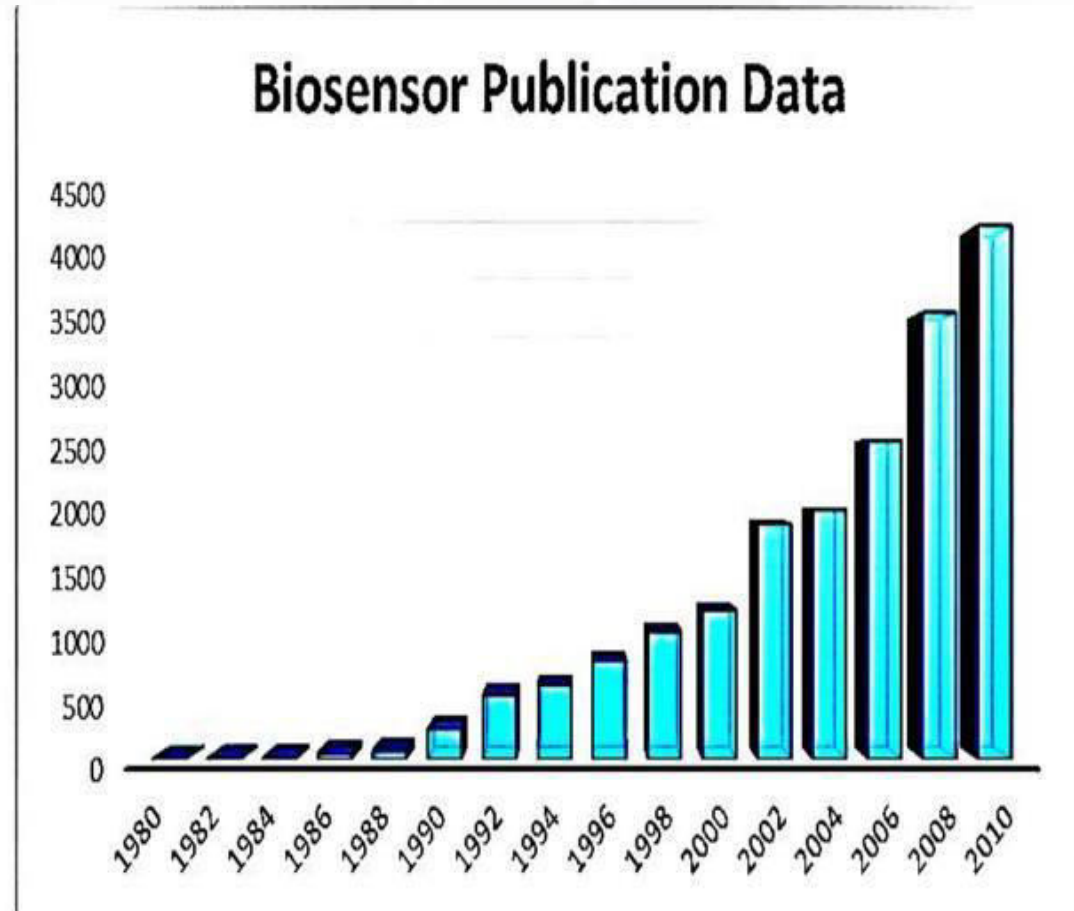
Market Pull

- i. Glucose sensors
- ii. pathogen detection
- iii. EKG Sensor measures cardiac electrical potential waveforms (voltages produced during the contraction of the heart).

This is currently the largest and most profitable area for the development and commercialization of biosensors

Economics and Impact

**Impact on
Academic
Research**



Biosensors

Applications



Applications

Applications of Biosensors

Biosensors has application in all fields of life

1. Health
 2. Drug designing
 3. Environment
 4. Process technology
 5. Food industry
 6. Agriculture
 7. Security
- And others

Applications

Demand of Biosensors in medical sciences

Because they Provides

1. Fast
2. Accurate
3. and Lab-quality results
4. handheld device
5. to Accelerate decision making process in the medical field.

Applications

Applications of Biosensors in Environment

The increasing number of potentially harmful pollutants in the environment demands the use of

1. Fast
2. Cost-effective
3. Analytical techniques

Environmental protection

- I. Heavy metals/Phenols
- II. Nitrogen compounds

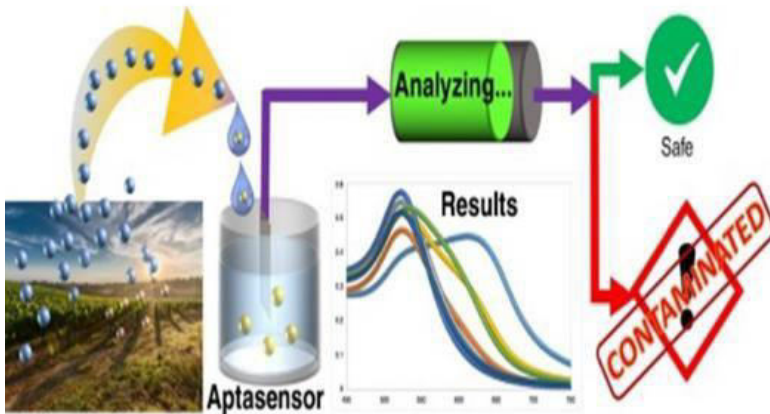
Applications

Advantages of Environmental biosensors

Environmental biosensors

Provide benefits over conventional analytical techniques.

1. Portability
2. Work on-site,
3. Measuring pollutants in complex matrices
4. with minimal sample
5. No pretreatment



Applications

Agriculture

Biosensors are widely used for

1. Testing the pesticides
2. Crop deterioration
3. Gases detection
4. Ripening of fruits
5. pH of Soil

The small size device is hooked up to a smartphone app that can allow to scan a piece of fruit or a vegetable to determine the pesticide levels.



Applications

Food Industry

There is an increasing demand for biosensors in the food industry.

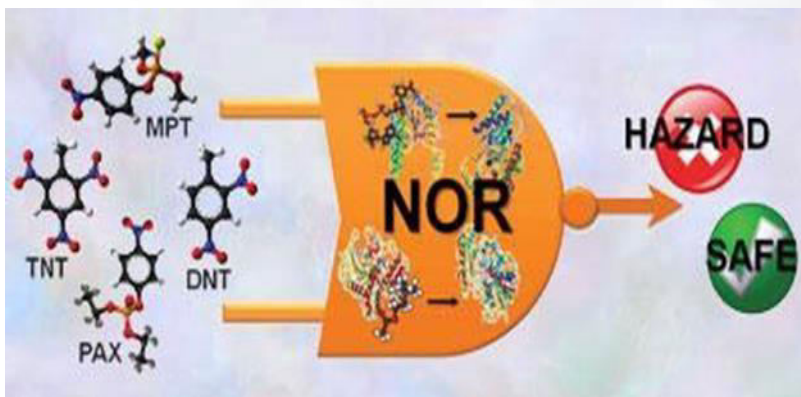
1. To examine food quality.
2. Detection of pathogens sensor that can accurately and quickly determine if food is contaminated or not., such as *Salmonella* and *Listeria*



Applications

Enzyme based biosensor for security surveillance

1. A system that rapidly detects both explosives and nerve agents, is developed, providing a simple yes-no response.
2. The technique could replace two time-consuming tests that are currently used to assess these threats.



The explosives and nerve gases are fed through an enzyme-based logic gate system, in which the depletion of hydrogen peroxide detects their presence.

Receptors

Receptors

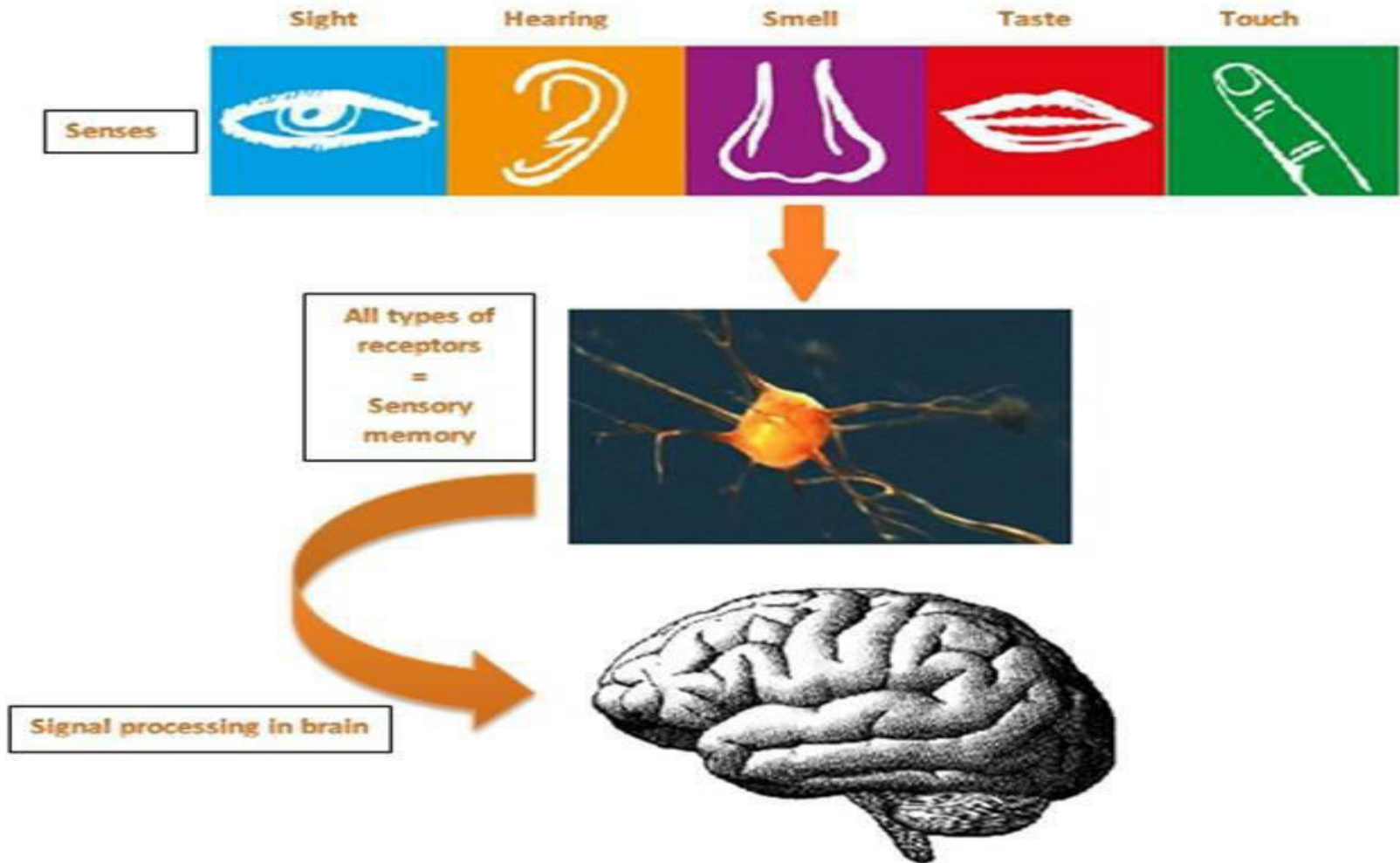
An abstract graphic on the right side of the slide. It features a stylized, tilted circuit board or microchip with various components and lines. Overlaid on and around the board are molecular structures, represented by blue and grey spheres connected by lines, suggesting a connection between technology and biology or chemistry.

Receptors in a Living System

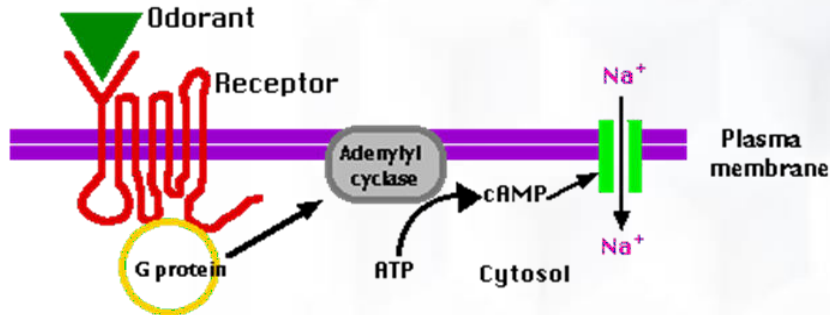
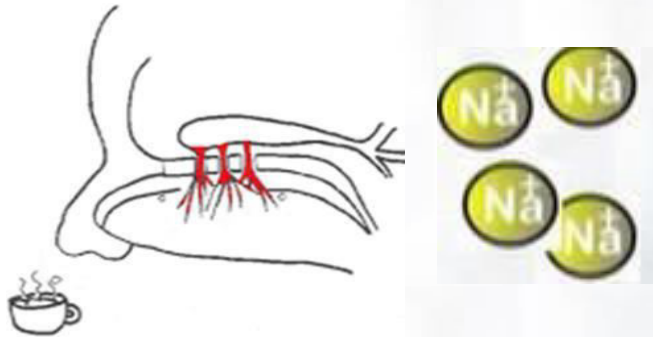
Receptors for General Senses

1. There are five general senses.
2. Receptors are located on sense organs.
3. Receptors are specific to stimuli.
4. A sensory receptor is a specialized cell
5. Ionic movement is involved in generating a CNS response .

Receptors in a Living System



Receptors in a Living System

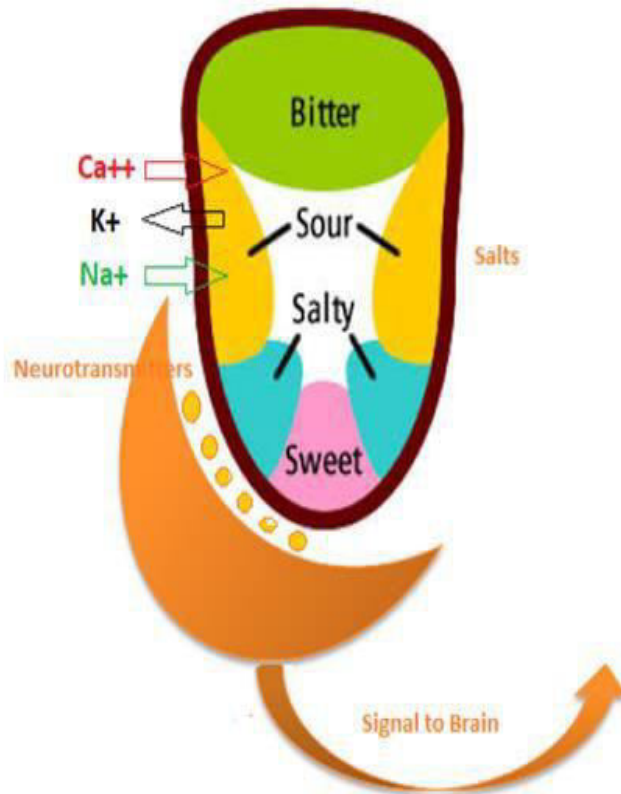


Olfaction

Sense of smell **p**rovided by paired olfactory organs on either side of the nasal septum

- 10 – 20 million olfactory receptors are packed in nasal area.
- A complex of Odorant receptor leads to activation of enzymatic mechanism involving ATP, for opening of sodium calcium gate channels leading to Na⁺ influx

Receptors in a Living System

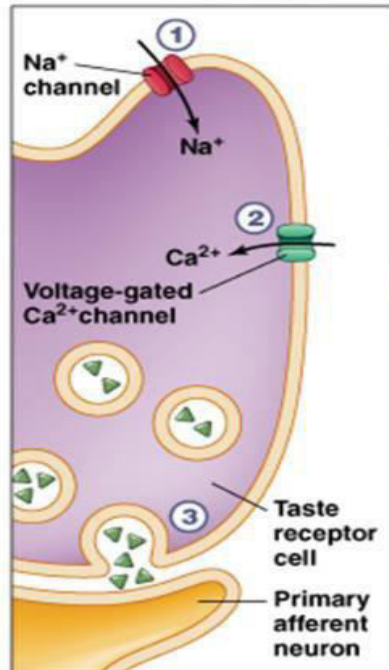
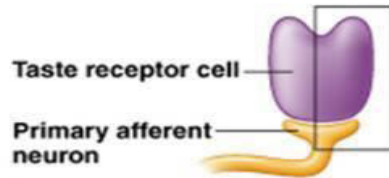


Gustation (Sense of taste)

Provides information about the taste foods and liquids

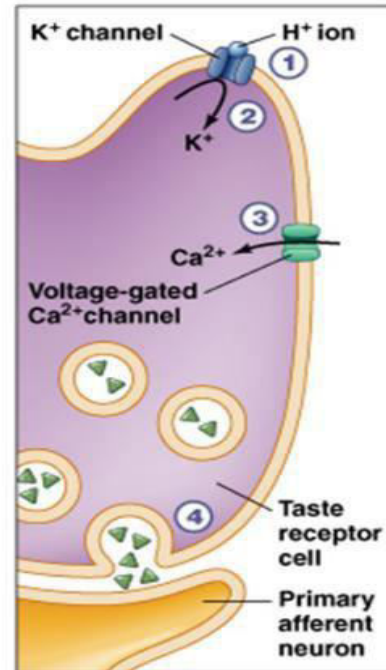
1. Gustatory receptors are about 3000 are called taste buds distributed over tongue.
2. Include superior surface, pharynx and larynx.
3. Movement of Ca^{++} , K^{+} and Na^{+} involved in the initiation of action potential.

Receptors in a Living System



(a) Salty

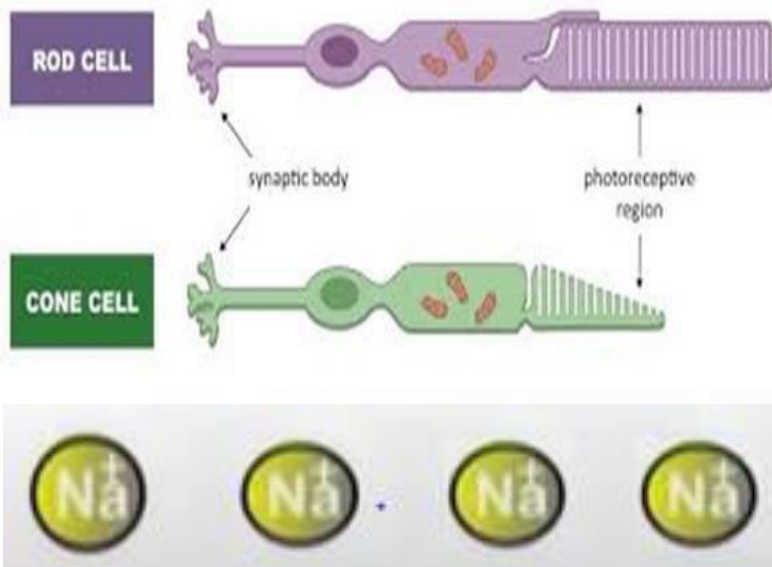
- 1 Na⁺ from salty food enters through a Na⁺ channel.
- 2 The resulting depolarization opens voltage-gated Ca²⁺ channels.
- 3 The influx of Ca²⁺ causes neurotransmitter release.



(b) Sour

- 1 H⁺ ions from sour foods block the K⁺ channel.
- 2 This blockage prevents K⁺ from leaving the cell.
- 3 The resulting depolarization opens voltage-gated Ca²⁺ channels.
- 4 The influx of Ca²⁺ causes neurotransmitter release.

Receptors in a Living System



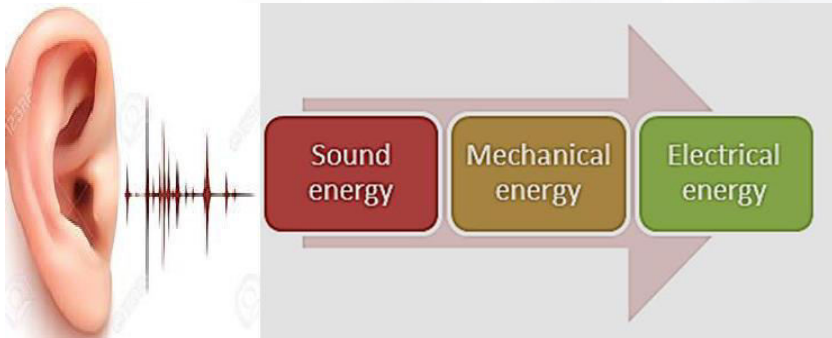
Vision

1. The sense through which a person or animal becomes aware of light Color, etc using the eyes.
2. Eye Receptors (Rods and Cones) transfer light energy into nerve impulses.
3. They transmit the information to the cortex of the brain's occipital lobe
4. Flow of Na ions play an important role in vision.

Receptors in Living System

Hearing

1. The hearing receptors are present in ear and are topped by hair bundles
2. The hair bundle moves in response (back and forth) to the sound waves.
3. Hair cells transform the mechanical forces of sound into the electrical impulses of hearing.
4. Ear acts like a transducer.

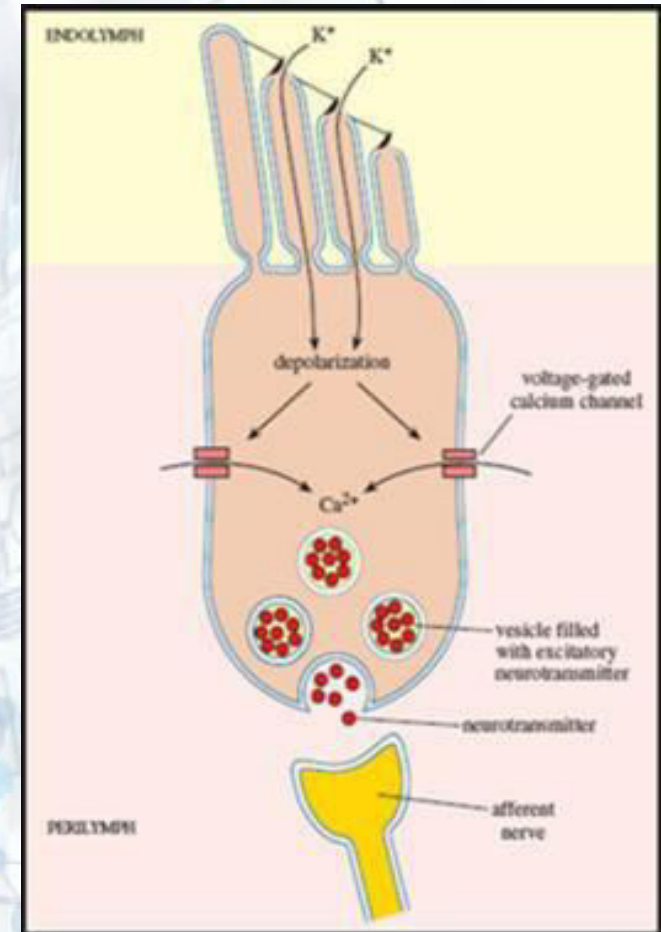


Receptors in a Living system

Hearing Mechanism.

The back and forth mechanical movements of the hair alternately opens and closes ion channels for K^+ and Ca^{++} ions in the hair cell.

It leads to the alternate depolarization hyperpolarization leading to potential changes, at the same frequency as the original sound.



Receptor in a living System

General Sense Receptors

1. three types of cell receptors
 - i. Exteroceptors:
 - A. Receive external stimuli
 - B. Provide information
 - C. About external environment.

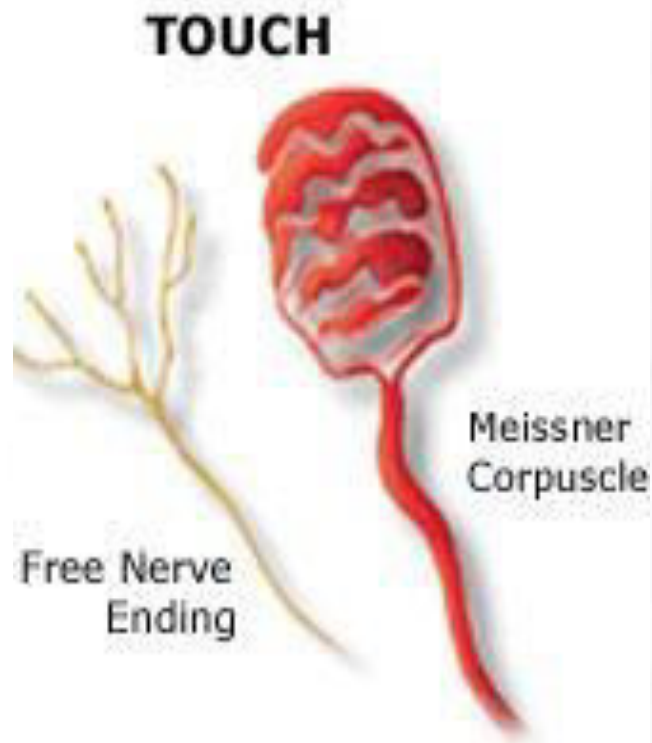
Receptor in a living System

Thermoreceptors (temperature sensors)

1. A type of exteroceptors in which free nerve endings located in:
 - i. Dermis of the skin
 - ii. Skeletal muscles
 - iii. Liver
 - iv. Hypothalamus

Cold receptors are high in number than warm receptors

Receptor in a living System



Mechanoreceptors

Sensitive to stimuli that distort the cell membrane

There are 3 classes of mechanoreceptors:

1. Tactile Receptors: provide the sensation of

i. touch

ii. pressure

iii. vibration

Receptor in a living System

Mechanoreceptors

1. Baroreceptors
2. detect pressure changes on walls of blood vessels in portions of the
 - i. Digestive
 - ii. Reproductive
 - iii. Urinary tracts

Receptor in a living System

Proprioceptors

1. a sensory receptor which receives stimuli from within the body,
2. to position and
3. movement.
4. provide info about skeletal muscles & joints

Receptors in a Living System

Chemoreceptors

Can detect

- i. small changes in the conc. of specific chemicals or compounds
- ii. commonly used as O_2 & CO_2 detectors in blood vessel

Receptors in a Living System

Five Special Senses

1. Olfaction (smell)
2. Gustation (taste)
3. Vision (sights)
4. Equilibrium (balance)
5. Hearing

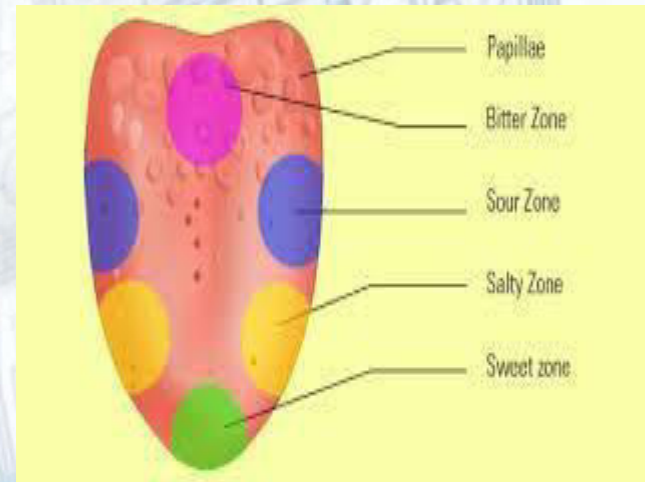
Receptors for these senses are located in specialized areas called

- i. Sense organs.

Receptors

Primary Tastes Sensations

1. Sweet
2. Salt
3. Sour
4. Bitter



Receptors in Living Cell

other Two taste sensations

1. Umami

Pleasant taste
characteristics of beef
& chicken broth

2. Water

Research shows the
presence of water
receptors at the back
of

i. tongue

ii. pharynx

Receptor in a Living System



Two types of photoreceptors

Rods

1. Very light sensitive
2. Enable us to see in DIM LIGHT

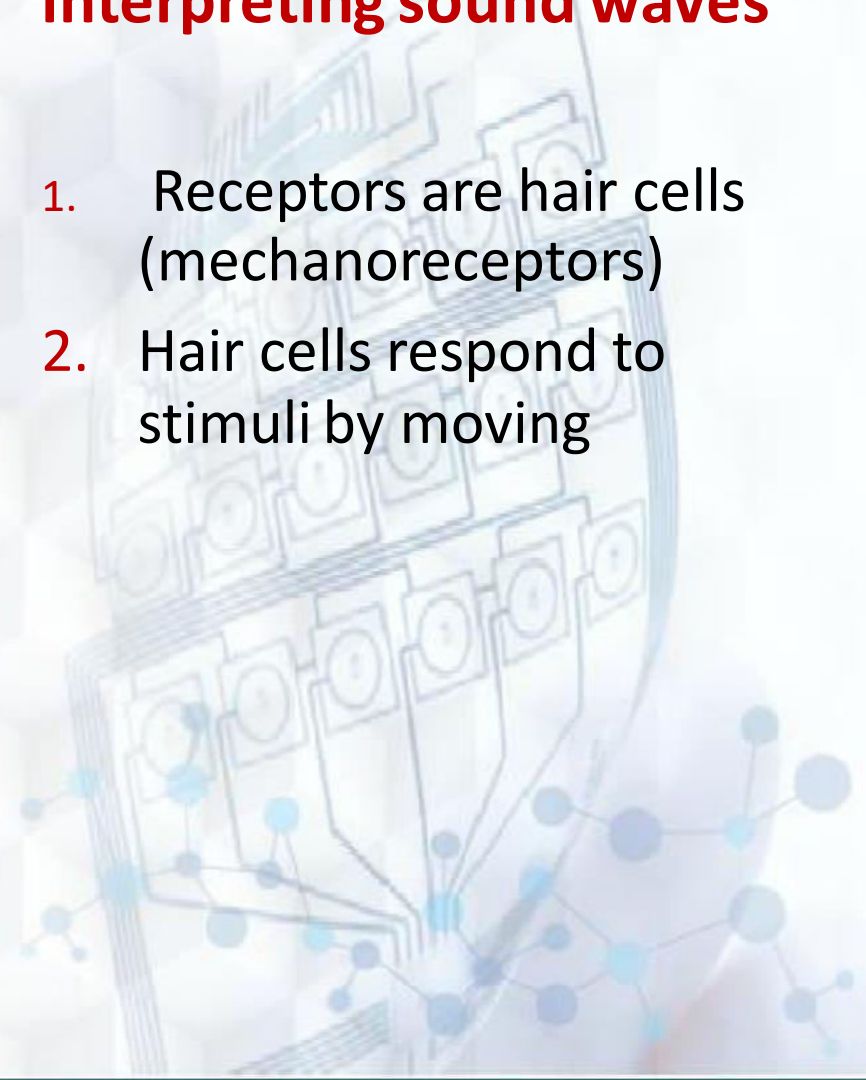
Cones

1. Provide us with color vision
2. their pattern of stimulation provide color vision .

Receptor in a living cell

Hearing is Detecting & interpreting sound waves

1. Receptors are hair cells (mechanoreceptors)
2. Hair cells respond to stimuli by moving



Receptor in a living System

Hearing

HEARING

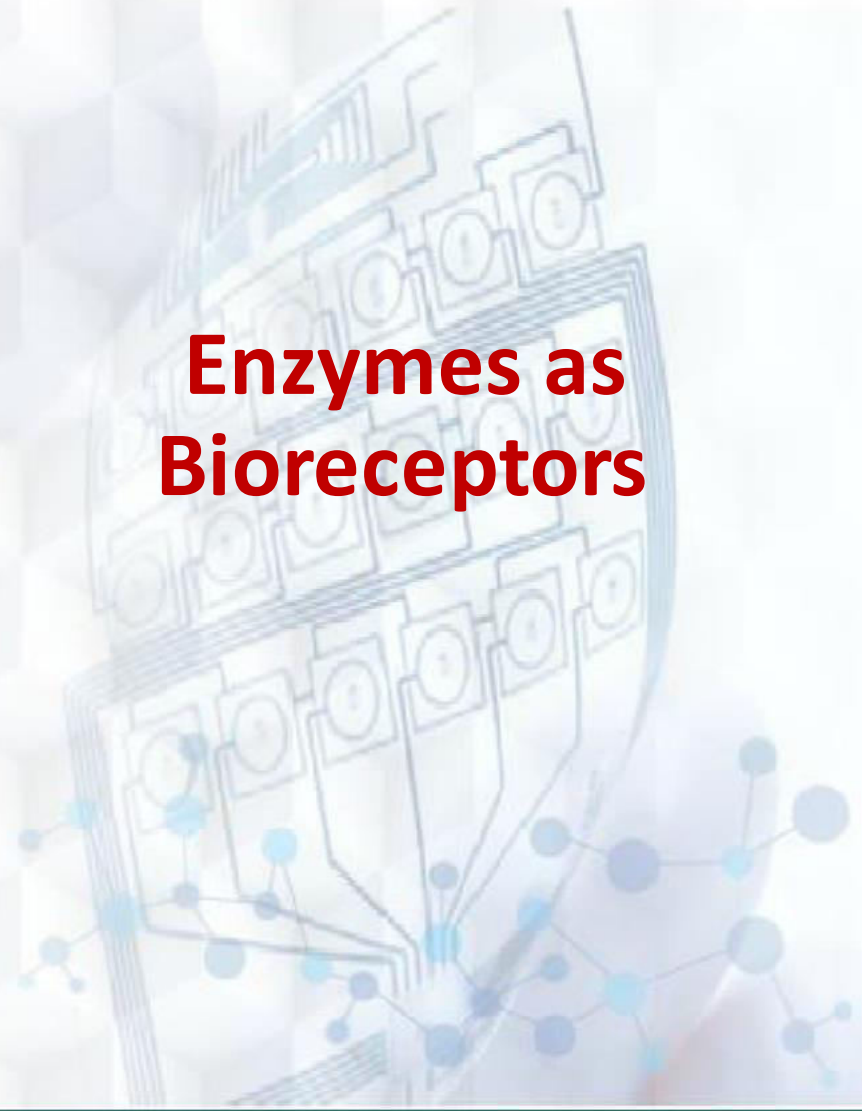


the ear's receptor neurons are topped by hair bundles that move in response to the vibrations of sound.

Source: Illustration, Eade Creative Services Inc. / George Eade illustrator (rod and cone cells adapted from Scientific American Vol. 256, No. 2, page 42, 1987). Edited by Dr. Hugo Heyrman

Biosensors

Enzymes as Bioreceptors



Enzymes a Bioreceptors

Enzymes as Bioreceptor

Enzymes can be divided into three different types, depending on the detection of molecules, as signal.

1. End product.
2. Side product
3. Detection of electron
4. Inhibition of enzyme

Enzymes as Bioreceptors

Enzymes

1. Enzymes can act on a variety of substrate (target molecule/Analyte) and can generate signal for their detection.
2. A diverse form of sensors are available to detect that specific signal.
3. Various Biosensors are designed using different classes of Enzymes.

Enzymes as Bioreceptors

Enzyme immobilization on different interfaces and possible effects on the enzyme conformation.

1. High charge density or a hydrophobic

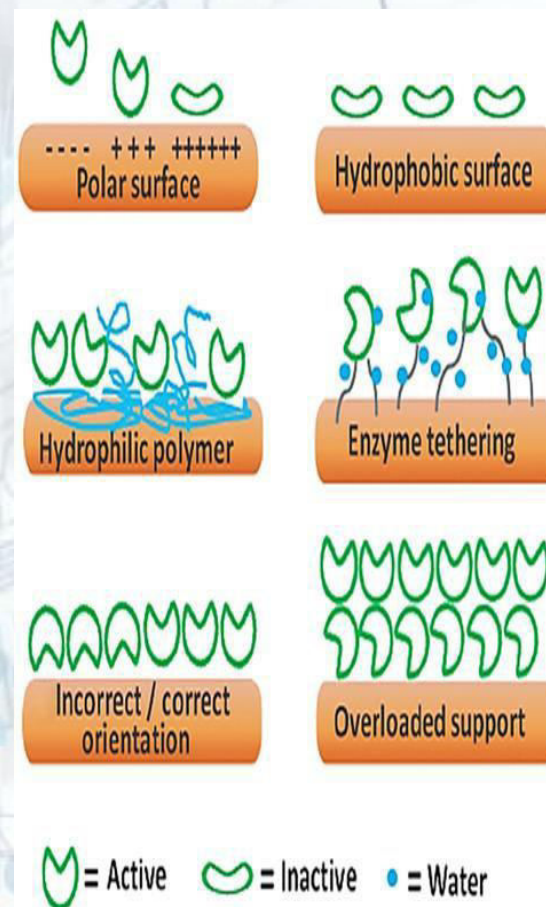
surface are possible causes of enzyme conformational changes and inactivation.

2. Enzyme co-immobilized with hydrophilic

polymers or tethered can reduce unfolding and inactivating support–enzyme interactions.

3. Incorrect orientation and multilayer

formation may cause reduction of specific activity.



Enzymes as Bioreceptors

Detection of Phenols by the Enzyme

1. Used in Biosensors for the detection of Phenols in
2. Food
3. Medicine
4. Environment

A variety of sensors are used in these Biosensors including

1. Optical
2. Thermal
3. Electrochemical and others

Enzymes as Bioreceptors

Detection of Pesticides by Enzymes

There are several enzymes used for the detection of Pesticides by Biosensor including.

1. Peroxidase
2. Alkaline Phosphatase
3. Acetylcholine esterase

Using a variety of sensors

- i. Optical
- ii. Potentiometric and others

Enzymes as Bioreceptors

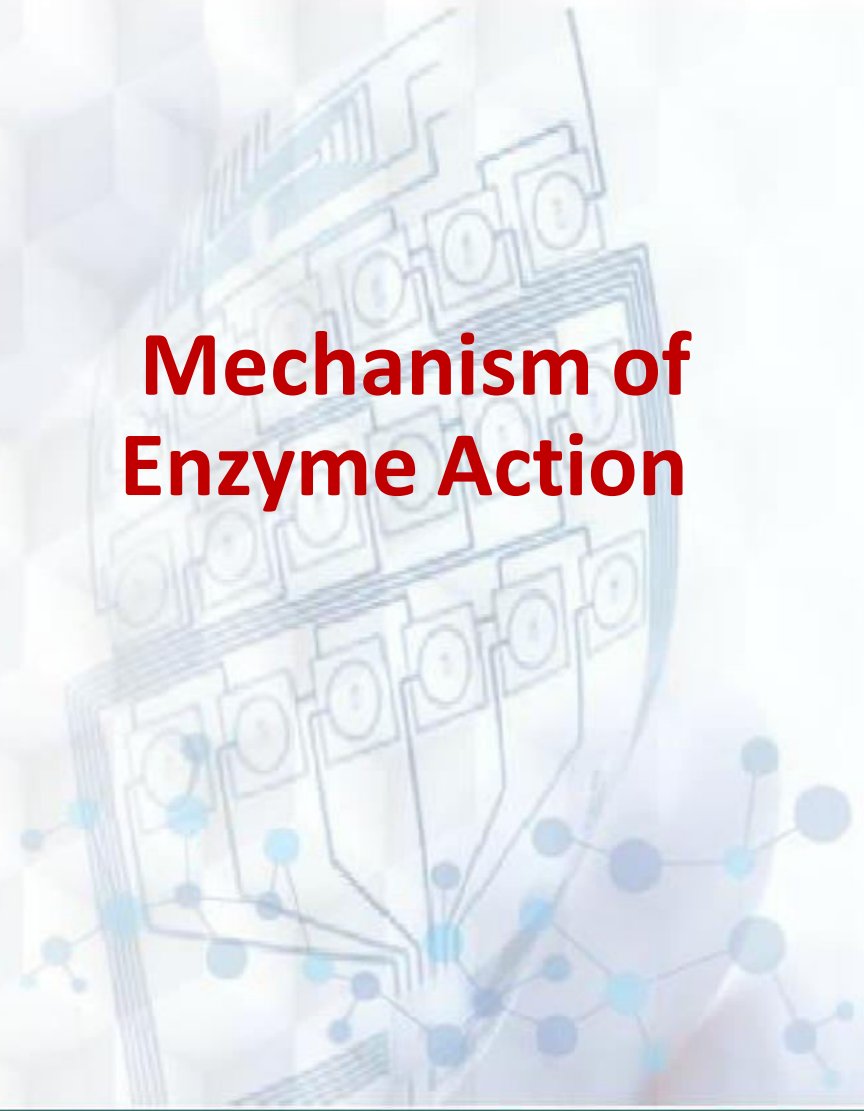
Enzymes Used in Glucometer

There are several enzymes that use glucose as a substrate. There are two types of enzymes that are used in Glucometer for the detection of glucose.

1. Glucose Oxidase
2. Glucose Dehydrogenase

Biosensors

Mechanism of Enzyme Action



Mechanism of Enzyme Action

Enzymes are Catalysts

1. Enzymes increase the rate of biochemical reactions.
2. Enzymes have catalytic and substrate binding sites.
3. The functional groups of the amino acid side chains that participate in catalysis
4. Enzymes activity is inhibited by the inhibitors

Mechanism of Enzyme Action

Enzymes mechanism and Biosensors

1. Reaction depends on the physical and chemical structures of the enzyme and the substrate.
2. Different classes of Enzymes display specific mechanism of action.
3. Designing of biosensor is dependent on the mechanism of enzyme action.

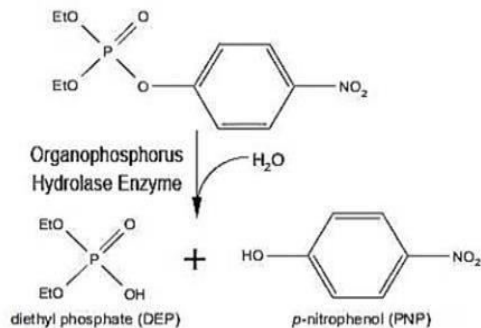
Enzymes

Removal or addition of Water Molecule.

Cleavage or formation of bonds with the addition or removal of H₂O

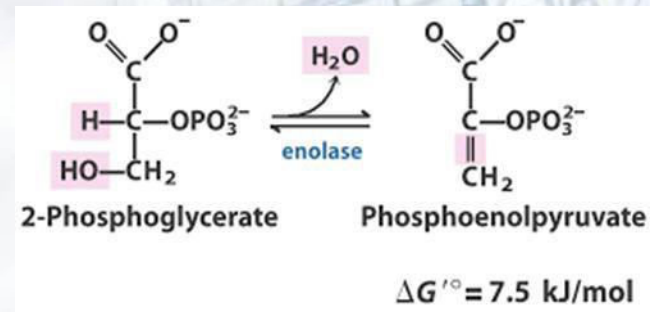
1. Hydrolases

Organophosphorus Hydrolase



2. Hydrases

Enolase



Biosensors for hydrolase enzyme acetylcholine esterase are based on the inhibition of enzyme for the detection of organophosphorus chemical pesticides.

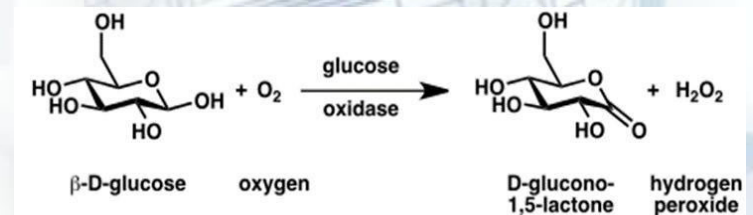
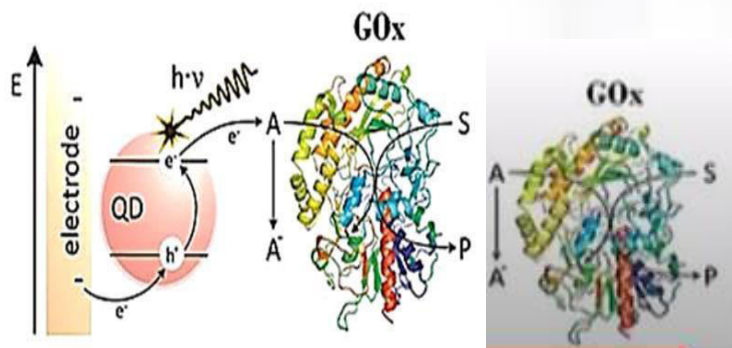
Introduction

Mechanism involves transfer of Electrons

Enzyme catalyze oxidation-reduction reactions where electrons are transferred. Examples of their sub classes are

1. Oxidases 2. Dehydrogenase 3. Oxygenase 4. Peroxidase

Application in Biosensor:, Release of electrons by glucose oxidase is detected by the Biosensor that is a step in mechanism of action



Chapter: Solid-state colorimetric biosensors
Sep 2018 Advances in Biosensors; Reviews
Book Series, Vol. 2

Introduction

Transfer of Radicals

Transferase enzymes catalyze the transfer of groups from one reactant to another.

Addition or removal of Carbon Bonds.

Lyases catalyze the reactions where functional groups are added to break double bonds in molecules or where double bonds are formed by the removal of functional groups.

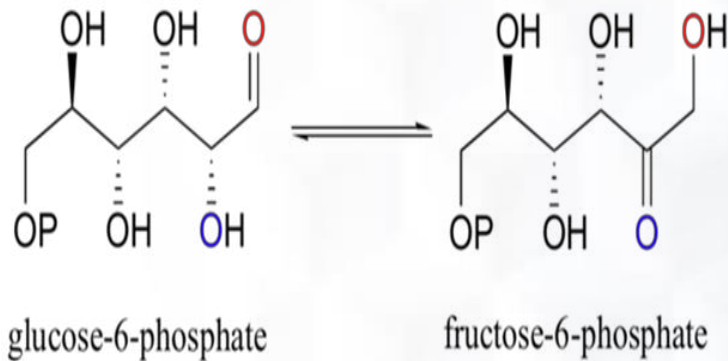
Introduction

3. Change of Geometry within a molecule

Isomerase enzymes catalyze reactions that transfer functional groups within a molecule so that isomeric forms are produced.

Example.

1. Recemase
2. Mutase



Enzymes

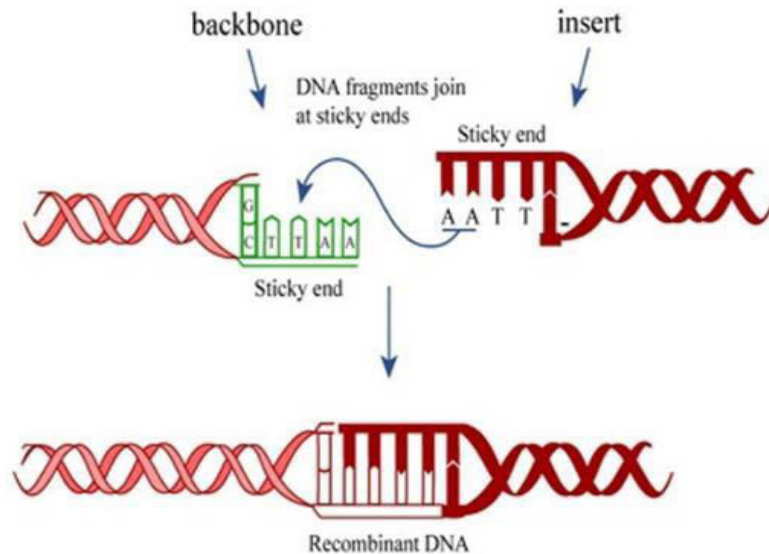
Joining two macro molecules

Ligase enzyme join two molecules through hydrolysis of pyrophosphate in ATP or other Triphosphates.

Example

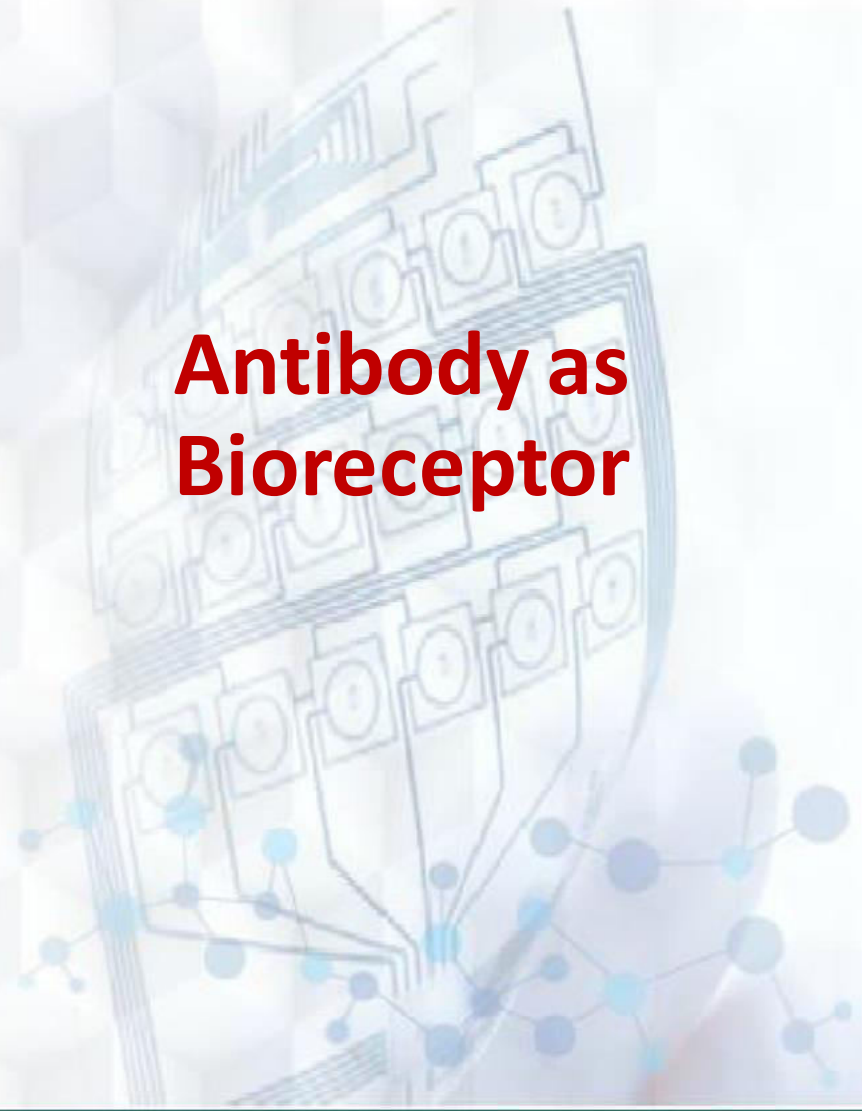
T4 DNA Ligase is a single polypeptide with a M.W of 6800 Dalton requiring ATP as energy source.

It has the unique ability to join sticky and blunt ended fragments.



Biosensors

**Antibody as
Bioreceptor**



Antibody as Bioreceptor

Antibody

1. Antibodies are large Y-shaped
2. proteins
3. produced by plasma cells
4. utilized by the immune system
5. to identify and target pathogens/ foreign body
such as
 - i. Bacteria
 - ii. Virus

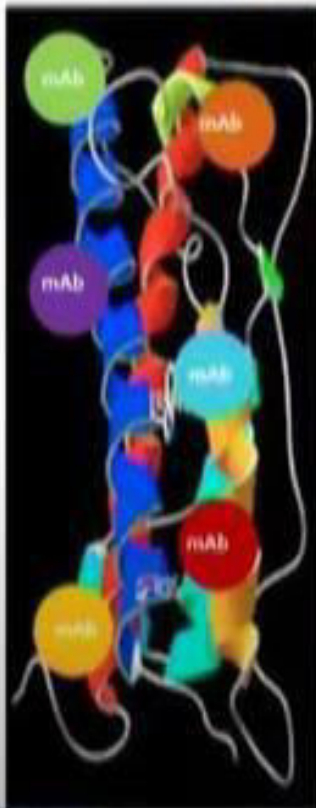
Antibody as Bioreceptor

Immuno sensors

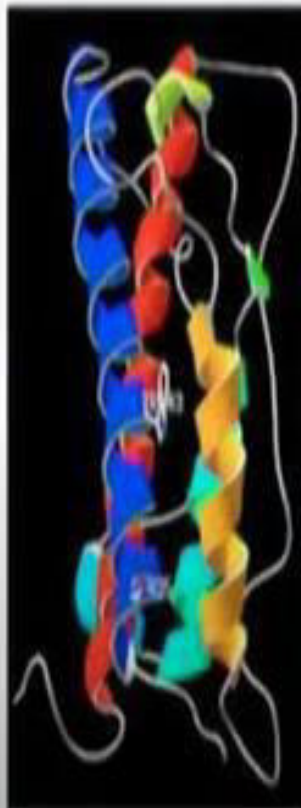
1. Antibodies are produced by plasma cells. Their function is to identify and target pathogens/ foreign body.
2. Antibodies are valuable and robust tools for the fabrication of immunosensors.
3. Antibody when binds with it specific antigen, a signal is generated, that is detected by the transducer.

Antibody as Bioreceptor

Binding of different
Monoclonal antibodies
on Growth hormone



Growth hormone
molecule



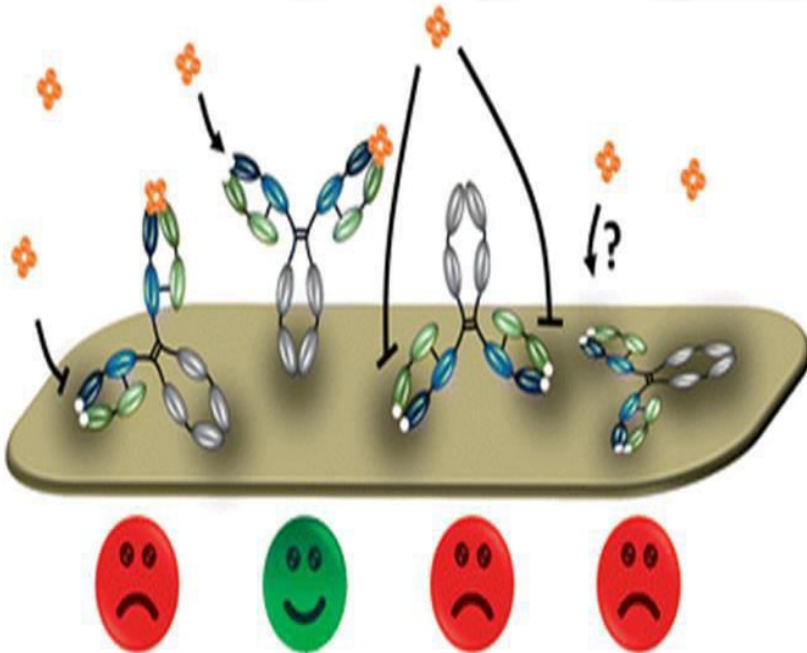
Antibody as Bioreceptor

1. There are two types of antibodies. Monoclonal and Polyclonal.
2. Monoclonal antibody are produced by the progeny of a single B cell against single epitope.
3. An epitope is five to six amino acid long region.
4. Polyclonal antibody are generated against multiple epitope.

Antibody as Bioreceptor

Immobilization of Antibody

1. Orientation of antibodies on the surface is important for sensitivity.
2. Covalent cross linking or oxidation of carbohydrate groups can be used to control the interaction between specific reactive groups on the surface and on the antibody.

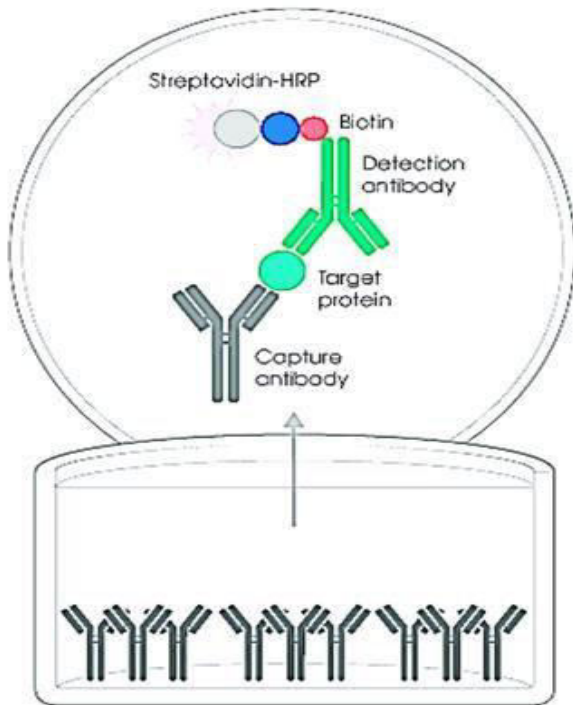


Antibody as Bioreceptor

Immuno biosensor with ELISA Assay

1. ELISA (enzyme-linked immunosorbent assay) is a plate-based assay technique.
2. It is designed to detect and quantify substances like peptides, proteins, antibodies and hormones.
3. It is also known as enzyme immunoassay (EIA).

Antibody as Bioreceptor



ELISA immunosensors

1. In assay, the antigen is immobilized on to a solid surface.
2. An antibody (grey) is immobilized on multi well plate, to detect the antigen of interest.
3. The antigen (turquoise) recognizes the detection antibody (green) that is conjugated to biotin and streptavidin-HRP, for signals.

Antibody as Bioreceptor

Immobilization of Antibody to sensor surface is required

1. without altering their specificity
2. Immunological activity

It is important in the designing and fabrication of a successful immunosensor.

Antibody as Bioreceptor

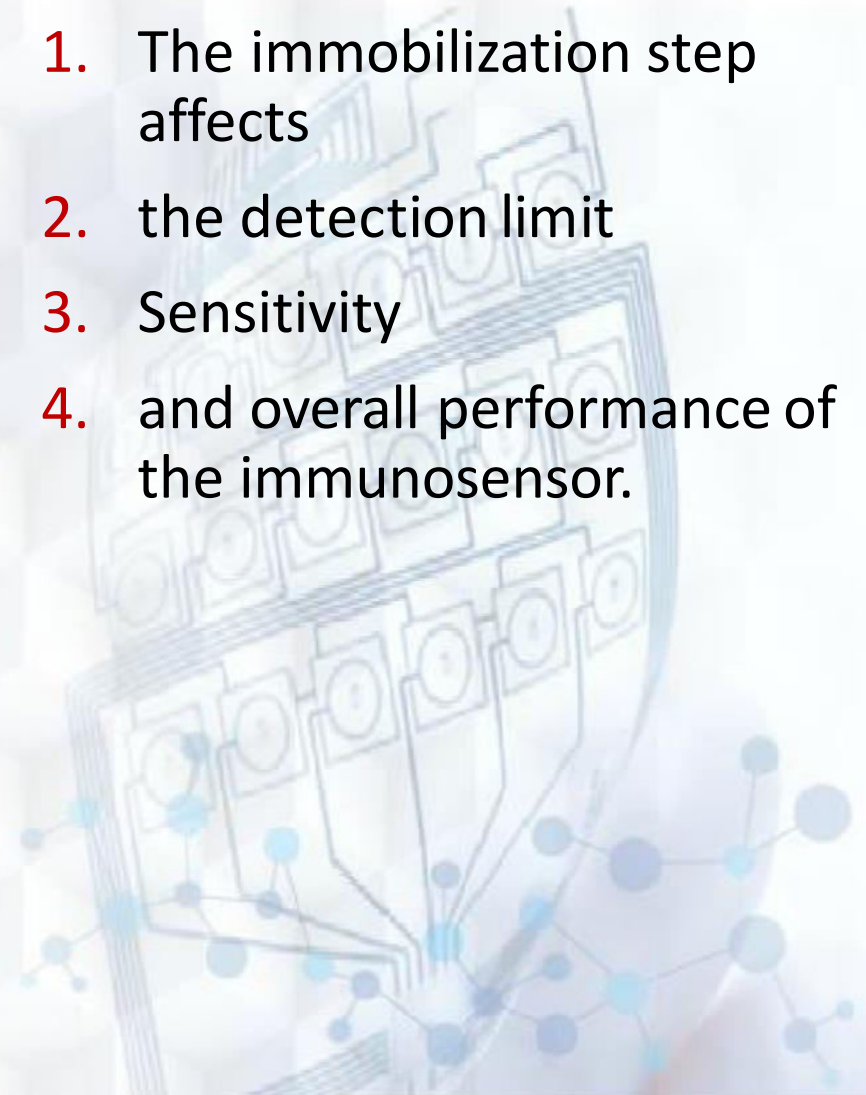
Methods Immobilization of Antibody to sensor

1. Covalent cross linking using NH_2 and COOH groups
2. Carbohydrate group modification by Oxidation
3. Fab fragments of antibodies are obtained by enzymatic digestion

Antibody as Bioreceptor

Immobilization of Antibody

1. The immobilization step affects
2. the detection limit
3. Sensitivity
4. and overall performance of the immunosensor.



Antibody as Bioreceptor

Immobilization of Antibody

1. Orientation of antibodies on the surface is important for sensitivity
2. It can be controlled by the interaction between specific reactive groups on the surface and on the antibody.

Antibody as Bioreceptor

Recombinant DNA Technology For Antibody Engineering

1. A library generate a library by PCR using spleen DNA.
2. The resultant PCR products for one heavy chain and one light chain are cloned separately and expressed.
3. It is a protein with antibody like activity.

Antibody as Bioreceptor

Engineering of Antibody

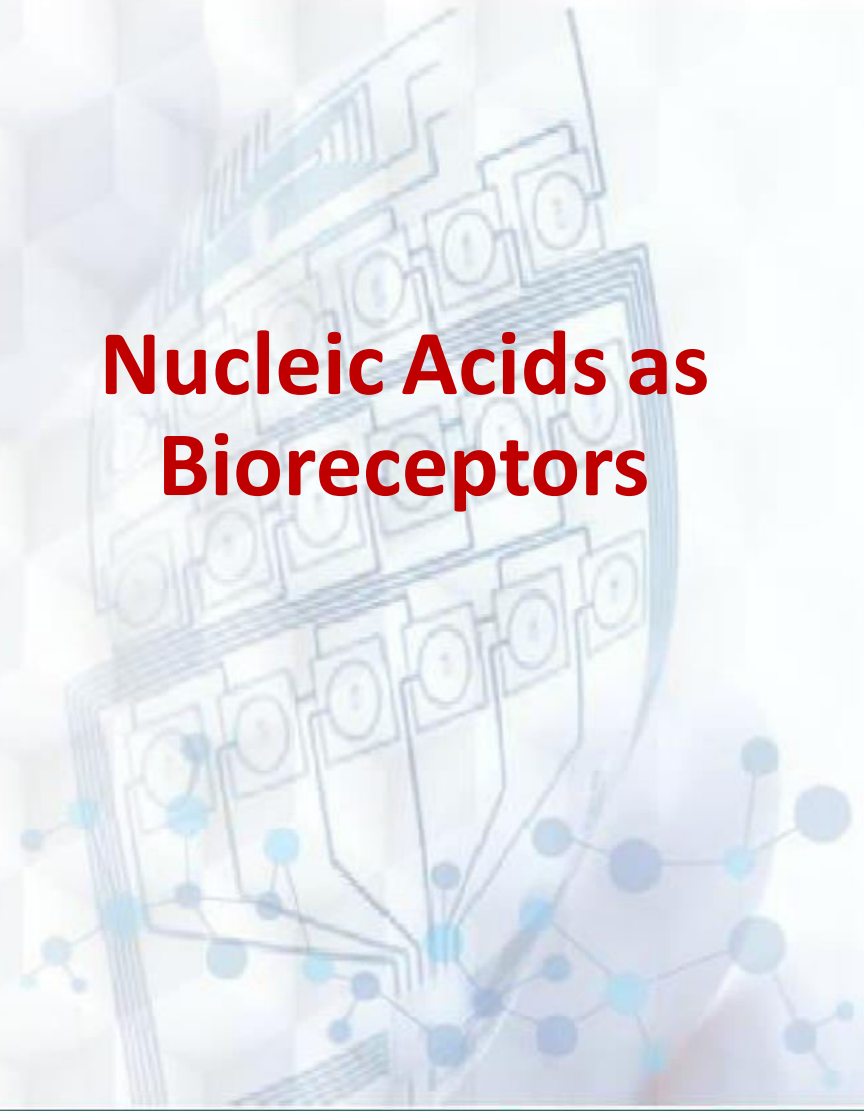
The resultant PCR products

1. one heavy chain
2. one light chain
3. Are cloned separately
4. And Expressed

A surface fusion protein with antibody like properties.

Biosensors

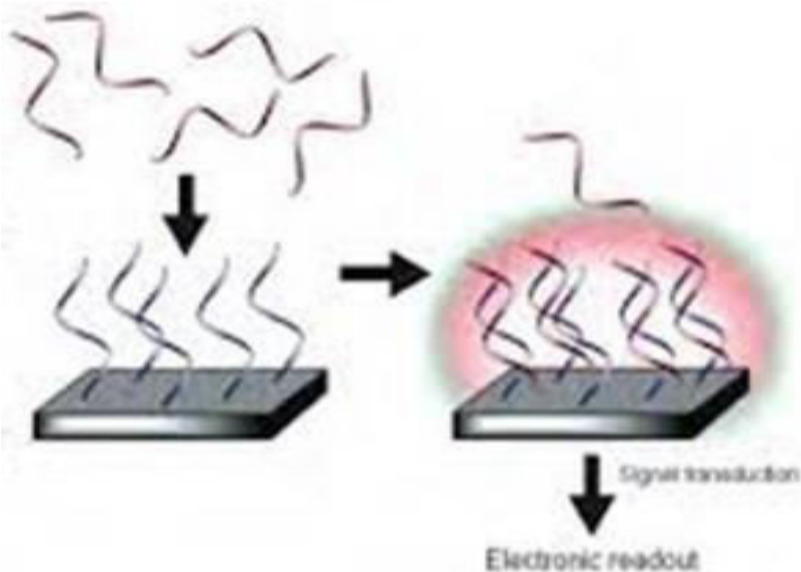
Nucleic Acids as Bioreceptors



Nucleic Acids as Bioreceptors

DNA Biosensor

1. DNA or RNA are the Biorecognition elements in Nucleic acid Biosensors.
2. The detection methods are based on recognition of complementary nucleic acid sequence
 - a) Rapid
 - b) Simple and
 - c) Economical testing for clinical, environmental, food analysis and other fields



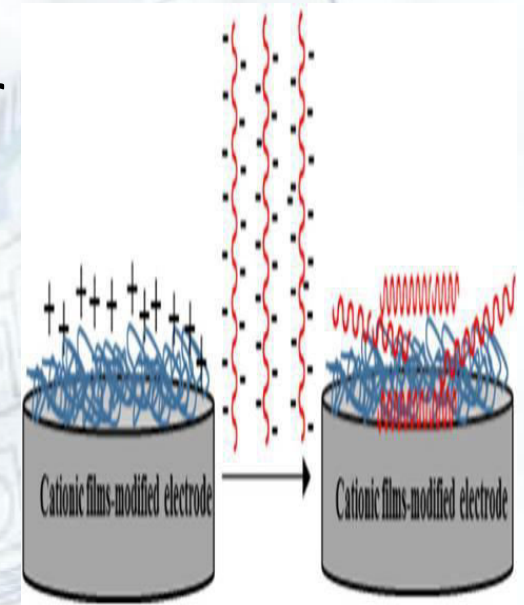
Nucleic Acids as Bioreceptors

Nucleic Acids immobilization

The DNA fragments have to be immobilized in a way that retains their

1. Stability
2. Reactivity
3. Accessibility

To target analyte for hybridization.



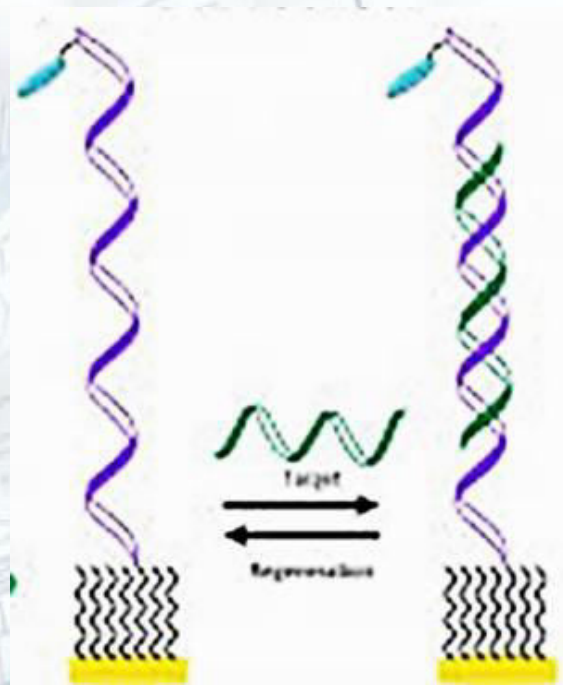
Adsorption

DNA probes are immobilized via electrostatic adsorption between a negatively charged phosphate group of DNAs on the positively charged surface

Nucleic Acids as Bioreceptors

Hybridization and Biorecognition

1. DNA hybridization is the process of joining two single strands of complementary DNA sequence and forming dsDNA.
2. Hybridization of complementary DNA base pairing is the basis for the biorecognition process.
3. An electrical signal is produced when target DNA binds to the complementary sequence.



Nucleic Acids as Bioreceptors

Sensing Methods

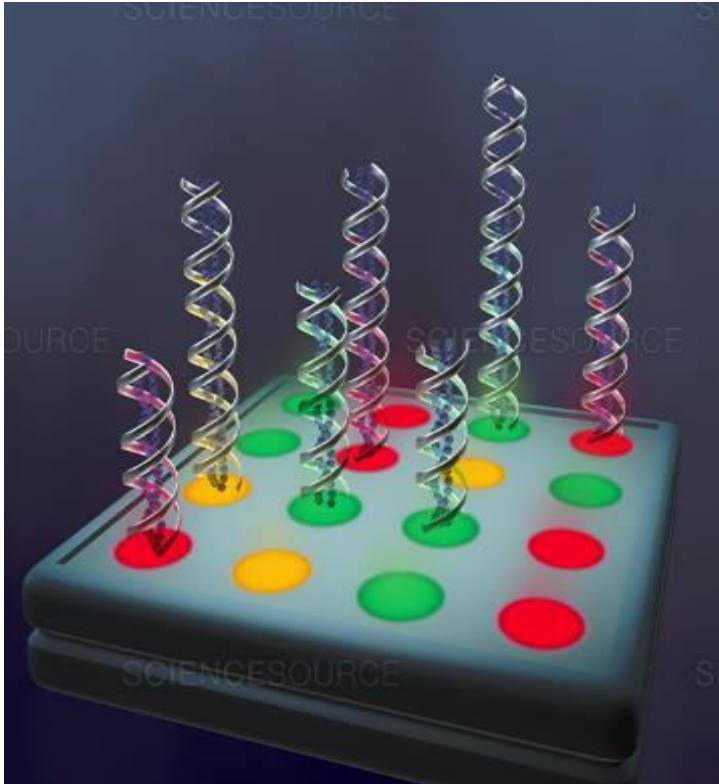
Label based method

specific organic dyes, metal complexes and enzymes are used for hybridization detection.

Examples

1. Methylene blue binds at minor groove of DNA.
2. Horseradish Peroxidase and alkaline phosphatase
3. Ferrocenyl naphthalene diimide (FND)

Nucleic Acids as Bioreceptors



DNA Microarray

1. A DNA microarray is a collection of microscopic DNA spots attached to a solid surface.
2. DNA microchips detect the presence of specific DNA sequences.
3. It can be used as a rapid indicator in clinical diagnostics, research and food safety testing.

Nucleic Acids as Bioreceptors

Nucleic Acids:

1. DNA probe sequence (surface bound or not)
Recognition of complementary nucleic acid sequence (Analyte or target).
2. are stretches of single-stranded **DNA**

Nucleic Acids as Bioreceptors

Nucleic Acids:

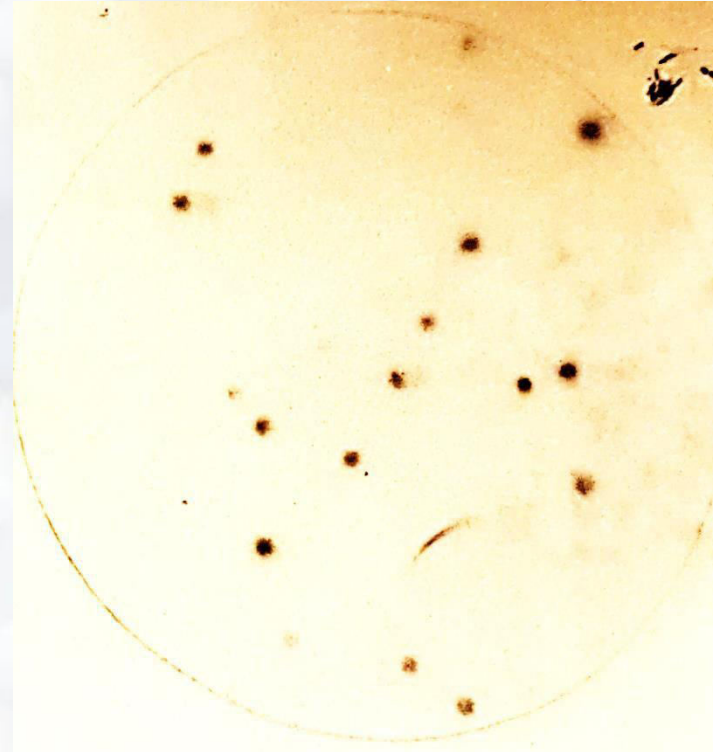
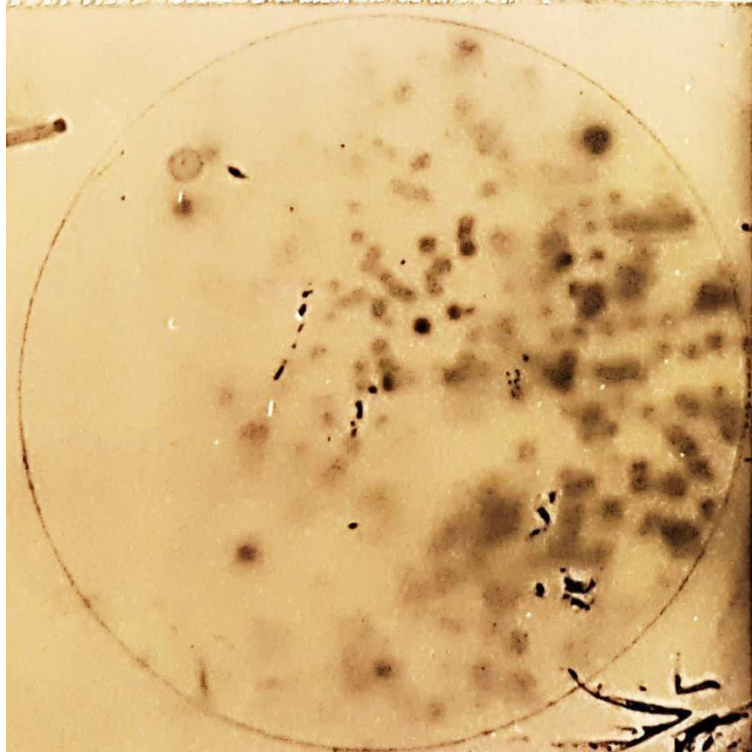
1. ssDNA (8-70 bases) for genetic analysis
2. used to detect the presence of complementary nucleic acid **sequences** (target **sequences**) by hybridization.
3. mRNA concentration indicates metabolic activity of a given gene (gene expression monitoring)

Nucleic Acids as Bioreceptors

Nucleic Acid Probe :

1. **DNA probes** are usually labelled, for example with
 2. Radioisotopes
 3. Epitopes
 4. Biotin or
 5. Fluorophores
- to enable their detection.

Nucleic Acids as Bioreceptors



Nucleic Acids as Bioreceptors

Steps involved in DNA biosensor working

1. Formation of the DNA recognition layer
2. Actual hybridization event
3. Transformation of the hybridization event into an electrical signal
4. Detection of Signal
5. Binding event converted into a signal (optical or electronic)

Nucleic Acids as Bioreceptors

Molecules: DNA & mRNA

1. DNA more stable than RNA, used more often
2. mRNA concentration indicates metabolic activity of a given gene (gene expression monitoring)
3. All use hybridization (A:T , G:C)
4. Melting" temperature is broad for small sequences, also depends on sequence, ionic strength.

Nucleic Acids as Bioreceptors

Immobilization Nucleic Acids in Biosensors:

Nucleic acids are immobilized on a support material, using different groups for covalent attachment

1. Carboxyl
2. Aldehyde
3. Epoxy
4. Isothiocyanate
5. Maleimide
6. Mercaptosilane

Nucleic Acids as Bioreceptors

Nucleic Acids in Biosensors:

1. ssDNA (8-70 bases) for genetic analysis for Biorecognition
2. Sensor is used measure change in
3. Mass
4. Optical
5. Electronic
6. Electrochemical property

Nucleic Acids as Bioreceptors

Applications

1. Gene expression
2. Regulatory pathways
3. Confirm mechanism of action
4. Validate drug targets
5. Diagnostic procedures
6. Genomic dna analysis
7. Whole or subset (SNP's)

Nucleic Acids as Bioreceptors

DNA biosensors

Optical Detection:

DNA-coated Au Particles

Quantum dots replace
fluorescent dyes

Types of DNA Sensors

1. Electrodes
2. Chips
3. Crystals

Biosensors

**Cells/tissues,
Virus as
Bioreceptors**

The background of the slide is a light gray with a subtle hexagonal grid pattern. On the right side, there is a stylized illustration of a circuit board with various components and lines. At the bottom right, there are molecular structures represented by blue spheres connected by lines, suggesting a biological or chemical context.

Cell as Bioreceptor

Cells as Bioreceptor

- Cell-based biosensors use cells as BRE
- apply the principles of cell-based assays by employing living cells for detection of different analytes from
 1. Environment,
 2. Food
 3. Clinical
 4. or other sources.

Cells as Bioreceptor

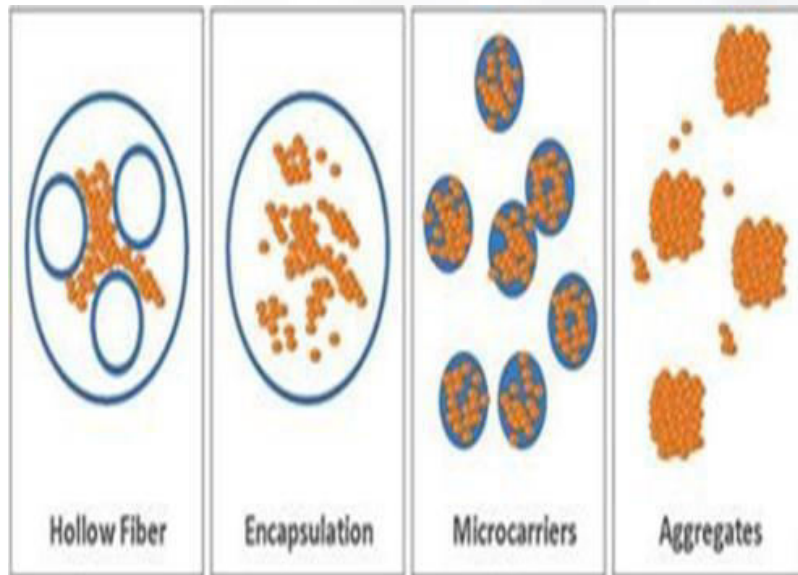
Cell-based biosensors use cells as Biological Receptor element

Prokaryotic (bacteria)

1. Eukaryotic
 - i. yeast
 - ii. invertebrate
 - iii. vertebrate.

Living cells are directly integrated onto the biosensor platform.

Cells as Bioreceptor



Immobilization of Cells

Entrapment is the ideal method to immobilize the cells/ tissues on support material. For example

1. Cellulose based polymer
2. Chitosan
3. Polyacrylamide
4. Porous silica
5. Polyvinyl chloride based polymers
6. others

Cells as Bioreceptor

Application for detecting Toxins Bio-toxins

1. Cells report the cellular responses upon exposures to toxins in a

Physiological conditions.

2. It reveal functional information about

i. mode of action

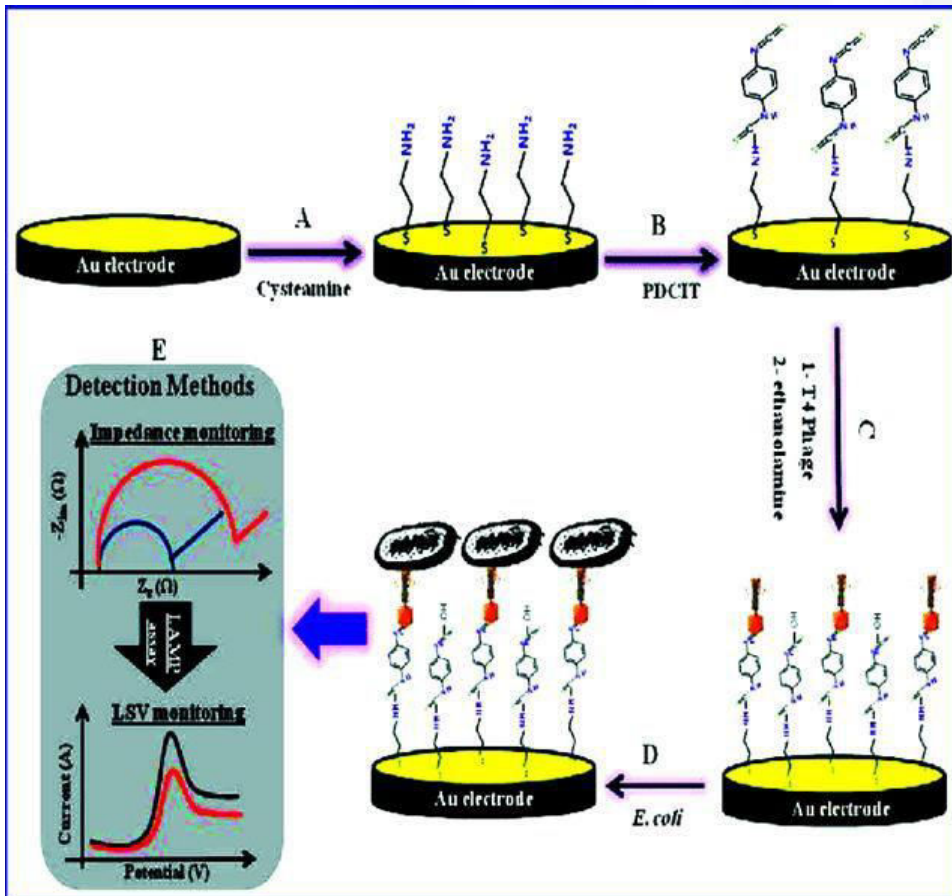
ii. toxic potency

iii. bioavailability

iv. Target tissue or organ

Cells as Bioreceptor

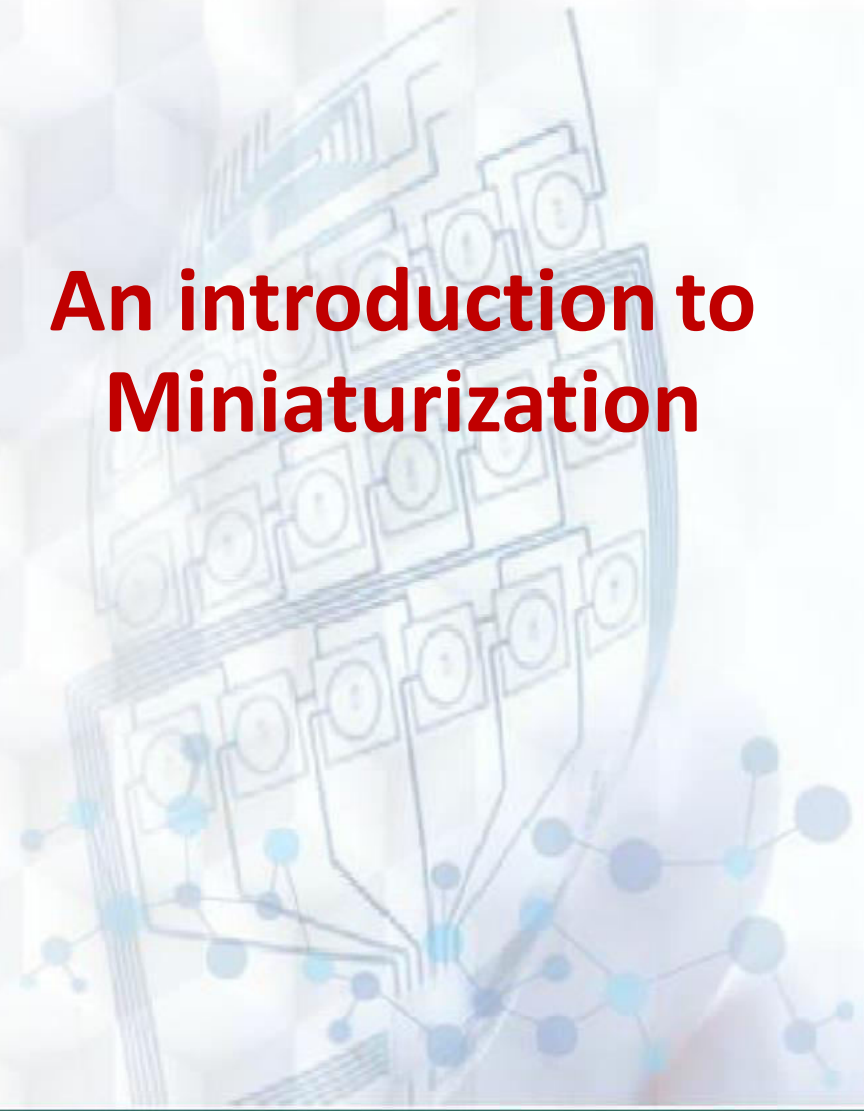
Bacteriophage for detection of pathogen



1. Bacteriophage is a virus that infects and replicates within a bacterium cell.
2. Bacteriophage specifically attack only to alive cells and they called as "Reporter Phage".
3. Schematic representation of the T4- bacteriophage biosensors for the detection of E.coli.

Biosensors

An introduction to Miniaturization



An introduction to Miniaturization

Miniaturization

The process of making something very small using modern technology.

1. Every object has a set of size dependent physical properties.
2. Size dependent physical properties change with the size. **Examples:** Mass, Volume
3. Advantages

An introduction to Miniaturization

Scaling and Scaling Forces

The study involves the sizing of the object is called scaling.

1. Force due to surface tension scales.
2. Force due to electrostatics with constant field scales
3. Force due to certain magnetic forces scales
4. Gravitational forces scale.

Scaling Factors inertia, speed resistance, heat capacity

An introduction to Miniaturization

Scaling Laws

Scaling law is a law that describes the variations of physical quantities with the size of the system.

There are two scaling laws.

1. Scaling in Geometry

Involves **Volume** and **Surface**

2. Scaling of Phenomenological behavior

Involves **size** and **material characterization.**

An introduction to Miniaturization

Reduction in 10 times size would leads to

1. 100 times reduction in heat flow.
2. 100 times greater power loss.
3. linear size of electrodes 1000 time reduction in Potential energy.
4. in radius 10000 times reduction in volumetric flow.
5. 10,000 times reduction Electromagnetic force.
6. 100 times reduction in Electrostatic force.
7. would result in 64 times reduction in torque.

Torque is the twisting force that tends to cause rotation.

An introduction to Miniaturization

Miniaturization in Biosenor Technology

1. Lab Based Biosensors

requires a proper setup, specialized user, large in size longer time processing time and expensive.

2. Commercially available

Biosensor are portable, cost effective. Easy to use , Immediate readout. No pre-treatment of sample required.

END

Biosensors

Microsystems

A stylized illustration of a microsystem chip, showing a grid of components and molecular structures. The chip is depicted with a grid of square components, each containing a circular element. The background features a light blue and white hexagonal pattern, and the bottom right corner shows a network of blue and white nodes connected by lines, representing a molecular or biological structure.

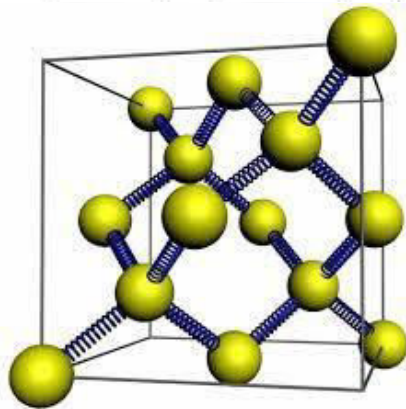
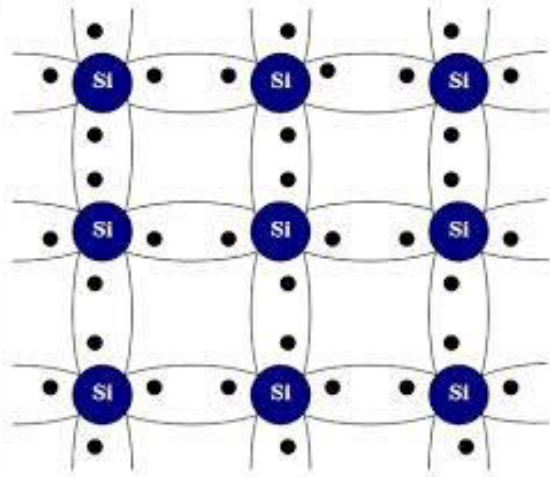
Microsystems

Microsystems (MEMS)

Microelectronic Mechanical system.

1. It is a process technology for biosensor fabrication.
2. MEM devices are made up of **components** 1-100 micrometers in size.
3. **MEMS devices** generally range in size from micrometers to millimeters.

Microsystems



<http://www.physics-and-radio-electronics.com/blog/wp-content/uploads/2016/12/Si.png>

Crystalline Silicon

1. Each silicon atom is bonded to four neighbouring atoms. Tetrahedral crystal structure like Diamond.
2. Crystalline Silicon (c-Si) is the crystalline forms of silicon, multi-crystalline Silicon, or mono-crystalline silicon.

Microsystems

Other Materials

Silicates, polysilicon, Silicon dioxide are also used in the manufacturing.

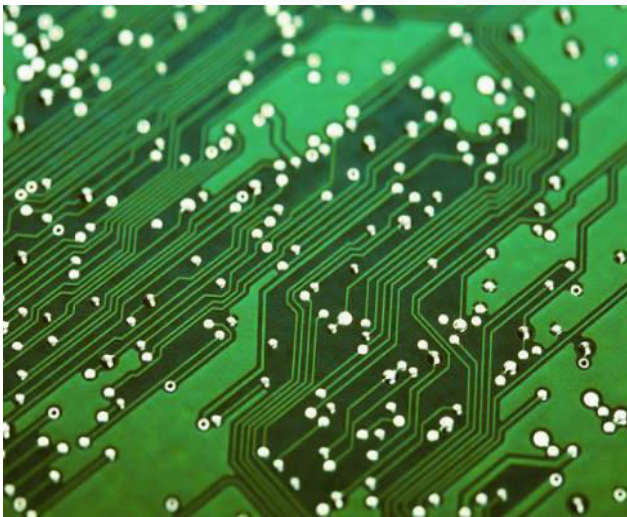
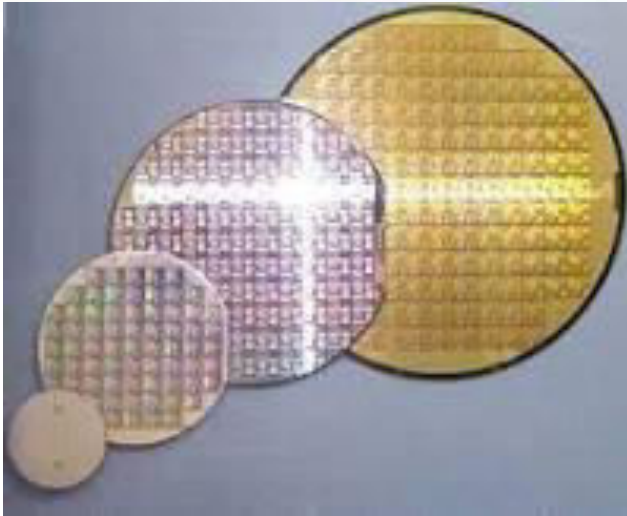
1. **Metals** are used in process by electroplating/ Gold, Nickel, Platinum

2. **Ceramics**

are inorganic non-metallic solid, they are **hard, corrosion-resistant, brittle, and cost effective.**

Example: Silicon Carbide

Microsystems



Basic Units of MEMS

1. Silicon wafers are made up of very thin slices of c-silicon (several hundred microns).

2. Integrated circuits are fabricated on the silicon chip. Silicon can hold millions of transistors, resistors, and capacitors.

Integrated circuit design, or IC design, is a subset of electronics engineering.

Biosensors

**Sensing
Techniques**



Sensing Techniques

Sensing Techniques

Sensing methods are used to detect, any change that happened in the physical quantity of a system.

1. **Electrochemical Techniques** generate an electrical signal.
2. **Optical methods** use light to probe the signal.
3. **Acoustic Methods** detects the mechanical, or acoustic, wave

Sensing Techniques

Electrochemical Methods

Electrochemical techniques could be mainly divided into

1. **Potentiometric** that measures the variations in open circuit potential, in volts.
2. **Amperometric** that measures measuring currents due to REDOX.
3. **impedimetric** method that measures the impedance of the system.

Sensing Techniques

Sensing Methods

Sensing methods rely on two types of detection methods to detect the target molecule by.

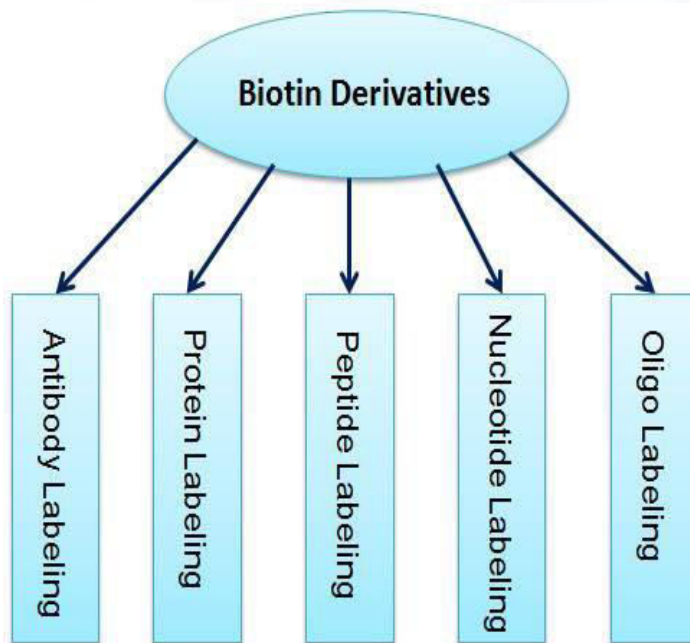
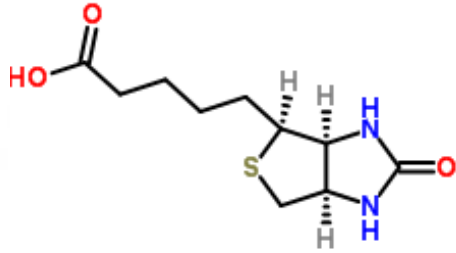
1. Label based Methods

Label-based techniques rely on the specific properties for detecting a particular target.

2. Label-free Methods

Relying solely on the intrinsic physicochemical properties of the system.

Sensing Techniques

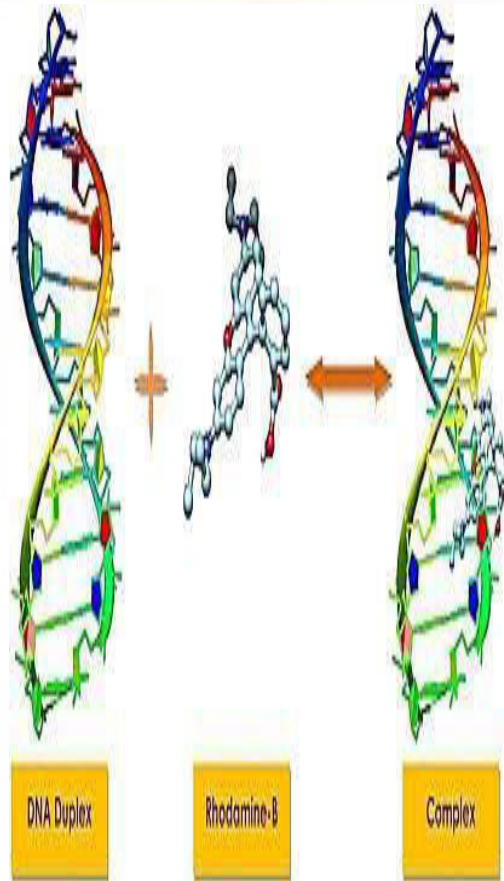


<https://www.genecopoeia.com/wp-content/uploads/2015/11/Biotin-derivative.jpg>

Biotinylation

1. Biotin is a vitamin B complex and classified (MW = 244.31) .
2. It is required in carboxylation reactions.
3. Biotinylation is the process of covalently attaching biotin to a molecule.
4. It is rapid, specific and is unlikely to disturb the natural function

Sensing Techniques



Fluorescent Probes

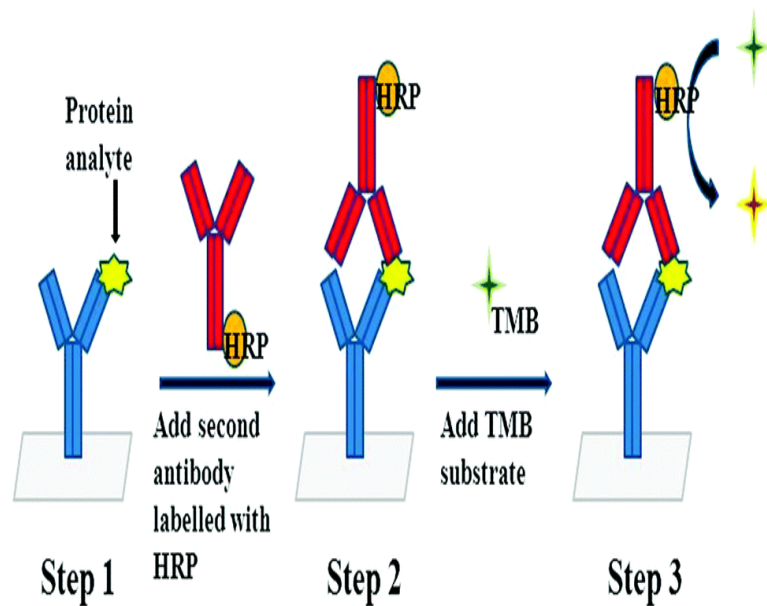
A fluorophore is a chemical compound that can reemit light upon light excitation.

It contain several combined aromatic groups, or planar or cyclic molecules with several π bonds.

Rhodamine dye is suitable for most biological molecules and respond distinctly to light by producing a detectable signal.

<https://ars.els-cdn.com/content/image/1-s2.0-S014372081300199X-fx1.jpg>

Sensing Techniques



Enzyme Label based Sensing Techniques

Enzymes can function as highly sensitive probes ideal for the detection of proteins in whole cells, tissues and/or lysates such as

1. Alkaline phosphatase
2. Glucose oxidase
3. Beta galactosidase
4. Horseradish peroxidase

http://pubs.rsc.org/services/images/RSCpubs.ePlatform.Service.FreeContent.ImageService.svc/ImageService/Articleimage/2014/AN/c3an01835k/c3an01835k-f1_hi-res.gif

Sensing Techniques

Label Free Techniques

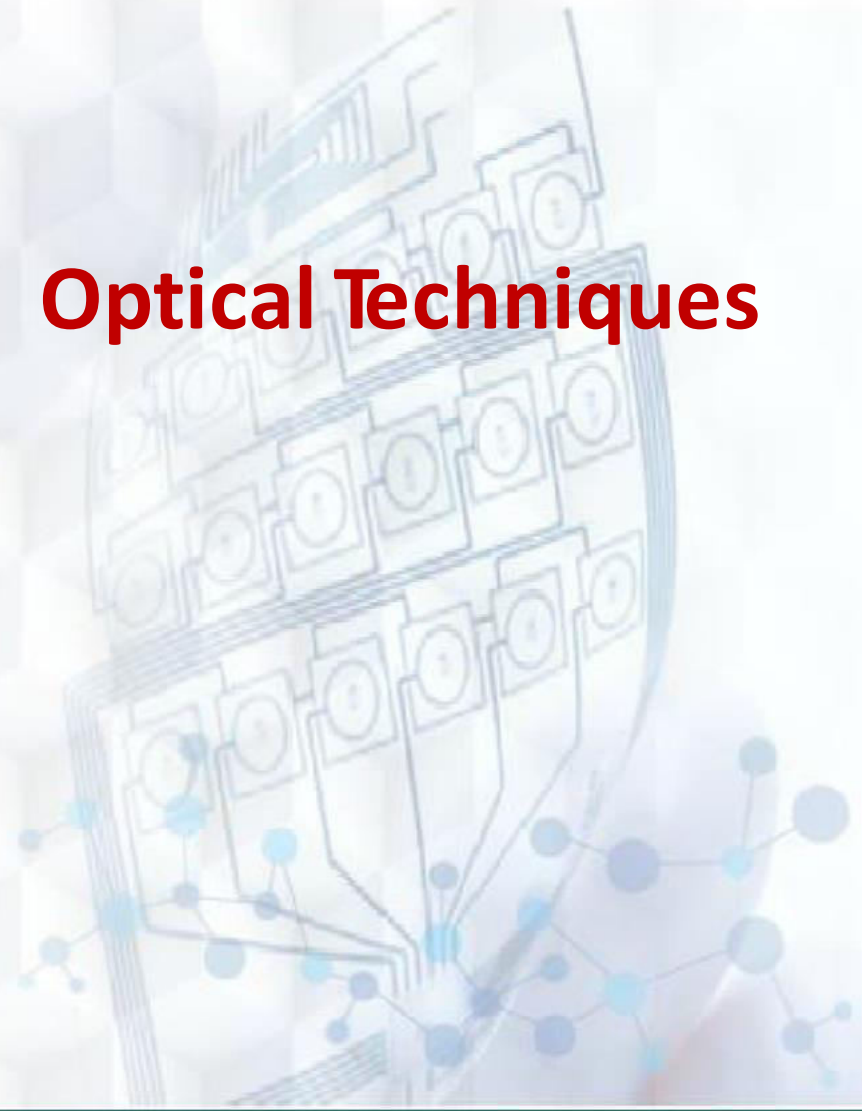
A number of sensing techniques allow the direct detection of the target molecule.

1. Surface Plasmon Resonance
2. Electrical technique
3. Mass spectrometry techniques
4. Acoustic Techniques
5. Optical techniques

END

Biosensors

Optical Techniques

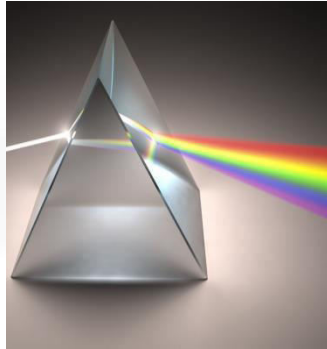


Optical Techniques

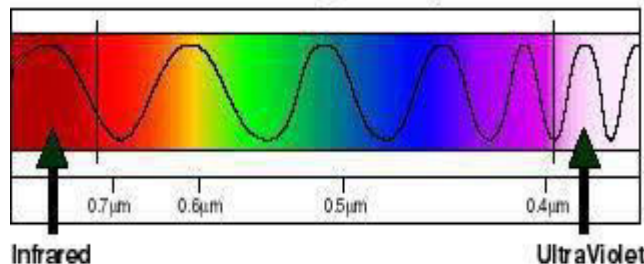
Optical Techniques

1. Optical techniques employ light rays as a probe for the detection of target molecules.
2. An optical sensor measures the physical quantity of light and then converts into electronic signals.
3. The electronic signals are then translate into a readable form.

Optical Techniques



Visible Light Region
of the Electromagnetic Spectrum



Source of Light

White light is composed of seven colors, they are visible after being dispersed by a prism.

Monochromatic light is used in optical techniques.

Example **Computer Mouse.**

Light source

Lens

Camera

Mounted Wheel

Optical Techniques



LASER Light

1. Light Amplification by Stimulated Emission of Radiation.
2. Lasers produce a narrow beam of light with very similar wavelengths.
3. Coherent, monochromatic, unidirectional with high intensity.
4. Types of LASER

<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSlp98pWeGcl7O-bKEwy0yt2AAZ749TnAUo2qHjqO2IJuiJbrNi-Q>
https://images-na.ssl-images-amazon.com/images/I/61AaGJBhBBL._SL1250_.jpg

Optical Techniques

Optical Techniques

1. Mass Spectrometry
2. Surface plasmon resonance (SPR)
3. Fiber-optics sensing technique
4. Reflectometric interference spectroscopy technique

Optical Techniques

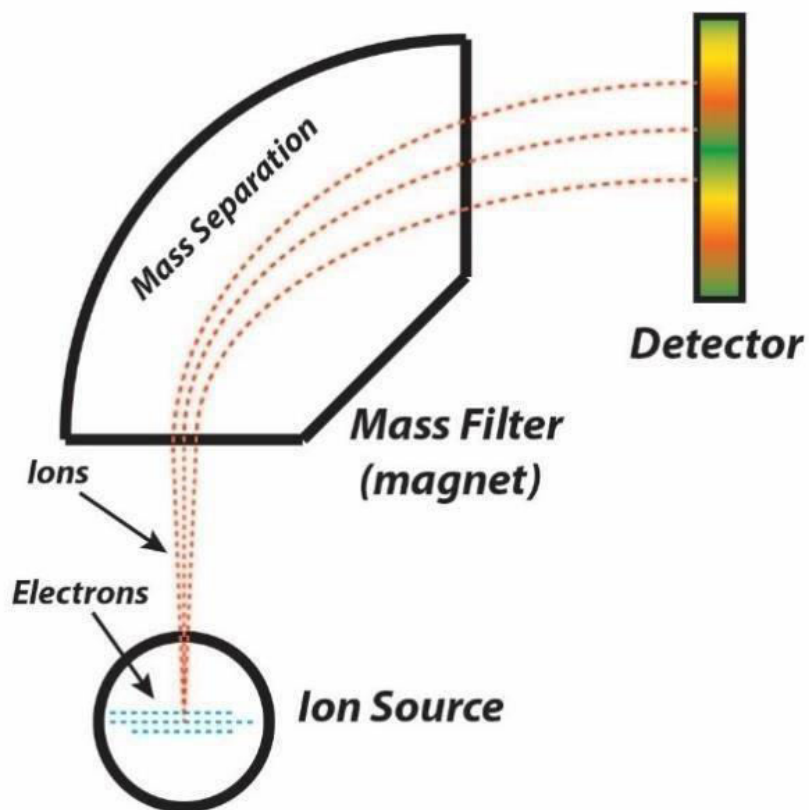
Mass Spectrometry (MS)

Mass spectrometry is an analytical tool useful for measuring the mass-to-charge ratio (m/z) of one or more molecules present in a sample.

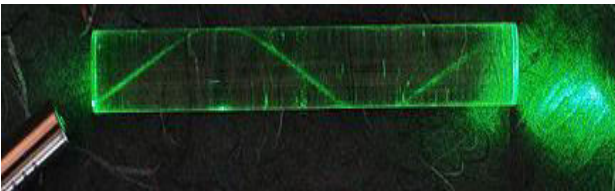
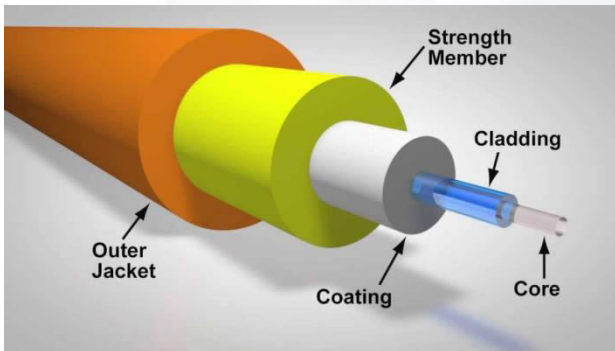
There are three key stages in the process/

1. Ionization.
2. Acceleration and Deflection.
3. Detection.

<https://images.app.goo.gl/bnW7w2pKDYscqNaD9>



Optical Techniques



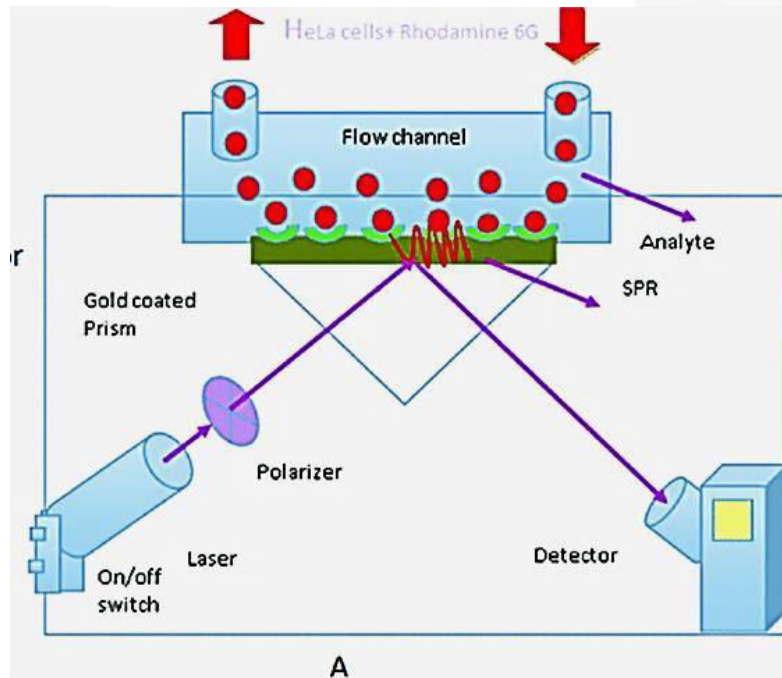
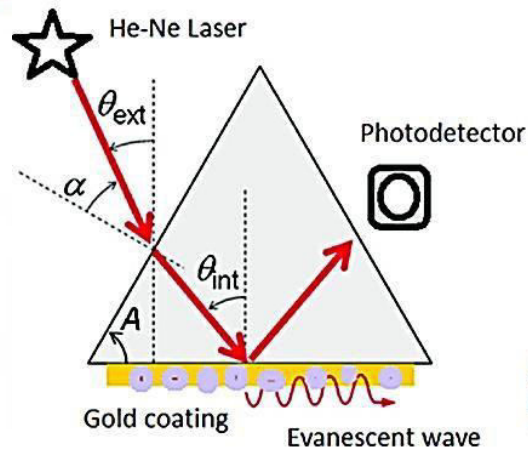
Fiber Optic Sensing

1. Fiber optic sensors are thin fibers, made up of glass or polymers, about the thickness of human hair.
2. An optical fiber consists of three parts; the core, the cladding, and the coating or buffer.
3. A LASER beam is sent to detect the Biorecognition process.

• <https://upload.wikimedia.org/wikipedia/commons/thumb/4/49/Fibreoptic.jpg/800pxFibreoptic.jpg>
• [https://upload.wikimedia.org/wikipedia/commons/thumb/b/b0/Laser_in_fibre.jpg/440px-Laser in fibre.jpg](https://upload.wikimedia.org/wikipedia/commons/thumb/b/b0/Laser_in_fibre.jpg/440px-Laser_in_fibre.jpg)

<https://www.technobyte.org/wp-content/uploads/2016/11/General-structure-of-all-types-of-optical-fibers-1024x576.jpg>

Optical Techniques

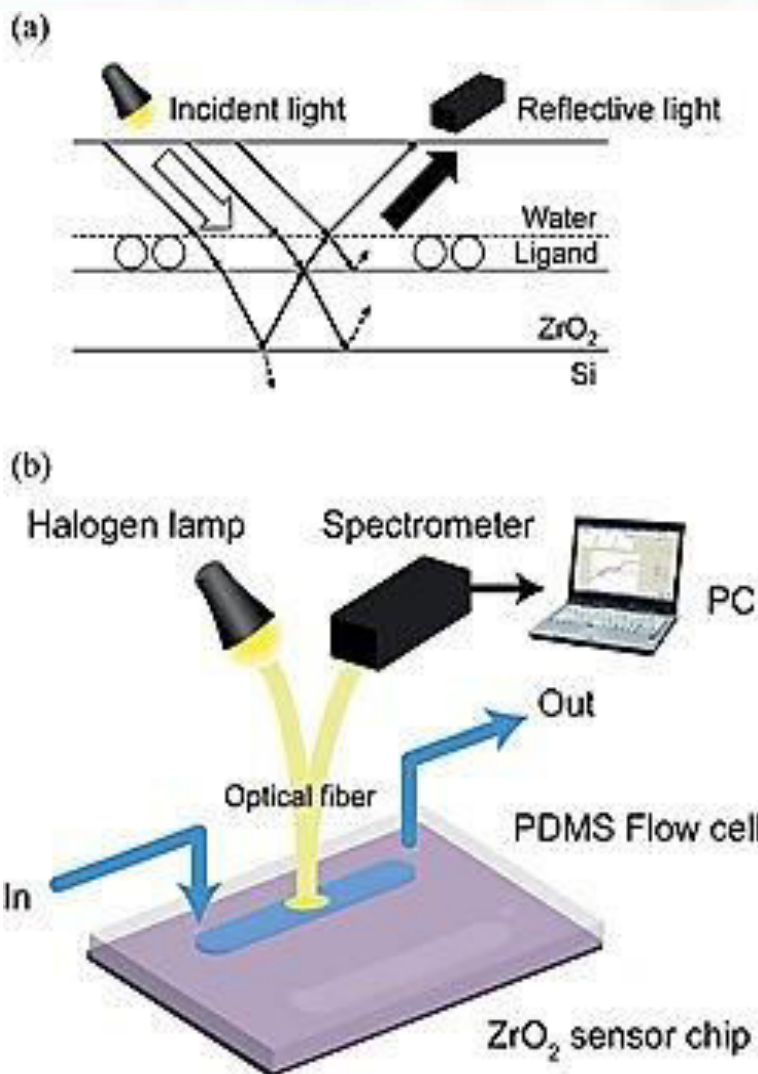


Surface Plasmon Resonance

1. SPR is an optical technique for the detection of interaction between different molecules in which one is mobile and one is fixed on a thin metal film.
2. Plasmon are density waves of electrons created when light hits metal surface under precise circumstances.

<https://www.researchgate.net/publication/324782715/figure/fig1/AS:619716565532678@1524763351980/Experimental-setup-of-surface-plasmon-resonance-SPR-biosensor-Kretschmann.png>

Optical Techniques

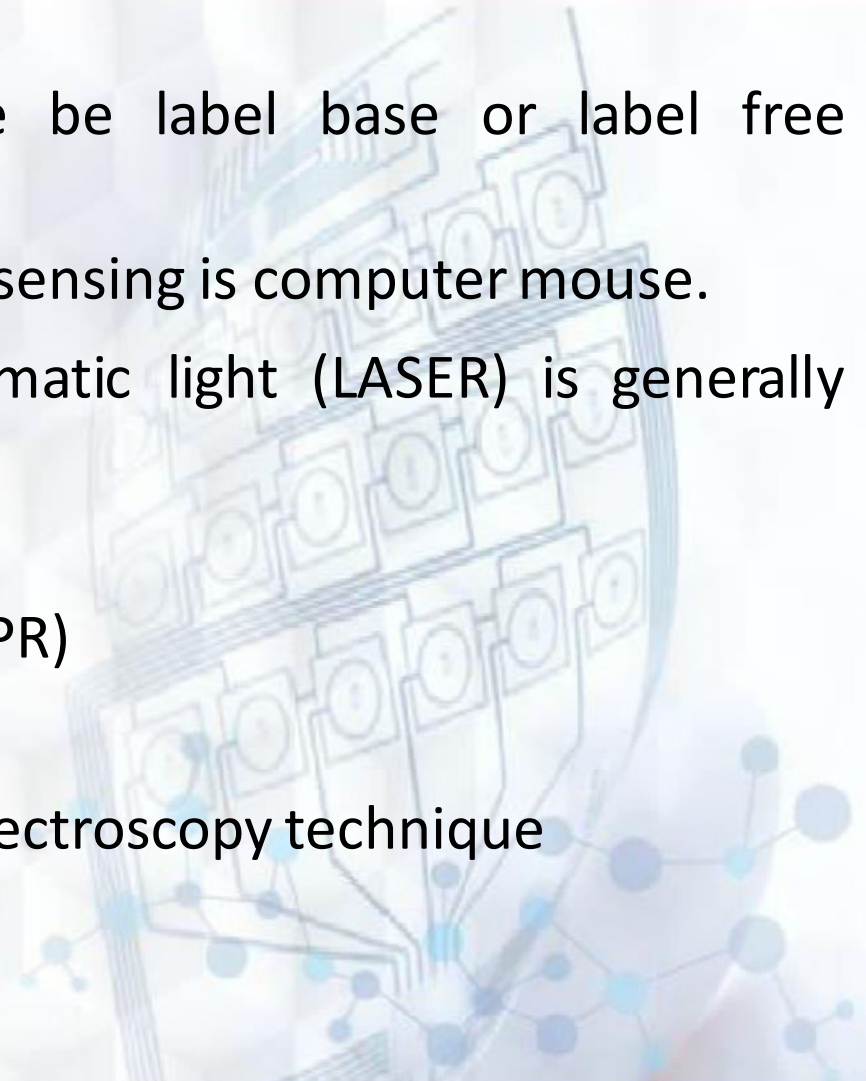


Reflectometric interference spectroscopy (RIFS)

1. RIFS is a physical method that uses white light.
2. The interference of white light with the multiple thin films, is used to investigate molecular interaction.
3. White light is directed vertically onto a multiple-layer system of a silicon oxide or other metals.

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRRSaoVaVD9YBKTD9AqRPXLR255kgbtYPm9YGQXJL3aPXSzW_5q

Optical Techniques

1. Optical sensing techniques allowed direct, real-time and label-free detection.
 2. These techniques could use be label base or label free methods.
 3. A common example of optical sensing is computer mouse.
 4. In optical sensing monochromatic light (LASER) is generally used.
 5. Mass Spectrometry
 6. Surface Plasmon resonance (SPR)
 7. Fiber-optics sensing technique
 8. Reflectometric interference spectroscopy technique
- 
- A faint background graphic on the right side of the slide. It features a blue-toned circuit board with various components and a molecular structure with blue and white spheres connected by lines, suggesting a scientific or technological theme.

Biosensors

Transistors



Transistors

Definition

1. A transistor is a miniature electronic component.
2. A transistor is a semiconductor device which transfers a weak signal from low resistance circuit to high resistance circuit.
3. It can control how much current goes through – by controlling the voltage across two of the transistor leads.

Transistors



Hearing aid

1. Hearing Aid was devised by using transistors.
2. It is a tiny microphone. it that picks up sounds from the world around and turns them into fluctuating electric currents. The fluctuating current is fed into a transistor.
3. The signals are amplified and send to a tiny loudspeaker.

<https://www.google.com/url?sa=i&source=images&cd=&ved=2ahUKEwjCk6fJ9NvjAhVjAWMBHTM5DUAQjRx6BAgBEA>

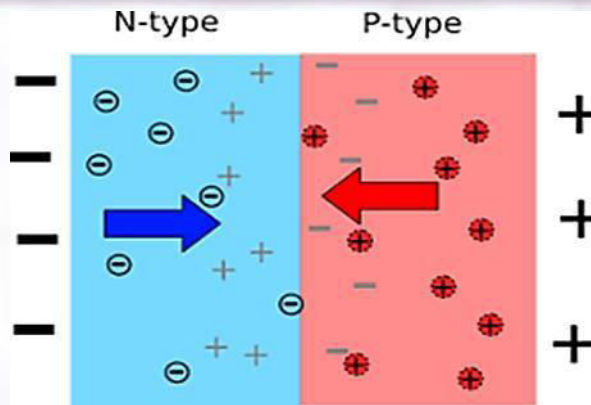
https://www.earq.com/images/PartsofaHA_BTE.jpg

Transistors

Doping

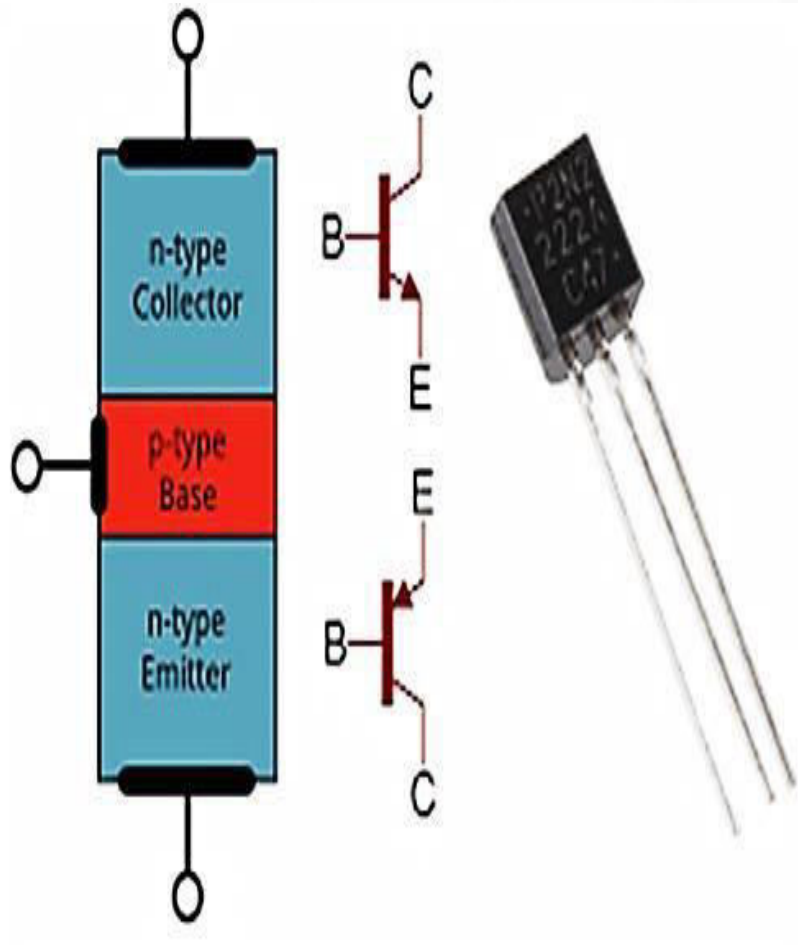
1. Doping is the addition of an impurity (called dopant) into a pure semiconductor
2. If the dopant is from 5th group of periodic table we get p-type (hole) extrinsic semiconductor P, Sb.
3. If the (dopant) is from 3rd group of periodic table we get n-type extrinsic semiconductor B, Ga, Al.

Transistors



Periodic Table of the Elements																		18	
1 1IA 11A		2 IIA 2A												13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	VIIIA 8A
1 H Hydrogen 1.0079		2 He Helium 4.0026																	
3 Li Lithium 6.941	4 Be Beryllium 9.0122											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.0054	8 O Oxygen 15.9994	9 F Fluorine 18.9984	10 Ne Neon 20.1797		
11 Na Sodium 22.989769	12 Mg Magnesium 24.305	13 Al Aluminum 26.981539	14 Si Silicon 28.0855	15 P Phosphorus 30.973762	16 S Sulfur 32.06	17 Cl Chlorine 35.4527	18 Ar Argon 39.948												
19 K Potassium 39.0983	20 Ca Calcium 40.078	21 Sc Scandium 44.95591	22 Ti Titanium 47.88	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.938	26 Fe Iron 55.847	27 Co Cobalt 58.9332	28 Ni Nickel 58.6934	29 Cu Copper 63.546	30 Zn Zinc 65.39	31 Ga Gallium 68.72	32 Ge Germanium 72.64	33 As Arsenic 74.9216	34 Se Selenium 78.96	35 Br Bromine 79.904	36 Kr Krypton 83.60		
37 Rb Rubidium 85.4678	38 Sr Strontium 87.62	39 Y Yttrium 88.90585	40 Zr Zirconium 91.224	41 Nb Niobium 92.90638	42 Mo Molybdenum 95.94	43 Tc Technetium 98.9062	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.9055	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.411	49 In Indium 114.818	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.6	53 I Iodine 126.90447	54 Xe Xenon 131.29		
55 Cs Cesium 132.90545	56 Ba Barium 137.327	57-71 Lanthanide Series	72 Hf Hafnium 178.49	73 Ta Tantalum 180.9479	74 W Tungsten 183.85	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.96655	80 Hg Mercury 200.59	81 Tl Thallium 204.3833	82 Pb Lead 207.2	83 Bi Bismuth 208.98037	84 Po Polonium [209]	85 At Astatine [210]	86 Rn Radon 222.01758		
87 Fr Francium [223]	88 Ra Radium 226.0254	89-103 Actinide Series	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [265]	109 Mt Meitnerium [268]	110 Ds Darmstadtium [271]	111 Rg Roentgenium [272]	112 Cn Copernicium [285]	113 Nh Nihonium [284]	114 Fl Flerovium [289]	115 Uup Ununpentium [288]	116 Lv Livermorium [293]	117 Uus Ununseptium [294]	118 Uuo Ununoctium [294]		
Lanthanide Series			57 La Lanthanum 138.9055	58 Ce Cerium 140.115	59 Pr Praseodymium 140.90765	60 Nd Neodymium 144.24	61 Pm Promethium 144.9127	62 Sm Samarium 150.36	63 Eu Europium 151.9655	64 Gd Gadolinium 157.25	65 Tb Terbium 158.92534	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93032	68 Er Erbium 167.26	69 Tm Thulium 168.93421	70 Yb Ytterbium 173.04	71 Lu Lutetium 174.967		
Actinide Series			89 Ac Actinium [227]	90 Th Thorium [232]	91 Pa Protactinium [231]	92 U Uranium 238.02891	93 Np Neptunium [237]	94 Pu Plutonium [244]	95 Am Americium [243]	96 Cm Curium [247]	97 Bk Berkelium [247]	98 Cf Californium [251]	99 Es Einsteinium [252]	100 Fm Fermium [257]	101 Md Mendelevium [258]	102 No Nobelium [259]	103 Lr Lawrencium [262]		
			Alkali Metals	Alkaline Earths	Transition Metals	Basic Metals	Semi-Metals	Nonmetals	Halogens	Noble Gases	Lanthanides	Actinides							

Transistors



Components of Transistor

Each transistor has three parts

1. Emitter
2. Base
3. Collector

The Collector, collects the current and the emitter 'emits' the current, while the base is the region between them.

<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTvBi7yEySijuwM-RQMixfKMBOjR0c5OVkT7i2Vv7iHhxrHrU6k>

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSHJbB_yZ5MgmA3MYhS6jMdqimJVCmufEpyefam-Eq8iPIZeOcl

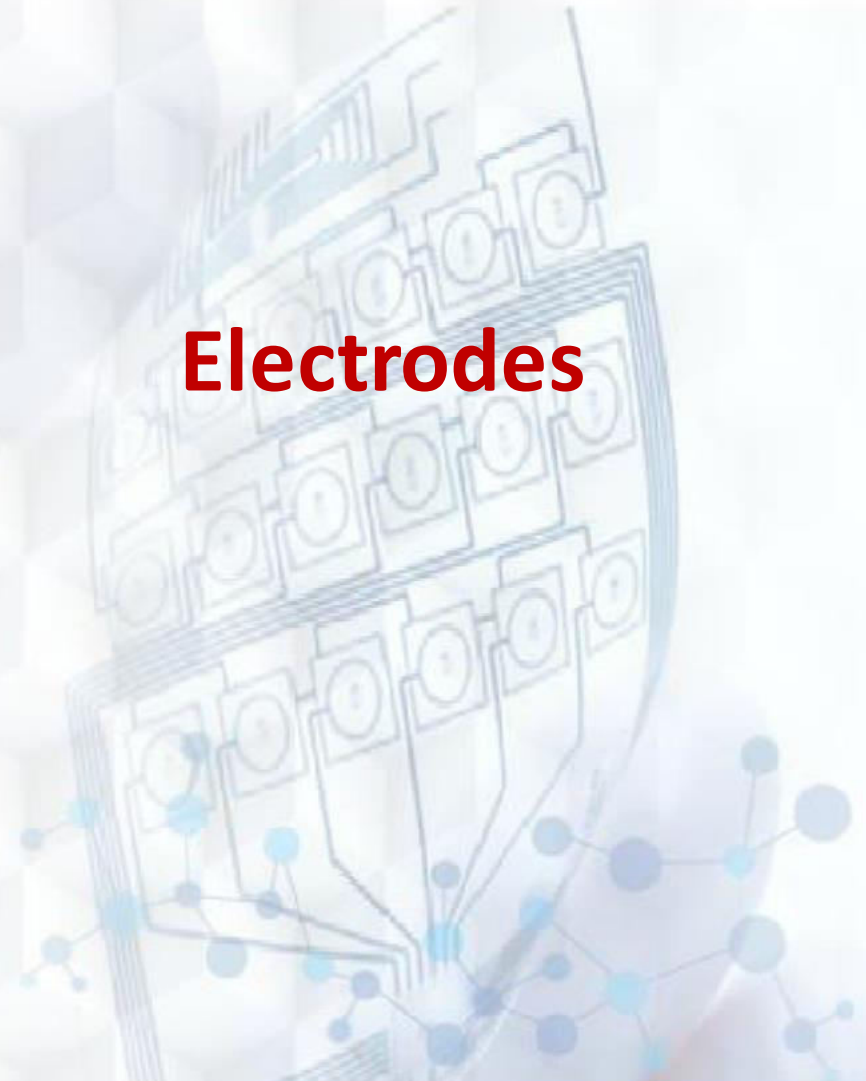
Transistors

Summary

1. Transistors are the building blocks of the modern electronics
2. They are composed of three parts called Emitter, Base and collector 'collects' the current and the emitter 'emits' the current, while the base is the region between them.
3. The pure elements of group 4 (Silicon and Gallium) are used as semi conductors and doped with the elements of group 3 and 5.
4. The process of doping is used improve the electrical property of the semiconductor. There are two types of transistors P-type and N-Type.
5. There are millions of transistors in a computer.

Biosensors

Electrodes



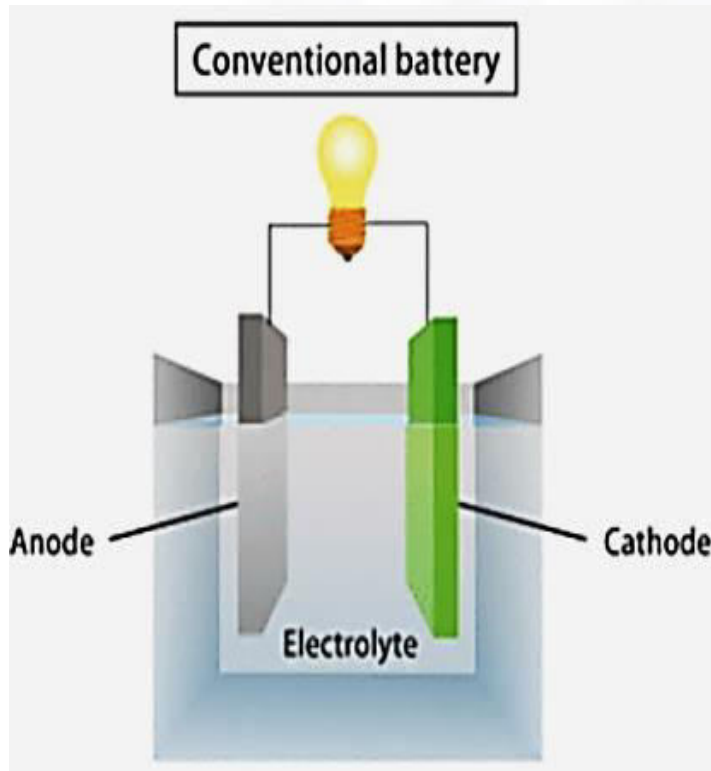
Electrodes

Electrode

1. Electrodes are electrical conductors.
2. Electrodes are made up of reactive materials such as metals (Cu, Ag, Zn) and non reactive materials such as gold or graphite.
3. They come in various shapes and sizes, and are used to connect and monitor different systems.

Electrodes

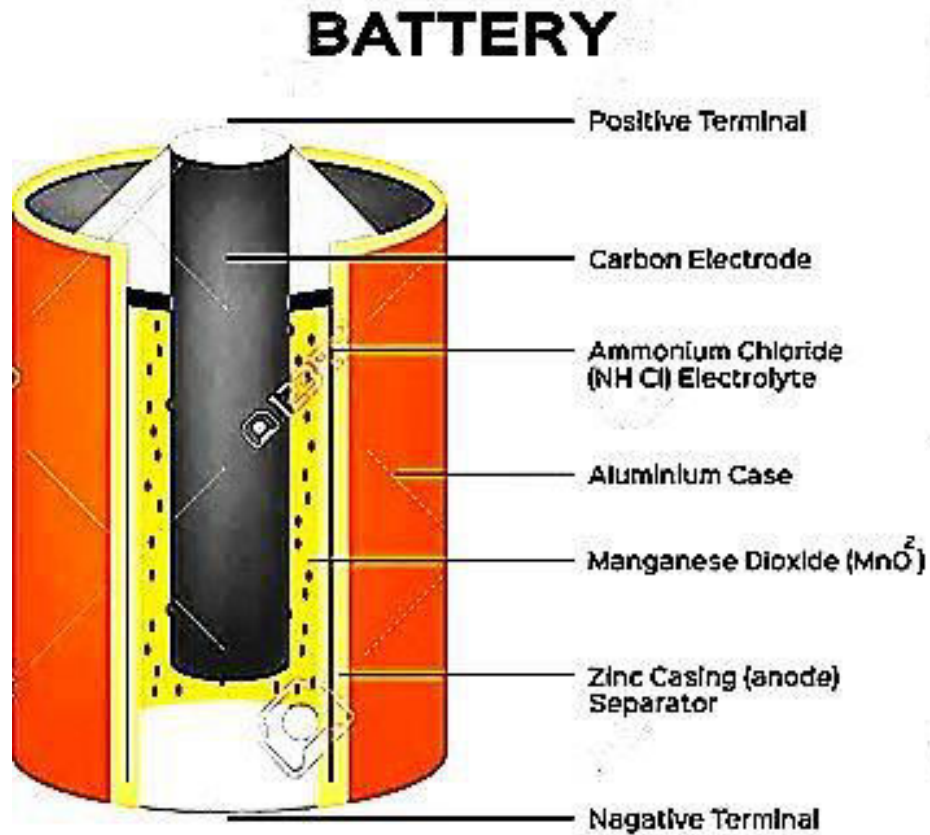
Cathode and Anode



<https://cdn57.androidauthority.net/wp-content/uploads/2016/10/Solid-state-battery-technology.jpg>

1. Cathode is the negatively charged electrode.
2. Cathode attracts cations or positive charge. It is an electron donor.
3. The anode is the positively charged electrode.
4. It attracts electrons or anions.

Electrodes



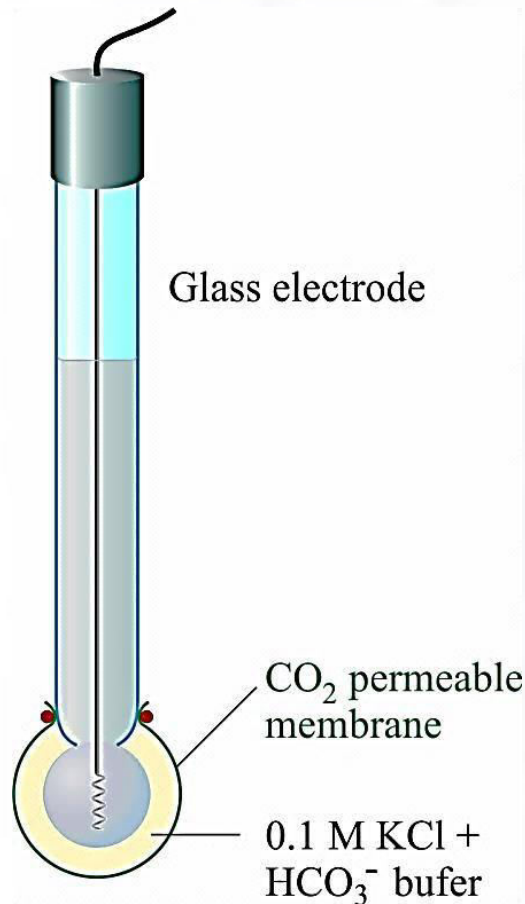
<https://previews.123rf.com/images/gritsalak/gritsalak1702/gritsalak17020004/70974074-dry-cell-battery-vector-diagram.jpg>

Electrodes

Electrodes for Medical Devices

1. Use of electrodes in diagnostics and health equipment has changed the pace of patient care.
2. The electrodes are the main components of the sensors used in Biosensors.
3. Special electrodes are designed for a number of diagnostic purposes.

Electrodes

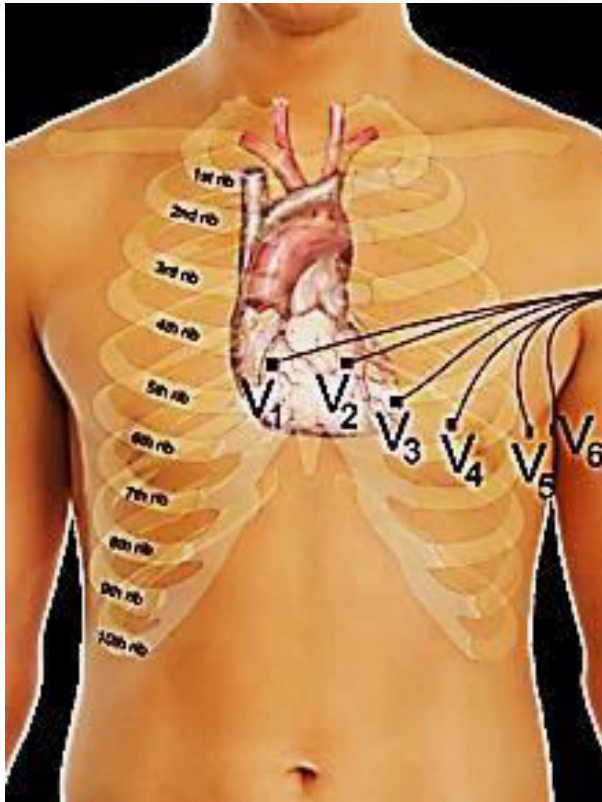


https://www.periodni.com/gallery/co2_electrode.png

CO₂-Sensitive Glass electrode

1. A CO₂-sensitive glass electrode is immersed in HCO₃ buffer and 0.1M KCl solution.
2. It is covered with a membrane, permeable to CO₂.
3. The glass electrode measures the resulting pH of the bicarbonate solution.

Electrodes



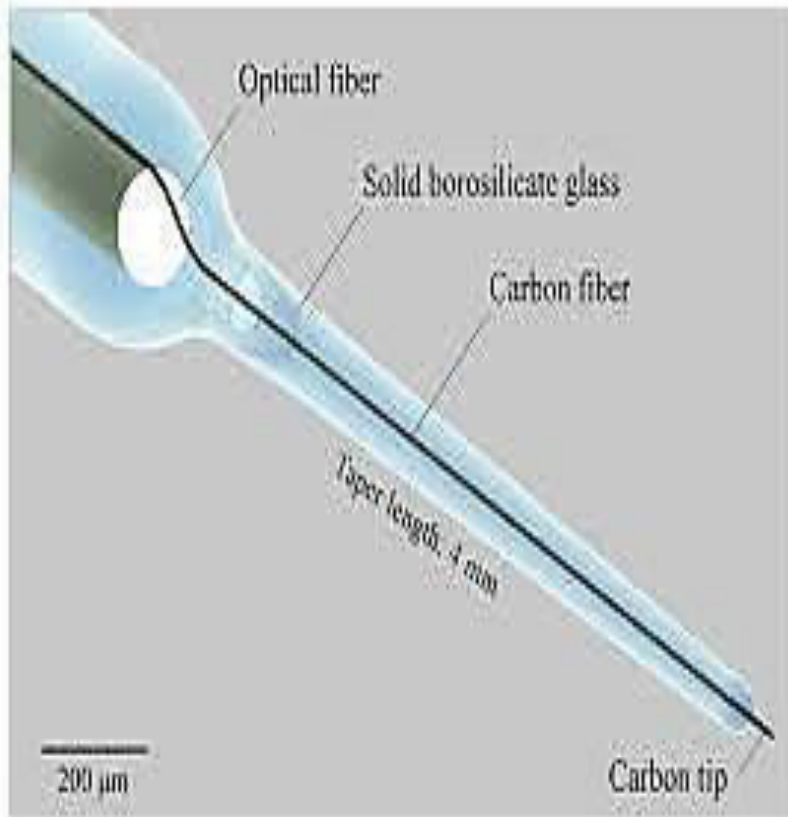
ECG Electrodes

1. Electrocardiogram is used to detect any coronary artery disease.
2. A typical surface electrode contains Ag/AgCl.
3. The signals are converted into electrical voltage by electrodes.
4. The disposable electrodes are attached to the patients' skin and can be easily removed.

<https://encryptedtbn0.gstatic.com/images?q=tbn:ANd9GcQOLQulpGbJlKO8cAEdr8EEtXjSsmpqdcIK7LCL0iYOTdkg9nE>

https://upload.wikimedia.org/wikipedia/commons/thumb/4/41/Precordial_leads_in_ECG.png/220px-Precordial_leads_in_ECG.png

Electrodes



<https://journals.plos.org/plosone/article/figure/image?size=large&id=10.1371/journal.pone.0193836.g001>

Micro Electrodes

1. They are small size terminal and used in electrophysiology.
2. It is made up of glass, optical fiber, metal or alloy wires, carbon fiber.
3. It is are usually filled with 3 M KCl as the electrical conductor
4. The tip diameters of about $0.5 \mu\text{m}$.

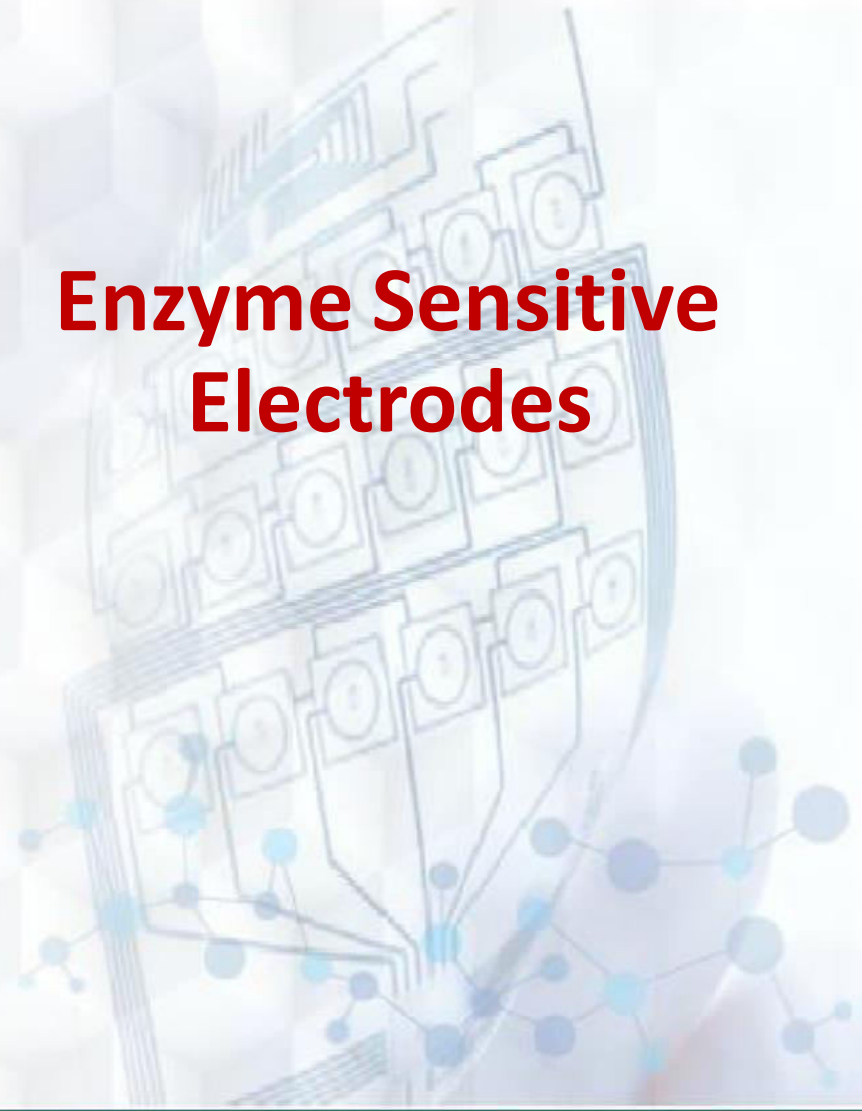
Electrodes

Summary

1. An electrode is a conductivity device.
2. Gas sensitive electrodes are used for the detection of CO₂ in bio samples.
3. ECG electrodes are used to monitor the cardiac functions.
4. Microelectrode is used in physiological analytical methods.

Biosensors

Enzyme Sensitive Electrodes

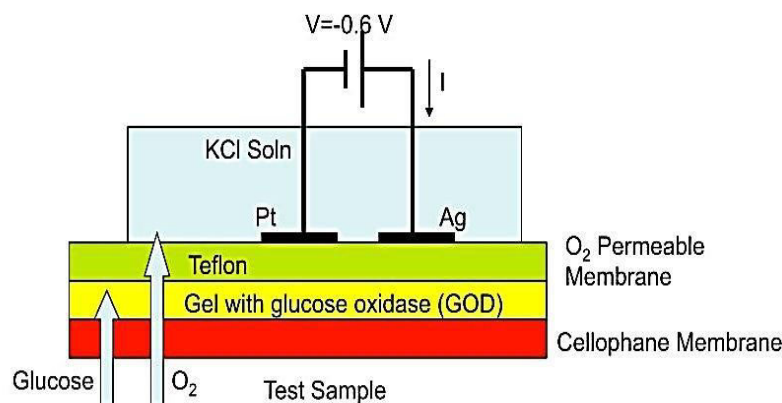


Enzyme Sensitive Electrodes

1. The enzyme electrode is a miniature chemical device.
2. It functions by combining an electrochemical procedure with immobilized enzyme activity.
3. The first reported model uses glucose oxidase immobilized on a gel to measure the conc. of glucose in sample.

Enzyme Sensitive Electrodes

Clark Glucose Sensor - Generation 1

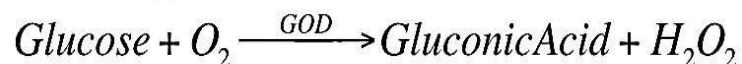


Test sample contains Glucose and O_2

$[O_2] = \text{CONSTANT}$

$[\text{Glucose}] = \text{VARIABLE}$

In presence of ENZYME (GOD)



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTWEFrdtgqQ1A5MILC-Bq2rlydo4J8y7cAPWsB2rVAfhu-xUrtr7g>

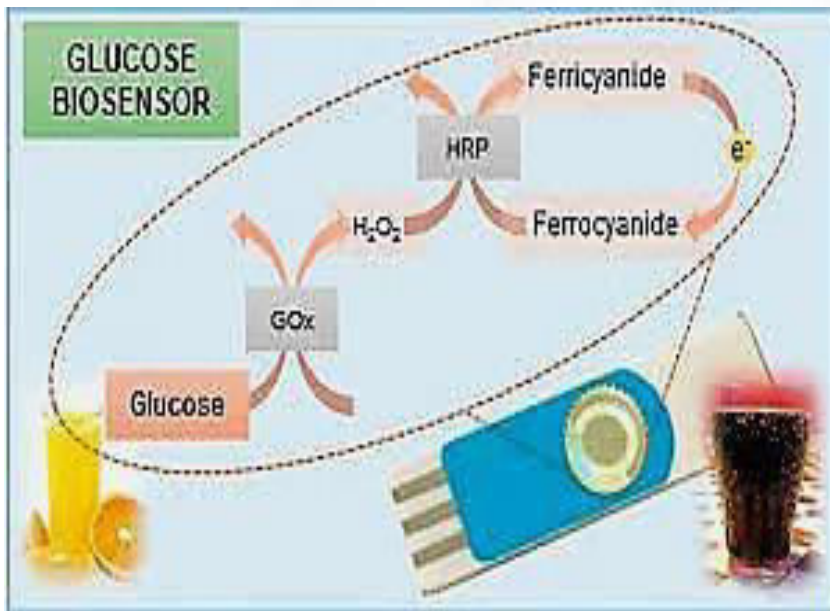
Amperometric ESE

1. The method is based on the physiological electron acceptance by Oxygen
2. A negative potential is applied to the platinum cathode for a reductive detection of the oxygen consumption.
3. It is subject to errors from fluctuation in the oxygen tension.

Enzyme Sensitive Electrodes

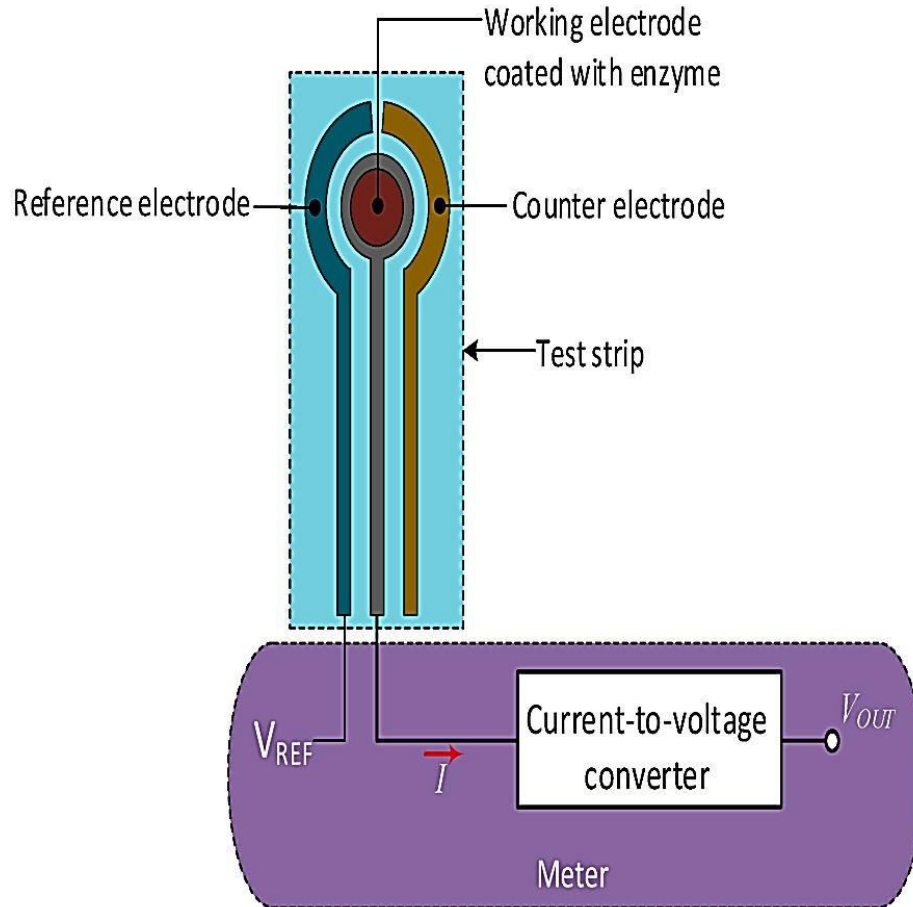
Enzyme Electrode with a mediator

1. Glucose is oxidized by Glucose oxidase and produced Gluconic acid and hydrogen peroxide.
2. The second enzyme Horse reddish peroxidase is used to convert ferricyanide to ferrocyanide.
3. Producing current signal that is proportional to the conc. of glucose.



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRj05UlWScalZwSe7dM02KCjIW7pcSZDJE70aA-dEd4VxqqvHvA>

Enzyme Sensitive Electrodes



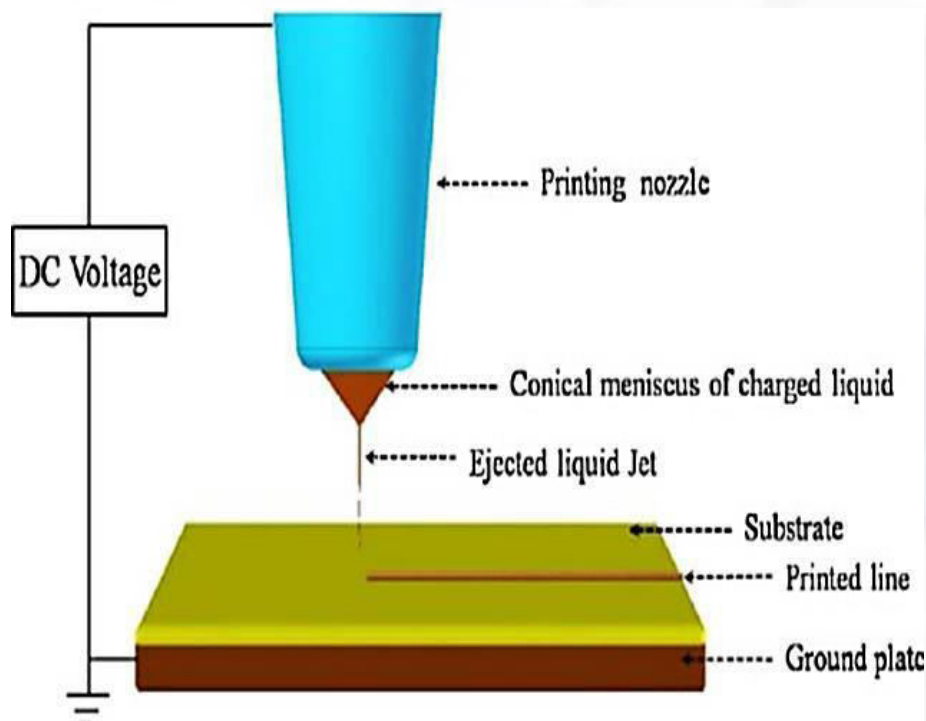
https://www.mdpi.com/sensors/sensors-19-00800/article_deploy/html/images/sensors-19-00800-g004.png

Three Electrode System

In biosensors 3 electrode systems are used.

1. Working electrode contains the sample which is being studied.
2. Reference Electrodes sets the potential of the system.
3. Counter electrode is a current source.

Enzyme Sensitive Electrodes

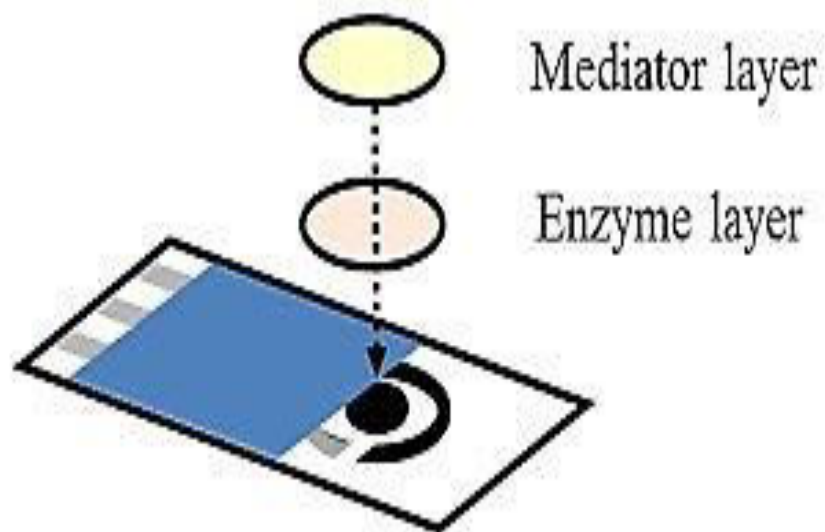


https://www.researchgate.net/profile/Arshad_Khan27/publication/251609794/figure/fig1/AS:284915545063424@1444940567255/Schema-tic-illustration-of-electrohydrodynamic-inkjet-printing-process_W640.jpg

Screen printed Electrodes

1. Screen printed electrode methodology is used for the fabrication of electrodes.
2. Screen-printed electrodes are based on carbon, gold, platinum, silver inks.
3. Ink paste is used to print, a previously marked plastic sheet substrate using a printing nozzle.

Enzyme Sensitive Electrodes



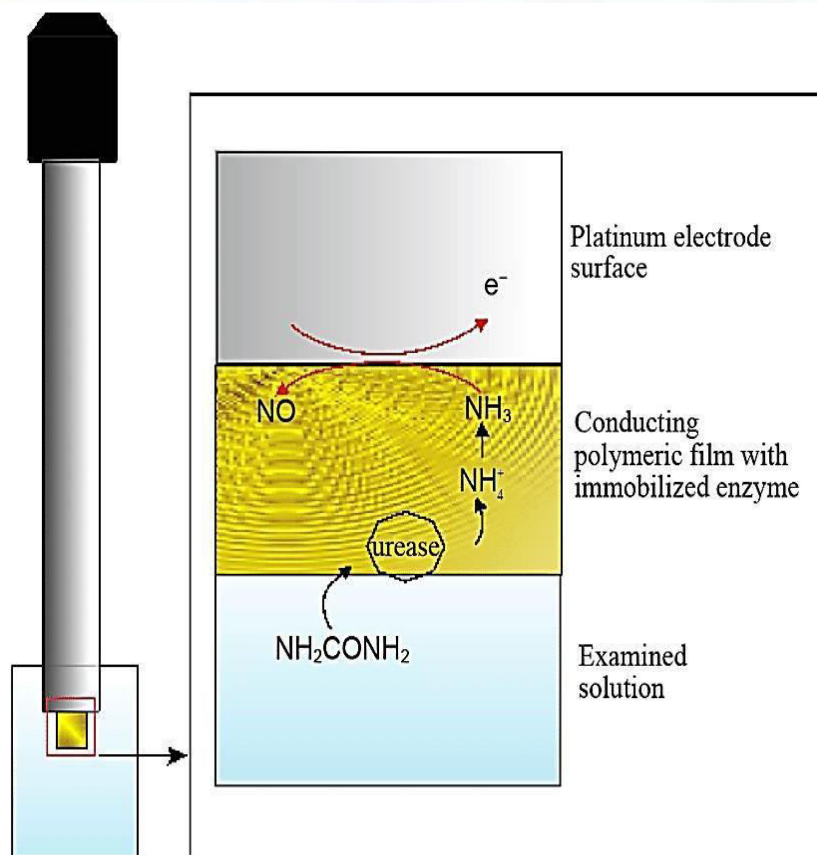
Screen-Printed Carbon Electrodes

Enzyme Printed Electrode

1. In Biosensors enzyme printed electrodes are used.
2. The enzyme layer is placed between the carbon electrode (being inert) and mediator layer.
3. A mediator dielectric layer is placed for the proper functioning of the electrode.

<https://www.researchgate.net/publication/276491989/figure/fig4/AS:612940453535750@1523147800412/Urea-detection-by-conducting-polymer-based-amperometric-biosensor.png>

Enzyme Sensitive Electrodes



Urease enzyme Electrode

1. Urease, an enzyme that catalyzes the hydrolysis of urea, forming ammonia and carbon dioxide.
2. Urease biosensor is used to determine human serum urea amount.
3. A liquid layer of urease trapped in a cellophane layer over a cation electrode, reacts with urea and increase the pH.

<https://www.researchgate.net/publication/276491989/figure/fig4/AS:612940453535750@1523147800412/Urea-detection-by-conducting-polymer-based-amperometric-biosensor.png>

Enzyme Sensitive Electrodes

Summary

1. The enzyme electrode is a transducer.
2. It couples the enzymatic reaction with the electrochemical method for the detection..
3. Enzyme screen printed electrodes are used in Biosensors;
4. A number of enzymes are used in Enzyme Sensitive Electrodes.

Biosensors

**Field Effect
Transistors**



Field Effect Transistors

Field Effect Transistor

1. A field-effect Transistor is a types of Transistor.
2. It is called as unipolar transistor, uses either electrons (in N-channel FET) or holes (in P-channel FET) for conduction
3. There are three terminals: source, gate, and drain.
4. FETs control the flow of current by the application of a voltage to the gate.

Field Effect Transistors

BIOFET

1. Biosensors containing a FET are known as Bio-FET
2. Bio-FETs, provides an effective, label-free and real-time bio-sensing platform, for the detection of a targets.
3. Bio-FETs couple a **transistor device** with a bio-sensitive layer containing the **Biological recognition element**.

Field Effect Transistors

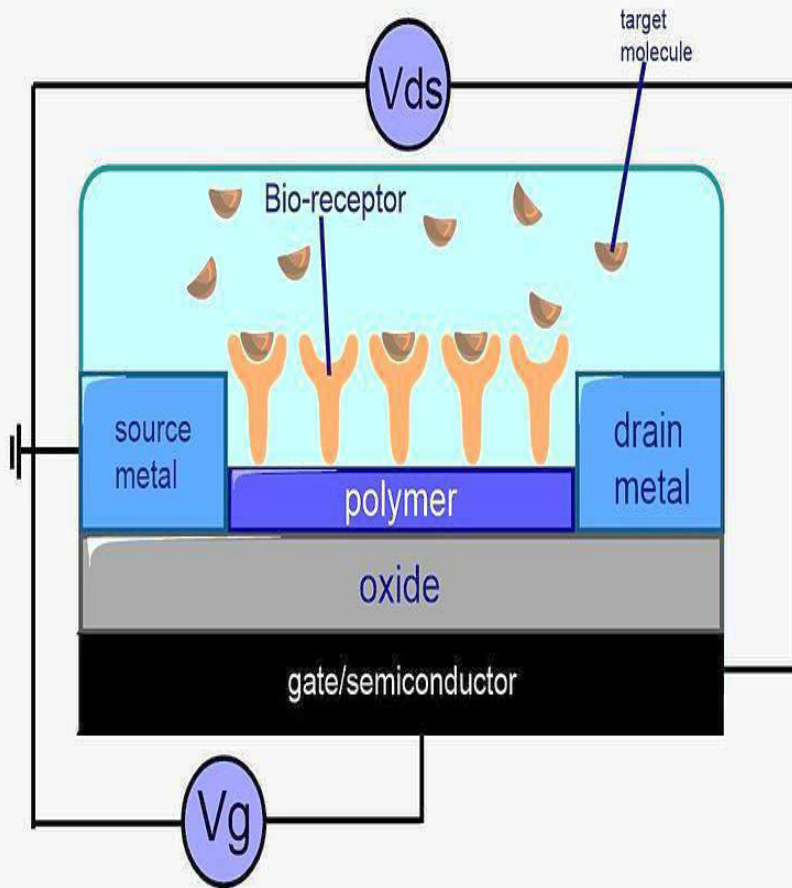
Structure and BioFET

A Bio-FET consists of two main components:

1. Biological recognition element
2. Field-effect transistor.

A chemically and electrically insulating layer is used as a bio-functionalized surface.

Generally made up of (3-Aminopropyl) triethoxysilane is used to fabricate the membrane layer.



[https://upload.wikimedia.org/wikipedia/commons/t
humb/2/2b/BioFET.jpg/800px-BioFET.jpg](https://upload.wikimedia.org/wikipedia/commons/thumb/2/2b/BioFET.jpg/800px-BioFET.jpg)

Field Effect Transistors

Classification

Bio-FETs are classified based on the bio recognition element used for detection:

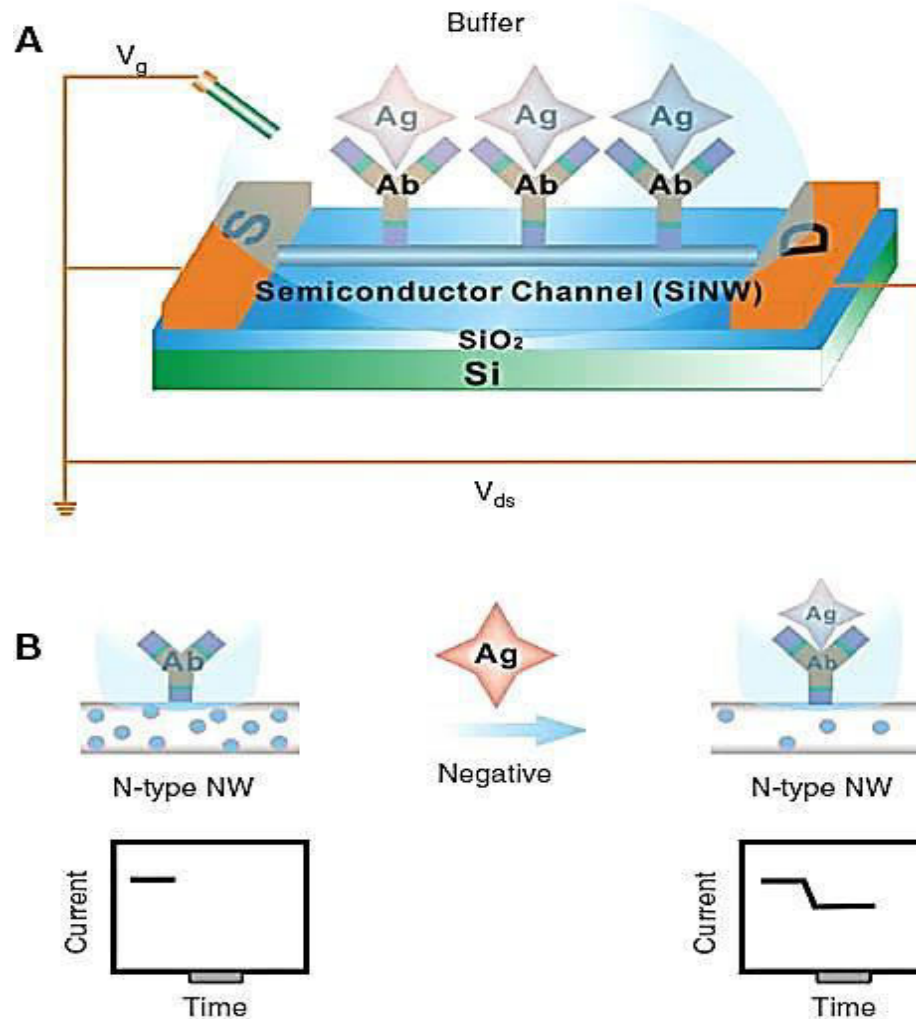
1. En-FET
2. Immuno-FET
3. DNA-FET
4. CPFET
5. Beetle/chip FET
6. Artificial BioFET-based.

Field Effect Transistors

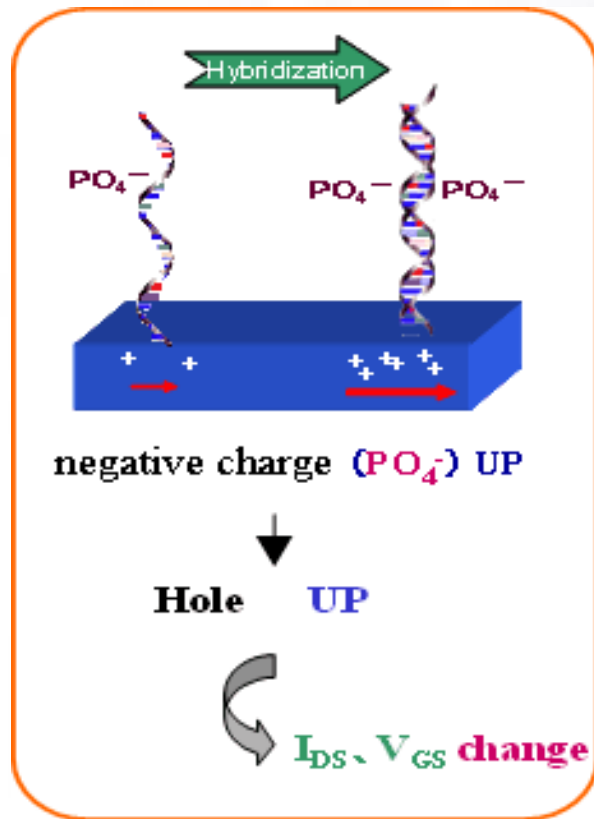
Working of Bio-FET

1. In a BioFET an Analyte binds with the BRE, on a bio-functionalized surface.
2. This introduce changes in the electrostatic potential of the surface .
3. The measurable change in current between electrodes is recorded, as a detecting signal.

Field Effect Transistors



Field Effect Transistors

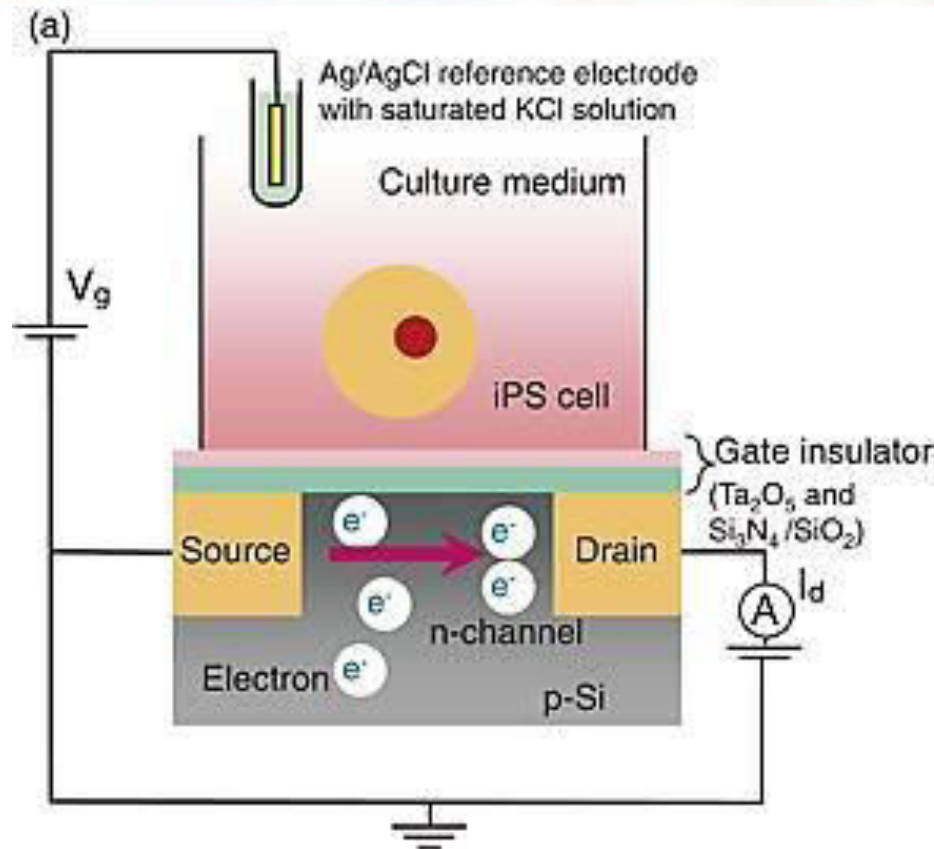


http://www.kawarada-lab.com/english/research/Application_of_Diamond/images/miniaturized_diamond_electrolyte_solution_02.gif

DNA field effect Transistor

1. A DNAFET is a field-effect transistor which uses the field-effect due to the partial charges of DNA molecules.
2. In DNAFETs, a layer of immobilized ssDNA acts as surface receptors.
3. After analyte/BRE hybridization, the charge distribution changes and signals are generated.

Field Effect Transistors

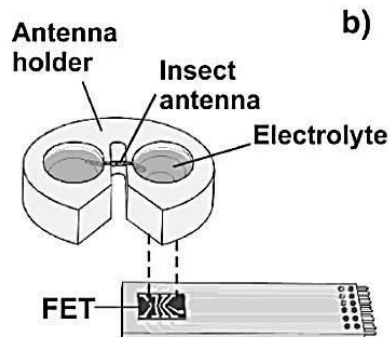
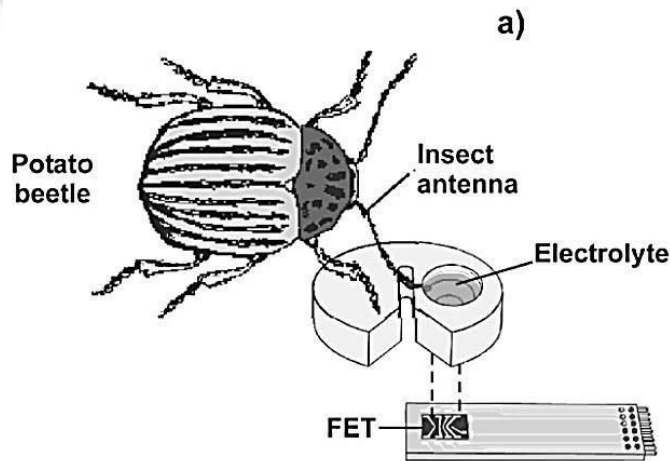


Cell-Transistor-Hybrid

1. A Biological cells are able to receive, process, and transmit information.
2. The platform based on the bio-FET would provide substantial information about different biological phenomenon.
3. Bio-FET is used in the shown slide to detect the molecular charges at cell membrane in an induced system.

https://cdn.iopscience.com/images/1347-4065/53/5S1/05FS02/Full/MR13072fig01a_online.jpg

Field Effect Transistors



BioFET using Beetle antenna to detect Crop Damage

1. The antenna is highly sensitive towards different odours.
2. The antenna of a beetle is coupled via an electrolyte to bioelectronic field-effect sensor.
3. Plant damages by the attack can be identified, due to the presence of odour.

https://www.researchgate.net/profile/Arshak_Poghossian/publication/11083921/figure/fig8/AS:394626209992712@1471097628359/Two-general-configurations-of-coupling-an-insect-antenna-as-biorecognition-component-to-a_W640.jpg

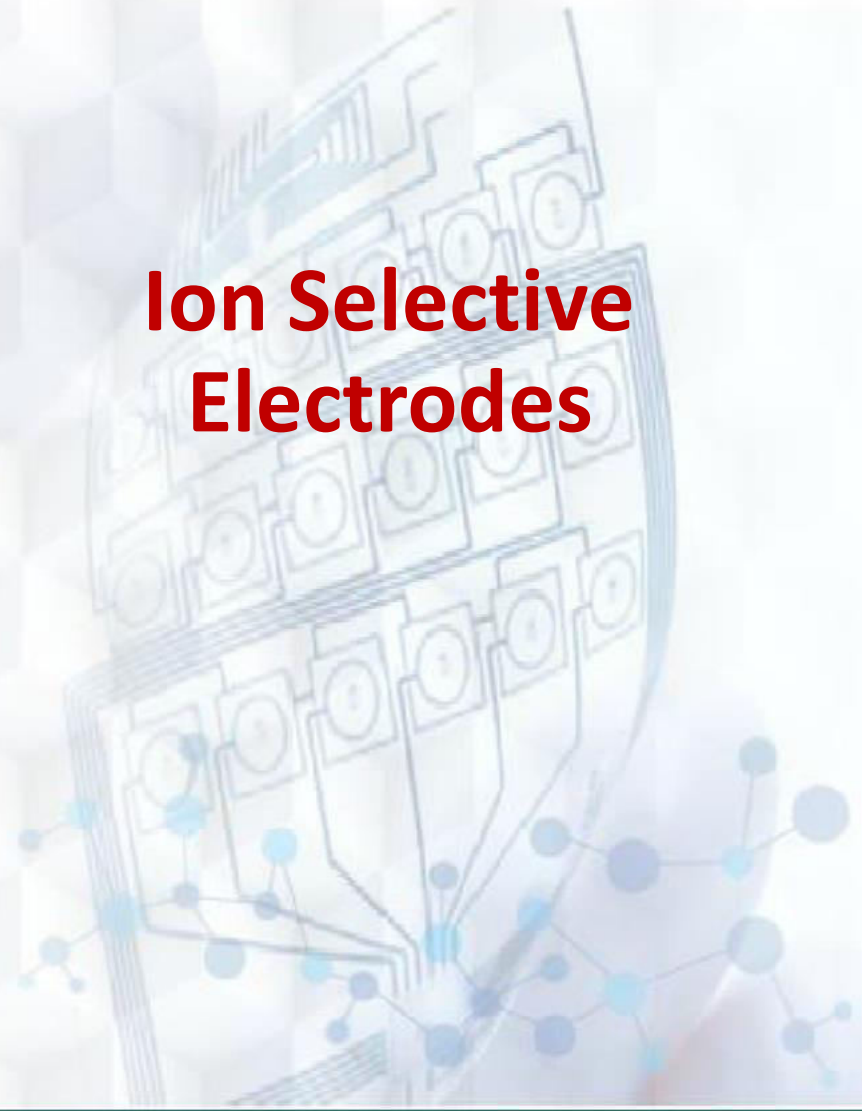
Field Effect Transistors

Summary

1. Field effect transistor is a type of Transistor.
2. FETs are also known as unipolar transistors since they involve single-carrier-type operation.
3. Biosensor in which a FET is used are known as BioFET.
4. There are several applications of the BioFET in Biosensor industry.

Biosensors

Ion Selective Electrodes



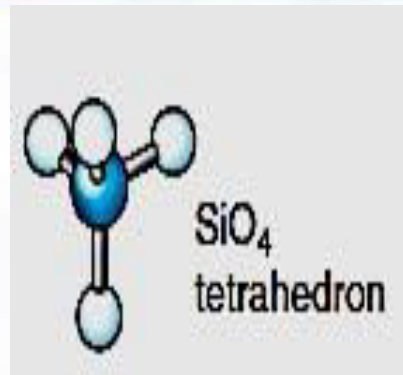
Ion Selective Electrodes

Ion Selective Electrode

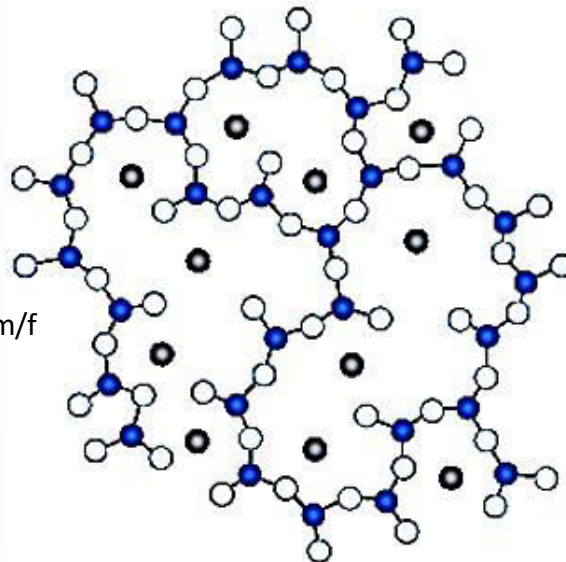
1. The basic structure of the ion-selective electrode requires a membrane which excludes all ions except for the ion of interest.
2. Ion selective electrode is composed of three parts. A membrane, an electrode and a volt meter to determine the potential difference.

Ion Selective Electrodes

<http://www.phadjustment.com/images/gif/phprobe.gif>



<https://i.stack.imgur.com/fhkWj.png>



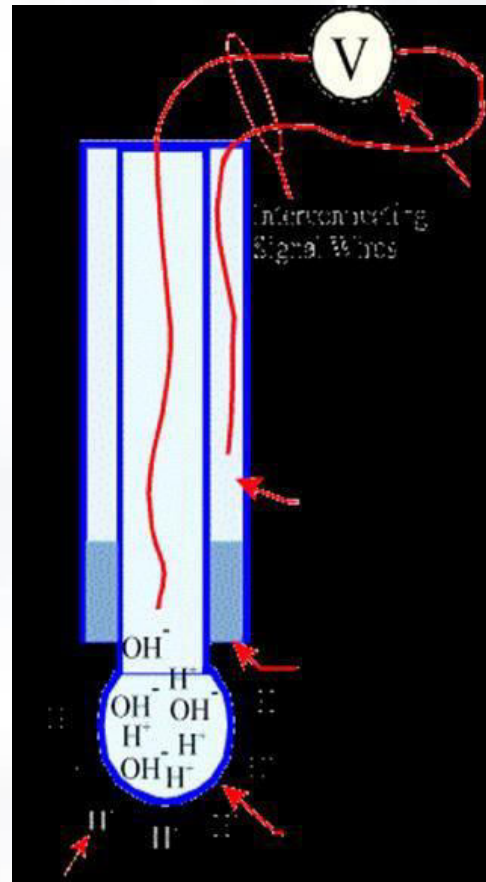
○ = O ● = Si ● = cation

(a) Silicate glass

Glass Electrode

1. The glass electrode is made of amorphous silicate glass.
2. Glass is made up of irregular network of silicate's tetrahedron.
3. It contains cations of metals like Na, K, Ca and Li that are in coordination with Oxygen atoms.
4. Silicate network is not planar.

Ion Selective Electrodes

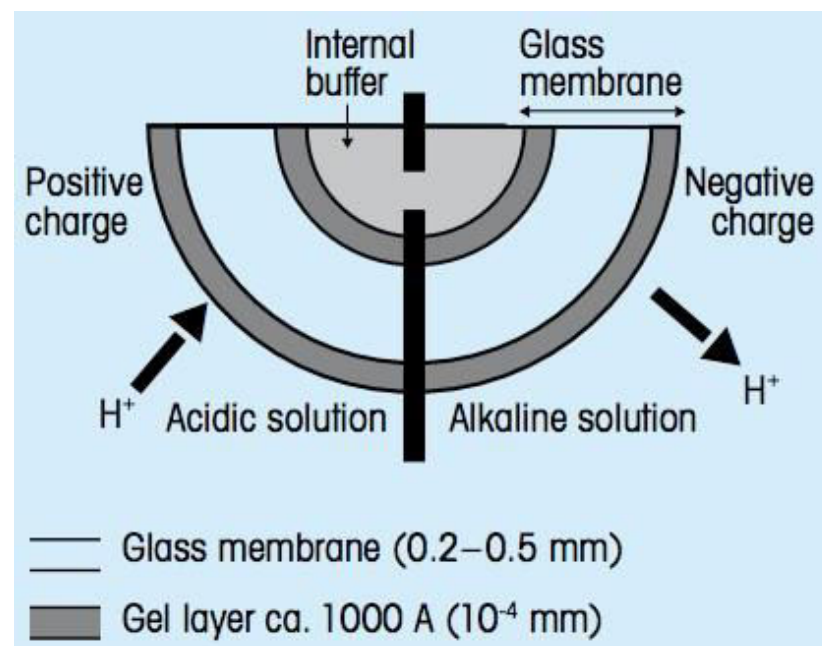
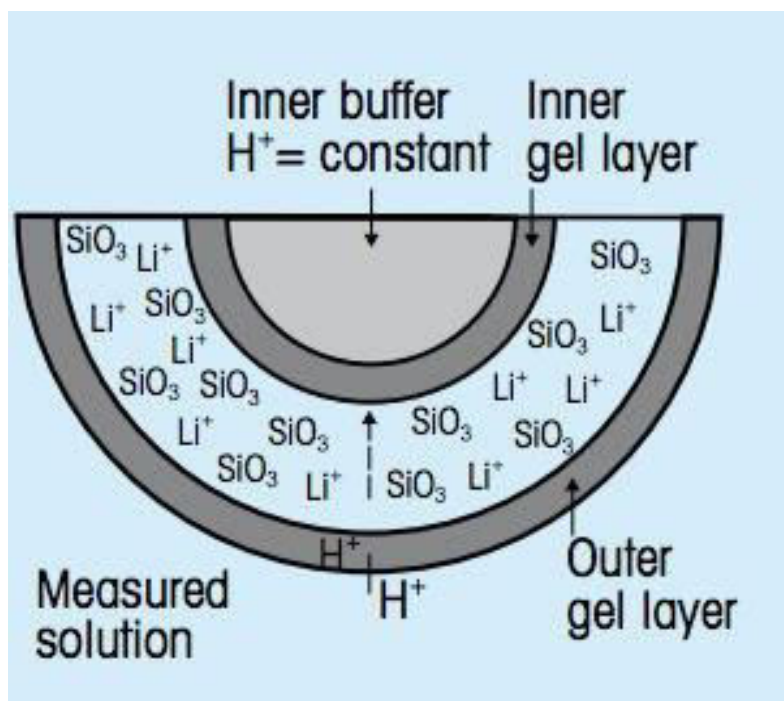


Ion Selective Electrodes

Bulb of Glass Electrode

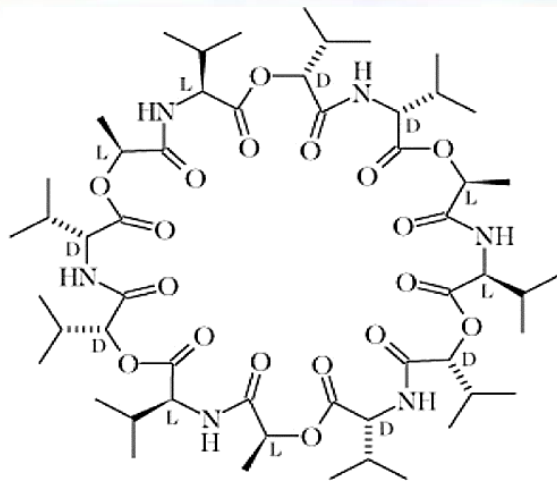
1. Glass electrode is made up of very thin membrane of glass.
2. Inner and outer layers of glass electrode bulb are known as gel layers.
3. Ion exchange equilibria takes place on surface of glass membrane as hydrogen ions replaced metal ions bound to Oxygen.

Ion Selective Electrodes

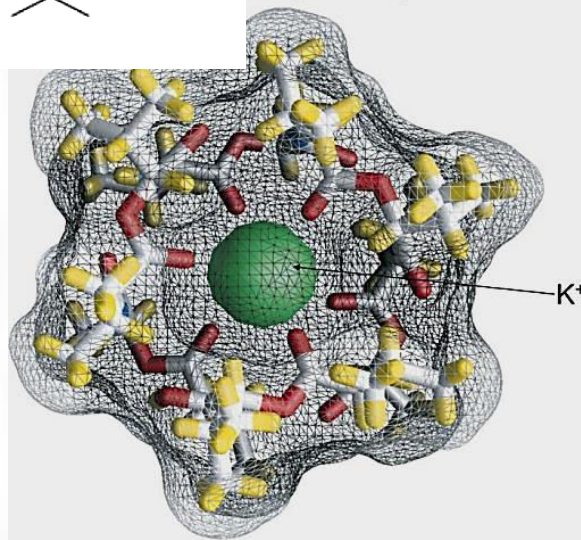


https://www.google.com/url?sa=i&source=images&cd=&cad=rja&uact=8&ved=2ahUKewi_v9K89o3kAhXNfFAKHytCC5gQjRx6BAgBEAQ&url=https%3A%2F%2Fchemistry.stackexchange.com%2Fquestions%2F72222%2Fph-probe-bulb-what-is-happening-within-the-glass&psig=AOvVaw3E0e6VTHVBi-KjdaCxNPQ4&ust=1566268782342513

Ion Selective Electrodes



Valinomycin

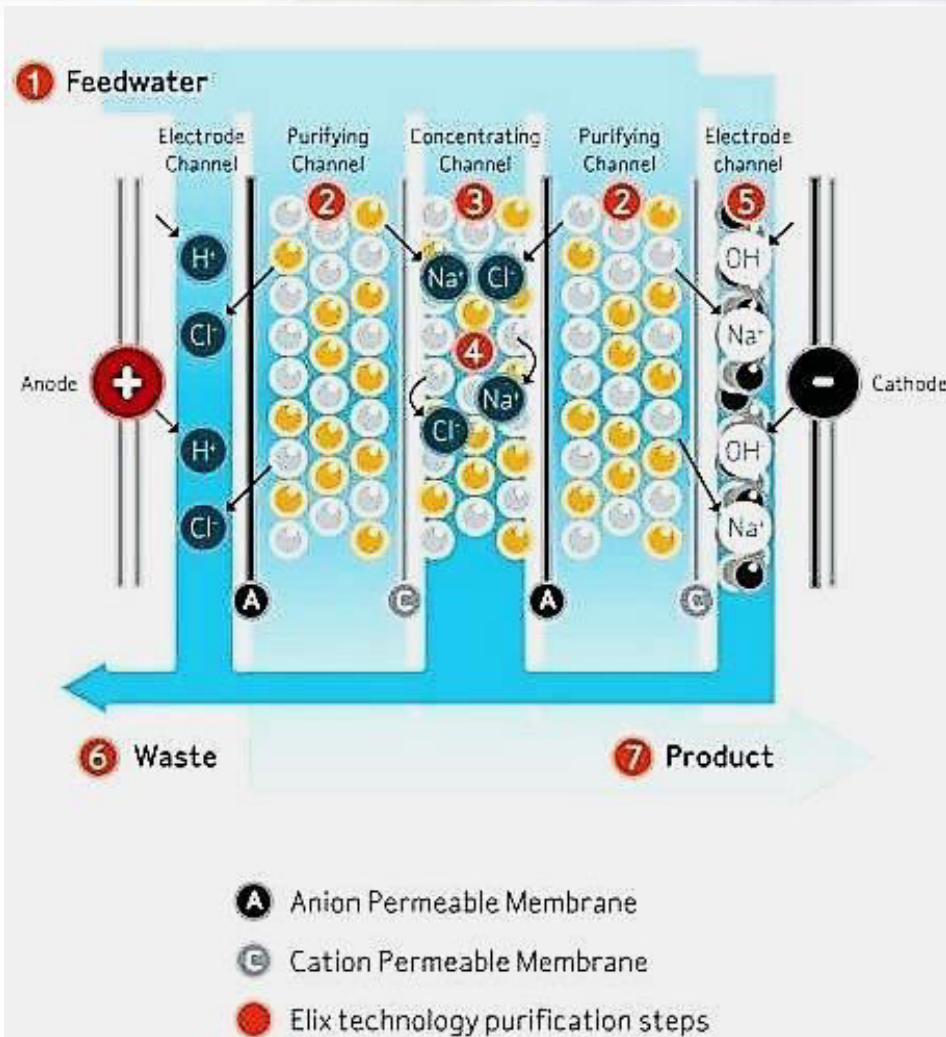


K-sensitive Electrode

1. A K-sensitive electrode is made up of a ionophore- (Valinomycin).
2. Ionophore is substance which facilitates the transport of particular ions across a membrane.
3. Valinomycin is a naturally occurring peptide and is used in the transport of K ions.

https://www.spektrum.de/lexika/images/arznei/fff1142_w.jpg

Ion Selective Electrodes



Ion exchange Electrodes

1. Ion-exchange resins are special organic polymer membranes containing a specific ion-exchange substance.
2. Ion-specific electrodes are the most widespread electrodes with anionic selectivity.

<http://idm-group.co/wp-content/uploads/2018/10/water-treatment-ion-exchange-freestyle-lab-water-water-lab-ion-exchange-water-treatment-system-pdf.jpg>

Ion Selective Electrodes

Summary

1. Ion selective electrodes are sensors that use the principle of selected transport by using special membranes.
2. Different ions, molecules and resins are used to make the membrane selectively permeable.
3. There are many applications of ISE in almost all fields of life.

Biosensors

**Latest research
Trends**

The background of the slide features a light gray hexagonal grid pattern. Overlaid on the right side is a stylized, blue-toned illustration of a circuit board or sensor array. This illustration includes various rectangular components, some with circular centers, and a network of lines connecting them. At the bottom right, there are molecular-like structures represented by small blue circles connected by lines, suggesting a biological or chemical interface.

Latest Research Trends

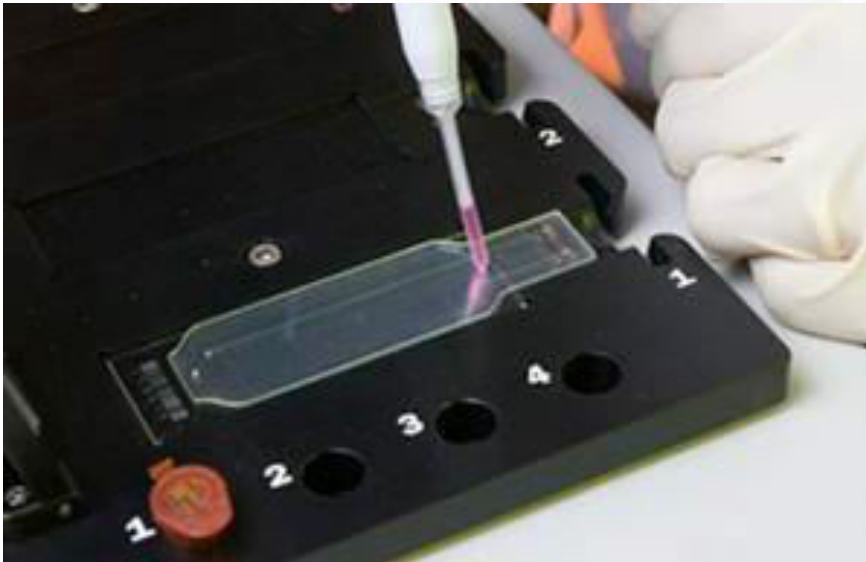
Latest trends in Miniaturization

1. The latest trends in miniaturization of devices are noted in the health sector.
2. The small devices are produced for diagnostics and wearable Biosensors.
3. A huge variety of products including smart watch and tattoos are available in market

Latest Research Trends

Microbial Detection Array

1. It is capable of identifying microbes in a sample devices.
2. This 3-inch long can identify approximately 2,000 viruses and 900 bacteria.
3. The can identify thousands of pathogens in just 24 hours.



<https://singularityhub.com/wp-content/uploads/2010/06/minature-biosensors-LLMDA.jpg>

Latest Research Trends



HIV Detection Chip

Microfluidics-based detection chip which functions to detect HIV.

1. It also determine viral load in patients.
2. It could test diagnose 100 samples at a time.
3. The chip can determine their viral load within 4 hours.

<https://singularityhub.com/wp-content/uploads/2010/06/miniature-biosensors-LLMDA.jpg>

Latest Research Trends



Tattoo Ink

A special tattoo ink that can sense blood sugar levels, pH, and sodium.

When blood sugar goes up, for example, the glucose sensing ink changes from blue to brown.

When salt levels increase, the sodium sensing ink becomes a more vibrant green under UV light.

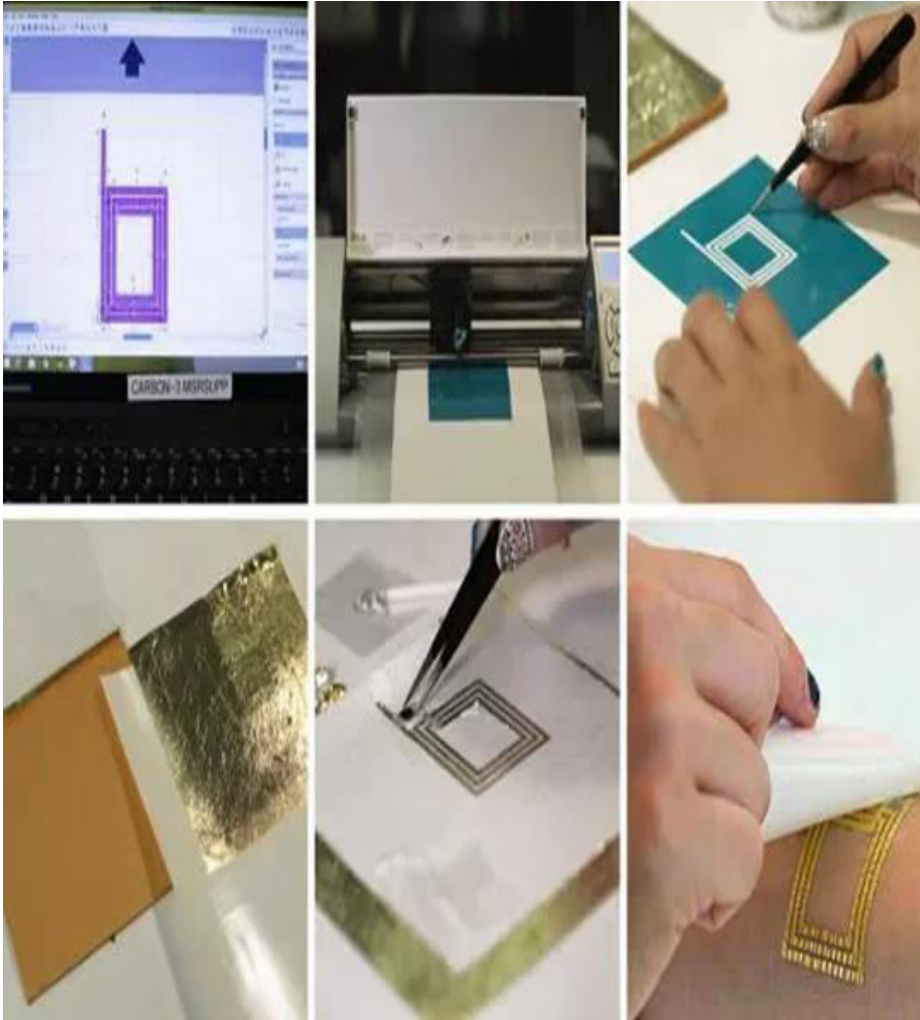
https://www.sciencealert.com/images/articles/processed/dermal-abyss-mit-screenshot-vimeo_1024.jpg

Latest Research Trends

Wearable Tattoo

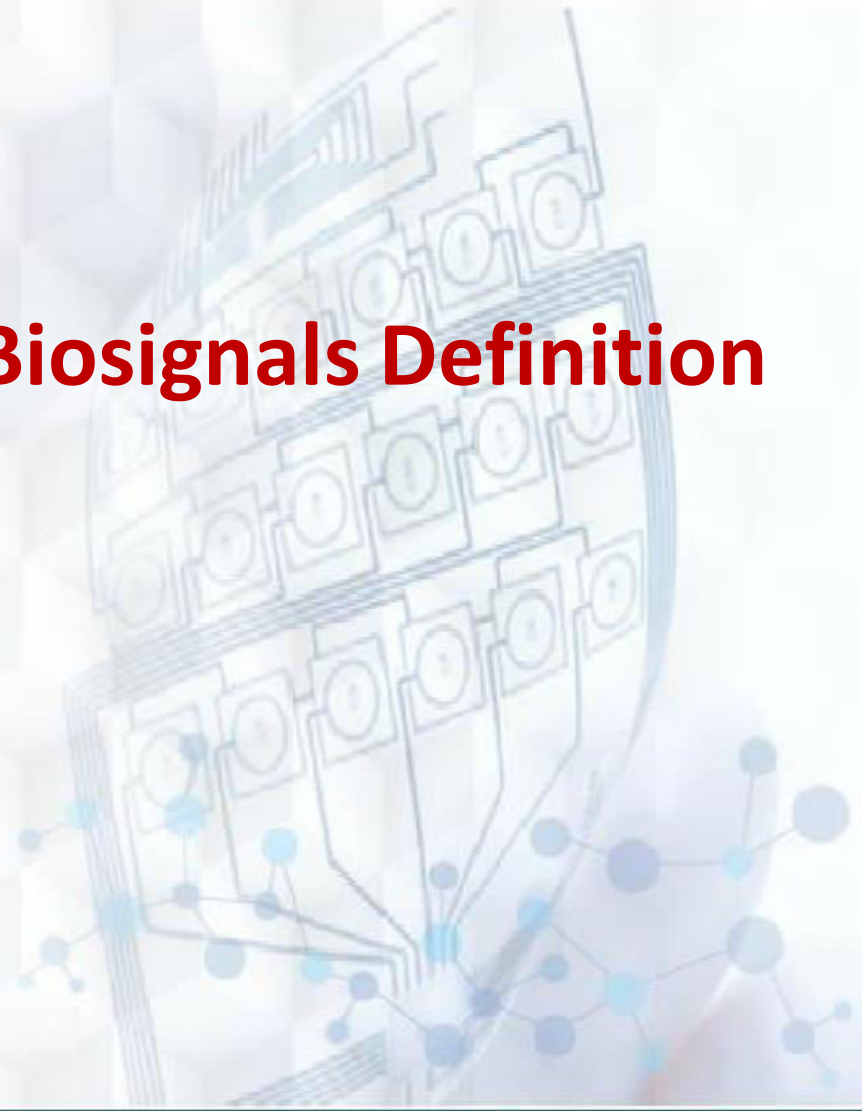
1. First the sketch of the circuit is design with a graphic design software.
2. It is fabricated on conductive gold leaf on top with spray adhesives.
3. The next step is mounting the electronics. The DuoSkin device is planted on the user's skin like a regular temporary tattoo.

https://cms.qz.com/wp-content/uploads/2016/08/9_duoskin_fabrication-process_2.jpg?quality=75&strip=all&w=620&h=374&crop=1



Biosensors

Biosignals Definition

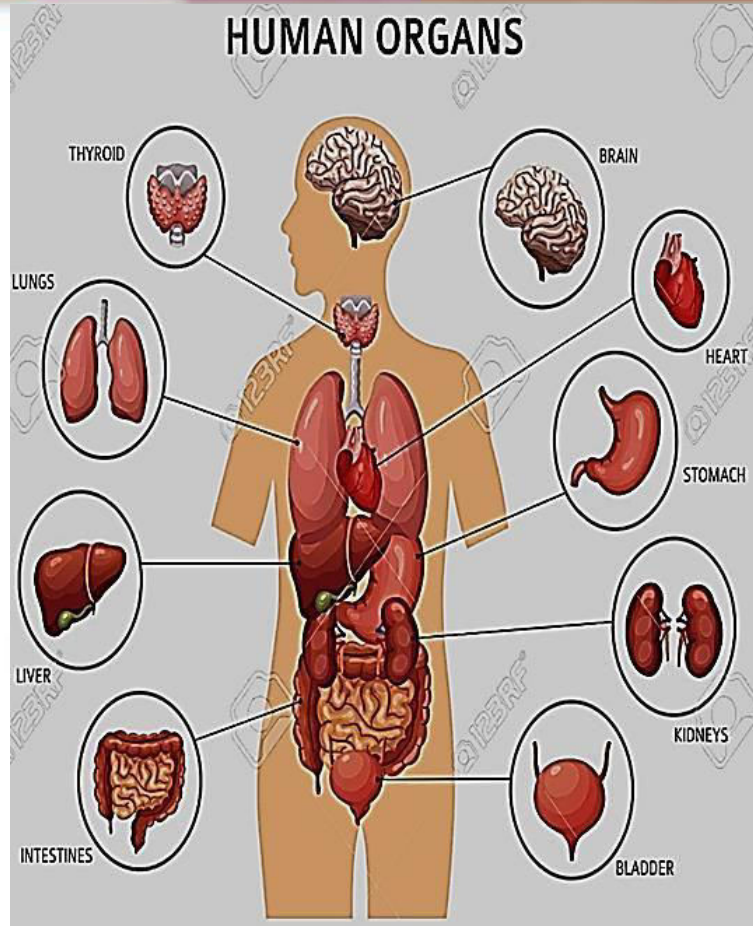


Definition

Signals and Biosignals

1. A signal is a function that conveys information about a phenomenon.
2. The term biosignal is used for living organisms and their function that carries information about their behavior or state”.
3. Biosignals are the key to determine the functions in a living organism.

Definition



<https://previews.123rf.com/images/microone/microone1811/microone181101636/112775864-human-body-internal-organs-stomach-and-lungs-kidneys-and-heart-brain-and-liver-medical-anatomy-vecto.jpg>

Biosignals in Human

1. Biosignals, are record of a biological event in a time frame such as a beating heart, body temperature, hormone level.
2. There are at least eleven organ system in human.
3. Biosignals are the only source of information describing the functioning of the human body in healthy and disease conditions.

Definition



Biosignals in Animals

1. Octopus have richly colored **ink stores** that are used for chemical deterrence, phagomimicry, sensory distraction, and evasion when attacked.
2. Cuttlefish have the ability to change their appearance, **camouflaging** themselves from predators and enabling them to surprise their prey.

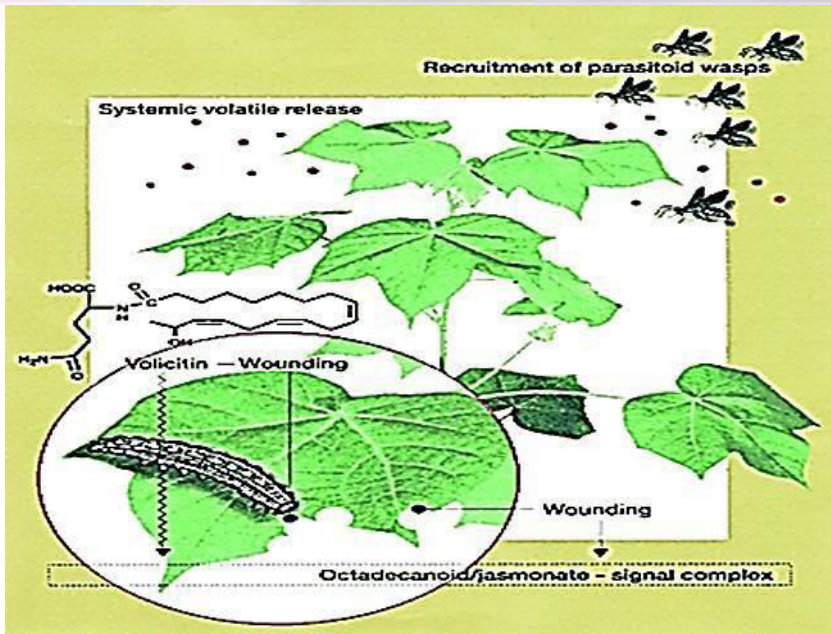


<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQb25JOO-8mHeya-m6-UDnY397j9RwxYFb67xj37biuvkJm4j3oA>

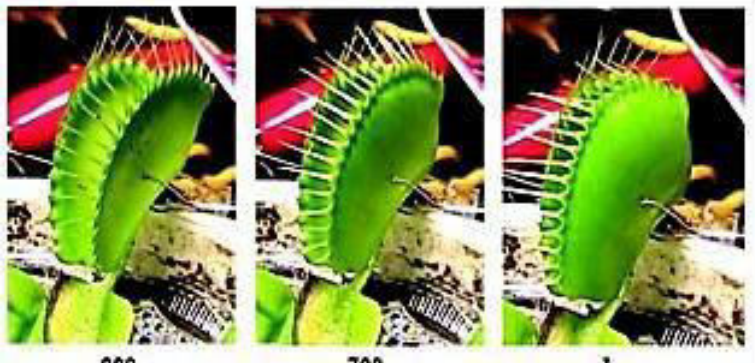
https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcS13fzist9ICpezL3cFmroWYQABWfZ75IDVj6u5-yI9te_nbvCRcw

3. <https://www.google.com/url?sa=i&source=images&cd=&ved=2ahUKEwii5YD6r3jAhWB34UKHXmIBwsQjRx6BAgBEAU&url=https%3A%2F%2Fwww.aqua.org%2Fblog%2F2017%2FOctober%2FCephalopods-Ink-Sacs&psig=AOvVaw2rwukaUMmLric3zcgY9y9Q&ust=1565150962416940>

Definition



<http://www.plantphysiol.org/content/plantphysiol/121/2/325/F1.medium.gif>



Biosignals in Plants

Plants produce certain volatile compounds and can also generate electrical current.

1. Wounded plants produce the volatile odour compounds.
2. The Biological electrical stimulation, is observed in venus flytrap plant.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2634039/bin/psb0203_0139_fig005.jpg

Definition

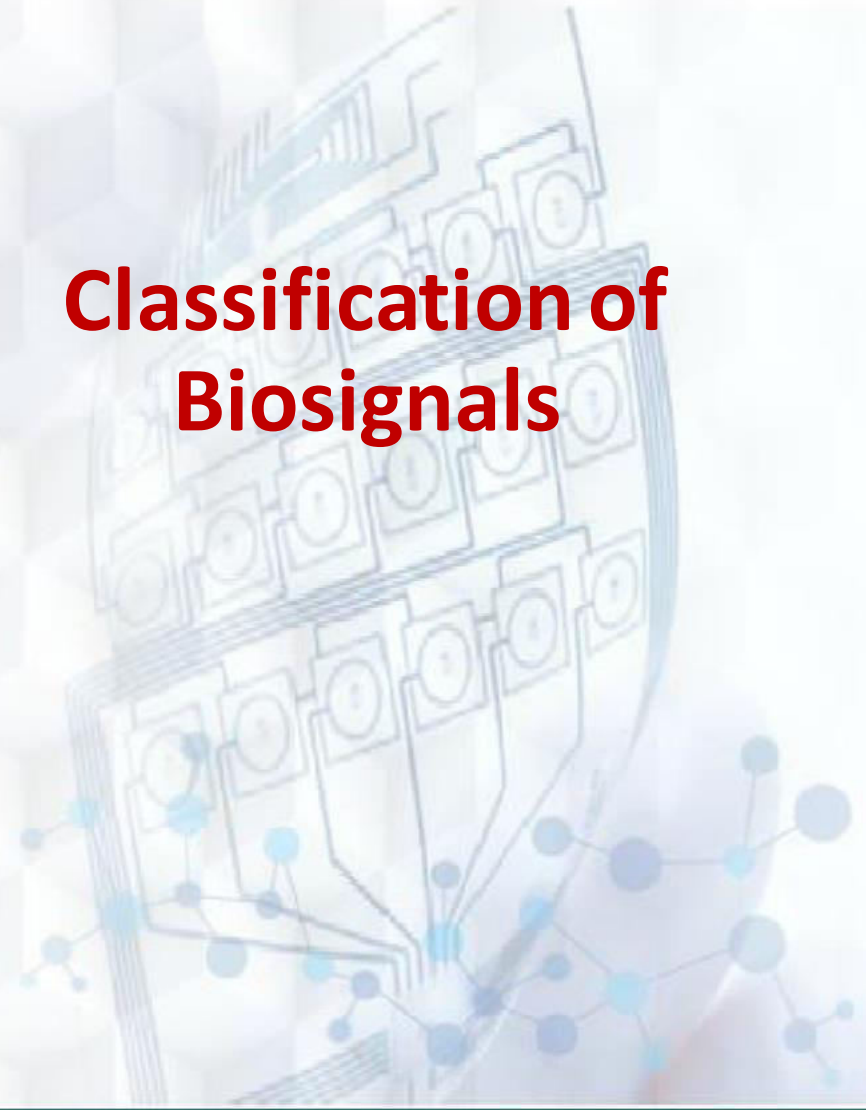
Summary

1. Biosignal are produced by all living organisms. They carry all information about the system.
2. Biosignals are analyzed to gather the information about functioning of the system.
3. Biomedical equipment, are used to measure the generated signals that influence the human body.

END

Biosensors

Classification of Biosignals



Classification of Biosignals

Classification

The types of biosignals are nearly unlimited.

It is difficult to classify them in a single way. There are at least four ways in which we can classify the Biosignals. These classification overlap.

1. Permanent biosignals
2. Induced Biosignals
3. Nature of Biosignal
4. Origin of Biosignals

Classification of Biosignals



Parmanant Biosignals

1. Permanent biosignals exist without any artificial impact, trigger, or excitation from outside the body.
2. They are available at any time. The source of the biosignal is already inside the body.

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRrs8VzoiOiYRekIEYA0TWmV_i3BjqA3FZTwYIGKu6Y-5l2hgEUjA

<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRFZJHO2ivZ3LgzGFGPGoZfzlrVEgae40ggCuKSeqphDWIIT4fA>

Classification of Biosignals



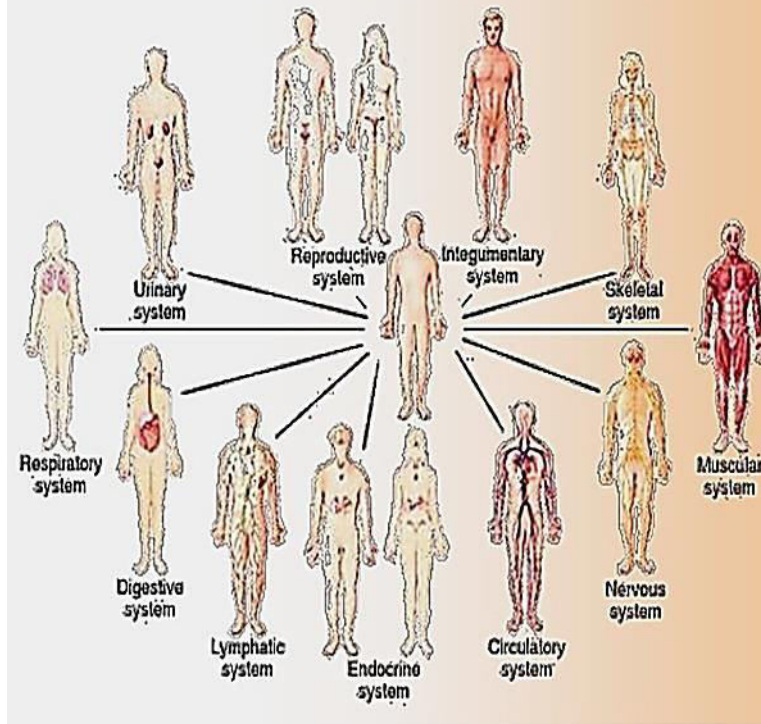
https://www.picclickimg.com/d/l400/pict/202736977188_/3MHZ-Ultrasound-Microcurrent-Spa-Electrotherapy-Beauty-Machine-Facial.jpg

Induced Biosignals

1. The induced biosignals considers are artificially triggered, excited, or induced .
2. Induced biosignals exist roughly for the duration of the excitation.
3. Cosmetic electrotherapy low electric currents passed through the skin to produce several beauty effects.

Classification of Biosignals

Human Organ Systems



<https://image.slidesharecdn.com/08humanorgansystems-131026171958-phpapp02/95/08-human-organ-systems-1-638.jpg?cb=1382808102>

Origin of Biosignals

The classification according to the system of origin of biosignals

1. Endocrine system
2. Nervous system
3. Cardiovascular system
4. Auditory system
5. Musculoskeletal system
6. Respiratory system
7. Gastrointestinal System

Classification of Biosignals

Nature of Biosignals

The classification is based on the physical or chemical nature of the signal.

1. Electrical biosignals
2. Magnetic biosignals
3. Mechanical biosignals
4. Optic biosignals
5. Acoustic biosignals
6. Chemical biosignals
7. Thermal biosignals

Classification of Biosignals

Summary

1. Biosignals are heterogeneous in nature.
2. There is not a single possible way to classify the Biosignals.
3. Several classifications for Biosignals are reported in literature
4. The most convenient classification is based on nature of Biosignals

END

Biosensors

Optic Biosignals



Optic Biosignals

Optic Biosignals

Optical signal is an electromagnetic signal.
Produced by a living cell

1. It has electric and magnetic fields that are orthogonal to each other.
2. These signals are produced by human, animal, plants and bacterial cell.

Optic Biosignals

Bioluminance in Firefly.

Firefly produce optical biosignals called Bioluminance, Using enzyme Luciferase And substrate luciferin.



https://www.researchgate.net/profile/Virgile_Adam/publication/200173355/figure/fig3/AS:669487699333123@1536629714240/4-Mechanism-of-the-firefly-bioluminescence-involving-Luciferase-PDB1LCI-2D1R.jpg

<https://www.creative-enzymes.com/images/Luciferase-2.png>

Optic Biosignals



[https://upload.wikimedia.org/wikipedia/commons/thu
mb/3/32/Marrus_orthocanna_crop.jpg /308px-
Marrus_orthocanna_crop.jpg](https://upload.wikimedia.org/wikipedia/commons/thumb/3/32/Marrus_orthocanna_crop.jpg/308px-Marrus_orthocanna_crop.jpg)

Bioluminescence in marine Animals

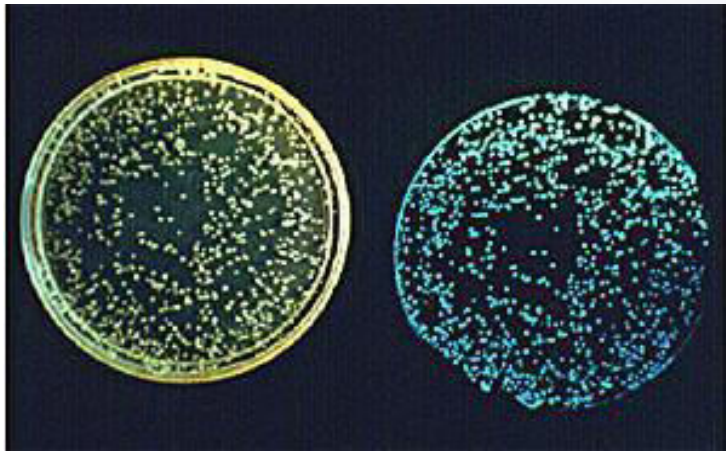
Most jellyfish bioluminescence is used for defense against predators.

Jellyfish such as comb jellies produce bright flashes to shock a predator.

The siphonophores produce a twinkling red lights to attract the small fish as prey.

Luciferase is used for the production of optical signals

Optic Biosignals



Bioluminescence in Bacteria

In cyanobacteria biochemical reaction involves the oxidation of an aliphatic aldehyde by a reduced flavin mononucleotide.

The products of this oxidation reaction produced a blue-green visible light.



https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRNbmG4TUrD3LdmufIxBHooV_IG799DvXntagwgCA4-5cqdUNic-2Q

<http://ncdnaday.org/wp-content/uploads/2016/12/mosquitobay.jpg>

Optic Biosignals

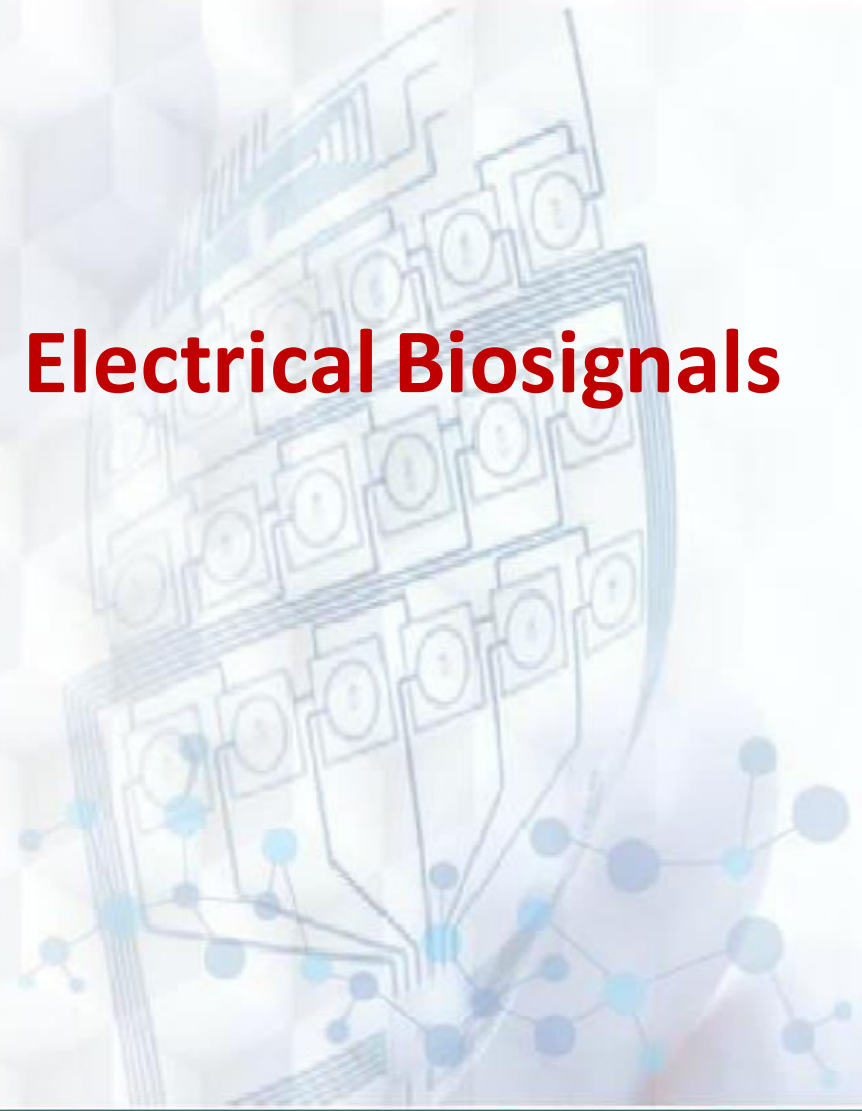
Application of Bioluminescence in Biotechnology

1. The luciferase reporter assay is commonly used as a tool to study gene expression at the transcriptional level.
2. It is widely used because it is convenient, relatively inexpensive, and gives quantitative measurements instantaneously.

END

Biosensors

Electrical Biosignals

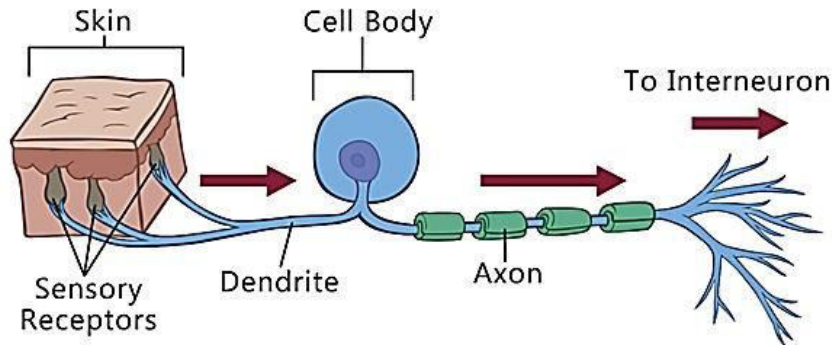


Electrical Biosignals

Bioelectrical Signals

1. Electric field is generated in cells (nerve and muscle) and organs because of intra- and extracellular ionic currents.
2. Typical values of membrane potential are in the range -40 mV to -70 mV .
3. Our body generates electrical signals and control the body functions.

Electrical Biosignals



Internal & External Ion Concentrations

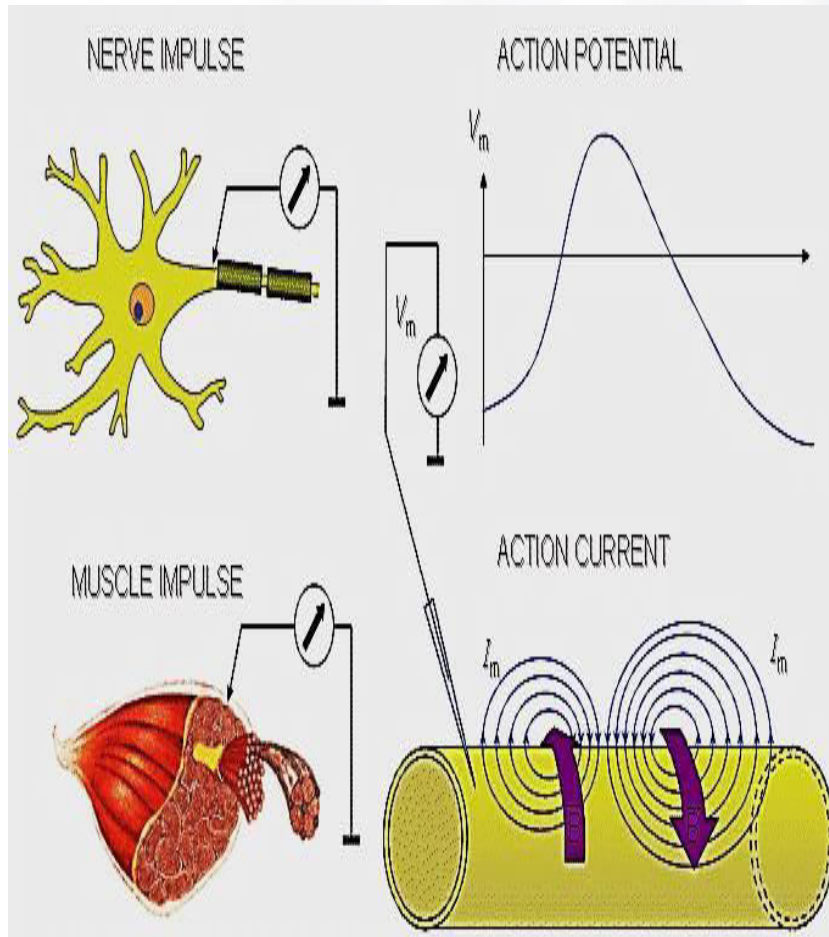
Cell Membrane		
Potassium+	37.5%	1.25%
Sodium+	12.5%	48.75%
Chlorine-	1.66%	50%
Anions-	48.3%	0%

<http://philcatbioflow.com/wp-content/uploads/2019/06/cell-1.jpg>

Bioelectrical Signals

1. Bioelectrical signals are generated from the complex self-regulatory system in human.
2. These signals are very low in frequency and can be measured.
3. Differences in the concentrations of ions on opposite sides of a cellular membrane lead to a voltage called the membrane potential.

Electrical Biosignals



<http://www.bem.fi/book/02/fi/0211.gif>

Types of Electrical signals

1. Neural cells

- a) Electroneurogram ENG
- b) Electroencephalogram EEG
- c) Electroretinogram ERG

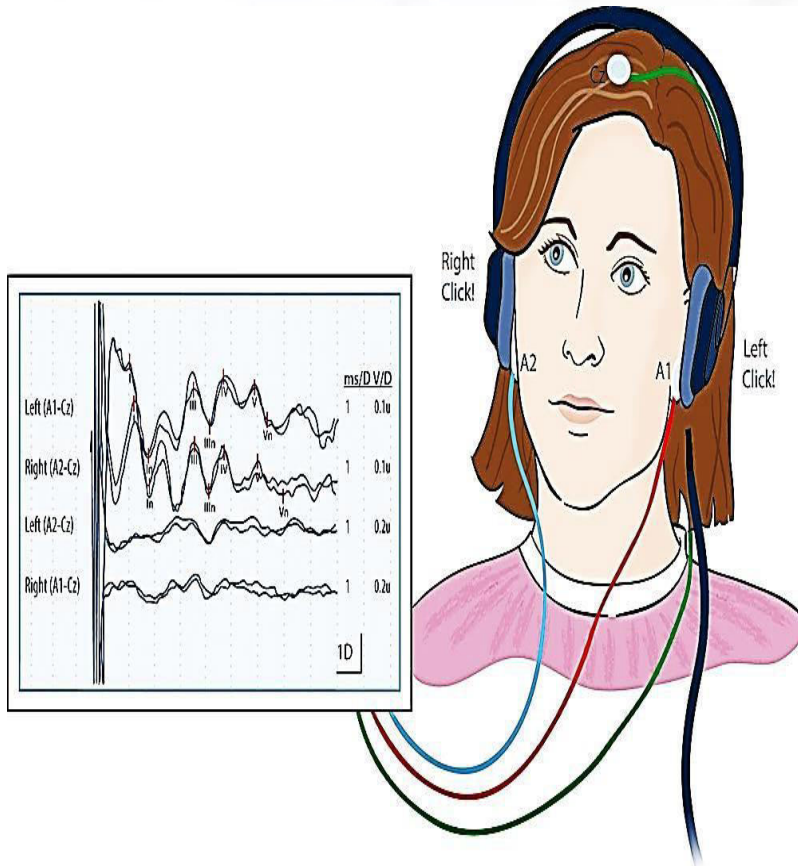
2. Muscle cells

- a) Electrocardiogram ECG
- b) Electromyogram EMG

3. Other cells

- a) Electrooculogram EOG
- b) Galvanic skin response GSR

Electrical Biosignals

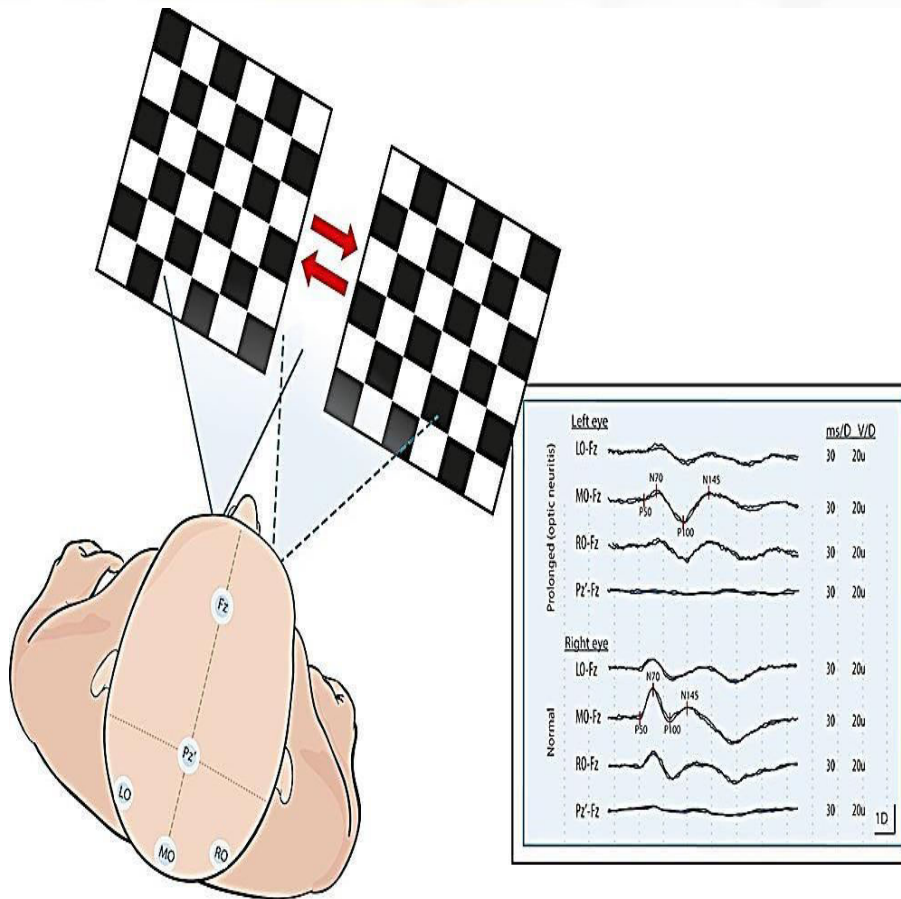


https://tidsskriftet.no/sites/default/files/styles/inline_original_custom_user_desktop_1x/public/2013--T-12-1176-01-ENG-Over.jpg?itok=kTaXLW2d

The brainstem auditory evoked potential (BAEP)

1. The brainstem auditory evoked potential (BAEP), or brainstem auditory evoked response (BAER), measures the functioning of the auditory nerve and auditory pathways in the brainstem.
2. The auditory paths are stimulated by means of clicks delivered through headphones.

Electrical Biosignals

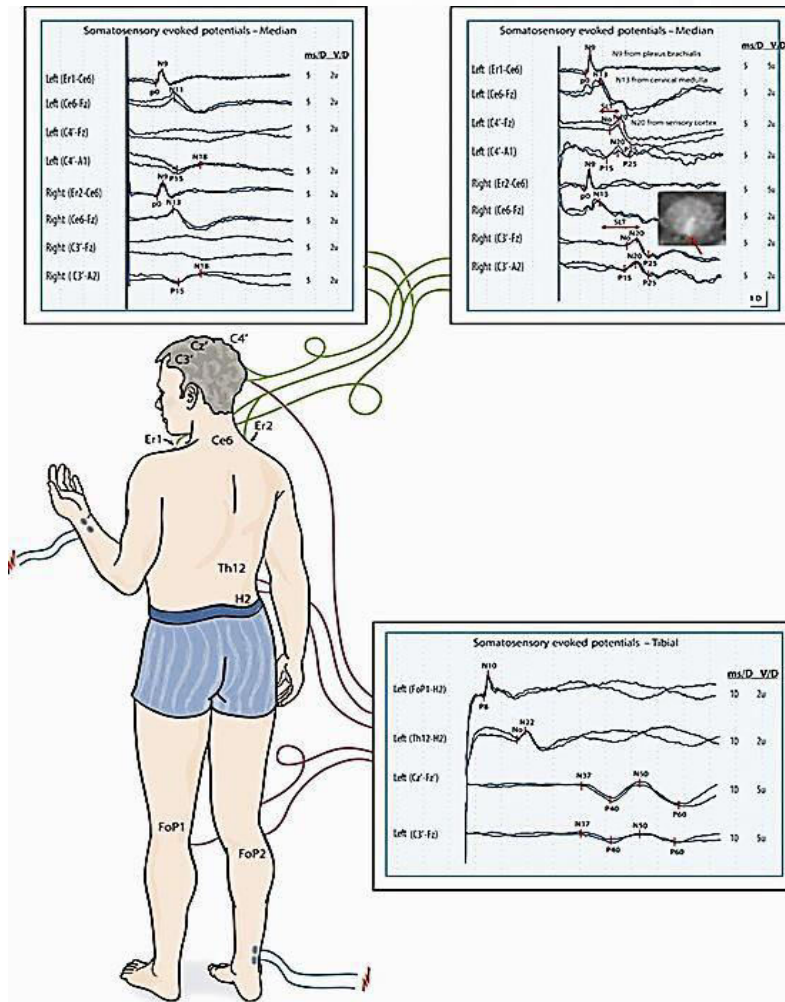


https://tidsskriftet.no/sites/default/files/styles/inline_original_custom_user_desktop_1x/public/2013--T-12-1176-01-ENG-Over.jpg?itok=kTaXLW2d

Visual Evoked Potential

1. A visual evoked potential is an evoked potential caused by a visual stimulus, such as an alternating checkerboard pattern on a computer screen.
2. It detects the mechanical or neural abnormalities related to vision.
3. These responses usually originate from the occipital cortex.

Electrical Biosignals



SOMATOSENSORY EVOKED POTENTIALS (SEP)

1. An somatosensory evoked potentiation test (SEP) studies the relay of body sensations to your brain and how the brain receives those sensations.
2. A stimulating electrode is placed on the arm or leg, and it generates an electrical signal.

https://tidsskriftet.no/sites/default/files/styles/inline_original_custom_user_desktop_1x/public/2013--T-12-1176-01-ENG-Over.jpg?itok=kTaXLW2d

Electrical Biosignals

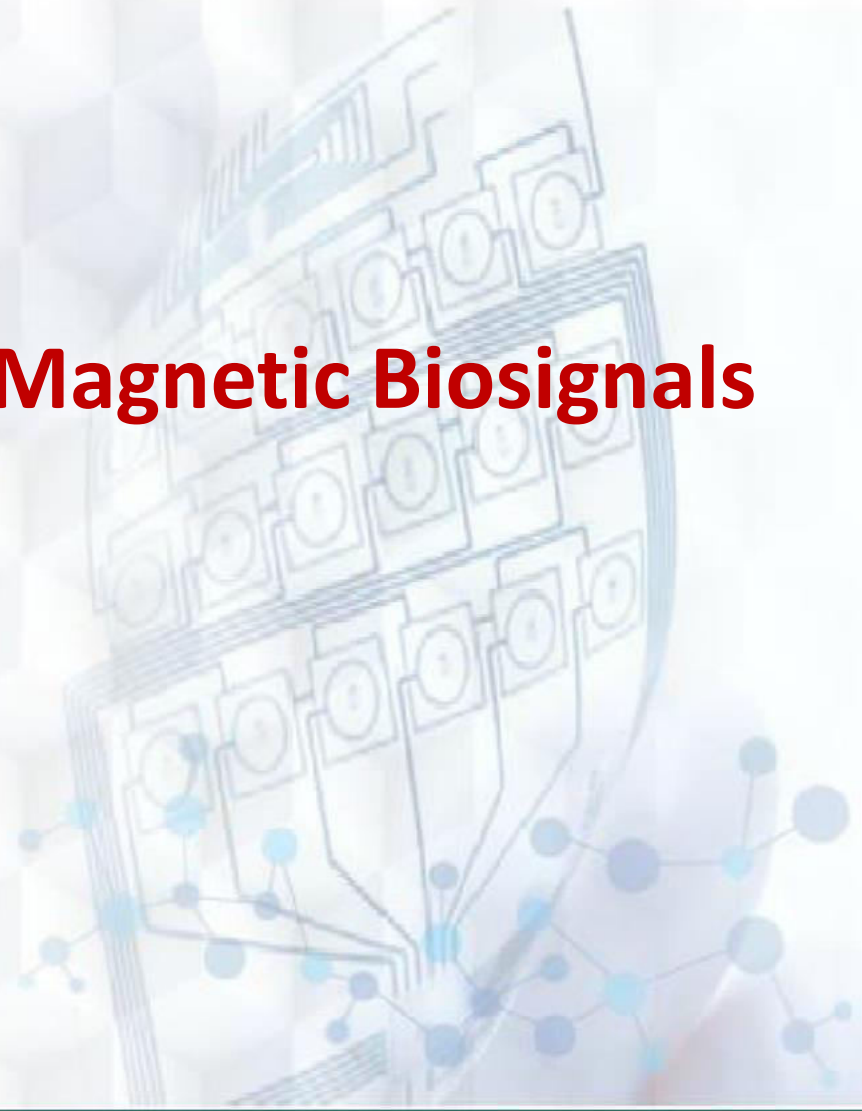
SUMMARY

1. Electrical biosignals are generated in our body by different systems .
2. The electrical Biosiganals are determined by highly sophisticated latest instruments, to check the body function.
3. Body functions are checked by natural or induced electrical Biosignals.

END

Biosensors

Magnetic Biosignals



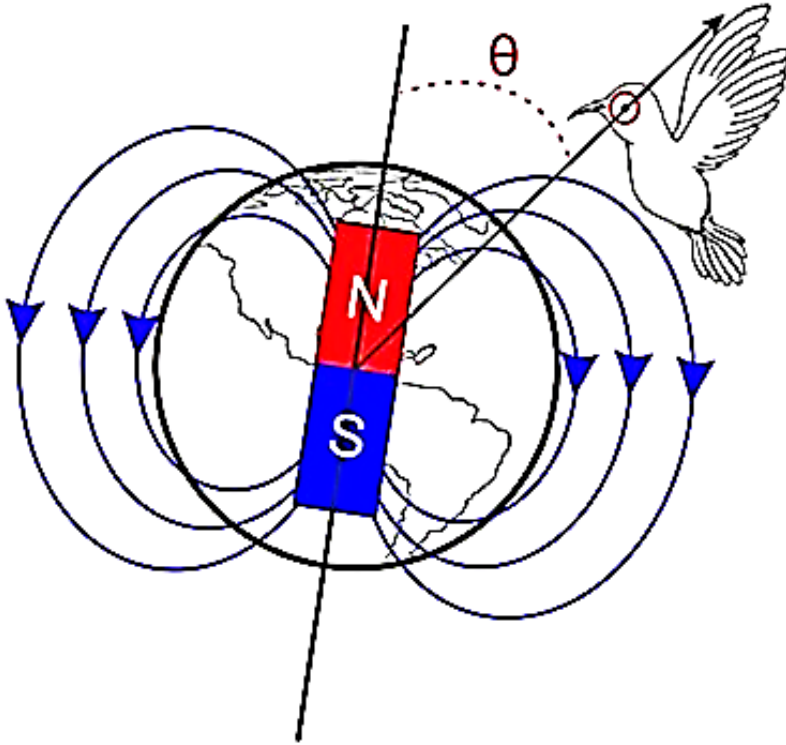
Magnetic Biosignals

1. Magnetic forces is one of the fundamental forces.
2. It plays a role in the life of organism on the earth planet.
3. Migration of birds is an important phenomenon in which magnetic forces play a role.

Magnetic Biosignals

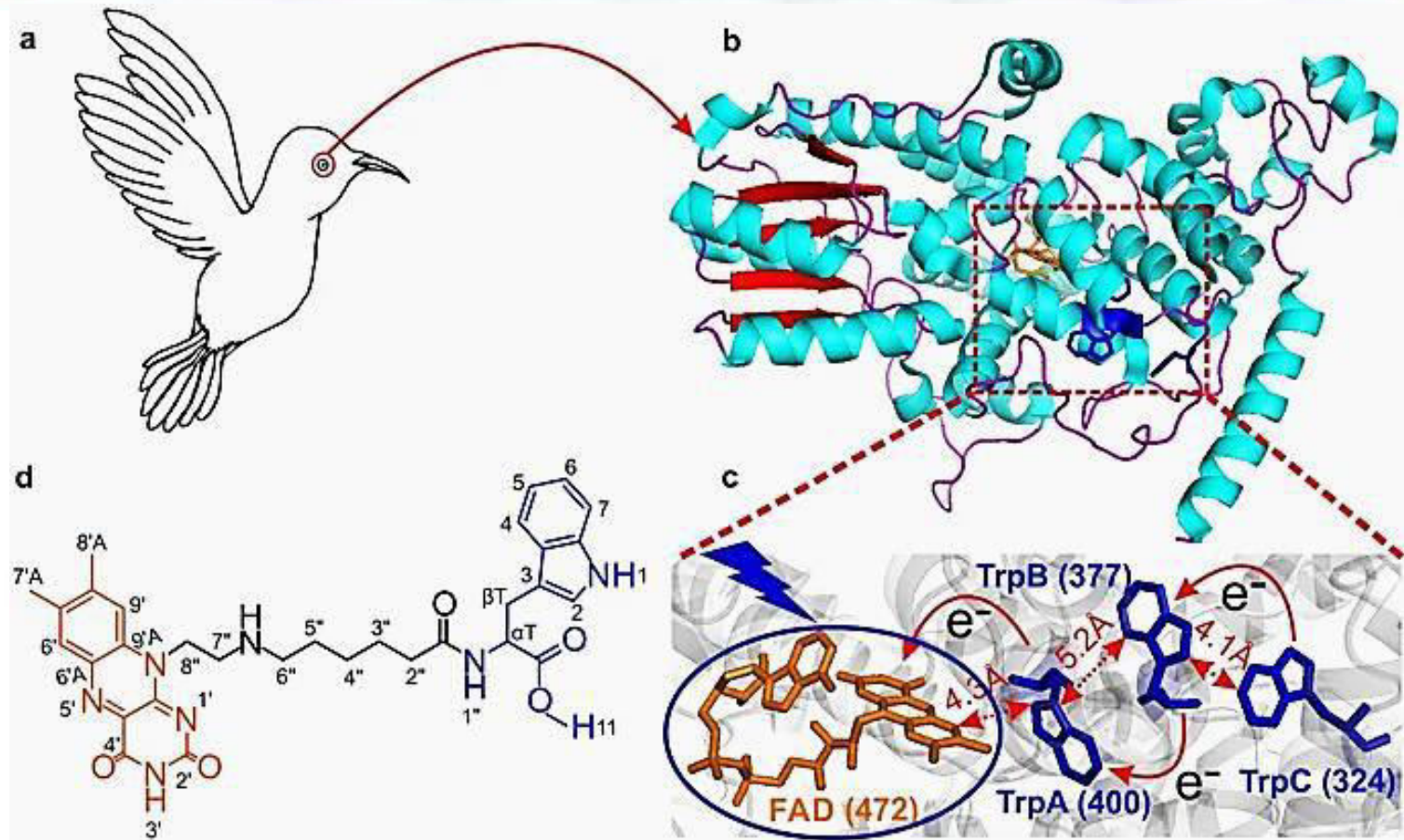
1. Humans are not believed to have a magnetic sense.
2. Human CRY2, which is heavily expressed in the retina, is supposed to have magnetic properties.
3. The results show that human CRY2 has the molecular capability to function as a light-sensitive magnetosensor.

Magnetic Biosignals



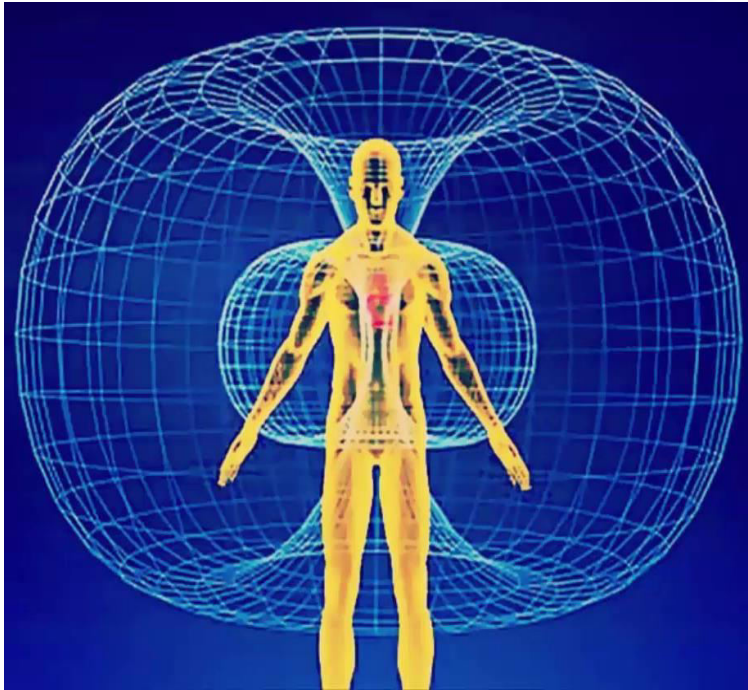
1. Birds sense the Earth's magnetic-field and use it for navigation.
2. A light-dependent quantum effect in cryptochrome proteins is produced.
3. They are present in retina.
4. It allows for an iron-free spin-chemical compass.

Magnetic Biosignals



https://media.springernature.com/lw685/springer-static/image/art%3A10.1038%2Fs41598-017-10356-4/MediaObjects/41598_2017_10356_Fig1_HTML.jpg?as=webp

Magnetic Biosignals



https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQPQXYS_WwxBvmvY6MoAkuSsoxrUwGZizraaMi21apSND0k_wjJ

Electromagnetic biosignals in Human

1. The human body's own internal magnetic fields are generated by the internal electrical activity.
2. Nerve impulses are electrical energy signals; and, they create energy-fields around the body.
3. The electro-magnetic energy waves that can travel away from the body.

Magnetic Biosignals

Magnetic biosignals Applications

Magnetocardiography (MCG)

Magnetocardiography (MCG)

1. Is a technique to measure the magnetic fields produced by electrical currents in the heart.
2. It uses extremely sensitive devices such as the superconducting quantum interference device (SQUID).

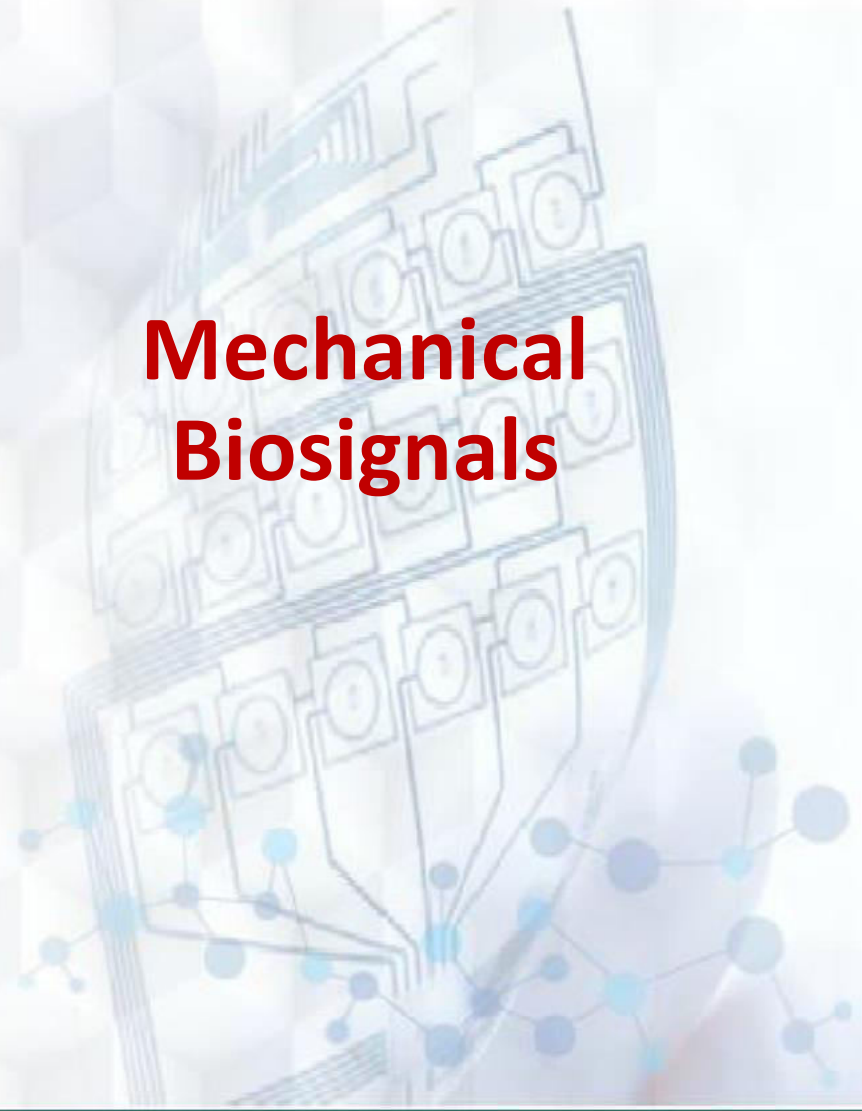
Magnetic Biosignals

Summary

1. Electromagnetic field is generated in humans under the influence of electrical biosignals.
2. In birds and humans, a light-dependent quantum effect in cryptochrome proteins, expressed in the retina, acts as an iron-free spin-chemical compass.
3. Tryptophan residues play an important role in their functioning.

Biosensors

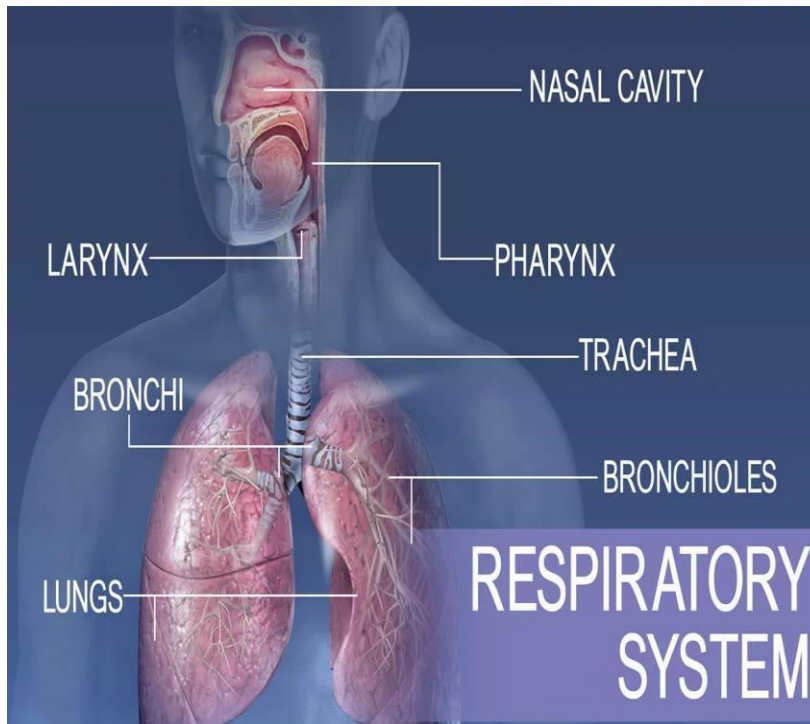
Mechanical Biosignals



Mechanical Biosignals

1. Mechanical biosignals are those which are related to displacement and change of position or vibration using a mechanical force by the organism.
2. Mechanical signals could effect be expansion as in case of respiration.
3. The mechanical signal could lead to a specific displacement as in case of honey bee.

Mechanical Biosignals

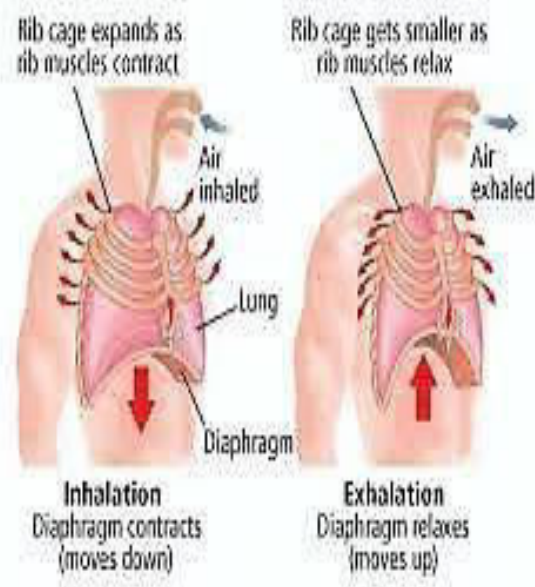


https://www.visiblebody.com/hs-fs/hub/189659/file-2545156127-jpg/Learn_Articles/Respiratory_System/Set_1_Five_Functions_of_Respiratory_Sys/1C-Respiratory-System.jpg

Respiratory system

1. Respiratory system is an excellent example of mechanical Biosignals, that are produced by inhalation and exhalation.
2. Lungs are mainly composed of an elastic functional protein Elastin.
3. The system facilitates the exchange of gases between the air and the blood .

Mechanical Biosignals

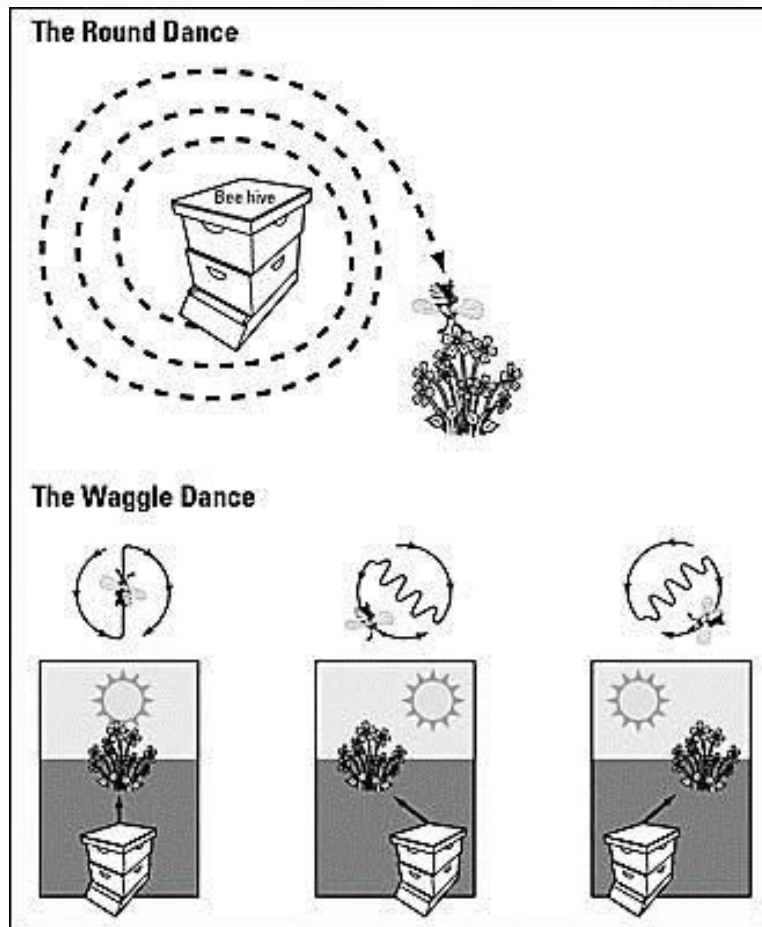


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Inspiration and Expiration

1. During inspiration, the diaphragm contracts and pulls downward.
2. The muscles between the ribs contract and pull upward.
3. During expiration, the diaphragm relaxes.
4. The lungs contract and air is forced out.

Mechanical Biosignals



<https://www.dummies.com/wp-content/uploads/248640.image0.jpg>

Honey bee signals.

1. Honey bee workers perform a series of movements (shape of 8) the "waggle dance".
2. It tells the location of food sources more than 150 meters from the hive.
3. A round dance in which she moves in close circles, alternating right and left.
4. It shows that the food source close to the hive.

Mechanical Biosignals



https://www.sciencenews.org/sites/default/files/2017/11/main/articles/110917_HT_pigeon-feathers_main.jpg

Feather movement in Birds

1. The pigeons flap their feathers use their feather as an alarm signal. This is a warning signal for the presence of predators.
2. Male peafowl perform a complicated dance in which they tilt, they shake their feathers and make dazzling moves.

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQefc8PsZGbRI2GZegJbQeZResqBQ-504tE01r1Hb4hV8_HeS6D

Mechanical Biosignals



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTSbKhhlZkOxuzV3y-pYSZ-FP45u-8KpDC51Dp1XHsfoOHXRNzn>



https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRlzM1puq5fWV0du4xbSZ60HWL1YawfXsKsZWbfV92_2EP3tmr_9A

Mechanical Biosignals in plants

1. **Sunflower Flowers** are tuned to sense changes in light and temperature, as the day progresses. It is related to their growth.
2. **Night blooming flowers** are heavily scented. They use their fragrance as signals to attract the pollinator at night.

Mechanical Biosignals

Summary

1. Mechanical biosignals are the signals produced by the living organisms using a mechanical force.
2. Respiration is an excellent example of mechanical biosignals.
3. The specific movements, indicates a the involvement of bio clock.
4. Animals show specific body moves.

Biosensors

Acoustic Biosignals



Acoustic Biosignals

Acoustic biosignals

Biological sounds, such as

1. Lung sounds
2. Heart sounds
3. Bowel sounds
4. Joint sounds

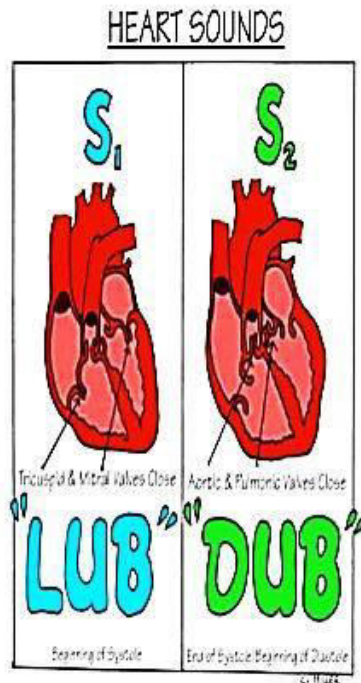
emitted from the human body during its functioning provide an easy and noninvasive way of indicating existing pathology.

Acoustic Biosignals

Heart Sound

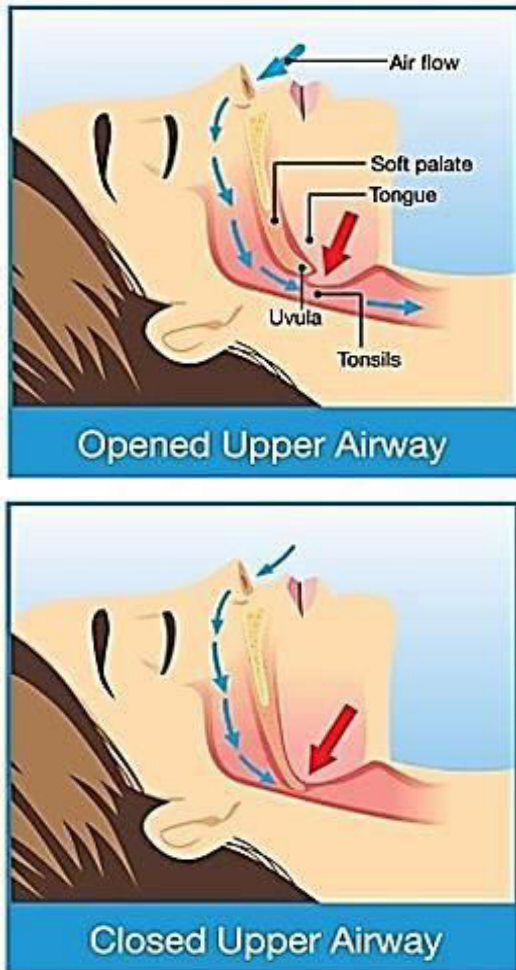
In healthy adults, there are two normal heart sounds, often described as a lub and a dub (or dup), that occur in sequence with each heartbeat.

These are the first heart sound (S1) and second heart sound (S2), produced by the closing of the atrioventricular valves and semilunar valves, respectively.



<http://www.sandipointe.com/im/baby/heart-sounds-5.jpg>

Biosignals

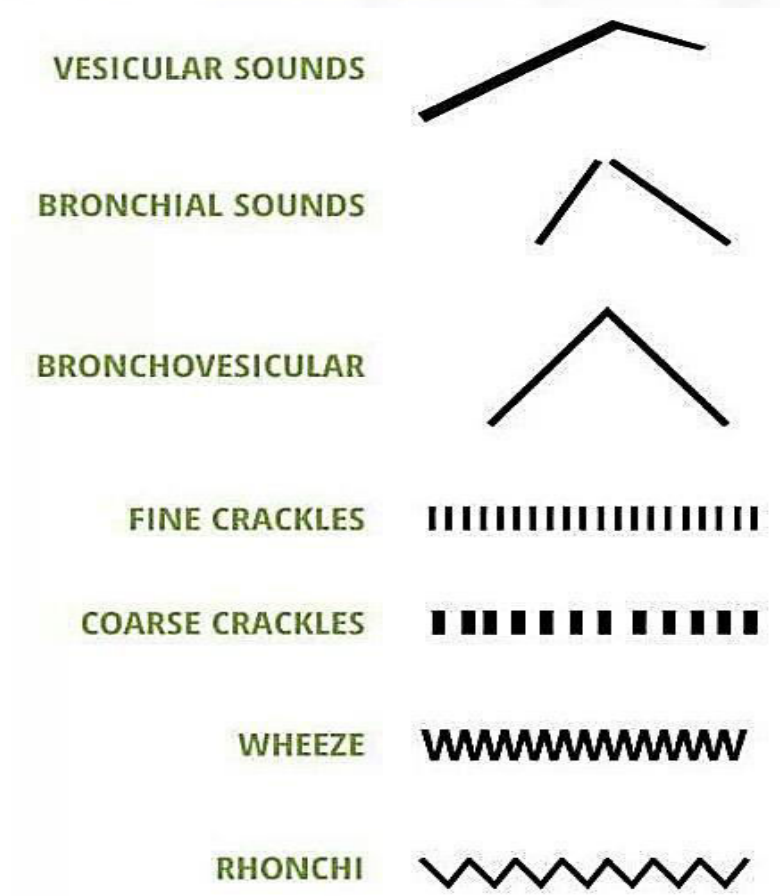


Snoring Sounds

1. Snoring is the vibration of respiratory structures and the resulting sound due to obstructed air movement during breathing while sleeping.
2. Snoring is the vibration of respiratory structures and the resulting sound due to obstructed air movement during breathing while sleeping.

<https://www.news-medical.net/health/Types-and-Causes-of-Snoring.aspx>

Acoustic Biosignals



Respiratory sounds

Normal breath sounds are classified as tracheal, bronchial, broncho vesicular, and vesicular sounds.

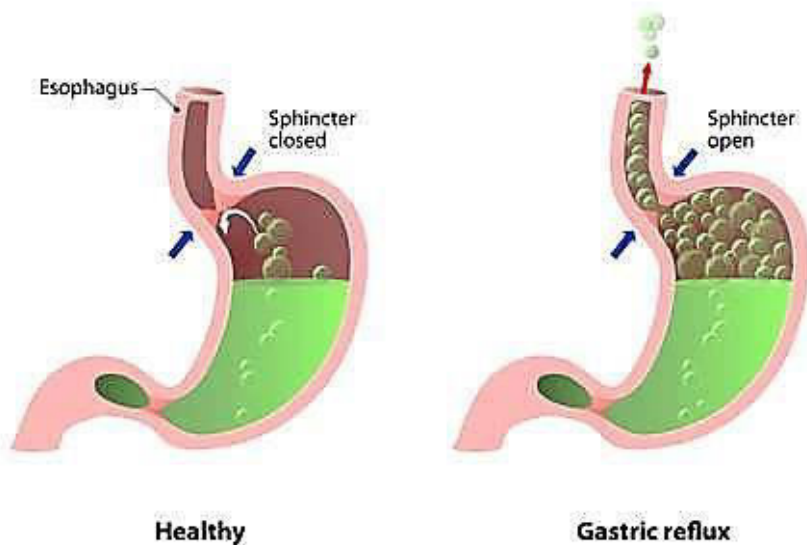
1. Breath sounds are described by: duration)
2. Intensity
3. Pitch timing (when the sound occurs in the respiratory cycle).

<https://www.pinterest.com/pin/404549978999931193/?lp=true>

Acoustic Biosignals

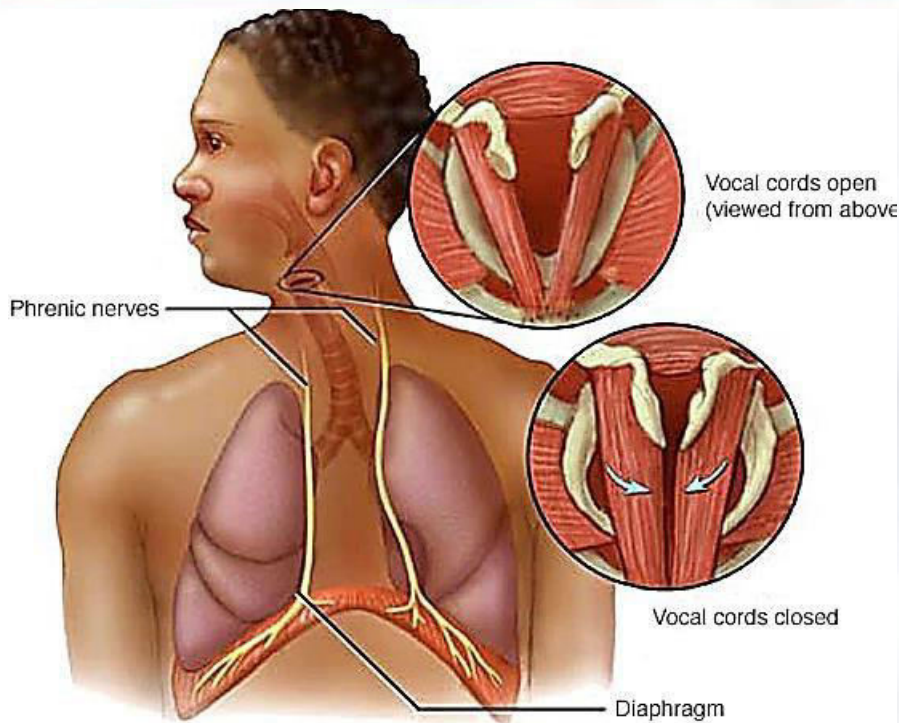
Burp Sound

1. A burp is a belch, or the loud release of trapped air loudly through your mouth.
2. A reflex is initiated, leading to relaxation of the lower esophageal sphincter, upward movement of the air through the esophagus.
3. It passes through the upper esophageal sphincter, during which an audible belch can sometimes be heard.



https://www.123rf.com/photo_59288388_stock-vector-stomach-disease-belching-heartburn-or-reflux-the-illustration-stomach-esophagus-and-sphincter-human-.html

Acoustic Biosignals



Hiccups

1. Hiccups are sudden, involuntary contractions of the diaphragm muscle.
2. As the muscle contracts repeatedly, the opening between the vocal cords snaps shut to check the inflow of air and makes the hiccup sound.
3. Irritation of the nerves that extend from the neck to the chest can cause hiccups.

https://www.mayoclinic.org//media/kcms/gbs/patientconsumer/images/2013/08/26/10/59/ds00975_im03070_r7_hiccupsthu.jpg

Acoustic Biosignals

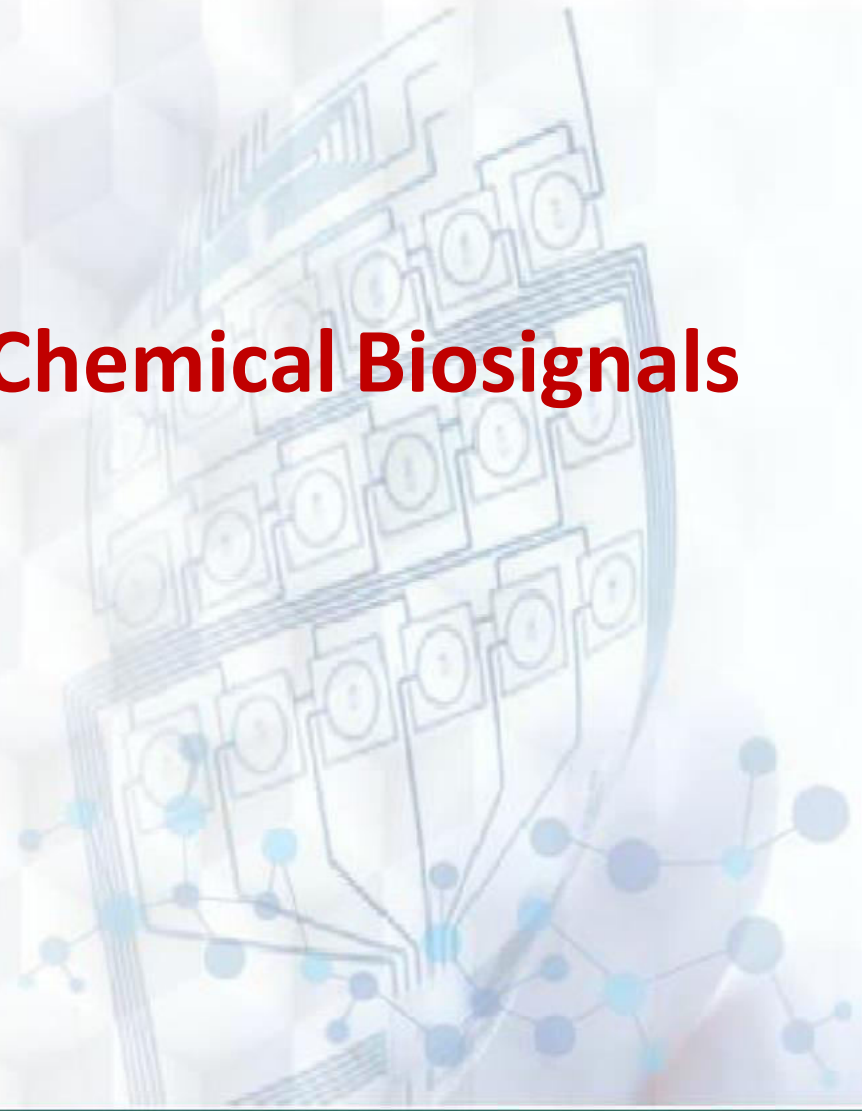
Advantages

1. The current knowledge of Acoustic Biosignals has led to the development of many techniques.
2. Several instruments are developed for diagnostic which are very helpful in for advancement of medical bioengineering

END

Biosensors

Chemical Biosignals



Chemical Biosignals

Chemical signals could be divided into two types.

Chemical signals produced inside the cells in response to an internal stimuli. For example

1. Level of glucose - Blood oxygen level (heart and kidney failure)
2. Gases in blood and breathing airflow
3. -pH

Chemical Biosignals

1. Second types of Biosignals are Chemical signals produced inside the cells in response to an external stimuli.
2. These chemical are volatile compounds and are released in the environment.
3. These signals are produced by all type of organisms bacterial, fungal, plants, animals.

Chemical Biosignals

Chemical biosignals produced by Plants

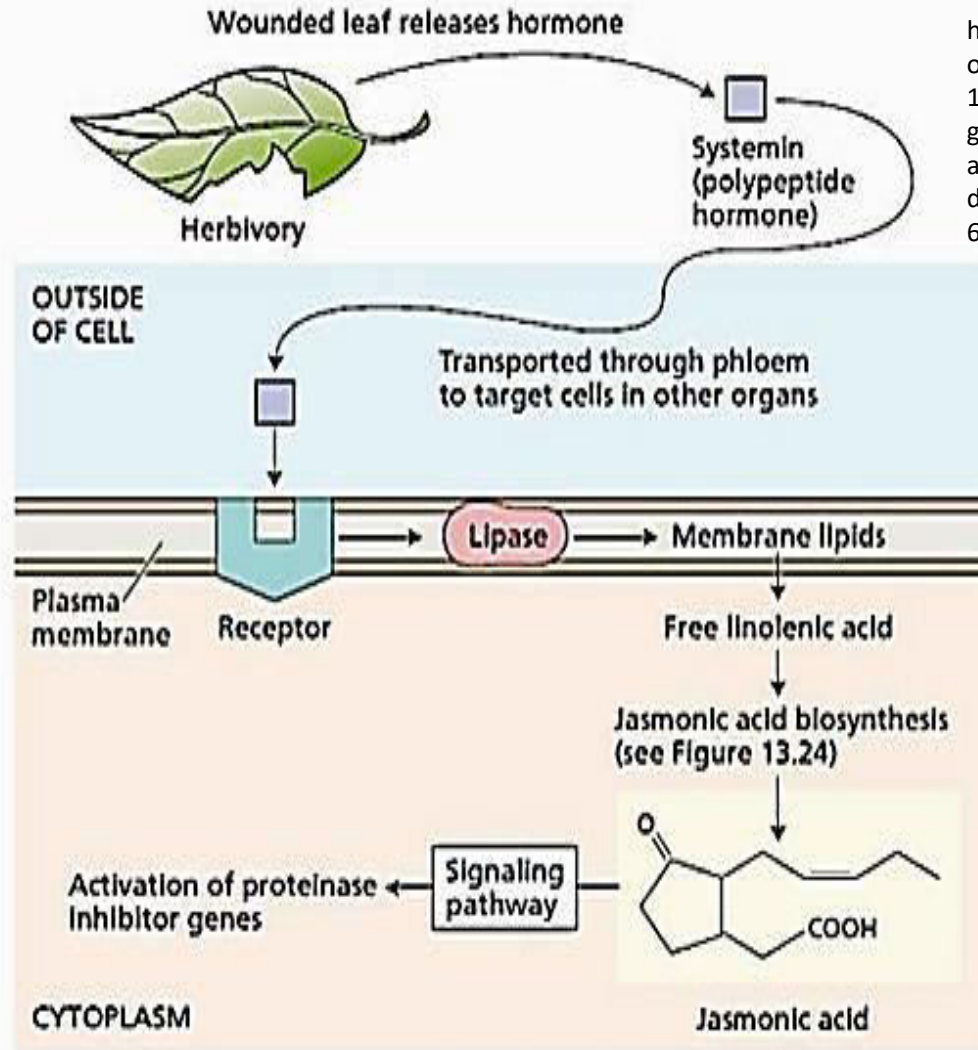
When a plant is damaged by herbivorous insects, volatiles are released in defense.

These volatiles attack both parasitic and predatory insects that are natural enemies of the herbivores. They may also induce defense responses in neighboring plants.

Chemical Biosignals

1. In response to insect feeding, volatile compounds are released by plants.
2. Jasmonic acid and its derivatives are released under stress and activate the genetic expression of protease inhibitors in the plants.
3. Jasmonic acid is also released into the environment.

Chemical Biosignals



<https://image.slidesharecdn.com/seminarmodified-150305132223-conversion-gate01/95/role-of-jasmonic-acid-in-plant-development-defence-mechanism-11-638.jpg?cb=1457590130>

Chemical Biosignals



<https://www.theguardian.com/science/2017/aug/09/why-do-cephalopods-produce-ink-and-what-on-earth-is-it-anyway#img-3>

1. The cephalopods release ink to create floating glowing blobs to create a distraction to predators.
2. It is Mucus-rich ink is a dangerous or annoying substance that interferes with fish gills.
3. The blue-ringed octopus has deadly toxin named tetrodotoxin, is present in its ink.

Chemical Biosignals



https://i.kinja-img.com/gawker-media/image/upload/s--4IMapg_x--/c_scale,f_auto,fl_progressive,q_80,w_800/18it0b0r1so0djpg.jpg

Chemical biosignals by Animals

1. Sea hares are known for the colorful, sticky ink they let loose when knocked around by hungry predators.
2. The substance is a mixture that prevents the odor molecules from getting to the sensory receptors,".

Chemical Biosignals

Chemical biosignals **Pheromones**

They are secreted outside the body, and they influence the behavior of another individual.

Pheromones may be present in all bodily secretions but most attention has been geared toward axillary sweat which contains the odorous 16-androstenes.

Pheromones are similar to hormones but work outside of the body.

Chemical Biosignals

Summary

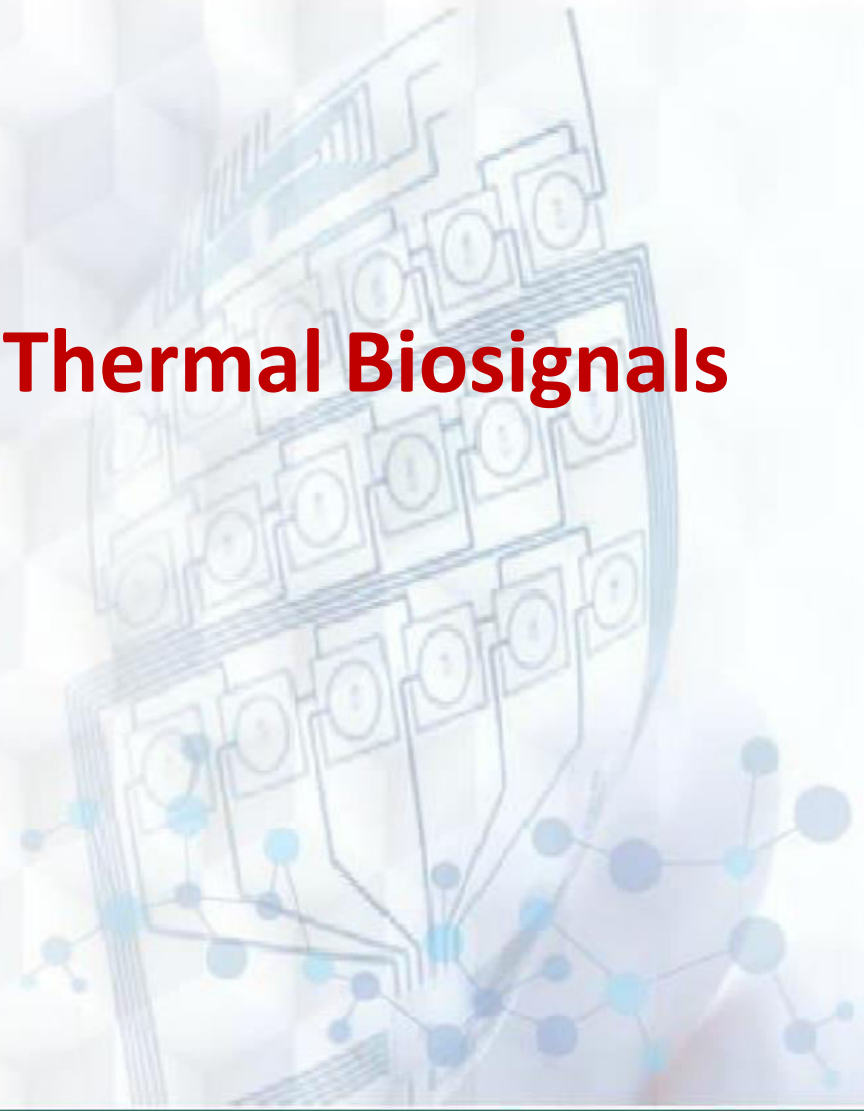
Different types of chemicals are produced by the living organisms, under different environmental and physiological conditions.

These molecules includes

1. Pheromones
2. Volatile organic compounds
3. Ink

Biosensors

Thermal Biosignals



Thermal Biosignals

Thermal Biosignals

Thermal biosignals are related to heat absorption or the release of heat.

Both of these actions leads to increase or decrease in temperature of the system.

The phenomena are the indication of a physiological change.

A change in the temperature is a signal for the change.

Thermal Biosignals

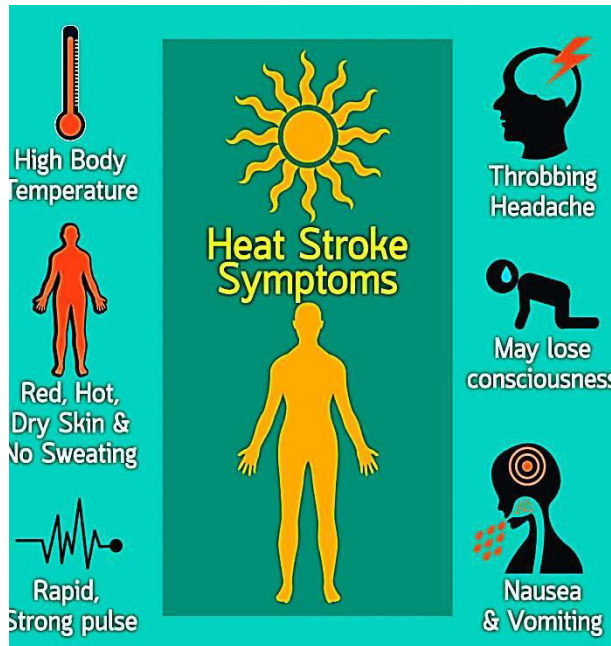
Thermoregulation

Thermoregulation is a process that allows our body to maintain its core internal temperature.

All thermoregulation mechanisms are designed to return human body to homeostasis.

The average person has a baseline temperature between 98°F (37°C) and 100°F (37.8°C).

Thermal Biosignals



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSckqFpMugkGIZ4OHkPC6f5kD7LtIX8MPji-MeOvtpoqDBJRVCiKQ>

Increase in temperature

Factors that can raise your internal temperature include:

1. Fever
2. Heat stroke

Body temperature could also be increased by keeping an onion in armpit.

Onion contains sulfur compounds which then by touching our body and produces heat.

<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQ8fIQlwJREoxDg3qpWoVnmYckAZFVmsGaC25ZSoOM4ZKBP-sEF>

Thermal Biosignals



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSXlgJ4H1I9wsDD0VOZ3M7rQiSQJApSvjvevzau7XmDmoGirZc5xg>

Lowering of temperature

Factors that can lower the body temperature include

1. Metabolic conditions, including under-functioning thyroid gland.
2. Cold sponging
3. Hypothermia occurs as body temperature falls below 35 °C.
4. After death body begins to cool at a set rate: about 1.5 degrees ° F an hour.

https://www.google.com/url?sa=i&source=images&cd=&ved=2ahUKEwjku3M9OrjAhXOAGMBHTn4D_0QjRx6BAgBEAQ&url=https%3A%2F%2Fwww.wikihow.com%2FTreat-a-Fever&psig=AOvVaw17X4HGLAIPV28Lt3XwSIDf&ust=1565065126530178

Thermal Biosignals

Applications of Thermal Signals

1. The normal temperature of skin is about 33 °C or 91 °F.
2. The temperature of the skin varies slightly throughout the body.
3. Inflamed part of the body has higher temperature.
4. There is a relationship between inflamed area and the temperature.

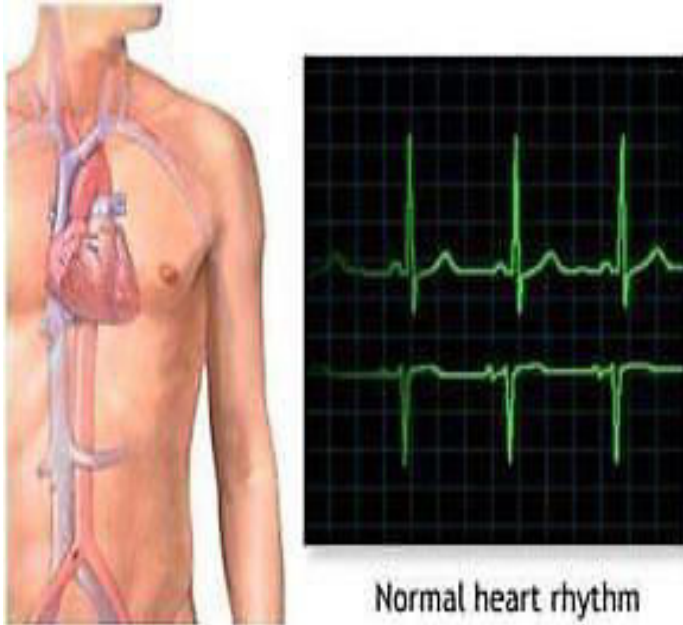
END

Biosignal

**Latest Trend and
Current Research**

The background of the slide features a light gray hexagonal grid pattern. Overlaid on the right side is a faint, blue-toned illustration of a circuit board or microchip, showing various components and connecting lines. At the bottom right, there is a network diagram consisting of blue circles of varying sizes connected by thin lines, resembling a molecular structure or a data network.

Latest Trends and Current Research



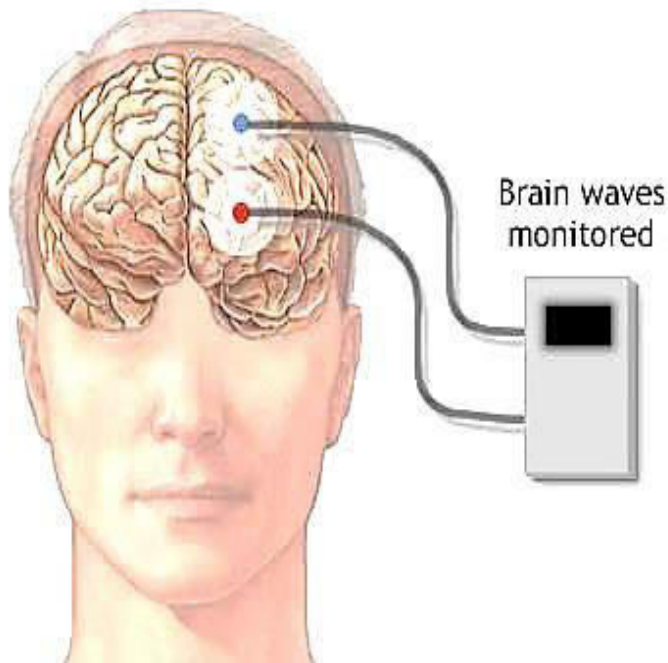
<https://www.ashpreetpro.com/wp-content/uploads/2018/08/ECG-300x189.jpg>

Measurements of electrical Biosignals

Electrocardiogram (ECG)

1. The graphical recording of the electrical activity of the heart and measured on the skin surface of the body
2. One of the main methods for assessing heart functions.
3. P wave: Atrial depolarization
QRS complex: Ventricular

Latest Trends and Current Research



<http://www.leehealth.org/healthinformation/HIE%20Multimedia-TextOnly/1/003931.htm>

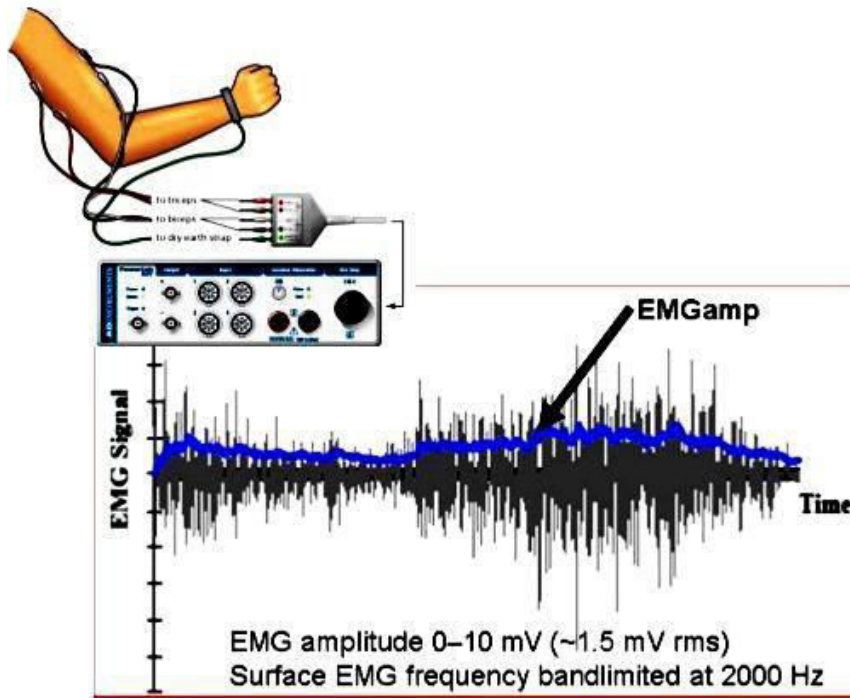
Measurements of Electrical Biosignals

Electroencephlogram (EEG)

1. The electrical potential fluctuations of the brain generate signals
2. Used for studying nervous system, monitoring sleep stages, diagnosing diseases like epilepsy.
3. The analysis is based on external frequency

Latest Trends and Current Research

Electromyogram (EMG)



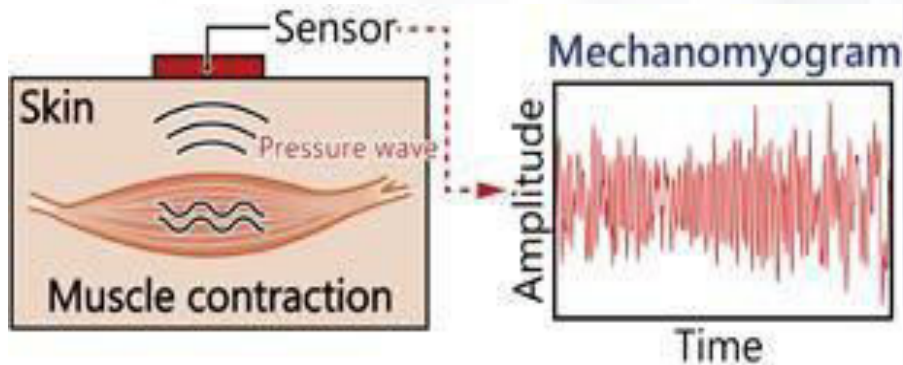
<https://www.google.com/imgres?imgurl=https%3A%2F%2Fimage.slidesharecdn.com%2Felectromyogram-emg-140630030825->

Measurements of Biosignals

Electromyogram (EMG)

1. The muscle function contraction and relaxation produces the electrical signals emanated during muscular contractions.
2. Can be employed for clinical and biomedical applications

Latest Trends and Current Research



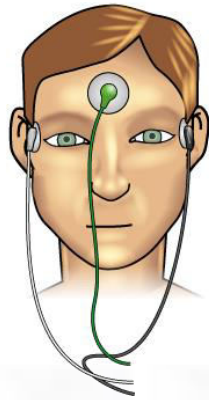
<https://iopscience.iop.org/article/10.7567/JJAP.57.11UD09>

Measurements of Biosignals

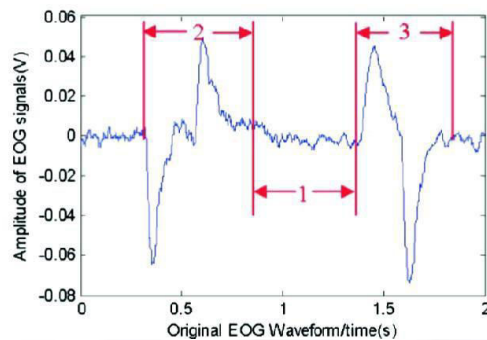
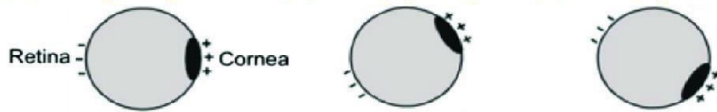
Mechanomyogram (MMG)

1. Low-frequency mechanical signal can be detected from the surface of a muscle when the muscle is contracted.
2. For monitoring activity of muscles present superficially
3. Gross changes in the muscle shape cause a large peak to appear in an MMG

Latest Trends and Current Research



(1) looking straight ahead (2) rolling eyes upward (3) rolling eyes downward



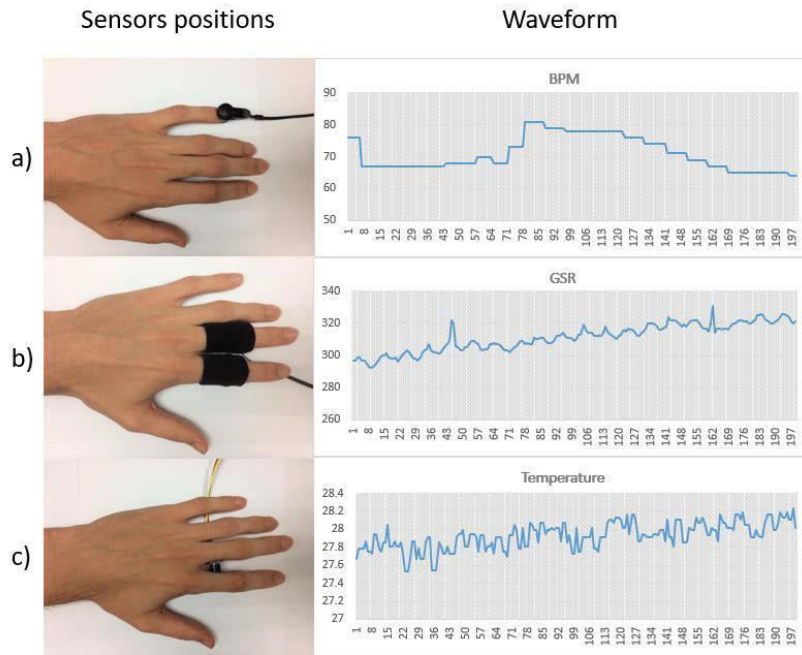
Measurements of Biosignals

Electrooculography (EOG)

1. Based on the principle that Eye acts as an electrical dipole between the cornea (+ potential) and the retina (- potential).
2. Rotation of the eye causes a change in dipole and thus in the electric potential field

https://doi.org/10.1007/978-3-030-14118-9_18

Latest Trends and Current Research



<http://www.ikelab.net/dspr-pdf/vol1-2/dspr-paper2.pdf>

Measurements of Biosignals

Galvanic Skin Respons (GSR)

1. A change in the electrical resistance of the skin caused by emotional stress is measurable with a sensitive galvanometer.
2. Commonly used for measuring autonomic nervous system activity

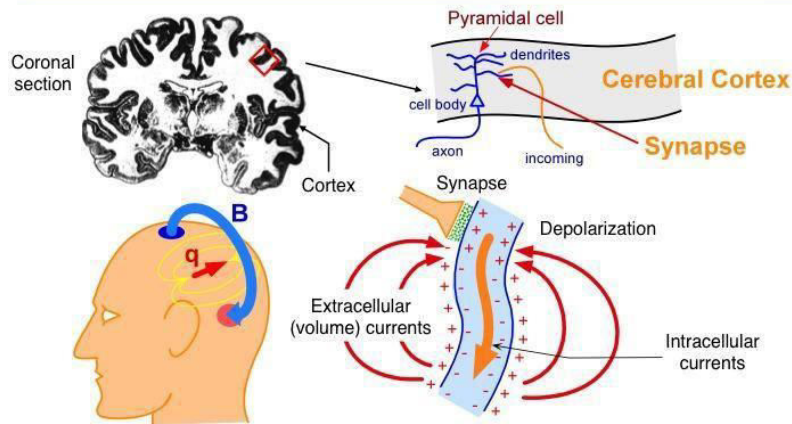
Latest Trends and Current Research

Measurements of Biosignals

Magnetoencephalogram (MEG)

1. Neuromagnetic signals generated in the brain by the neurons during brain activity
2. MMG provides timing and spatial information about brain activity by using these signals
3. For diagnosing perceptual and cognitive brain disease

Origins of MEG Signals



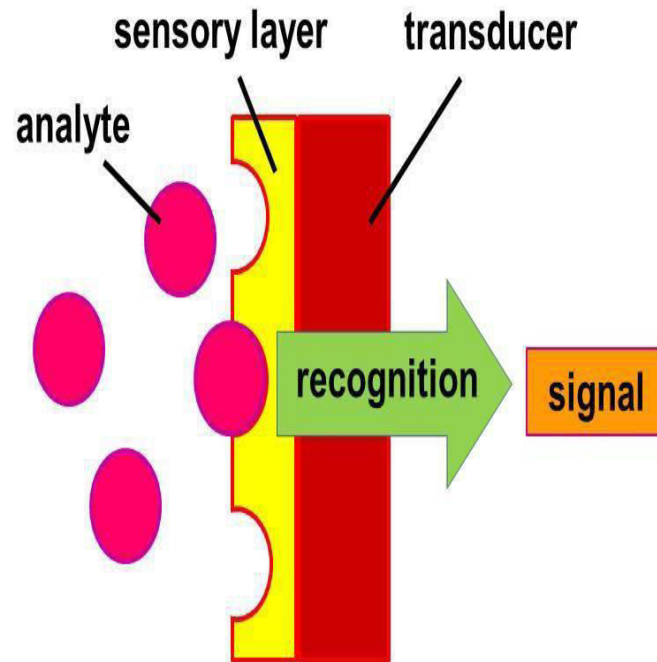
<http://ilabs.washington.edu/what-magnetoencephalography-meg>

Biosensors

An Introduction to Support Materials



An Introduction to Support Materials



<https://www.elprocus.com/wp-content/uploads/Bio-Sensor.jpg>

Support Material

1. The support material immobilizes the enzyme for a short period or permanently.
2. The ideal support is cheap, inert, physical strong and stable.
3. It should discourage microbial growth.
4. Ideally there should be no non-specific absorption.

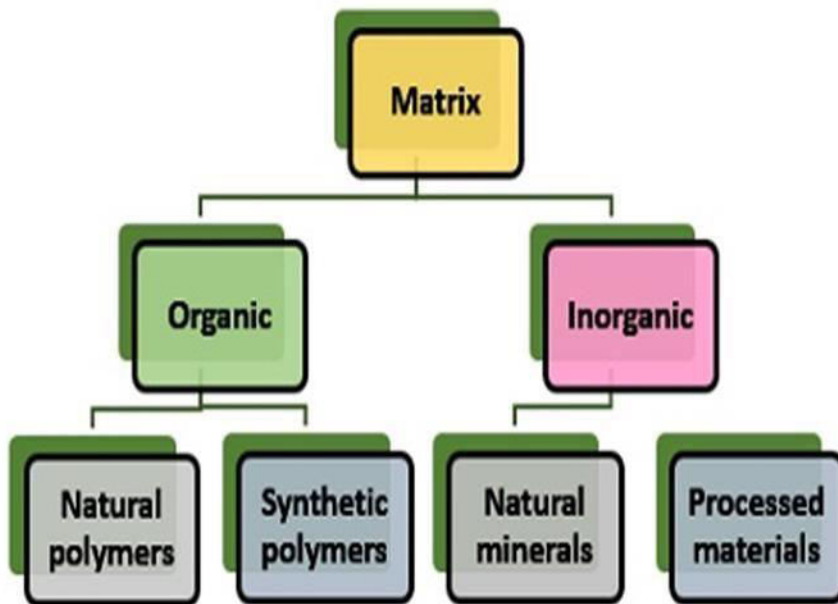
An Introduction to Support Materials

Physical characteristics of Support material

1. **Mean particle size**: It is the parameter which decides the porosity of a carrier.
2. **Hydrophilic character**: It determines the level of activity of an immobilized enzyme.
3. **Mechanical strength**: A mechanically strong material is required as a support material for immobilization.

An Introduction to Support Materials

Classification of Support material

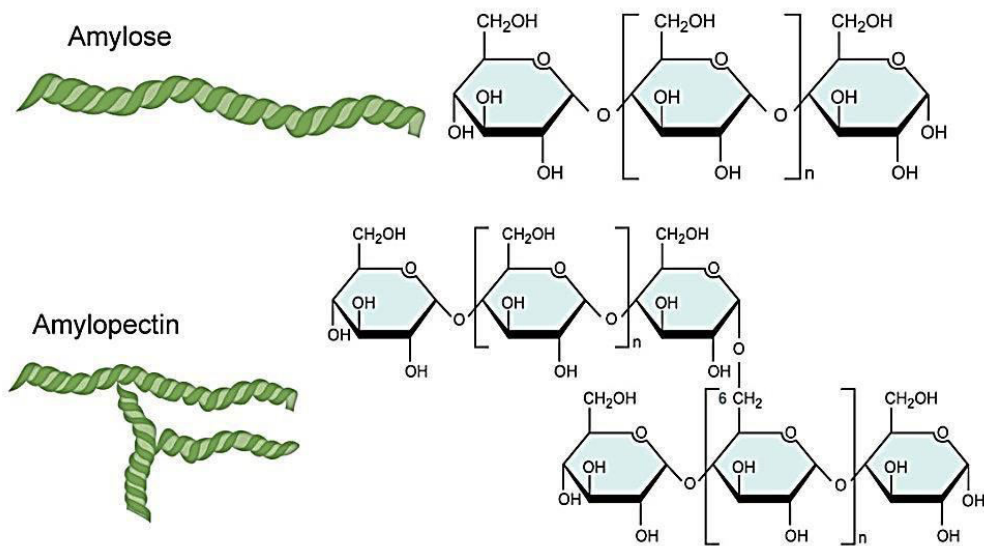


<https://biologyreader.com/wp-content/uploads/2018/11/entrapment-type.jpg>

1. **Synthetic polymers** show high chemical and mechanical stability. Polystyrene.
2. **Inorganic matrix**: It subdivides into Natural and Processed minerals.
3. **Natural minerals**: Zeolite, celite, silica.
4. **Processed materials**: Porous glass, metal oxides, nano materials

An Introduction to Support Materials

Starch



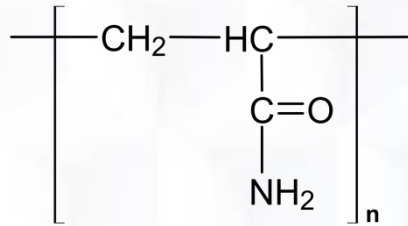
<https://futurefoodchemist.weebly.com/amylose-or-amylopectin.html>

Natural Support Materials.

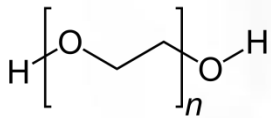
The natural support materials are the organic support materials from animal and plant sources such as.

1. Alginate
2. Cellulose
3. Starch
4. Gelatin
5. Collagen
6. Chitosan and chitin

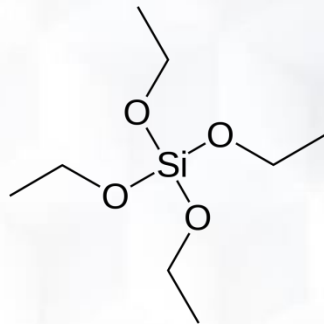
An Introduction to Support Materials



Polyacrylamide



Polyethylene glycol



TEOS

https://en.wikipedia.org/wiki/Polyethylene_glycol

<https://en.wikipedia.org/wiki/Polyacrylamide>

https://en.wikipedia.org/wiki/Tetraethyl_orthosilicate

Synthetic Polymers

1. Polyacrylamide
2. Polyaniline
3. DEAE Cellulose
4. Poly vinyl chloride
5. Polyethylene glycol
6. Tetraethylorthosilicate (TEOS) derived silicates
7. Sepharose

An Introduction to Support Materials



Zeolite



Silica

<https://encryptedtbn0.gstatic.com/images?q=tbn:ANd9GcRWwFV39L1jbYB3t2foWc3qYdOZJ0xHYOfHDwRFdX9odYtgrfb>

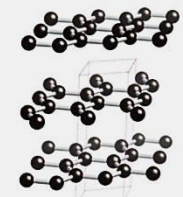
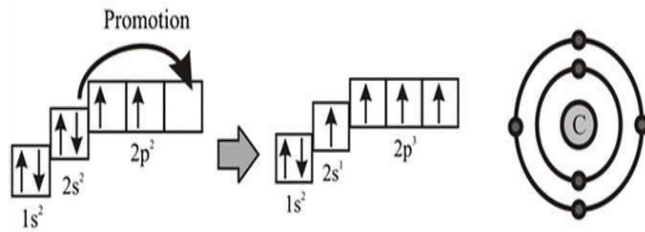
<https://upload.wikimedia.org/wikipedia/commons/thumb/2/2d/Stilbite-Ca-Natrolite-Laumontite-247898.jpg/220px-Stilbite-Ca-Natrolite-Laumontite-247898.jpg>

Inorganic materials

A number of inorganic materials are used for the matrix to immobilize BRE. These materials are porous

1. Zeolite
2. Ceramic
3. Silica
4. Glass
5. Activated Charcoal
6. Carbon based nano materials

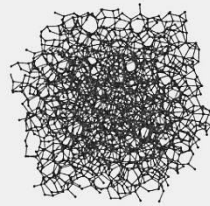
An Introduction to Support Materials



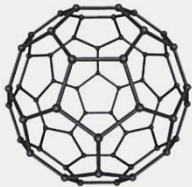
graphite



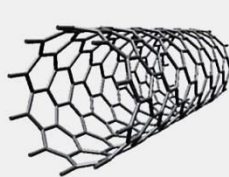
diamond



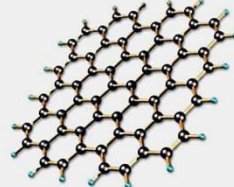
amorphous
carbon



fullerene



nanotube



graphene

Carbon Based Nano Materials

1. Carbon has an ability to polymerize at the atomic level, thus can form long carbon chains.
2. Carbon has four electrons in the outer electron and can be linked via single, double or triple covalent bonds, or also with other elements.

https://media.springernature.com/lw785/springer-static/image/art%3A10.1186%2Fs40538-016-0070-8/MediaObjects/40538_2016_70_Fig1_HTML.gif

<https://www.shungite.com/images/type-s-of-carbon.jpg>

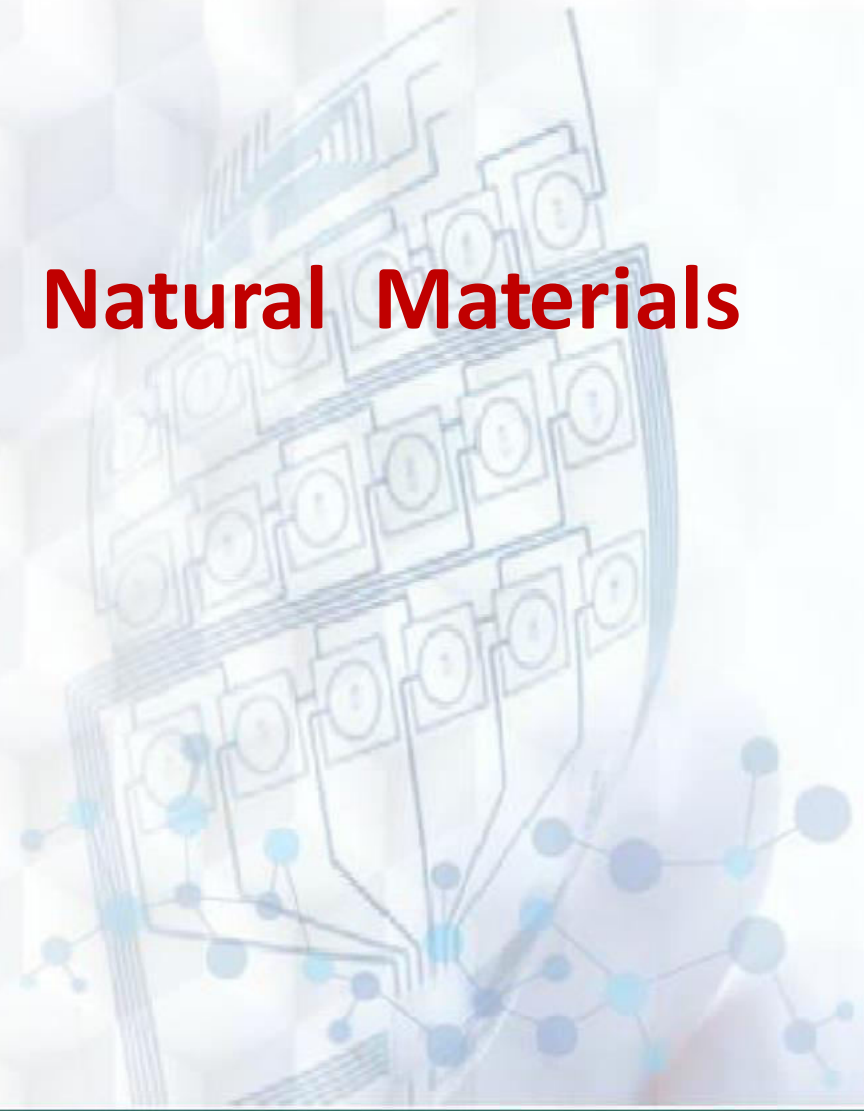
An Introduction to Support Materials

Summary

1. Support matrix in a Biosensor is a sensitizing platform on which the Biorecognition process take place. It is required to be, stable, semi conductor and inert.
2. There is a huge variety of organic materials is available including plant based, animal based, microbial based .
3. Synthetic and inorganic and allotropic forms of Carbon based materials are also used.

Biosensors

Natural Materials



Natural Materials

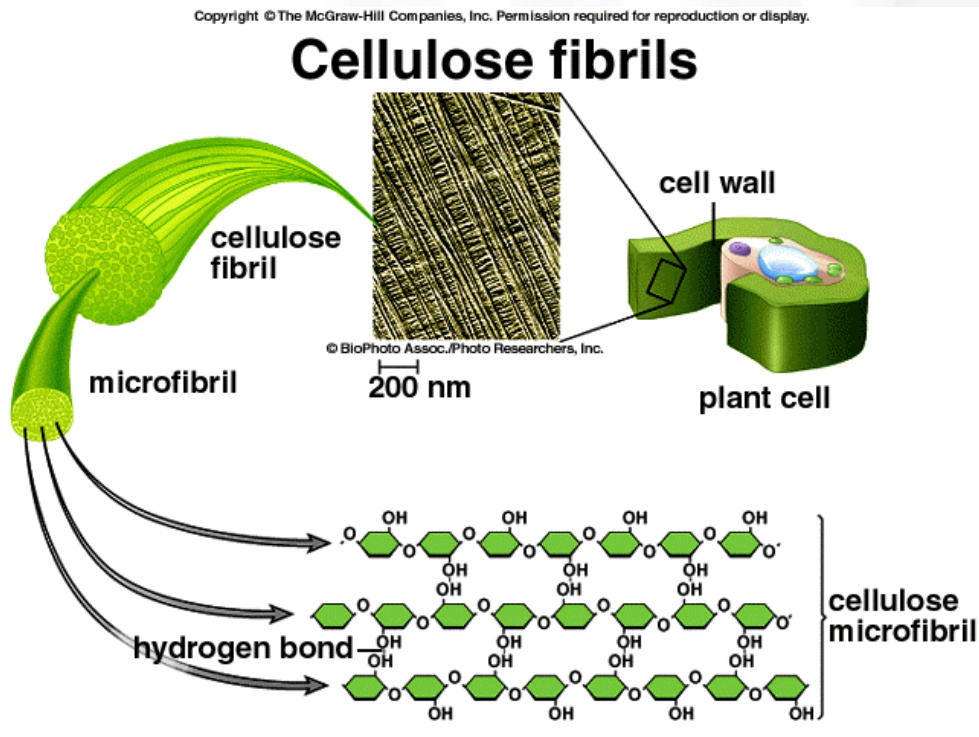
Types of Natural Material Organic and Inorganic

There are at least four types of materials that are being used for the platform for biosensors.

ffort By: AARISH ALI

1. Plant derived materials
2. Animal derived materials
3. Carbon Based materials
4. Silicone based materials

Natural Materials

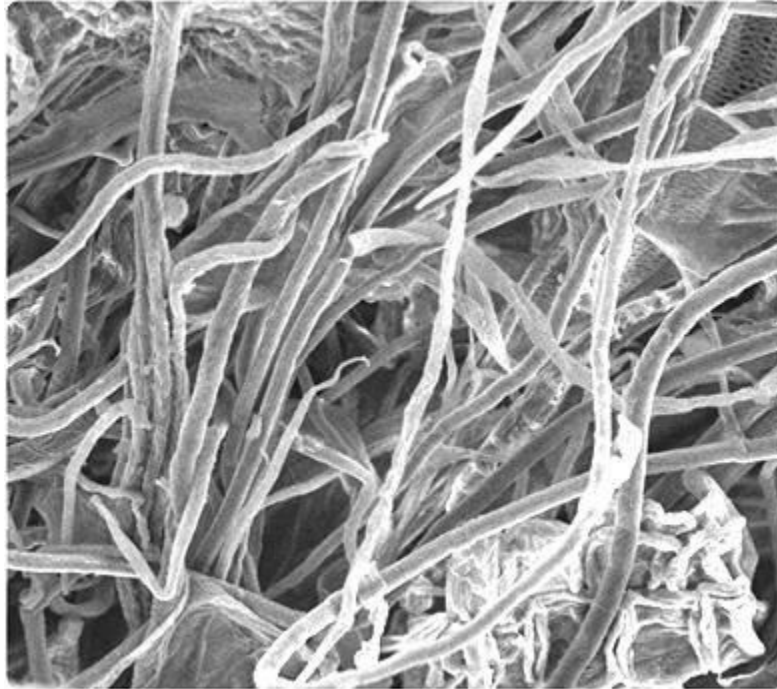


http://desertbruchid.net/4_GB_Lecture_figs_f/4_GB_03c_OrgChem_Fig/3_8_mad7_Cellulose_Polys.GIF

Cellulose

1. Cellulose is a plant based polymer. It is composed of sub units of glucose joined through beta linkages.
2. It is an important structural component of the primary cell wall of green plants, many forms of algae.
3. **Cellulose** is the most abundant organic polymer on Earth.

Natural Materials



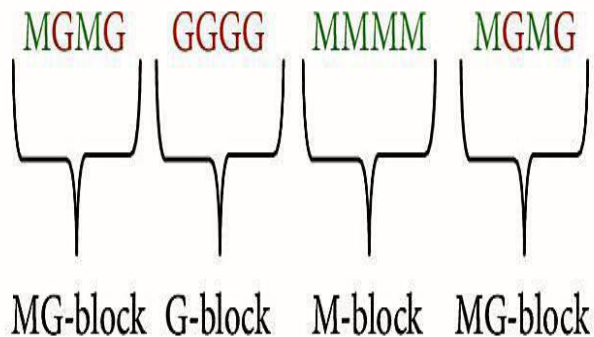
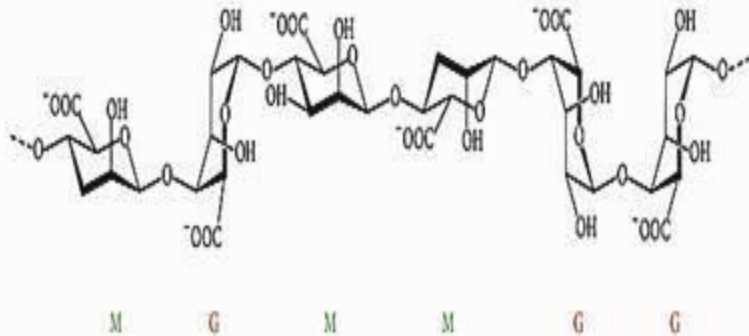
[https://www.azonano.com/images/Article/Images/ImageForArticle_3139\(2\).jpg](https://www.azonano.com/images/Article/Images/ImageForArticle_3139(2).jpg)

A Transmission Electron Micrograph image of a nanocellulose membrane shows the individual cellulose nanofibers

Nano Cellulose

1. Nanocellulose Light weight
2. Stiff
3. Electrically conductive
4. Non-toxic
5. The crystalline form is transparent, and gas impermeable.
6. It has a very high tensile strength - 8 times that of steel
7. It is highly absorbent

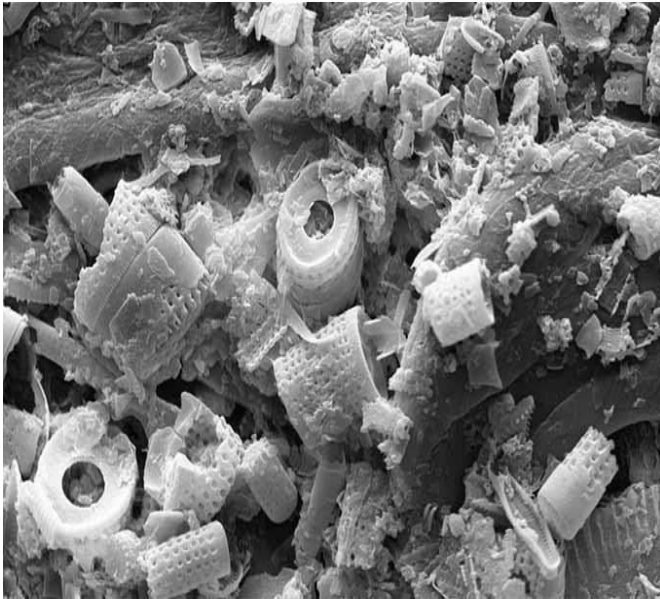
Natural Materials



Alginate

1. Alginate is a naturally occurring anionic gelatinous polysaccharide.
2. It is present in cell wall of algae and seaweed,
3. Alginate is an anionic linear polymer of linear copolymers of β -(1-4) linked **D-mannuronic acid** and β -(1-4)-linked **L-guluronic acid** units.

Natural Materials



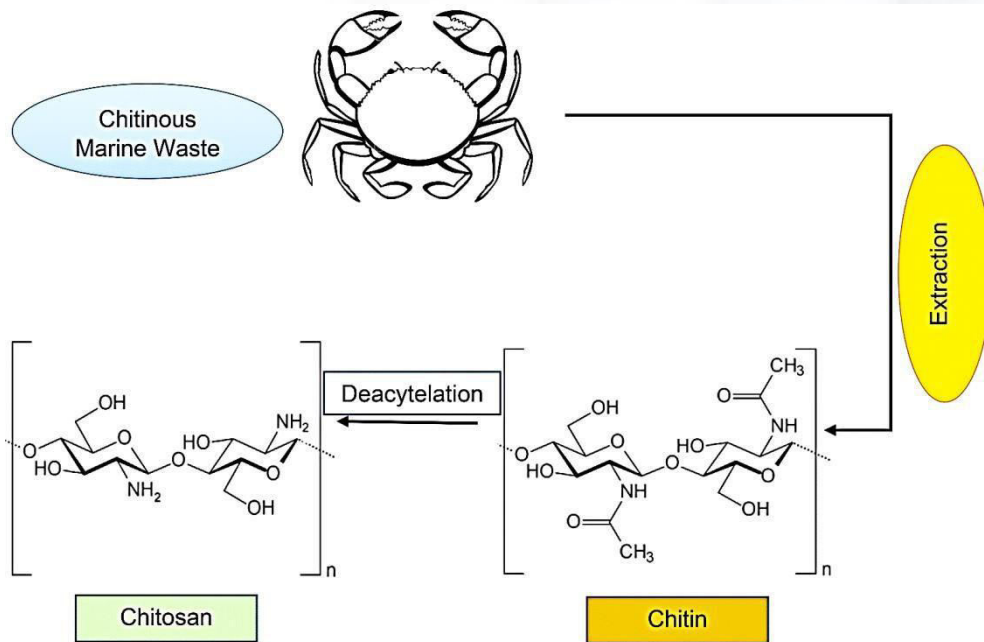
Highly magnified diatomaceous earth shows the intricate fossils that make up the substance

<https://gardenerspath.com/wp-content/uploads/2017/04/Diatomites.jpg>
g

Diatomaceous earth

1. Diatomaceous earth is a naturally occurring, soft, siliceous sedimentary rock forms a fine white to off-white powder.
2. It has a particle size ranging from 10 to 200 μm .
3. Diatomite forms by the accumulation of the amorphous silica (opal, $\text{SiO}_2 \cdot n\text{H}_2\text{O}$) remains of dead diatoms

Natural Materials

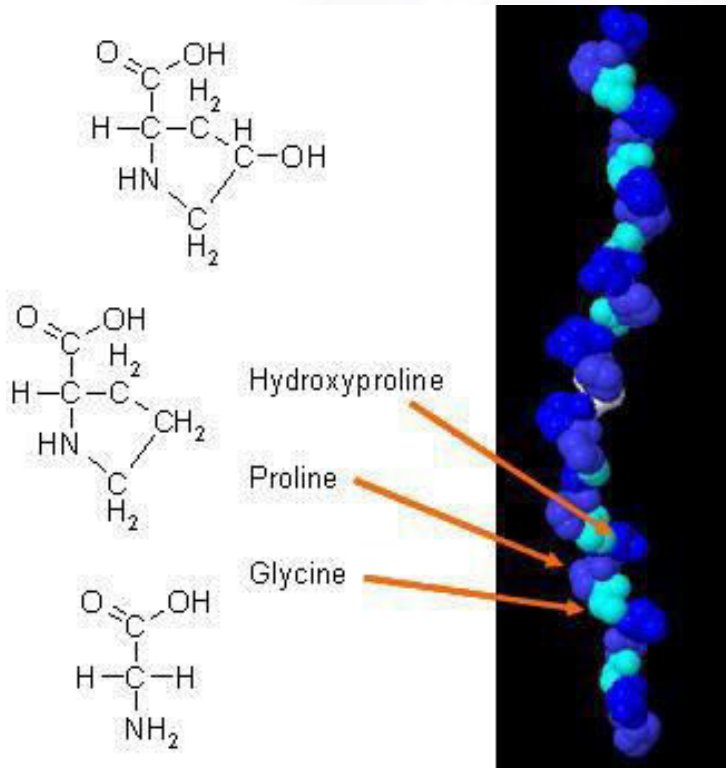


Chitosan

1. Chitosan is a linear polysaccharide.
2. It is composed of randomly distributed β -(1 $\rightarrow$ 4)-linked **D-glucosamine** (deacetylated unit) and **N-acetyl-D-glucosamine** (acetylated unit).
3. It is made by treating the chitin shells of shrimp and other crustaceans with an alkaline solution.

Yadav, M., Goswami, P., Paritosh, K. et al. Bioresour Bioprocess. (2019) 6: 8. <https://doi.org/10.1186/s40643-019-0243-y>

Natural Materials



https://proteopedia.org/wiki/images/e/e5/Collagen_%28alpha_chain%29.jpg

Collagen

1. **Collagen** is the main **structural** protein in the extracellular space in the various connective tissues in the body.
2. **Collagen** consists of amino acids bound together to form triple-helices of elongated fibrils.
3. It is mostly found in fibrous tissues such as tendons, ligaments, and skin.

Natural Materials

Summary

1. Different type of materials are present in nature from biological and Inorganic sources.
2. The biological compounds include cellulose, nano cellulose, collagen.
3. The inorganic and processed materials include silicon based and nano carbon based materials.

Biosensors

Synthetic Materials



Synthetic Materials

Introduction and Types

An number of synthetic materials are used as platform for the immobilization of Biorecognition element. For example

1. Ion exchange resins
2. Aniline based polymers
3. Carbon based materials
4. Silicone based materials

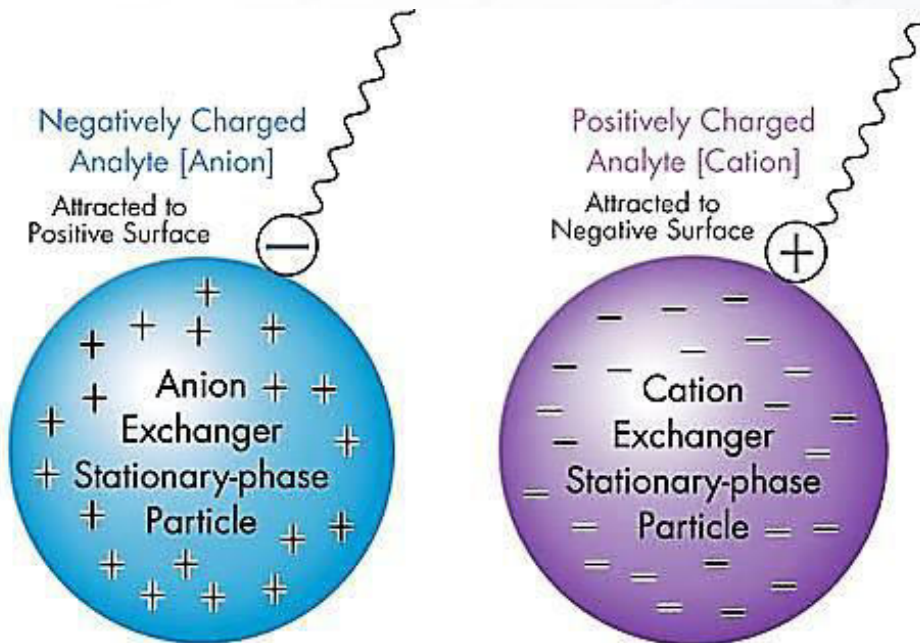
are used as platform for immobilization.

Synthetic Materials

Ion Exchanger Resins

Ion Exchange resins are small beads that are composed of organic polymer chains.

1. Cationic resin has a negative functional group and therefore attracts positively charged ions.
2. Anionic resin has a positive functional group and attracts negatively charged ions.



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vc/s1600/primer_T_Ion_Exchange_CHromatography.j
pg

Synthetic Materials

Polyaniline (PANI)

1. The basic subunit PANI is aniline.
2. It forms discrete long threads with an average diameter in the range of 30 nm - 100 nm.
3. PANI is a semi-flexible.
4. It has different colors in oxidized and reduced forms.

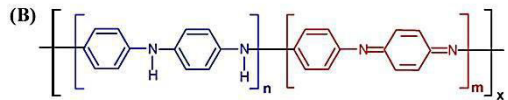
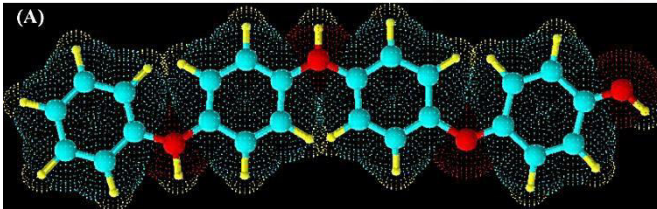
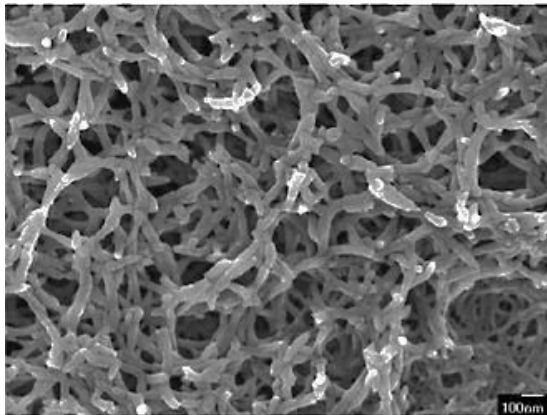


Fig. 2. (A) 3D and (B) 2D structure of polyaniline



<https://ai2-s2-public.s3.amazonaws.com/figures/2017-08-08/195d8928332e90ddbba7ab475462b2fa5fc08598/4-Figure2-1.png>

https://upload.wikimedia.org/wikipedia/commons/thumb/b/3/3c/Polyaniline_synthesis.jpg/330px-Polyaniline_synthesis.jpg

Synthetic Materials

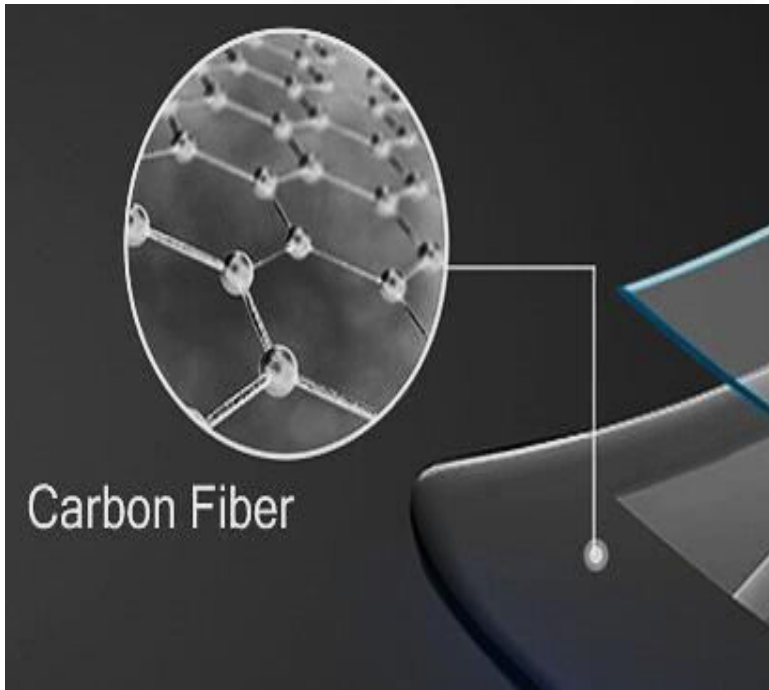
Carbon based Materials

Carbon paste has successfully used in Biosensor as a platform.

1. It requires simple construction procedures.
2. It exhibits low background current.
3. It has an ability for surface regeneration.
4. It has a very low cost.

On the other hand it suffers short life time.

Synthetic Materials



Glassy Carbon

Glass-like carbon, (non-graphitizing), combines glassy and ceramic properties with those of graphite.

1. Mechanically stable, Impermeable to gases and fluids
2. chemically resistant
3. high porosity
4. low background current.

its low sensitivity to peroxide, as well as to other mediators.

https://mail.google.com/mail/u/0?ui=2&ik=3e9207c5bd&attid=0.1&permmsgid=msg-a:r-2841920254573458713&th=16ce5b6db8630fdb&view=fimg&sz=s0-l75ft&attid=ANGjdJ8t_UwiOhaTa6X6oyWGBWK_rnGB5yLOpSBiYj3ziGpSNOFnmntwQ3xUNlcWF980zaA934Af37CM531aOnObYDFuM3f7icX6ztvX1GSPBP8YhypWEbn5MAQ-g&disp=emb&realattid=ii_jzyziuku0

Synthetic Materials

Summary

1. Biological recognition elements are immobilized on especially designed platforms.
2. Different types of materials include carbon based materials, aniline, ion exchange materials.
3. A number of novel materials including Glassy carbon , nano carbon , silicone based polymers are used..

Biosensors

Hydrogels



Hydrogels

Definition

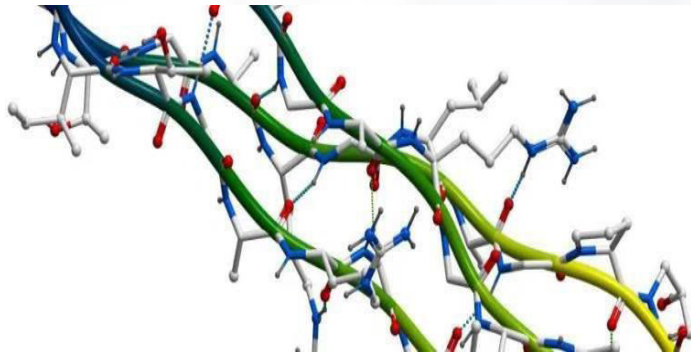
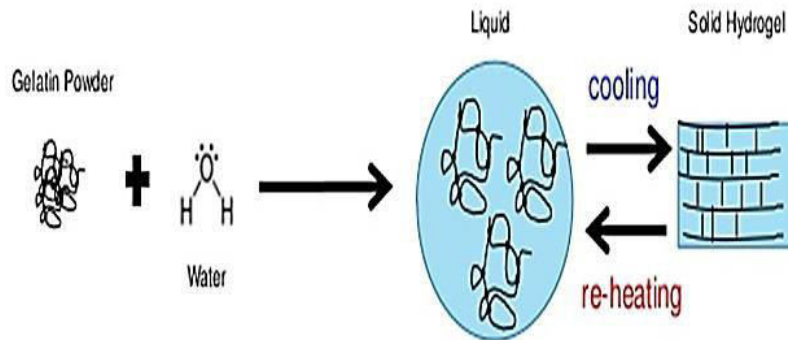
1. Hydrogels are regarded as the three dimensional network of natural or synthetic polymers.
2. They have an ability to absorb a large amount of water or biological fluids.
3. Hydrogels do not dribble water due to the presence of a novel exclusive zone (about 100micrometer).

Hydrogels

Types

1. Hydrogels can be produced from natural polymers include proteins such as collagen and gelatin
2. The polysaccharides include starch, alginate, and agarose.
3. Synthetic polymers that form hydrogels are prepared using chemical polymerization methods.

Hydrogels



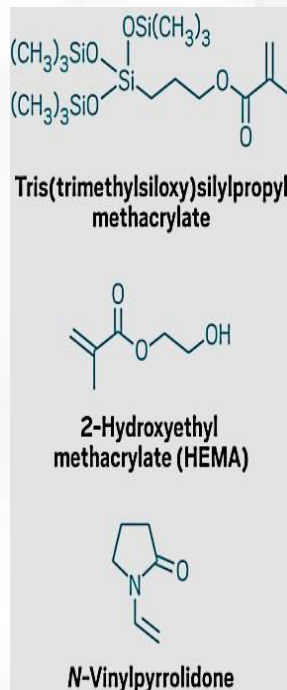
https://i0.wp.com/www.nomnomenon.com/wp-content/uploads/2015/07/braise-collagen-triple-helix-protein-Depositphotos_12083419_original-11-e1438127594169-1024x307.jpg?resize=900%2C270

<https://image.slidesharecdn.com/426ceef3-5fc6-428b-82e0-c02e30d3920d-151007082427-lva1-app6891/95/hydrogel-9-638.jpg?cb=1444206299>

Gelatin

1. Gelatin is a heterogeneous mixture of single or multi-stranded polypeptides
2. Glycine, proline, and valine most abundant amino acids.
3. Each strand with extended left-handed helix conformations.
4. A helix contains between 50 - 1000 amino acids.

Hydrogels



Hydrogels in Eye lens

Artificial or synthetic eye lens are used wearing.

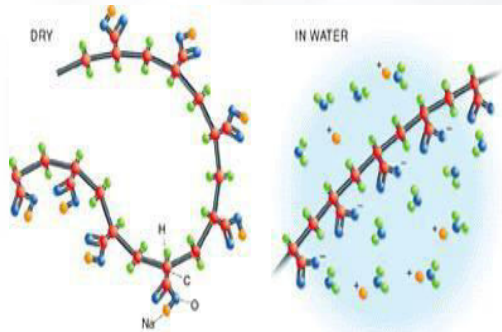
Modern soft contact lenses are made of hydrogel and silicon hydrogel polymers.

Hard contacts are made of polymethyl methacrylate (PMMA) or Plexiglas.

Soft contacts are mass-produced, but hard contact lenses are made to fit the wearer.

https://cen.acs.org/content/cen/articles/95/i13/Making-better-contactlenses/_jcr_content/articlebody/subpar/articlemedia_0.img.jpg/1490616024512.jpg

Hydrogels



<https://www.pnas.org/content/pnas/113/47/13295/F4.large.jpg>

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Hydrogels in Diapers

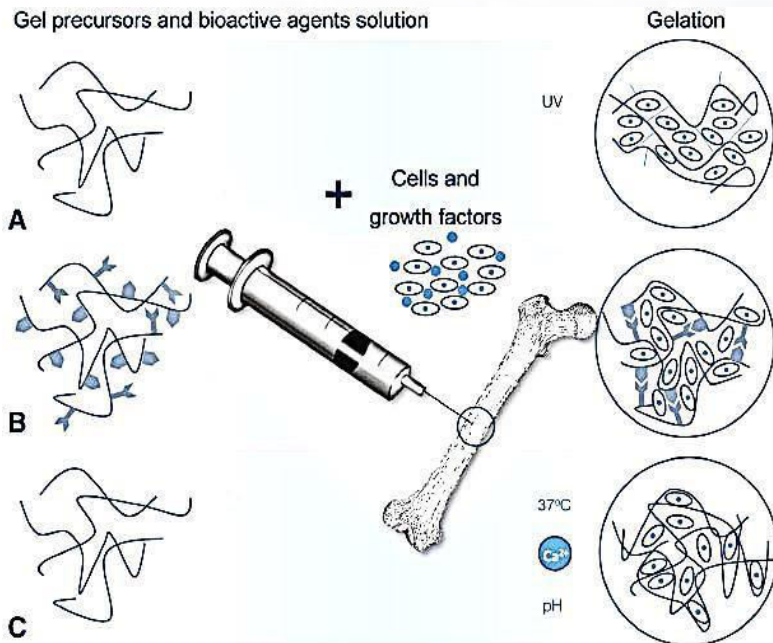
1. Sodium polyacrylate is used in the production of disposable diapers.
2. When exposed to water, the sodium polyacrylate swells and adsorbs up to 1000 times its own weight in water.
3. The diapers soak up urine and keep moisture away from a baby's skin, helping to prevent diaper rash.

Hydrogels

Applications

Hydrogels can be regarded as highly valuable biomaterials for human-beings. There are numerous applications of hydrogels in modern day life.

1. Tissue engineering
2. Dressing of wounds
3. Diapers
4. Cosmetic products skin and hair gel
5. Biosensing.



<https://www.degruyter.com/view/j/revce.2017.33.issue-1/revce-2015-0074/revce-2015-0074.xml>

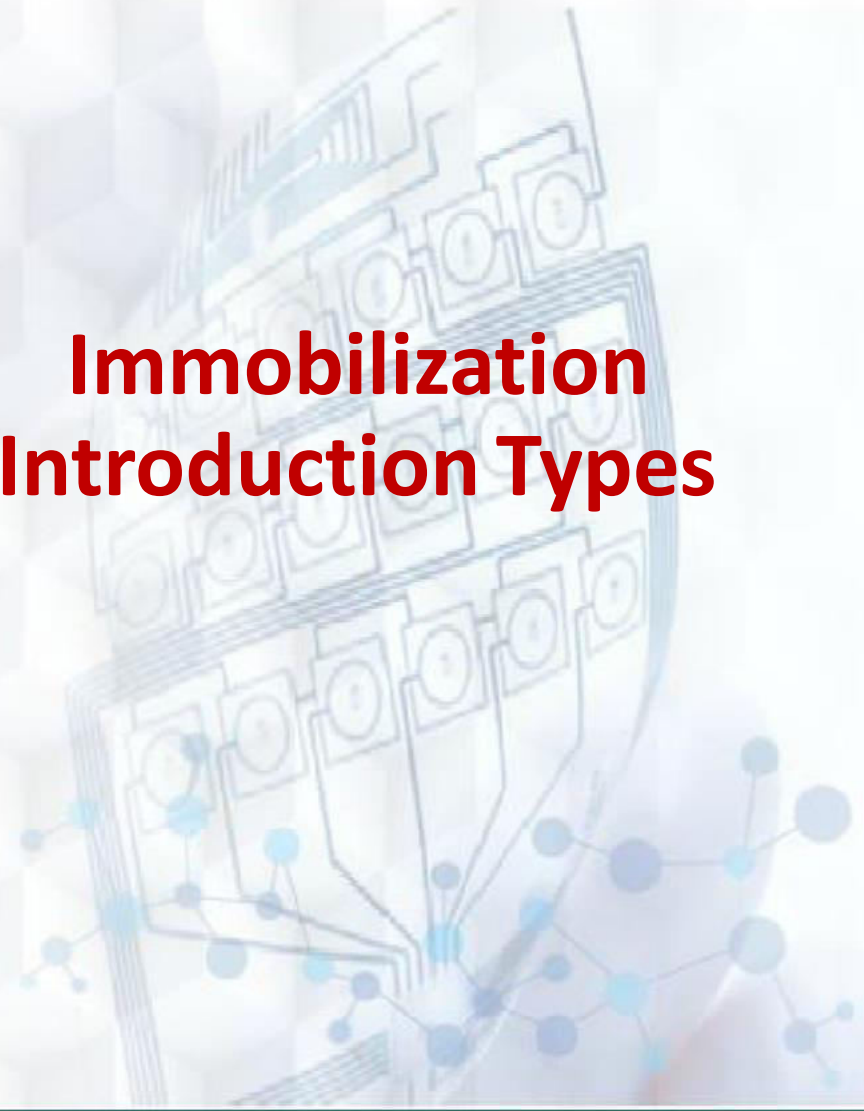
Hydrogels

Summary

1. Hydrogels are polymers and are able to retain maximum quantity of water.
2. Hydrogels could be made from natural and synthetic materials.
3. There are several applications of hydrogels in different fields including medicine, drug delivery, agriculture, environment.

Biosensors

**Immobilization
Introduction Types**

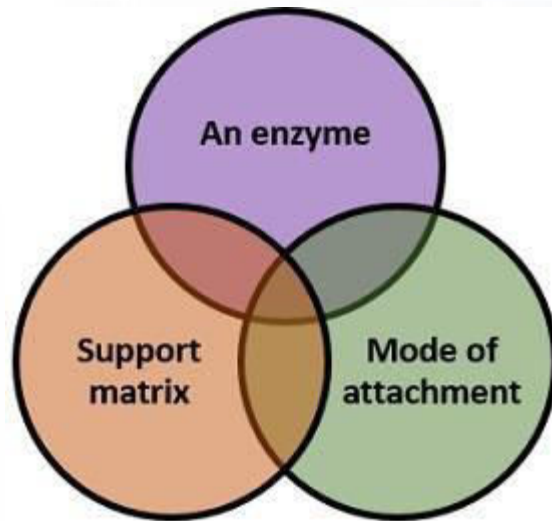


Immobilization Introduction Types

Introduction

1. Immobilization is the binding of biomolecules on to a solid support with functional activity.
2. It gives a stable environment to the biomolecules.
3. In Biosensors a close proximity of the biomolecules with the transducer is required.

Immobilization Introduction Types

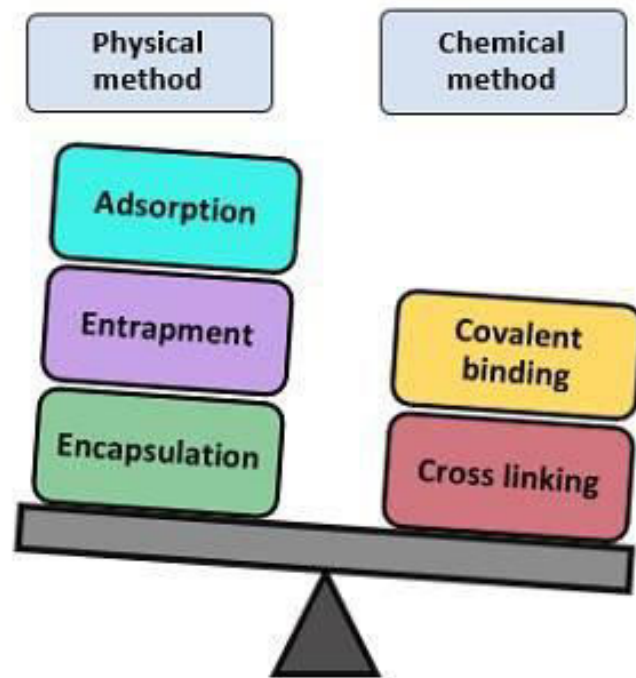


<https://biologyreader.com/wp-content/uploads/2018/11/components.jpg>

Components of Immobilization

1. The main components of an enzyme immobilization include a Biological recognition element , a support matrix and mode of attachment of a catalyst to the carrier.
2. **Support matrix** is a material that uses for the imprisonment of an enzyme.

Immobilization Introduction Types



<https://biologyreader.com/wp-content/uploads/2018/11/methods-of-immobilization.jpg>

Types

There are two methods of for immobilization.

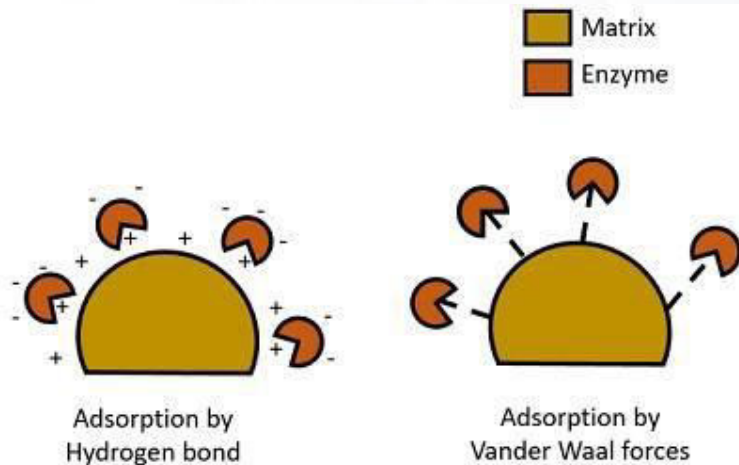
Chemical Methods

- I. Covalent Binding
- II. Copolymerization

Physical methods.

- I. Adsorption
- II. Entrapment
- III. Encapsulation

Immobilization Introduction Types

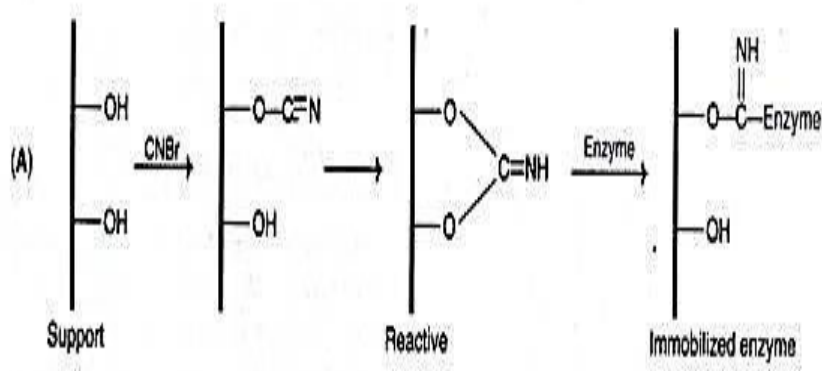


<https://biologyreader.com/wp-content/uploads/2018/11/adsorption-type.jpg>

Adsorption

1. In adsorption Weak bonds are involved such as Hydrogen bond and Vander Wall forces.
2. The matrix particle size must be small.
3. It is simple and easy process. There is limited loss of enzyme activity.
4. Disadvantages include desorption and Non-specific adsorption.

Immobilization Introduction Types



http://cdn.biologydiscussion.com/wp-content/uploads/2015/09/clip_image008_thumb19.jpg

Covalent coupling

Chemical binding between functional groups of the enzyme and the support.

1. No diffusion barrier
2. Stable
3. Short response time
4. High enzyme activity loss

Disadvantage

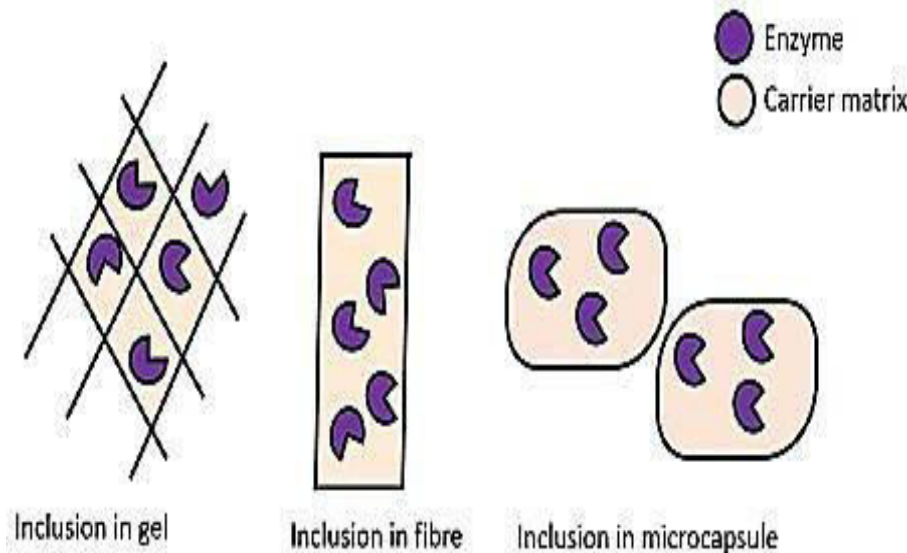
Matrix not regeneratable

Coupling with toxic product.

Immobilization Introduction Types

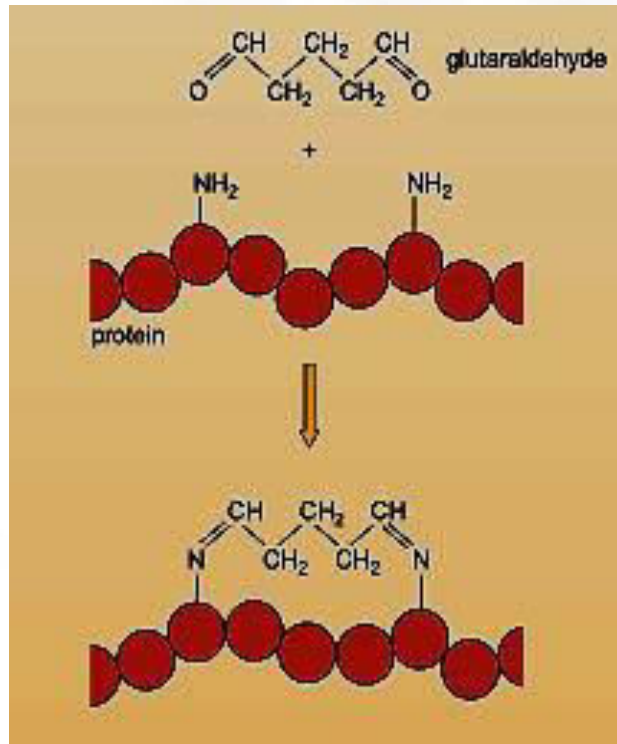
Entrapment

1. Entrapment is the Incorporation of the enzyme within a gel or a polymer.
2. Several types of enzymes can be immobilized within the same polymer Diffusion barrier.
3. Disadvantages are Enzyme leakage, High conc. of monomer and enzyme needed for electro-polymerization.



<https://biologyreader.com/wp-content/uploads/2018/11/entrapment-type.jpg>

Immobilization Introduction Types



Fixation reaction of glutaraldehyde. The amine groups ($-\text{NH}_2$) on amino acid residues in the protein, react with aldehyde groups ($-\text{CHO}$) in the fixative.

Cross-linking

The cross linking of Biological recognition element with the support material involves bonding between the amino group of the matrix and amino group of an enzyme. Example: Glutaraldehyde.

1. Advantages are Low or no enzyme leakage
2. Higher enzyme stability
3. Simple and cheap method.

https://www.open.edu/openlearn/ocw/pluginfile.php/65622/mod_oucontent/oucontent/440/fcc69e0b/dc7151ff/s210_1_f5.jpg

Immobilization Introduction Types

Summary

1. In Biosensors the BRE (Protein, DNA, Cell) is required to be attached on a support material.
2. A solid, stable and inert support is required to entrap the protein molecule.
3. In Biosensing adsorption, entrapment, chemical binding, encapsulation and crosslinking are used.

Biosensors

**Reversible
Immobilization**

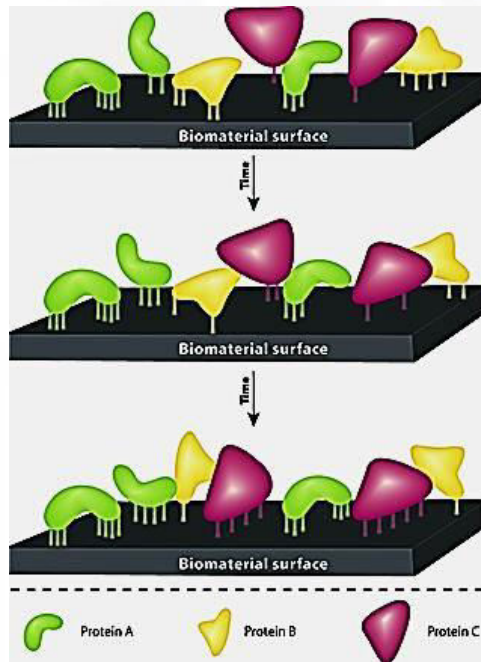


Reversible Immobilization

Explanation

1. Reversible immobilization is the method in which the biological recognition element can be removed from the matrix. It is temporarily fixed on support material.
2. Generally no drastic conditions are required for the immobilization.
3. Reversible immobilization techniques allow the multiple time use of BRE.

Reversible Immobilization



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Methods of immobilization by adsorption

Static method: immobilization of enzyme and carrier molecule without agitation.

Dynamic process: The mixing of an enzyme with the carrier under constant agitation.

Reactor loading: transferring of both enzyme and carrier in the reactor with the agitation

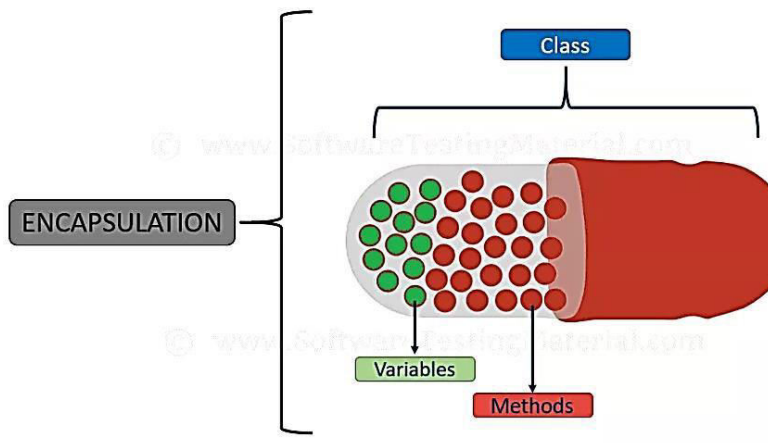
Electro-deposition: Due to electric current the enzyme gets deposited on the surface.

Reversible Immobilization

Encapsulation

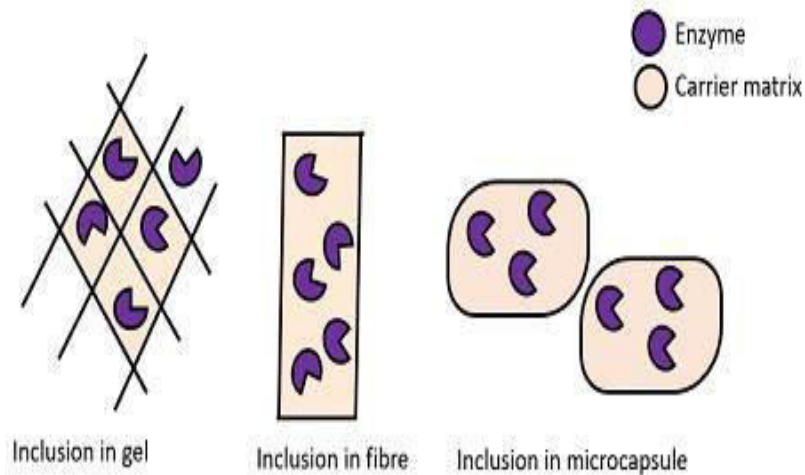
There are four methods for encapsulation.

1. Encapsulation in reaction vessel.
2. Encapsulation by hollow fibre membrane.
3. Microencapsulation by chemical polymerization, by using 1-6-diaminohexane.
4. Encapsulation by liposomes.



<https://i2.wp.com/www.softwaretestingmaterial.com/wp-content/uploads/2018/03/Encapsulation.png?ssl=1>

Reversible Immobilization



<https://biologyreader.com/wp-content/uploads/2018/11/entrapment-type.jpg>

Entrapment .

There are different types of entrapment.

1. Inclusion in gels: Involves entrapment of enzyme inside the gel matrix.
2. Inclusion in fibres: Entrapment of enzyme inside the fibre matrix.
3. Inclusion in microcapsules: Involves entrapment inside a microcapsule

Reversible Immobilization

Summary.

1. Reversible immobilization is based on physical forces.
2. The binding sites of the Biorecognition element remain protected. The methods include adsorption, ionic binding, entrapment.
3. There are some latest methods such as solid phase reverse immobilization are reported.

Biosensors

**Irreversible
Immobilization**

A stylized illustration of a biosensor chip. The chip is depicted as a rectangular grid of circular elements, each containing a small number. The grid is tilted and has a 3D effect. At the bottom right, there is a molecular structure with blue and grey spheres connected by lines. The background is a light blue and white hexagonal pattern.

Irreversible Immobilization

Covalent bonding

1. In covalent bonding , enzyme molecule binds to the carrier by a covalent bond.
2. A complex form through this bonding is stable.
3. There is no enzyme loss during the process.
4. Covalent binding occurs between the active part, the, i.e. functional group of an enzyme and carrier molecule.

Irreversible Immobilization

- Carboxyl group at C term
- Amino group at N terminal
- Carboxyl group of Asp/Glu
- Amino group of Lys/Arg
- Phenol ring of Tyrosine
- Thiol group of Cysteine
- OH group of Ser/Thre
- Imdazole group of His
- Indole group of Tryp

Chemical groups

1. There are several groups in protein structure which could participate in chemical bonding with the support material.
2. The order of reactivity of these functional group to the carrier depends upon their charged status:

$-S- > -SH > -O- > -NH_2 > -COO- > -OH \gg -NH_3+$.

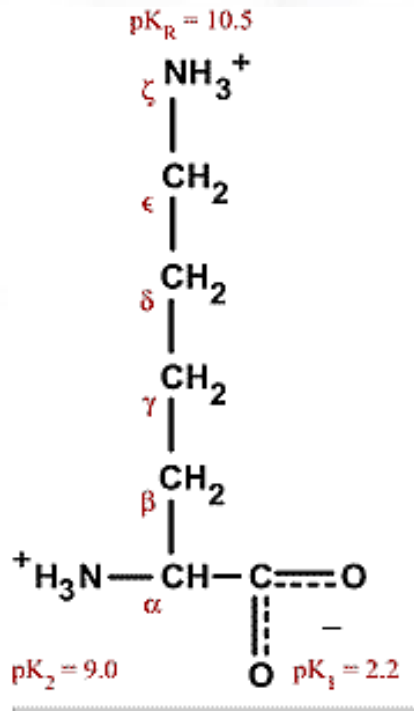
Irreversible Immobilization

Lysine

Lysine residues are found to be the most useful groups for covalent bonding of enzymes to insoluble supports due to their widespread surface exposure and high reactivity, especially in slightly alkaline solutions.

Epsilon amino group of lysine take part in the reaction.

Lysine residues appear to be only very rarely involved in the active sites of enzymes.



Irreversible Immobilization

Supports Used for Covalent Binding

A huge variety of support materials are used for immobilization, including.

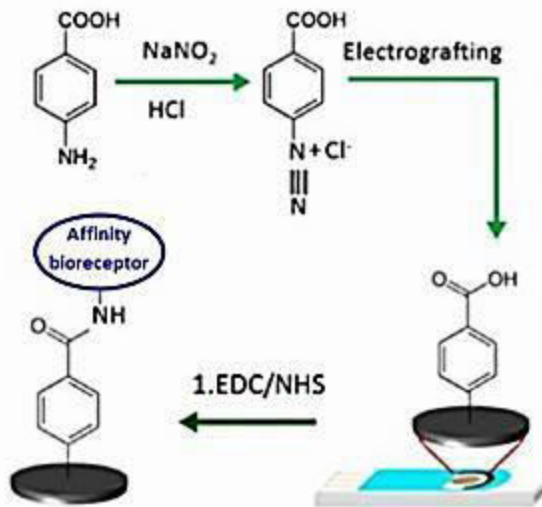
1. Cellulose
2. Polyacrylamide
3. Collagen
4. Gelatin
5. Porus glass
6. Silica
7. CNBr- Agarose

Irreversible Immobilization

Methods for Irreversible Immobilization

1. Diazotization Reaction between a primary aromatic amine and nitrous acid to give a diazo compound.
2. Peptide bond formation between carboxyl and amino group of carrier and protein.
3. Bond between amino groups of carrier and support.

Irreversible Immobilization

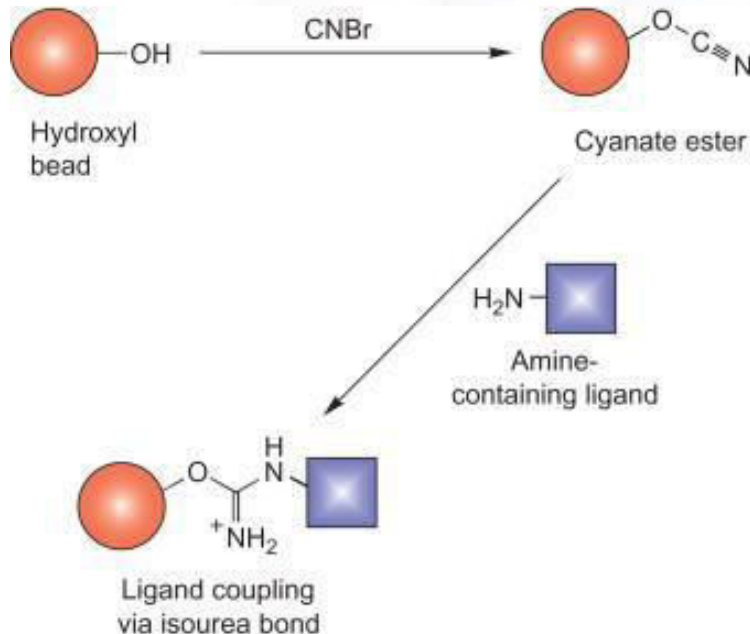


Diazotization

1. The process of conversion of primary aromatic amines into its diazonium salt is called diazotization.
2. This process involves bonding between the amino group of the matrix and tyrosyl or histidyl group of an enzyme on reaction with NaNO_2 and HCl .

<https://www.ncbi.nlm.nih.gov/corecgi/tileshop/tileshop.fcgi?p=PMC3&id=588817&s=75&r=1&c=3>

Irreversible Immobilization



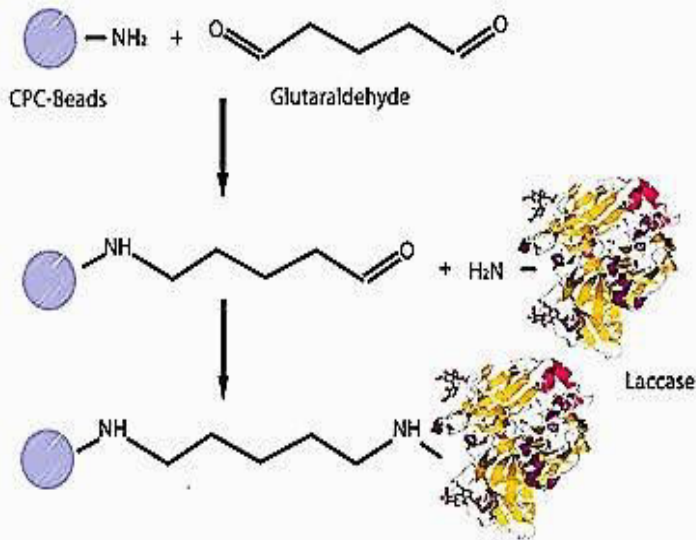
Cyanogen Bromide

1. Cyanogen bromide is used for the activation of Sepharose or Agarose hydroxyl groups.
2. The activate hydroxyl groups on carbohydrate creates reactive cyanate esters.
3. $CNBr$ hydrolyzes peptide bonds at the C-terminus of Met residues. Met is the first amino acid residue of a protein.

Cyanogen bromide can be used to activate a hydroxyl particle to a reactive cyanate ester, which can then be used to couple

amine-[https://ars.els-cdn.com/content/image/3-s2.0-B9780123822390000145-f14-17-9780123822390.jpg?](https://ars.els-cdn.com/content/image/3-s2.0-B9780123822390000145-f14-17-9780123822390.jpg?_)
containing ligands.

Irreversible Immobilization



http://2012.igem.org/wiki/images/thumb/f/f8/Bielefeld2012-Overview_Immobilization.jpg/400px-Bielefeld2012-Overview_Immobilization.jpg

Glutaraldehyde

1. Glutaraldehyde reacts with the support material containing primary amino groups.
2. It requires the low-ionic strength.
3. It acts as a link between the protein and support material.
4. The amino group of glutaraldehyde reacts with the protein and support material.

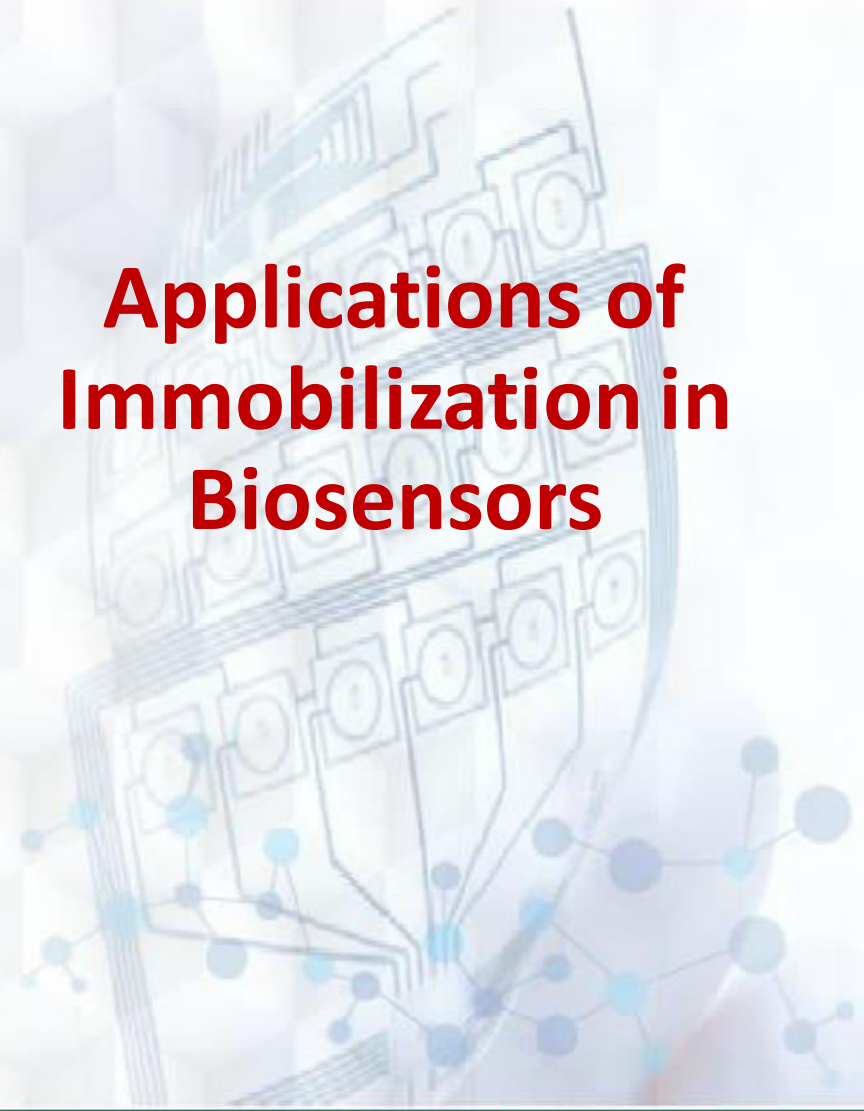
Irreversible Immobilization

Summary

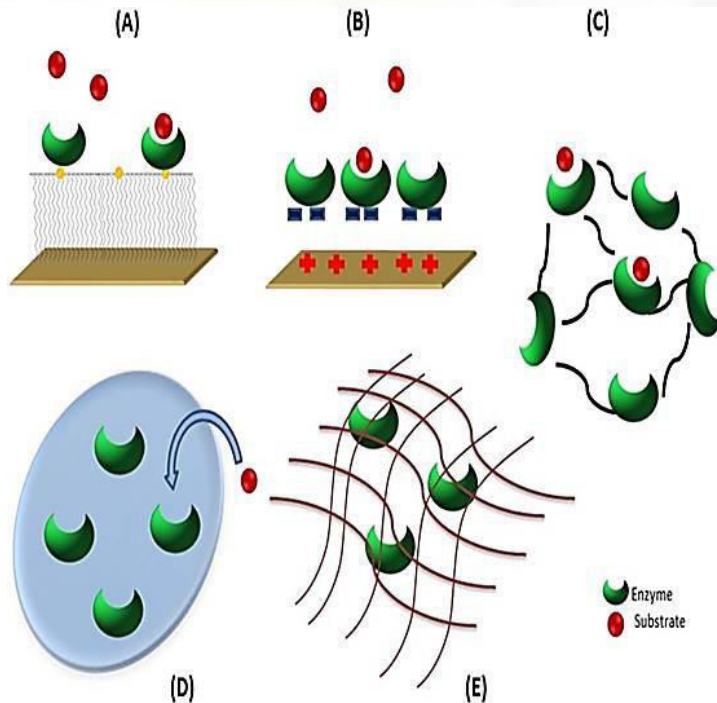
1. This is a permanent method for protein immobilization.
2. It involves formation of chemical bonds between the biological recognition element and the support.
3. Different types of reagents such as Glutaraldehyde and CNBr acts as a linker between the support material and the BRE.

Biosensors

Applications of Immobilization in Biosensors



Applications of Immobilization in Biosensors



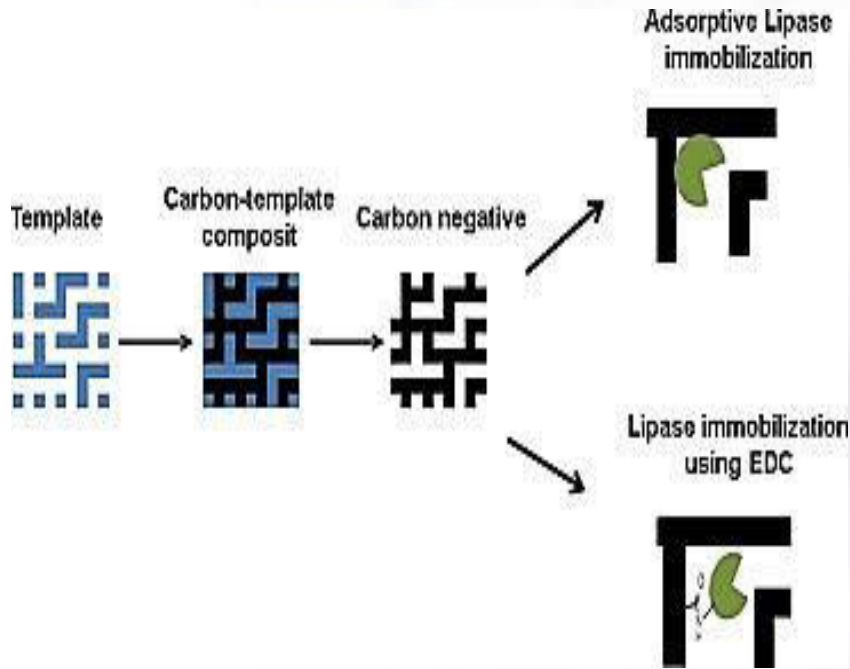
https://www.researchgate.net/profile/Roberto_Parra5/publication/283873093/figure/fig3/AS:296590020497415@1447723979243/Common-methods-of-immobilization-used-in-biosensing-A-covalent-binding-B.png

Common methods of immobilization used in biosensing: (A) covalent binding; (B) adsorption; (C) cross-linking; (D) encapsulation; and, (E) entrapment.

Introduction

1. Immobilization is the most important step in the fabrication of biosensor.
2. It forms the foundation of the Biosensor.
3. Immobilization of BRE provides a a working platform for the bio-recognition process.
4. Several novel approaches are used for fabricate novel working platform .

Applications of Immobilization in Biosensors



Porous carbon material synthesis followed by *Thermomyces lanuginosus* lipase (TLL) immobilization.

<https://ars.els-cdn.com/content/image/1-s2.0-S1369703X18302213-fx1.jpg>

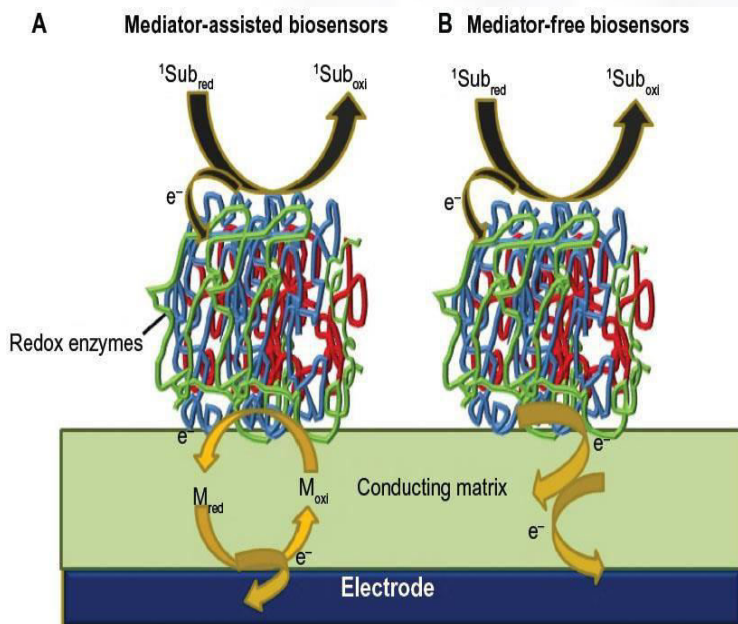
Porous Carbon Material

1. Porous carbon materials (PCM) were synthesized by sucrose infiltration into template material.
2. Carbon particles from 125 to 200 μm were generated.
3. Tailor-made porous carbon material is a promising support for lipase immobilization, which is adaptable in shape and dimension.

Applications of Immobilization in Biosensors

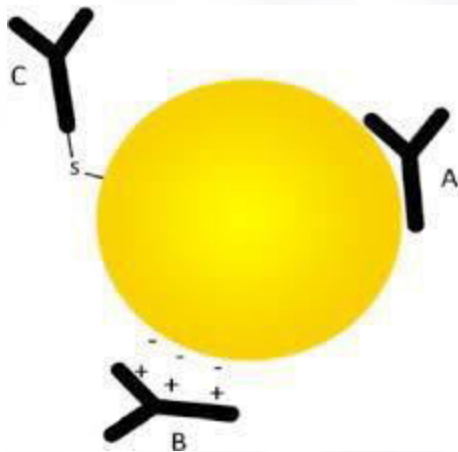
Immobilization on PANI

1. Polyaniline (PANI) is a conducting polymer of the semi-flexible rod polymer family.
2. PANI is used to develop mediator-less biosensors.
3. It is used as support matrix and insulating material.
4. PANI is an efficient electrode material due to low redox potential and high conductivity.



https://www.dovepress.com/cr_data/article_fulltext/s64000/64841/img/fig1small.jpg

Applications of Immobilization in Biosensors



Hydrophobic and ionic interactions between antibody and gold.

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTWq5-Qbn-xZ_YpiKheKBhCyHi4ULAM7jRuuy2llp_va45KJ1kR

Immobilization on NPG

1. Nanoporous gold (NPG) provides a high surface area form of gold that is suitable for covalent modification by self-assembled monolayers.
2. The material can be used as a high surface area electrode and with immobilized enzymes can be used for amperometric detection schemes..

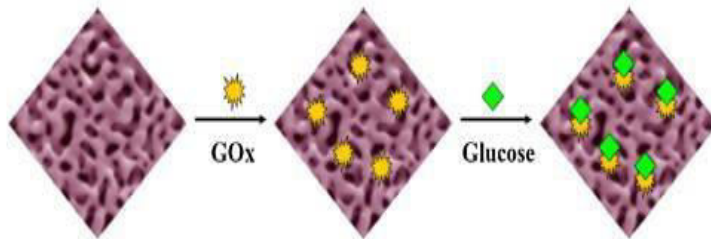
Applications of Immobilization in Biosensors

Immobilization on NPG

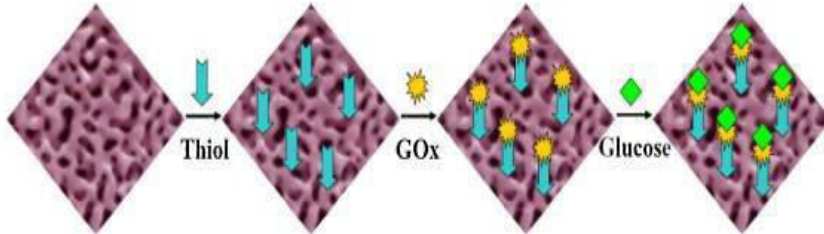
NPG films with tunable pore sizes have been successfully developed for enzyme-modified electrochemical biosensors.

1. By immobilizing GOx on the surface of gold ligament via thiol linking
2. The bio-functionalized NPG exhibits remarkable glucose sensitivity, detection limit, selectivity and response time.

Method 1: GOx-NPG



Method 2: GOx-Thiol-NPG



<https://ars.els-cdn.com/content/image/1-s2.0-S0013468612001703-gr3.jpg>

Applications of Immobilization in Biosensors

Summary

Modern approaches are used to immobilize the enzyme on novel designed fabricated platforms for the efficient working of biosensor.

1. PANI is used to increase the performance of the Biosensors.
2. Charged Porous carbon, Nano gold particle and porous gold is successfully used in Biosensor.

Potentiometric biosensor

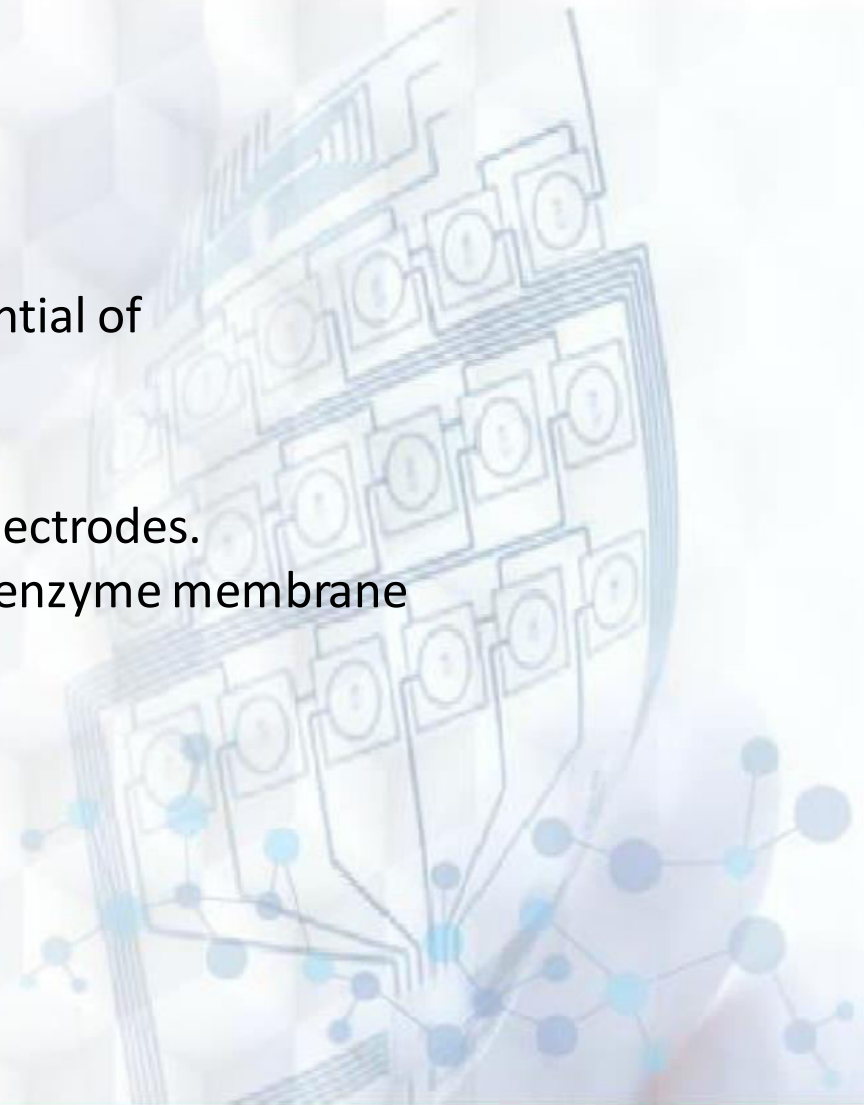
In this type of sensor
measured parameter

Oxidation or reduction potential of
an electrochemical reaction

Potentiometric biosensors

Make use of ion-selective electrodes.

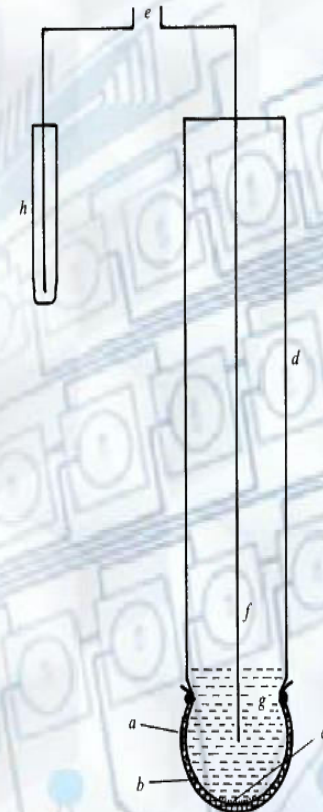
Consists of an immobilised enzyme membrane



potentiometric Biosensor

A simple potentiometric Biosensor

- (a) A semi-permeable membrane
- (b) surrounds the biocatalyst
- (c) Entrapped next to the active glass membrane
- (d) pH probe
- (e) electrical potential
- (f) Ag/AgCl electrode (g) bathed in dilute HCl
- (h) An external reference electrode



potentiometric Biosensor

- **Three types of ion-selective electrodes used**

Glass electrodes for cations

- e.g. normal pH electrode

- **Glass pH electrodes**

- coated with a gas-permeable membrane selective for CO_2 , NH_3 or H_2S

- **Solid-state electrodes**

- glass membrane is replaced by a thin membrane



Virtual University

-

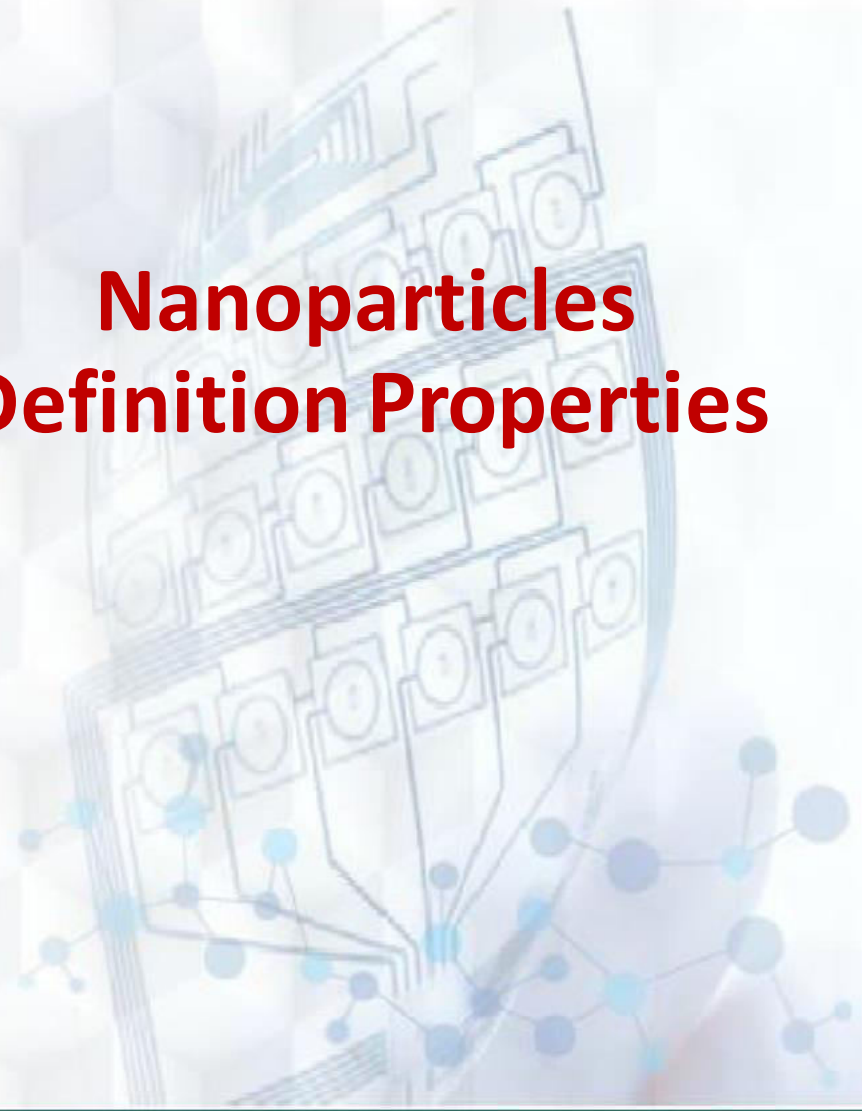
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Thank you



Biosensors

Nanoparticles
Definition Properties

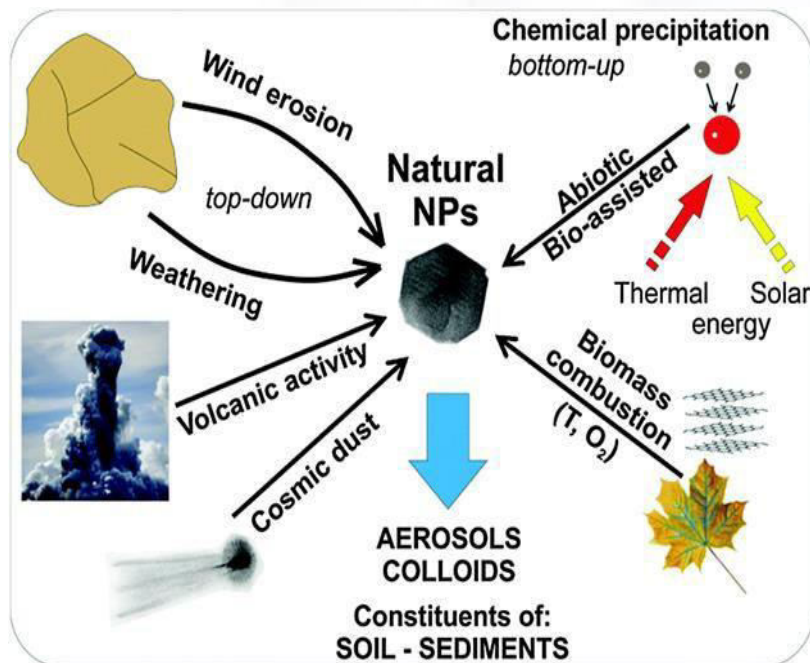


Nanoparticles Definition Properties

Definition

1. Nanoparticle, is an ultrafine unit with dimensions in nm (1-100).
2. A nanoparticle is a nano-object with all three external dimensions in the nanoscale.
3. For a NP the longest and shortest axes do not differ significantly.
4. NPs are produced naturally and artificially.

Nanoparticles Definition Properties

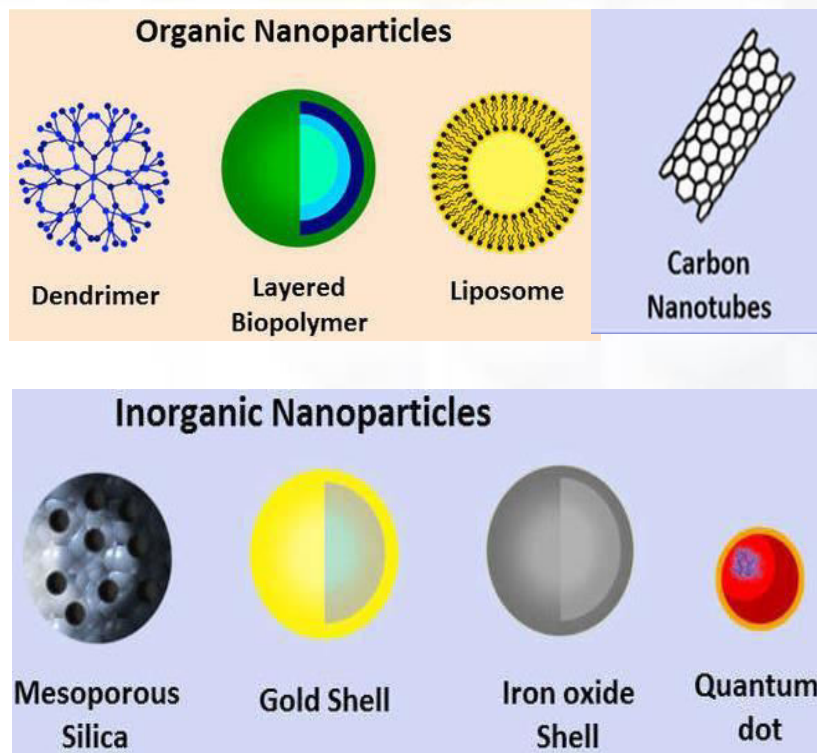


https://pubs.rsc.org/image/article/2015/CS/c5cs00236b/c5cs00236b-f3_hi-res.gif

Source

1. Natural sources of nanoparticles include combustion products forest fires, volcanic ash, ocean spray, and the radioactive decay of radon gas.
2. Nanoparticles are also produced artificially from metals, metal oxides, non metals, biological molecules and different polymers.

Nanoparticles Definition Properties



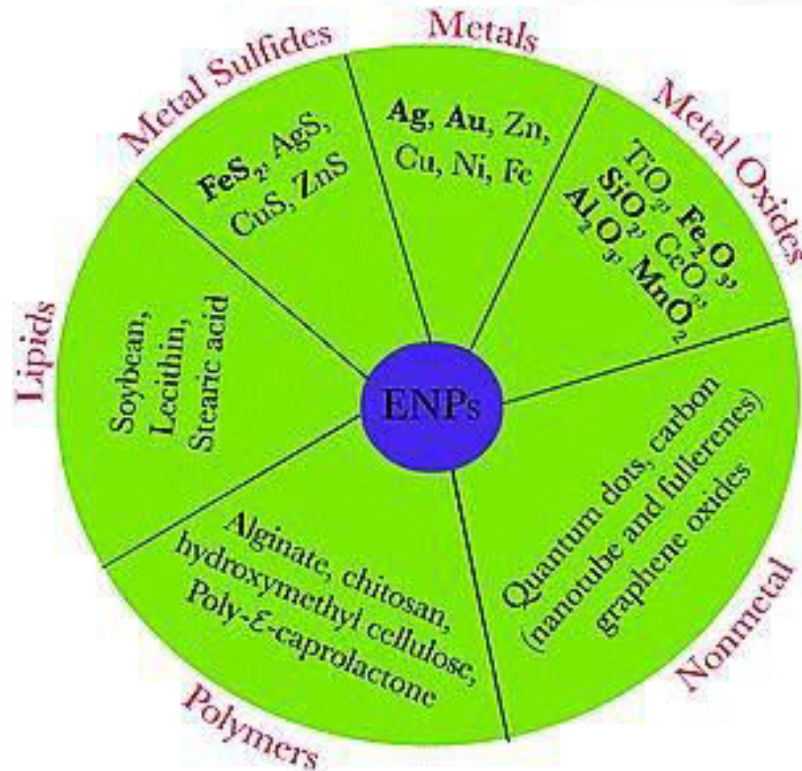
Classification

NPs are heterogeneous in nature. On the basis of chemistry they are classified into two types.

1. **Organic Nanoparticles** includes dendrimers, liposomes, carbon based nanoparticles and polymeric nanoparticles.
2. **Inorganic Nanoparticles** include Mesoporous silica, Gold nano particles, Iron oxide, Quantum dots.

<https://www.researchgate.net/publication/330292807/figure/fig1/AS:713867797745664@1547210754228/Scientific-representation-of-different-types-of-nanoparticles-NPs-divided-into-organic.jpg>

Nanoparticles Definition Properties



<https://ai2-s2-public.s3.amazonaws.com/figures/2017-08-08/c479294e808d7c2b2351e322d4eb6cf2ed02a975/3-Figure1-1.png>

Engineered NPs

A Nanoparticles engineered for shape, size, and surface properties. They impart

1. Special functions catalytic behavior
2. Improved strength
3. Enhanced thermal conductivity and
4. Electrical conductivity
5. Controlled release of host molecules.

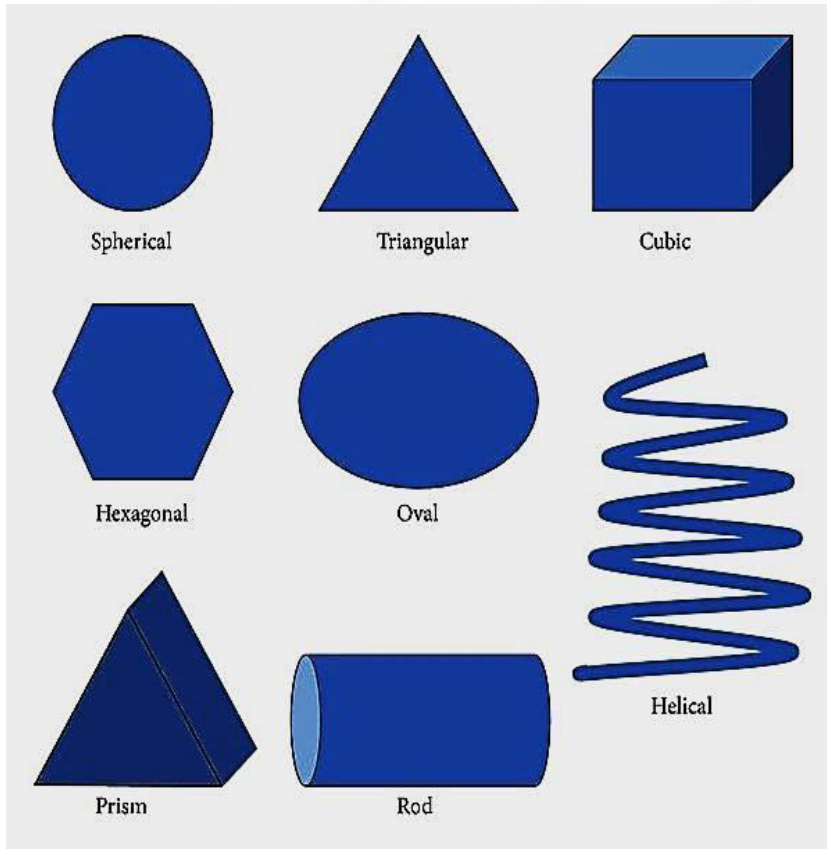
Nanoparticles Definition Properties

Properties

Nanoparticles have four distinctive properties.

1. Nanoparticles are ultrafine units having a size in the range of 1-100nm.
2. Nanoparticles are highly **mobile** in the free state.
3. Nanoparticles have **enormous** specific **surface areas**.
4. Nanoparticles may exhibit **quantum effects**.

Nanoparticles Definition Properties



https://www.researchgate.net/profile/Mir_Arfat3/publication/265133668/figure/fig1/AS:202880964403211@1425381999808/Various-shapes-of-nanoparticles_W640.jpg

Shape of Nanoparticles

1. Nanoparticles are very heterogeneous in shape.
2. Depending on the production method and manufacturing conditions, nanoparticles exhibit different forms and structures..
3. NPs come in different shapes including spherical, Triangular, Cubical, Hexagonal, Oval, Helical, Pyramidal and rod like.

Nanoparticles Definition Properties

Summary

1. NPs are found in nature and are also produced artificially.
2. NPs are heterogeneous in chemical nature .
3. NPs has size in the range of 1-100 nm with various shapes, with large surface area.
4. NPs are highly mobile and may show quantum effect.

Biosensors

Organic and Inorganic Nanoparticles

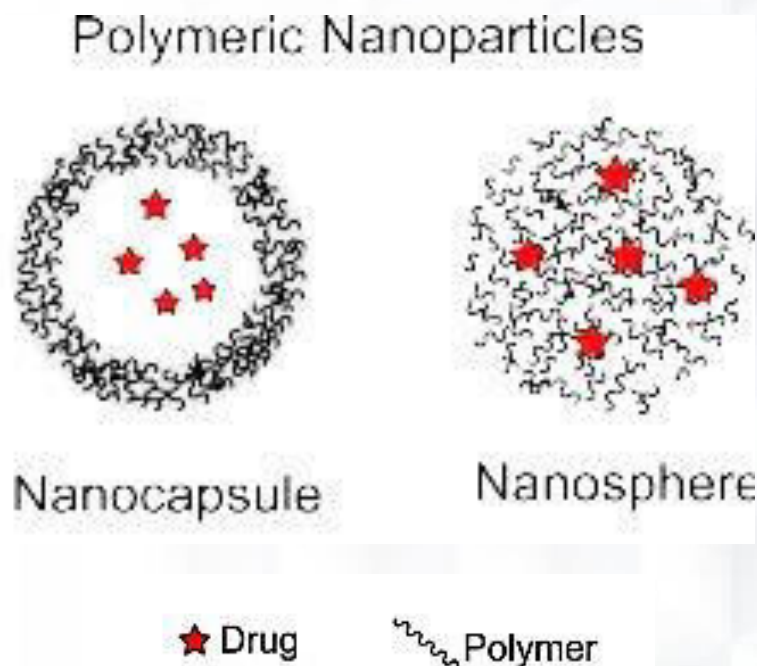


Organic and Inorganic Nanoparticles

Organic Nanoparticles

1. Organic NPs are derived from biological material and processed materials such as lipids and biological polymers.
2. The processed material based NPs includes carbon NPs.
3. The organic NPs are biocompatible and are widely used in Biomedical applications.

Organic and Inorganic Nanoparticles

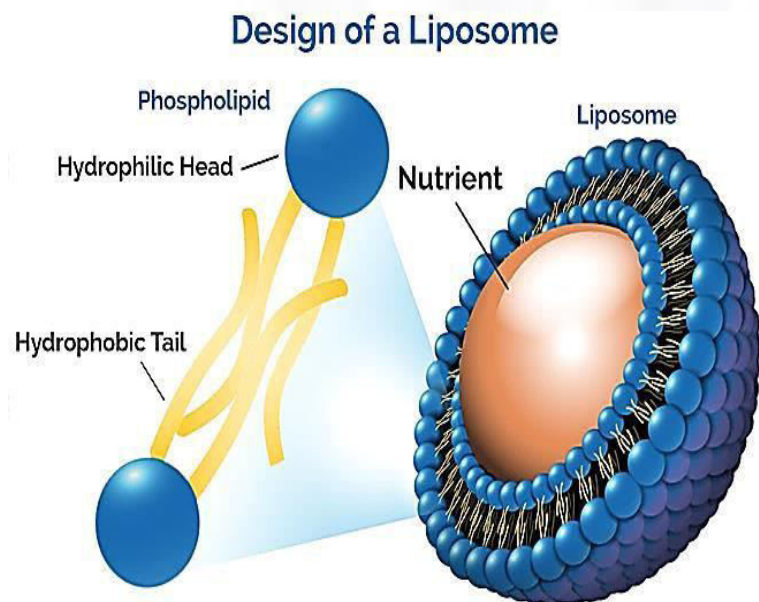


<https://ars.els-cdn.com/content/image/1-s2.0-S092849311732163X-fx1.jpg>

Polymeric Nanoparticles

1. Polymeric nanoparticles are nano sized solid particles.
2. They are prepared from natural and synthetic polymers and are used in drug delivery.
3. Natural material includes Chitosan, Gelatin.
4. Synthetic materials include Polylactides and Polymalic acid.

Organic and Inorganic Nanoparticles



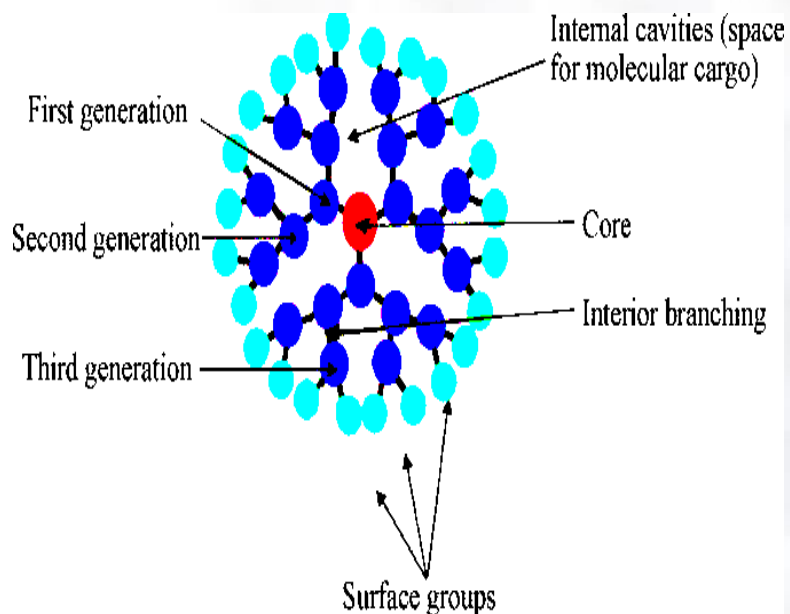
<http://www.integratedhealth.com/media/wysiwyg/liposome-diagram.jpg>

Liposomes

Liposomes are drug delivery vehicles. They can be formulated with a wide variety of natural, synthetic, and modified lipid species to deliver drugs and contrast agents.

1. Drug delivery
2. Gene delivery/transfection
3. Precursors for lipid coatings (eg: biosensors)
4. Model cell membranes

Organic and Inorganic Nanoparticles



https://www.researchgate.net/profile/Keerti_Jain/publication/215702385/figure/fig1/AS:692768582139904@1542180309079/General-structure-of-dendrimers.jpg

Dendrimer

Dendrimers are repetitively branched molecules. They have three main components.

1. A central core
2. An interior the branches.
3. The terminal functional group.

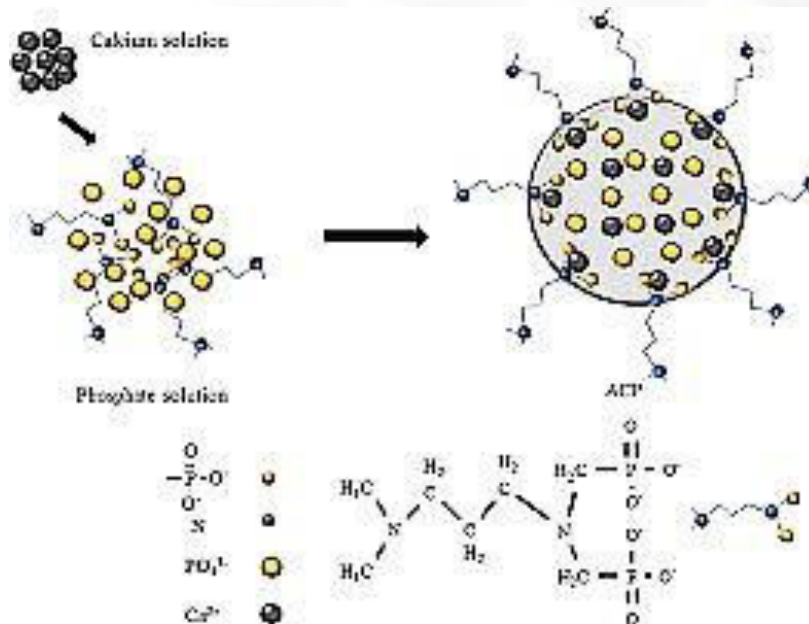
The terminal functional group is present at the outer surface of the molecule, used for *in vivo* and *in vitro* transport of DNA.

Organic and Inorganic Nanoparticles

Inorganic Nano particles

1. Inorganic nanoparticles cover a broad range of substances including elemental metals, metal oxides, and metal salts.
2. Inorganic nanoparticles are non-toxic, hydrophilic, biocompatible and highly stable compared to organic materials.
3. They are widely used in drug delivery systems.

Organic and Inorganic Nanoparticles



Calcium phosphate nanoparticles

<https://ars.els-cdn.com/content/image/1-s2.0-S1001841713004993-fx1.jpg>

Metallic NPs

1. Metal oxides of, Ti, Al, Zr, Mn and Ag, Au, are used for the synthesis of Inorganic NPs.
2. Functionalized Calcium Phosphate NPs are used for drug delivery.
3. Silica nanoparticles (SiO_2) exhibit intrinsic surface reactivity, for chemical modifications.

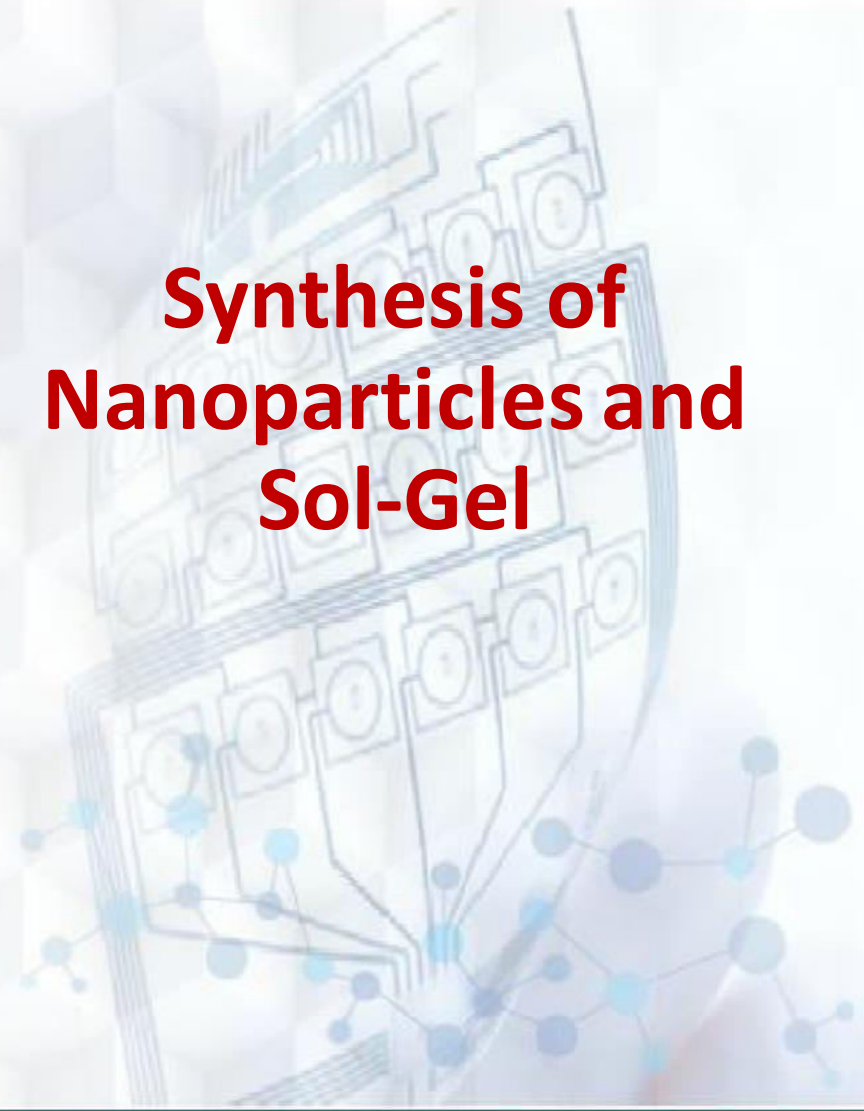
Organic and Inorganic Nanoparticles

Summary

1. Organic and Inorganic NPs are two main types of NPs.
2. In medical applications and drug delivery Organic NPs have an upper edge in terms of biodegradability over the inorganic nanoparticles.
3. NPs are used in the fabrication of Biosensors. Their main application is in the optical biosensors, due to quantum effect.

Biosensors

Synthesis of Nanoparticles and Sol-Gel

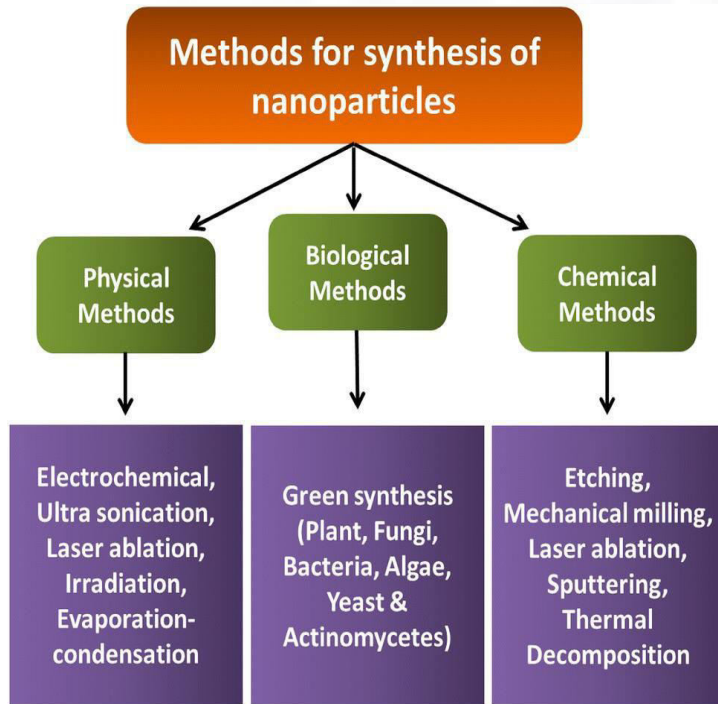


Synthesis of Nanoparticles and Sol-gel

Introduction

1. Free nanoparticles are formed through either the breaking down of larger particles or by controlled assembly processes.
2. Natural phenomena and many human industrial and domestic activities, such as cooking, manufacturing or road and air transport release nanoparticles into the atmosphere.

Synthesis of Nanoparticles and Sol-gel



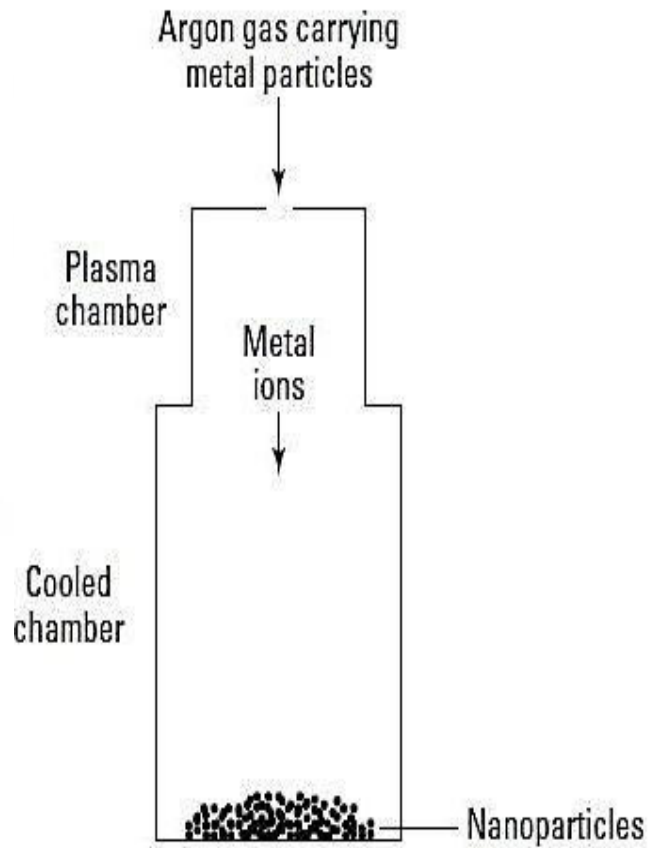
https://media.springernature.com/original/springer-static/image/art%3A10.1007%2Fs40097-018-0267-4/MediaObjects/40097_2018_267_Fig1_HTML.png

Methods

There are three main methods that are used to produce nanoparticles.

1. Biological method involved living organism for the synthesis.
2. Physical method employed different physical techniques.
3. Chemical methods used different chemical reagents and techniques.

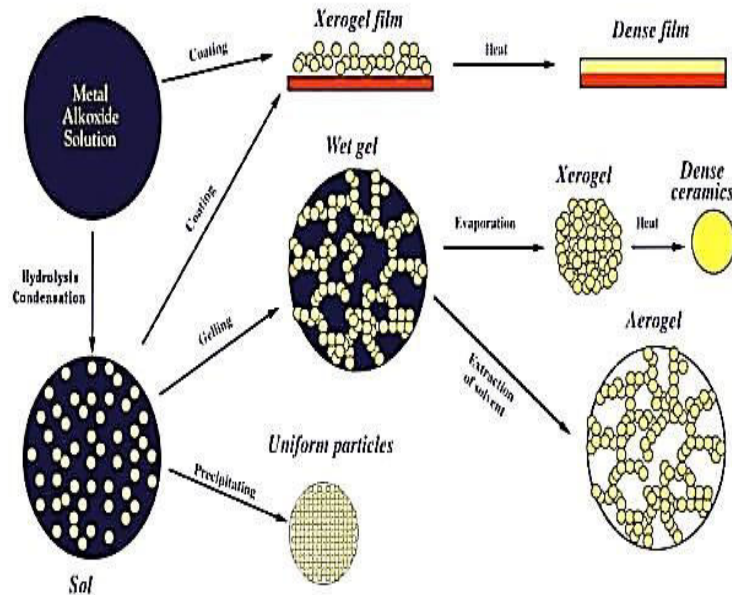
Synthesis of Nanoparticles and Sol-gel



Plasma source system

1. In a **plasma source system**, an inert gas, such as argon, flows into a chamber.
2. Gas carries **macroscopic particles** for nanoparticles production.
3. The **plasma is produced** by high-power radio frequency signal applied to the carrier gas.
4. The **ions then condense** into nanoparticles.

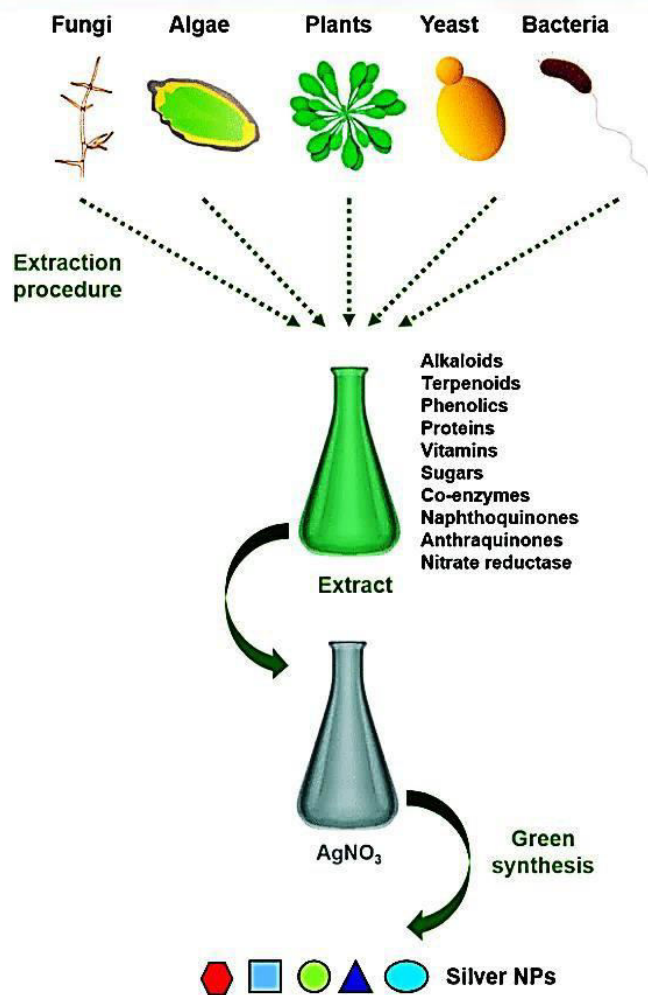
Synthesis of Nanoparticles and Sol-gel



Sol gel

1. **Partial hydrolysis** of metal alkoxides to form reactive monomers,
2. **Condensation** of these monomers to form colloid-like oligomers (sol formation).
3. **Additional hydrolysis** to promote polymerization and cross-linking leading to a three-dimensional matrix (gel formation).

Synthesis of Nanoparticles and Sol-gel



<https://pubs.rsc.org/image/article/2019/RA/c8ra08982e/c8ra08982e-f1.gif>

Green Synthesis of NPs

1. Plant leaf extracts are considered to be an excellent and benign source for metal as well as metal oxide nanoparticle synthesis.
2. Additionally, plant leaf extract play a dual role by acting as both reducing and stabilizing agents in nanoparticles synthesis process to facilitate nanoparticles synthesis .

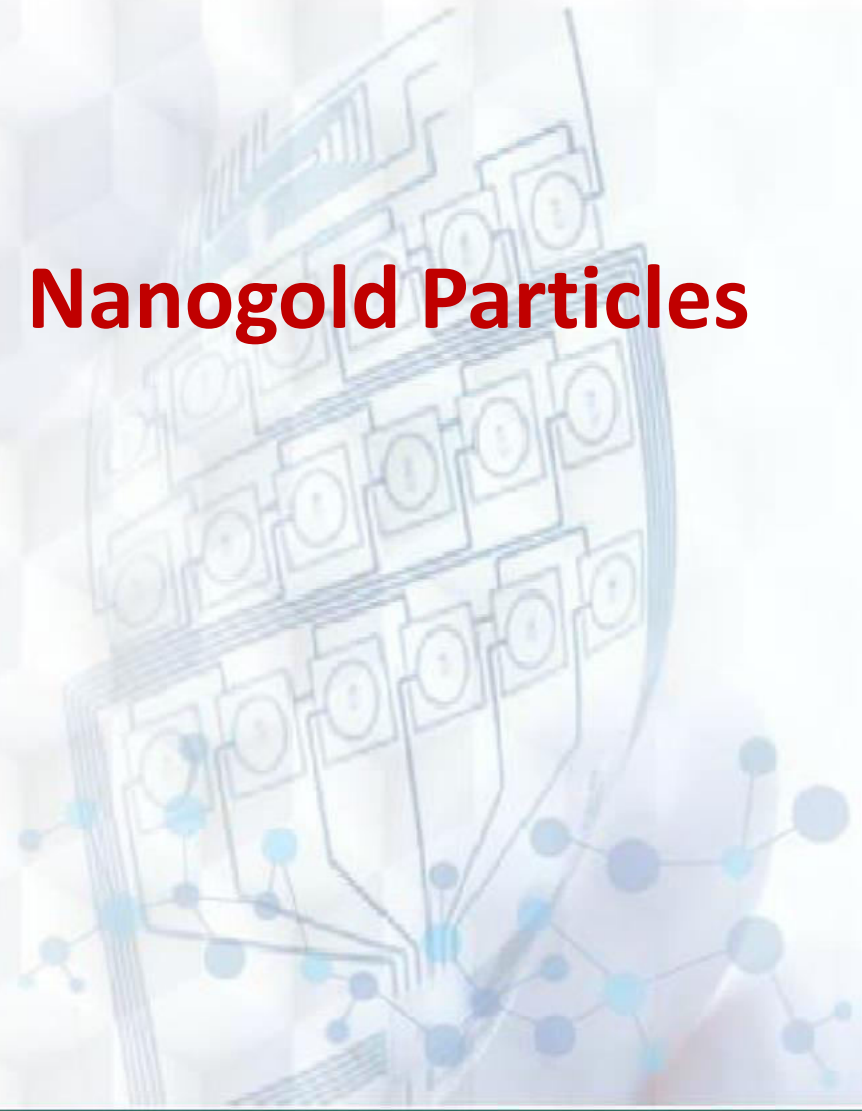
Synthesis of Nanoparticles and Sol-gel

Summary

1. Nanoparticles are produced naturally in the Eco system.
2. There are three different methods including Physical, chemical and green method for the synthesis.
3. A plasma source system for the efficient production of NPs using Argon.

Biosensors

Nanogold Particles



Nanogold Particles

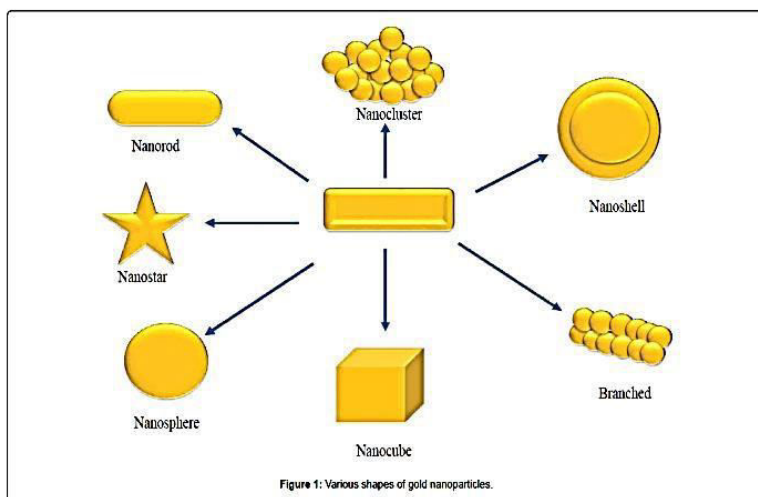


<https://lifeboat.com/blog.images/from-medicine-to-nanotechnology-how-gold-quietly-shapes-our-world3.jpg>

Introduction

1. Nanogold particles also known as colloidal gold.
2. AuNPs ranges from 1-50 nm in size.
3. AuNPs do not have any adverse effect on BRE and are highly stable.
4. They can be easily conjugated using thiol chemistry.
5. Large surface area for accommodation of BRE.

Nanogold Particles



<https://www.omicsonline.org/articles-images/Analytical-Toxicology-Variou-shapes-gold-nanoparticles-6-384-g001.png>

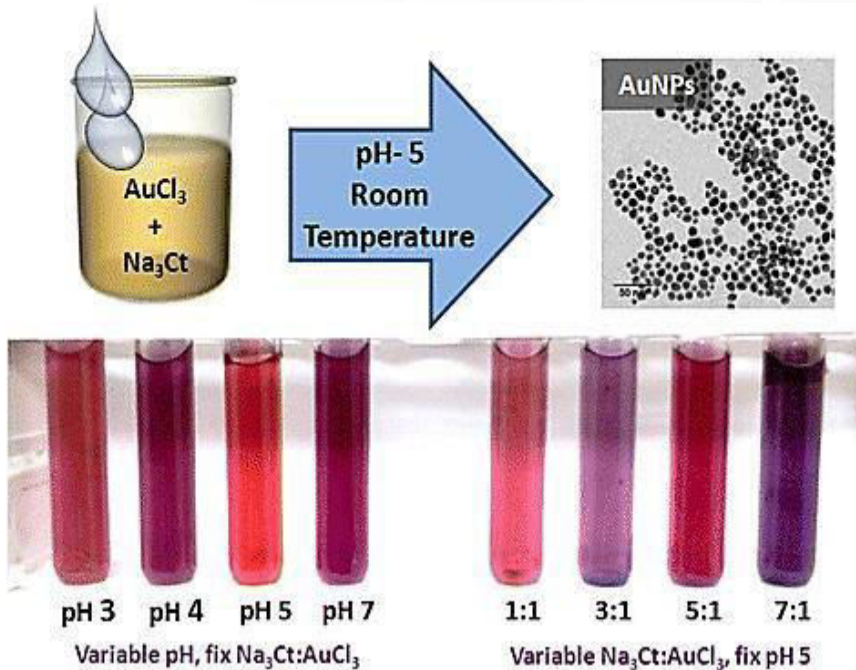


<https://www.sigmaaldrich.com/content/dam/sigma-aldrich/materials-science/nanomaterials/gold-nanoparticles/figure-1.jpg>

Shapes and color emission

1. AuNPs come in different shapes such as, star, rods, sphere and branch. depending upon the mode of synthesis.
2. For small ($\sim 30\text{nm}$) mono-disperse gold NPs, there is an absorption of light in the range $\sim 450\text{ nm}$.
3. For larger AuNPs light is absorbed in the range of $\sim 700\text{ nm}$ is reflected, yielding a rich red color.

Nanogold Particles



https://media.springernature.com/lw685/springer-static/image/art%3A10.1186%2Fs11671-016-1576-5/MediaObjects/11671_2016_1576_Sch1_HTML.gif

Synthesis

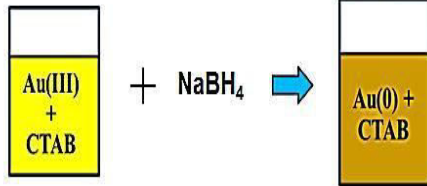
AuNPs can be synthesized by different ways. The most common method is to reduce gold salt.

1. Typically Hydrogen tetrachloro-Aurate or potassium tetrachloro-Aurate in the presence of Tri sodium citrate.
2. Citrate will be converted into citric acid reducing the Au into AuNPs with a spherical shape.

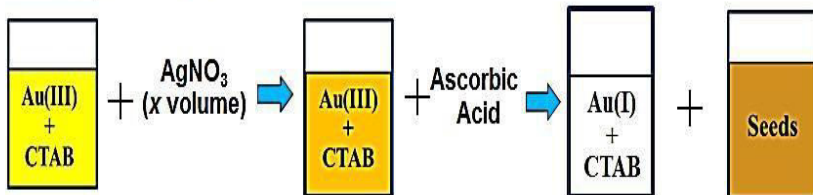
Nanogold Particles

Synthesis

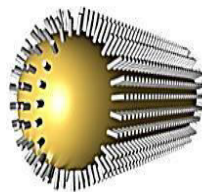
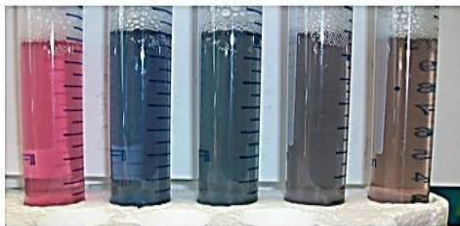
Step 1 - Seeds



Step 2 - Particles



GOLD NANOPARTICLES



Synthesis

An other method for the synthesis of AuNPs is the reduction of gold salt by Sodium tetra borate. This is a two step reaction.

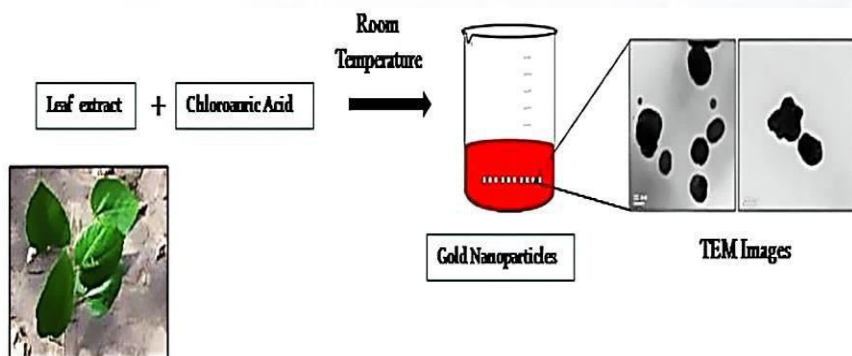
1. First step is the seeding in the presence of sodium tetra borate.
2. Second step is the separation of gold particles in the presence of Ascorbic Acid.

Nanogold Particles

Green Synthesis

A Green method for the isolation of AuNPs is also reported.

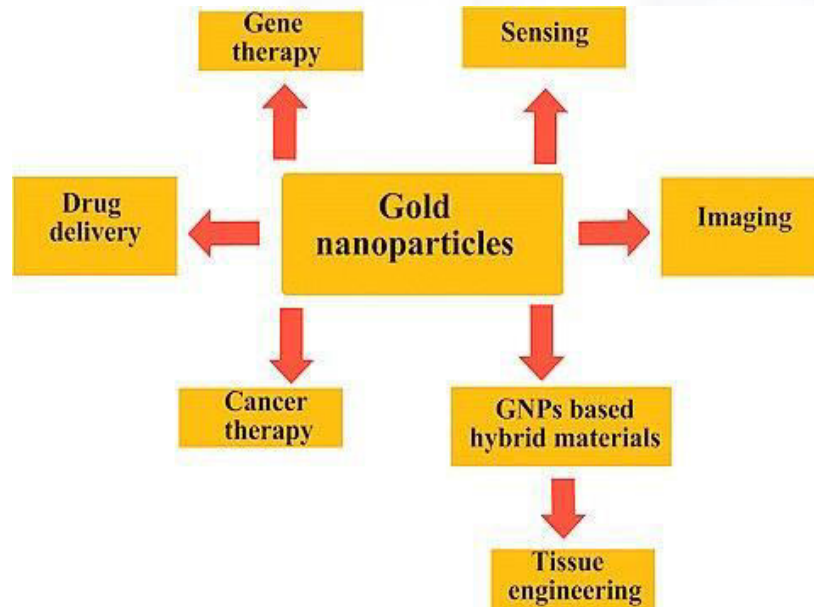
1. A gold compound chloroauric acid (HAuCl_4) is reduced with plant extract.
2. Size of Au nanoparticles was found to be around 20 nm and spherical in shape.
3. This is a simple method.



Croton Caudatus Geisel leaf

https://ars.els-cdn.com/content/image/1-s2.0-S0167577X1831588X-ga1_lrg.jpg

Nanogold Particles



https://media.springernature.com/original/springer-static/image/chp%3A10.1007%2F978-3-319-76090-2_10/MediaObjects/462172_1_En_10_Fig1_HTML.gif

Application

1. Electronics
2. Therapeutic Agent Delivery.
3. Sensors
4. Probes
5. Diagnostics
6. Biosensors
7. Catalysis
8. Tissue Engineering

Nanogold Particles

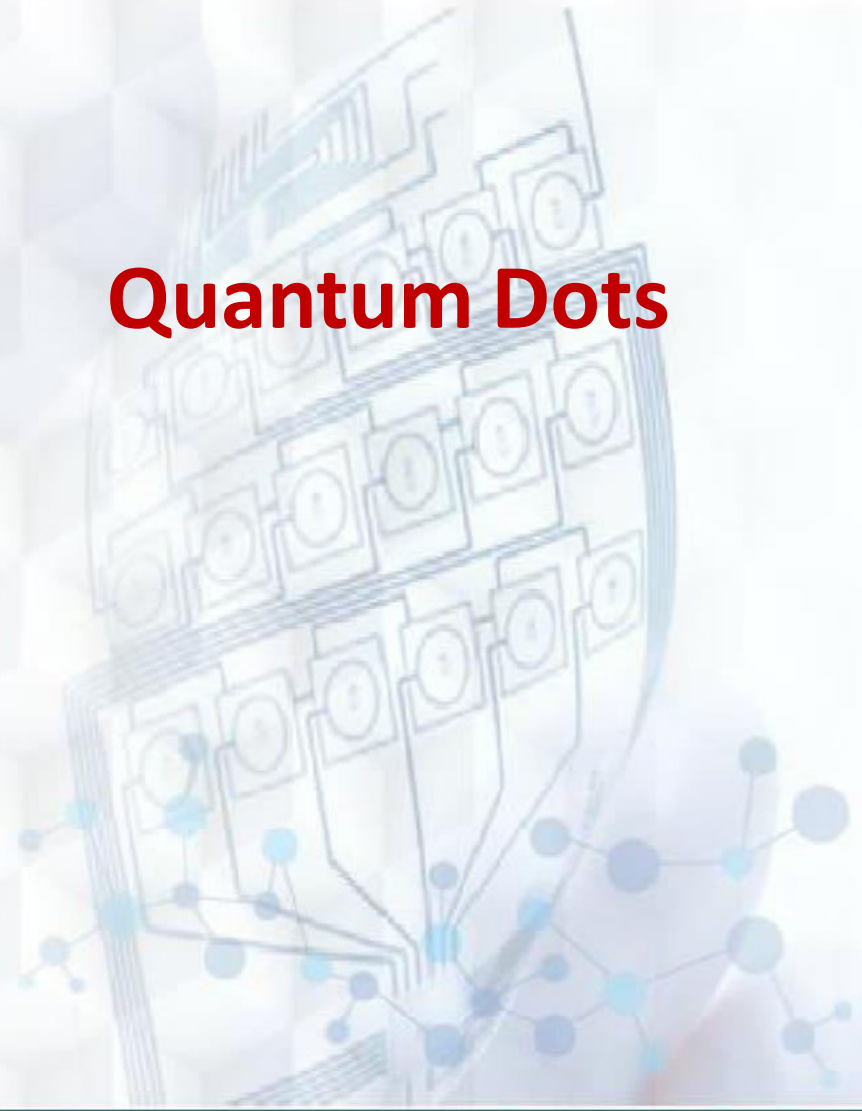
Summary

AuNPs are most suitable nano materials for Biosensor fabrication due to their

1. Stability
2. Biocompatibility,
3. Conductive properties
4. Surface plasmon effect
5. High surface area
6. AuNPs can be prepared by chemical methods and Green method.

Biosensors

Quantum Dots

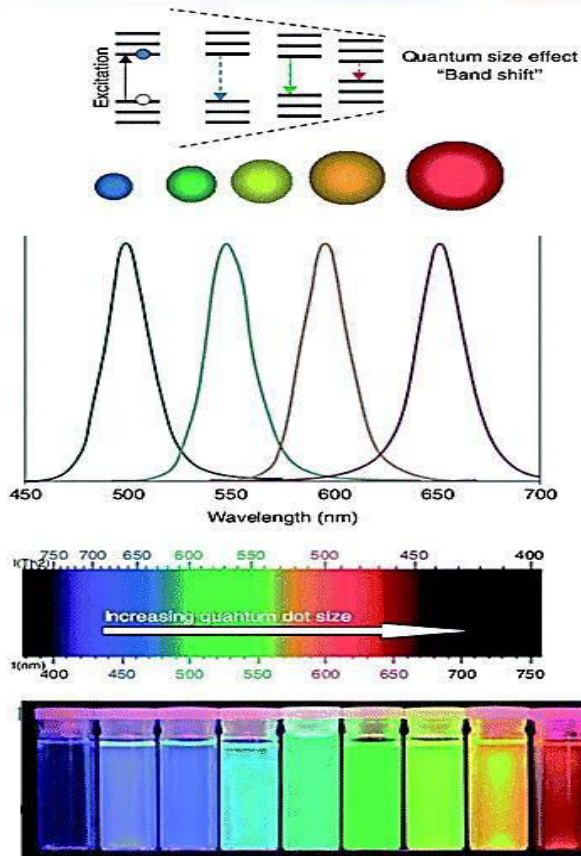


Quantum Dots

Introduction

1. Quantum dots are nanocrystals of semiconductor.
2. They have a diameter within the range of 2-10 nm (10-50 atoms).
3. QDs are fluorescent in nature.
4. QDs are comprised of two semiconductor materials. Example lead sulfide and lead selenide.

Quantum Dots

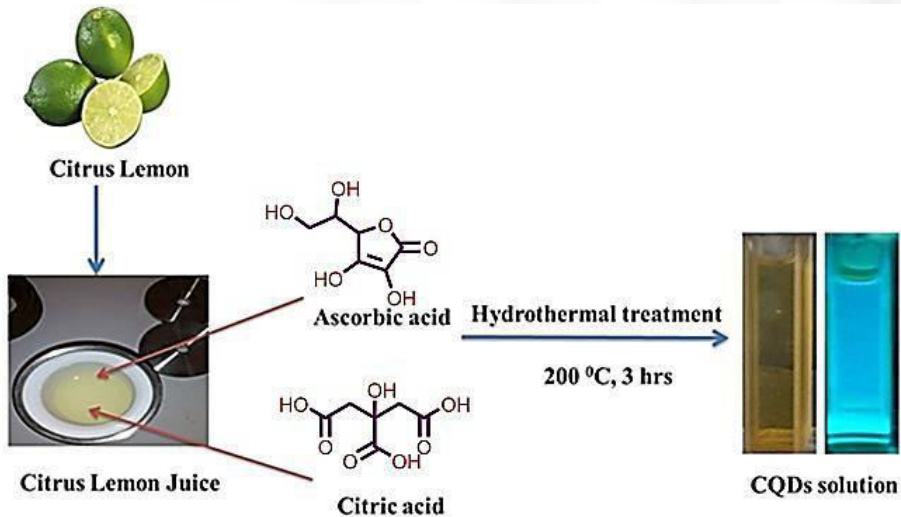


Color Emission

1. A small Quantum dots require more energy to excite it.
2. Smaller dots with higher energy produce higher frequencies and shorter wavelengths.
3. The smaller emit blue light and intermediate emit green light.
4. The larger dots produce red color.

https://encryptedtbn0.gstatic.com/images?q=tbn:ANd9GcRpyf1nPaLYmuUv_bHdBvNsxwsQl379ihW_rB3BMss_V8vGwTCh

Quantum Dots



<http://www.ajnanomat.com/data/ajnm/coversheet/951519968777.jpg>

Synthesis

There are several ways to fabricate quantum dots.

1. **Colloidal synthesis** QDs are synthesized from solutions, like a chemical processes.
2. **Plasma synthesis**, is the gas-phase approaches for the production of QDs.
3. **Green method** plant derived material is reduced to produce QDs.

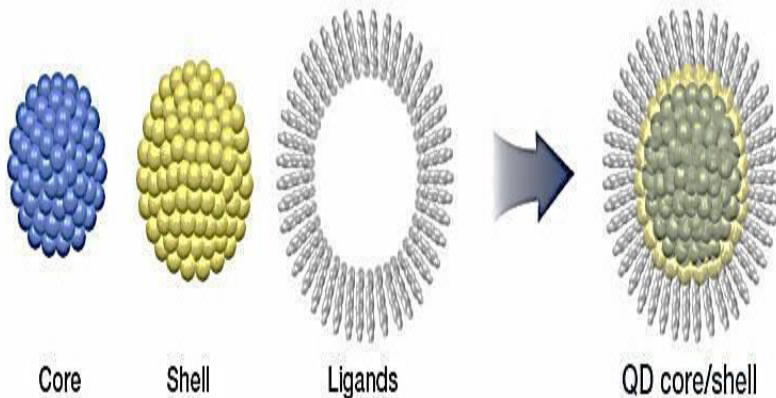
Quantum Dots

Classification

Quantum dots can be classified into different types based on their composition and structure.

1. Core-Type Quantum Dots
2. Core shell quantum dots
3. Alloyed Quantum Dots

The core and the shell are semiconductors, such as CdS/ZnS, CdSe/ZnS and CdSe/CdS, have **low fluorescence quantum** yield.



<https://assets.pid.samsungdisplay.com/s3fs-public/qd-shell-ligands.jpg>

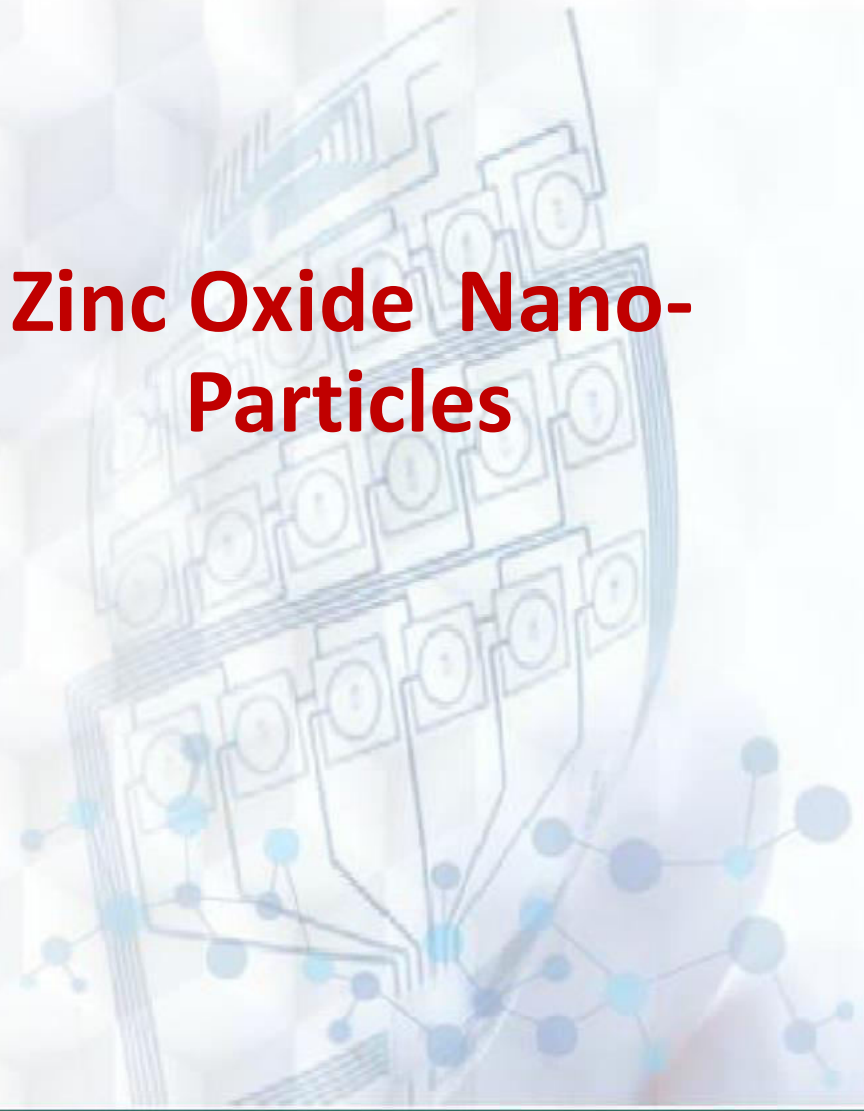
Quantum Dots

Summary

1. QDs are tiny particles of a semiconducting material with diameters in the range of 2-10 nms (10-50 atoms).
2. They can emit any color of light from the same material simply by changing the dot size.
3. QDs are comparable to nanogold particles.

Biosensors

Zinc Oxide Nano-Particles

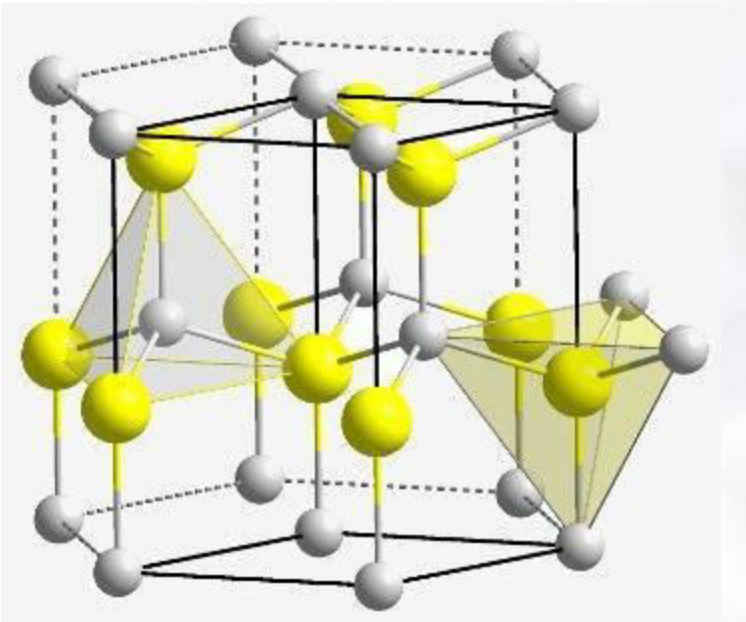


Zinc Oxide Nanoparticles

Introduction

1. Zinc oxide is a metal oxide.
2. Zinc is a chemical element with the symbol Zn and atomic number 30.
3. Zinc is a slightly brittle metal at room temperature.
4. It is the first element in group 12 of the periodic table.
5. Zinc oxide is used for the synthesis of nanoparticles.

Zinc Oxide Nanoparticles

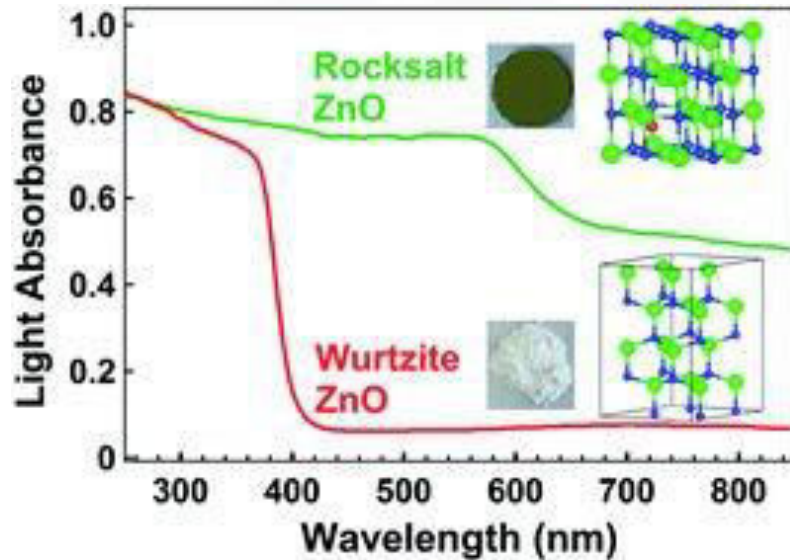


<https://whatiszincoxide.weebly.com/uploads/9/8/9/7/9897238/722846068.jpg>

Crystal Structure of Nano particle

1. Zinc oxide crystallizes in two main forms Hexagonal Wurtzite crystals and cubic zinc blend crystals.
2. The Wurtzite structure is most stable at ambient conditions.
3. The zinc oxide nano particles exist in wurtzite crystal form.
4. The atomic position of crystal is same as hexagonal diamond.

Zinc Oxide Nanoparticles



<https://pubs.rsc.org/en/Content/Image/GA/C7TA05262F>

Electrical Properties

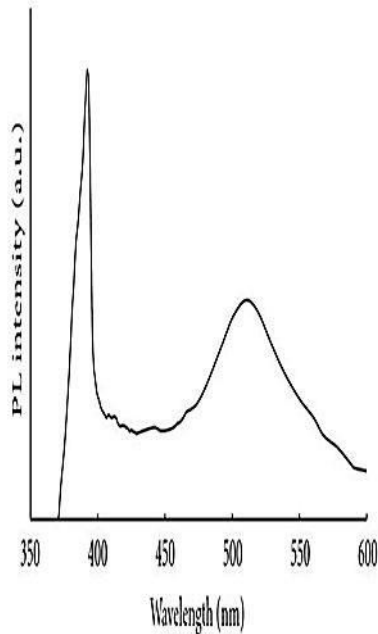
oxide has a relatively large direct band gap of around 3.3 eV at room temperature. The advantages associated with large band gap include:

1. Higher breakdown voltage
2. Ability to sustain large electrical field
3. Lower electronic noise
4. High temperature operation
5. High power operation

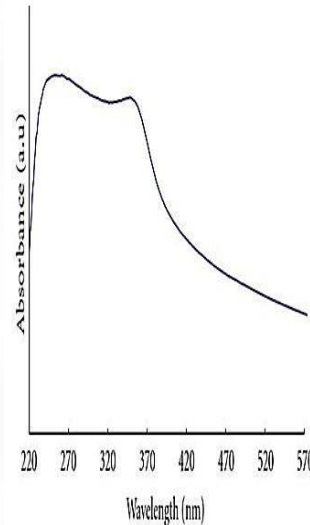
Zinc Oxide Nanoparticles

Quantum properties

1. The UV-vis absorption spectrum shows an absorption band at 355 nm due to ZnO nanoparticles.
2. The photoluminescence spectrum exhibits two emission peaks
 - First at 392 nm
 - Second located at 520 nm.

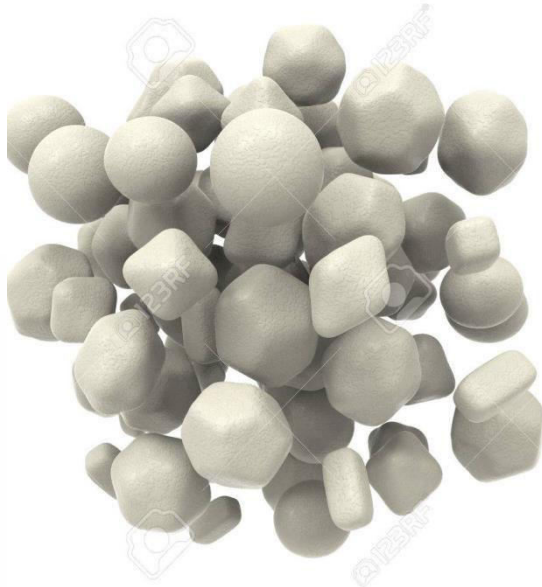


Photoluminescence spectrum of prepared ZnO nanoparticles



UV-vis absorption spectrum of ZnO nanoparticles.

Zinc Oxide Nanoparticles



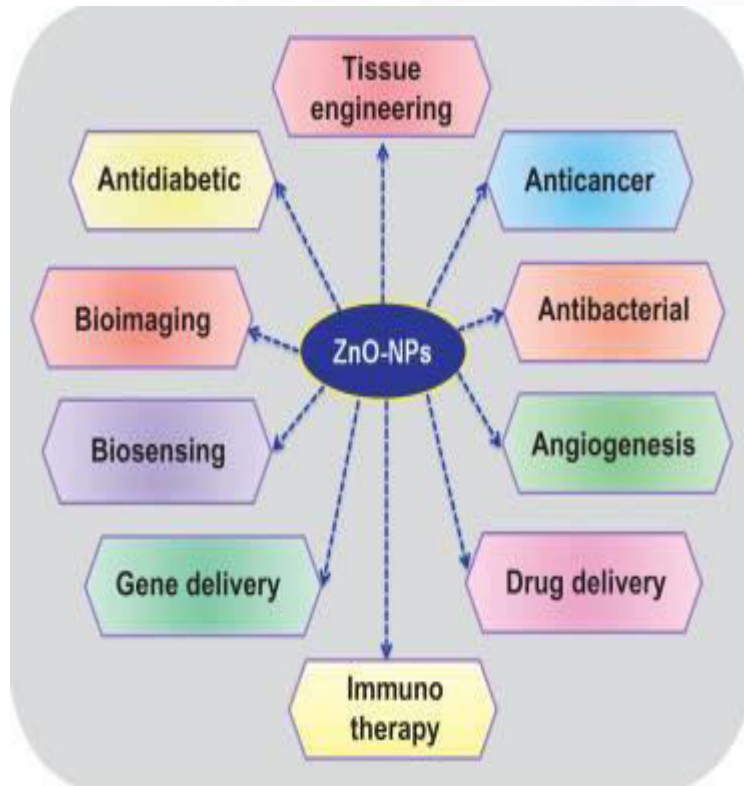
<https://previews.123rf.com/images/drmicrobe/drmicrobe1709/drmicrobe170900037/85616531-zinc-oxide-nanoparticles-3d-illustration-zno-nanoparticles-have-application-as-biosensors-in-drug-de.jpg>

Properties

Zinc oxide is a metal oxide. Zinc oxide nano particles are within the range of 1-100 nano-meters.

1. They are chemically stable.
2. Thermal resistant
3. High catalytic activity
4. Long shelf life
5. Anti-bacterial activity
6. Zn is an essential mineral

Zinc Oxide Nanoparticles



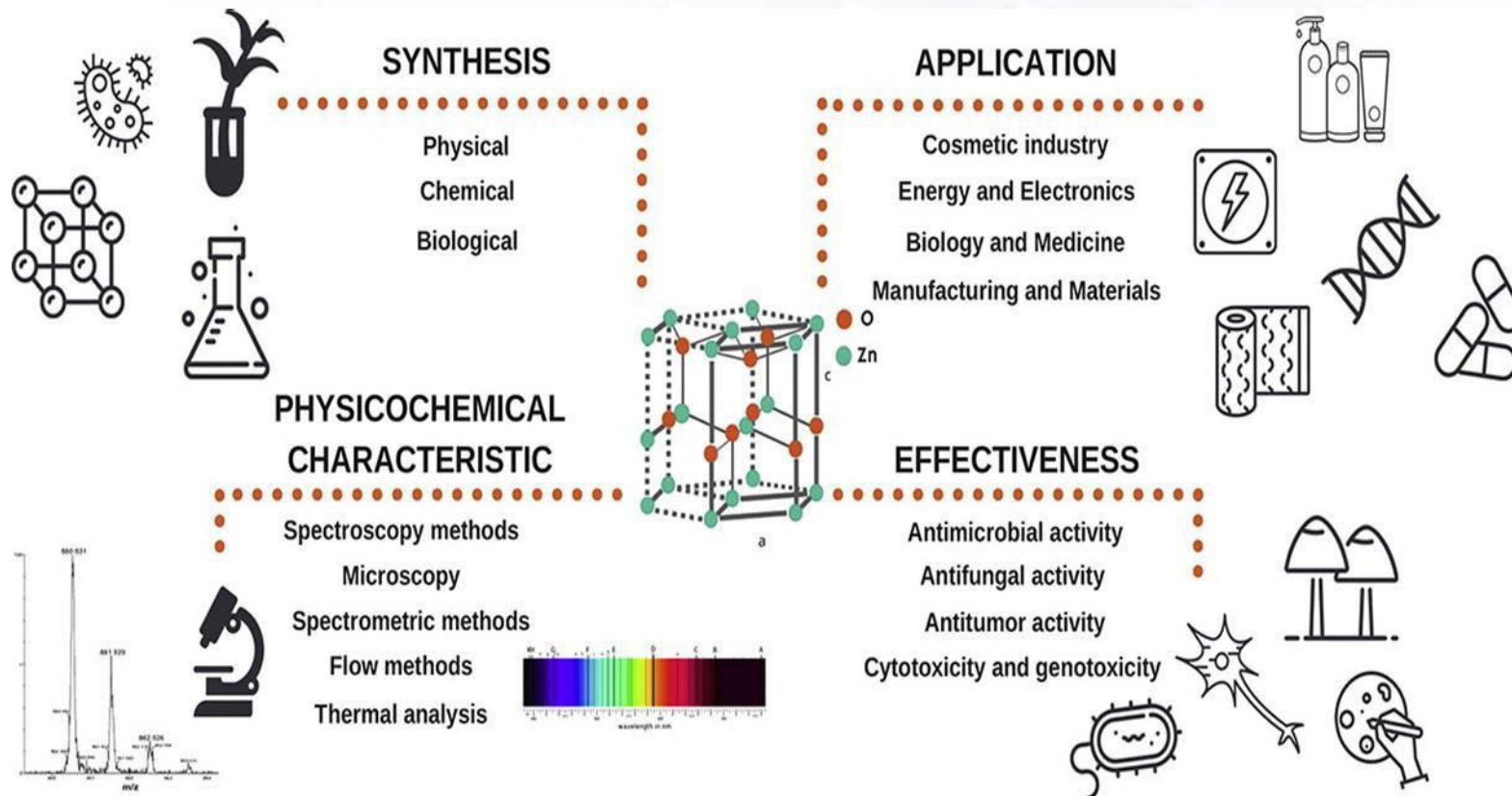
<https://ars.els-cdn.com/content/image/3-s2.0-B9780128136614000067-f06-sc01-9780128136614.jpg>

Applications

ZnO nano particles has application in many fields. Large surface area relative to their size

1. Medical Science
2. Nano generator
3. Gas sensor
4. Biosensor
5. Solar cells
6. Photodetector
7. Biosensors

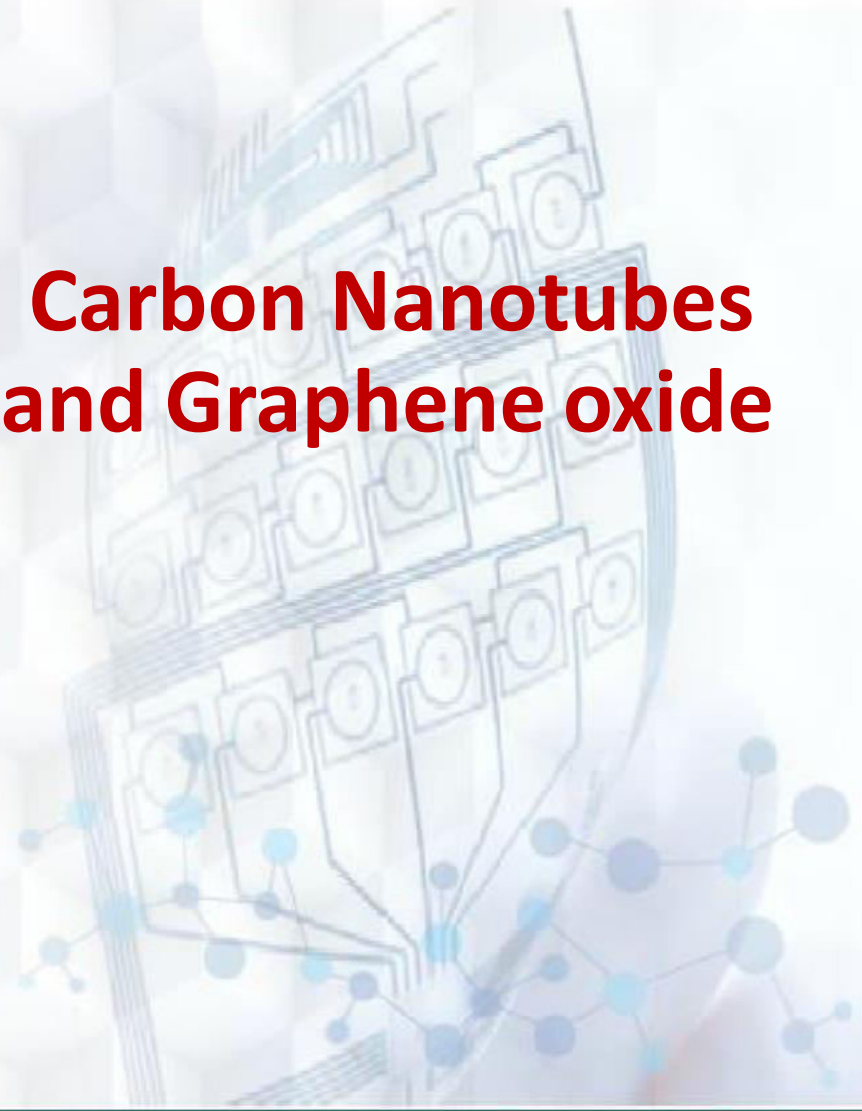
Zinc Oxide Nanoparticles



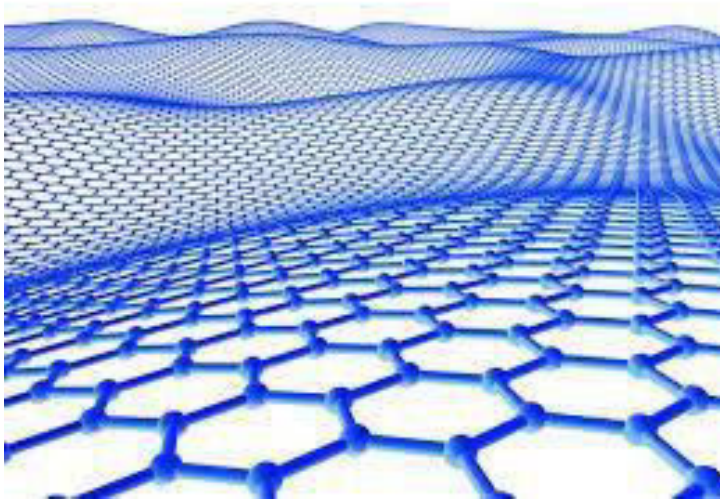
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Biosensors

Carbon Nanotubes and Graphene oxide



Carbon Nanotubes and Graphene Oxide

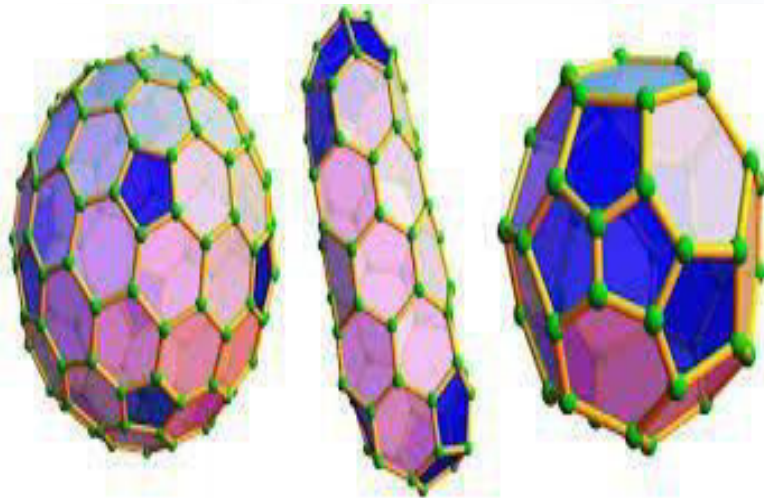


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Graphene

1. Graphene is an allotrope of carbon in the form of a single layer of atoms
2. It has in a two-dimension and has hexagonal lattice.
3. In the structure one atom forms each vertex.
4. It is the basic structural element of other allotropes, including graphite, charcoal, carbon nanotubes and fullerenes.

Carbon Nanotubes and Graphene Oxide

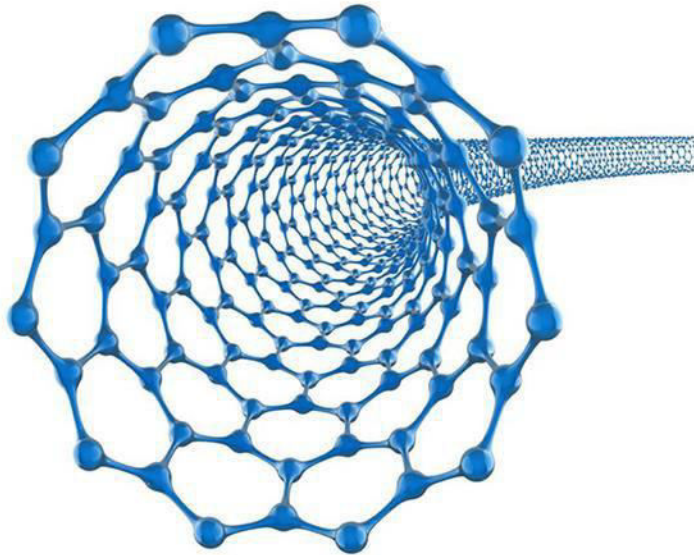


<http://i2.wp.com/blog.zacharyabel.com/wp-content/uploads/2012/07/big-fullerenes.png?resize=600%2C250>

Fullerene

1. Twisting and warping of the graphene sheet results in formation of 0-dim Fullerenes.
2. Fullerene is single and double bonds with fused rings of five to seven atoms.
3. The molecule may be a hollow sphere, ellipsoid, tube, or many other shapes and sizes.

Carbon Nanotubes and Graphene Oxide

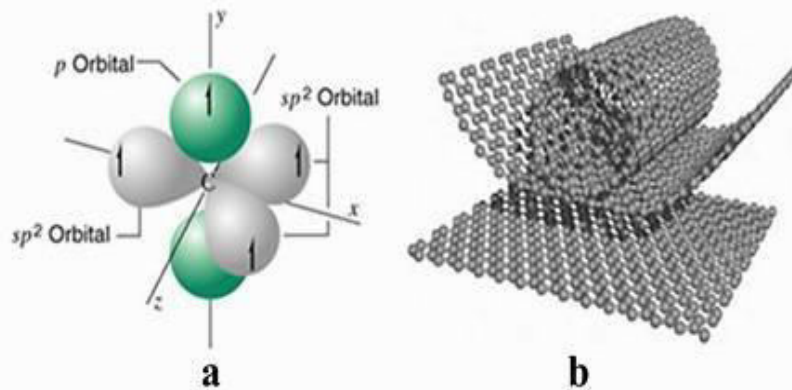


<https://knowledge.ulprospector.com/media/2015/02/NanotubeFeatured.jpg>

Carbon Nanotubes

1. Rolling of the sheet along an axis results in formation of 1-dim CNTs.
2. Carbon nanotubes (CNTs) are cylindrical molecules that consist of rolled-up sheets.
3. The tensile strength is approximately 100 times greater than that of steel of the same diameter.
4. Nanotubes are elastic.

Carbon Nanotubes and Graphene Oxide

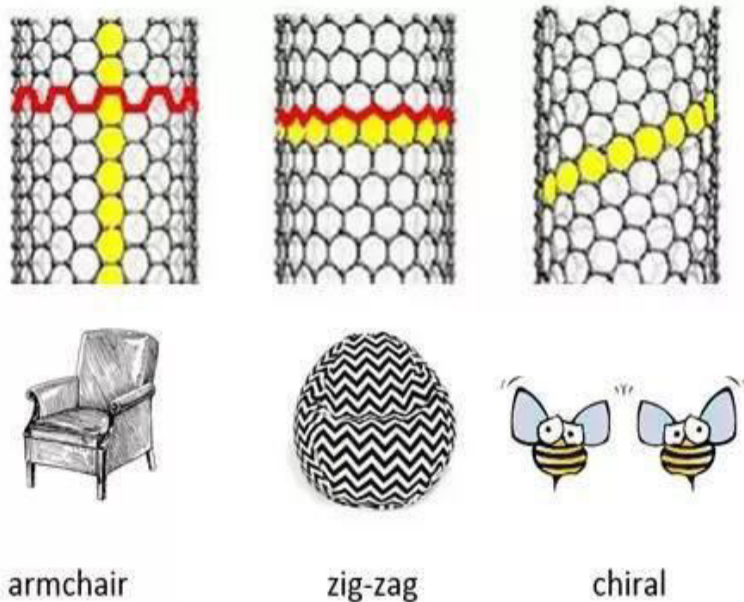


<https://steemitimages.com/640x0/https://s14.postimg.org/89arqev41/fig2.png>

Structure and stability

1. The unique strength of carbon nanotubes is due to orbital hybridization.
2. It causes the bonds between adjacent carbon atoms to be of the sp^2 type.
3. These bonds, which are similar to those of graphene, are stronger than the sp^3 bonds in alkanes and diamond.

Carbon Nanotubes and Graphene Oxide

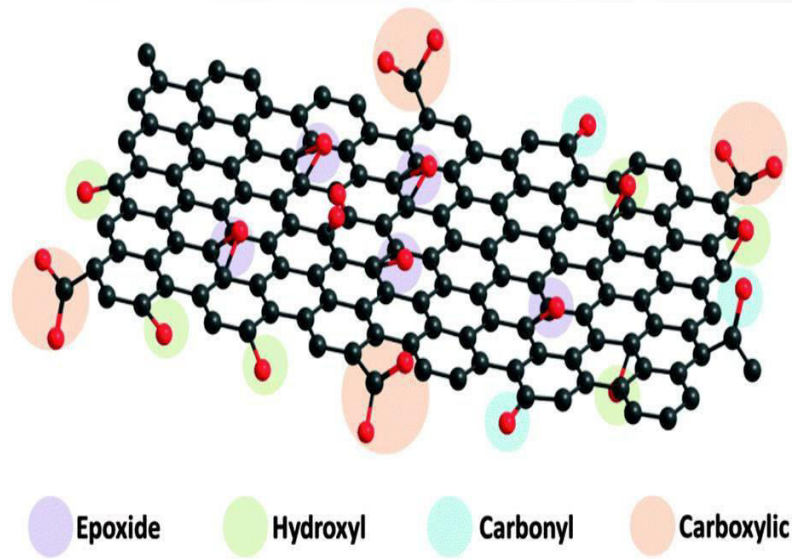


<https://qph.fs.quoracdn.net/main-qimg-76c1b2d3ee94624c370f23e72fb4e0df.webp>

Single Walled CNTs

1. The structure of an ideal (infinitely long) single-walled carbon nanotube is a regular hexagonal lattice.
2. It is drawn on an infinite cylindrical surface, whose vertices are the positions of the carbon atoms.
3. There are three types of CNTs. The zigzag, chiral and armchair configurations.

Carbon Nanotubes and Graphene Oxide



https://nanografi.com/product_images/uploaded_images/graphene-oxide-market.gif

Graphene Oxide

1. Graphene oxide (GO) is a single monomolecular layer of graphite.
2. It has various oxygen-containing functionalities such as epoxide, carbonyl, carboxyl, and hydroxyl groups.
3. It has optical and electronic properties enable graphene oxide to be used in many fields.

Carbon Nanotubes and Graphene Oxide

Summary

1. Graphene is the basic unit for the production of carbon based different nanostructures.
2. Graphene, fullerene and carbon nanotubes are the allotropic forms of the carbon.
3. Graphene is used sensors, faster and efficient electronics,, faster DNA sequencing, drug delivery, and more.

Biosensors

Porous Gold

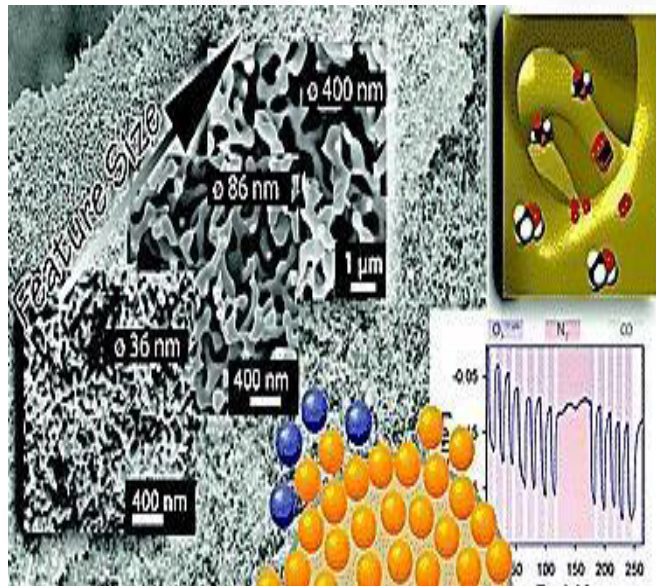


Porous Gold

Introduction

1. Nanoporous gold (np-Au), a three-dimensional nanoporous bulk material.
2. Made by selective corrosion of Ag from Ag–Au alloys.
3. The composition is Ag_{0.03}-Au_{0.97}.
4. It has a catalytic activity and act as a functionalize platform for a number of reactions.

Porous Gold

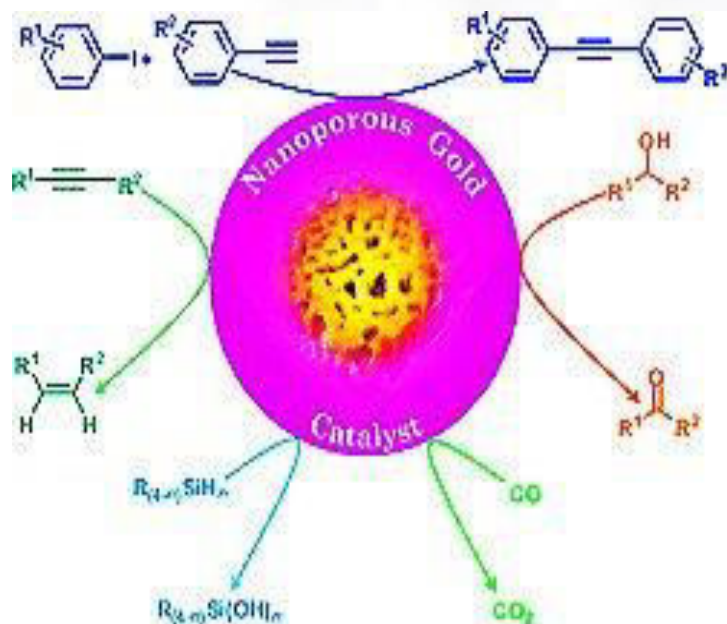


<https://pubs.rsc.org/en/Image/Get?imageInfo.ImageType=GA&imageInfo.ImageIdentifier.ManuscriptID=C0CP00757A&imageInfo.ImageIdentifier.Year=2010>

Structure.

1. Based on a self-organisation process.
2. It is a 3 dimensional sponge like gold structure evolves with ligaments in the range of only a few tens of nanometers.
3. Due to its continuous porosity, the material can be penetrated by gases which then adsorb and interact with the surface.

Porous Gold



<https://pubs.rsc.org/en/Image/Get?imageInfo.ImageType=GA&imageInfo.ImageIdentifier.ManuscriptID=C3CY00241A&imageInfo.ImageIdentifier.Year=2013>

Unique properties

NPG is a heterogeneous catalyst due has unique structural properties, including

1. Open nanoporosity
2. High surface area
3. Excellent electrical conductivity,
4. Nontoxicity nature
5. Easy recyclability
6. tunable surface chemistry.

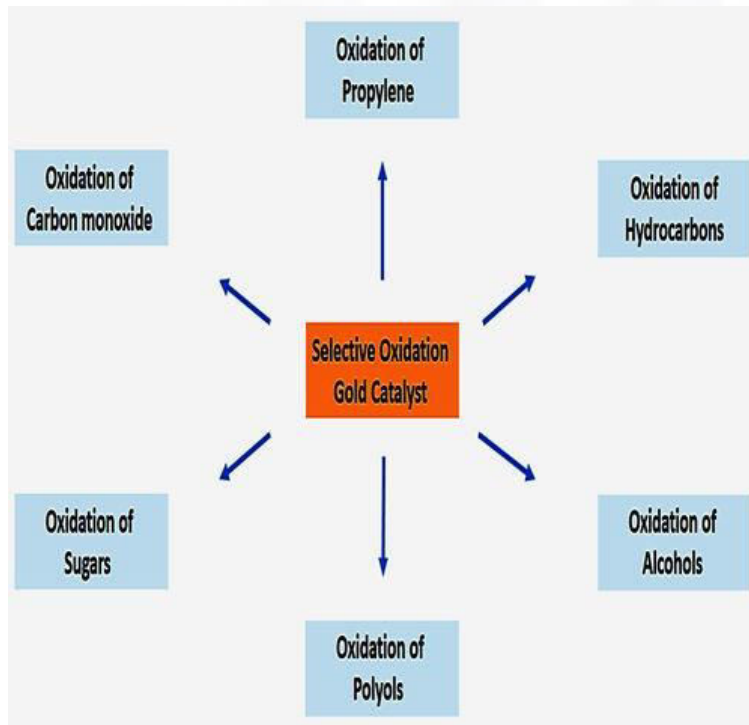
Porous Gold

Types of Reactions

The catalytic performance of nanoporous gold

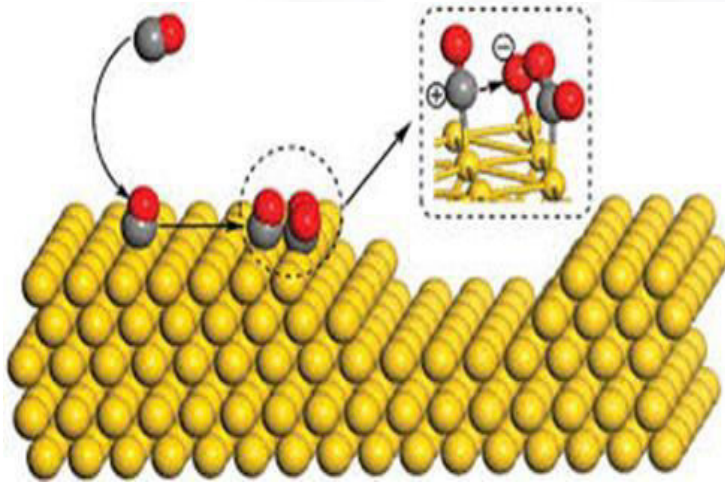
1. CO oxidation
2. Hydrogen oxidation and production
3. Oxidation of alcohols
4. Reactions in liquid phase

Reduction reaction or hydrogenation is more difficult because H-H dissociation is not easy on gold.



https://www.mdpi.com/catalysts/catalysts-09-00402/article_deploy/html/images/catalysts-09-00402-g004.png

Porous Gold



Aerobic Methanol Oxidation over
Unsupported Nanoporous Gold

https://media.eurekalert.org/multimedia_prod/pub/web/174572_web.jpg

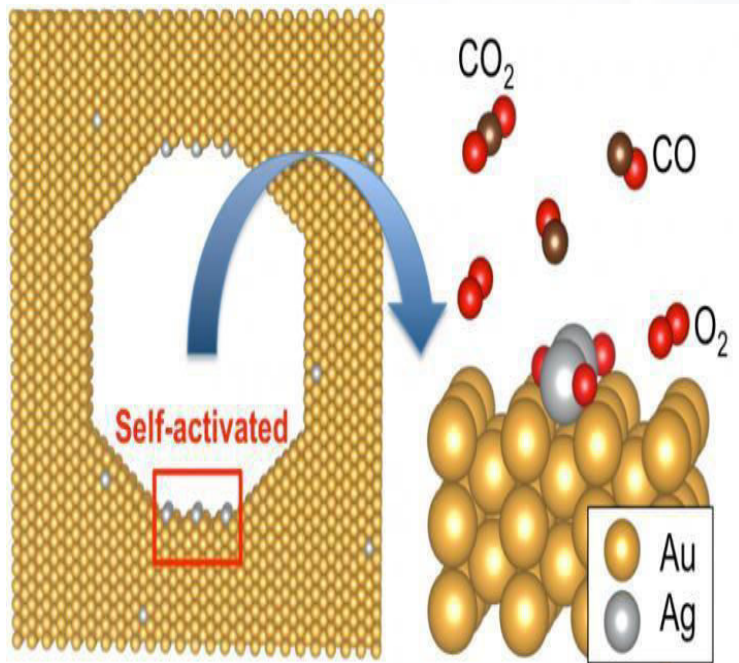
Catalytic Activity

The catalytic performance of nanoporous gold

1. CO oxidation
2. Hydrogen oxidation and production
3. Oxidation of alcohols
4. Reactions in liquid phase

Reduction reaction or hydrogenation is more difficult because H-H dissociation is not easy on gold.

Porous Gold



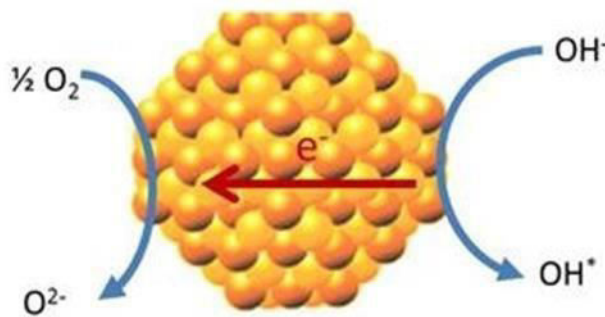
https://media.eurekalert.org/multimedia_pro/d/pub/web/174572_web.jpg

CO Oxidation

The catalytic performance of nano particulate gold strongly depends on size and support.

1. CO oxidation has been the representative model reaction for investigating the catalytic efficiency.
2. It is found to be very effective at temperatures as low as 30°C when nanoporous gold was used as catalyst.

Porous Gold



Aerobic Methanol Oxidation over Unsupported Nanoporous Gold

https://www.mdpi.com/catalysts/catalysts-09-00416/article_deploy/html/images/catalysts-09-00416-ag-550.jpg

Selective Oxidation of Alcohol

Selective oxidation of alcohol is one of the important catalytic conversions in petrochemical industry.

1. In Gold-based catalysts molecular oxygen can be used as an oxidant for the selective oxidations.
2. It works under mild conditions, that is, at low temperature ($<100^\circ\text{C}$) and under an ambient pressure

Porous Gold

Summary

1. Nano Porus Gold is a sponge like with is an interconnected three-dimensional network.
2. The network formed with ligament and pore sizes typically in the range of 20–50 nm.
3. It has emerged as a promising solid catalyst for reactions It allows to work at moderate conditions.

Biosensors

Latest Research and Current Trends



Latest Research and Current Trends

Nanoparticles for bone regeneration

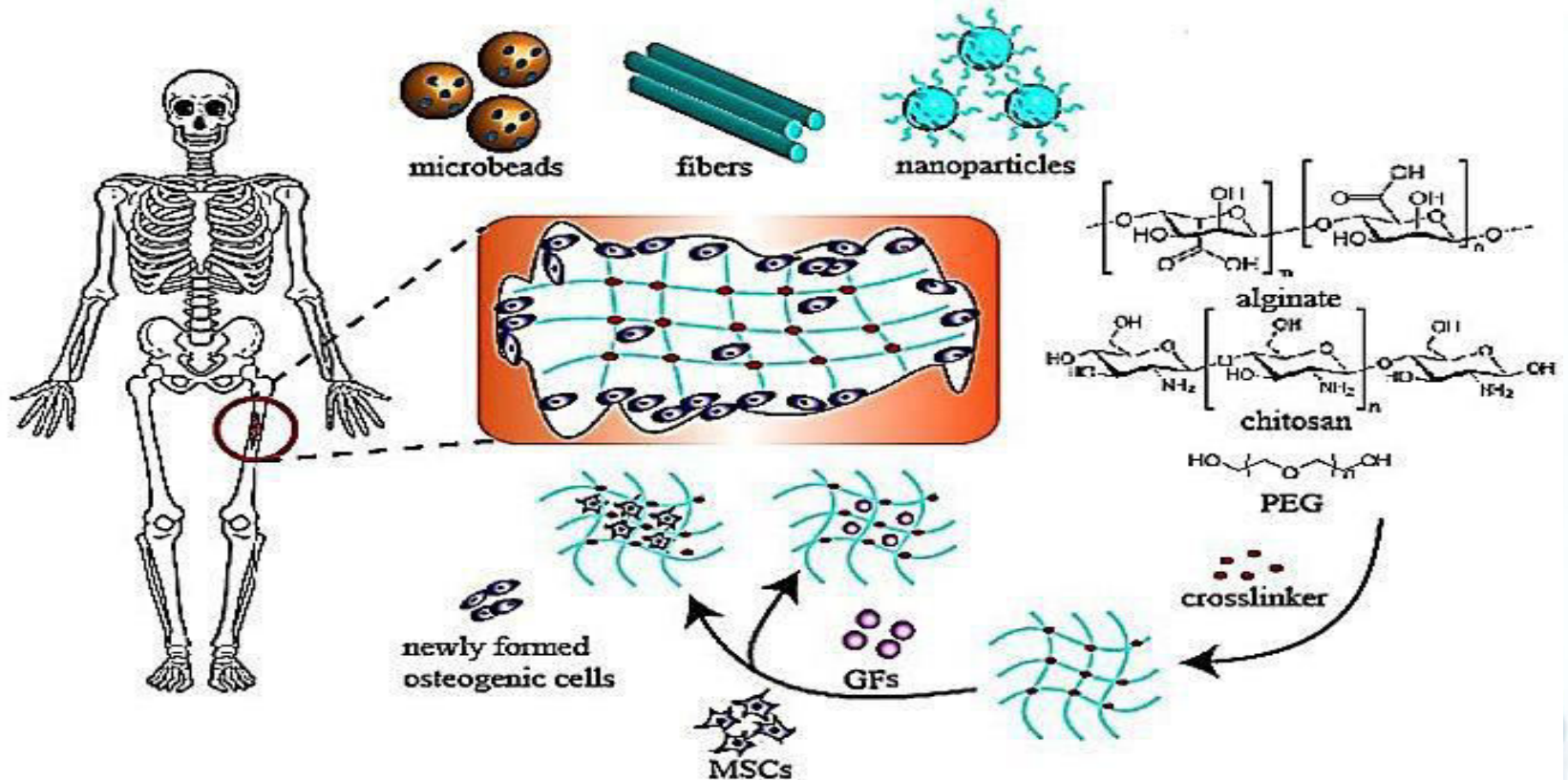
1. Nano particles are ultra small particles within the range of nano scales.
2. They have high loading capacity due to large surface area;
3. They can easily travel into the target cell due to their small size, shape, and crystallinity.

Latest Research and Current Trends

Bioactive hydrogels for bone regeneration

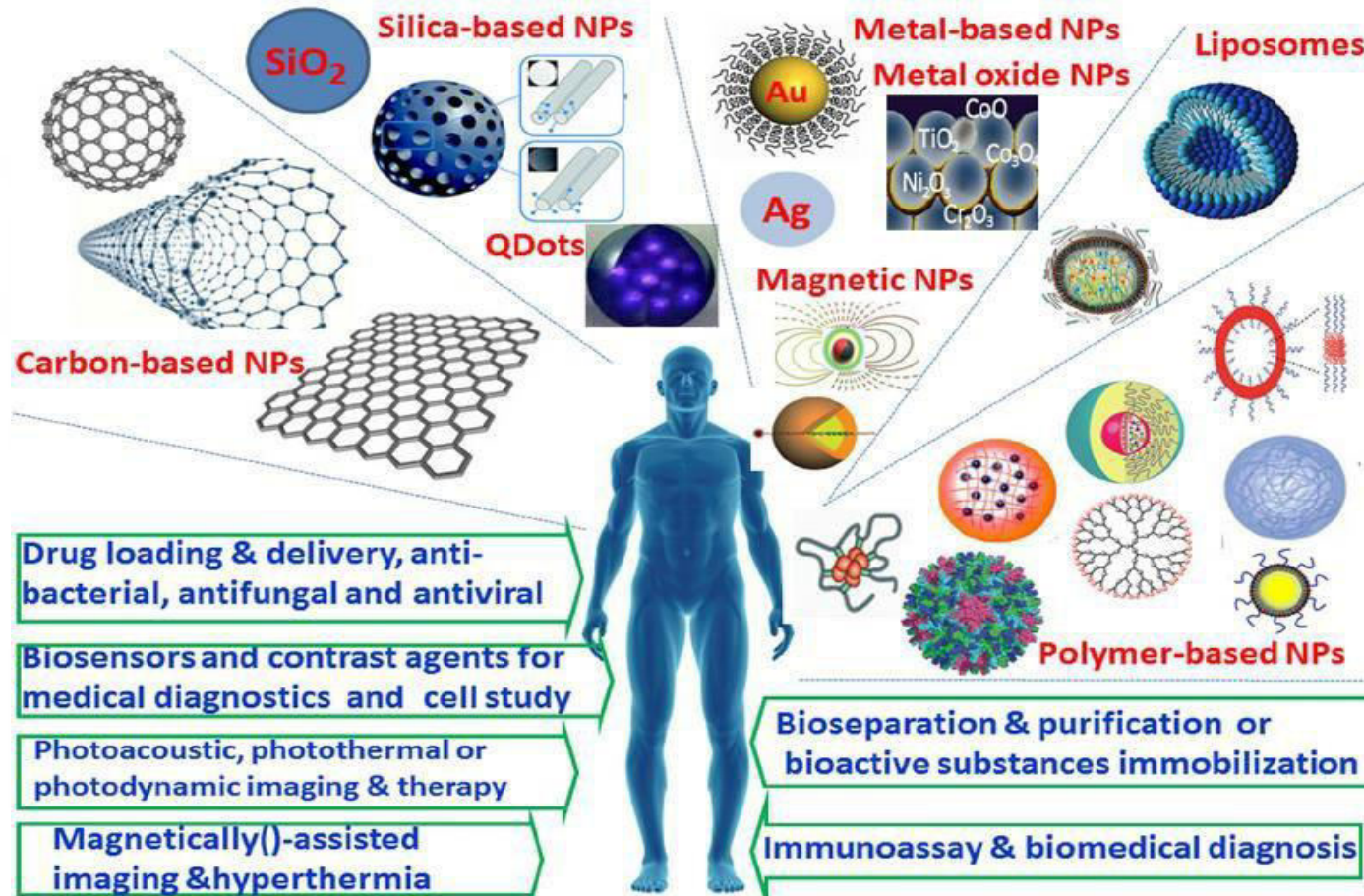
1. The human skeleton is the internal framework of the human body.
2. In elderly people bone regeneration is a problem due to decrease in level of certain hormone and growth factors.
3. To address the issue scientists have introduced a Bioactive hydrogel for bone regeneration.

Latest Research and Current Trends



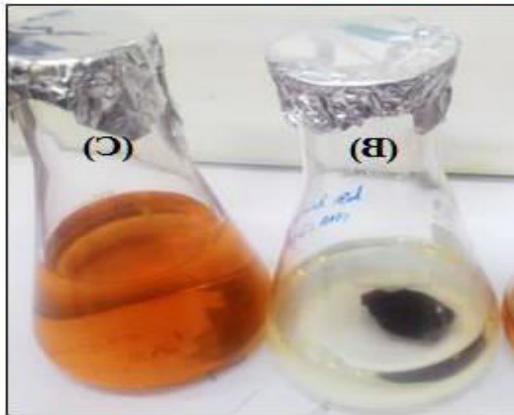
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6038268/bin/fx1.jpg>

Latest Research and Current Trends



https://www.researchgate.net/profile/Jianhua_Zhang27/publication/285628437/figure/download/fig2/AS:303838159491073@1449452070544/Examples-of-typical-nanoparticles-and-their-applications-in-biomedical-fields.png

Latest Research and Current Trends



Graphene Oxide for Textile wastewater Treatment

1. Textile waste water contained a considerable amount of carcinogenic dyes.
2. Graphene oxide is an oxidized form of Graphene.
3. GO is helpful in efficiently removing the organic dyes from wastewater..

<http://researcherslinks.com/current-issues/Synthesis-Application-Chitosan-Starch-Nanocomposite-Wastewater-Treatment/20/1/241/html>

Latest Research and Current Trends

Summary

1. The main application of nanomaterials is in medicine and diagnostics
2. Nanoparticles and nanomaterials are used for the production of smart materials like bioactive gels.
3. A number of products are introduced to treat industrial wastewater economically.

Biosensors

Electrochemical Sensor

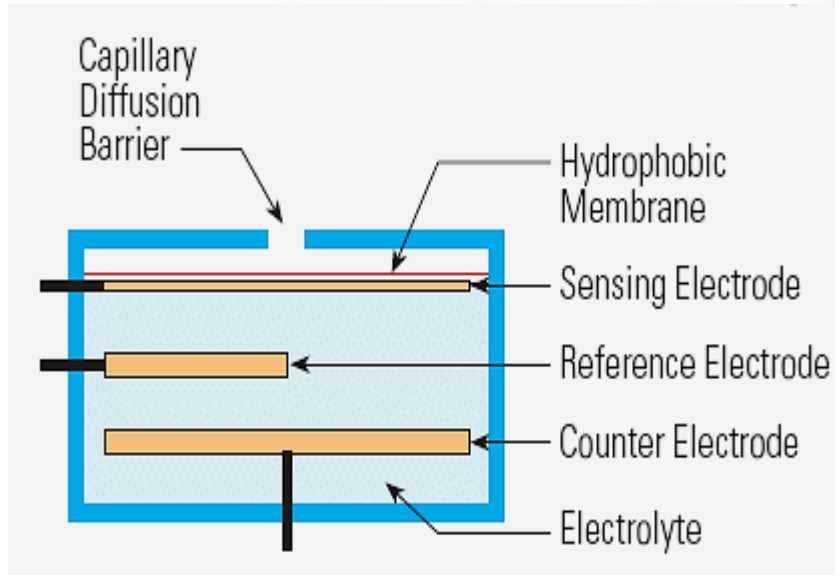


Electrochemical Sensors

Introduction

1. Electrochemical sensors are based on the principle of Electrochemistry.
2. Oxidation and reduction reactions are monitors in the Electrochemical Sensor.
3. Oxidation and reduction reactions involved the transfer of electrons between two electrodes, leading to the generation of current.

Electrochemical Sensors



<https://www.researchgate.net/profile/Boris-Stoeber/publication/255660220/figure/fig7/AS:669434851127304@1536617114456/Schematic-of-a-typical-electrochemical-gas-sensor-17.ppm>

Components

A typical electrochemical sensor is composed of the following components.

1. Gas Permeable Membrane
2. Sensing Electrodes
3. Reference Electrode
4. Counter electrode
5. Electrolyte
6. Filter
7. Transducer

Electrochemical Sensors

Working

1. Electrochemical sensors are operated based on the **diffusion of gas of interest** into the sensor (Physical Process).
2. The **diffused gas** in turn gets **oxidized or reduced** at the sensing electrode.
3. This results in the **production of an electrical signal** that is proportional to the gas concentration.

Electrochemical Sensors

Factors effecting

1. **Pressure changes.** The differential pressure within the sensor can cause sensor damage.
2. **Temperature.** Electrochemical sensors are also quite sensitive to temperature.
3. **Humidity.** As the sensor is dependent on REDOX reaction humidity can alter the catalytic properties of the system.

Electrochemical Sensors

Types

1. **Potentiometric**: Use Ion Selective Electrodes to determine the conc. of chosen ions (H ion conc.).
2. **Amperometric**: Measure current resulting from redox reactions.
3. **Conductometric** Measure changes in ionic strength composition resulting from an enzyme reaction.

Electrochemical Sensors



Alcohol Sensor

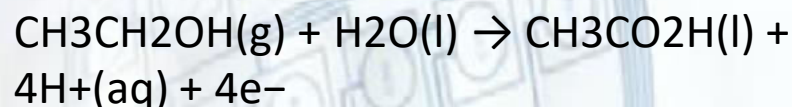
1. Ethanol is oxidized to acetic acid in the presence of Oxygen.
2. A layer of Titanium oxide on alumina act as a catalyst and reaction occurs.
3. The electric current produced by this reaction is measured by a microcontroller, is proportional to blood alcohol content (BAC).

<https://a.pololufiles.com/picture/0J2254.1200.jpg?bfece8ab31f247b19d1e279394beceef>

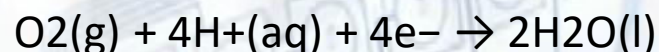
Electrochemical Sensors

Reaction

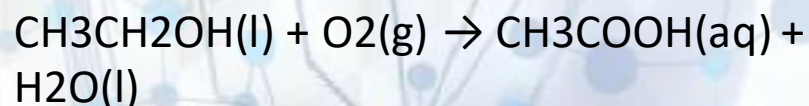
When the user exhales into a breath analyzer, any ethanol present in their breath is oxidized to acetic acid at the anode:



At the cathode, atmospheric oxygen is reduced:



The overall reaction



Electrochemical Sensors

Summary

1. The electrochemical sensors are based on the principle of Electrochemistry.
2. The main components involve the electrodes, a membrane, a transducer and a catalyst (Chemical or Biological) .
3. There are many applications of the electrochemical Biosensors.

Biosensors

Mass Sensitive Sensor



Mass Sensitive Sensors

Introduction

1. Mass is a physical property of the matter.
2. If there is a change in the quantity of one state of matter, it is an indication of Physical or chemical change.
3. Mass Sensitive Sensors determines the change.
4. Piezoelectric technique
5. Surface Plasmon Résonance technique

Mass Sensitive Sensors

Piezo electric Sensor

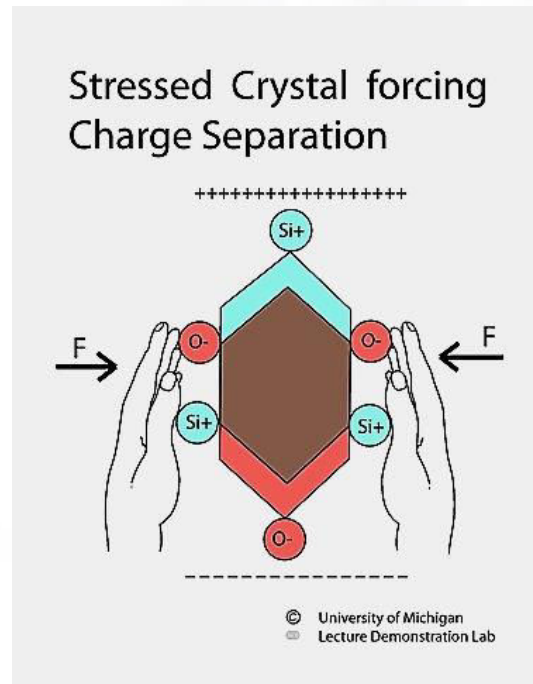
Piezoelectric sensor converts **pressure or mechanical stress** into an **alternating electrical force**.

It is used for measuring the physical quantity like

1. Force
2. Pressure
3. Stress

which is directly not possible to measure.

Mass Sensitive Sensors

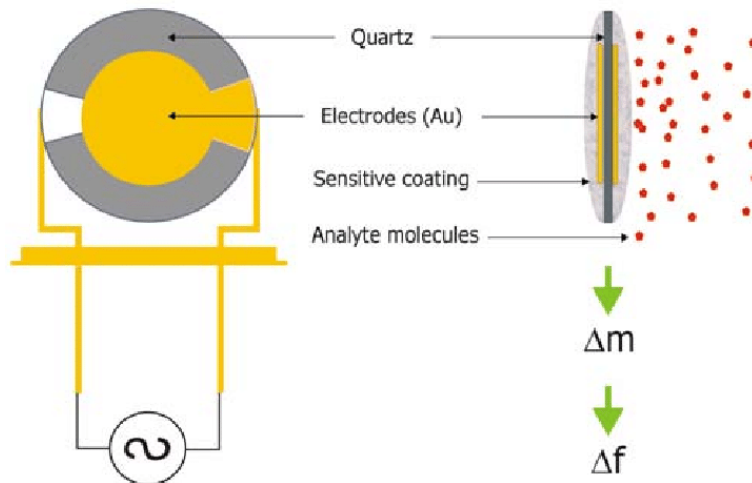


https://sharepoint.umich.edu/lsa/physics/demolab/DemoLab%20Asset%20Library/Diagram%20of%20Stressed%20PiezoElectric%20Crystal%20forcing%20Charge%20Separation_5A10.01.jpg

Piezo electric Technology

1. Piezoelectric materials are used in Piezo electric technology.
2. Piezoelectric materials are able to generate internal electrical charge from applied mechanical stress.
3. Piezoelectric material is used for mechanical and acoustic devices.
4. Quartz is a mineral with a framework of Si and O.

Mass Sensitive Sensors



https://www.researchgate.net/profile/Arief_Yuwono/publication/252415670/figure/fig4/AS:667616867794951@1536183673296/Basic-working-principles-of-a-quartz-crystal-microbalance-QCM-sensor.png

Quartz Crystal Microbalance

Quartz crystal microbalance is a mass sensitive piezoelectric device.

1. It consists of a gold electrode, Quartz crystal, sensitive coating and volt meter.
2. Quartz crystal oscillates at a defined frequency by applying an appropriate voltage usually via metal electrodes.

Mass Sensitive Sensors

Surface Plasmon Resonance Technique

1. It is an optical phenomena generated by light.
2. The technique is based on **plasmonic effect**.
3. The plasmonic effect takes place on the surface.
4. The plasmonic effect is the interaction between **free electrons** in metal **nano particles** and **incident light**

Mass Sensitive Sensors

Summary

In mass sensor technology two different types of techniques are used.

1. Piezoelectric technique
2. Surface plasmon resonance technique

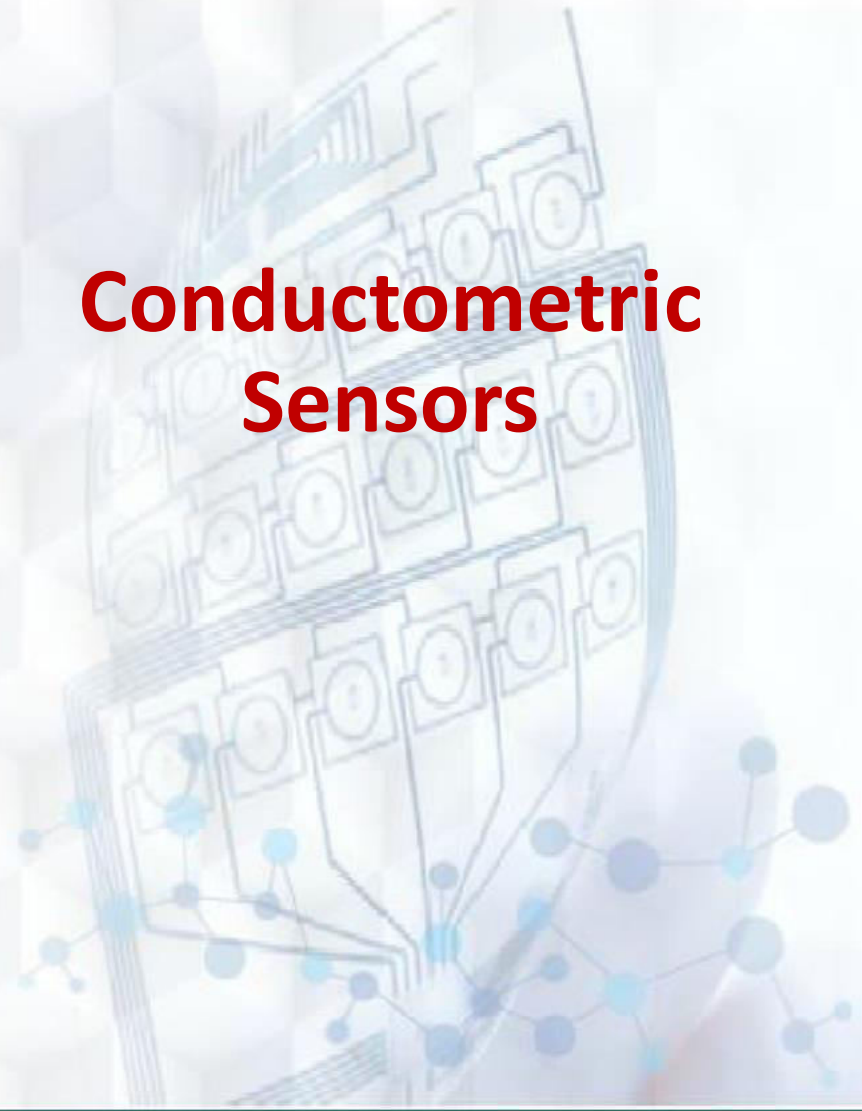
Application of these two techniques allowed the mass determination at nano scale.

Example.

Quartz crystal microbalance

Biosensors

Conductometric Sensors



Conductometric Sensors

Introduction

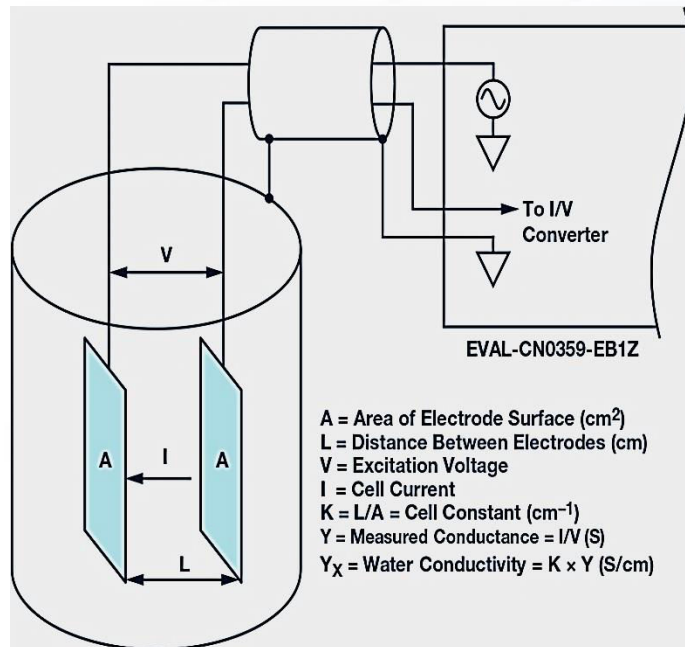
1. Conductivity is the measure of a solution's ability to pass or carry an electric current.
2. Conductometry is to monitor **the changes in the electrical conductivity** of the test sample.
3. This method is of great importance in the detection of toxic, harmful gases, like ammonia.

Conductometric Sensors

Conductivity measurements

Conductivity is measured by conductivity cell.

1. The simplest kind of measuring cell used consists of two similar electrodes.
2. An alternating voltage applied to one of the conductivity electrodes causes the ions in the solution to migrate towards the electrodes.



<https://www.analog.com/-/media/images/analog-dialogue/en/volume-50/number-4/articles/fully-automatic-self-calibrated-conductivity-measurement-system/136516-fig-01.png?la=en>

Conductometric Sensors

Working

1. Conductometric methods are based on the chemical reactions.
2. In a solution gas dissociation in solution or chemically reacts.
3. The conductivity before and after the reaction is measured by a differential method.

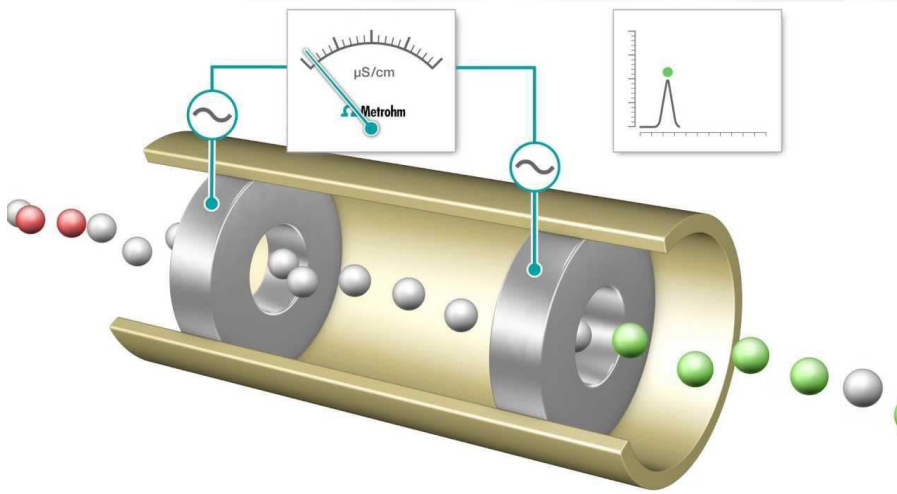
Conductometric Sensors

Gas Sensors

Conductometric gas sensors are widely used for the environmental monitoring; these possess

excellent sensitivity,

1. Excellent sensitivity
2. short response time,
3. Low cost
4. Device feasibility.
5. Portable.



https://i.ytimg.com/vi/fA4p8_Idyaw/maxresdefault.jpg

Conductometric Sensors

Conductometric Methods in Biosensors

1. Conductometric sensors can be used effectively in Enzyme Biosensors.
2. The conductivity can be used to quantify the extent of in REDOX Enzyme reaction.
3. If the change is too small to be detected resistivity can be measure the change, by use a conductivity bridge.

Conductometric Sensors

Summary

1. Conductometric sensors determine the concentrations of the molecules depending upon their ionic movement.
2. Conductivity is inverse to resistivity, therefore the sensor could use a conductivity bridge for weak signals in Biosensors.
3. One important application is to determine the quality of water.

Biosensors

Calorimetric Sensor



Calorimetric Sensors

Introduction

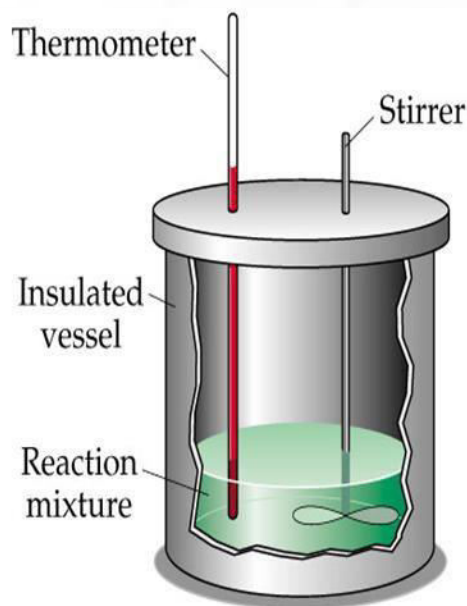
Calorimetric sensors are based on measurement of the heat produced by the molecular recognition reaction.

The amount of heat released/absorbed is related to the reactant concentration.

- Exothermic Reactions
- Endothermic Reactions

A calorimeter is used to determine the rate of reaction.

Calorimetric Sensors



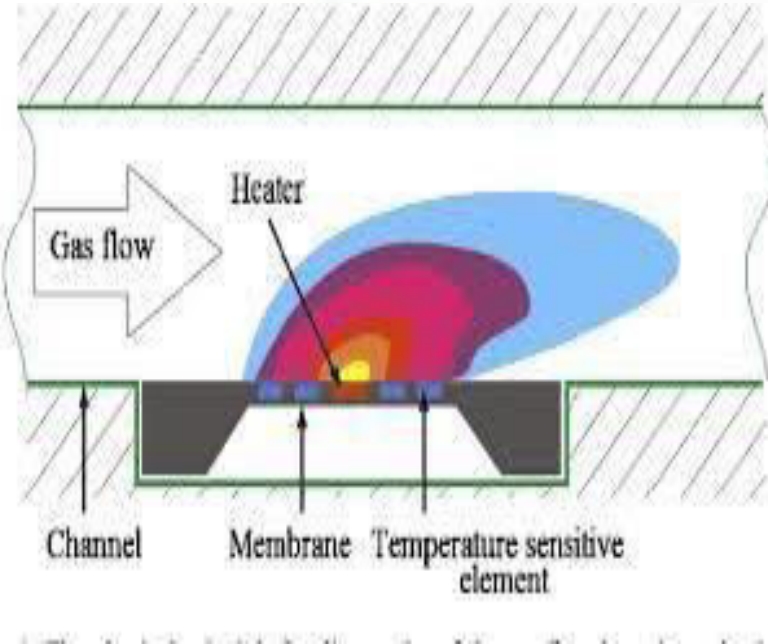
<http://wps.prenhall.com/wps/media/objects/4678/4790699/images/aabjvgia.jpg>

Principle

1. The principle of the calorimeter is the determination of the presence or concentration of chemical species.
2. It is the measurement of the enthalpy change produced by any chemical reaction or physical process that releases or absorbs heat.

Examples. Neutralization, Ammonium Chloride solubility

Calorimetric Sensors



Gas Calorimetric Sensor

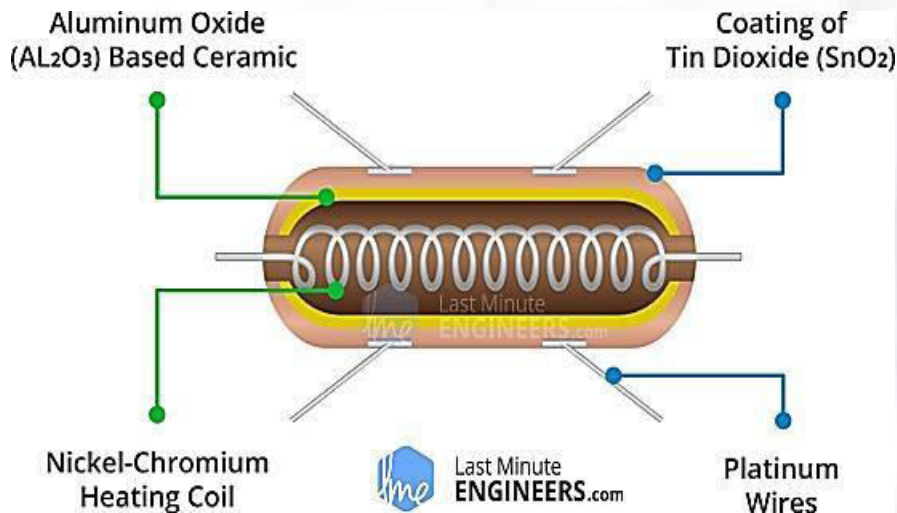
1. Gas calorimetric sensors are built on the basic principal of calorimetry.
2. Their main function is to detect gases.
3. The reaction is carried out on the surface of sensor.
4. The sensor coil is coated with the catalyst like Pt, Pd, silicon di oxide on alumina or other supporting metal.

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSW hHelmPpOX_6OFPR9fEYhWpaUmgvsRTuCQi t5jd3a5f-kuhn1

Calorimetric Sensors

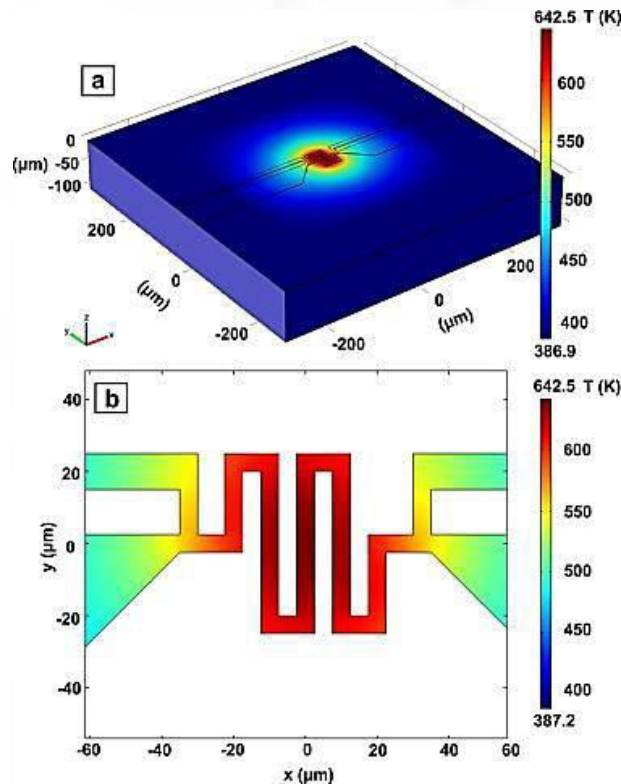
Working

1. The sample is allowed to pass through a coiled column of to ensure uniform temperature.
2. The sample is then passed through a coil coated with a catalyst.
3. The heat generated during the reaction is recorded .
4. Sensor are cost effective, small and reliable.



<https://lastminuteengineers.com/wp-content/uploads/2018/08/MQ2-Gas-Sensor-Internal-Structure-Sensing-Element.jpg>

Calorimetric Sensors



<https://ars.els-cdn.com/content/image/1-s2.0-S0925400514010545-gr5.jpg>

Working in Biosensor

1. Calorimetric biosensors use enzymes as Biological Recognition element.
2. Many enzyme catalyzed reactions that are exothermic, and generate heat.
3. The temperature range is selected, to accommodate the thermophilic and mesophilic Biological recognition elements.

Calorimetric Sensors

Summary

1. Calorimetric sensors are mainly used for the detection of change in temperature.
2. Their major application is in detection of gas.
3. These sensors use catalyst to increase the rate of reaction.
4. Enzymes, generating heat during reaction, are used in calorimetric Biosensors.

Biosensors

Amperometric Sensor



Amperometric Sensors

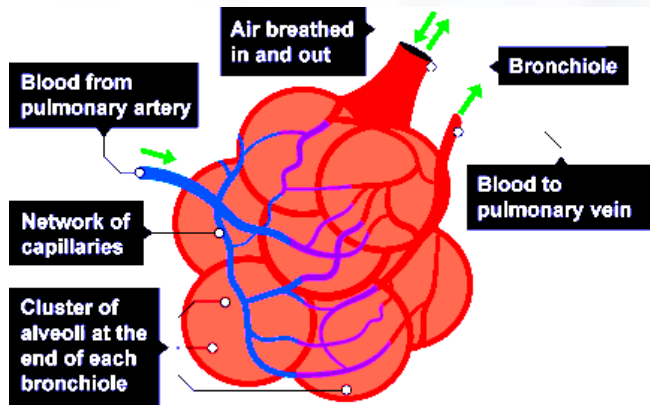
Introduction

1. Amperometric sensor is a type of electrochemical sensor.
2. It relies on oxidation and reduction reaction.
3. The movement of electron generates current.
4. There is also change in the ionic concentration.
5. Amperometric sensors use ion selective electrodes.

Amperometric Sensors



Large, flat leaves like this green taro leaf have an effective exchange surface



The exchange of gases occurs between the alveoli and blood in the capillaries that supply the lungs.

Fick's Law of Diffusion

1. The rate of diffusion is proportional to **surface area** and **concentration difference**. It is **inversely proportional** to the **thickness of the exchange membrane**.
2. The rate of diffusion will **double** if: **surface area** or **conc. difference** is **doubled** or **thickness of the membrane** is **halved**.

Amperometric Sensors

Terminology

1. **Diffusion current**: It is the current in the **semiconductor** caused by the **diffusion of charge carrier**. It is **proportional** to the **conc. of ions** at the electrode.
2. **Residual current**: It flows briefly into the **circuit**, due to the **movement of charge carrier** when **voltage** is reduced to **zero**.

Amperometric Sensors

Terminology

3. **Limiting current**: Depends upon the diffusion of ions to the electrode. It is the **difference** between the **residual current** and the **diffusion current**.
4. **Migration current**: It is **additional current**, produced **on the surface of electrode**, due to the **movement of charge carrier**.

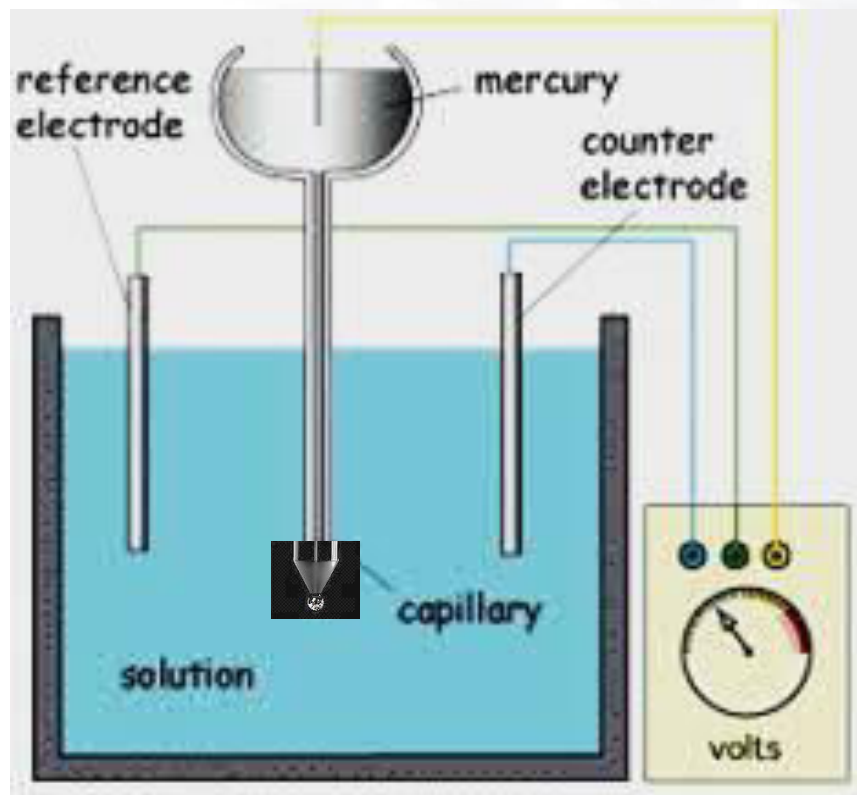
Amperometric Sensors

Electrodes in Amperometric Methods

There are three types of different electrodes that can be used in amperometric sensors.

1. Dropping Mercury Electrode (DME).
2. Rotating Platinum Electrode (RPE).
3. Twin Polarized Micro Electrode (TPME).

Amperometric Sensors



[http://soft-](http://soft-matter.seas.harvard.edu/images/c/cf/DroppingMercuryElectrode.png)

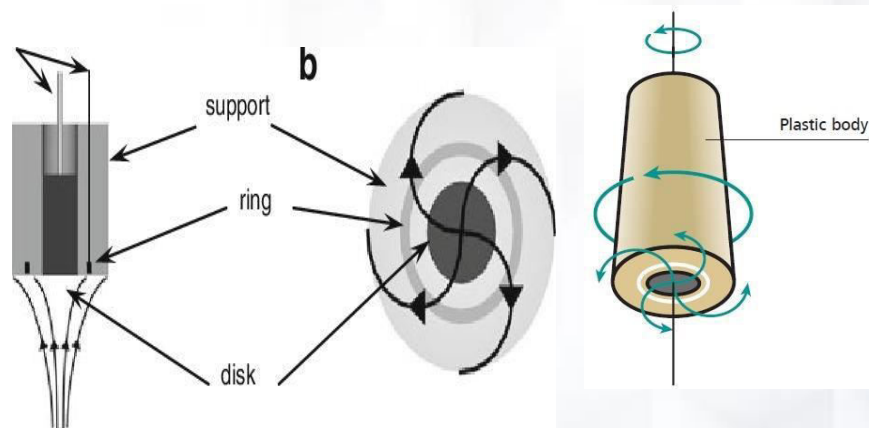
[matter.seas.harvard.edu/images/c/cf/DroppingMercuryElectrode.png](http://soft-matter.seas.harvard.edu/images/c/cf/DroppingMercuryElectrode.png)

https://upload.wikimedia.org/wikipedia/commons/f/f5/DME_2013-11-20_19-44.jpg

Dropping Mercury Electrode (DME).

1. The dropping mercury electrode (DME) is a **working electrode**.
2. **Diameter** of Hg droplet **0.03-0.05 mm** at the intervals of **2-5 sec**.
3. **Advantage** is the constant **renewal of electrode surface**. **Eliminating** the effect of electrode **poisoning**.

Amperometric Sensors



[https://www.azom.com/images/Article_images/ImageForArticle_14363\(5\).jpg](https://www.azom.com/images/Article_images/ImageForArticle_14363(5).jpg)

https://www.researchgate.net/profile/TONDA-MIKIELA_Pradel/publication/224969555/figure/download/fig2/AS:667683267817480@1536199504911/Scheme-of-a-rotating-ring-disk-electrode-the-arrows-indicate-the-path-of-electrolyte.png

Platinum rotating Electrode

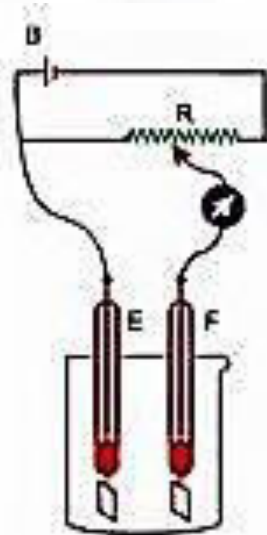
1. It consists of a glass tube with bent Pt wire at its tip. Platinum wire sealed in a **soft-glass tubing**
2. It is attached to an **electric motor**. The motor has very fine control of the **electrode's rotation rate**.
3. Diffusion current is **20 times larger than DME** which allows to measure the **small concentration of ion**.

Amperometric Sensors

Twin Polarized Micro Electrode

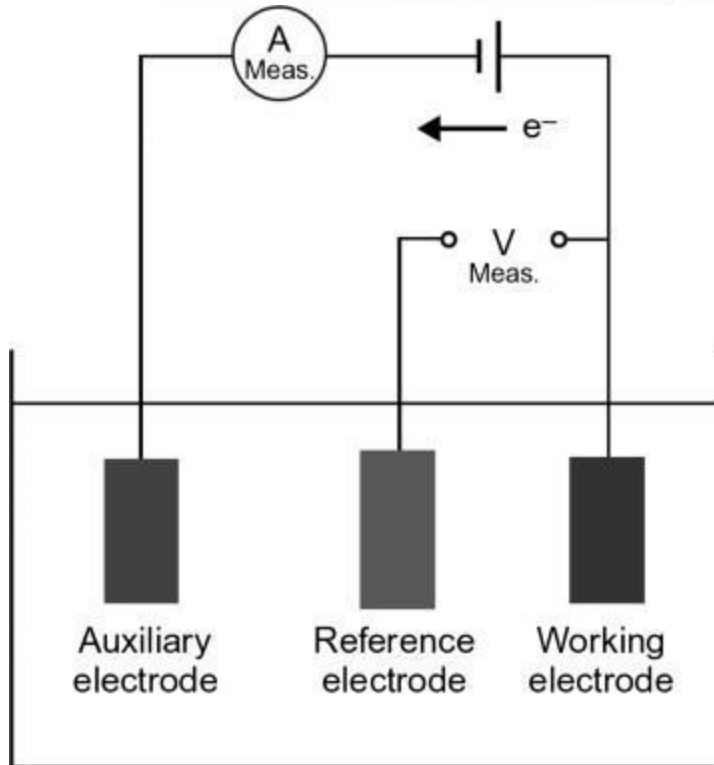
It is also known as dead stop end point or bi-amperometric method.

1. This method is performed by using two platinum electrodes.
2. This technique is specifically applied when the oxidation reduction exists both before as well as after the equivalent point.



<https://tse1.mm.bing.net/th?id=OIP.MYUu8Gd3V6ddVLK42XbBYgHaDI&pid=Api&P=0&w=321&h=156>

Amperometric Sensors



<https://ars.els-cdn.com/content/image/3-s2.0-B9780128148624000041-f04-07-9780128148624.jpg>

Working

1. Amperometry is one of an electrochemical methods.
2. The potential applied to a sensing electrode is controlled instrumentally;
3. As a result current is generated due to REDOX reactions.
4. The current occurring at the electrode surface is recorded as analytical signal.

Amperometric Sensors

Summary

1. Amperometric sensors are based on REDOX reaction and measure the diffusion of current between the two electrodes.
2. Fick's Law of Diffusion.
3. In the techniques three different types of electrodes are used, depending upon the reaction conditions.
4. Example. **Glucometer.**

END

Biosensors

Potentiometric Sensor



Potentiometric Sensors

Introduction

1. These sensors measure the electrical potential of an electrode when **no current is present.**
2. The signal is measured as the **potential difference** (voltage) between the **working electrode** and the **reference electrode.**
3. The reference electrode provides a defined **reference potential.**

Potentiometric Sensors

Salt Bridge

It maintains electrical neutrality within the internal circuit.

1. It prevents the cell from rapidly processing its reaction to attain equilibrium.
2. Usually potassium chloride or sodium chloride is used.
3. The anion and cation have similar conductivity, and hence similar migratory speed.

Potentiometric Sensors

Types of Electrodes

There are many types of electrodes which are used in potentiometric sensors. Some of them are mentioned here.

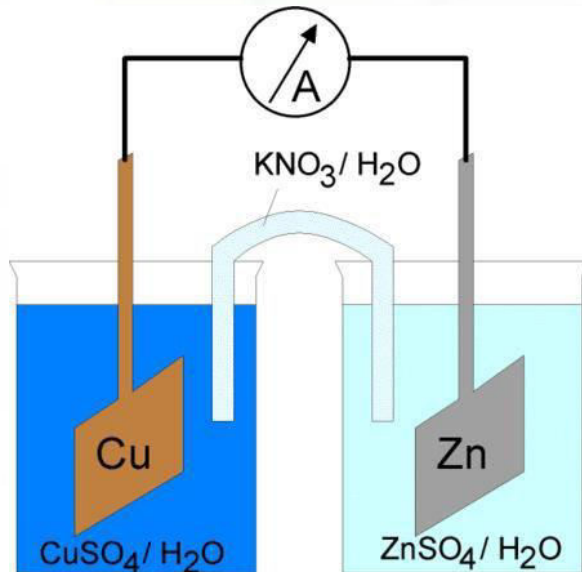
1. Chemically modified electrodes
2. Solid state devices
3. Membrane based ion selective electrodes
4. Screen printed Electrodes
5. Ion selected field effect transistors

Potentiometric Sensors

Glass Salt Bridge

1. The glass salt bridge consists of a U-shaped glass tube filled with an inert electrolyte.
2. Potassium chloride or sodium chloride is used.
3. The electrolyte is often gelified with agar-agar to help prevent the intermixing of fluids which might otherwise occur.

Potentiometric Sensors



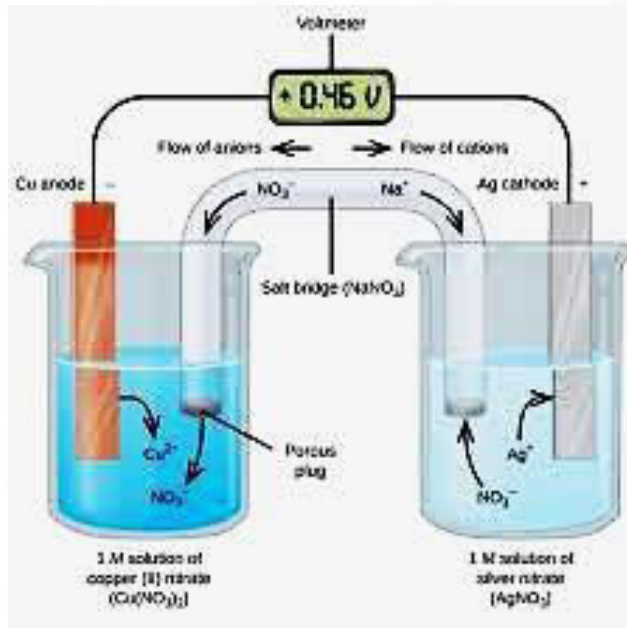
An electrochemical cell (resembling a Daniell cell) with a filter paper salt bridge. The paper has been soaked with a KNO_3 solution

https://upload.wikimedia.org/wikipedia/commons/thumb/3/3b/Electrochemical_element_with_salt_bridge.png/450px-Electrochemical_element_with_salt_bridge.png

Paper salt Bridge

1. The salt bridge consists of a filter paper soaked in a solution .
2. Filter paper provides a solid medium for conduction.
3. It is soaked with a relatively inert electrolyte.
4. Generally nitrates or chlorides of potassium or sodium are used because they are chemically inert.

Potentiometric Sensors



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Working

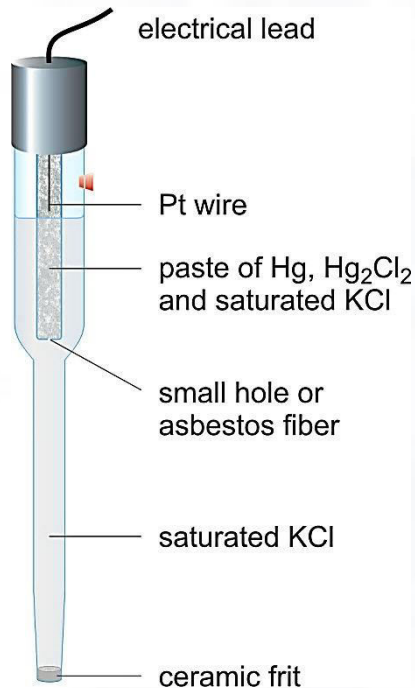
A potentiometer determines the difference in potential.

1. Cathode is the sensing electrode. (right half-cell)
2. Anode is the reference electrode. (left half-cell)
3. $E_{\text{cell}} = E_{\text{c}} - E_{\text{a}}$

E_{c} is the reduction potential at the cathode

E_{a} is the reduction potential at the anode.

Potentiometric Sensors



https://www.periodni.com/gallery/saturated_calomel_electrode.png

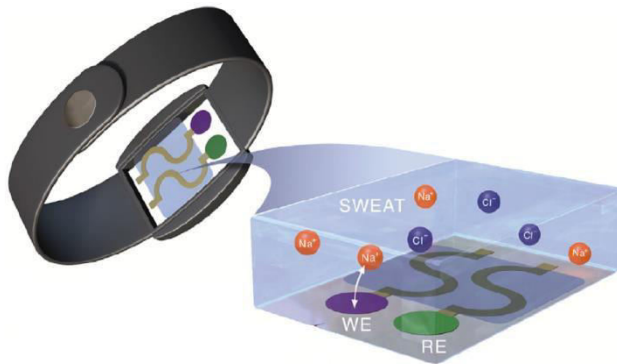
Calomel electrode

1. Calomel is the common name for the compound Mercurous chloride.
2. The electrode consists of an inner tube packed with a paste of Hg, Hg Salt.
3. This tube is situated within a second tube containing a saturated solution of KCl.
4. A small hole connects the tubes and a porous wick serves as a salt bridge.

Potentiometric Sensors



c) WEARABLE POTENTIOMETRIC ION SENSORS



Applications

1. Illustration of a wearable potentiometric ion sensor based on modified Working Electrode and Reference Electrode.
2. Electrodes are in contact with the sweat.
3. The exchange of the target ion (in this case Na^+) with the selective membrane in the WE is responsible for the potentiometric response.

https://www.mdpi.com/sensors/sensors-19-00363/article_deploy/html/images/sensors-19-00363-g001-550.jpg

Potentiometric Sensors

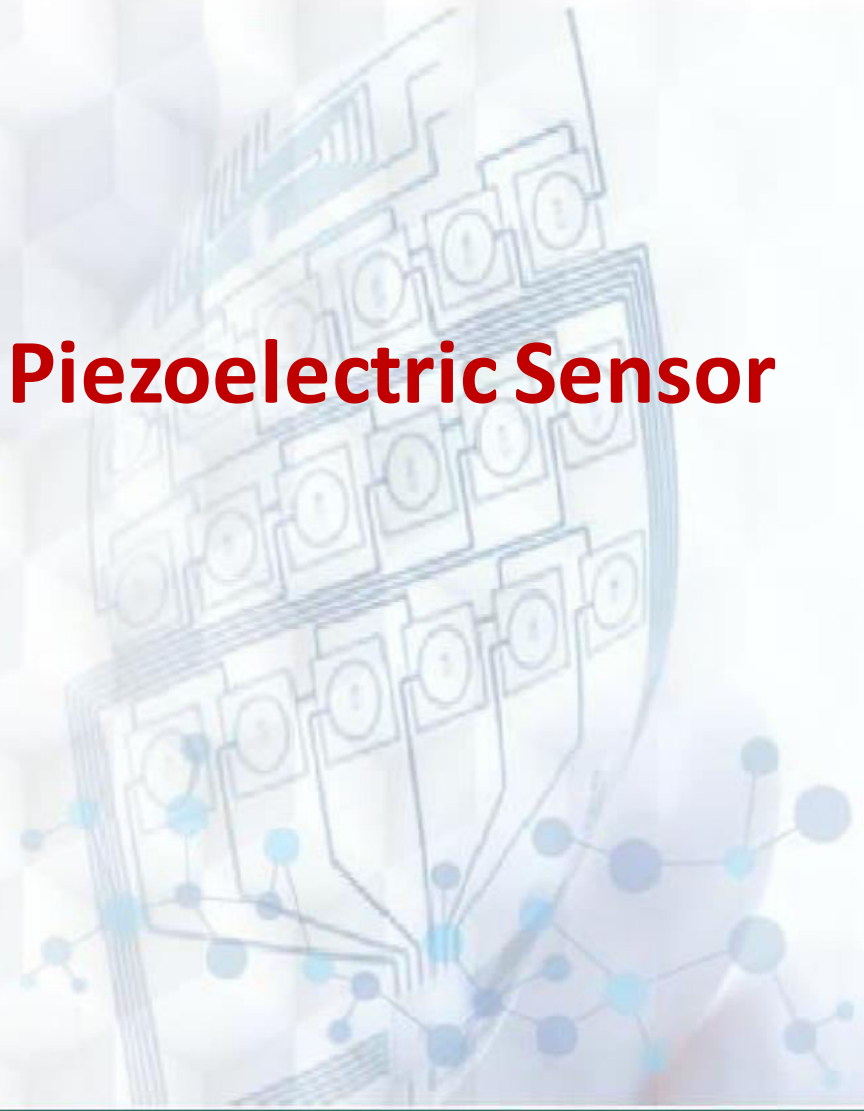
Summary

1. Potentiometric sensors are ion selective electrodes.
2. They are used to determine the conc. of ions such as pH meter, in which the concentration of hydrogen ion is measured as pH of the solution.
3. The sensors have many applications in the fields of, health, agriculture and Environment.

END

Biosensors

Piezoelectric Sensor

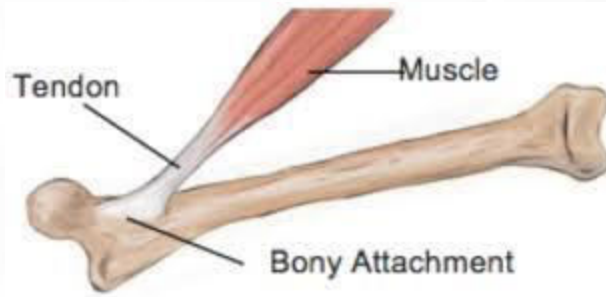


Piezoelectric Sensors

Introduction

1. Piezoelectric sensors are based on piezoelectric materials.
2. Piezoelectric materials has Piezoelectric property.
3. Piezoelectricity is defined as the **generation of electrical potential** in response to applied **mechanical stress**.
4. The effect alter the dimensions of an object.

Piezoelectric Sensors



Piezoelectric Materials

Piezoelectric are found in nature and are also synthesized. In nature there are two types of materials

- 1. Biological matters** e.g. DNA, bone table sugar, collagen, tendons.
- 2. Minerals** including Quartz, Topaz. Tourmaline.
- 3. Synthetic materials** ceramics, **lead zirconate titanate**. GaPO_4 .

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<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSElC9dsN8HtbmpZXWStr13aQyNvAuM2iZYFsCy0IAPvpekNzx>

Piezoelectric Sensors



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<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQm32HsOwquKWoR5x9jyGW-beby92XR-oi4plckUhjPZL913i6C>

Examples and Curie point

Tourmaline is a colorful crystalline boron silicate with Al, Fe, Mg, Na, Li, or K.

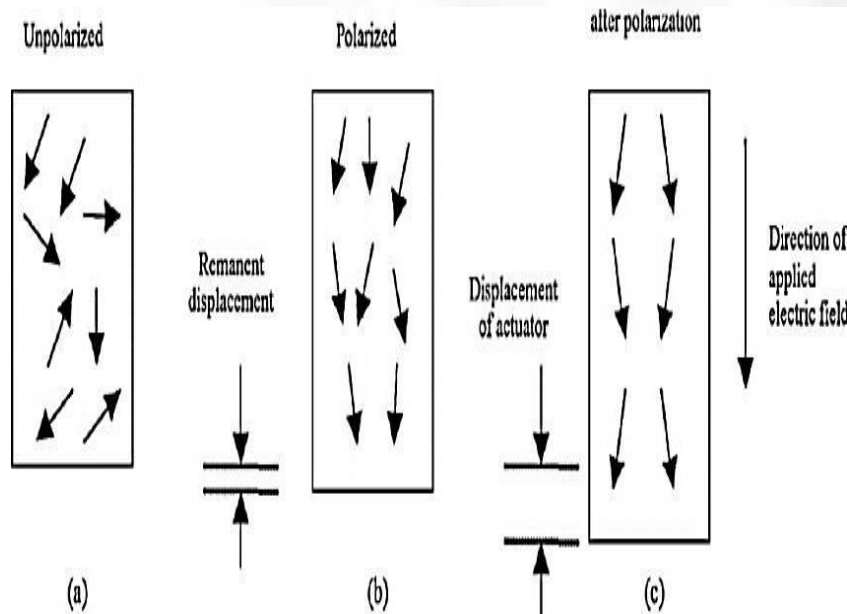
Rochelle Salt has a Greatest piezoelectric effect.

Curie point is a temperature at which certain magnetic materials undergo a sharp change in their magnetic properties.

Rochelle Salt

Curie Point -18°C and 24°C

Piezoelectric Sensors



Piezo technology

On the basis of polar axis there are two types of crystals, **Poly crystals** (random polar axis) and **mono crystals** (single polar axis).

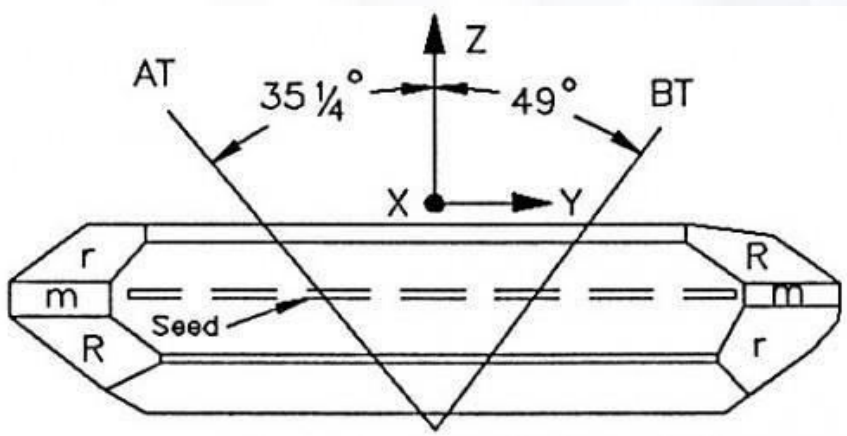
1. The voltage is applied to the poly crystal to align the dipole.
2. Once the dipoles are aligned parallel to the electrode piezoelectric effect can be observed.

https://www.researchgate.net/profile/Ben_Cazzolato/publication/222551413/figure/download/fig1/AS:305227002597406@1449783196998/Polarization-process-for-a-piezoelectric-actuator-4.png

Piezoelectric Sensors

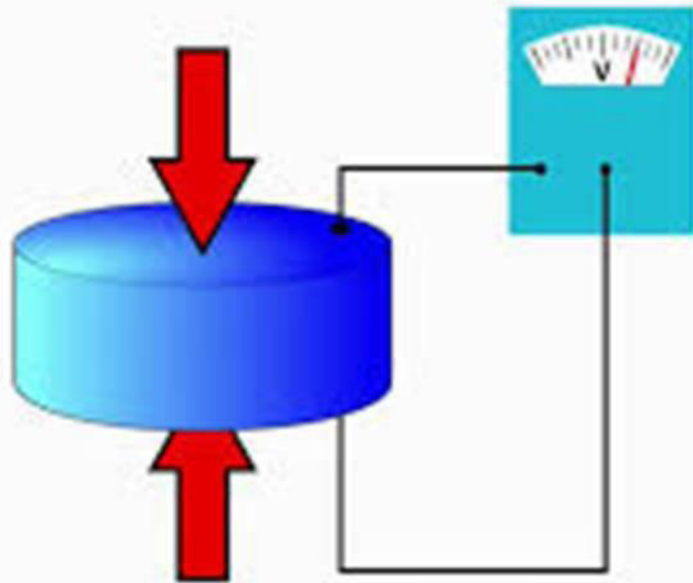
AT-cut Quartz Crystals

1. There are many ways in which a crystal can be cut, relative to the x, y, and z, axes.
2. The **angle** to the main crystal axes at which a crystal blank is cut has a major impact on its performance.
3. The AT crystal cut at an angle of **35 ° 15'** from the Z-axis.



<https://www.iqdfrequencyproducts.com/media/c/blg/404/1435566839/rc/500/251/90/cut.jpg>

Piezoelectric Sensors



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQPv3tYfQhQMlQrB2xKf1y6Dr1sly-9DpthDHo1tDL0QkLloyVO>

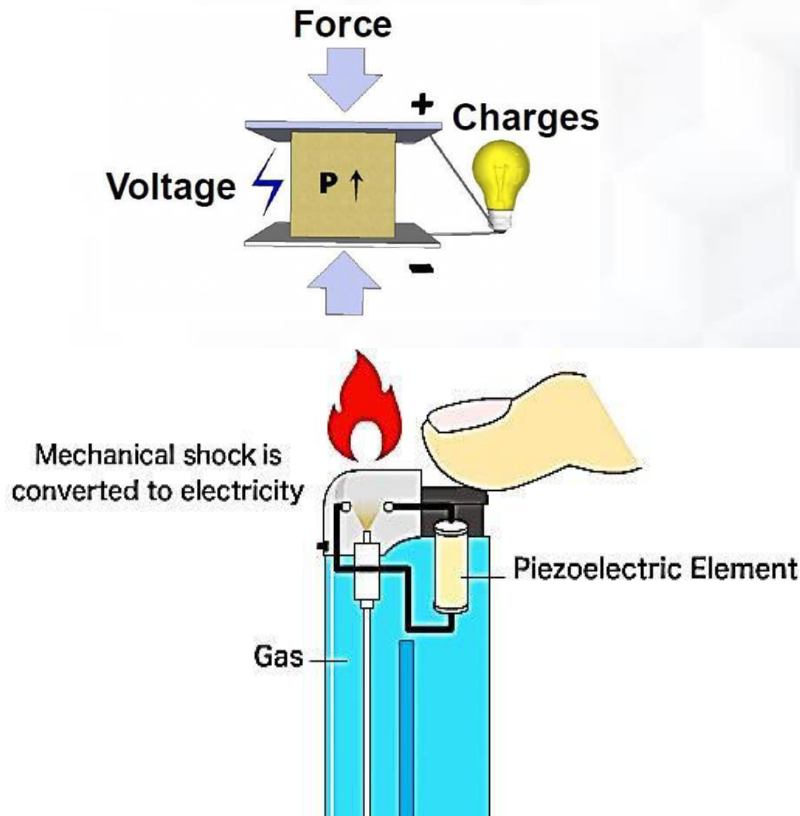
Piezoelectric Sensor

A piezoelectric sensor is a device that uses the piezoelectric effect, to measure changes in

1. Pressure
2. Acceleration
3. Temperature
4. Strain
5. Force

by converting them to an electrical charge.

Piezoelectric Sensors



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http://4.bp.blogspot.com/_O2ADmGeMSTY/TOk4cn9VWTI/AAAAAAAU/Z5rD8t7RtR0/s1600/piezo.gif

Applications

1. Piezo electricity material is used on push button cigarette lighters.
2. Piezo electrical ignition creates enough energy to lit a flame.
3. Quartz is typically the material used to generate this spark.

Other common applications include Microphones, Headsets and Loud-Speakers.

Piezoelectric Sensors

Summary

1. Piezoelectric sensors are based on piezoelectric material.
2. Piezoelectric materials are found in nature and synthesized as well.
3. The piezoelectric material converts the physical quantity into an electrical voltage.
4. Piezo sensors are used in many technologies.

END

Biosensors

Surface Plasmon Sensor



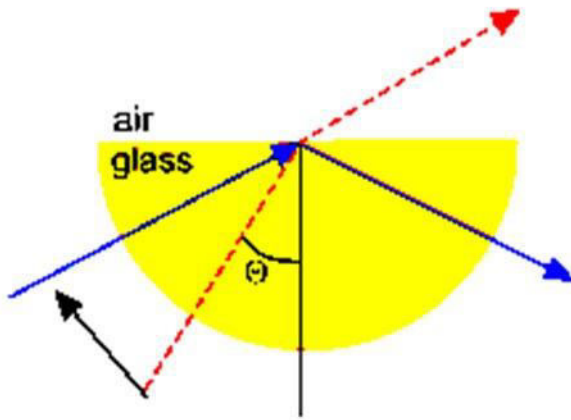
Surface Plasmon Sensor

Plasmon

Plasmon are describe as the **vibration** of the **electron cloud** in a metal.

1. These plasmon oscillate at particular frequency **characteristic** the material.
2. Surface plasmons are oscillations that are confined to be surface.
3. Velocity and momentum of the plasmon changes when the composition of the medium changed.

Surface Plasmon Sensor



<https://www.sprpages.nl/images/SPRPages/Theory/TIR.png>

Surface Plasmon Resonance

1. Surface Plasmon Resonance (SPR) is a **physical process**.
2. It is an optical technique and involves **total internal reflection**.
3. When light hits from a **denser medium** to a **lesser medium** there is a **change in angle of incidence**, at the **critical angle**.

Surface Plasmon Sensor

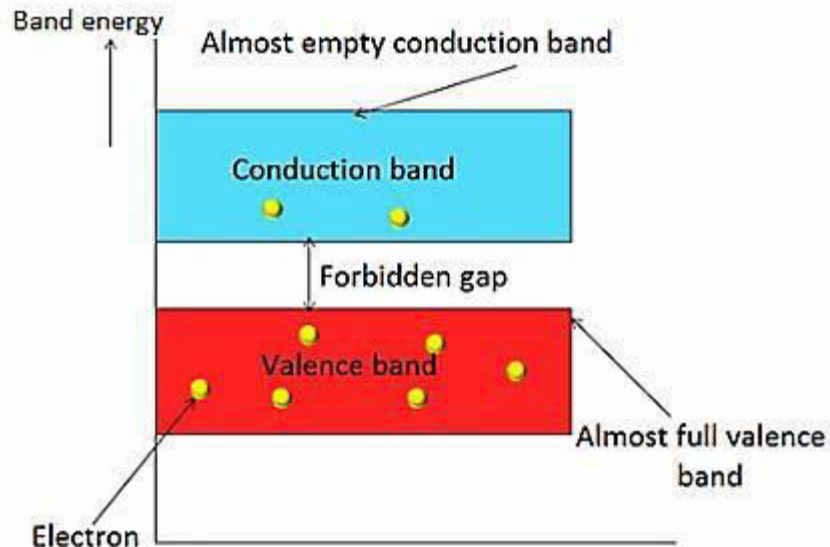
Surface Plasmon Resonance (SPR) Technology

SPR Technology is mainly **dependent** on the

1. Properties of **the metal film**
2. **Wavelength** of the **incident light**
3. **Refractive index** of the **medium**

Refractive index of the medium is temperature sensitive.

Surface Plasmon Sensor



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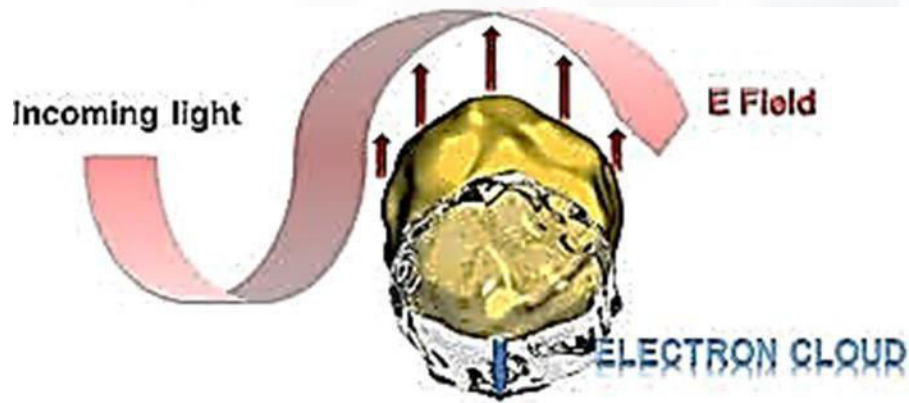
<https://www.physics-and-radio-electronics.com/electronic-devices-and-circuits/semiconductor/images/heateffectonsemiconductor.png>

Characteristics of Metal

Metals must have a conduction band electrons capable of resonating with the incoming light at a suitable wave length.

1. Metal on the sensor must be **free of Oxides, Sulfides**.
2. Must **not react** with other **molecules** on exposure to atmosphere.
3. **Ag, Au, Cu, Al, Na, In.**
4. Most suitable metal **GOLD**

Surface Plasmon Sensor

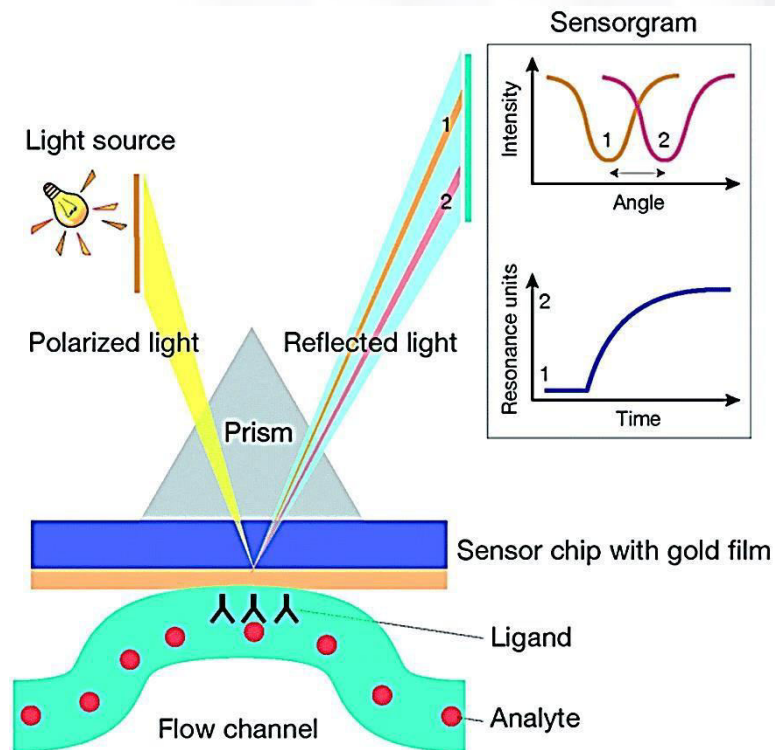


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Factors Effecting SPR

1. The optimal **thickness** of gold metal is about 50 nm.
2. The light should be **polarized** in the **plane of surface** to obtain a sharp dip.
3. Incident light and temperature should remain **constant**.
4. Signals are **dependent** on change of **refractive index**.

Surface Plasmon Sensor

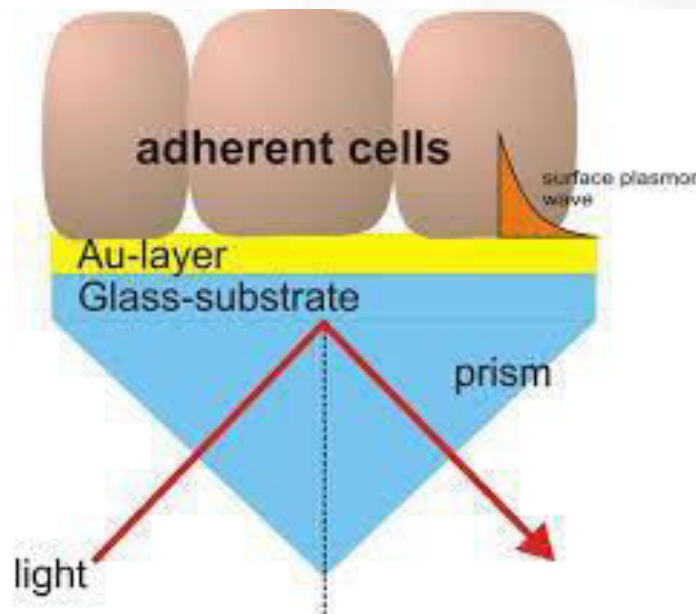


SPR in Biosensors

1. The biological recognition element is immobilized on the surface.
2. When a **target molecule is detected** by a SPR biosensor there is a **variation of the optical properties**.
3. It leads a **change in refractive index** of this dielectric layer will cause a **measurable shift in the SPR angle**.

https://pubs.rsc.org/image/article/2014/md/c4md00088a/c4md00088a-f1_hi-res.gif

Surface Plasmon Sensor



<http://www-analytik.chemie.uni-regensburg.de/wegener/robelek/Fig2.jpg>

Application

1. To monitor the different physiological effects in cells.
2. SPR sensors are used for culturing layers of living cells, adherent to gold surface.
3. If the cells undergo a physiological change, it can be monitored by SPR signals.
4. Such sensors are valuable tools for cell biology.

Surface Plasmon Sensor

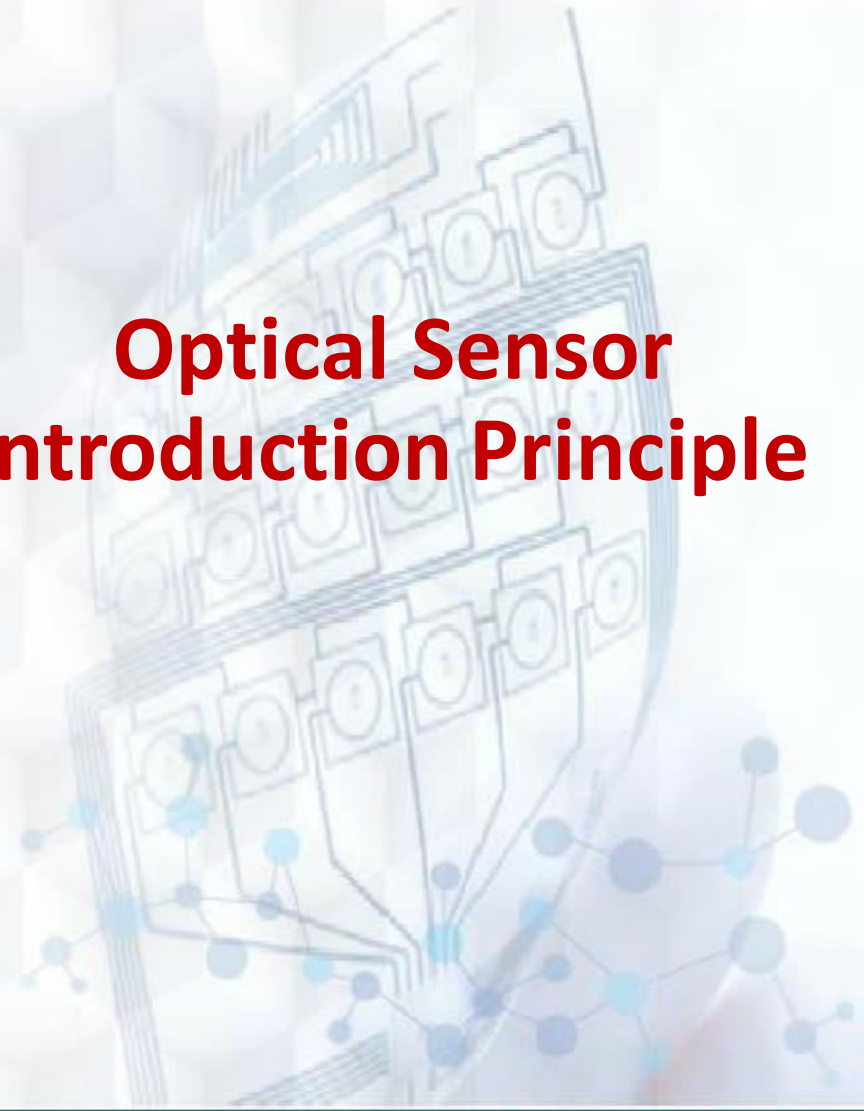
Summary

1. SPR sensors are based on optical principal of Total internal reflection.
2. Gold is an ideal metal for SPR sensors.
3. There are different factors including thickness of the **surface metal, wave length and refractive index** that monitor the performceof SPR sensors.
4. SPR is a valuable in Biosensor technology.

END

Biosensors

Optical Sensor Introduction Principle



Optical Sensors Introduction Principle

Introduction

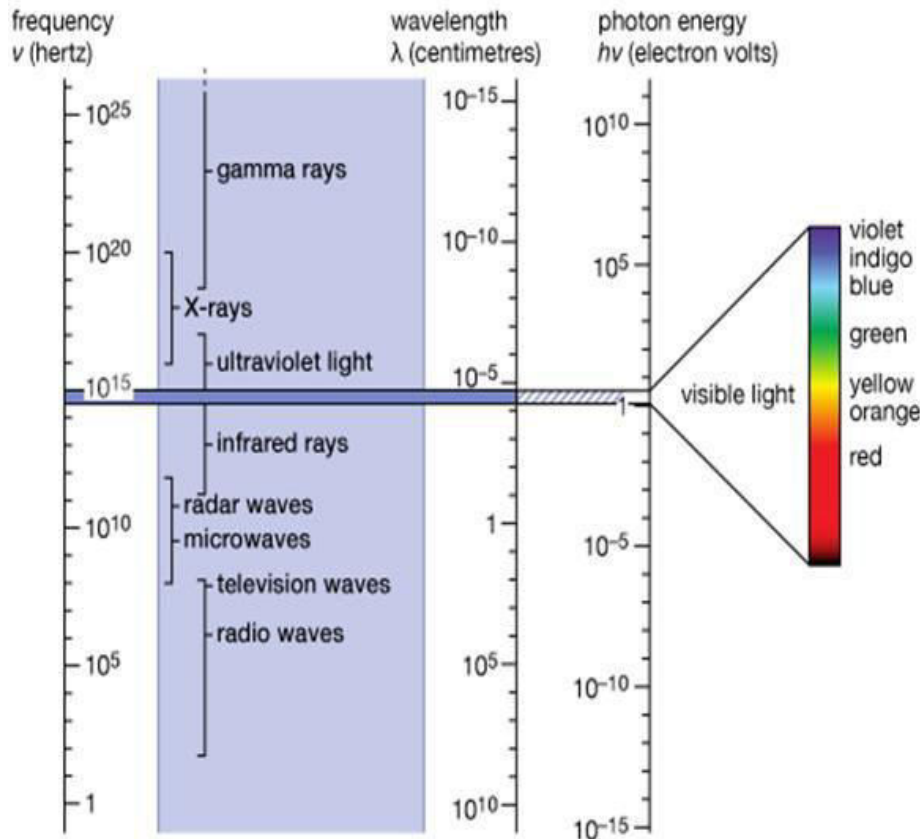
1. The optical sensors are employed to **measures a physical quantity**, using a **light source**.
2. Generally a **monochromatic light** source is used in optical sensors.
3. The optic sensors are based on **optic law**.
4. Optical Sensors has in numerous applications.

Optical Sensors Introduction Principle

Source of Light

There are two sources of light are used in optical sensors.

1. **Natural sources** include all the electromagnetic radiations such as visible, UV, IR and others
2. **Man made** light sources include LASER and Light emitting Diodes.
3. The light source is used according to the requirement.



<https://cdn.britannica.com/76/276-004-404D8CFF.jpg>

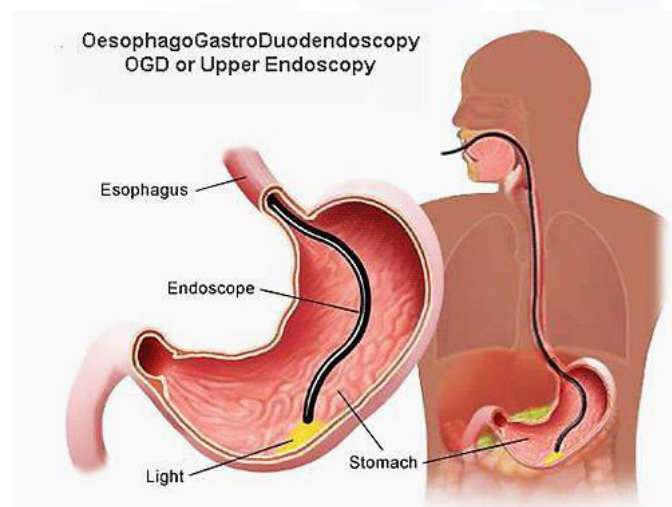
Optical Sensors Introduction Principle

Working of optical sensors

Optical Sensors are used for **contact-less detection, counting or positioning of parts.**

1. They can measure rotation, pressure, strain, volume.
2. During sensing, Light may change its optical properties **intensity, phase, polarization, wave length or spectral distribution.**
3. Optical sensors can be **intrinsic or extrinsic.**

Optical Sensors Introduction Principle



Intrinsic Optical sensors

1. Intrinsic optical sensors are placed internally to sensors are most often **measures the internal changes.**
2. They are **more sensitive** to extrinsic sensors.
3. **Endoscope** It has a thin, long **flexible tube**, a lens, a **light transmitting system** and an **eyepiece.**
4. **Capsule Endoscopy**

<http://www.gastrosurgery.co.uk/images/ogd.jpg>

<http://www.norwichgi.com/wp-content/uploads/2014/04/BestGIDoctorCT221.jpg>

Optical Sensors Introduction Principle

Extrinsic Optical sensors .

1. Extrinsic optical sensors can measure different physical parameters such as **temperature, pressure, liquid level, rotation, Force Radiation, pH Humidity, Strain, Velocity Electric fields.**
2. As compared to intrinsic optical sensors they are **less sensitive, easily multiplexed and Easier to use.**

Optical Sensors Introduction Principle



Europium is a luminescent material the bank uses in their currency. This chemical makes the Euro notes illuminate when exposed to UV light.

<https://s4.scoopwhoop.com/anj/euro/325163838.jpg>

Bank Note Counter

1. **Visible Light** sees the banknote images to check the No. and detect fake note .
2. **Infra Red Light** sees the banknote images **with low frequency light**, to acquire both **reflected and penetrated** images.
3. **UV Light** sees the banknote images with **high frequency** light. Usually banknotes **do not reflect** any UV light.

Optical Sensors Introduction Principle

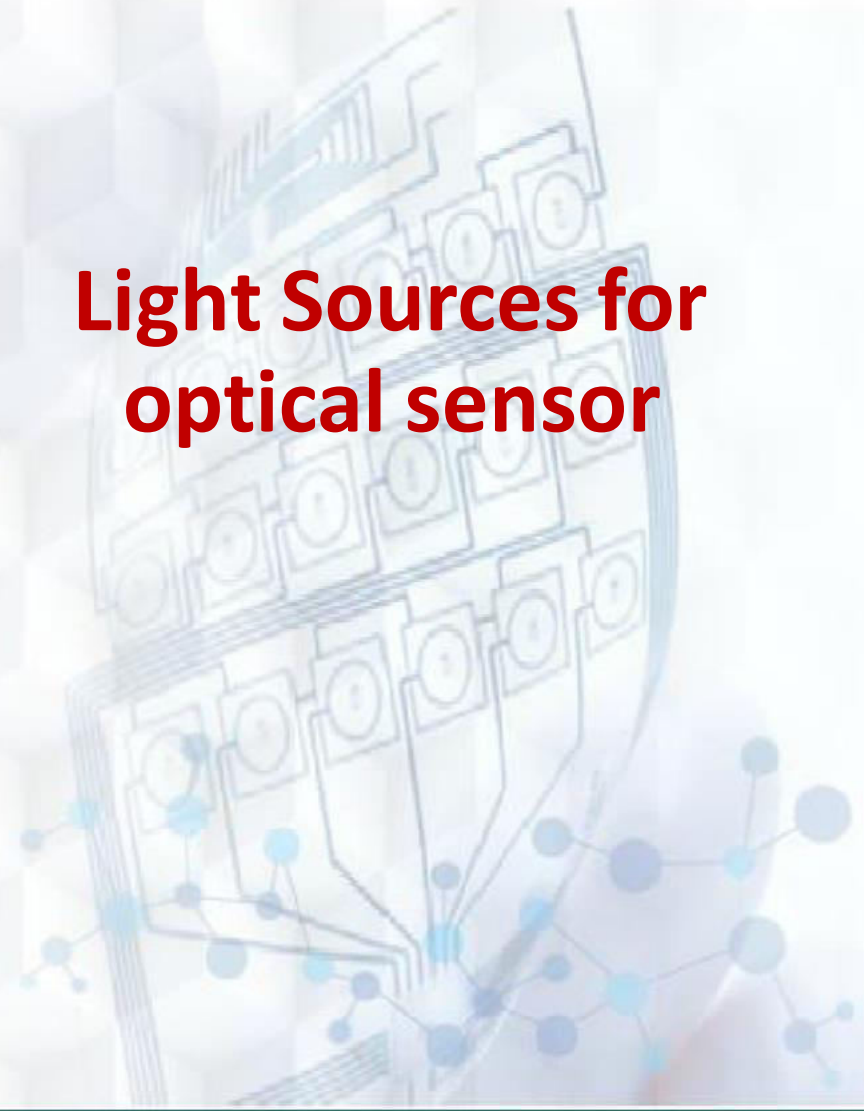
Summary

1. Optical sensors use light and its application.
2. Rules governing the light are applicable on optical sensors.
3. The optical sensors can measure Temperature, Velocity Liquid level, etc.
4. There are two types of optical sensors.
5. Optical sensors are used in many fields of life.

END

Biosensors

**Light Sources for
optical sensor**



Light Sources for Optical Sensors

Introduction

1. Optical sensors use light for sensing techniques.
2. The light source are extremely important for the performance of optical sensors.
3. A focused light source is required with well known properties.
4. There is a set of properties for the light sources, that are used in optical sensors.

Light Sources for Optical Sensors



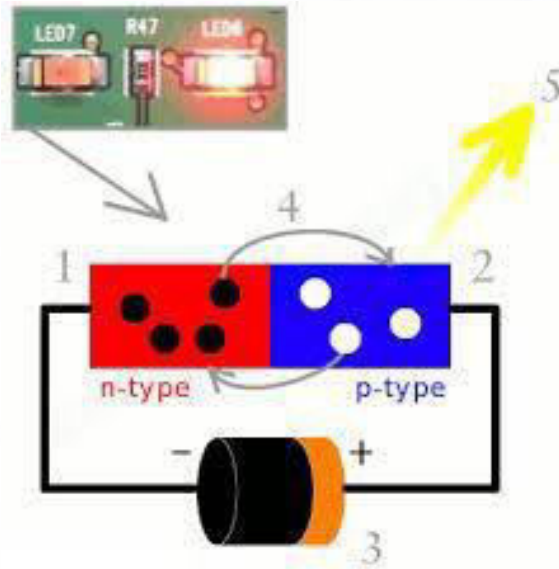
https://ae01.alicdn.com/kf/HTB1mLb7QpXXXXXPXpXXq6xXFXXXV/1Pcs-5mW-532nm-Green-Laser-Pen-Powerful-Laser-Pointer-Presenter-Remote-Lazer-Hunting-Laser-Bore-Sighter.jpg_640x640q70.jpg

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Properties and Types

1. **Monochromatic light** source is required, consists of extremely **narrow wave length**.
2. The two most commonly used **optical sources** are
 - **Light-Emitting Diodes (LED)**
Example Traffic lights
 - **LASER diodes** that is light Amplification by Stimulated Emission of Radiation.
Example LASER pointer

Light Sources for Optical Sensors



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Light Emitting Diodes (Transistors).

1. **Electrons (n-type)** in the semiconductor recombine with **electron holes (p-type)**, releasing energy in the form of **photons**.
2. **Color** of the emitted light is dependent on the material used for **doping**.
3. In LEDs **current** can only flow in **one direction**.
4. LEDs never burn out unless their current limit is passed.

Light Sources for Optical Sensors



https://images-na.ssl-images-amazon.com/images/I/51VXtQcjThL._SX342_.jpg

Types of Light Emitting Diodes

1. Gallium Arsenide **infra-red**
2. Aluminium Gallium Phosphide – **green**
3. Gallium Indium Nitride (GaInN) – near **ultraviolet**, **bluish-green** and **blue**
4. Aluminium Gallium Nitride (AlGaN) – **ultraviolet**
5. To **white LED** light blue LED is coated with phosphorous.

Light Sources for Optical Sensors

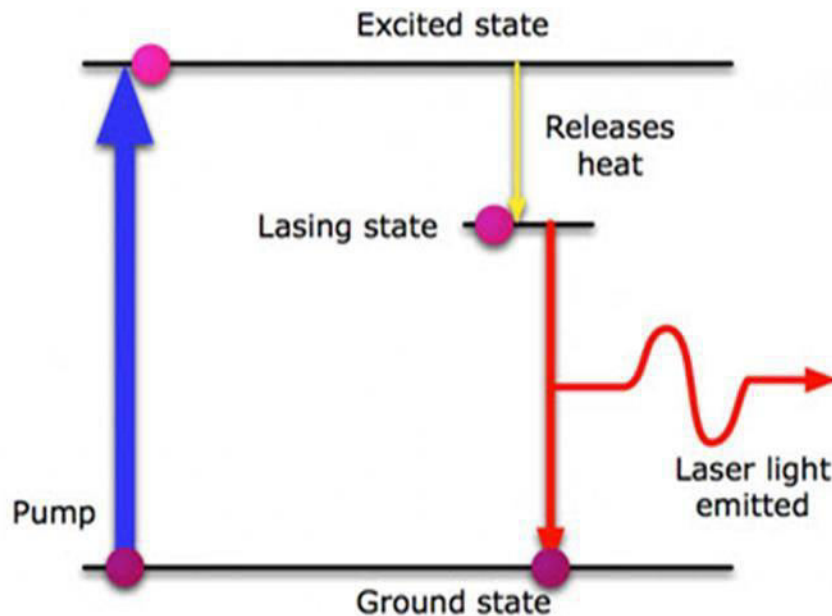


<https://image.dhgate.com/0x0s/f2-albu-g5-M01-FA-15-rBVal1jjYBmASl7HAABPKRm9O3g830.jpg> /8-fat-beam-red-green-blue-moving-head-laser.jpg

LASER gain Medium

1. **Atom** such as He-Ne , He-Cd.
2. **Molecule** such as carbon dioxide, Nitrogen Gas.
3. **Liquids** involving organic dyes dissolved in various soln.
4. **Dielectric solids** involving Neodymium.
5. **Semiconductor materials** such as Gallium crystals and ruby crystals.

Light Sources for Optical Sensors

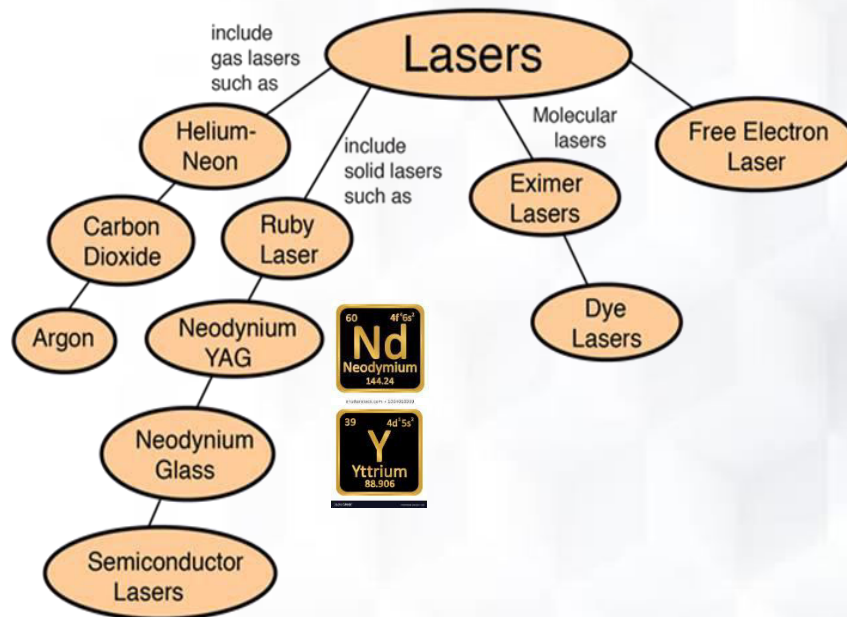


<https://www.elprocus.com/wp-content/uploads/2017/03/10.jpg>

Creation of a LASER Beam

1. A LASER is created, when the **electrons** in the atoms of **LASER gain medium**, become **excited** by absorbing energy and move to **higher energy state**.
2. Before returning to ground state, **heat is released** and electron reached at **Lasing state**.
3. Between the Lasing and ground state Laser light is emitted.

Light Sources for Optical Sensors



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[http://hyperphysics.phy-](http://hyperphysics.phy-astr.gsu.edu/hbase/optmod/imgopm/lastype.png)

[astr.gsu.edu/hbase/optmod/imgopm/lastype.png](http://hyperphysics.phy-astr.gsu.edu/hbase/optmod/imgopm/lastype.png)

Types of LASER

LASER are commonly typed by the **LASING material**.

1. Gas LASER
2. Solid LASER
3. Molecular LASER
4. Free Electron LASER

There are many other types of LASER **classification** based on their medical, industrial and commercial **applications**, **wavelength** and **safety**.

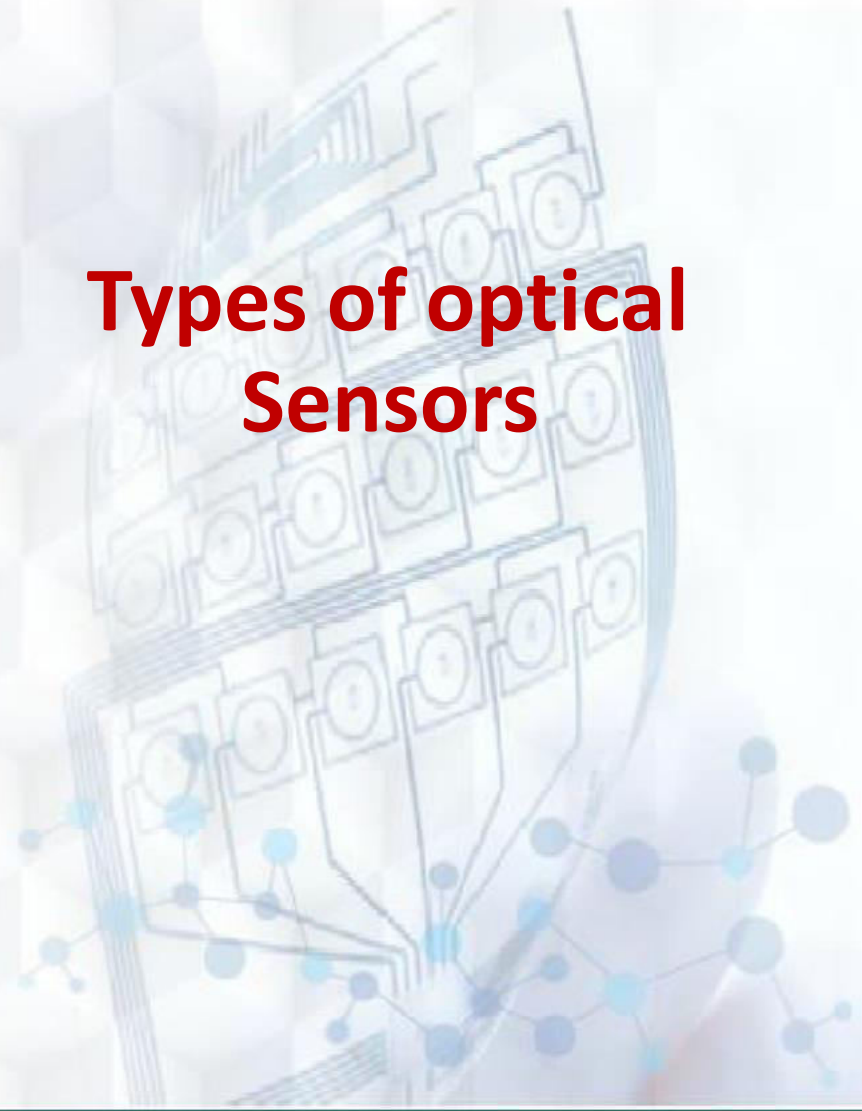
Light Sources for Optical Sensors

Summary

1. There are two types of artificial light sources for optical sensor with unique properties LED and LASER diodes
2. LED are based on transistors and the color of light depends upon the dopants.
3. LASER waves are produced after exciting the LASING medium. Beam color is dependent of gain medium used.

Biosensors

Types of optical Sensors



Types of Optical Sensors

Classification

Optical sensors could be classified in different ways. At least three different classes of Optical sensors are accepted.

1. Classification on the basis of location (extrinsic intrinsic)
2. Classification on the basis of output (resistance Current, Voltage)
3. Classification on the basis of applications.

Types of Optical Sensors

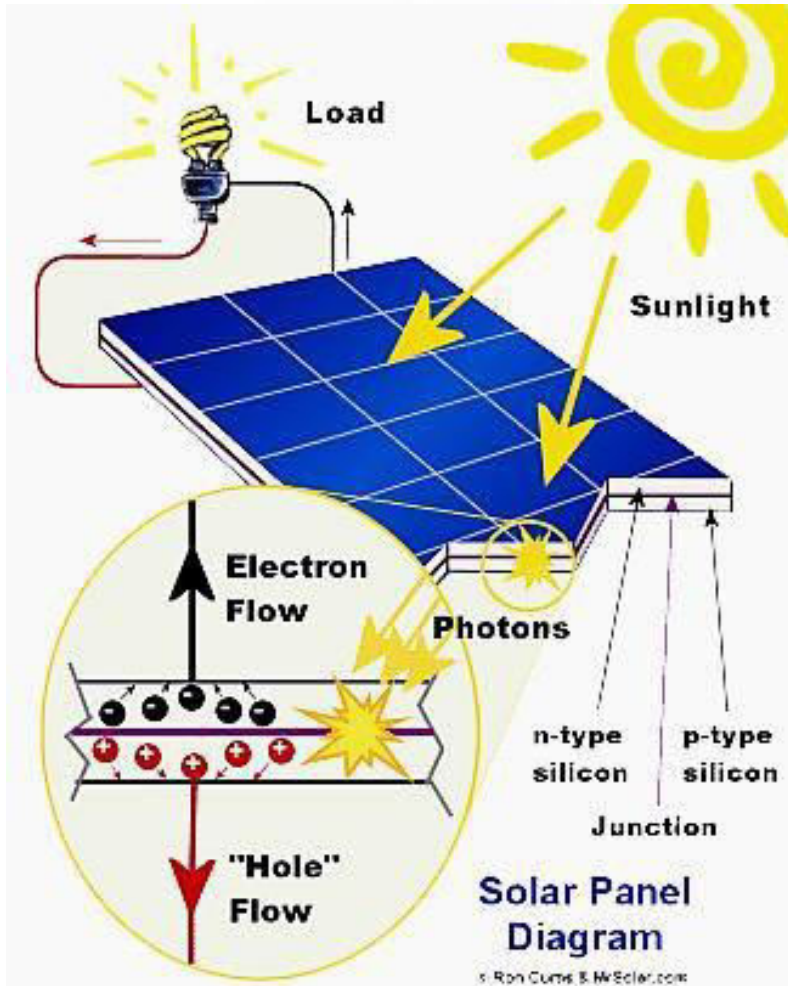
Types

1. **Photovoltaic cell** (solar cell) **converts** an amount of **incident light** into an **output voltage**.
2. **Photodiodes** convert an amount of **incident light** into an **output current**.
3. **Photoconductive devices** measure the resistance by converting a change of incident light into a change of resistance.

Types of Optical Sensors

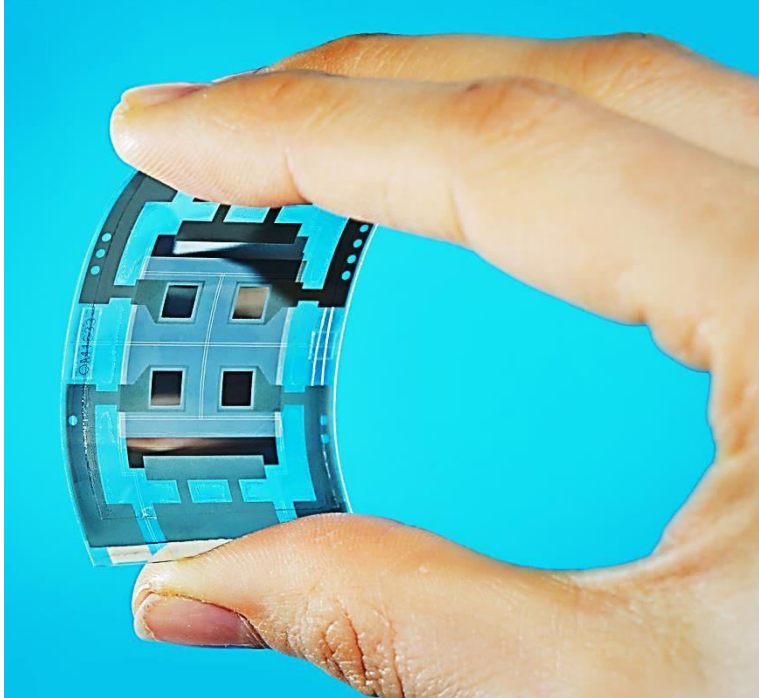
Photovoltaic cell

1. A photovoltaic cell is composed of positive and negative junctions of Boron doping (p-type) and phosphorus doping (n-type).
2. The common single junction silicon solar cell can produce a maximum open-circuit voltage of approximately 0.5 to 0.6 volts.



<https://images.squarespace-cdn.com/content/v1/58e54675cd0f680e6d88336e/1495050760893-ZETD2V4OPZN26F4ZVLO6/ke17ZwdGBTodd18pDm48kEVvD7qVFad-P5KGQB1CsnVZw-zPPgdn4jUwVclE1ZvWEtT5uBSRWt4vQZAgTJucoTqqXjS3CfNDSuuf31e0tVHfxRDyb3cqdc4gmCYiqMXq3zk1uHLAix626oPFhd2KJuG45vQwBxdpDrCGUSSl5w/solar-panel-diagram.gif?format=500w>

Types of Optical Sensors

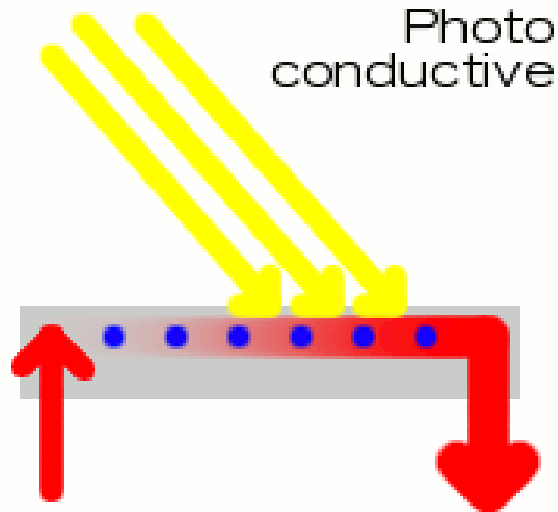


Organic photodiodes are extremely lightweight and also inexpensive to produce
<https://scx2.b-cdn.net/gfx/news/hires/2014/organicphoto.jpg>

Photodiodes Sensors

1. It is a Photodiodes may contain optical filters, built-in lenses,
2. Material used in the synthesis includes **Silicon, Germanium, Indium Gallium, Mercury**, etc.
3. Silicon based photodiodes are very popular due to low noise.
4. Organic material based are also in market.

Types of Optical Sensors



Applications for Photoconductive cells are in Light meter, in camera Daylight sensor, Elevator, safety stop ,Garage door

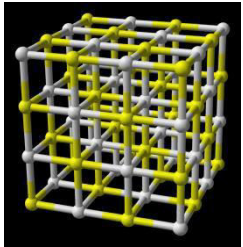
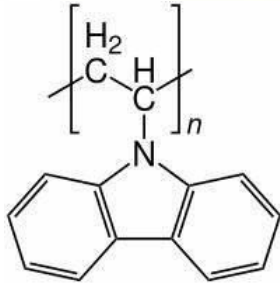
<https://cdn4.explainthatstuff.com/photoconductive-diagram.gif>

Photoconductives Sensors

Photoconductivity is an **optical** and **electrical phenomenon**.

1. The material becomes **more electrically conductive** due to the absorption of electromagnetic radiation.
2. Visible light, UV, IR,, or gamma radiation
3. **Photoconductive** materials increases the flow of electrons and reduces the resistance.

Types of Optical Sensors



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<https://i.ytimg.com/vi/omcme8er-7o/maxresdefault.jpg>

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Materials for Photo-conductives Sensors

1. **Polyvinylcarbazole** is a temperature-resistant thermoplastic conductive polymer used in copiers and laser printers.
2. **Lead sulfide**, used in infrared detection applications, such as **heat-seeking missiles**.
3. **Selenium**, employed in early television and xerography.

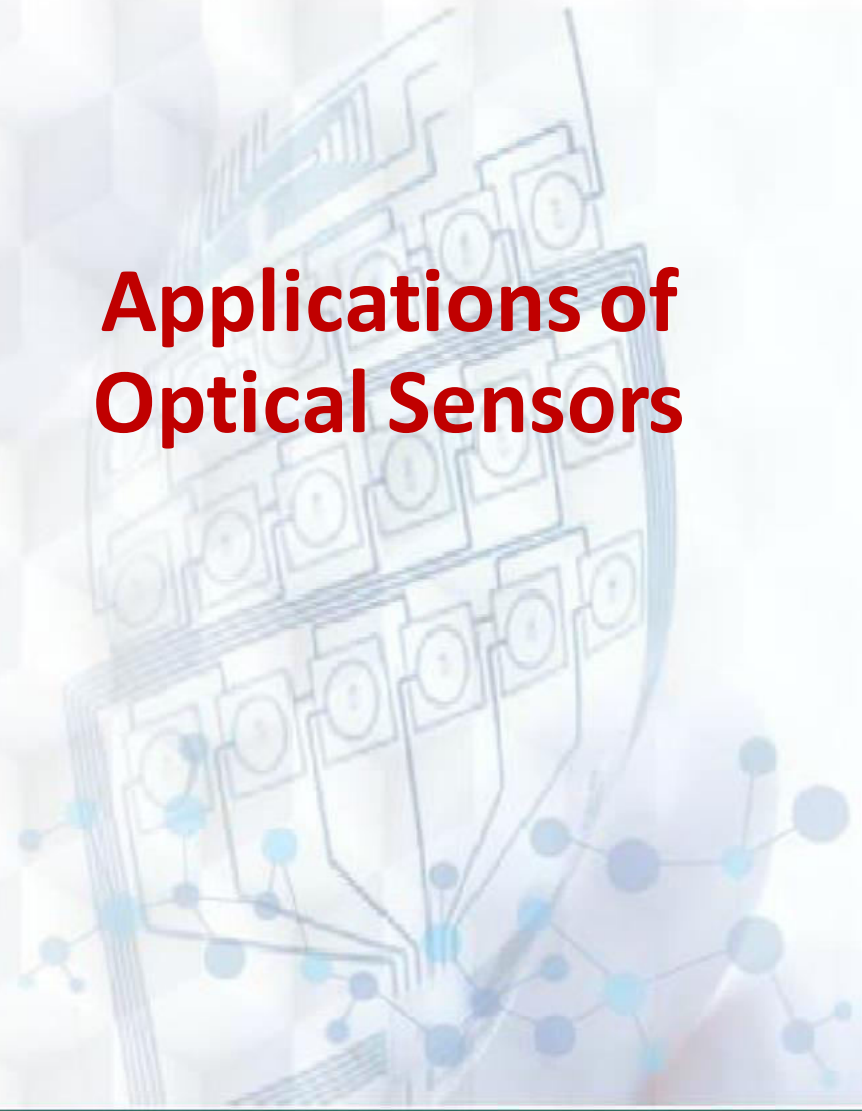
Types of Optical Sensors

Summary

1. Photoconductive device output is **resistance**.
2. Solar cell output is **voltage**
3. Photodiode output is **current**.
4. Optical sensors are extremely useful and have a number of applications.
5. Heat sensitive missiles.
6. Power generation by solar cells.

Biosensors

Applications of Optical Sensors

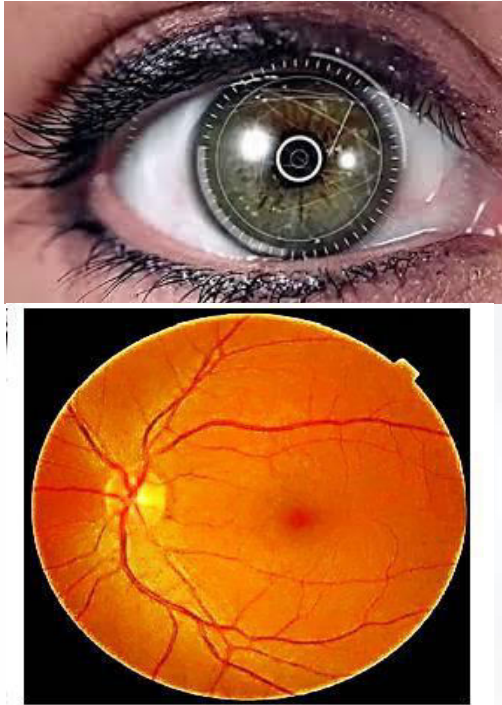


Applications of Optical Sensors

Introduction

1. Optical sensors have many applications including **health, diagnostics, defense, research, surveillance, environment and process control.**
2. Optical sensors address approximately all aspects of **detection technology.**
3. The optical sensors ranges from **micro-probes to large devices.**

Applications of Optical Sensors



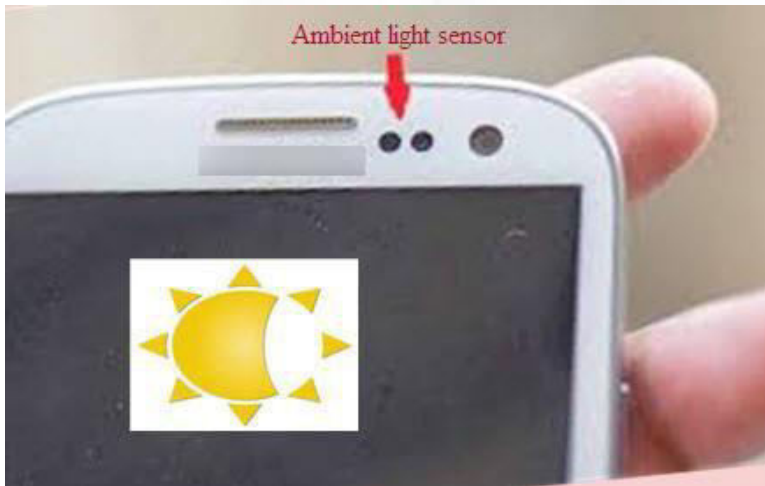
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[https://www.verywellhealth.com/thmb/hVYMF3Zzaioso0qzSOtgFSkov04=/768x0/filters:no_upscale\(\):max_bytes\(150000\):strip_icc\(\)/Gettyimages-308783-003-56acdc85f9b58b7d00ac8e8.jpg](https://www.verywellhealth.com/thmb/hVYMF3Zzaioso0qzSOtgFSkov04=/768x0/filters:no_upscale():max_bytes(150000):strip_icc()/Gettyimages-308783-003-56acdc85f9b58b7d00ac8e8.jpg)

Biometric Verification

Biometric verification is any means by which a person can be uniquely identified by evaluating one or more distinguishing biological traits.

1. A biometric identifier known as a retinal scan is used to map the unique patterns of a person's retina.
2. This beam traces a path on the retina.

Applications of Optical Sensors



<https://www.elprocus.com/wp-content/uploads/2017/03/3-1.jpg>

Ambient Light Sensor

1. Ambient light is that light which already present in a scene, lights.
2. Mostly we have seen this sensor on our mobile handsets.
3. It is a photodetector that is used to sense the amount of ambient light present
4. It appropriately dim the device's screen to match it.

Applications of Optical Sensors



Laser-guided bombs such as the Paveway II and III can home in on a coded pulse laser light which is indicated onto a target.

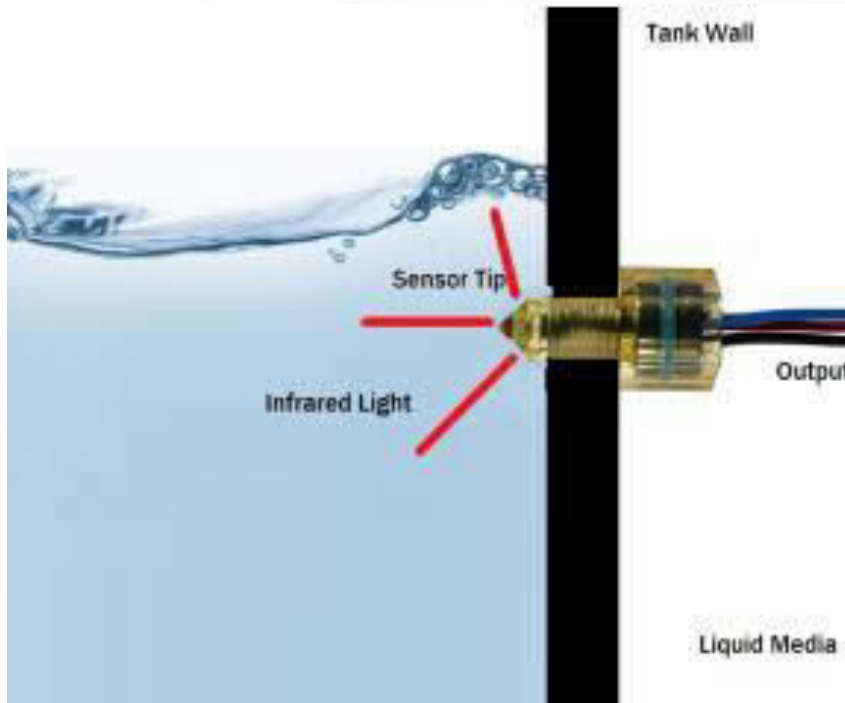
http://news.bbc.co.uk/1/hi/world/02/military_fact_files/cps_mini_guides/smart_bombs/img/lasergif

LASER Guidance

Laser guidance directs a robotics system to a target position by means of a laser beam.

1. Laser guided technology is used by military **LASER guided precision targeting**.
2. Laser guided electronics are **available in market** including , video games, toys, communication and smart home technologies.

Applications of Optical Sensors



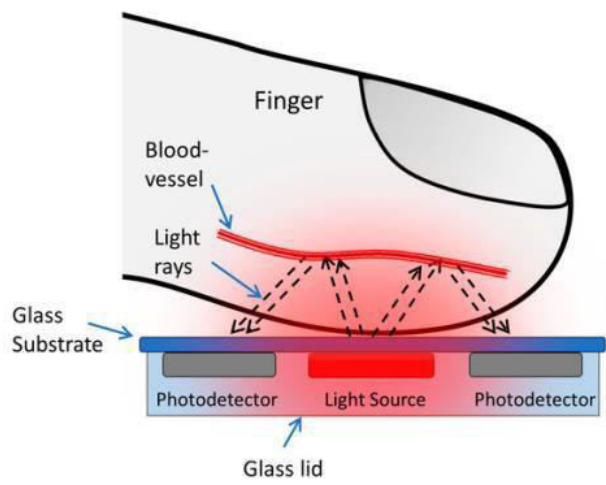
<https://www.elprocus.com/wp-content/uploads/2017/03/8.jpg>

Optical Sensor Based Liquid Level Indicator

Optical Sensor Based Liquid Level Indicator consist of three components.(1).**infrared Light Emitting Diode** coupled with (2). **a Light transistor**, and a (3). **Transparent prism tip**.

1. Sensor is dipped in liquid
2. Light disperses throughout
3. Less is returned to the transistor, depending on the depth/level of liquid.

Applications of Optical Sensors



https://www.mdpi.com/biosensors/biosensors-09-00087/article_deploy/html/images/biosensors-09-00087-g001-550.jpg

Pulse Oximeter

1. The oxygenated hemoglobin and deoxy hemoglobin have different rates of absorption of red and infrared light.
2. Oxy-hemoglobin absorbs more infrared light (990 nm) and deoxy-hemoglobin absorbs more red light (660 nm).
3. Difference is detected and oxygen saturation.

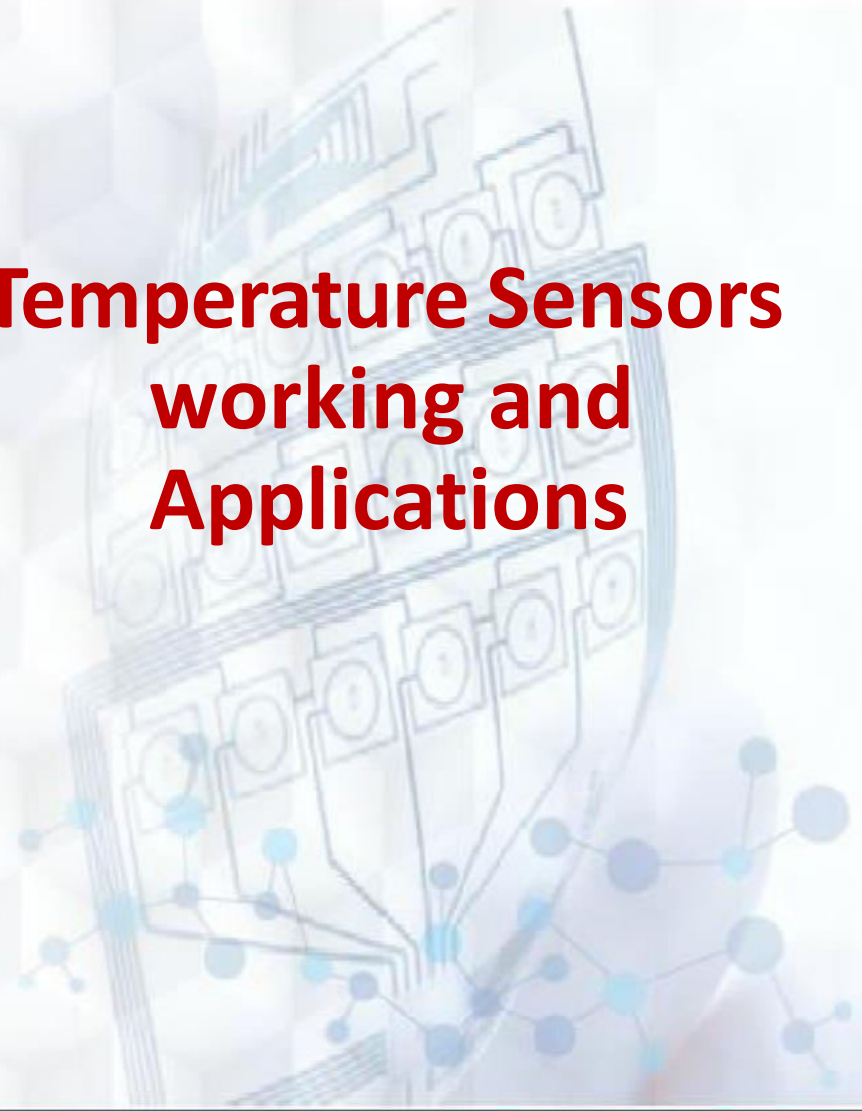
Applications of Optical Sensors

Summary.

1. Applications based on security include biometric identification of individuals.
2. LASER guided technology is heavily used in war against terrorism.
3. Oximeter is an important sensor that determines the level of oxygen saturation in blood, in an individual, without taking blood sample.

Biosensors

Temperature Sensors working and Applications



Temperature Sensors Working and Applications

Introduction

Temperature sensor is a device, to measure the **temperature** through an electrical signal.

Common scales of temperature measured in degrees:

1. Celsius ($^{\circ}\text{C}$)
2. Kelvin (K)
3. Fahrenheit ($^{\circ}\text{F}$)
4. Rankine ($^{\circ}\text{R}$ or $^{\circ}\text{Ra}$)

Temperature Sensors Working and Applications

Classification

1. **Contact Temperature Sensor** They required to be in physical contact with the object being sensed. They use conduction to monitor changes in temperature.
2. **Non-contact Temperature Sensor.** They use convection and radiation to monitor changes in temperature. They can be used to detect liquids and gases that emit radiant energy

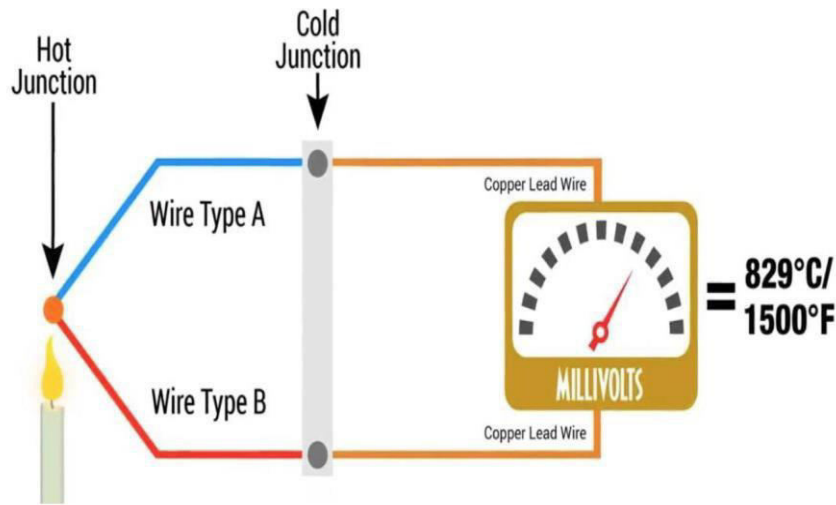


https://www.google.com/search?q=non+contact+non+touch+thermometer+dr+trust&client=firefox-b-d&source=lnms&tbn=isch&sa=X&ved=0ahUKEwjF-6GeirLIAhVBbFAKHQ5MDsMQ_AUIESgB&biw=1025&bih=460&dpr=1.33#imgsrc=DKi2hFxAetudtM:

Temperature Sensors Working and Applications

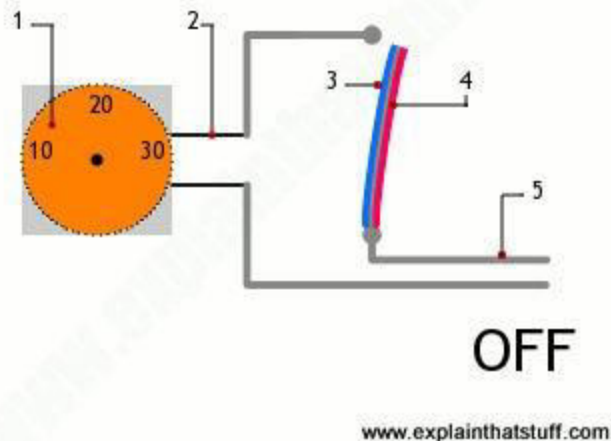
Thermocouples

1. A thermocouple is a sensor that measures the temperature.
2. It consists of two different types of metals, joined together at one end.
3. When the junction of the two metals is heated or cooled.
4. A voltage is created that can be correlated back to the temperature.



<https://assets.omega.com/resources/how-thermocouples-work-1.jpg>

Temperature Sensors Working and Applications

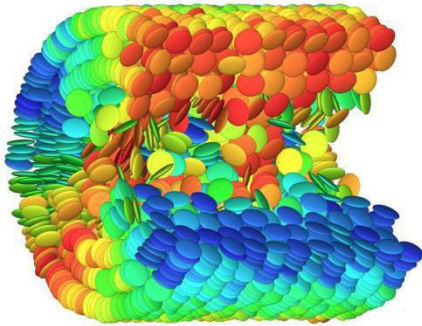


<https://cdn4.explainthatstuff.com/how-bimetal-thermostat-works.gif>

Thermostat

1. It senses the temperature of a physical system and performs actions so that the system's temperature is maintained near a desired setpoint.
2. A Bi-metallic strip (brass and iron) acts a bridge to through the electric circuit when it is cool.
3. When it is hot iron expands less the brass it bends and circuit breaks.

Temperature Sensors Working and Applications



Cutaway-view into a largely self-organised liquid crystal in a nanopore.

Liquid crystal Infra red thermometer

Liquid crystals has an ability to map out thermal regions of different temperature.

1. An infrared thermometer to make approximate temperature measurements.
2. As the temperature changes, their color also changes from red to orange, yellow, green, blue, and purple.



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http://www.desy.de/e409/e116959/e119238/media/5412/wedge_thumbnail_thumbnail.jpg

Temperature Sensors Working and Applications



<https://petapixel.com/assets/uploads/2015/06/FLIR-ONE-iOS-Android-copy-e1435257947413.jpg>

Thermal Image Camera (TIC)

1. Thermal cameras detect temperature by recognizing and capturing different levels of **infrared light**.
2. Thermal cameras help to see the hot areas that are masked by smoke, darkness, barriers such as tissues.
3. These images are shown by thermal image cameras.

Temperature Sensors Working and Applications



Yellow patch in breast is shown by thermal image camera .

https://i.dailymail.co.uk/1s/2019/10/22/14/20035712-7600337-image-m-2_1571749447211.jpg

Thermal Imaging

Thermal Imaging is used for Diagnostics.

1. Thermal imaging is used to study a broad number of diseases, using thermal image camera.
2. Skin temperature can reflect the presence of inflammation in underlying tissues.
3. Blood flow is increased or decreased due to a clinical irregularity.

Temperature Sensors Working and Applications

Summary

1. A **temperature sensors**, **sense** a change in the physical characteristic.
2. There are four types of different scales, for measurements.
3. Liquid crystals play an important role in advancement of thermal sensors.
4. Thermal image camera uses IR light to measure temperature for sensing.

Biosensors

Pressure Sensor Definition Working

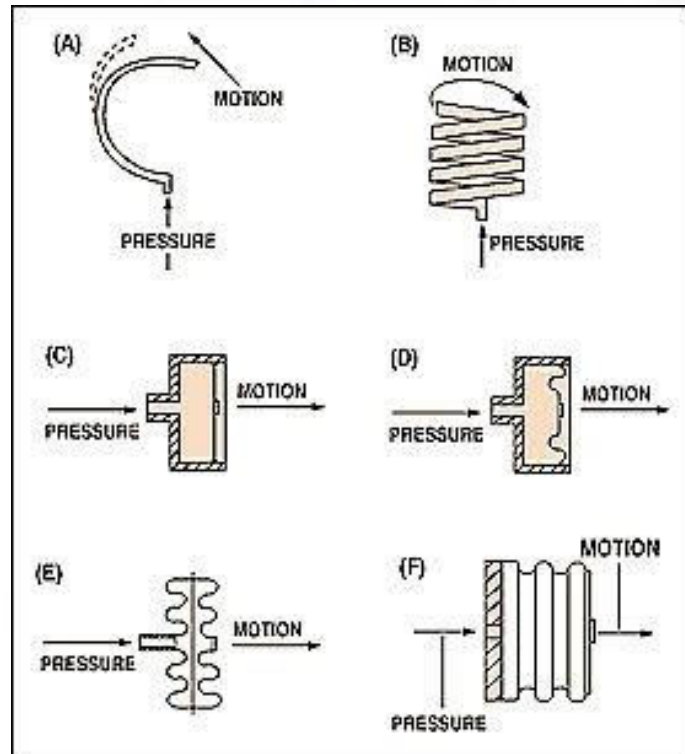


Pressure Sensors Definition and Working

Introduction

1. A pressure sensor is a device for pressure measurement of gases or liquids and piezo electric materials (solids).
2. Pressure sensors can measure other variables such as fluid/gas flow, speed, water level, and altitude.
3. Also called as pressure indicators, piezometers and manometers, etc.

Pressure Sensors Definition and Working

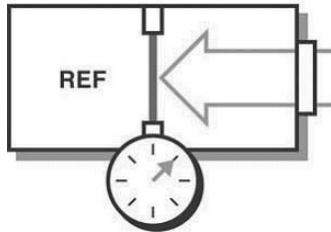


The basic pressure sensing element can be configured as a C-shaped Bourdon tube (A); a helical Bourdon tube (B); flat diaphragm (C); a convoluted diaphragm (D); a capsule (E); or a set of bellows (F).

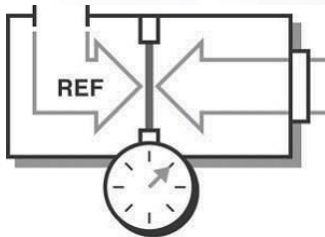
Pressure sensing Elements and Basic Mechanism

1. Pressure is sensed by mechanical elements such as plates, shells, tube and diaphragms.
2. These elements are designed and constructed to deflect when pressure is applied.
3. The basic mechanism in the pressure sensors is to convert the pressure to physical movement.

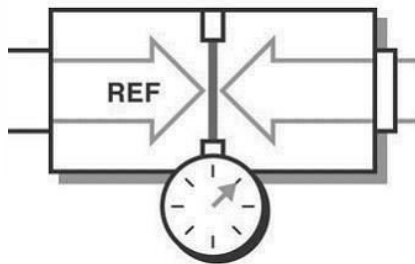
Pressure Sensors Definition and Working



Absolute Pressure Sensor



Gauge Pressure Sensor



Differential Pressure Sensor

Pressure Measurements

There are three types of pressure measurements:

- 1. Absolute Pressure** measurement is measured relative to a vacuum.
- 2. Gauge Pressure** is measured relative to ambient atmospheric pressure.
- 3. Differential Pressure** measurements are taken with respect to a specific reference pressure

Pressure Sensors Definition and Working



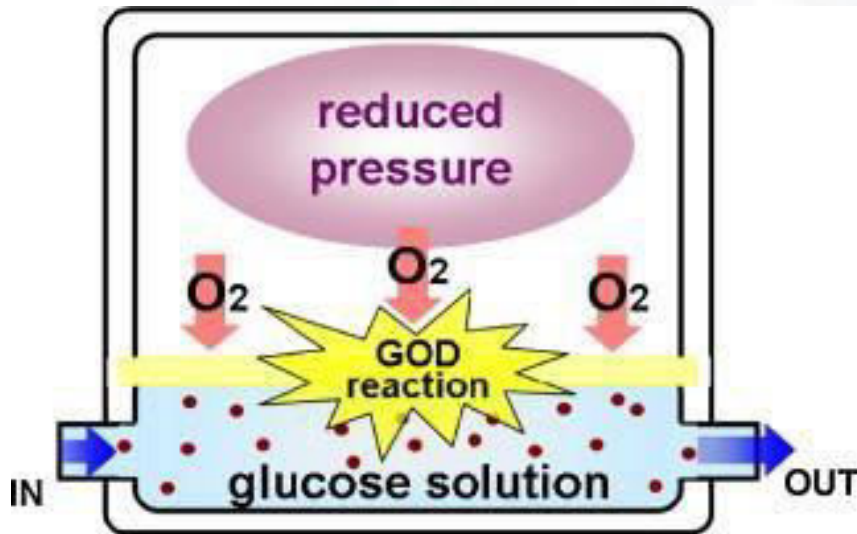
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https://media.springernature.com/full/springer-static/image/art%3A10.1038%2Fs41598-019-44348-3/MediaObjects/41598_2019_44348_Fig1_HTML.png?as=webp

Blood Pressure Apparatus.

1. **Sphygmomanometer**, containing an inflatable cuff and a Mercury manometer, to measure the BP.
2. **Wearable wrist Biosensor** is available in market for monitoring blood pressure, containing photo-plethysmography (PPG) and impedance-plethysmography (IPG) sensors.

Pressure Sensors Definition and Working

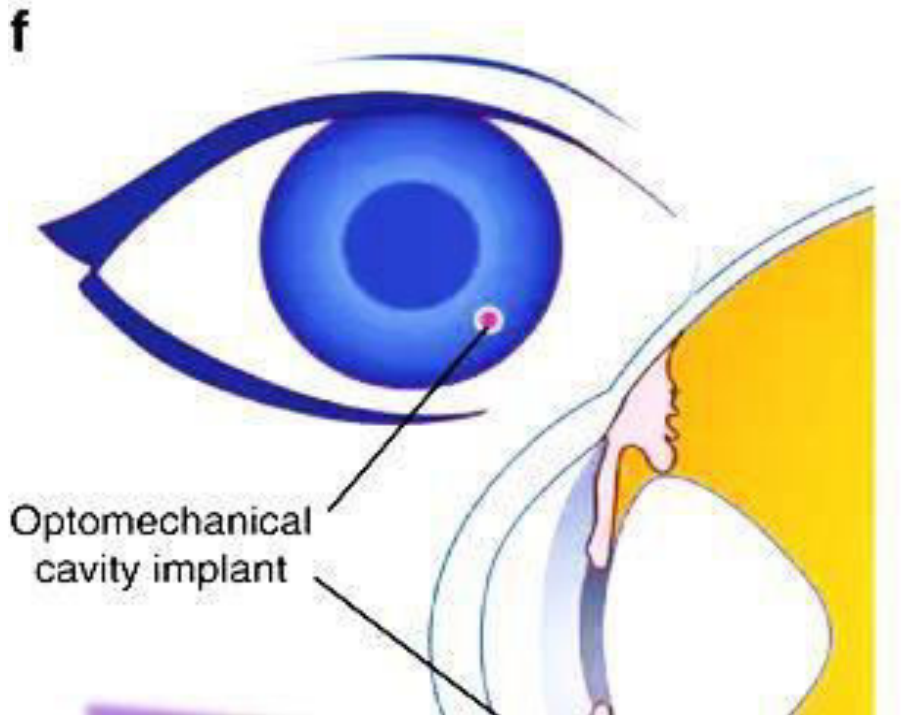


<https://ars.els-cdn.com/content/image/1-s2.0-S0956566308004399-gr1.sml>

Pressure Biosensor for Glucose measurement

1. A pressure biosensor is designed to measure differential pressure measurement of glucose level, using Oxygen levels.
2. Enzyme is immobilized on one side of the membrane
3. The pressure in the gas cell decreased continuously with consumption of Oxygen induced by (GOD) enzyme reaction.

Pressure Sensors Definition and Working



Sensor for Ocular Pressure

1. An implantable sensor is designed to provide convenient, accurate, on-demand Intra Ocular Pressure monitoring in the home environment.
2. The sensor is excited by near-infrared (NIR) light from a tungsten bulb.
3. The optical cavity of sensor reflects a pressure-dependent signal that can be converted to IOP.

https://media.springernature.com/lw685/springer-static/image/art%3A10.1038%2Fmicronano.2017.57/MediaObjects/41378_2017_Article_BFmicronano201757_Fig1_HTML.jpg?as=webp

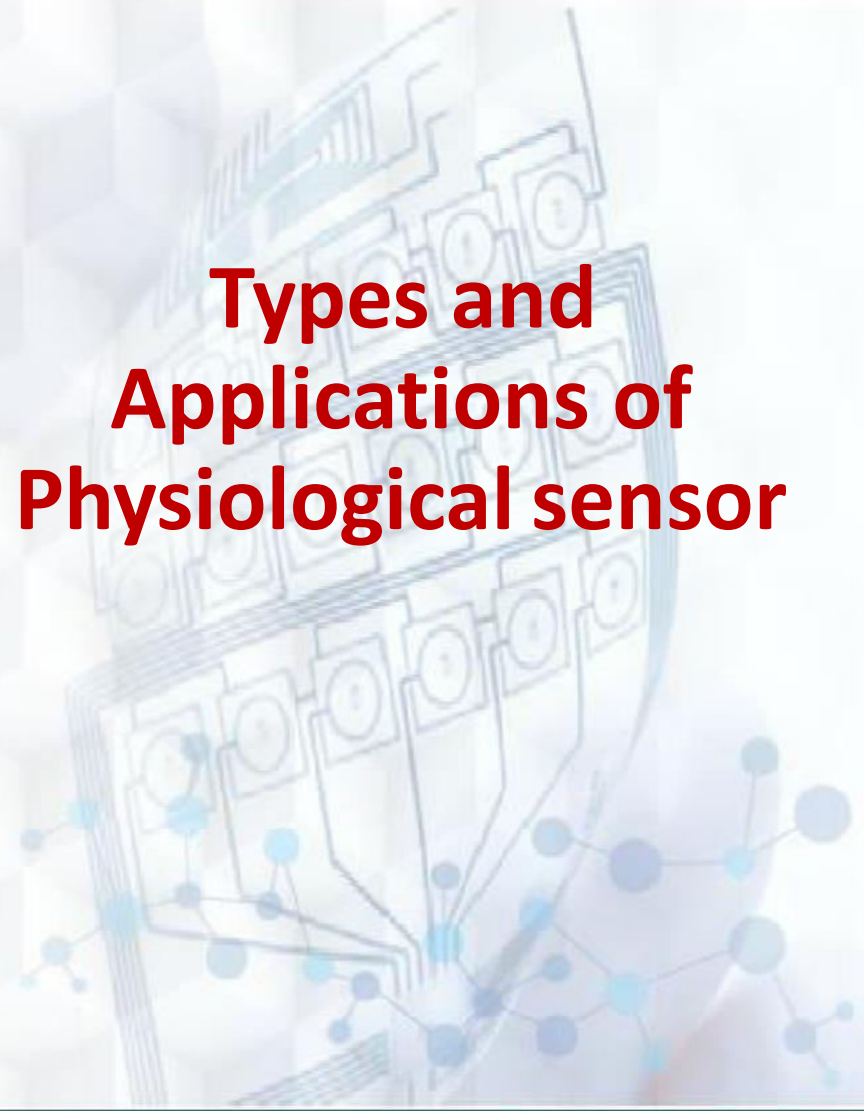
Pressure Sensors Definition and Working

Summary

1. Pressure sensors measures the pressure of a system.
2. Pressure is measured as differential pressure, gauge pressure and absolute pressure.
3. Applications of sensors in diagnostics include
 - A cuff less wearable sensor blood pressure.
 - Sensor for IOC in eye
 - Measurment of Glucose

Biosensors

Types and Applications of Physiological sensor



Types and Applications of Physiological Sensors

Introduction

1. Physiological sensors measures physiological parameters in our body.
2. There are four to six parameters which are considered as vital signs
3. These physiological parameters reflect the general **physiologic state** as well as **specific disease** states of a person.
4. Vital sign are monitors by physiological monitors

Types and Applications of Physiological Sensors

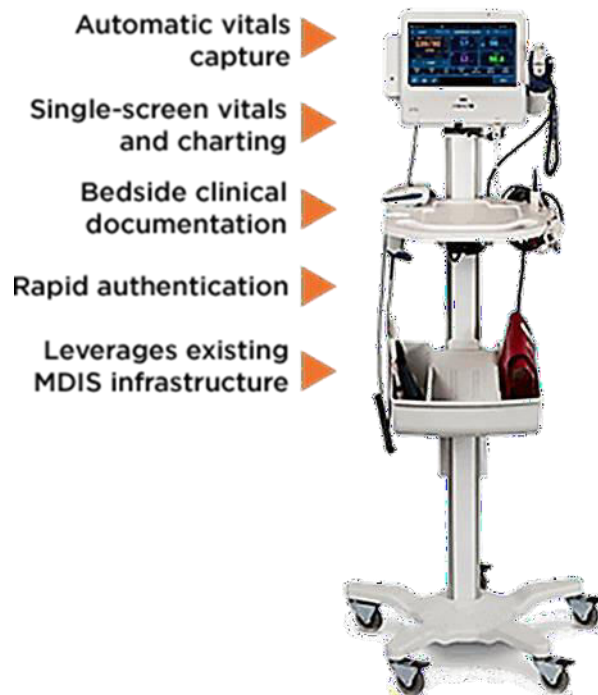
<i>Parameter</i>	<i>Range</i>	<i>Technology/Instrument</i>
<i>Rate of the heart</i>	<i>0.5 - 4.0 mV</i>	<i>Electrodes</i>
<i>Body temperature</i>	<i>32.0 - 45.0 °C</i>	<i>Temperature sensors</i>
<i>Blood pressure</i>	<i>10.0 - 400.0 mm Hg</i>	<i>Pressure sensors</i>
<i>Respiration rate</i>	<i>2.0 - 50.0 breaths/min</i>	<i>Pressure sensors Strain gauge</i>
<i>Glucose in blood</i>	<i>0.5 - 1.0 mM (millimoles per liter)</i>	<i>Electrochemical</i>
<i>Pulse oxygenation</i>	<i>80% -100%</i>	<i>Optical sensors</i>

Types and Applications of Physiological Sensors

Sensors/ Monitors

Physiologic monitoring systems monitor vital physiologic parameters so that clinicians can be informed of changes in a patient's condition. They typically consist of

1. A Central station
2. Bedside monitors
3. Telemetry transmitters
4. Receivers.



Types and Applications of Physiological Sensors

Types of Physiological sensors

Physiological sensors could be classified in different ways.

1. Classification on the basis of **Function** pressure sensor for determining BP.
2. Classification on the basis of **sensor type** used, for example optical sensor for Oxygen saturation.
3. **Smart Sensors**
4. **Wearable sensors.**

Types and Applications of Physiological Sensors

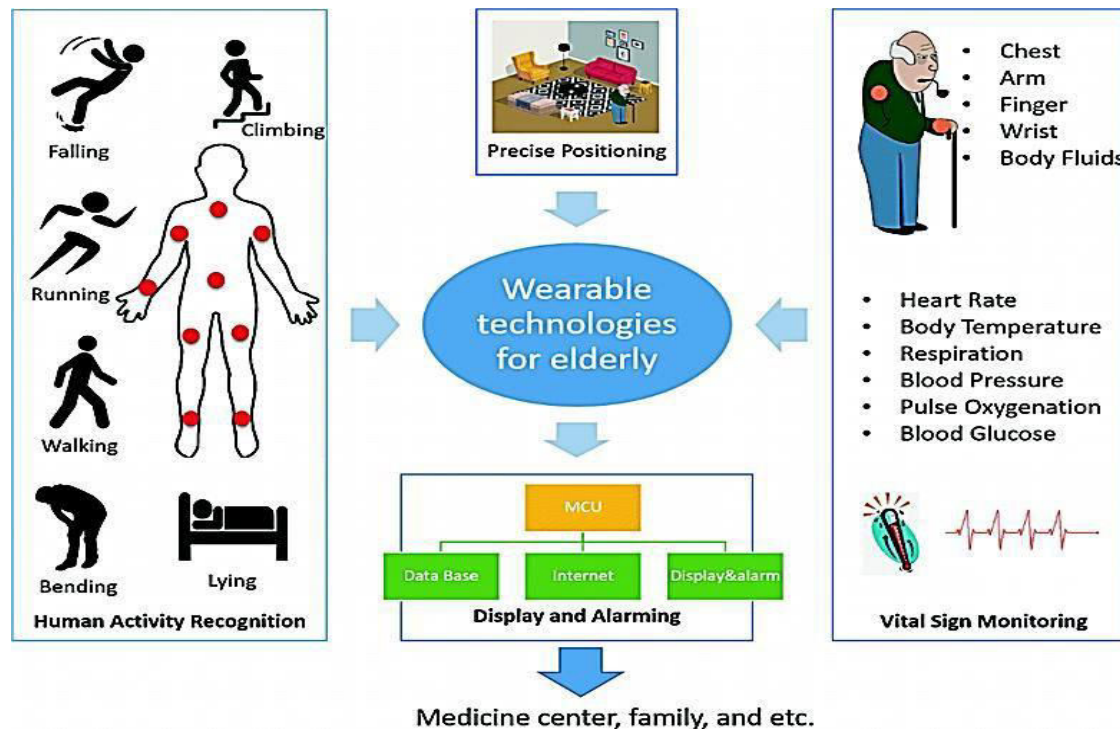


Figure demonstrates an example of a wearable elderly care system. The technologies are used for detection of accurate positioning, tracking the physical activity and monitoring the body signs data.

https://media.springernature.com/lw785/springer-static/image/chp%3A10.1007%2F978-3-030-10752-9_10/MediaObjects/477428_1_En_10_Fig5_HTML.png

Types and Applications of Physiological Sensors

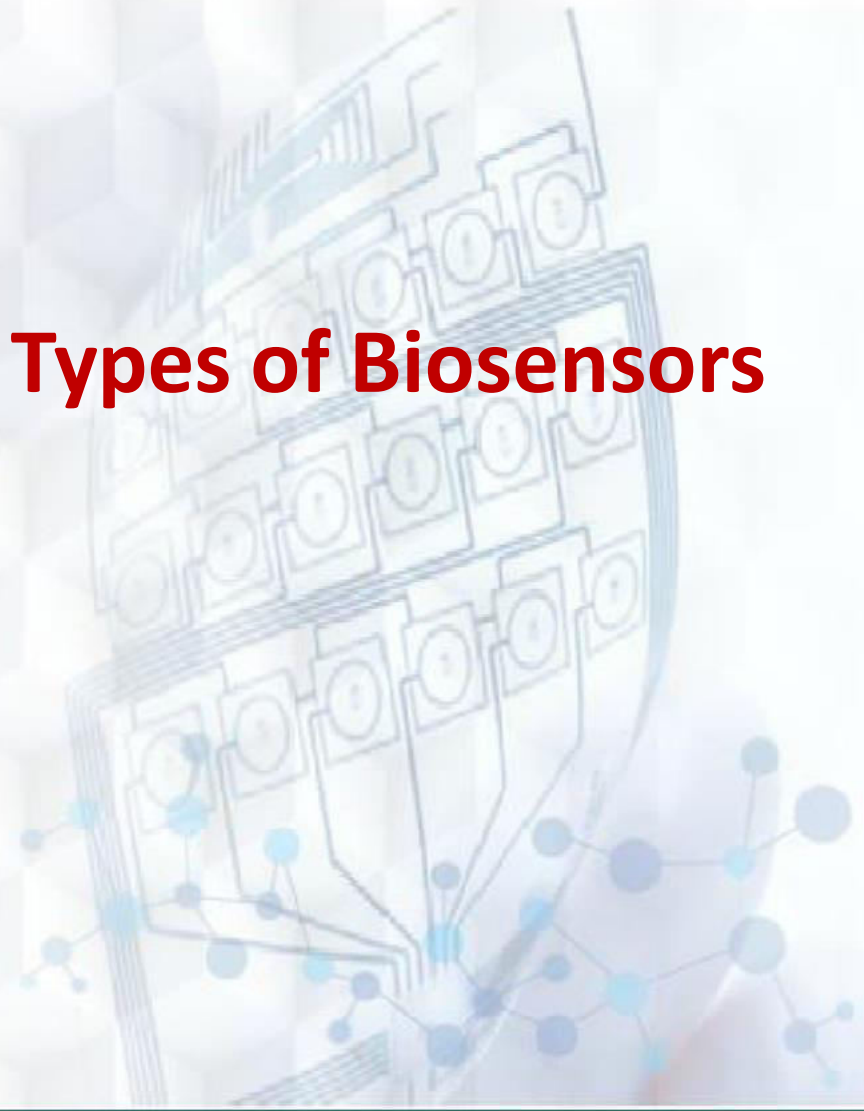
Summary

1. Physiological sensors are also commonly known as physiological monitors.
2. They monitor the vital signs associated with the health of a person.
3. The monitors are based on different sensing techniques, depending upon the **output of the physiological signal**.
4. Smart sensors, wearable sensors.

END

Biosensors

Types of Biosensors



Types of Biosensors

Introduction

Biosensors are classified on the basis of defend criteria.

1. **Generations** of Biosensors
2. **Biorecognition** Element
3. Type of **Sensor** Element
4. Type of **Application**
5. **Commercial** Biosensors

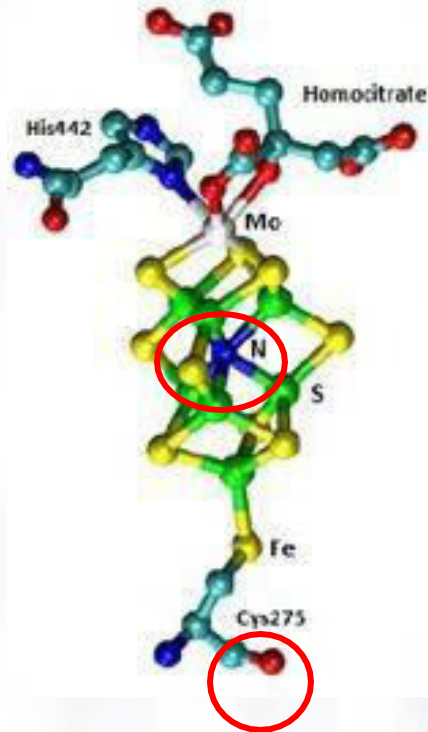
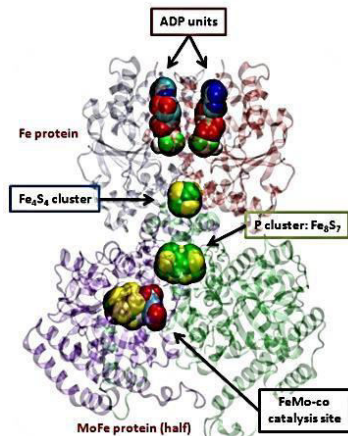
The most convenient classification is based on either the use of BRE or the transducer.

Types of Biosensors

Types on the basis of Bioreceptor element

1. Enzyme Biosensors
2. Immuno Biosensor
3. Cell Biosensor
4. DNA Biosensor
5. Aptamer Biosensor
6. Immuno fragment Biosensor
7. Biomimetic Biosensor

Types of Biosensors



A) Molybdenum containing nitrogenase enzyme.
B) Indicates the FeMo complex of the enzyme interacting with the key amino acid residues of the enzyme.

Reference: DOI 10.3390/molecules16010442

Biomimics strategy for detection

1. The approach is to synthesize Biomimetic enzymes, that contains the key amino acid residues of the active site.
2. These residues are involved in the catalytic and binding activity of the enzyme with the target molecule.
3. Nitrogenase enzyme is shown with Mo and Fe ions, involved in catalysis.

Types of Biosensors

Commercial Biosensors

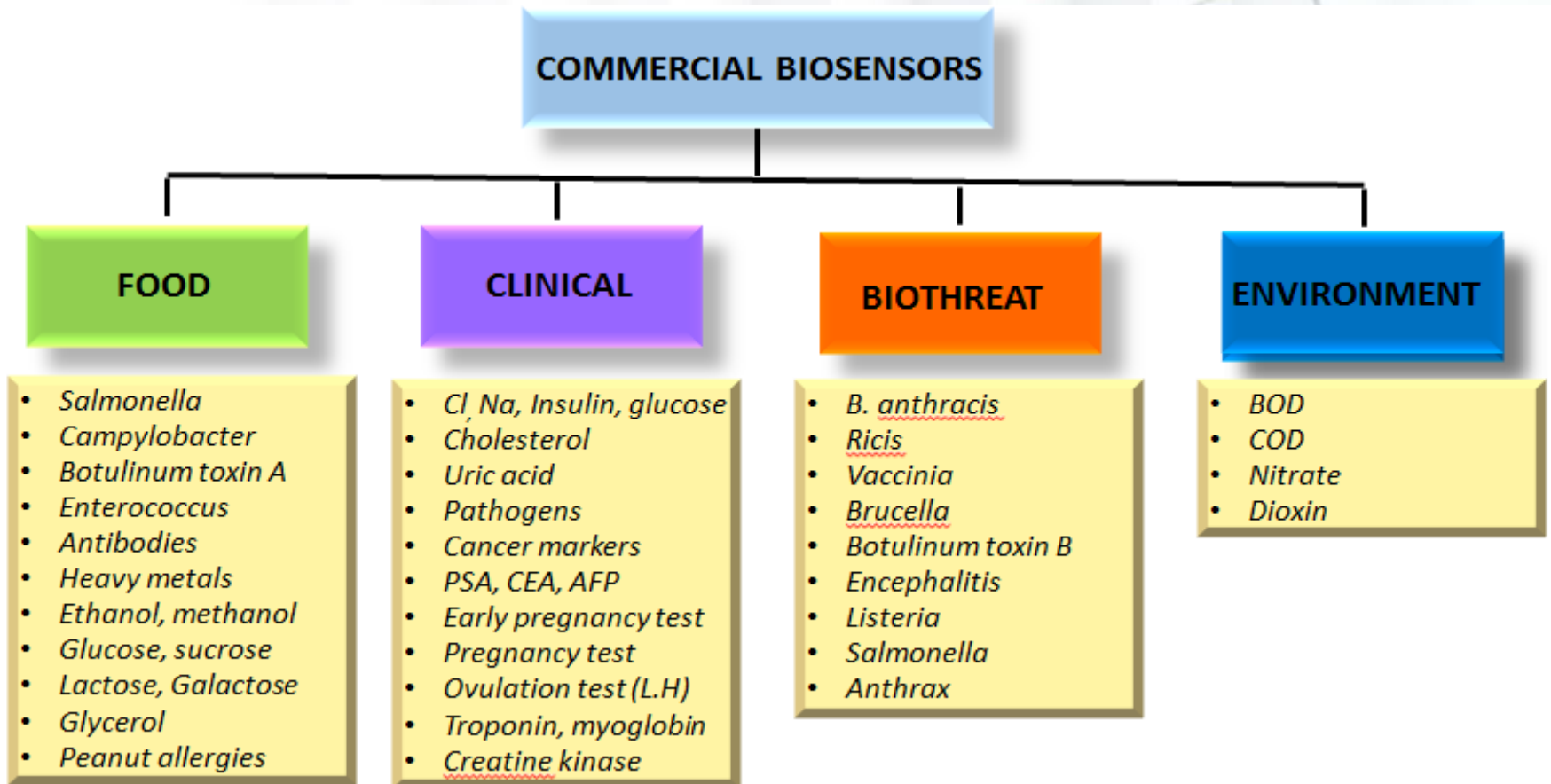
The commercial biosensors have small size and simple construction and they are ideal for point-of-care biosensing.

There are four main categories in commercial Biosensors.

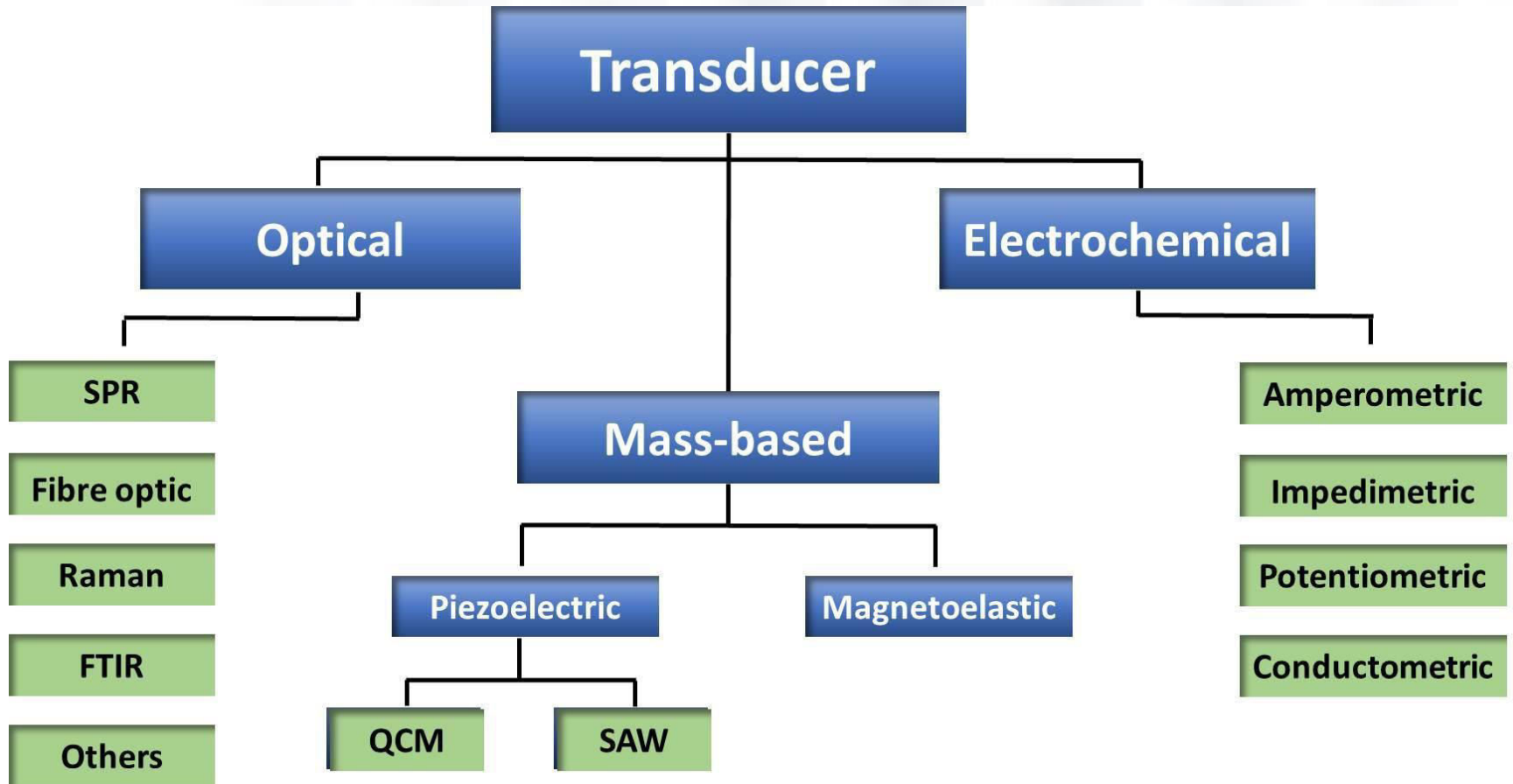
1. Clinical Biosensors
2. Food Biosensors
3. Bio threat
4. Environment

END

Types of Biosensors



Types of Biosensors



Types of Biosensors

Summary

1. Biosensors could be classified on different basis.
2. Use of Biomimetics in the Biosensors is the latest approach.
3. Biomimetic enzymes share the common binding abilities with the natural enzymes and are equally useful.

Biosensors

Latest Research and Current Trends



Latest Research and Current Trends

Introduction

A sensor is an input device that can detect a change after receiving a signal or stimulus. weight, pressure etc.

Sensors are becoming the biggest growing markets because they have applications in almost every field of life.

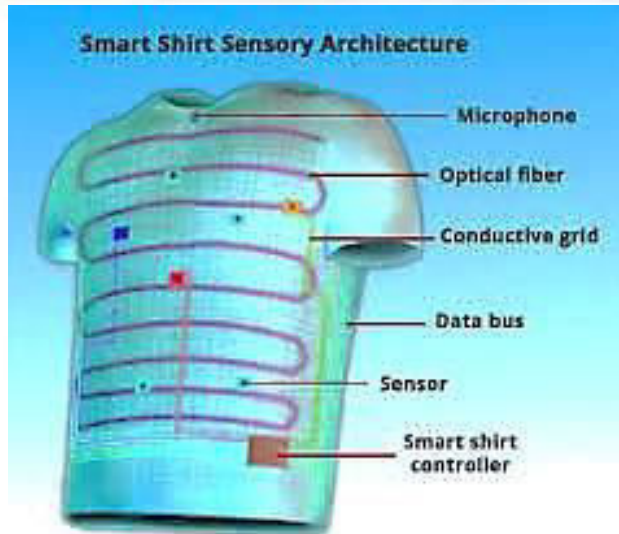
1. Smart sensors
2. Internet of things
3. Robotics

Latest Research and Current Trends

Smart Sensor

1. A smart sensor is a small **multi tasking device**.
2. It can **detect signals** from physical environment. It **saves time** and **energy**.
3. A smart sensor contain an embedded **Digital Motion Processor (DMP)**.
4. Components
 - i. **A Sensor**
 - ii. **A microprocessor**
 - iii. **Communication device**

Latest Research and Current Trends



Examples of Smart Sensor

Wearable T smart shirt, it uses optical fibers to detect wounds. It can monitor the vital signs like **heart rate**, **respiratory rate** and **temperature**.

Oximeter is a smart sensor. It can measure

1. Finger-based heart rate
2. Blood oxygen saturation level

<http://www.concordhealthsupply.com/v/vspfiles/photos/75006-2.jpg>

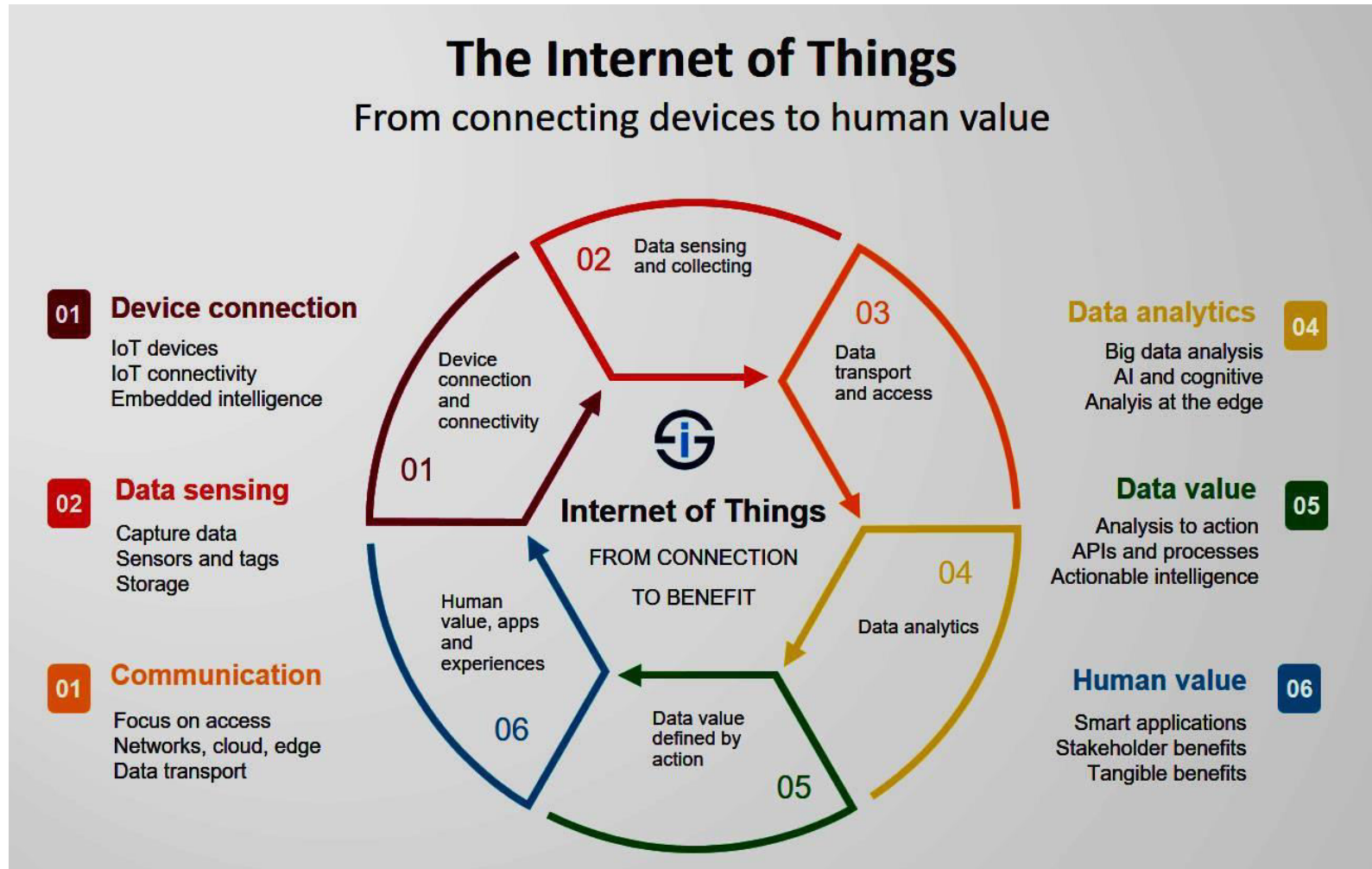
https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSsvVQQYyTe698CYf5awnydsr7PVNz_8GTsTOAwCtxE5VK-EtZ_&s

Latest Research and Current Trends

Internet of Things (IOT)

1. The Internet of Things (IoT) is a system of inter related things including **Computing Devices, Mechanical Machines, Digital Machines, Objects, People and Animals.**
2. They are provided with a **unique identity (UID)** to **access a facility, data, ability to transfer data** without requiring human – human interaction, .

Latest Research and Current Trends



Latest Research and Current Trends

Robotic Sensors

1. A robot is a programmable machine, to **carrying out a complex series of actions automatically.**
2. Robotic sensors are used to estimate a robot's condition and environment, for further actions.
3. Sensors in robots are based on the functions of human sensory organs.

Latest Research and Current Trends

Summary

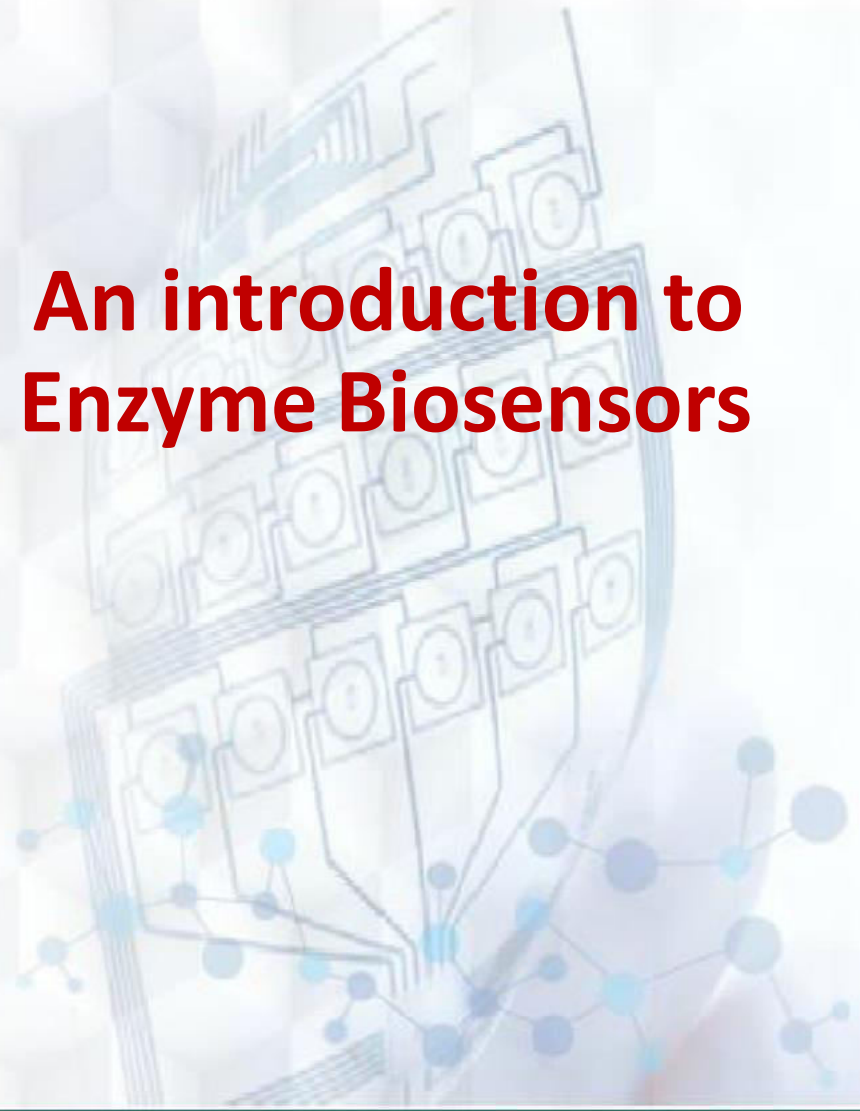
Sensors are the most common device that is used in every day life and is one of the highly growing markets in the world.

1. Advancement in the sensor technology leads to the production of smart sensors, internet of things and Robotics.
2. Wearable sensing technology is one of the fastest growing field around 60 Billion USD.

END

Biosensors

An introduction to Enzyme Biosensors



An introduction to Enzyme Biosensors

Introduction

1. Enzyme Biosensor used an Enzyme as a **Bioreceptor** element for *the detection*.
2. Enzymes are biological **catalysts**.
3. The first Biosensor introduced in **market** was an **enzyme Biosensor**.
4. The most successful product among Biosensor is an Enzyme Biosensor that is **Glucometer**.

An introduction to Enzyme Biosensors



<https://i.ytimg.com/vi/NABYn4wY84c/hqdefault.jpg>

Chemical method for Glucose Estimation

Chemical method using DNS for glucose estimation requires

- 1. Large volume** of sample
- 2. Pretreatment** of sample
- 3. Drastic conditions**
- 4. High temperature**
- 5. Proper lab setup**
- 6. Trained staff**
- 7. Time consuming**

An introduction to Enzyme Biosensors



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Advantages of Enzyme based Biosensor

1. Glucometer work by analyzing a **small amount** of blood.
2. Glucometer is a small **hand held** device.
3. Glucometer works at **ambient conditions** .
4. **No pretreatment** of blood sample is required.
5. **Commercially** available.

An introduction to Enzyme Biosensors

Detection methods in Enzyme Biosensors

1. There is a huge number of enzymes that can act on their **unique substrates** and produce respective **products and bi-products**.
2. Enzyme Biosensors could work either by **detecting the end product, bi-product** or any **step during catalysis** (flow of electrons) by using **different types of sensors**.

An introduction to Enzyme Biosensors

Sensors in Enzyme biosensors

The selection of sensor in a Biosensor is dependent on the type of signal being detected
detection of the signal

Optical sensor could be used if there is a change in the optical properties (**change in colour**).

Amperometric Sensors a could be used to detect the flow of current.

Pressure sensor could be used for a change in pressure.

An introduction to Enzyme Biosensors

Dependencies of Enzyme Biosensors

1. The catalytic activity of enzymes is strongly **dependent on pH, temperature and pressure.**
2. Membrane biodegradation by the enzymes could lead to Biocompatibility-based failure **membrane biodegradation.**
3. Example cellulase enzymes is **immobilized on a cellulose** membrane.

An introduction to Enzyme Biosensors

Summary

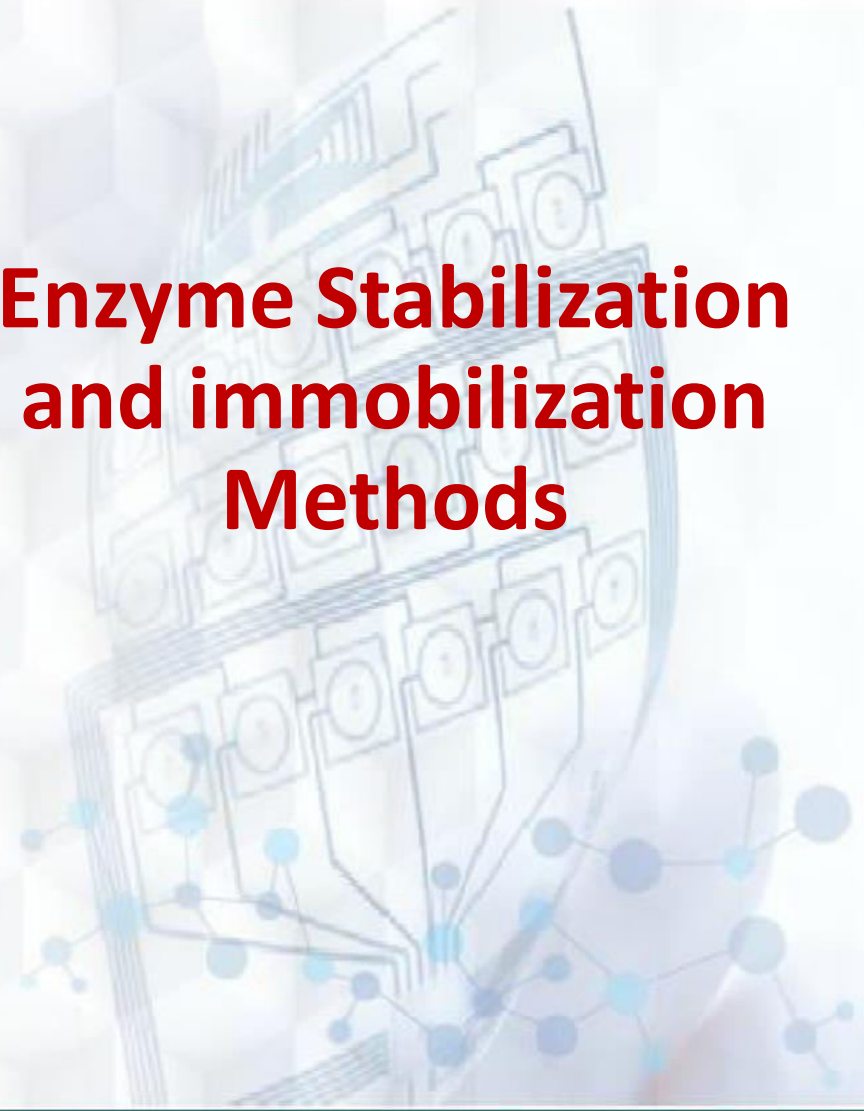
Enzyme based Biosensor are the most popular type of Biosensors (Glucometer).

1. A **huge variety of signals** could be **detected** by enzyme biosensors.
2. **Glucometer** has a big market for the product. **unique features and economical availability.**
3. Easy to use and **cater the need** of a common man.

END

Biosensors

Enzyme Stabilization and immobilization Methods



Enzyme Stabilization and Immobilization

Introduction

For the stabilization and immobilization of enzymes, biosensors involve, wiring of an enzyme to the electrode.

It involves the co-immobilizing the enzyme and mediator directly onto the electrode surface.

- 1. Carbon nanotubes**
- 2. Gold nanoparticles**
- 3. Polyelectrolyte**
- 4. Self assembled monolayer**

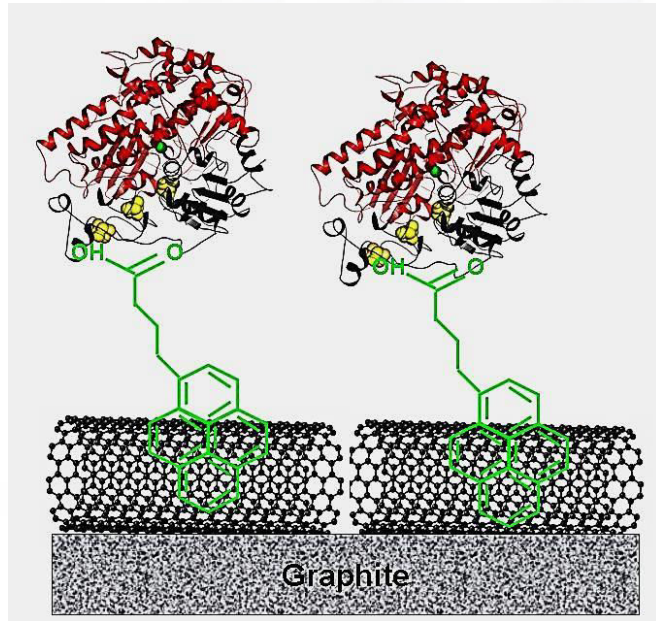
Enzyme Stabilization and Immobilization

Immobilization of Enzyme onto CNT

1. Carbon nanotubes are made up of **Graphene**. They have aromatic structure and are hydrophobic.
2. CNTs can act as a mediator for enzyme immobilization on electrode after treatment.
3. CNTs are used for
 - **Physical adsorption**
 - **Chemical cross-linking,**

Enzyme Stabilization and Immobilization

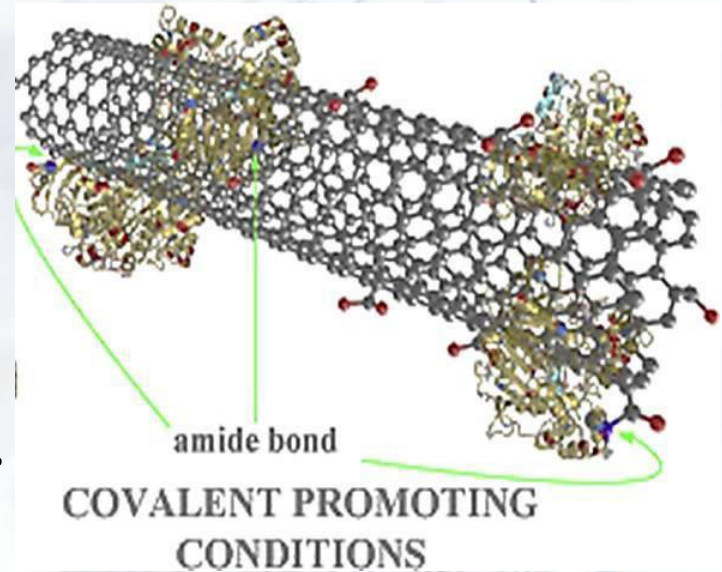
Physical adsorption on CNTS



CNTs are functionalized by reducing agent such as citrate to the positively charged amino acid residues of the enzyme.

<https://www.intechopen.com/media/chapter/38918/media/image4.jpg>

Cross Linking on CNTS



Chemical cross-linking involves covalent attachment of the enzyme onto CNT via a linker molecule using Glutaraldehyde'BSA or silica gels, sol grls.

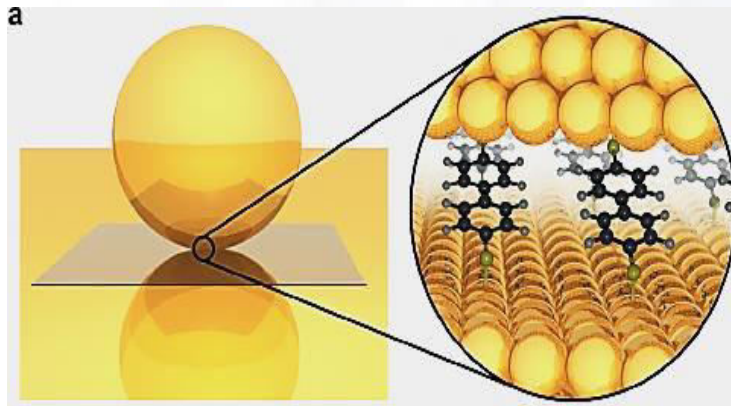
<https://ars.els-cdn.com/content/image/1-s2.0-S1631074815003239-fx1.jpg>

Enzyme Stabilization and Immobilization

Physical Adsorption on Gold Nano Particles

1. Physical adsorption is a **simple and quick** method.
2. It involves **reducing the gold nanoparticles** with a negatively charged ligand such as **citrate**.
3. Sulfur containing amino acid **Cysteine** is also used as a mediator or linker
4. The chemistry of **sulfur absorption on gold surface** is not clear yet.

Enzyme Stabilization and Immobilization



https://www.researchgate.net/profile/Christos_Tserkezis/publication/280919242/figure/fig1/AS:284473666097159@1444835215632/Conductive-and-nonconductive-self-assembled-monolayers-in-plasmonic-junctions-a.png

Self-Assembled Monolayer

1. A Self-Assembled Monolayer (SAM) is a one molecule thick layer of material that bonds to a surface.
2. **Thiol SAMs** on gold are the most popular molecular films because the resulting **oxide-free, clean, flat surfaces** can be **easily modified** both in the gas phase and in liquid media under **ambient conditions**.

Enzyme Stabilization and Immobilization

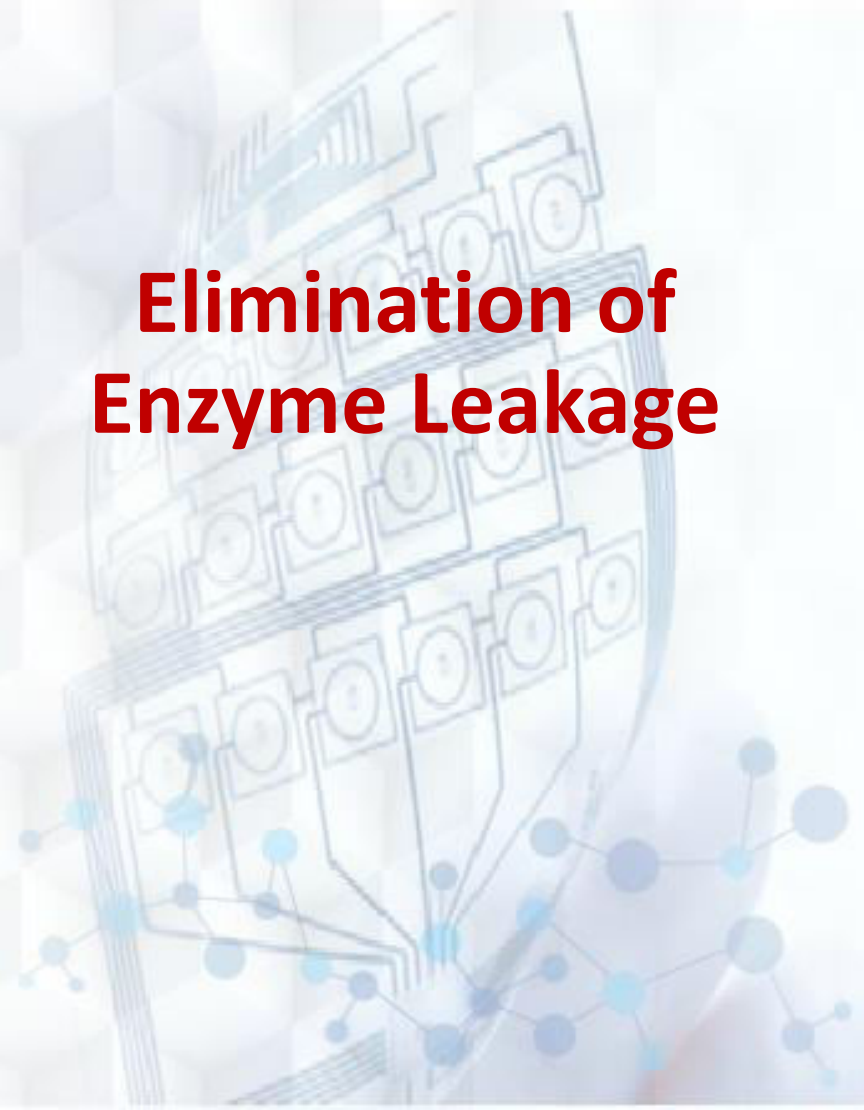
Summary

1. Enzymes are used BRE in biosensors.
2. Immobilization and stabilization is a crucial step in the fabrication of biosensor.
3. Use of different mediator is a successful approach to increase the stability and performance.
4. **Polyelectrolytes, Self-Assembled Monolayer, CNTs and GNP**

END

Biosensors

**Elimination of
Enzyme Leakage**



Elimination of Enzyme Leakage

Introduction

For artificial immobilization several methods are used.

Use of **chemical methods** for immobilization is expected to be **flaw less**.

It has been noted that the immobilized enzyme is eliminated, because the method **relied on**

1. The **completion of chemical reaction** and
2. Specific **chemical bonding**.

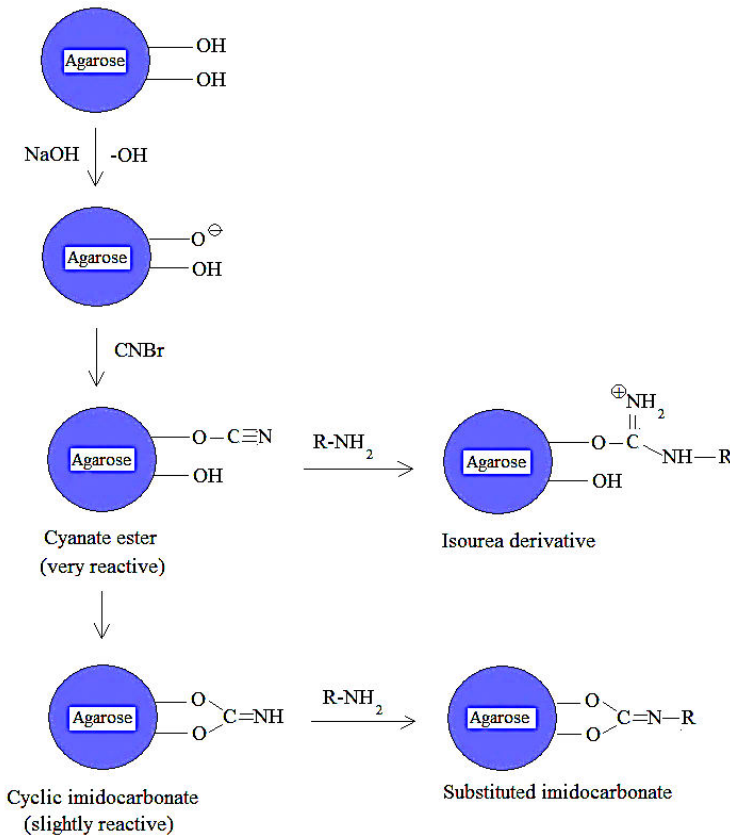
Elimination of Enzyme Leakage

Chemical methods

Activated Agarose gels are used for immobilization, Agarose is a polysaccharide it is linear polymer of **D-Galactose and 3,6 anhydro-L-galactopyranose**.

1. CNBr Agarose
2. Amino hexanoic acid Agarose
3. Carbonyldiimidazole Agarose
4. Epoxy activated Agarose

Elimination of Enzyme Leakage



https://upload.wikimedia.org/wikipedia/commons/9/9d/CNBr_activated_matrices_reaction.PNG

Cyanogen Bromide Immobilization

1. The method is **simple**, works at **mild pH and temperature** and is well documented.
2. It is a **rapid method** and biological molecule **retains the biological activity**.
3. Method offers **multiple points for the attachment** of the ligand molecules,
4. It is rich in ***cyanate esters and imido carbonates***.

Elimination of Enzyme Leakage

Leakage population

There are **at least two populations** of the covalently attached enzymes.

1. Liable to Leakage
2. Resistant to cleavage

The **proportion of the two population varies** with the **extent of cross linking** during CNBr activation.

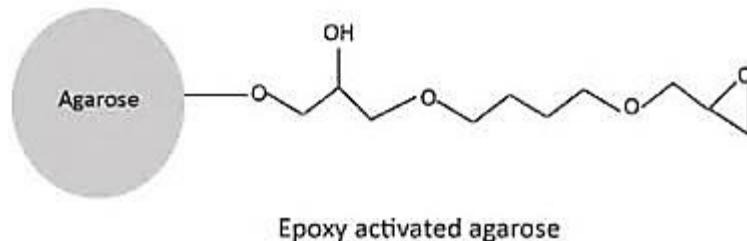
The process is not **completely controllable for multiple binding sites**.

Elimination of Enzyme Leakage

Enzyme leakage

1. The **bond between the cyanate ester and amine group of protein is unstable.**
2. The weak bond constant leakage of the ligand
3. Low molecular weight ligands are slowly released from the polymeric support in the **aqueous media above pH 5.0.**

Elimination of Enzyme Leakage

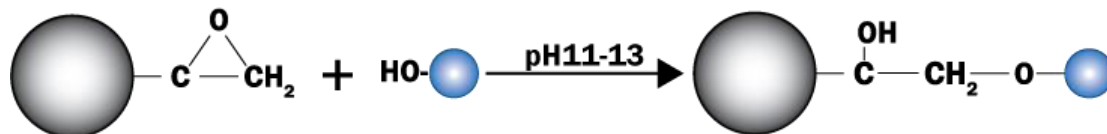
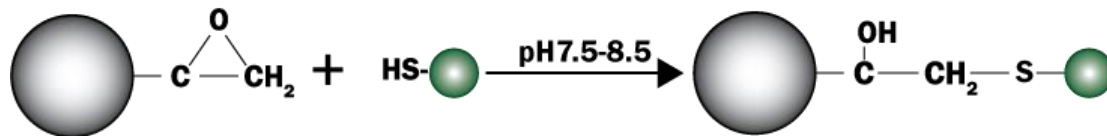
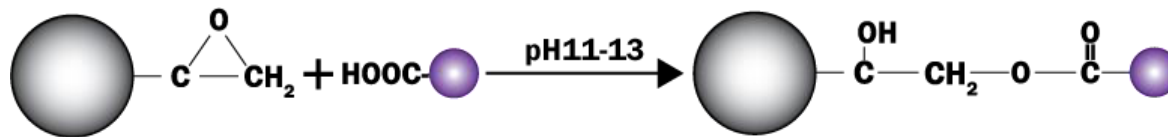
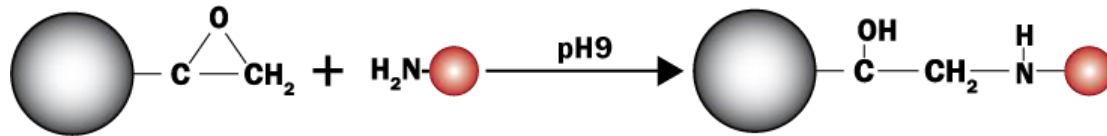


<https://info2.gbiosciences.com/hubfs/Epoxy-Activated-Agarose.jpg>

Epoxy Activated Agarose

1. Agarose is prepared by treatment with **Oxiranes**.
2. The pre-activated resin provides **hydrophilic 12 atom spacer arm**.
3. Epoxy activated Agarose can immobilize ligands via **amine, Thiol, and Hydroxyl** groups.
4. The immobilization is pH dependent.

Elimination of Enzyme Leakage



https://info.gbiosciences.com/hs-fs/hubfs/Epoxy-Activated_Resin_Interactions.png?width=777&name=Epoxy-Activated_Resin_Interactions.png

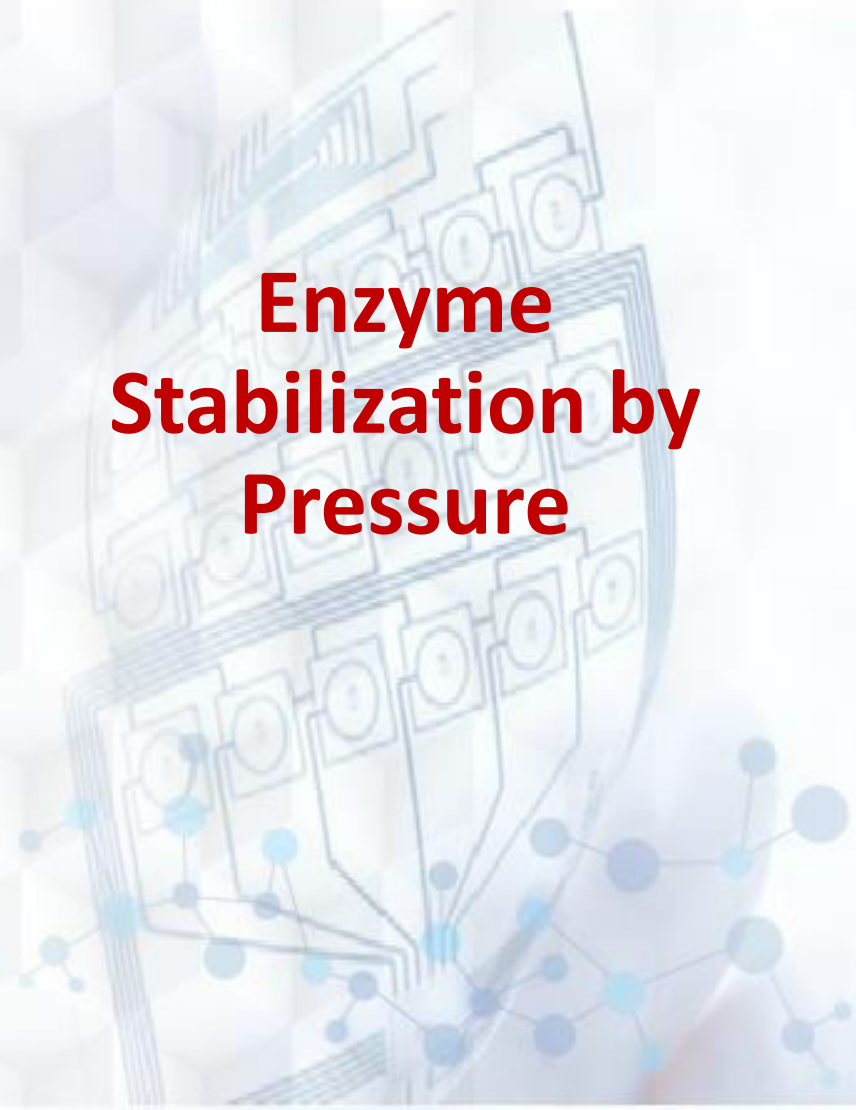
Elimination of Enzyme Leakage

Summary

1. Chemical methods are used for irreversible immobilization of proteins on the solid matrix for Biosensor fabrication.
2. Use of CNBr for immobilization is widely is used, is reported to be not a perfect method, as enzyme leakage is observed.
3. Use of Epoxy activate Agarose is another alternative.

Biosensors

Enzyme Stabilization by Pressure



Enzyme Stabilization by Pressure

Introduction

Enzymes are globular functional proteins, that are produced by the living organisms.

1. They catalyze the chemical reaction by the virtue of their unique catalytic and binding sites for their substrates, at standard conditions.
2. Effect of pressure on the biological activity of the enzymes is studied for long.

Enzyme Stabilization by Pressure

Pressure Effect

The effect of high pressure of enzyme is heterogeneous due to change in weak forces. It is dependent on the

1. Type of enzyme
2. Level of pressure
3. Effect of pH and temperature.

In some cases it can improve the biological activity and in some cases it is damaging for enzyme.

Enzyme Stabilization by Pressure

Impact on Hydrogen Bonds

The amide groups form hydrogen bonds either with carbonyls or with water.

1. These hydrogen bonds are shortened by pressure.
2. The estimated shortened distances varied from site to site (0–0.11 Å)
3. Larger reduction in the turn but smaller in the interiors of secondary structures (alpha helix, beta sheets).

Enzyme Stabilization by Pressure

Stabilization and Activation

There are two examples for the Stabilization and activation of enzyme activity, due to high pressure.

1. Glucose Oxidase
2. Polyphenol oxidase

The high pressure promoted **changes in the**

- **Biological activity**
- **Stability**
- **Kinetic parameters**
- **Structure.**

Enzyme Stabilization by Pressure



3D structure of Polyphenoloxidase

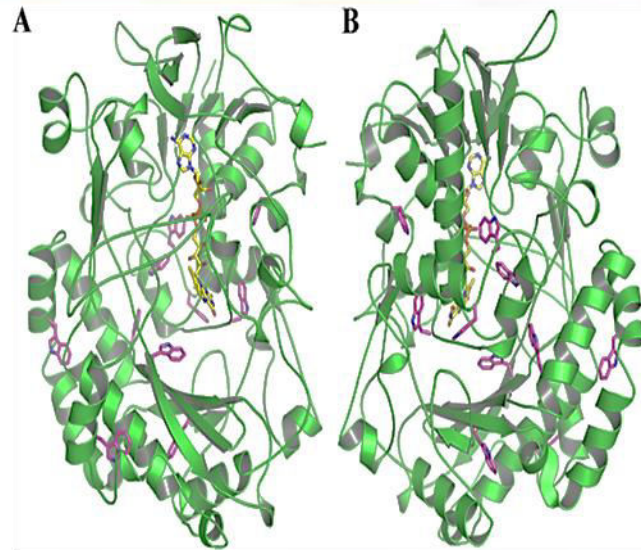
https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRvS3C1iHjiqkge_nnjGLTXiIOr9tbXJSZbGNI38KPxFYNlrDS1

Phenol Per oxidase

Activation and conformational changes of mushroom PPO after high pressure.

1. Loss of α -helix content
2. Change in β -sheet, β -turn and random coil
3. Trp and Tyr residues exposed to the less hydrophobic environment.
4. SH groups content on the surface increased.

Enzyme Stabilization by Pressure



Schematic representation of the GO structure from *A. niger* in two orientations (180°) indicating the positions of the tryptophan residues (carbon atoms in magenta).

<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0103410>

Glucose Oxidase

1. Pressure leads to a stable and new confirmation of the enzyme molecule.
2. Changes are observed in the 3-Dimensional structure of the protein at different levels.
3. Tryptophan residues was changed.
4. The helical content of the enzyme molecule was changed.

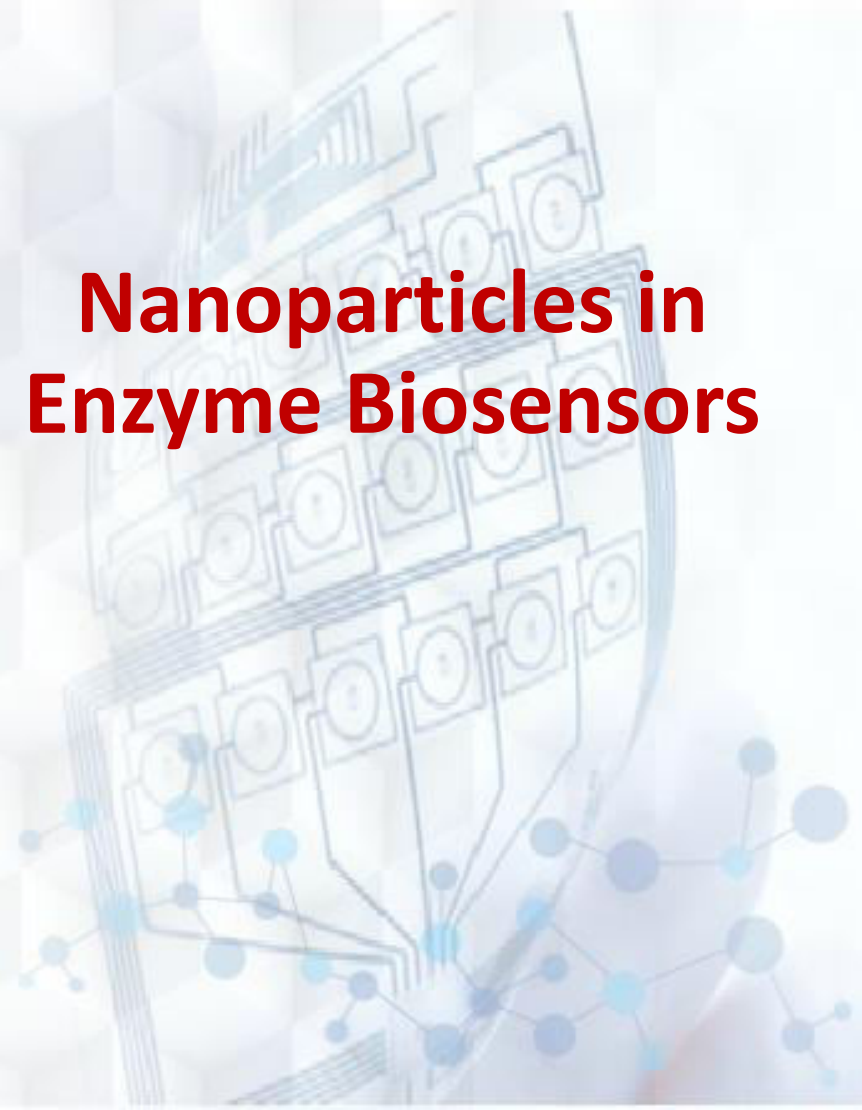
Enzyme Stabilization by Pressure

Summary

1. Treatment of enzyme at high pressure is a latest approach to increase their stability and activation.
2. Pressure has different effects on different enzymes depending on the type and conditions.
3. Optimum pressure has a positive impact on GO and PPO, with some changes in their 3D structure of the molecules.

Biosensors

Nanoparticles in Enzyme Biosensors



Nanoparticles in Enzyme Biosensors

Introduction

There are two types of nanoparticles which are used in Enzyme biosensors.

1. Organic nanoparticles
2. Inorganic nanoparticles.

For increasing the applications of nanoparticles it is required to modify the surface of nanoparticles. The surface could be modified by using physical methods and chemical methods.

Nanoparticles in Enzyme Biosensors

Introduction

The important functions provided by nanoparticles include the

1. **Immobilization** of biomolecules
2. The **catalysis** of electrochemical reactions,
3. The enhancement of **electron transfer** between electrode surfaces and proteins
4. **Labeling** of biomolecules

Nanoparticles in Enzyme Biosensors

Surface Modifications

For biological application **polar surface** modifications are required for the attachment of **biological molecules (Enzyme proteins)**, to increase the **solubility** and **prevent** them from **aggregation**.

As per the requirements, NP could be functionalized by

1. **Electrostatic adsorption**
2. **Hydrophobic Ligand**
3. **Hydrophilic Ligand**

Nanoparticles in Enzyme Biosensors

Electrostatic Adsorption

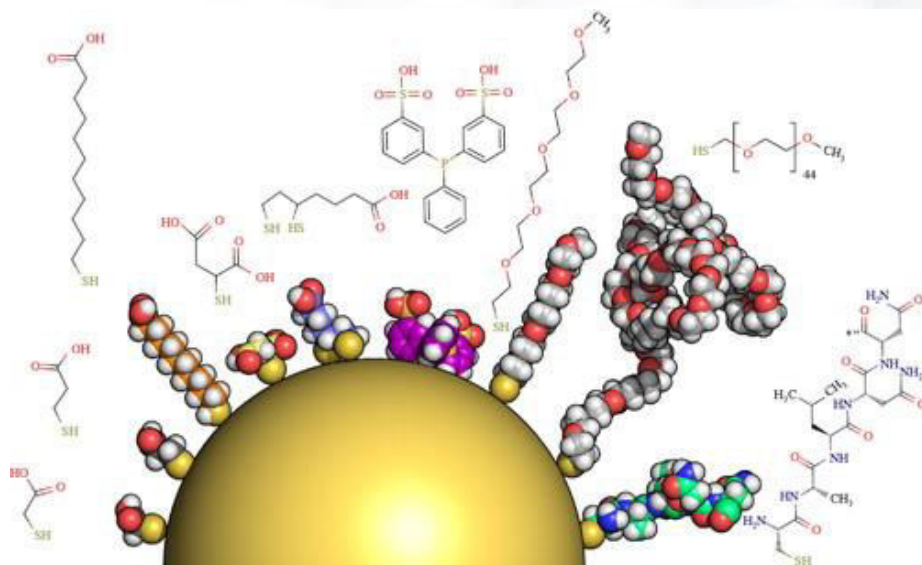
1. **Electrostatic Adsorption** is a widely used linkage approach, commonly used for carbons: amorphous activated carbon and carbon black.
2. **Strong electrostatic adsorption** is a method, in which metal (Pt) are strongly adsorbed over support surfaces by controlling the pH of the solution (NP size 1-2 nm).

Nanoparticles in Enzyme Biosensors

Hydrophilic Ligand

For hydrophilic surface modifications a number of hydrophilic compounds are used such as

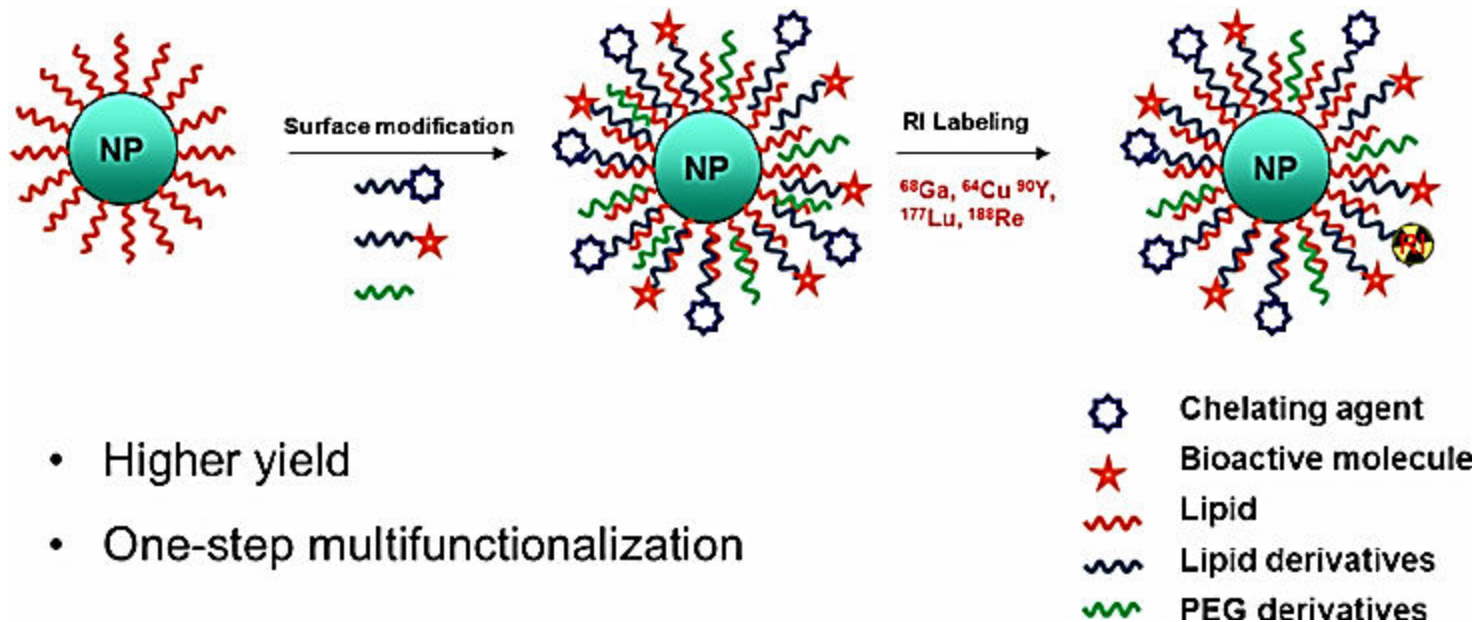
1. Mercaptoacetic acid
2. Mercaptopropionic acid
3. Mercaptosuccinic acid
4. Bis-sulphonated triphenylphosphine
5. Polyethylene glycol SH
6. short peptide CALNN.



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTxII7RcUfaq94OVnbAtmYeuq8FBExBlCaltw7hRDk-cSk6Sd54gQ>

Nanoparticles in Enzyme Biosensors

Surface modification of nanostructures with long alkyl tail and hydrophilic ligand head, which enable multi-functionality on nanoparticles in one step



https://www.researchgate.net/profile/Yong-il_Kim/publication/281143738/figure/fig1/AS:287046867406849@1445448714222/Schematic-figure-of-the-surface-modification-of-nanostructures-with-long-alkyl-tail-and.png

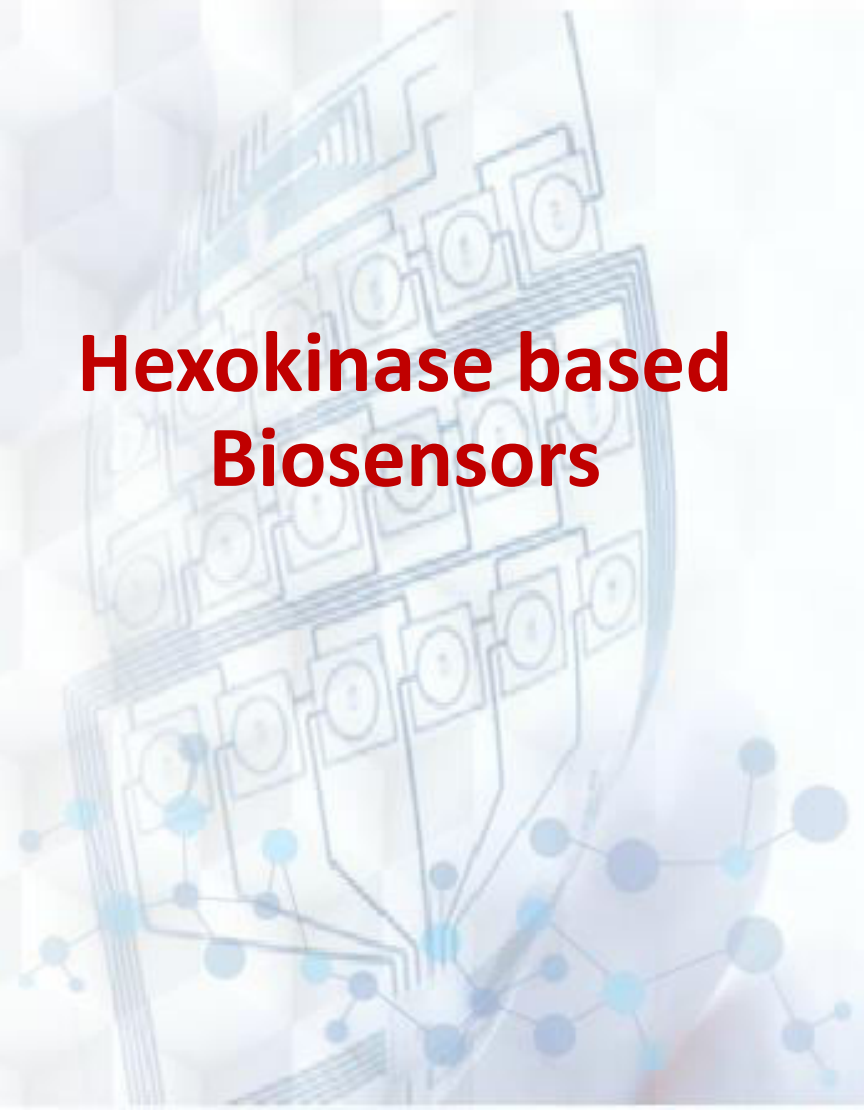
Nanoparticles in Enzyme Biosensors

Summary

1. The unique properties of nanoparticles are extremely suitable for designing biosensors.
2. For specific purposes surface modification is required.
3. One step modification by using long alkyl tail and hydrophilic ligand head for multiple functionality of NP.

Biosensors

Hexokinase based Biosensors

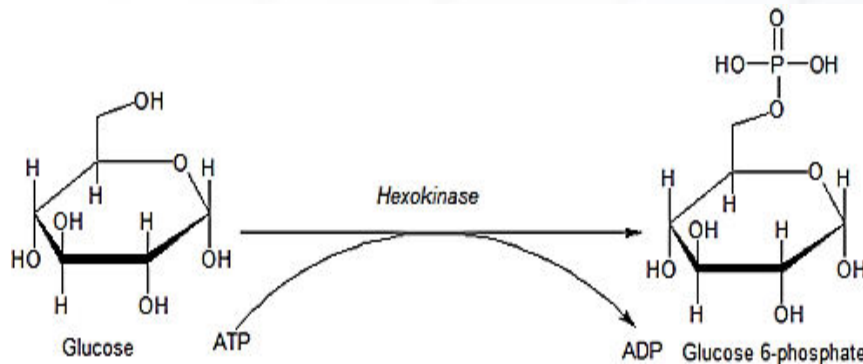


Hexokinase Based Biosensors

Hexokinase Enzyme.

1. It is a Metabolic Enzyme
2. It acts on all hexoses
3. Four mammalian types
4. MW 50 KD-100 KD
5. It requires ATP
6. It produces ADP
7. HK D is Glucokinase
8. Glucokinase has high K_m value and has low affinity for glucose.

Hexokinase Based Biosensors



<https://upload.wikimedia.org/wikipedia/commons/1/10/Hexokinase-glucose.png>

Hexokinase Action

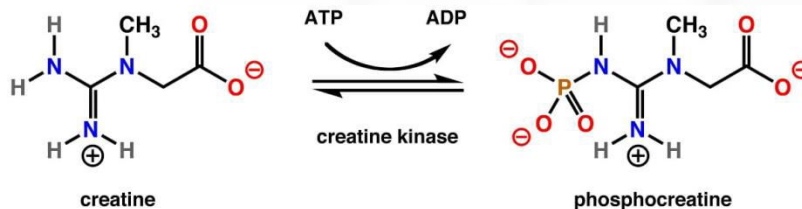
1. Hexokinase catalyzes the **phosphorylation** of hexoses and produced **hexose phosphate**.
2. Glucose kinase is produced by **liver, pancreas, hypothalamus, small intestine**.
3. Its **deficiency** is a genetic **autosomal recessive disease** that causes **Chronic Haemolytic Anaemia**.

Hexokinase Based Biosensors

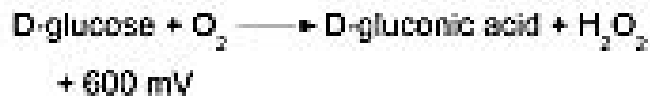
Hexokinase Based Biosensors.

1. Hexokinase enzyme is immobilized on the electrodes or the conductometric support membrane.
2. HK biosensors are **quick to response, simple, and low cost.**
3. Mainly biosensor are used for the determination of ATP, in the samples.

Hexokinase Based Biosensors



Glucose oxidase



Hexokinase



Creatine kinase



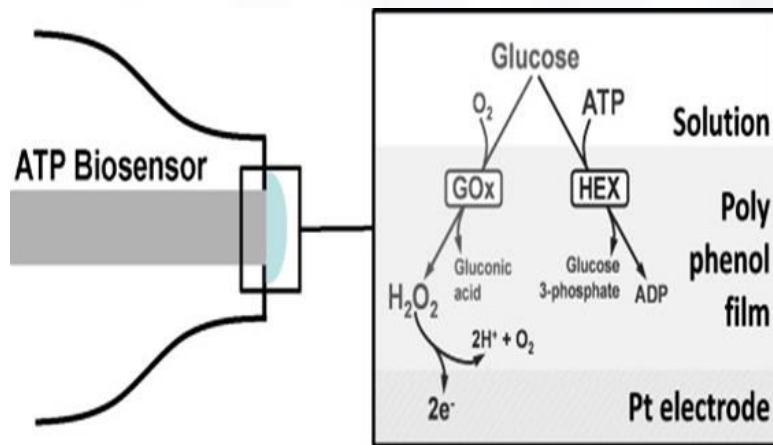
Creatine Kinase.

1. Creatine kinase is an important enzyme of **muscle cells**.
2. A biosensor for rapid and convenient determination of **CK activity by measuring the rate of ATP production** by this enzyme.
3. GO was also used in the biosensor to determine the glucose concentration in the sample.

<https://biologydictionary.net/wp-content/uploads/2018/05/Creatine-kinase.jpg>

<https://ars.els-cdn.com/content/image/1-s2.0-S0039914015301223-fx1.jpg>

Hexokinase Based Biosensors

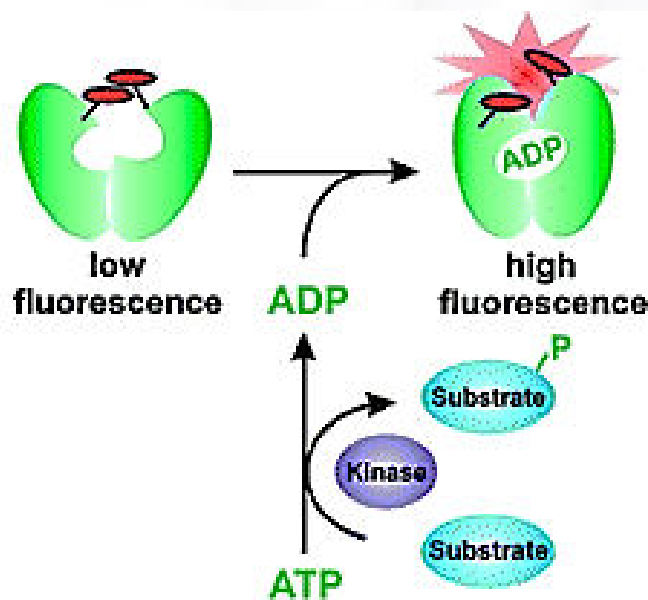


ATP determination in tissues/ Cells.

ATP is consumed during metabolic processes, by different enzymes. The level of enzymatic reactions could be determined indirectly.

1. ATP biosensor was designed for the detection in **ileum, colon**.
2. An ATP biosensor is designed to determine **tyrosine phosphatase** in osteoblastic cells.

Hexokinase Based Biosensors



Hexokinase Based Fluorescent Biosensor

1. Fluorescent method for the determination of ADP.
2. A fluorophore compound **diethylamino-coumarin** and genetic engineering techniques were used for the development of a biosensor for ADP.
3. Biosensor shows high fluorescent binding with ADP.

https://pubs.acs.org/na101/home/literatum/publisher/achs/journals/content/acbcct/2010/acbcct.2010.5.issue-4/cb9003173/production/images/medium/cb-2009-003173_0006.gif

Hexokinase Based Biosensors

Summary.

1. Hexokinase is a metabolic enzyme.
2. The enzyme converts ATP into ADP.
3. Biosensors are designed to detect the ATP in samples.
4. GO is used with HK to detect the CK enzyme activity in samples.
5. A fluorescent based HK biosensor for direct estimation of ADP.

Biosensors

Cholesterol Oxidase Biosensors



Cholesterol Biosensors

Cholesterol

1. Cholesterol is a **lipid** and synthesized by the animal involving **37 steps**.
2. **Serum cholesterol** level is required for the **diagnosis and treatment** of **heart disease, diabetes and liver diseases**.
3. **The Liebermann–Burchard** or acetic anhydride test is used for the detection of cholesterol, in the presence of sulfuric acid.

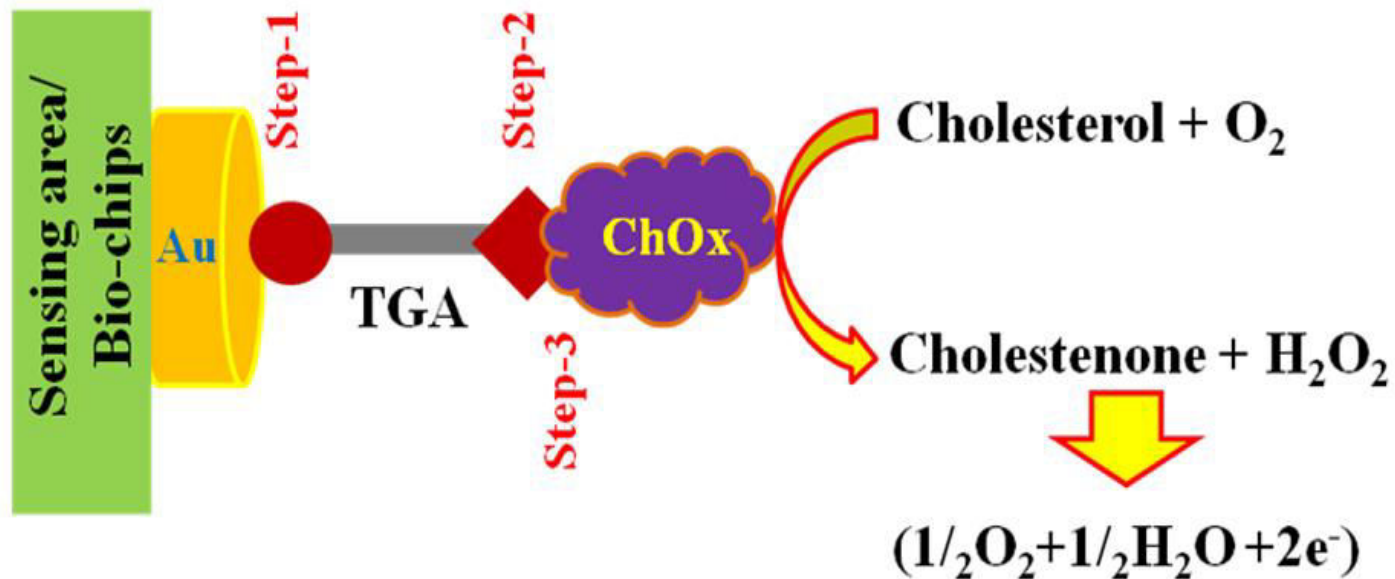
Cholesterol Biosensors

Enzymes

1. **Cholesterol esterase**
(ChoE) or bile-salt activated lipase or sterol esterase catalyzes the hydrolytic cleavage of cholesterol.
2. **Cholesterol oxidase**
(CH₂Ox), a FAD-dependent enzyme of the oxido-reductase family catalyzes the oxidation of cholesterol to cholestenone.

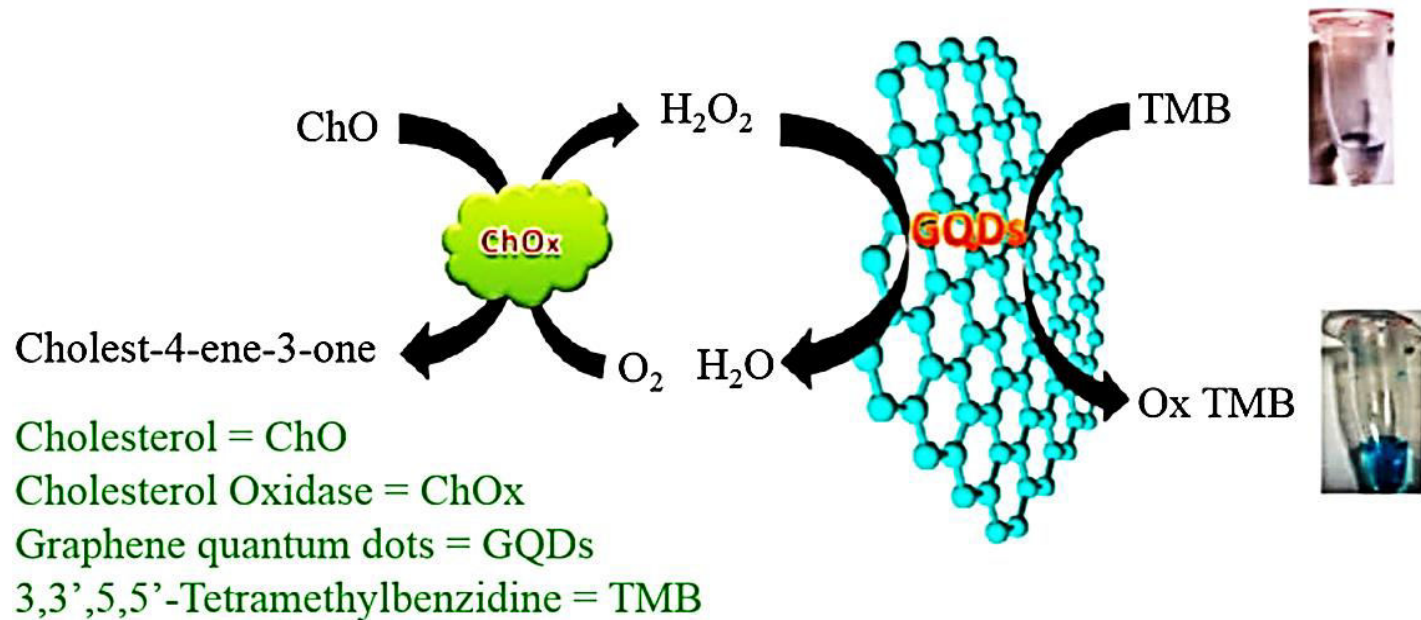
Cholesterol Biosensors

Fabrication of an amperometric cholesterol biosensor onto smart micro-chips based on **cholesterol oxidase (ChOx)** co-immobilized **thioglycolic acid self-assembled monolayer (TGA-SAM)**.



Cholesterol Biosensors

Biosensor based on Cholesterol Oxidase and Cholesterol Estrase TMB and GQDs Pt Nano dots



Wei, SC., Lin, YW. & Chang, HT. J. Anal. Test. (2019) 3: 191. <https://doi.org/10.1007/s41664-019-00107-x>

Cholesterol Biosensors

Summary

1. Cholesterol level is an important parameter in the diagnosis and treatment of several diseases.
2. Cholesterol oxidase is employed in Biosensors for the estimation of serum level of cholesterol.
3. A number of strategies are used to fabricate efficient biosensor for estimation cholesterol level.

Biosensors

76. Glucose Oxidase Biosensors



Glucose Oxidase Biosensors

Introduction.

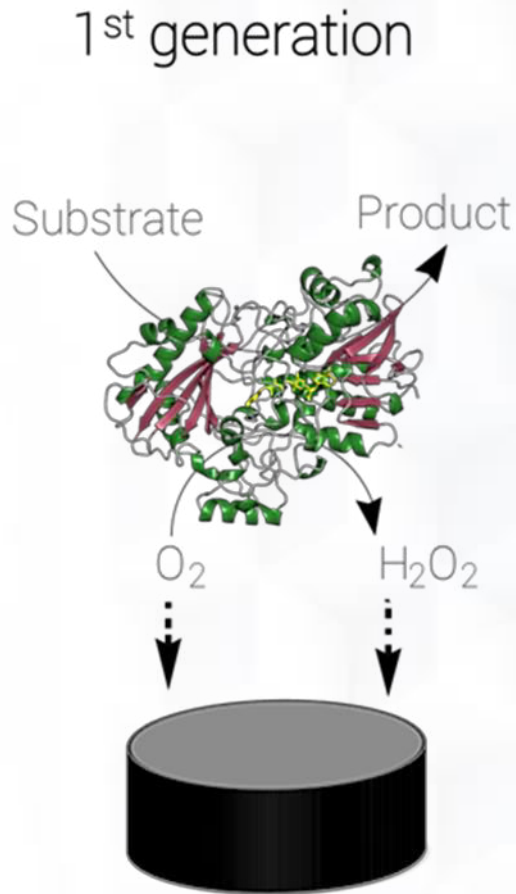
First ever Glucose Biosensor was composed of an

1. A platinum electrode,
2. An Oxygen permeable membrane
3. A thin layer of Gox enzyme
4. Outer dialysis membrane.

Principle

It was based on the principle of **measurement of Oxygen levels** during the enzymatic reaction.

Glucose Oxidase Biosensors



First Generation Biosensors

In first-generation biosensors work on the following principle.

1. The electrons generated by enzymatic REDOX reaction, are transferred to molecular oxygen.
2. The electrode was dipped into the biological sample.
3. The dissolved Oxygen was utilized for the production of hydrogen peroxide.

Problems were associated.

Glucose Oxidase Biosensors

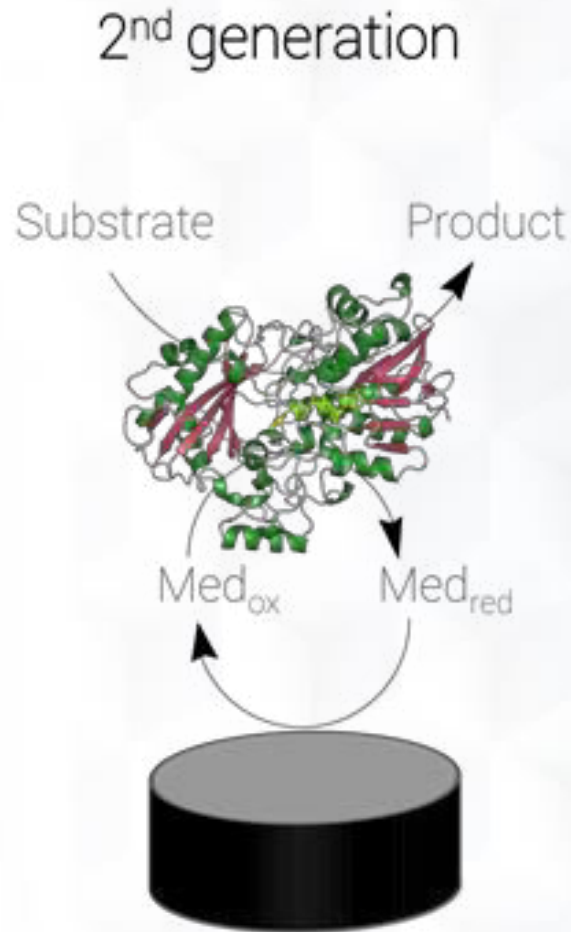
Problems

Interference was observed by **ascorbic acid**, **uric acid** and certain **drugs**.

For example Ascorbic Acid gets oxidized at the electrode and gives false high readings.

Oxygen deficit. The method is based on the availability of dissolved **oxygen** in the biological fluids. It could be affected by certain factors lead to **fluctuations** in the Oxygen tension.

Glucose Oxidase Biosensors

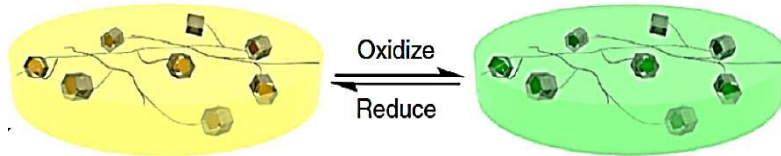


Second Generation

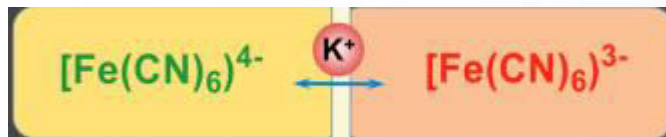
The second generation Biosensors are called mediated glucose Biosensors. Introduced as pen size Biosensor in 1987.

1. A reduced mediator is formed by the action of enzyme.
2. Reoxidation at electrode.
3. Signals are detected for a change in the chemistry of mediator.

Glucose Oxidase Biosensors



Ferrocene



Ferricyanide.



<https://ars.els-cdn.com/content/image/1-s2.0-S2211285517306614-fx1.jpg>

Mediators

A variety of mediators were used, to improve the sensors performance.

1. Ferrocene
2. Ferricyanide

Are the iron based complexes.

1. Methylene blue is an organic dye and used as REDOX indicator.

- <http://chembites.org/wp-content/uploads/2018/05/fig2.png><https://i.ytimg.com/vi/h-UgUybxyQ/maxresdefault.jpg>

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcS4B_E1Z-hUWTkvXk0NG913dTmJ6qxa1jIjv-lAxSuPEGrGGqFC&s

Glucose Oxidase Biosensors

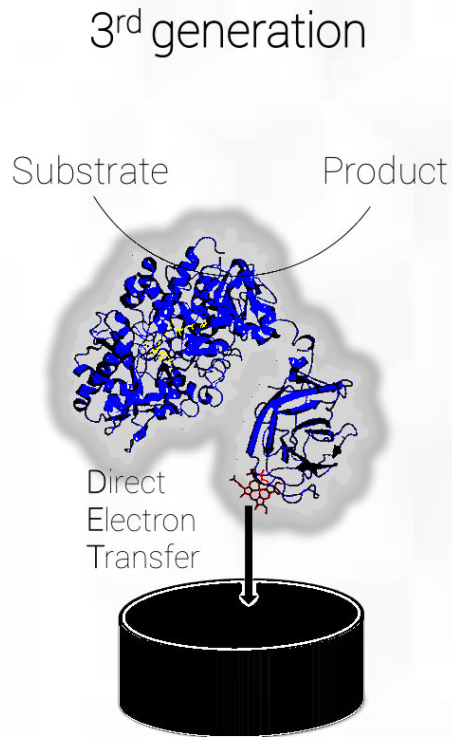
Qualities of Ferrocenes

A Ferrocenes fits all the criteria for good mediator.

1. **Not reacting** with Oxygen
2. **Independent of pH**
3. **Stable in oxidized and reduced form**
4. Shows **reversible electron transfer Kinetics**
5. **Rapid reaction with GOx**

Ferrocenes are used for e shuttling between **Gox** and **GDH-PQQ** and the **electrodes**.

Glucose Oxidase Biosensors



<http://www.directsens.com/wp-content/uploads/2015/12/3rd-gen.png>

Third generation

In third-generation biosensors, the electrons are transferred directly from the enzyme to the electrode without any intermediate stages or use of nanoparticles.

1. Reagent less
2. Mediator less
3. Direct transfer between enzyme and the electrode

Glucose Oxidase Biosensors

Latest Devices

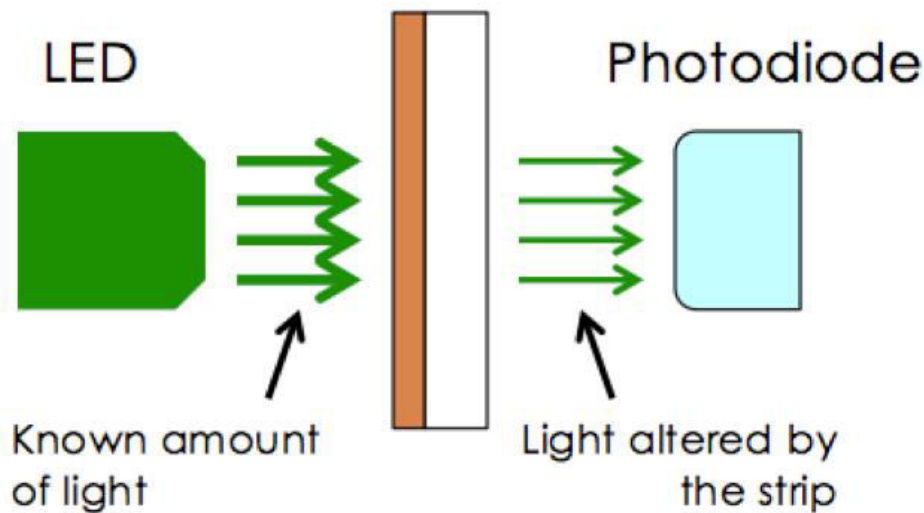
Two types of latest devices are introduced.

Continuous Monitoring glucose system (CGMS)

Needle type enzyme electrode for Subcutaneous for subcutaneous implantation are used for CGMS.

Non-Invasive Glucose Monitoring system

Optical or transdermal approaches are used for non invasive GMS.



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQNBoQtVpXAujUzbKPN3It9cly39tVMyTUTlhN31WbFCdgEtEWe&s>

Glucose Oxidase Biosensors

Summary

1. Glucose oxidase based Biosensors are mostly used for estimating the blood glucose levels.
2. There are at least three generations of the Glucometer, with improved functioning.
3. Latest devices include continuous system for BGL monitoring and non invasive optical devices used with smart phone.

Biosensors

77. Lactate Oxidase Biosensors



Lactate Oxidase Biosensors

Introduction

1. The enzyme lactate oxidase catalyzes the **oxidation of L-lactate** to **pyruvate** in the presence of dissolved oxygen.
2. It forms hydrogen peroxide.
3. Hydrogen peroxide is electrochemically active and can be either reduced or oxidized to give a current proportional to the L-lactate concentration.

Lactate Oxidase Biosensors



Lactate Oxidase Sources

LOD is a globular flavoprotein can be obtained from a variety of bacterial sources, such as

1. *Pediococcus*
2. *Aerococcus*
3. *Mycobacterium*

Lactate detection and quantification plays a vital role in healthcare, especially in conditions Hemorrhage , Sepsis, Tissue hypoxia including Respiratory failure.

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcS-wFTPpmlIdMeEg0p7pvcQgKrjmG1RflaOmjnaT9chul_mFG80cWA

Lactate Oxidase Biosensors

Significance

Lactate level is widely used in clinical diagnostics for

1. Continuous surveillance in surgery
2. Sports medicine
3. Shock
4. Trauma
5. Food industry.

The baseline lactate level in blood ranges **from 0.5 to 1.5 mmol/l at rest.**

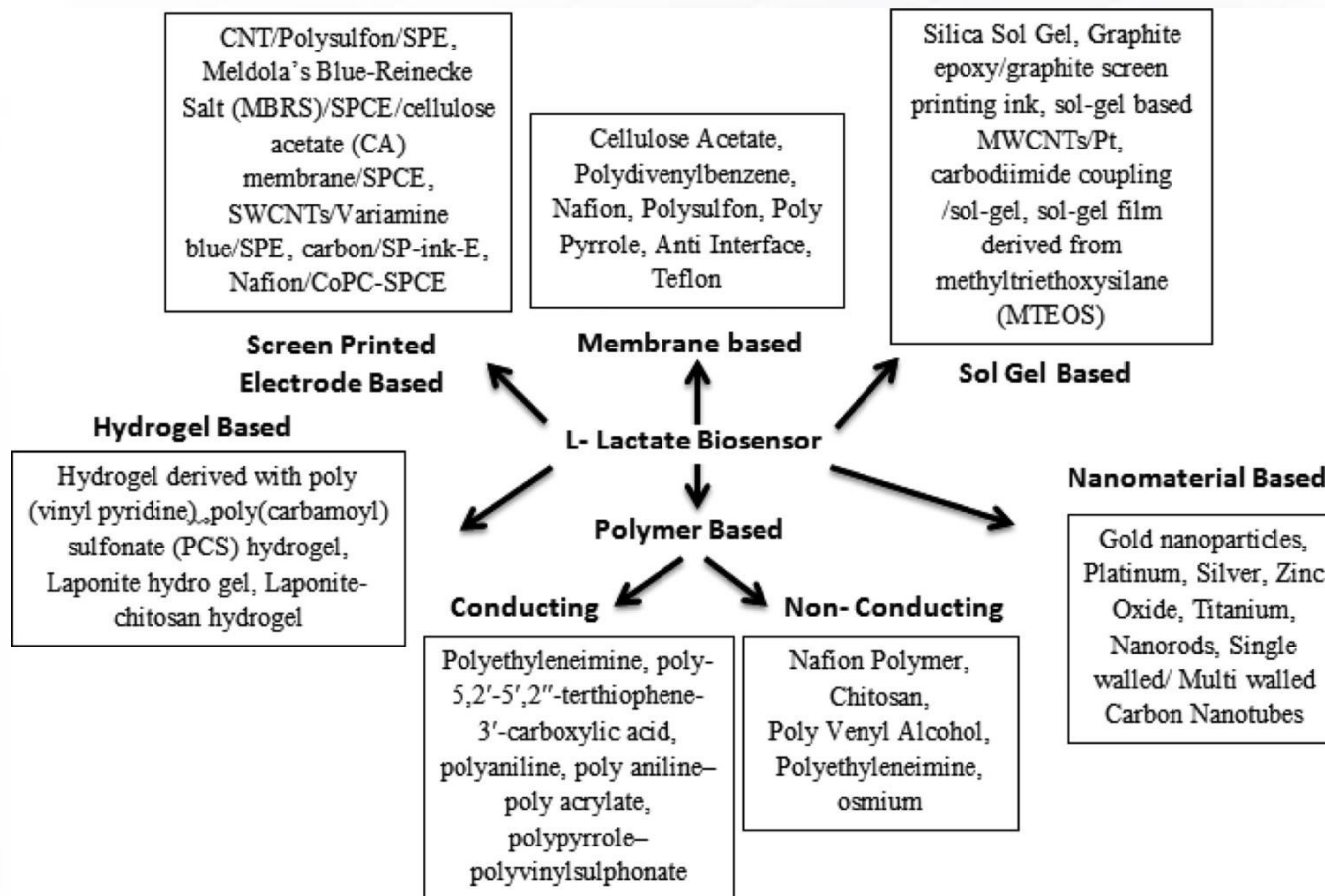
Lactate Oxidase Biosensors

Lab method for the estimation

In clinical practice blood plasma levels are estimated by chemical method.

1. Lactate oxidase catalyzes the oxidation of lactic acid to pyruvate and hydrogen peroxide.
2. It catalyzes the reaction with a hydrogen donor, in the presence of 4-aminophenazone, to form a coloured complex.

Lactate Oxidase Biosensors



Lactate Oxidase Biosensors



<https://www.ekfdiagnostics.com/res/lactate%20scout%204%20-%20Lactate%20testing%20for%20athletes%20-%20steps-2>

Lactate Oxidase Biosensor

1. It utilizes lactate oxidase and has applications in sports medicine.
2. It is used to monitor lactate production during exercise. Results in just 10 seconds.
3. Pocket-sized.
4. It is simple, accurate, reliable lactate analyzer for sports performance monitoring.

Lactate Oxidase Biosensors

Summary

Lactate Oxidase is an enzyme which is predominantly used in the detection of Lactate.

Estimation of lactate is important in health and diagnostics.

Applications of Lactate Biosensors are found in Health and sports medicine.

Recently a LO biosensor is introduced commercially for sportsmen.

Biosensors

78. Alcohol Oxidase Bopsensors



Alcohol Oxidase Biosensor

Introduction

Alcohol oxidase is responsible for the **oxidation of low molecular weight alcohols** to the corresponding **aldehyde or ketone**.

It uses **molecular oxygen** as the electron acceptor.

1. It is a **FAD dependent enzyme**.
2. Assay methods rely on **consumption of Oxygen** or **production of Hydrogen peroxide**.

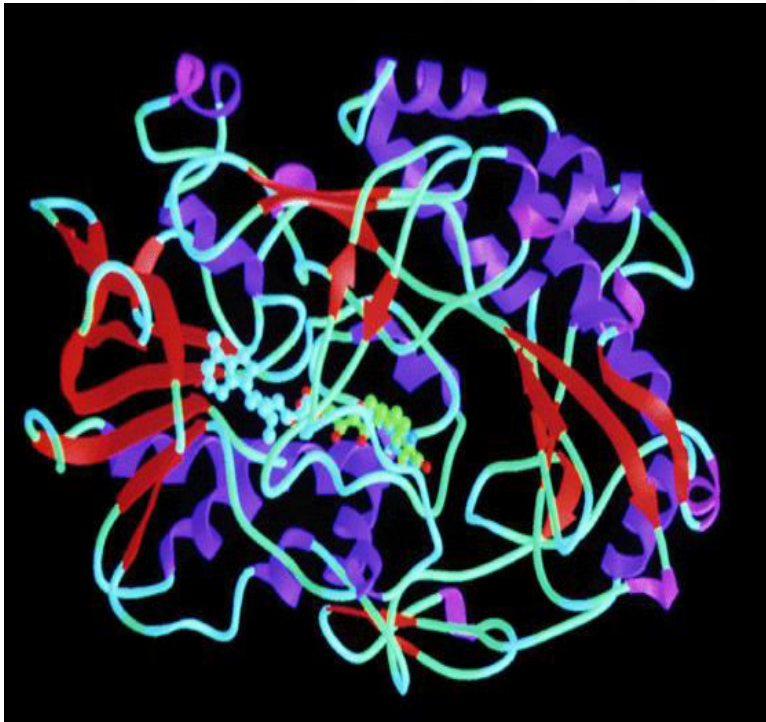
Alcohol Oxidase Biosensor

Types

Based on substrate specificity, alcohol oxidases may be categorized broadly into four different groups namely,

- 1. Short chain alcohol oxidase (SCAO)**
- 2. Long chain alcohol oxidase (LCAO)**
- 3. Aromatic alcohol oxidase (AAO) and**
- 4. Secondary alcohol oxidase (SAO).**

Alcohol Oxidase Biosensor



Ribbon representation of the 3D model of *H. polymorpha* AO monomer . α -helices are presented in purple, β -strands- in red. The ball-and stick model represents FAD

Alcohol Oxidase

1. Alcohol oxidase (AO) is the key enzyme of **methanol metabolism**.
2. AO is a molecule of high molecular mass (**600 kDa**)
3. It consists of **eight identical subunits**.
4. Each subunit carry **one non-covalently bound flavin adenine nucleotide (FAD) molecule as the prosthetic group**.

Alcohol Oxidase Biosensor

Ethanol Biosensor

1. Alcohol is produced during **metabolic pathways**.
2. Alcohol is an important **industrial product**.
3. **Health and diagnostics, Food industry, Medical science, Agriculture and Environment.**
4. Ethanol biosensors based on **alcohol oxidase** have been reported to be **stable and practical**.

Alcohol Oxidase Biosensor

Ethanol Biosensor

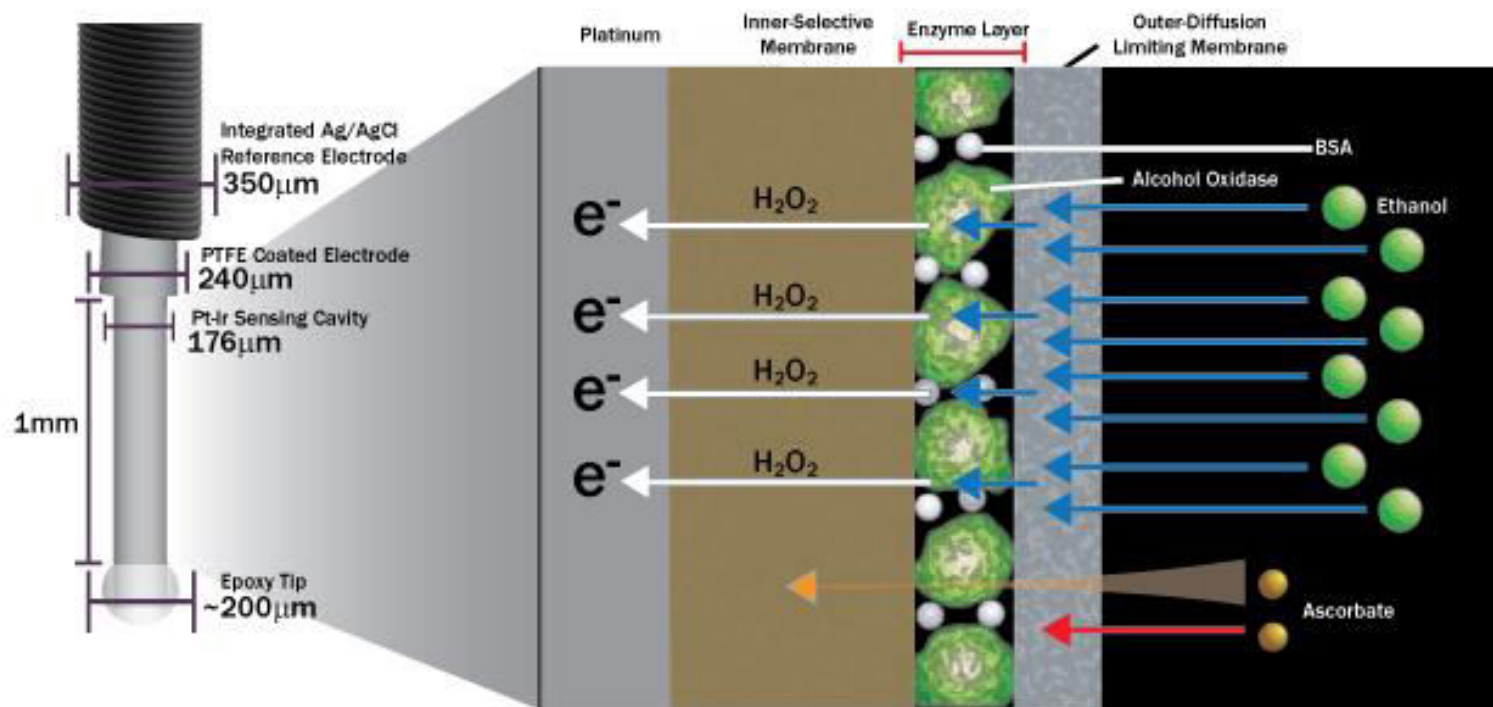
The enzyme-mediated processing of the analyte of interest.

This results in the production of hydrogen peroxide that is then detected by oxidation at a Pt-Ir electrode.

Electroactive interferents present in the brain are excluded via a passive selective membrane and through active removal when necessary.

Alcohol Oxidase Biosensor

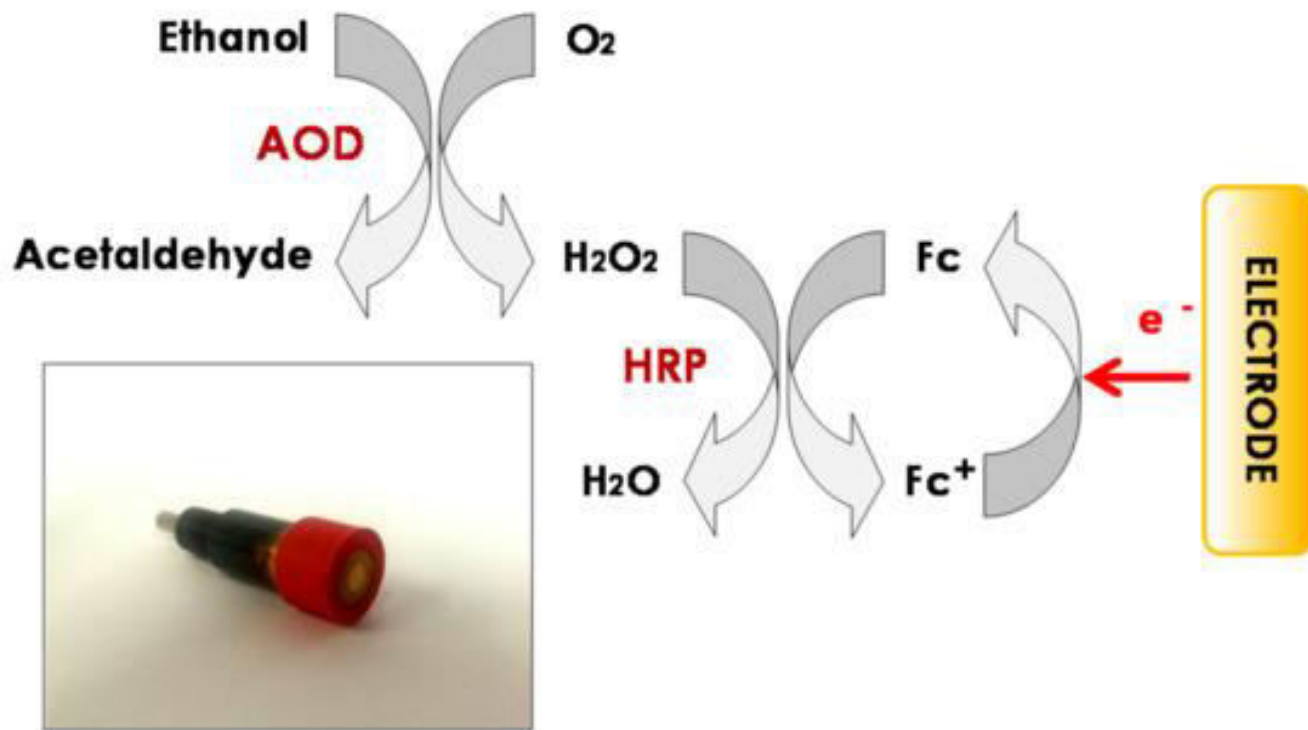
Alcohol Oxidase Biosensor for *invivo* applications



<https://www.pinnaclet.com/images/biosensors/ethanolBiosensorDiagram.jpg>

Alcohol Oxidase Biosensor

Automated Bioanalyzer Based on Amperometric Enzymatic Biosensors for the Determination of Ethanol



https://www.mdpi.com/beverages/beverages-03-00022/article_deploy/html/images/beverages-03-00022-g001-550.jpg

Alcohol Oxidase Biosensor

Summary

1. Alcohol oxidase is an FAD dependent important enzymes.
2. Alcohol oxidase biosensors are used for the detection of alcohol in diagnostics and industries, to control the process technology.
3. Sufficeticated biosensors are designed to invivo applications, such as to monitor brain activity.

Biosensors

79. Xanthine oxidase Biosensors



Xanthine Oxidase Biosensor

Xanthine Oxidase

Xanthine and **Hypoxanthine** are the substrates for Xanthine oxidase.

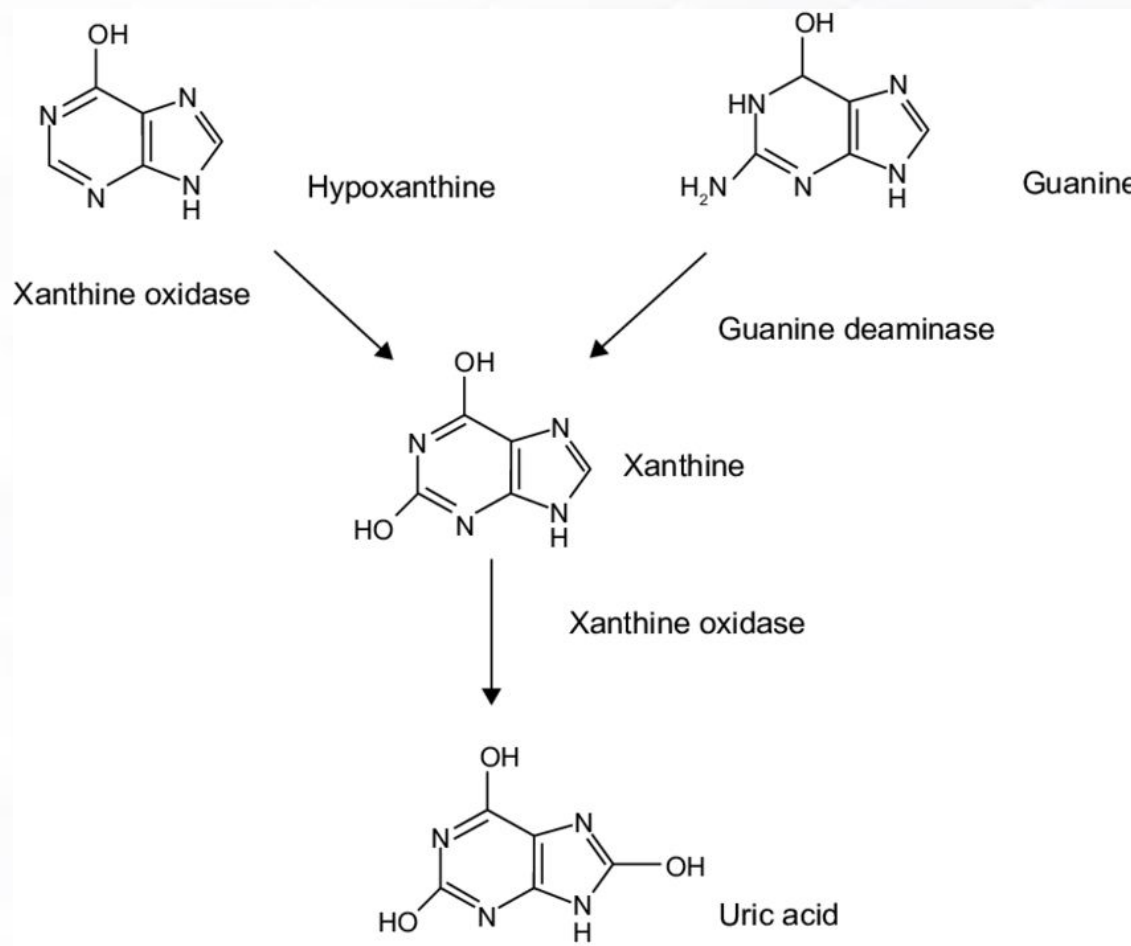
1. XO requires **two Flavin adenine di nucleotide** molecules for its activity.
2. It has **two sites** for binding with **Molybdenum ions**.
3. XO can bind with **eight FeS** clusters.
4. These metallic ions and co factor and take part in the reaction

Xanthine Oxidase Biosensor



Crystallographic structure (monomer) of bovine xanthine oxidase,. **The bounded FAD (red), FeS-cluster (orange), with molybdenum (yellow) are indicated.**

Xanthine Oxidase Biosensor



<https://www.researchgate.net/publication/281680962/figure/fig1/AS:613912655437859@1523379591938/Uric-acid-formation.png>

Xanthine Oxidase Biosensor

Clinical Significance

Xanthine oxidase is a **superoxide -producing** enzyme.

1. It is found normally in **serum and the lungs**, and its activity is increased **during influenza A infection**.
2. During severe **liver damage**, **xanthine oxidase** is released into the blood.
3. Blood assay for XO is a way to determine if **liver damage** has happened.

Xanthine Oxidase Biosensor

Xanthine Oxidase Biosensor

XO is a source of reactive oxygen species which can **amplify and potentiate inflammation** and **delay wound healing**.

1. For **tracking the healing progress**, a Xanthine oxidase biosensor is fabricated.
2. Inhibition of XO holds a potential for **improving wound healings**.

Xanthine Oxidase Biosensor



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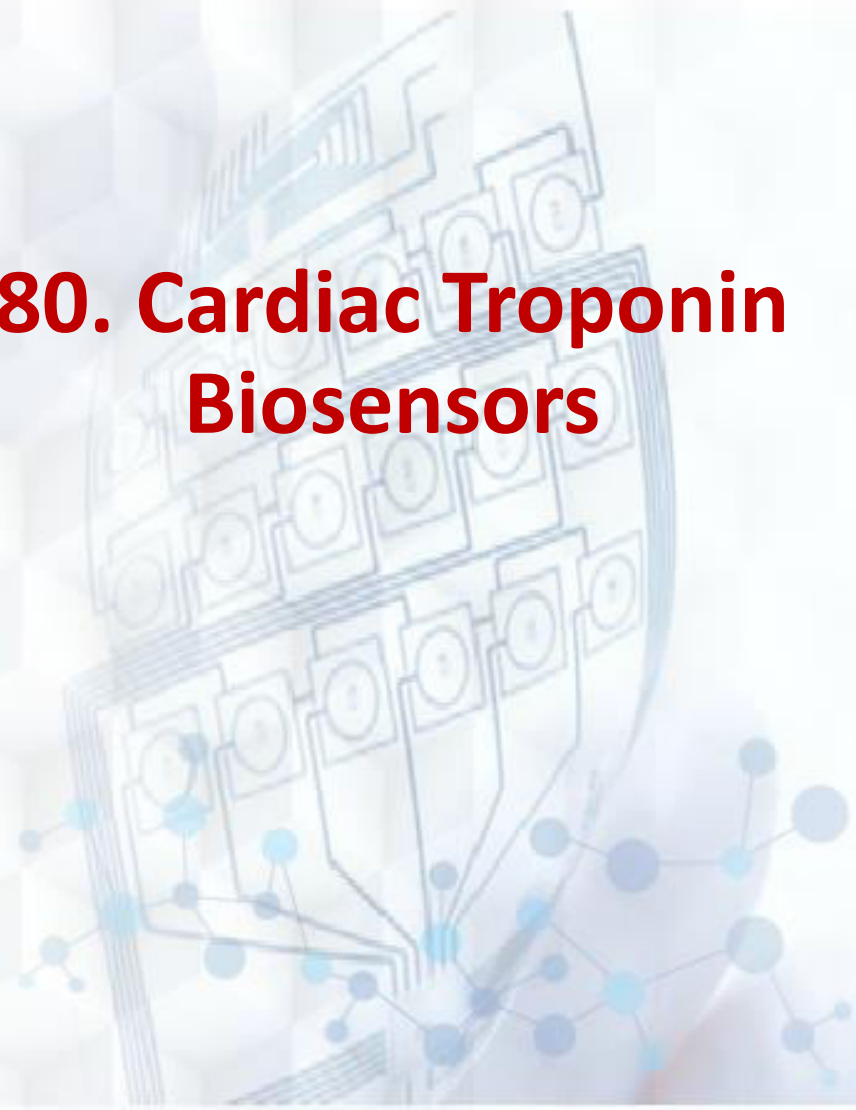
Xanthine Oxidase Biosensor

Summary

1. Xanthine oxidase is a REDOX enzyme.
2. Biosensors based on the enzyme are of great importance.
3. The enzyme biosensor can monitor the wound healing.
4. Xanthine oxidase Biosensors are also designed to monitor the fish and meat freshness.

Biosensors

80. Cardiac Troponin Biosensors



Cardiac Troponin Biosensors

Troponin Complex

Troponin, or the troponin complex, is a complex of three regulatory proteins

1. Troponin C
2. Troponin I
3. Troponin T

It is integral to muscle contraction in muscle.

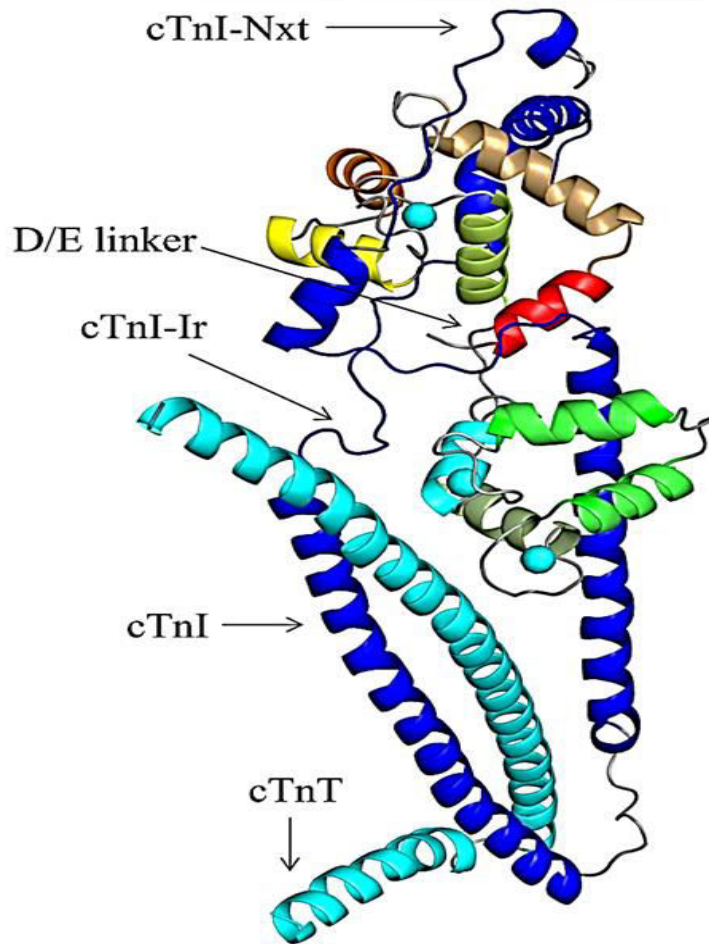
cTnI and cTnT are used as **diagnostic** and **prognostic indicators** in the management of AMI.

Cardiac Troponin Biosensors

Troponin Complex Function

1. **Troponin C** binds to calcium ions to produce a conformational change in TnI.
2. **Troponin T** binds to tropomyosin, interlocking them to form a troponin-tropomyosin complex
3. **Troponin I** binds to actin in thin myofilaments to hold the troponin-tropomyosin complex in place.

Cardiac Troponin Biosensors



Troponin Complex Function

1. cTnT and cTnI are colored cyan(blue green) and **blue** respectively.
2. The cTnC helices N, A, B, C, D in the N-domain are colored, red, mustard, orange, yellow and lime green.
3. The bound Ca^{2+} ions at sites 2, 3, and 4 are rendered as spheres and colored cyan.

Cardiac Troponin Biosensors

Troponin I

Cardiac Troponin I (cTnI) is a subtype of the **troponin family**, a marker for myocardial damage.

1. Cardiac troponin I is a very **sensitive** and **specific indicator** of myocardium **damage**.
2. Serum levels can be used to help differentiate between unstable angina and myocardial infarction.

Cardiac Troponin Biosensors

Earliest Cardiac Markers

The earliest cardiac biochemical marker employed includes

- 1. Lactate dehydrogenase isoenzymes**
- 2. Aspartate aminotransferase**
- 3. Lactate dehydrogenase.**
- 4. ECG (only 57% diagnosis)**

However, none of enzyme display high specificity to AMI.

Cardiac Troponin Biosensors

Troponin

1. Human cTnI single isoform.
2. is made up of **209 amino acid** residues,
3. It has a molecular weight of roughly 22.5 kDa.
4. It can **inhibit** the **actomyosin ATPase**
5. The function can be **enhanced** in the presence of tropomyosin.
6. It has also been produced by genetic engineering techniques.

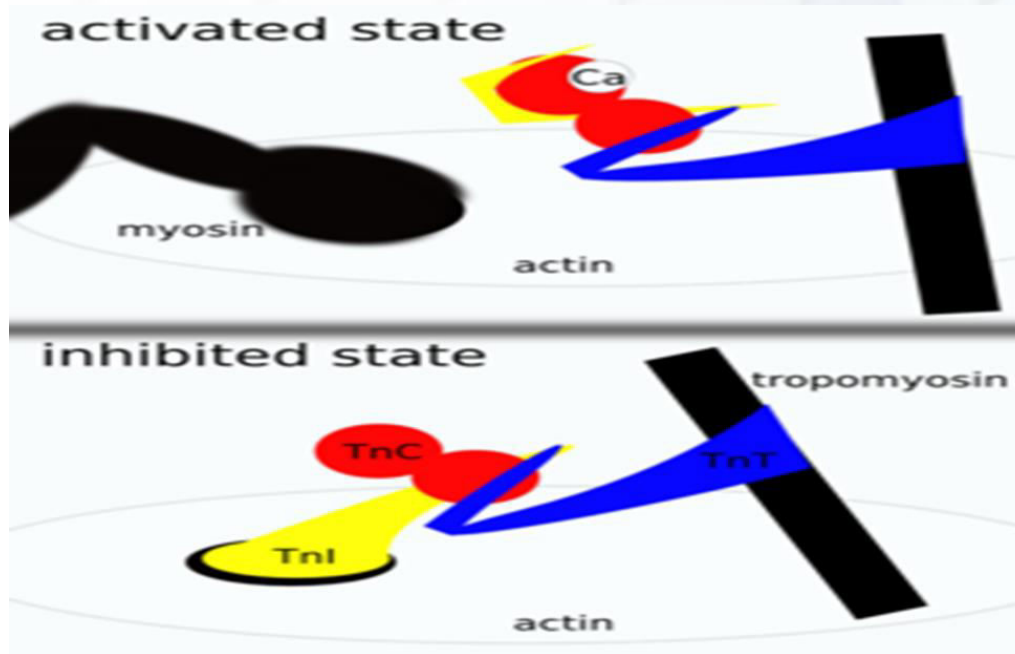
Cardiac Troponin Biosensors

Significance of cTnI

The isoform of cTnI in myocardial tissue differs from the sequence of those in skeletal muscles.

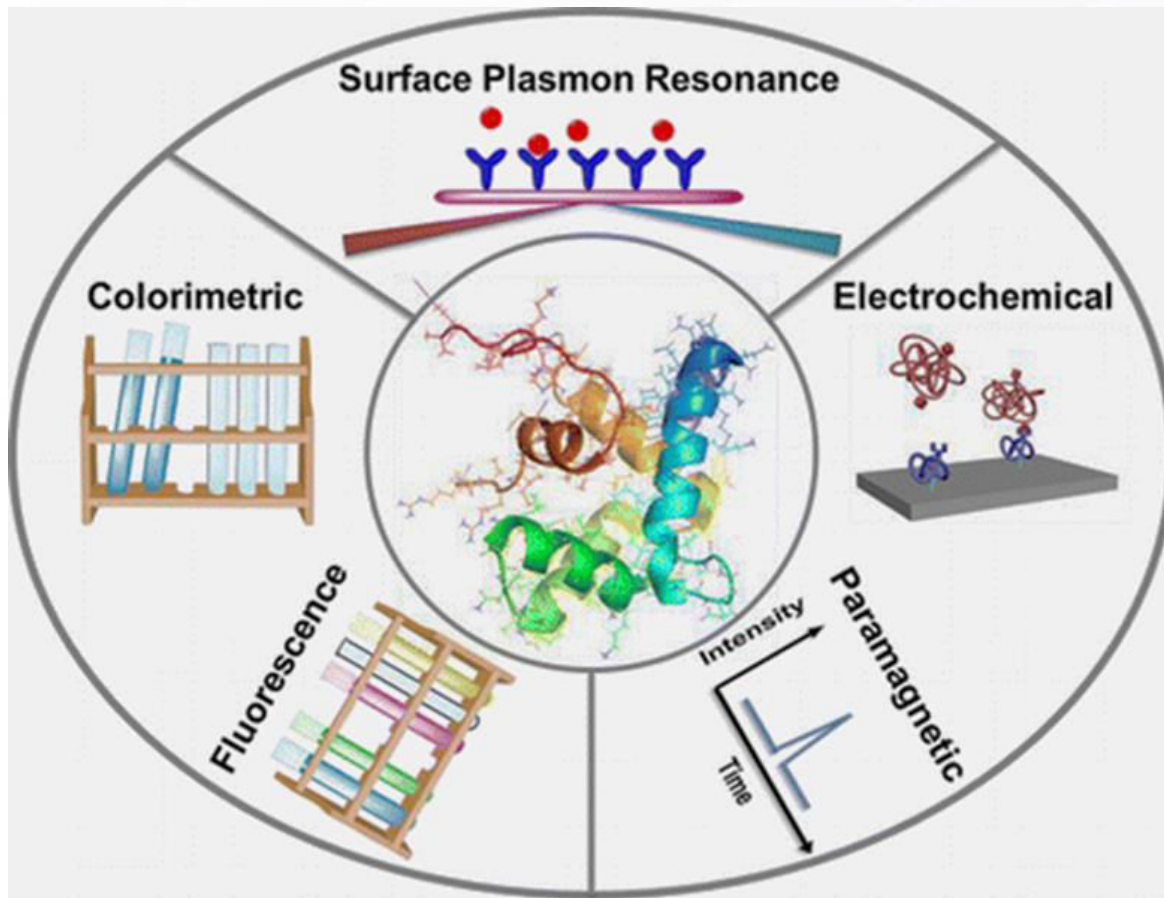
1. The cTnI conc. can go up to **50 ng/mL** within **3–6 h**, and finally to a level around **550 ng/mL**.
2. The cTnI level will go back to its standard status in **6–8 days**, such that the diagnostic window is long enough for detection.

Cardiac Troponin Biosensors



Troponin activation. Troponin C (red) binds Ca^{2+} , which stabilizes the activated state, where troponin I (yellow) is no longer bound to actin. Troponin T (blue) anchors the complex on tropomyosin

Cardiac Troponin Biosensors



https://pubs.acs.org/na101/home/literatum/publisher/achs/journals/content/ascefcj/2016/ascefcj.2016.1.issue-2/acssensors.5b00318/20160222/images/medium/se-2015-003187_0006.gif

Cardiac Troponin Biosensors

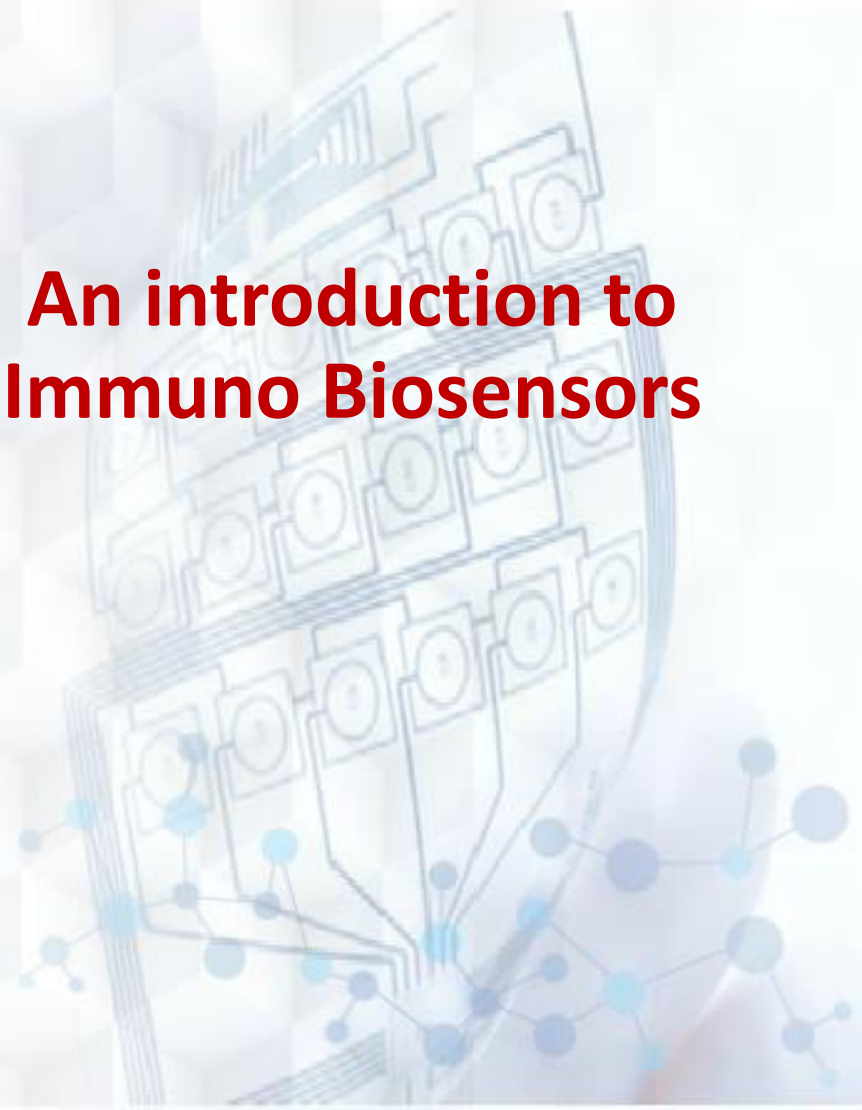
Summary

1. Troponin is a marker protein for cardiac disease.
2. Troponin levels are tested in blood for the detection of damage to heart tissues.
3. Laboratory tests are based on immunogenic properties of the protein.
4. Biosensors are also designed using monoclonal antibodies.

Prepared By: AARISH ALI

Biosensors

An introduction to Immuno Biosensors



An introduction to Immuno Biosensors

Introduction

1. The biochemical detection systems that use **antibodies** or **antigens** as the recognition element, known as immunosensors .
2. The **basis** of such **detection** is the **complex formation** between antibody and antigen.
3. Sometimes it is not required to use the whole antigen molecule, therefore AB fragments are used.

An introduction to Immuno Biosensors

Structure of an immunobiosensor

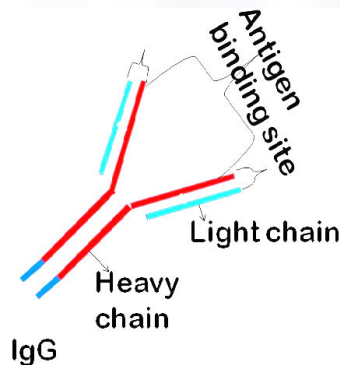
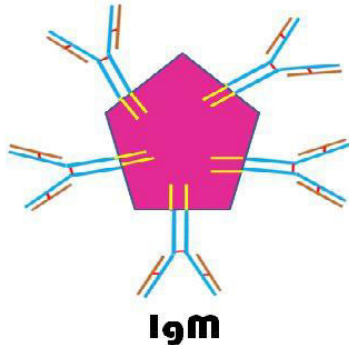
A typical immune biosensor is composed of two main components.

1. Biological Component
2. Physical Component

The **biological component** is composed of an immunogenic molecule, that is responsible for the detection of target molecule or analyte by binding with it.

The **physical component** detect the biorecognition signals.

An introduction to Immuno Biosensors



<https://biodifferences.com/wp-content/uploads/2017/03/igG-final.jpg>

Types of Antibody

There are five kinds of antibodies.

1. **IgM is the first antibody** to fight against, and attack.
2. IgM is a *pentamer* unit **with ten antigen binding sites**, but only five are functional.
3. **IgG reacts** later but helps in the permanent eradication of antigens.
4. IgG has **two high affinity binding sites**.

An introduction to Immuno Biosensors

Antibody Affinity

Antibody affinity is the strength with which the epitope binds to an individual paratope (antigen-binding site) on the **antibody**.

High **affinity antibodies** has **following properties**.

1. They bind quickly to the antigen.
2. They permit greater sensitivity in assays
3. They maintains this bond more readily under difficult conditions.

An introduction to Immuno Biosensors

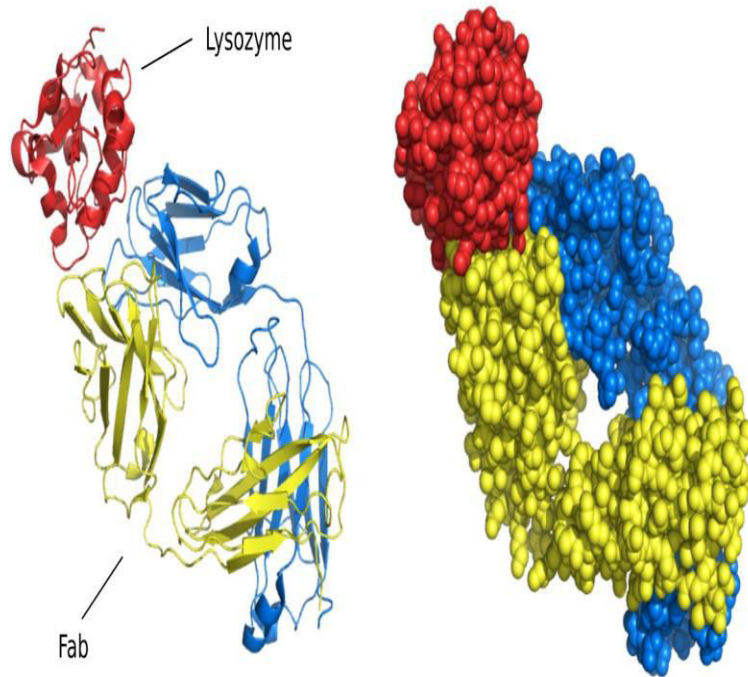
Forces involved in Binding

The formation of Ab-Ag complex is dependent on certain forces. All these forces are weak physical forces.

1. Hydrophobic forces
2. Hydrogen bonding
3. Salt bridge
4. Van der Waal forces
5. Electrostatic interactions

These forces could be altered by change in pH, salt conc.

An introduction to Immuno Biosensors

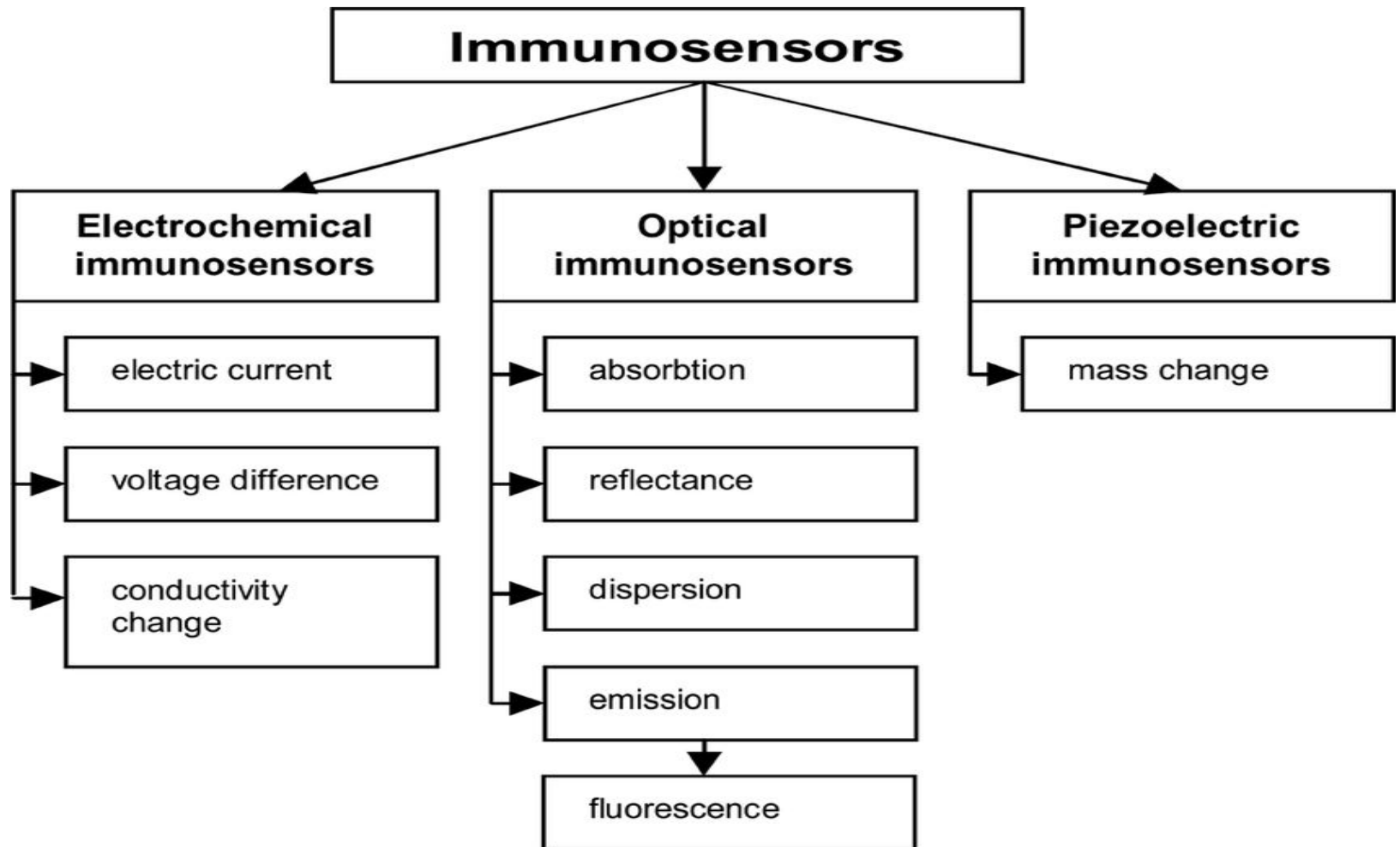


<https://www.creative-biostructure.com/images/Crystallization-of-Antibody-Antigen-Complexes.jpg>

Strength of complex

1. The strength of the binding of the **antigen-antibody complex** could be characterized by its **affinity constant** that is K .
2. The value of K in the order between 5×10^4 – 1×10^{12} per mole liter, is expected to be for high affinity antigen-antibody complex.

An introduction to Immuno Biosensors



An introduction to Immuno Biosensors

Summary

1. The strength of antigen-antibody complex is based on weak physical forces, that could be effected by pH, ionic strength and temperature.
2. The formation of **high affinity antigen antibody complex** is a requirement for a good quality sensor.
3. The Ab-Ag complex is detected by using a number of techniques.

Biosensors

ELISA



Enzyme Linked Immunosorbent Assays

ELISA

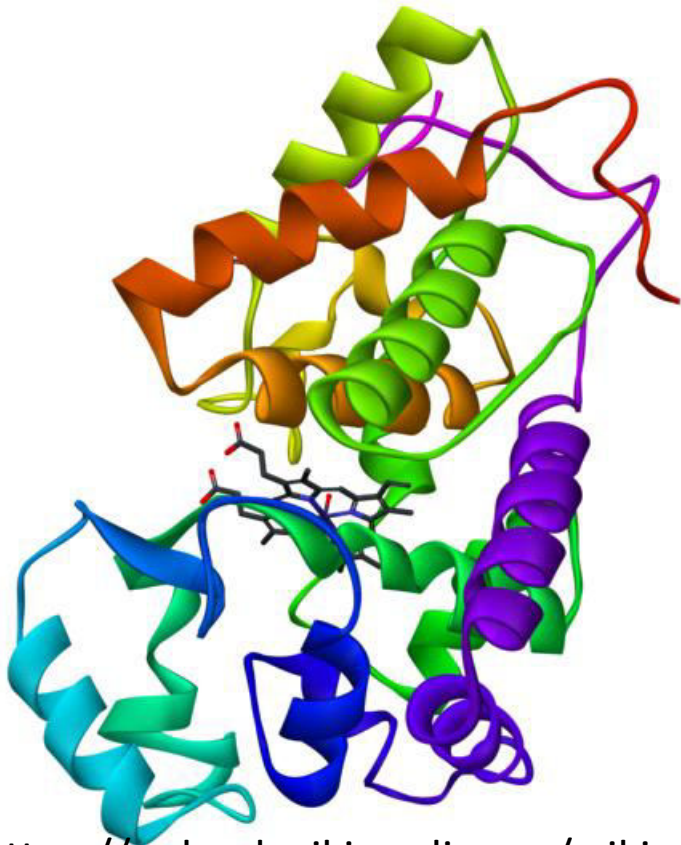
1. E Enzyme
2. L Linked
3. I Immuno
4. S Sorbent
5. A Assay

It is also known as enzyme immunoassay (EIA).

Two types of Biological molecules are involved.

- i. Antibody
- ii. Enzyme

Enzyme Linked Immunosorbent Assays



<https://upload.wikimedia.org/wikipedia/commons/thumb/1/10/HRP-xray.png/440px-HRP-xray.png>

Horse Reddish peroxidase

HRP belongs to the ferroprotoporphyrin group of peroxidases.

The optimal pH is 6.0-6.5 with stability. stable in the pH range of 5.0-9.

HRP is a **single chain** polypeptide containing **four disulfide bridges** with Mol Wt of about **44Kd** with **six Lysine residues** suitable for conjugation with a protein.

Enzyme Linked Immunosorbent Assays

Enzyme Conjugation

1. HRP can be conjugated to antibodies
glutaraldehyde, periodate oxidation, through disulfide bonds, and amino and thiol directed cross-linkers.
2. **HRP is smaller and more stable** than the enzyme labels **β -galactosidase and alkaline phosphatase.**
3. Its **glycosylation** leads to lower non-specific binding.

Enzyme Linked Immunosorbent Assays

Polystyrene as Sorbent Material

1. Polystyrene is a long carbon chain .
2. Surface flavoring are introduced to change the binding properties of the matrix.
3. Different functional groups are attached for binding with Bio molecules.
4. Surface chemistry can also be changed by UV treatment.

Enzyme Linked Immunosorbent Assays

Mechanism

1. An antigen is immobilized on a solid surface.
2. Antigen forms a complex with an antibody that is linked to an enzyme.
3. The enzyme is incubated with a substrate to produce a measurable coloured product.
4. Highly specific antibody-antigen is very important for interaction.

Enzyme Linked Immunosorbent Assays



<https://thumbs.dreamstime.com/z/well-plate-chemical-solutions-quantification-proteins-bicinchonic-acid-method-biochemical-test-elisa-plates-120786715.jpg>

Performance

1. ELISA kits are commercially available.
2. ELISAs tests are typically performed in 96-well (or 384-well) polystyrene plates, which bind antibodies and proteins.
3. On polystyrene it easy to separate bound from non-bound material during the assay.

Enzyme Linked Immunosorbent Assays

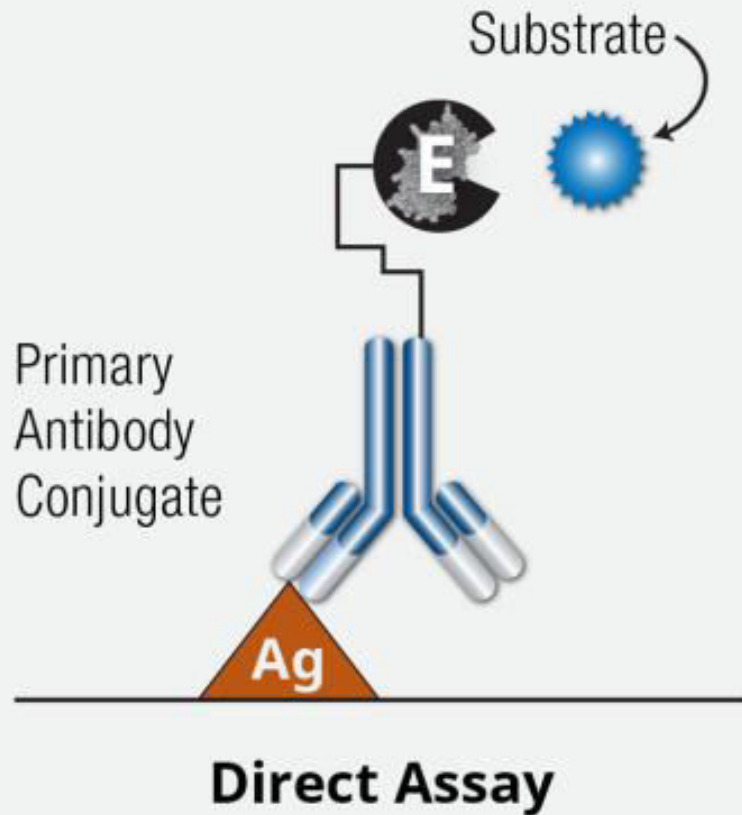
Types of ELISA

There are at least five types of ELISA assay,

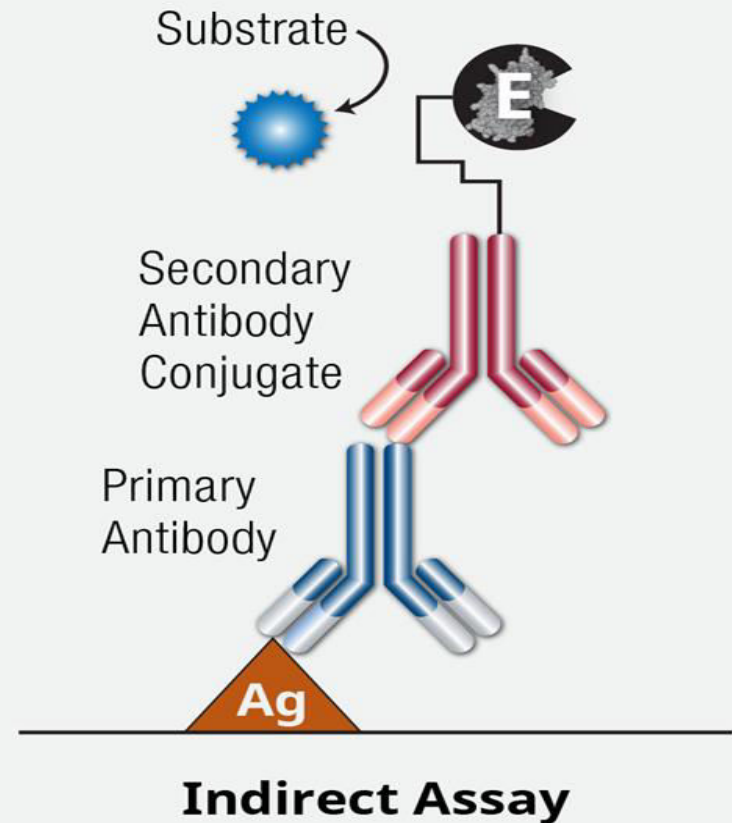
1. Direct ELISA Assay
2. Indirect ELISA Assay
3. Sandwich ELISA Assay
4. Competitive Assay
5. Multiplex ELISA Assay

Other variations are also available binder, and detection methodology.

Enzyme Linked Immunosorbent Assays

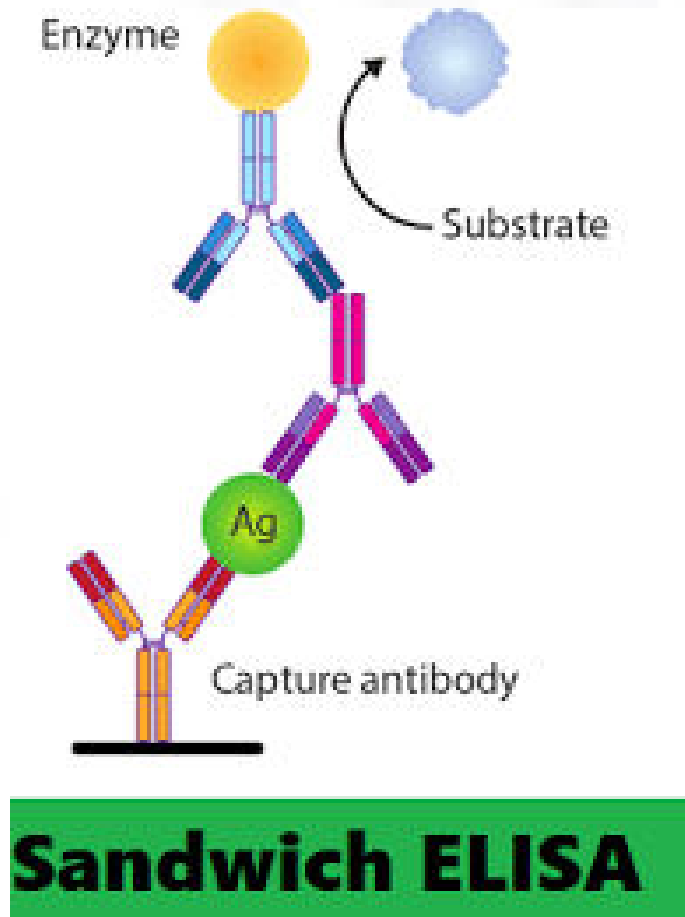


<https://media.cellsignal.com/www/images/resources/applications/elisa/elisa-direct-assay.jpg>



<https://media.cellsignal.com/www/images/resources/applications/elisa/elisa-direct-assay.jpg>

Enzyme Linked Immunosorbent Assays



Sandwich ELISA Test

The Sandwich ELISA is a sensitive and robust method which measures the antigen concentration in an unknown sample.

The Antigen of interest is quantified **between two layers of antibodies**:

1. **Capture Antibody**
2. **Detection Antibody.**

https://cdn.shortpixel.ai/client/q_glossy,ret_img,w_267/https://microbionotes.com/wp-content/uploads/2018/10/Sandwich-ELISA.png

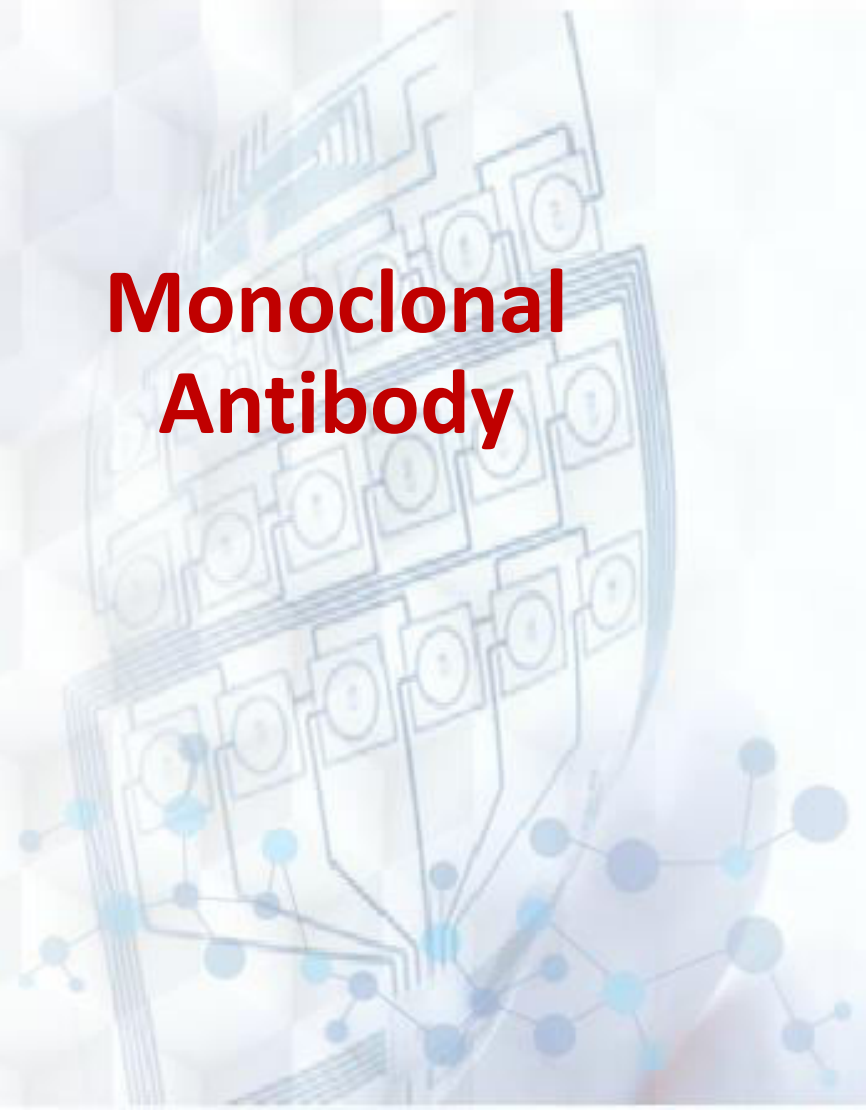
Enzyme Linked Immunosorbent Assays

Summary

1. ELISA is a useful method for the detection. The signals are amplified by the *enzyme HRP*.
2. The **enzyme is suitable** for conjugation due to the availability of lysine residues and thiol groups. .
3. The **modified form of polystyrene** is used for immobilization.
4. There are several types of ELISA methods.

Biosensors

**Monoclonal
Antibody**



Monoclonal Antibody

Introduction

Monoclonal antibodies (mAb) are important reagents used in biomedical research, in diagnosis of diseases, and in treatment of infections and cancer.

1. Each Monoclonal antibody is produced against a single epitope.
2. An epitope, is an antigenic determinant, is the part of an antigen that is recognized by the immune system.

Monoclonal Antibody

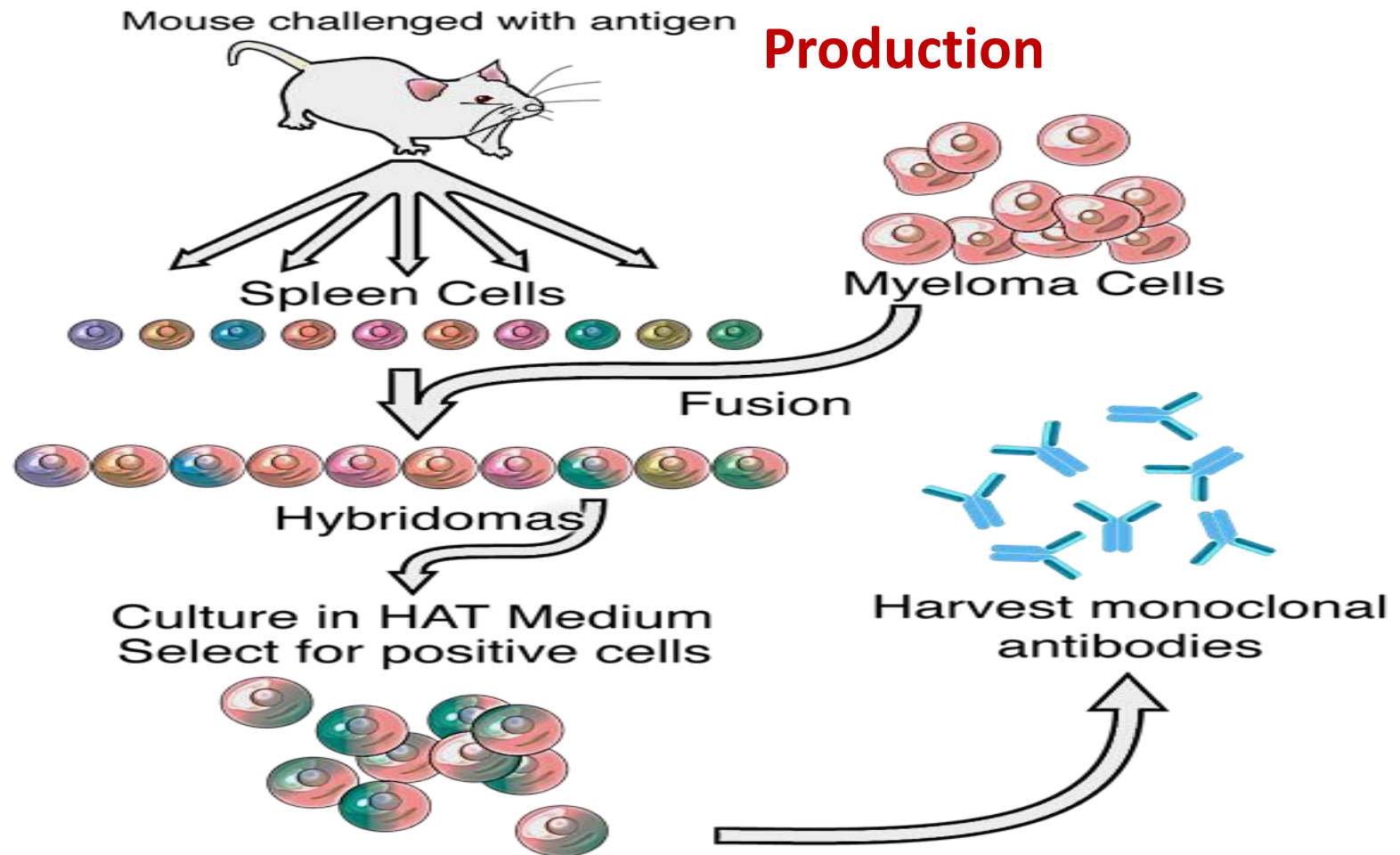
Production of Monoclonal Antibody

There are three general ways to produce monoclonal antibodies.

1. Hybridoma technique
2. Ascites method

Hybridoma antibody production method involves the *in vivo* and *in vitro* methods while Ascite method involve *in vivo* synthesis of MABs., which involve the use of animals.

Monoclonal Antibody



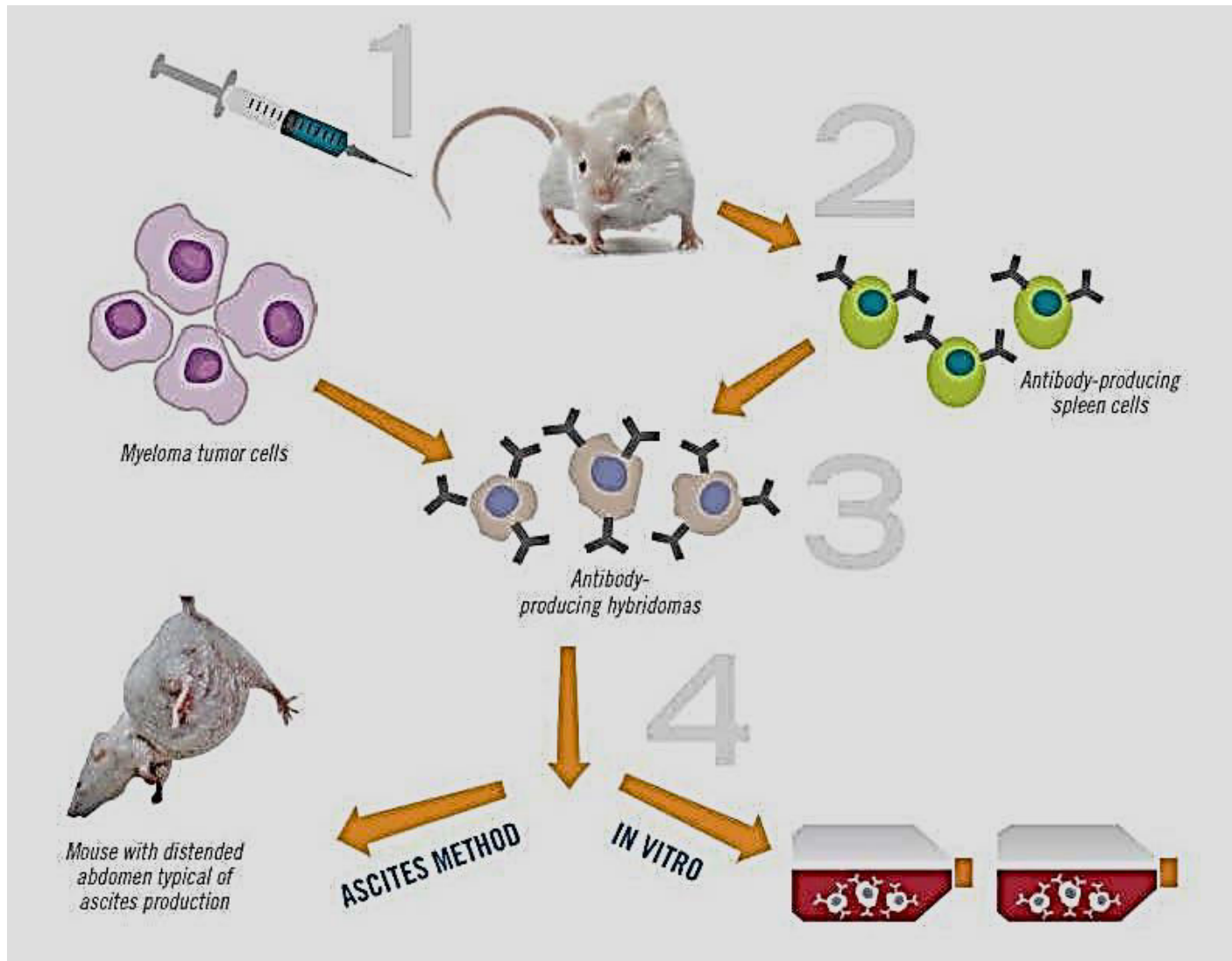
<https://upload.wikimedia.org/wikipedia/commons/9/9a/Monoclonals.png>

Monoclonal Antibody

Ascite Method

1. Monoclonal antibodies can be produced by growing Hybridoma cells within the peritoneal cavity of a mouse (or rat).
2. The Hybridoma cells are injected into the mouse, multiply and produce fluid (ascites) in its abdomen.
3. This fluid contains a high concentration of antibody which can be harvested.

Monoclonal Antibody



Monoclonal Antibody



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Diagnostic Applications

MAB are used RIA and ELISA in the laboratory.

1. Pregnancy Test . MABS produced against human chorionic gonadotropin are used for detecting the pregnancy, using urine sample.
2. Cancer diagnostics
3. Hormonal Disorders
4. Infectious Diseases.
5. Cardiac Troponin test

Monoclonal Antibody

Therapeutic Applications

Monoclonal antibodies have a wide range of therapeutic applications.

1. Treatment of **cancer**
2. **Transplantation of bone marrow**
3. Transplantation of **organs**
4. **Autoimmune diseases**
5. **Cardiovascular diseases**
6. **Infectious diseases.**

Monoclonal Antibody

Summary

1. Monoclonal antibodies are produced by a single types of cells. Each MAP is generated against a specific epitope.
2. There are two methods for the production of MABS.
3. Now a days Ascite method (*in vivo*) is used for MAB production.
4. MAB has major uses in the field of diagnostics and Therapeutics.

Biosensors

**Antibody
Fragments**



Antibody Fragments

Antibody Fragment

For many applications antibody fragments are preferable.

Antibody fragments can be generated by biological, chemical or genetic methods.

Chemical fragmentation utilizes reducing agents to break the disulfide bonds within the hinge region and digestion of the antibody with proteases including pepsin, papain, and ficin..

Antibody Fragments

Chemical and Biological methods.

Chemical and Biological fragmentation utilizes reducing agents to break the disulfide bonds within the hinge region.

Chemicals

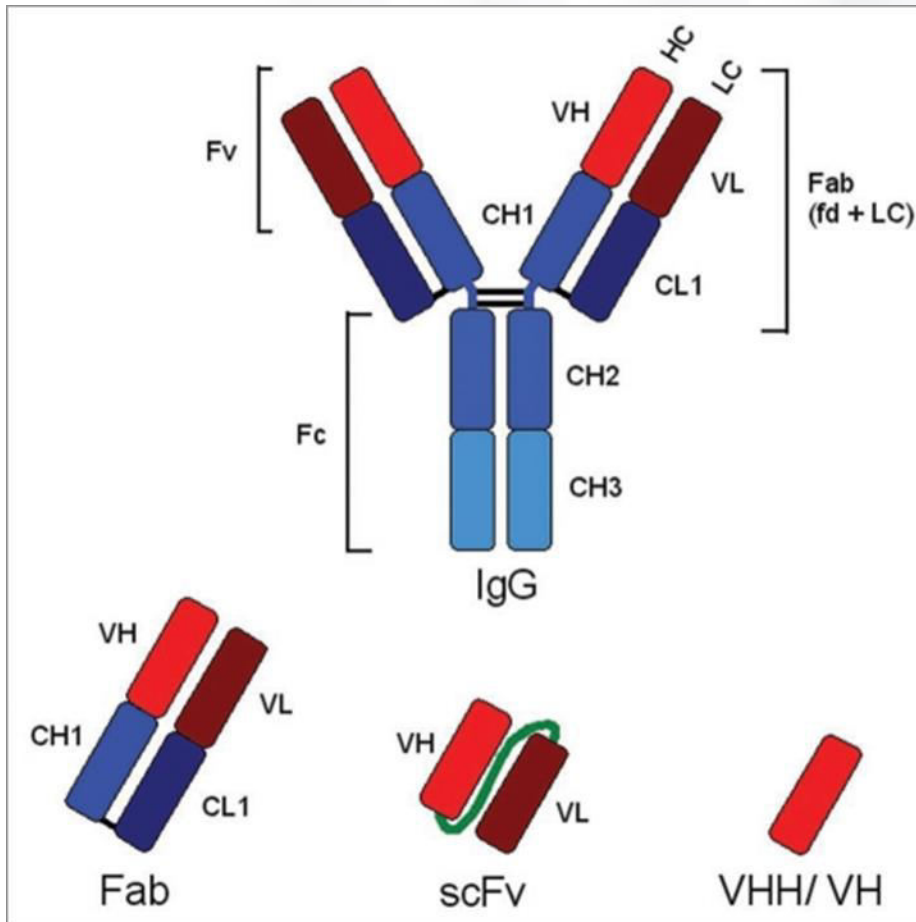
Dithiothreitol, Cyanogen bromide is also used for initially digesting the protein to get its fragments.

Protease enzyme digestion of the antibody with proteases pepsin, papain, and ficin.

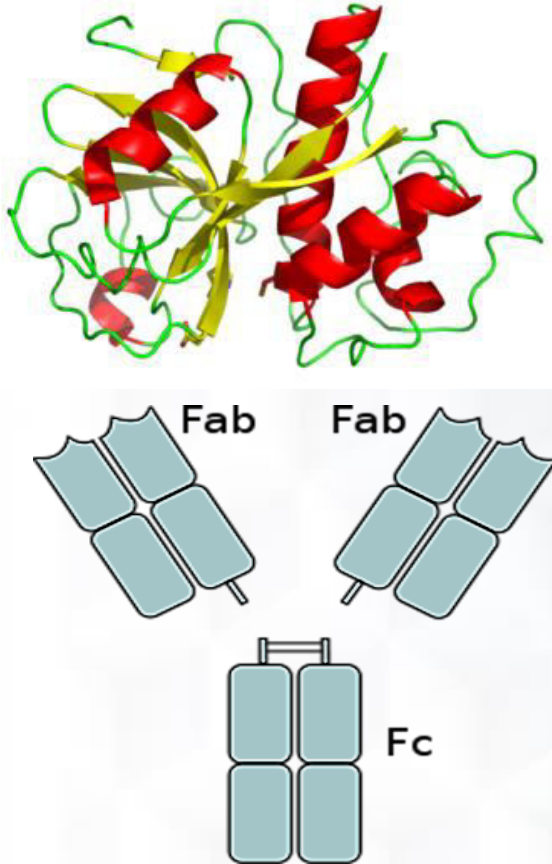
Antibody Fragments

Antibody fragment types.

1. CH constant heavy chain
2. CL constant light chain
3. Fab antigen binding fragment
4. scFv single chain variable fragment
5. VH variable heavy chain;
6. VL variable light chain.



Antibody Fragments



https://upload.wikimedia.org/wikipedia/commons/thumb/e/e2/2fab_fc.svg/330px-2fab_fc.svg.png

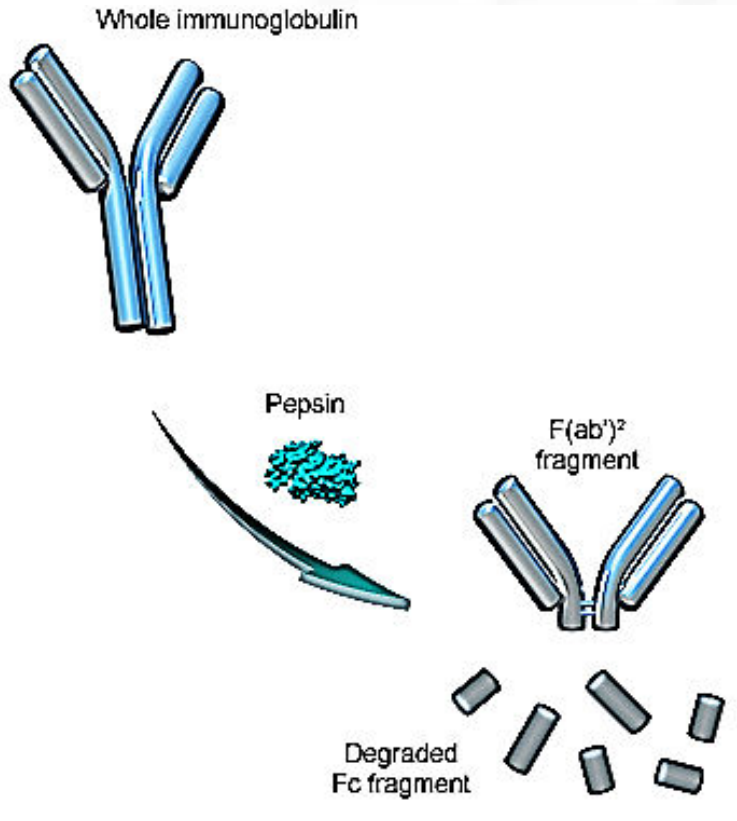
Papain Enzymatic digest

Papain is a Cys protease.

1. It cleavage results in reduction of **exposed disulfide bonds** in the **hinge region** of the antibody.
2. 2 Fabs consisting of equal parts heavy chain and light chain,
3. **1 Fc** region.

https://upload.wikimedia.org/wikipedia/commons/thumb/0/06/Papain_cartoon.png/330px-Papain_cartoon.png

Antibody Fragments



Fragmentation by Pepsin

Pepsin digestion results in

1. A **F(ab')₂** fragment composed of two Fab fragments still linked together by the remaining portion of their respective heavy chains at the hinge region.
2. The **Fc fragment** is **extensively degraded** into small fragments.

https://www.covalab.com/media/wysiwyg/diagrams/pepsin_digestion.jpg

Antibody Fragments

Fragment crystallizable region

1. Fragment crystallizable region (Fc region) is the tail region of an antibody.
2. It **interacts with cell surface receptors** called Fc receptors and some **proteins of the complement system**.
3. This property allows **antibodies to activate the immune system**.

Antibody Fragments

Recombinant Fragments of antibody

1. Well characterized and selected regions of antibody are produced by recombinant DNA technology.
2. This production method is totally animal free.
3. Target specific Antibody fragments are designed and produced for best efficiency.

Antibody Fragments

Advantages

Recombinant antibody fragment offers several advantages over both traditional monoclonal and polyclonal antibodies:

1. Improved consistency and reproducibility.
2. Improved sensitivity and specificity.
3. Ease of scalability
4. Animal-free high-throughput production

Antibody Fragments

Summary

1. In many application whole antibody structure is not required.
2. There are a number of techniques by which antibody can be fragmented and specific fragments required for the assay could be purified.
3. The required antibody fragment could be synthesized by recombinant DNA technology.

Biosensors

Lateral Flow Assay



Lateral Flow Assay

Introduction.

This is a very simple test and is widely used in many countries with varying common names

- 1. Lateral flow test (LFT)**
- 2. Lateral flow device (LFD)**
- 3. Lateral flow assay (LFA)**
- 4. Lateral flow immunoassay (LFIA)**
- 5. Dipstick**

Lateral Flow Assay

Introduction.

1. The test is a **simple** to use for the **presence or absence of an Analyte**.
2. This **test bypass the laboratory test** and is **economical** therefore widely used.
3. Minimal training is required to operate.
4. The test is qualitative and read **visually**.

Lateral Flow Assay

Lateral flow Assay

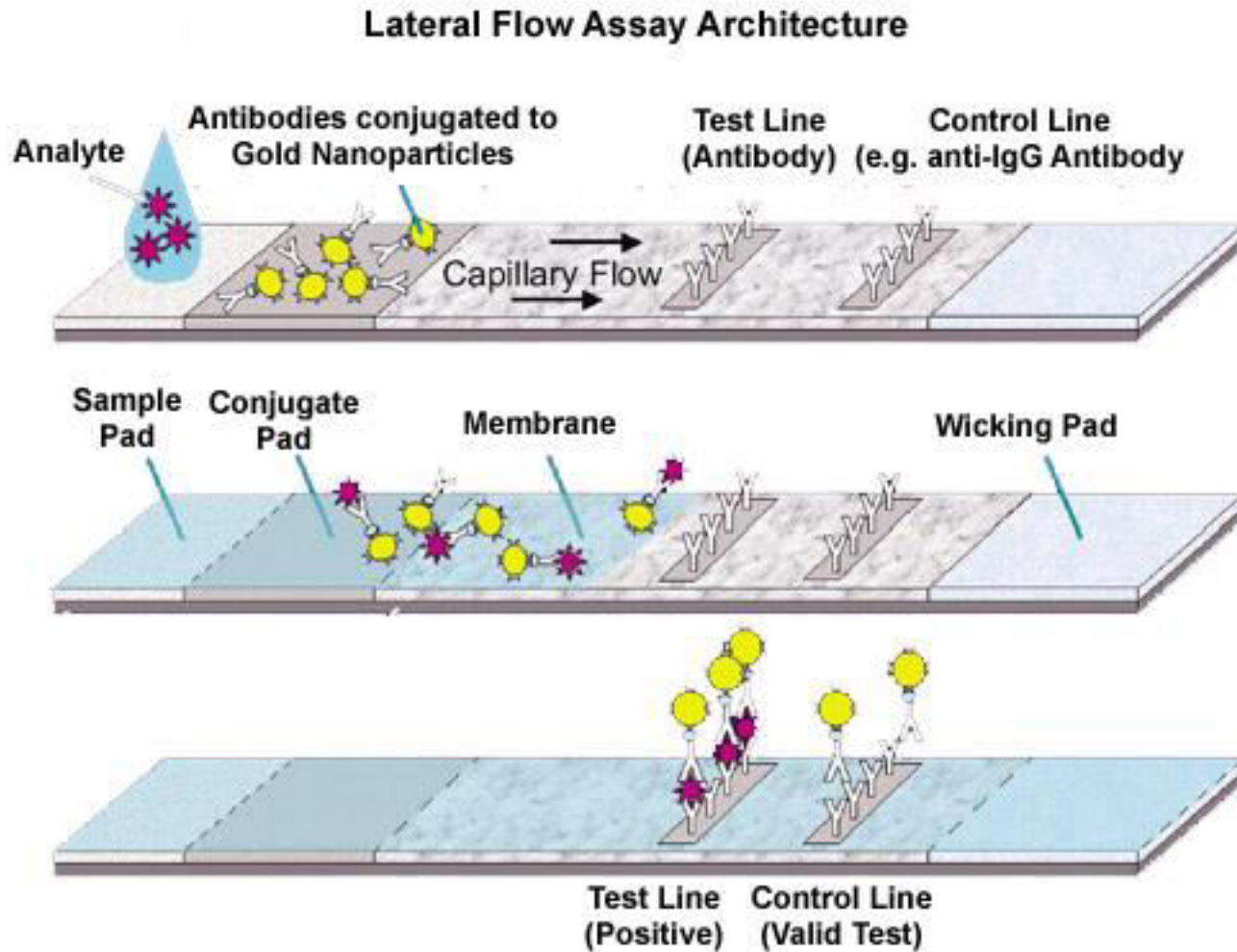
A typical lateral flow rapid test strip consist of the following components:

- **Sample pad** - an adsorbent pad onto which the test sample is applied.
- **Conjugate or reagent pad** – this contains **antibodies** specific to the target analyte **conjugated to coloured particles** (usually colloidal gold nanoparticles, or latex microspheres).

Lateral Flow Assay

- **Lateral flow Assay**
Reaction membrane
typically a **nitrocellulose**
or **cellulose acetate**
membrane onto which
anti-target analyte
antibodies are
immobilized.
- **Backing Material**. The
components of the strip
are usually fixed to an
inert backing material

Lateral Flow Assay



http://www.cytodiagnosics.com/store/pc/catalog/lateral_flow_assay.jpg

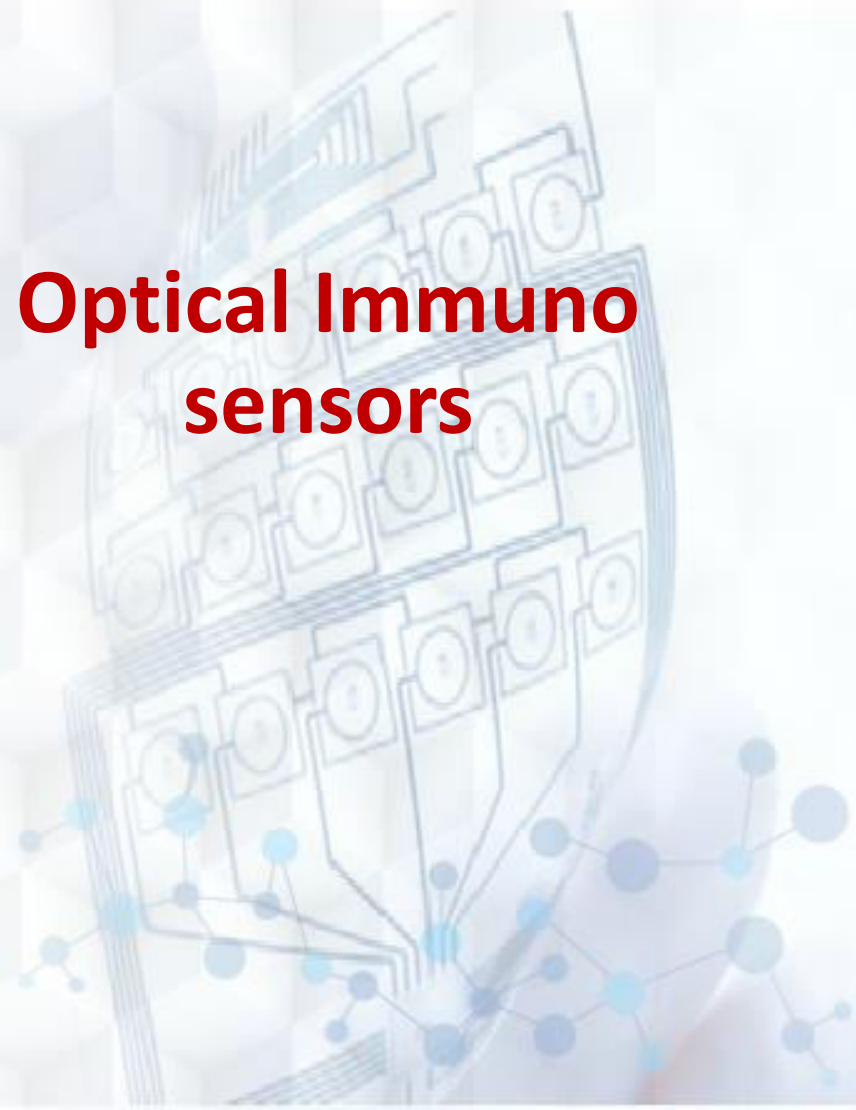
Lateral Flow Assay

Summary

1. Lateral flow assay is the simplest testing method.
2. It uses the immunological techniques to detect the molecule.
3. The dipstick designed is used for the test.
4. LFA is heavily used in industry, diagnostics, environment and other fields of life.
5. Pregnancy test .

Biosensors

Optical Immuno sensors



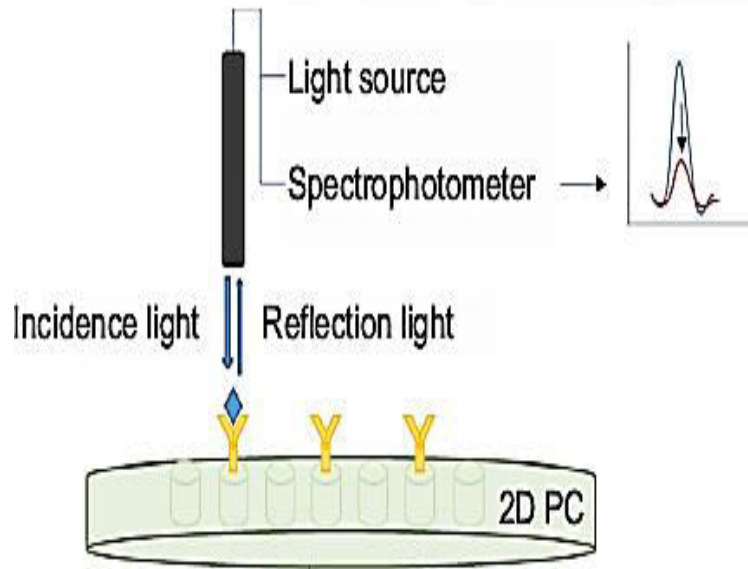
Optical Immunosensors

Introduction

In optical biosensors, the interaction between the AB (BRE) and AG (Analyte) is marked by formation of AB-AG complex, leading to a change in the optical properties of the wave.

1. Reflectance
2. Absorbance
3. Refractive index
4. Optical path
5. Emission of Light.

Optical Immunosensors



https://www.dovepress.com/cr_data/article_fulltext/s94000/94025/img/AHCT-94025_F010.jpg

Geometric Design of optical Platform

There are several geometric layouts, including

1. Strips
2. Waveguide fibers
3. Planar optical waveguide systems
4. Capillary sensors
5. Arrays.

Key point is the position of light source to detect the Biorecognition process.

Optical Immunosensors

Planar waveguide platforms

Planar wave guided platform is a popular design as planar geometry is **compatible with micro fabrication technologies** For example Lab on chip.

The planar waveguide platform is based on different materials

1. **Glass**
2. **Plastic**
3. **Silicon**

Optical Immunosensors

Some Methods or Techniques employed in optical immunosensors

<u>Optical Signal</u>	<u>Methods/Transducing technique</u>
1. Absorbance	Light intensity measurement (Different absorbance for different colors)
2. Reflectance	Light intensity measurement
3. Fluorescence	Total internal reflection fluorescence
4. Refraction index	(SPR) Total internal reflection

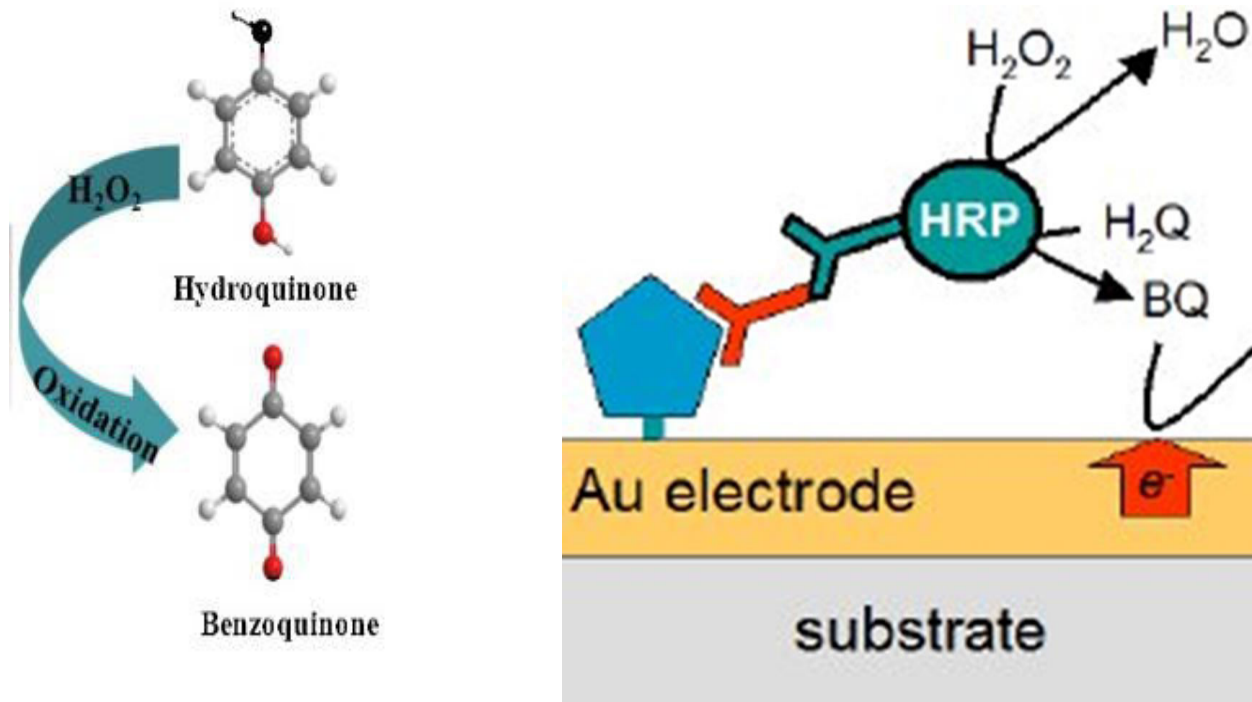
Optical Immunosensors

Use of mediator in Optical sensors

1. **Hydroquinone** is an organic compound and used as a **mediator** in optical Biosensors.
2. With reference to biological organisms, It is **present in abdomen of Bombardier beetle** and employed as defensive agent.
3. **Hydroquinone is reduced to 1,4-benzoquinone .**

Optical Immunosensors

Use of Hydroquinone as mediator in Optical sensors



To detect the signals a mediator is used. Hydroquinone is converted into **benzoquinone (bright-yellow crystals)** that is electrochemically reduced at the electrode surface.

<https://ars.els-cdn.com/content/image/1-s2.0-S0926337314003257-gr1.jpg>

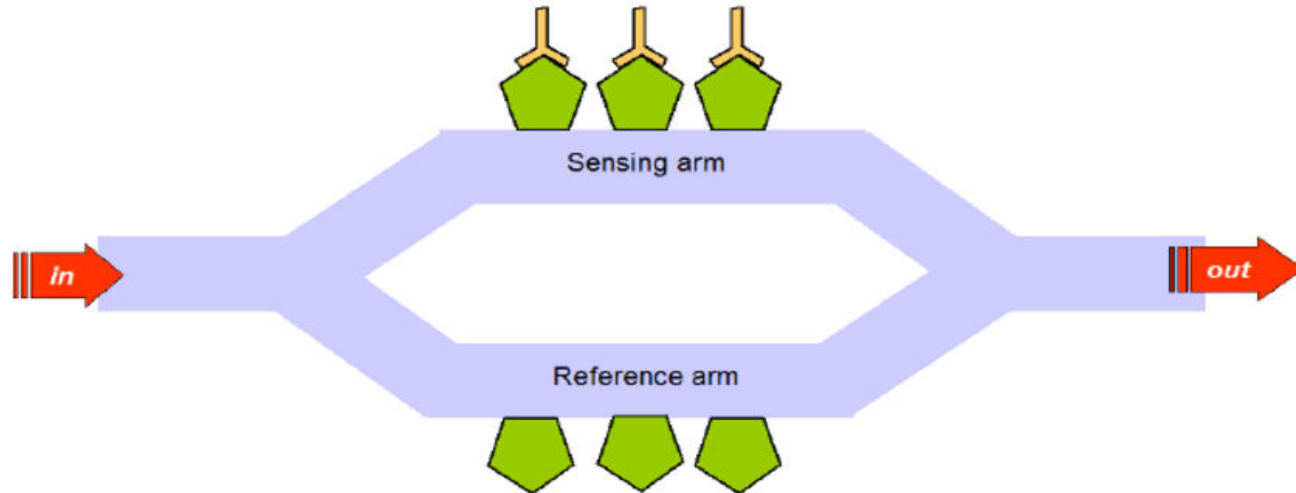
https://www.researchgate.net/profile/Gabriel_Ybarra/publication/221927052/figure/fig1/AS:305246183149572@1449787769379/Schematic-representation-of-the-electrochemical-detection-of-enzyme-linked-immunoassay

Optical Immunosensors

Refractometric sensors

1. The **refractive index** of a material is defined as the **ratio of velocity of light** in a **vacuum** to that in a defined material.
2. In a special **interferometer** the **optical power is transported by a single-input waveguide.**
3. It a **Y shape splitters.** both branches are coated with the biological sensitive element.

Optical Immunosensors



At the sensing layer the formation of Ab-Ag complex leads to a change in the refractive index. For reference layer there is no change in the refractive.

https://www.researchgate.net/profile/Gabriel_Ybarra/publication/221927052/figure/fig5/AS:305246183149580@1449787769709/Schematic-of-a-Mach-Zehnder-interferometer-used-for-immunosensing.png

Optical Immunosensors

Summary

1. The optical immunosensors looked for the change in the optical properties, after the bio-recognition process.
2. The Geometric Design of optical Platform is the pivotal in immunosensors.
3. Hydroquinone is widely used for change in color as an optical signal.
4. A number of techniques are used to detect the optical signals.

Biosensors

87. Surface Plasmon Resonance Immunosensors



Surface Plasmon Immunosensors

Surface Plasmon Resonance

1. Surface plasmon resonance (SPR) is a spectroscopic method.
2. The Biorecognition element is **immobilized** on the **surface of thin metal**.
3. A wave is used to **detect** the ligand on a **thin metal film**.
4. The change in **refractive index** upon binding of the analyte, is measured.

Surface Plasmon Immunosensors

pH and Type of Buffer solution

Antigen antibody complex formation is a physiological phenomenon and it occurs at pH 7.4 and 10mM strength.

1. PBS (Phosphate buffer saline) antigen antibody complex formation.
2. HEPES (4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid)

Surface Plasmon Immunosensors

Metals and protein interactions

The Protein-metal interaction depends on **functional groups of amino acids** and the **position of metal in the periodic table**.

1. Alkali metals interact weakly.
2. Alkaline earth Metals interact most strongly.
3. Transition earth Metals due to the availability of d-orbital stands a good chance for forming a protein metal complex.

Surface Plasmon Immunosensors

Amino acids in metal interaction

Four groups of amino acids.

1. **Aspartic acid and Glutamic acid** involve the carboxylate chain is involved.
2. **Cysteine and Methionine** involve the Thiol and Thioether groups.
3. **Histidine** in which the Imidazole group is involved
4. **Serine, Threonine and Tyrosine** involve Hydroxyl groups

Surface Plasmon Immunosensors

Immobilization of Fc Antibody fragment using Bacterial Proteins

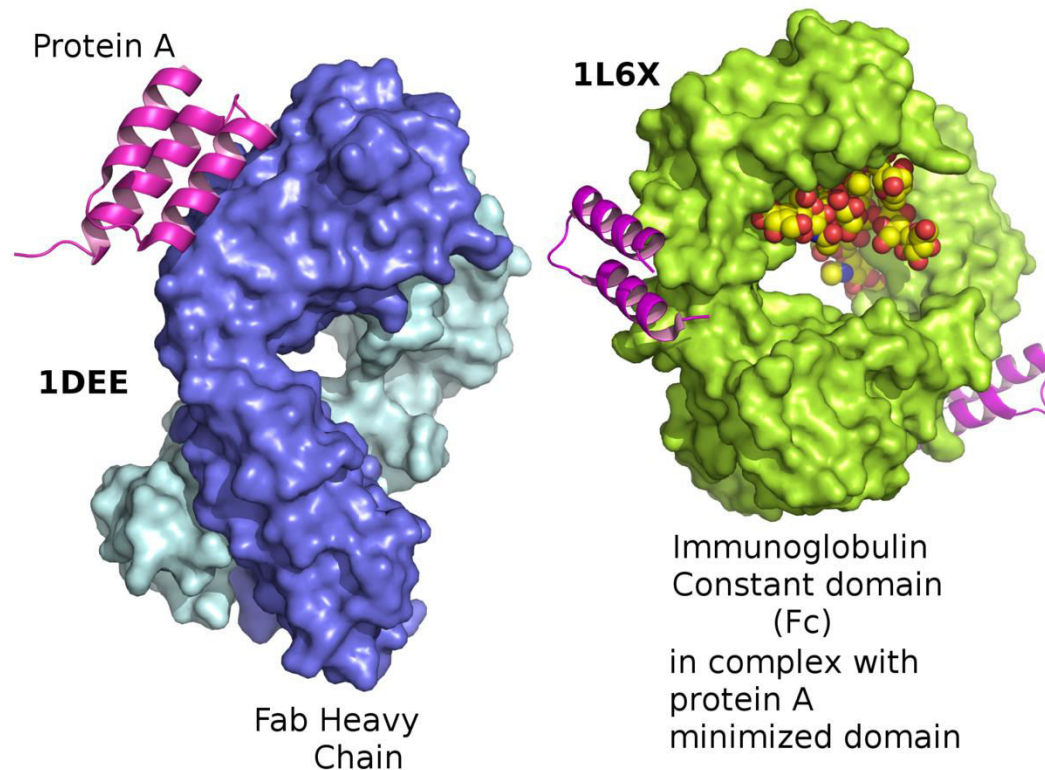
There are three types of proteins produced by bacteria *Staphylococcus aureus*.

1. Protein A is a 42 kDa
2. Protein G 65-kDa
3. Protein G 58 kDa

They have specific binding affinity has binding affinity with the **Fc region and Fab region** of the Immunoglobulins.

Surface Plasmon Immunosensors

Immobilization of Fc Antibody fragment using Proteins



https://upload.wikimedia.org/wikipedia/commons/thumb/2/2c/Protein_A_1DEE_1L6X.png/1200px-Protein_A_1DEE_1L6X.png

Surface Plasmon Immunosensors

Kinetic parameters for Antigen Antibody complex

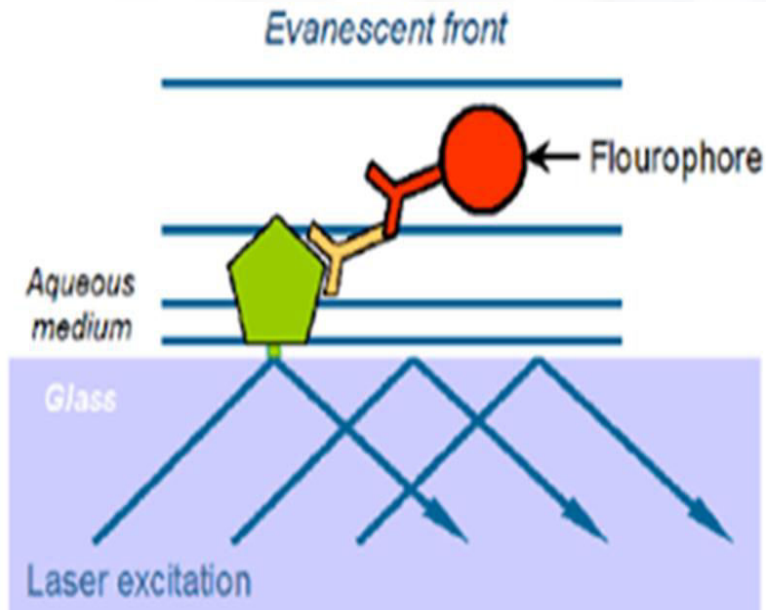
Kinetic parameters give information about **affinity of the complex formed**.

Association rate constant (k_a) is indicative of number of AG-Ab complex formed per second'

The **dissociation rate constant** (k_d) is the rate of decay of complex formation per second.

For a higher affinity complex the rate of complex formation is high.

Surface Plasmon Immunosensors



Use of fluorophore in SPR immune sensors

A **fluorophore** is a fluorescent chemical compound that can re-emit light upon light excitation.

The secondary antibody is conjugated with the fluorescence probe.

The fluorescence probe enhanced the sensitivity of the SPR Immuno Sensor.

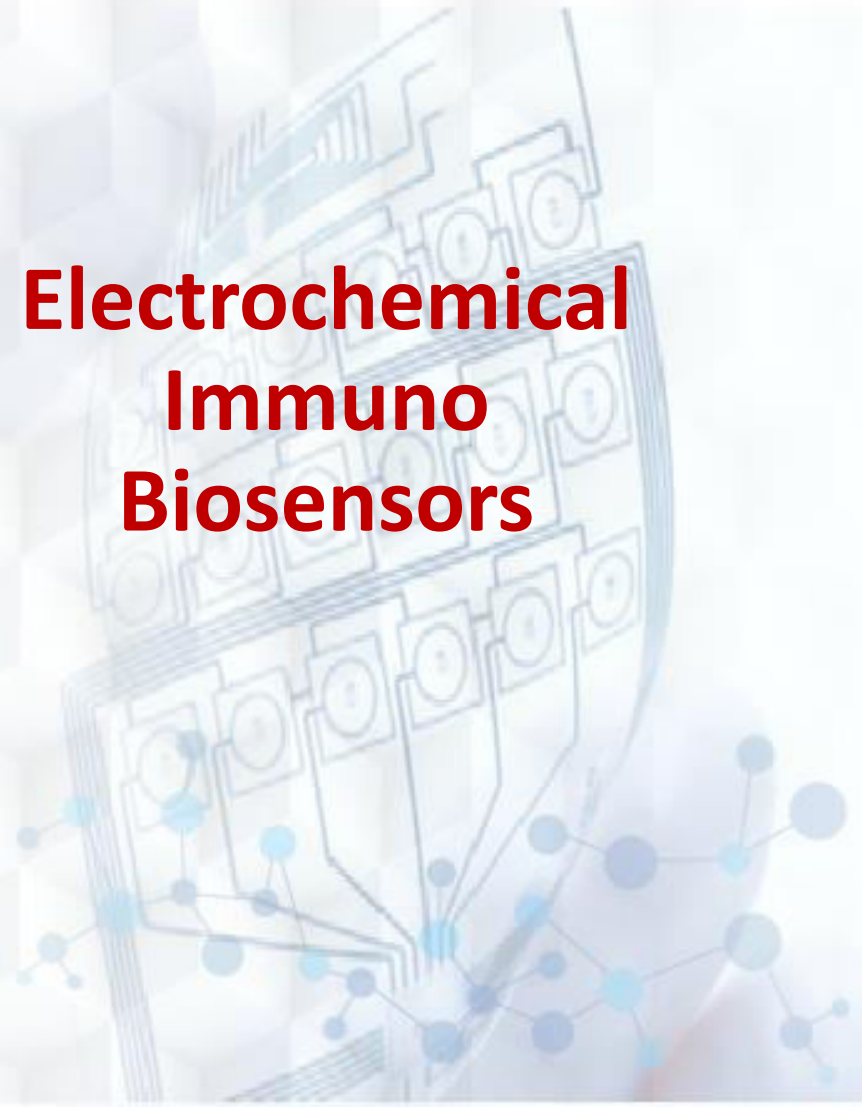
Surface Plasmon Immunosensors

Summary

1. SPR immune sensor uses a spectroscopic method for the detection of Analyte.
2. A thin metal film is used for the immobilization of the Biological element.
3. Proteins has a binding affinity for metals.
4. In SPR Fluorophore molecules are used to enhance the signals.
5. Physiological pH is used in SPR Immuno sensors.

Biosensors

**Electrochemical
Immuno
Biosensors**



Electrochemical Immuno Biosensors

Introduction.

In electrochemical immunosensor, the signals detected for the formation of Ab-AG complex are converted into an electrical signal. There are three types

1. Amperometric immunosensors
2. Potentiometric immunosensors
3. conductometric immunosensors

Electrochemical Immuno Biosensors

Strategies for Immobilization

1. Different strategies are employed for the immobilization **on surface of conductive electrode.**
2. Immobilization on GNP using molecular linkers such as **thiols groups.**
3. Thiols form a **strong bond onto the surface of gold** forming a **self assembled monolayer.**

Electrochemical Immuno Biosensors

Magnetic beads for immobilization

The magnetic beads for the immobilization of BRE.

1. The **magnetic beads** are fixed onto the surface of electrodes by applying the magnetic field.
2. **Antibody** can be **immobilized** onto the surface of magnetic beads.
3. After running a test, beads can be **removed from the electrode**, and electrode can be reuse.

Electrochemical Immuno Biosensors

Amperometric Sensors

The formation of **the antigen-antibody labeled antibody complex** is **detected by using a REDOX enzyme**.

1. A REDOX mediator (cofactor) is used to detect the reaction.
2. To detect the reaction organic compounds are used having different colours in oxidized and reduced stages.

Electrochemical Immuno Biosensors

Potentiometric Sensors

1. **Polyaniline based matrix** is used in potentiometric immune sensors
2. The color of PANI is pH dependent.
3. The color changes from green (acidic solution) to blue (basic solution).
4. **O-phenylenediamine (OPD)** is also used in potentiometric immuno biosensors.

Electrochemical Immuno Biosensors

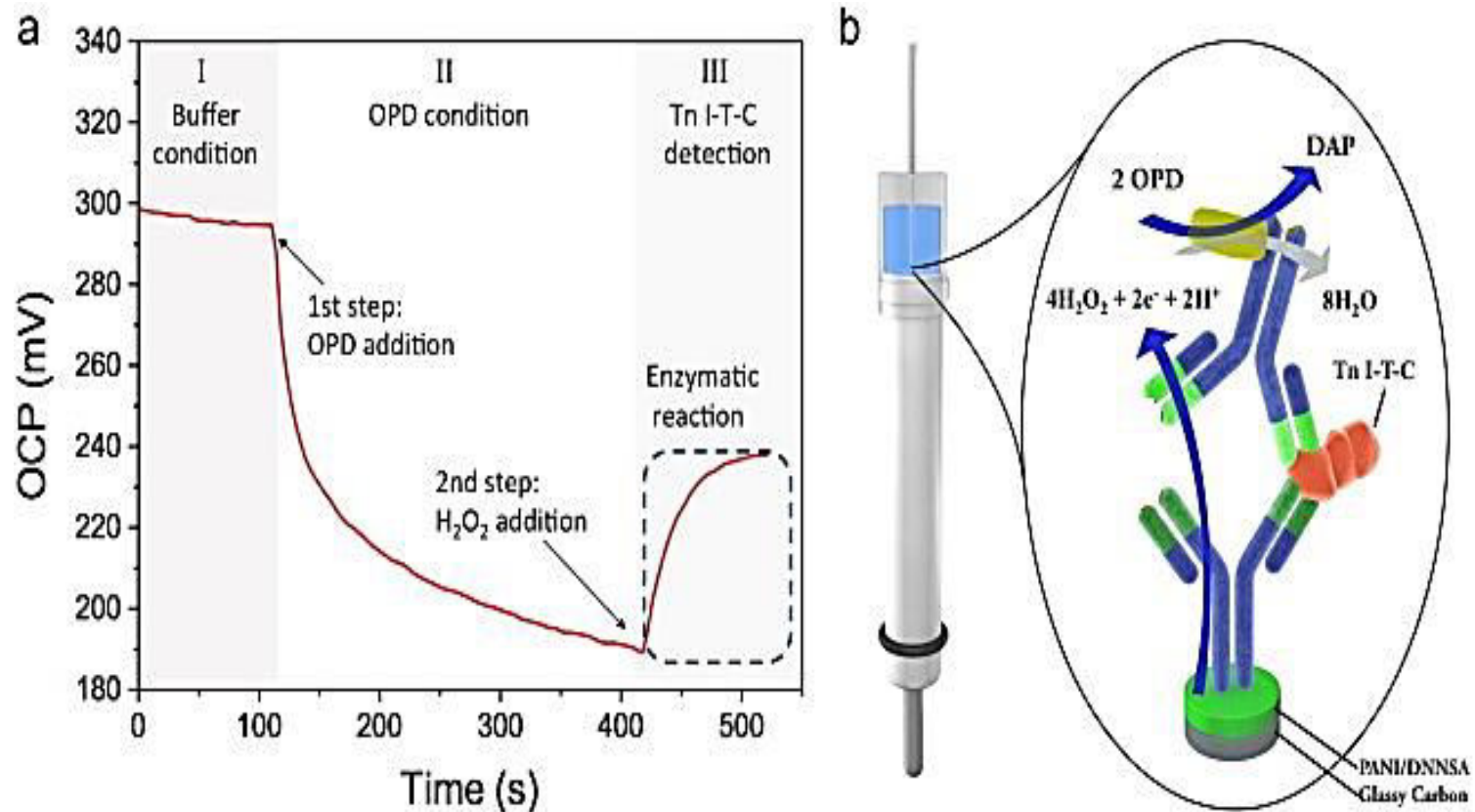
O-phenylenediamine based potentiometric sensor

O-phenylenediamine (OPD) is used as a mediator in the system.

1. Hydrogen peroxide catalyzes by HRP labeled on the secondary antibodies.
2. The electrons released during the reaction converts OPD into Di-amino benzedine (Brown in color).

Electrochemical Immuno Biosensors

Potentiometric Cardiac Immuno Biosensor



<https://ars.els-cdn.com/content/image/1-s2.0-S0956566315300853-gr1.jpg>

Electrochemical Immuno Biosensors

Summary

1. Electrochemical immune sensors detect the event of Antibody and Antigen complex by electrical signals.
2. The biological recognition event produced signals that are converted into **resistance, voltage or current**.
3. A number of organic molecules, forming color products are used as mediators.
4. It is a sensitive technique.

Biosensors

**Impedance
Immuno sensors**

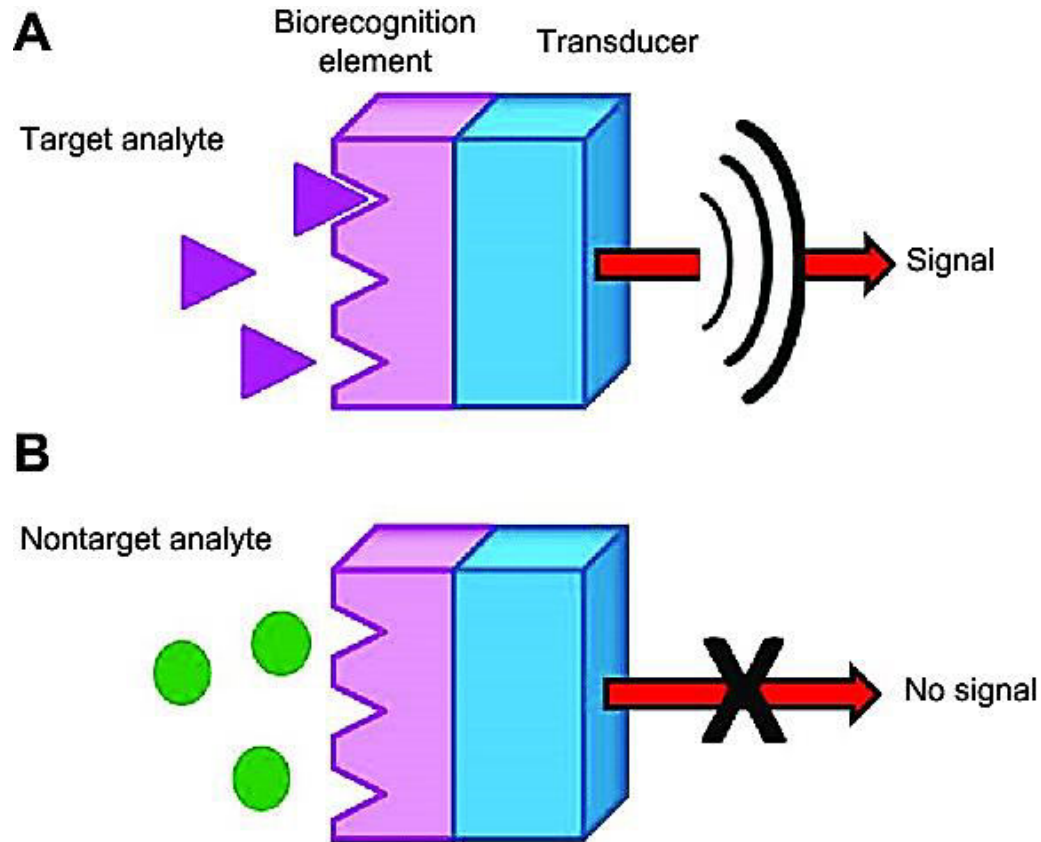
A stylized illustration of a biosensor chip, likely an impedance immuno sensor. The chip is depicted as a grid of circular elements, possibly representing individual sensor units or electrodes. It is overlaid with a network of blue lines and dots, suggesting a molecular or biological structure. The background features a faint, repeating pattern of hexagons, giving it a technical or scientific feel.

Impedance Immuno sensors

Introduction

1. Impedance is a measure of the opposition to electrical flow, that is measured in ohms.
2. The **nature of impedance** sensors is all electrical.
3. Impedance biosensors are **simpler than other** methods because they lack optical or acoustic components.
4. Therefore it offers significant **advantages for portable** applications.

Impedance Immuno sensors



https://www.researchgate.net/profile/Miguel_Rodrigo2/publication/281601468/figure/fig2/AS:667787139751937@1536224269244/Schematic-diagram-of-biosensor-with-target-analyte-A-and-nontarget-analyte-B.png

Impedance Immuno sensors

Immobilization

Immobilization of the BRE on the sensing platform is an important step.

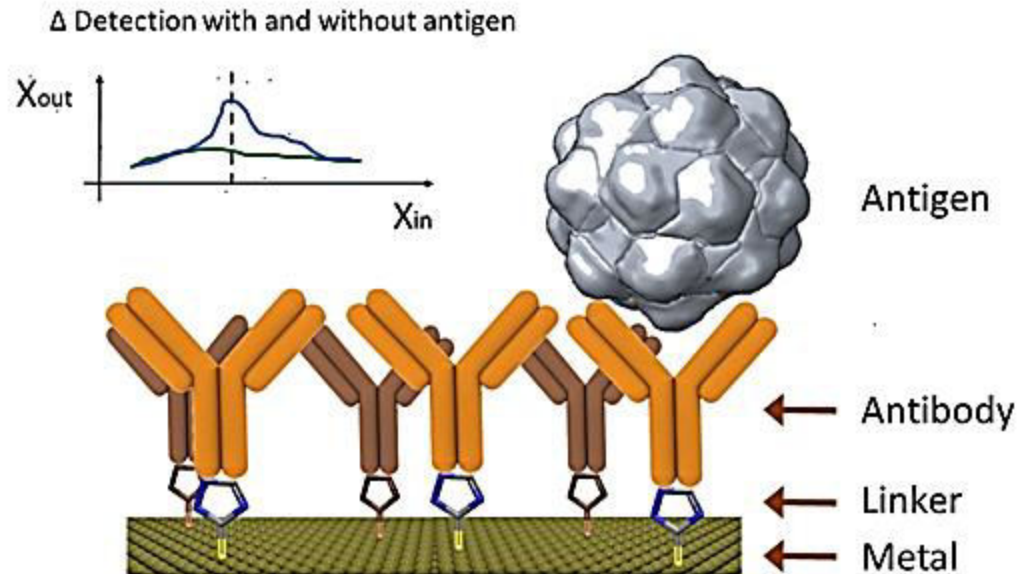
1. Gold Nano-materials are coated on copper electrodes, gold and silver layers for immobilization.
2. Magnetic particles
3. Hydroxyapatite powder
4. ion-exchange membranes
5. Polymeric films

Impedance Immuno sensors

Working

1. A BRE is immobilized on - **functionalized surface**, of the impedance Immuno sensor.
2. When an AG-AB complex formed changes occur in the electrical properties of the surface such as dielectric constant or resistance.
3. These changes could be detected by the transducer.
4. In immuno impedance sensors no label is required for impedance sensing

Impedance Immuno sensors



Most common configuration of electro-immuno sensor devices and their response with and without the presence of the antigen.

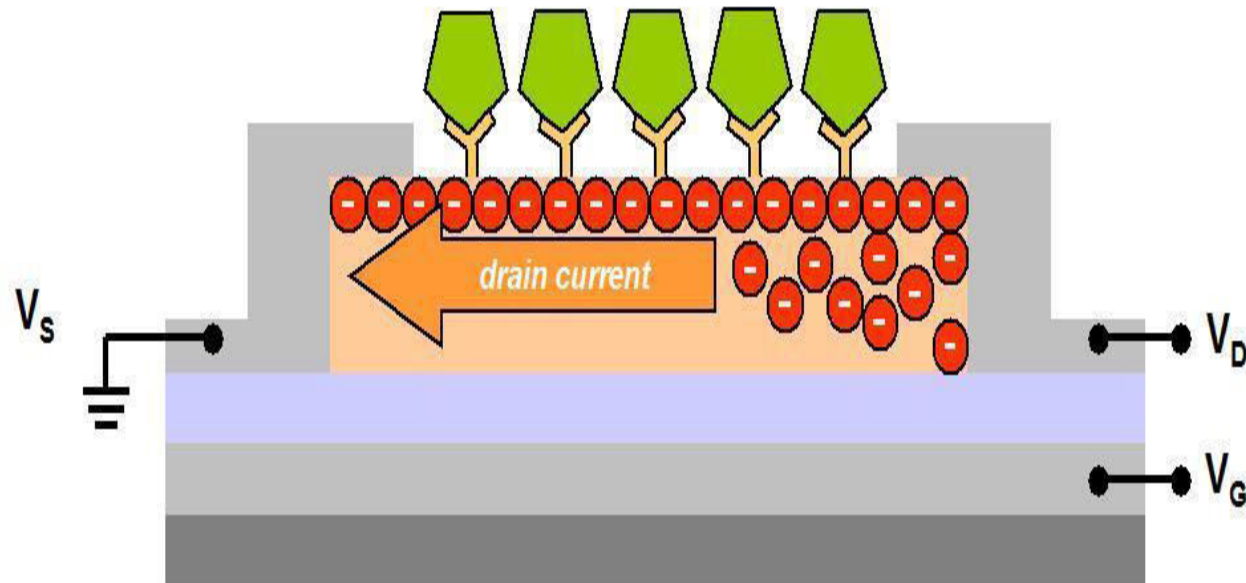
<http://medcraveonline.com/IJBSBE/images/IJB>
SBE-02-00010-g002.png

Impedance Immuno sensors

Field Effect Transistors

1. A FET consists of three terminals, called gate, drain, and source.
2. N-type transistors are used in impedance biosensors.
3. The gate controls the current between the source and the drain.
4. The BRE is immobilized on the gate.
5. The formation of antigen-antibody complex at the gate terminal.

Impedance Immuno sensors



<https://docplayer.net/docs-images/62/46507129/images/80-3.jpg>

Impedance Immuno sensors

Merits and Demerits

1. Advantages of Impedance immuno sensor are, **small instrument size**, low cost and **speedy**.
2. Suitable for use at a bedside, in an ambulance, or during a clinic visit.
3. Disadvantages are the use of **small analytes**, **complexity in detection**, **nonspecific adsorption**, and **problems stability of biomolecule immobilization**.

Impedance Immuno sensors

Summary

1. impedance immuno biosensors are based on the determination of resistance as a signal for the detection of Analyte.
2. FET are also used as impedance sensors when an immunogenic molecule is immobilized on the sensing platform of the sensor.
3. Impedance sensors are small in size and have several advantages and disadvantages.

Biosensors

Piezoelectric Immuno sensors



Piezoelectric Immuno Biosensors

Introduction

1. Piezoelectric materials are used in **the physical component** of the PZ immuno sensors.
2. PZ materials has a unique ability to **generate current** when **stress is applied**.
3. In all type of Immuno sensors **Ab-AG complex** is considered as a **Biorecognition process**.
4. This process is determined by the **transducers** , using a number of techniques.

Piezoelectric Immuno Biosensors

PZ material in sensors

1. Piezo electric **crystals** are used in Piezo electric immune sensors, in the **physical component** of the sensor.
2. In PZ immuno **sensor**, **mass changes** due to the **formation of antigen-antibody** complex, is detected by Piezo electric crystal (Quartz) present in the transducer.

Piezoelectric Immuno Biosensors

Immobilization

1. Antigens or antibodies can be **immobilized** onto the surface of **piezoelectric sensing platform**.
2. It is required to **modify** the **sensing platform**, for functionalize the surface to facilitate the immobilization.
3. A variety of **organic and inorganic materials** are used for coating the sensing platform. On the coated platform, BRE is immobilized.

Piezoelectric Immuno Biosensors

Coating materials

The coating materials include

1. Silane reagents
2. Modified silica/HRP based membrane
3. Chitosan-/Multi Walled Carbon Nanotubes
4. ZnO micromechanical membrane
5. Protein A based film

They are used for **covalent attachment** to the **PZ crystal** surfaces.

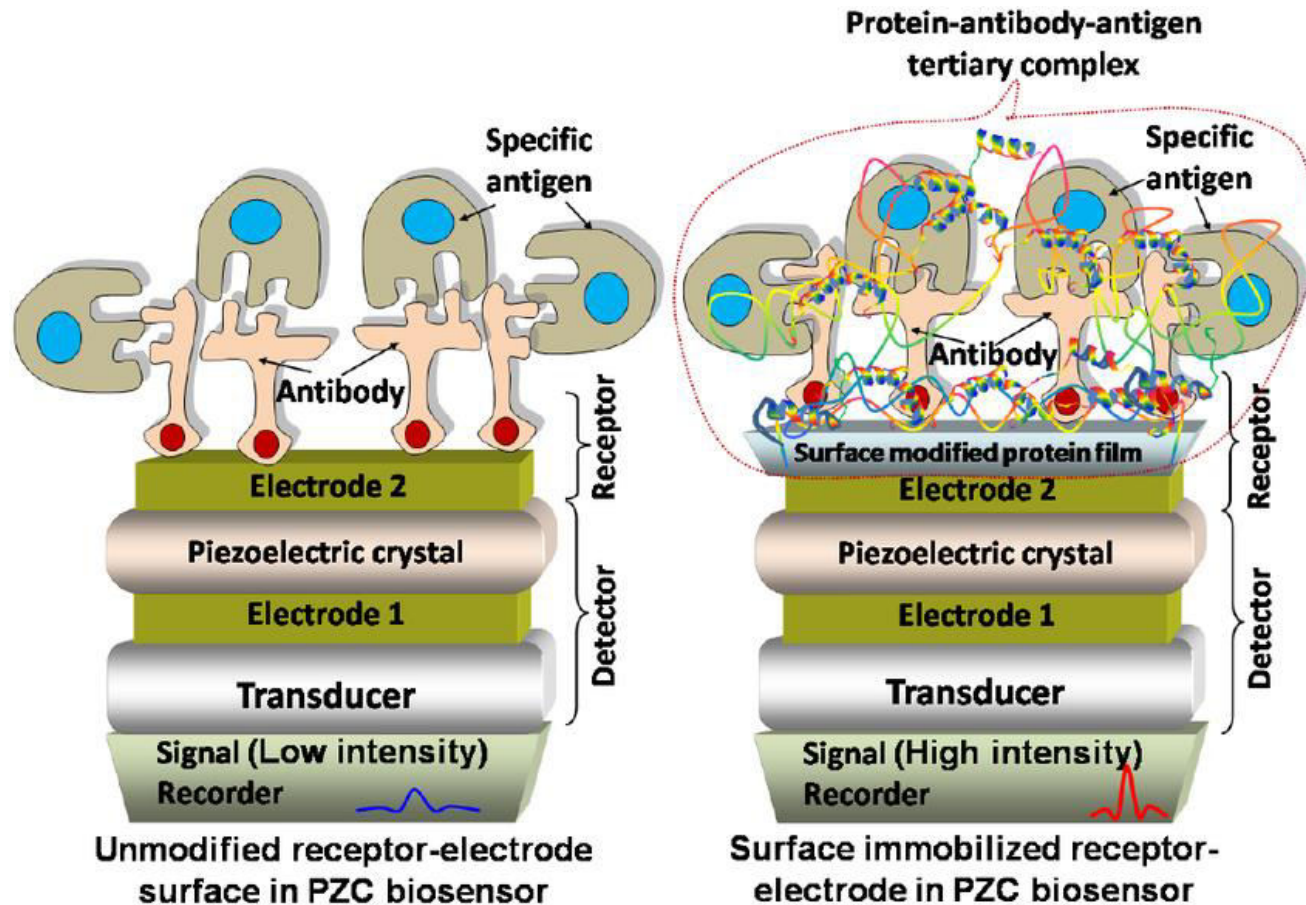
Piezoelectric Immuno Biosensors

Requirements for immobilization

There are certain requirements for the coating material for immobilization.

- 1. Layer should be thin**
- 2. Homogenous coating**
- 3. Able to incorporate the AA functional groups**
- 4. Contribute to high sensitivity**
- 5. Minimum interference**
- 6. Reproducible results**

Piezoelectric Immuno Biosensors



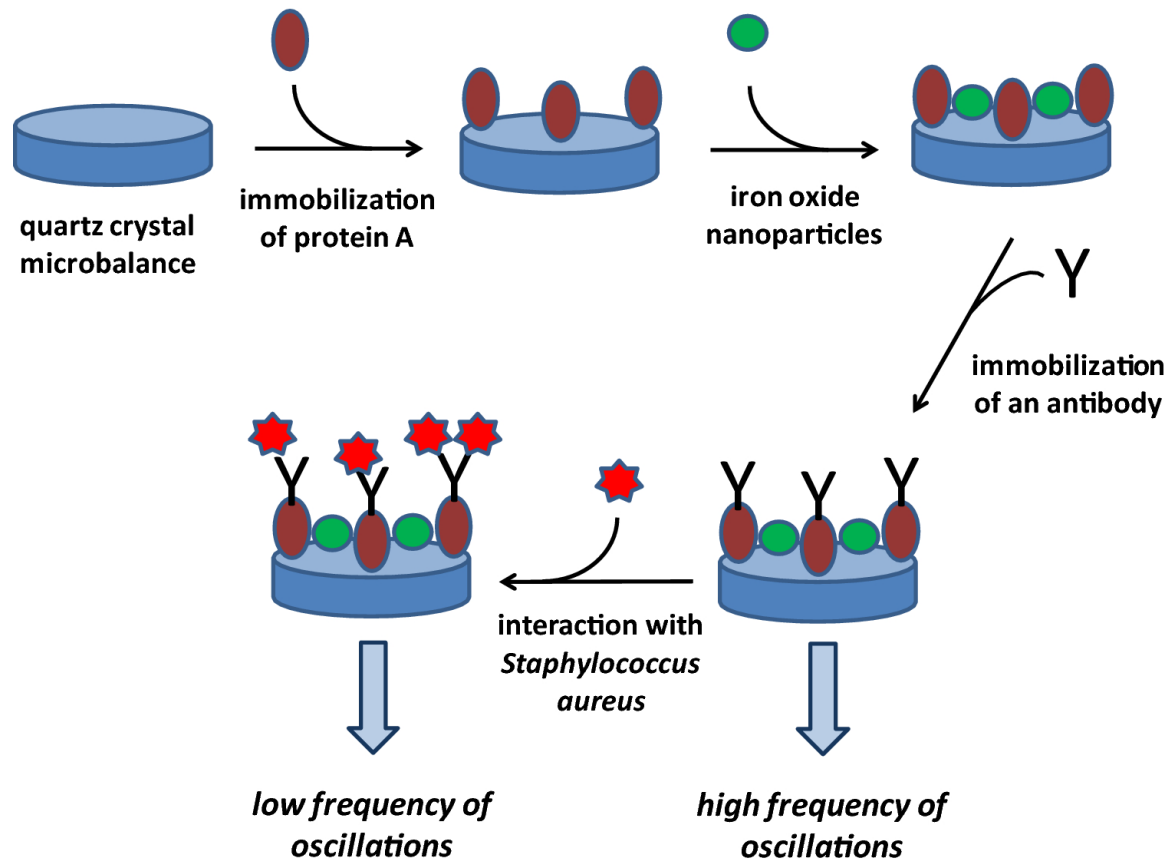
https://www.researchgate.net/profile/Noor_Azuan_Abu_Osman/publication/258052187/figure/fig8/AS:297464780345353@1447932538931/Typical-advantage-ie-high-intense-signal-of-surface-modified-right-piezoelectric.png

Piezoelectric Immuno Biosensors

QCM technology

1. QCM is a label free and extremely mass sensitive device.
2. It measures the mass variation per unit area by measuring the **resonance**.
3. The resonance is disturbed by the **addition or removal of a small mass**.
4. A **change in the resonance** is considered as a signal for target detection.

Piezoelectric Immuno Biosensors



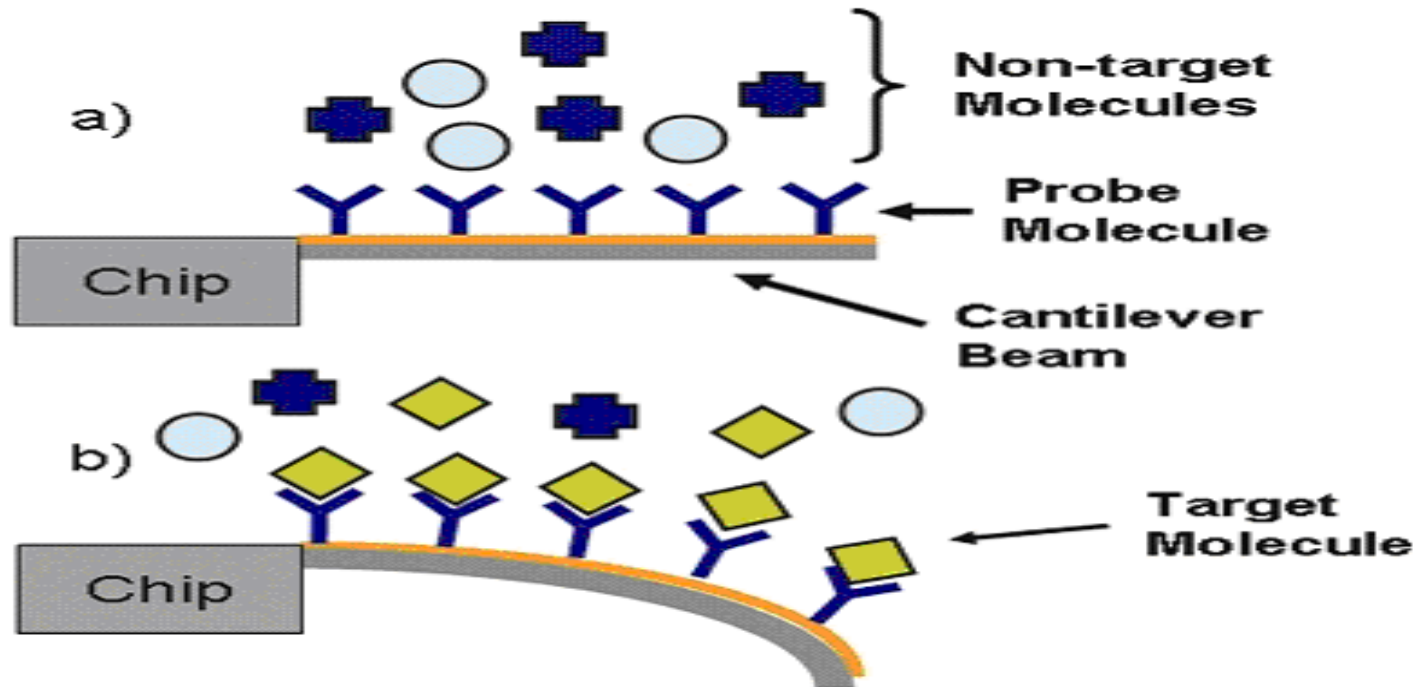
https://media.springernature.com/original/springerstatic/image/art%3A10.1007%2Fs11696-019-00889-5/MediaObjects/11696_2019_889_Fig1_HTML.png

Piezoelectric Immuno Biosensors

Microcantilevers

1. A cantilever is a rigid structural element, such as a **beam or a plate**.
2. It is attached at one end to a support and the other end is free.
3. For use in Biosensors cantilever are **commercially available** made up of different natural and synthetic PZ materials such as , **silicon nitride, polymers or quartz**.

Piezoelectric Immuno Biosensors



Microcantilever working in static mode. The formation of the antigen-antibody complexes provokes a surface stress and, consequently, a deflection of microcantilever, which is detected optically.

http://archive.nmji.in/archives/Volume_19_1_january_february2006/review%20series/3.gif

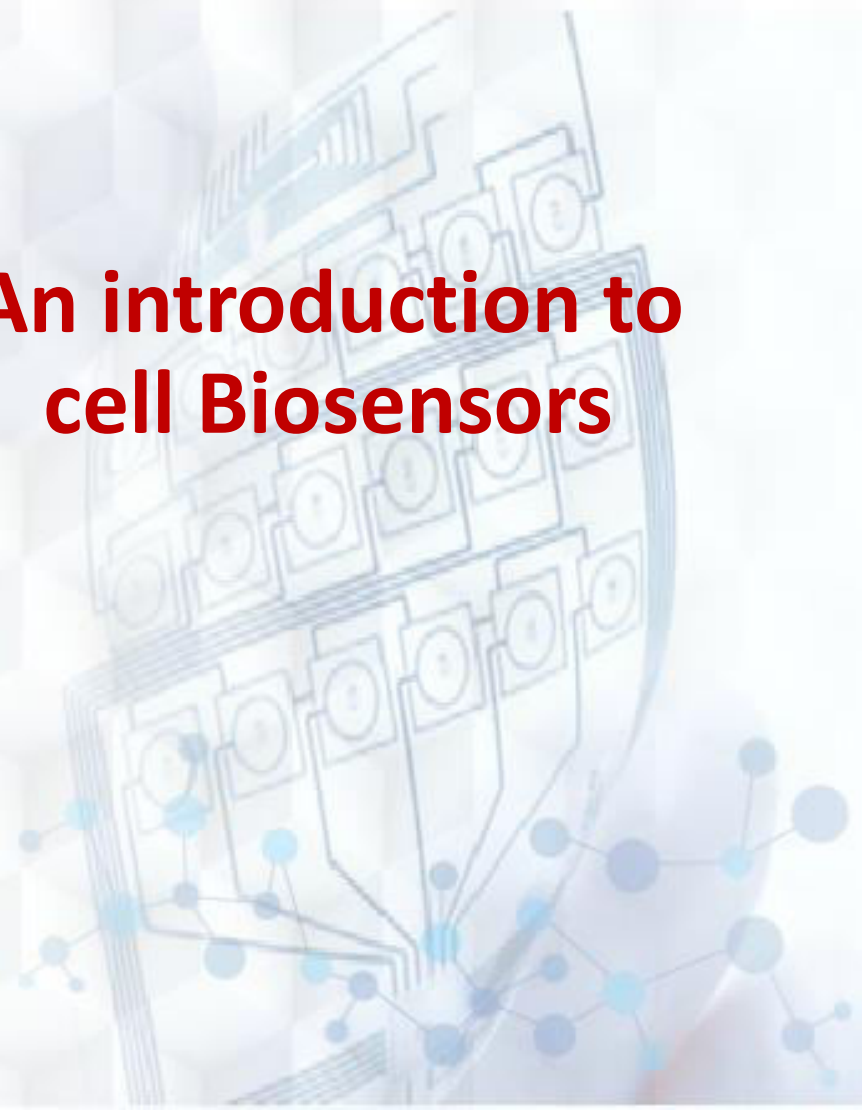
Piezoelectric Immuno Biosensors

Summary

1. Piezoelectric material is used in immune sensors for the detection of target molecule.
2. BRE is coated on the membrane attached with the piezo electric material.
3. The formation of Ab and AG complex **induce a vibration** that is detected by QMC balance or cantilever based physical component.
4. The technique is highly sensitive.

Biosensors

An introduction to cell Biosensors



Introduction to Cell Biosensors

Introduction

A biosensor that employs whole cell **in viable** or **non viable** form is known as a cell Biosensor.

There are at least four types of cells that can be used as a Biorecognition element in cell Biosensors.

1. **Microbial cell**
2. **Organelle**
3. **Tissue cells**
4. **Stem cells.**

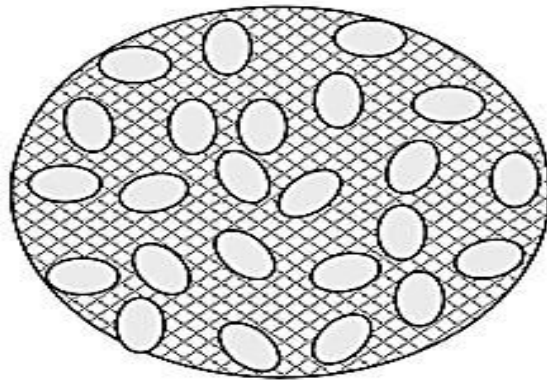
Introduction to Cell Biosensors

Immobilization

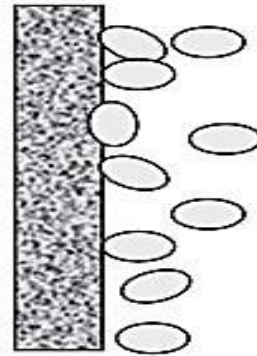
The most important step in the development of a biosensor is the immobilization of whole cells on a matrix. Mild conditions are used to attach the cells to the matrix.

1. **Entrapment**, it bypasses the use of drastic chemical conditions.
2. **Self Aggregation**
3. **Adsorption on solid support**
4. **Mechanical confinement**

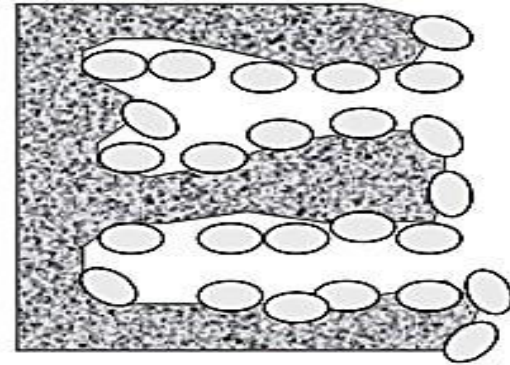
Introduction to Cell Biosensors



(I) Entrapment within a porous matrix

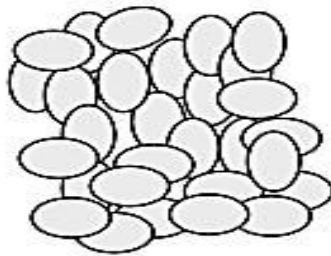


(a)

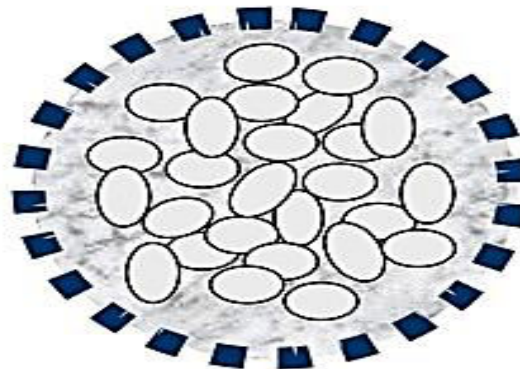


(b)

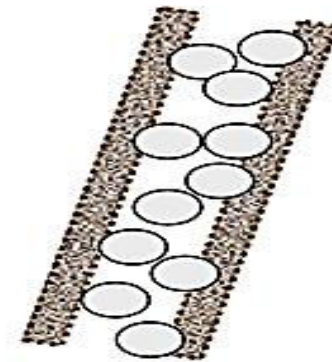
(II) Attachment or adsorption on solid carrier



(III) Self-aggregation



(a)



(b)

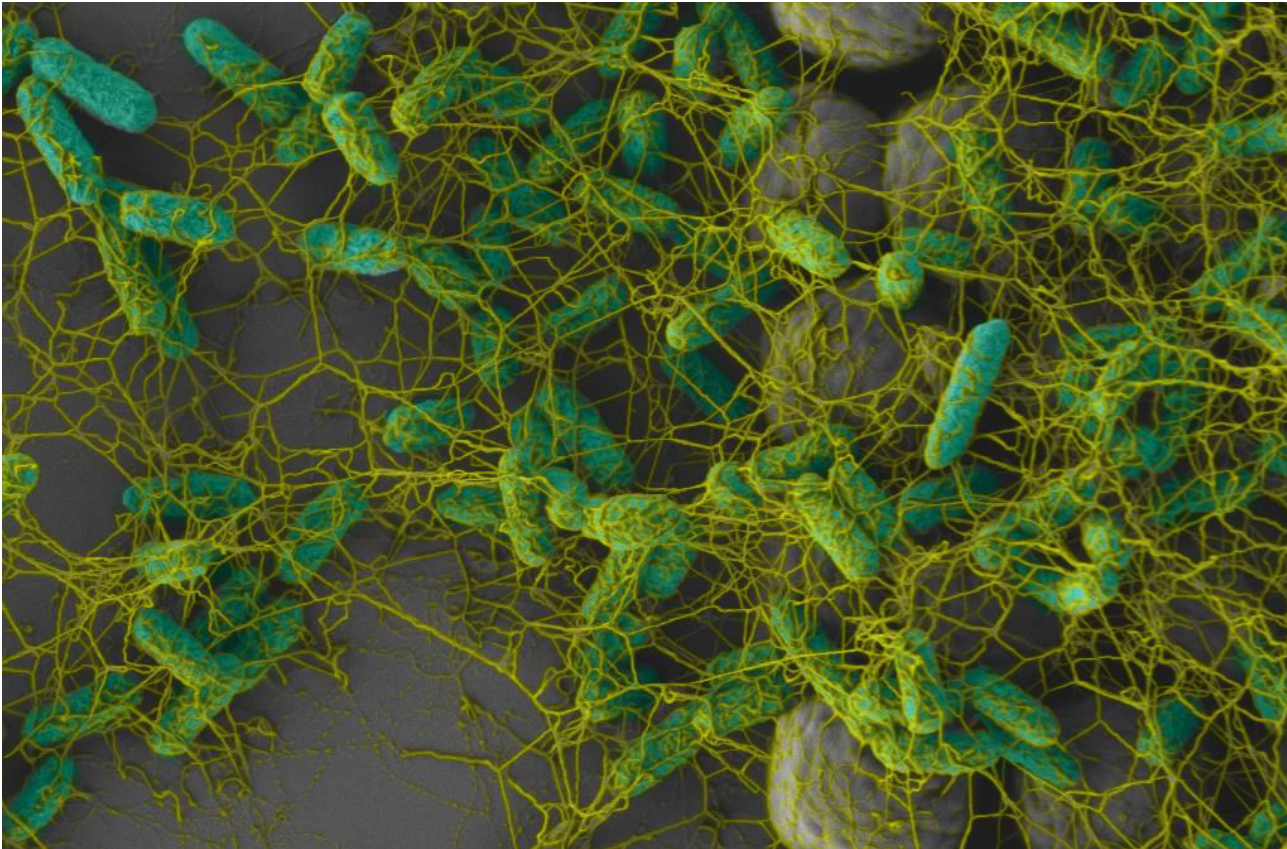
(IV) Mechanical containment behind a barrier

Introduction to Cell Biosensors

Biofilm for immobilization of biological Cells

1. Biofilms are formed by bacterial aggregates.
2. These **aggregates** held **together** by a **mucus like matrix** of **carbohydrate** which **adheres** to a surface.
3. A biofilm has a nutrient rich environment **complex 3 dimensional** structure that created an **internal protective environment** for bacterial cells.

Introduction to Cell Biosensors



E.coli produce biofilms and with the help of **pili** and **flagella** they form weak bonds and survive the harsh environment.

<https://scx1.b-cdn.net/csz/news/800/2013/clingingtoqr.jpg>

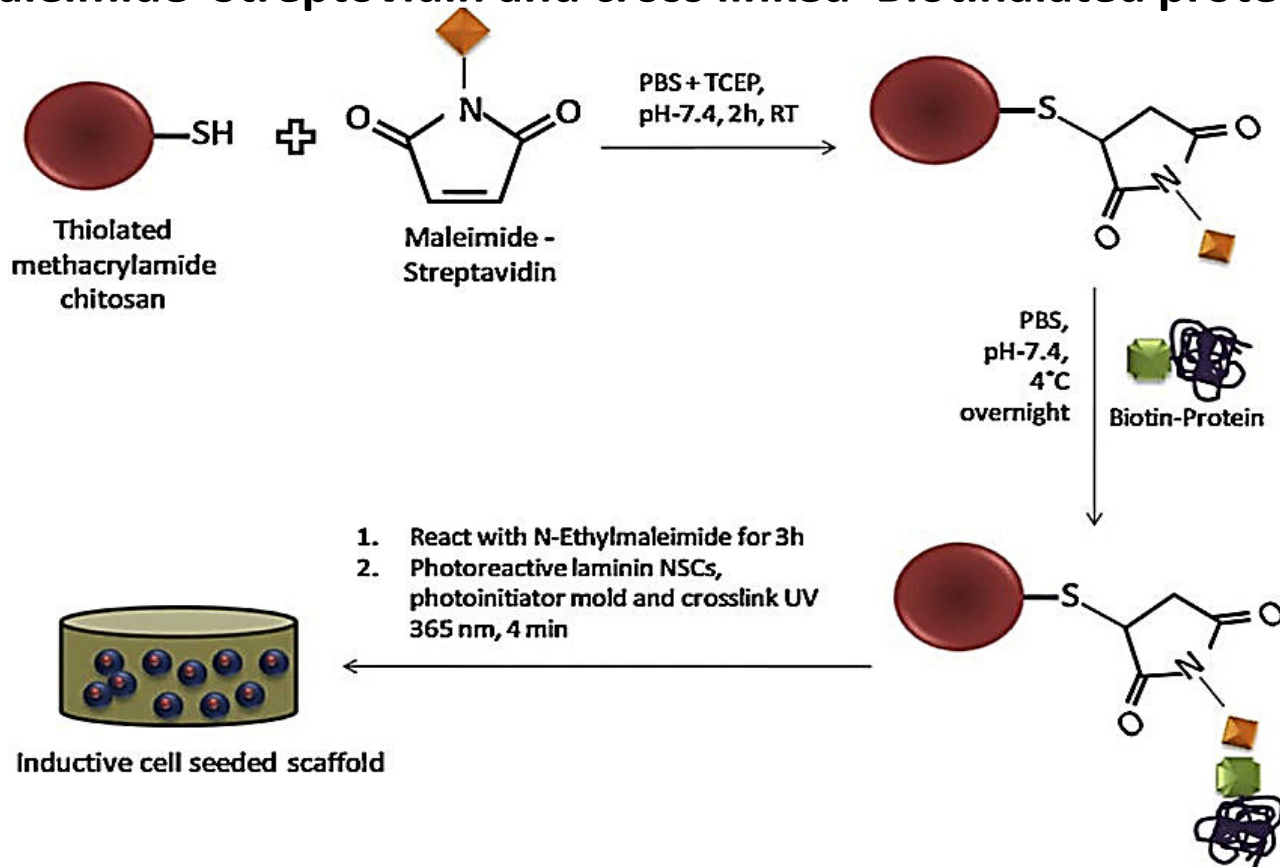
Introduction to Cell Biosensors

Scaffolds for Cell immobilization

1. In biology scaffold is a **structure providing support**.
2. In labs Scaffolds are specifically designed for the **immobilization of whole biological cells** for different purposes.
3. A variety of **natural and synthetic materials** is available and used for the fabrication of novel scaffolds.

Introduction to Cell Biosensors

Fabrication of Scaffold based on Thiolated methacrylamide chitosan and Maleimide Streptavidin and cross linked Biotinalated protein .



https://www.researchgate.net/profile/P_R_Sivashankari/publication/295244366/figure/fig4/AS:560116303962112@1510553541007/Schematic-representation-for-immobilization-of-growth-factors-onto-chitosan-hydrogel-57.png

Introduction to Cell Biosensors

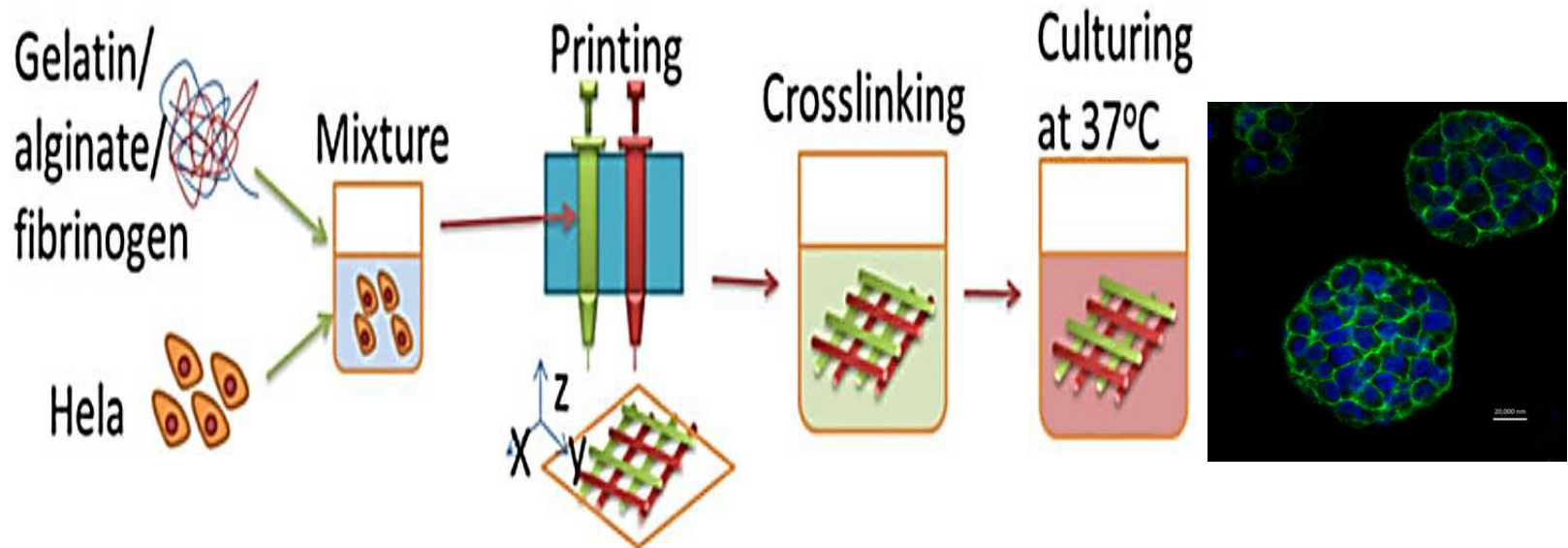
ffort By: AARISH ALI

Three D printing of Hella cells

1. **Culturing of cells** in petri plate is called **2D printing**.
2. 3 D model consists of a scaffold made up of three fibrous proteins (**gelatin, alginate, and fibrin**) corresponding to the **extracellular matrix** (support structure) of a tumor.
3. The matrix was **coated with Hela cervical** cancer cells.
4. This 3D model to **mimics the environment for tumor**.

Introduction to Cell Biosensors

3D printing of Hella cell lines



<https://www.kurzweilai.net/images/3D-cell-printing.jpg>

<https://www.kurzweilai.net/images/3D-cellular-morphology-on-day-8-512x512.png>

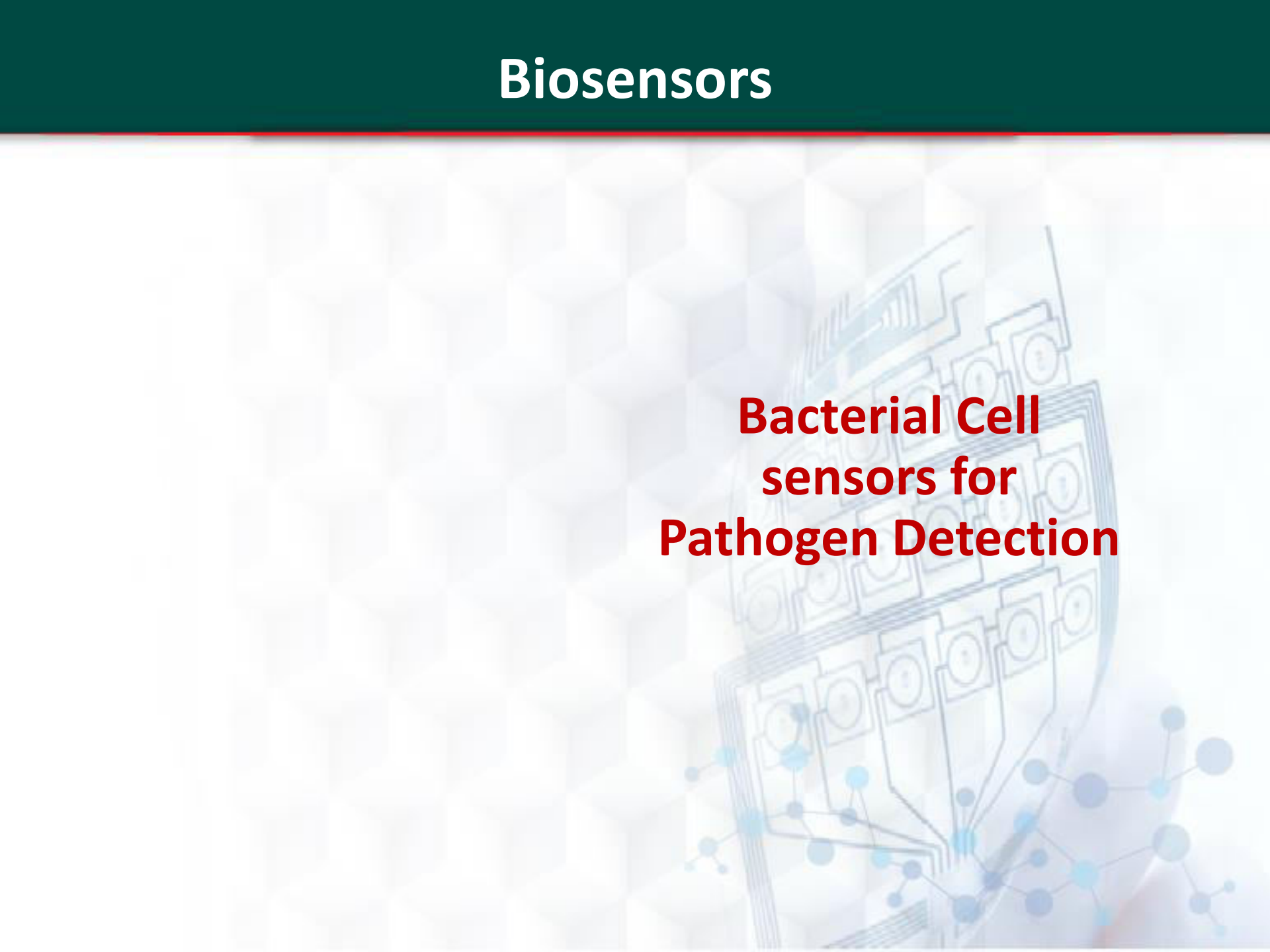
Introduction to Cell Biosensors

Summary

1. In cell biosensors whole Biological cells (**viable or non viable**) are used as Bio recognition element.
2. In a cell Biosensor Cell **immobilization is the physical confinement of intact cells** to a certain defined area, **with the of desired activity**.
3. **Successful immobilization is the first step** in the **development of biosensor**.

Biosensors

**Bacterial Cell
sensors for
Pathogen Detection**

The background of the slide features a light gray hexagonal grid pattern. Overlaid on this is a faint, stylized illustration of a circuit board with various components and lines. In the bottom right corner, there is a molecular structure diagram consisting of blue and gray spheres connected by lines, representing a network or a chemical structure.

Bacterial Cell sensors for Pathogen Detection

Introduction

Generally two types of Biosensors are used for Cell Biosensors.

1. Impedimetric Biosensors

2. Optical Biosensors

Pathogens associated cases cause an estimated (per year)

- I. **1.7 million hospital-associated infections**
- II. **3.0 Million hospitalizations**
- III. **0.1 Million deaths each year.**

Examples: *E. coli* , *Salmonella*

Bacterial Cell sensors for Pathogen Detection

Justification.

1. Most Bacterial infections are often **misdiagnosed** as they are treated without **identification of pathogen**.
2. Incase of identification there is an **unacceptable delay** in the diagnosis.
3. This delays leads to further complications.
4. Biosensors offer a **rapid and cost-effective** method of bacterial detection

Bacterial Cell sensors for Pathogen Detection

Current Methods

Current methods of bacterial detection rely upon laboratory-based techniques such as

1. Cell culture
2. Microscopic analysis
3. Biochemical assay.
4. These procedures are **time-consuming, costly** and require specialist equipment and trained users.
5. Biosensor can identify the pathogen in short time.

Bacterial Cell sensors for Pathogen Detection

Gram Positive Bacteria

Gram-positive bacteria retain the violet Gram stain due to their **thick peptidoglycan** layer on the outside of the cell membrane

1. Lipids
2. Surface proteins
3. Glycoproteins

Examples *Streptococcus*,

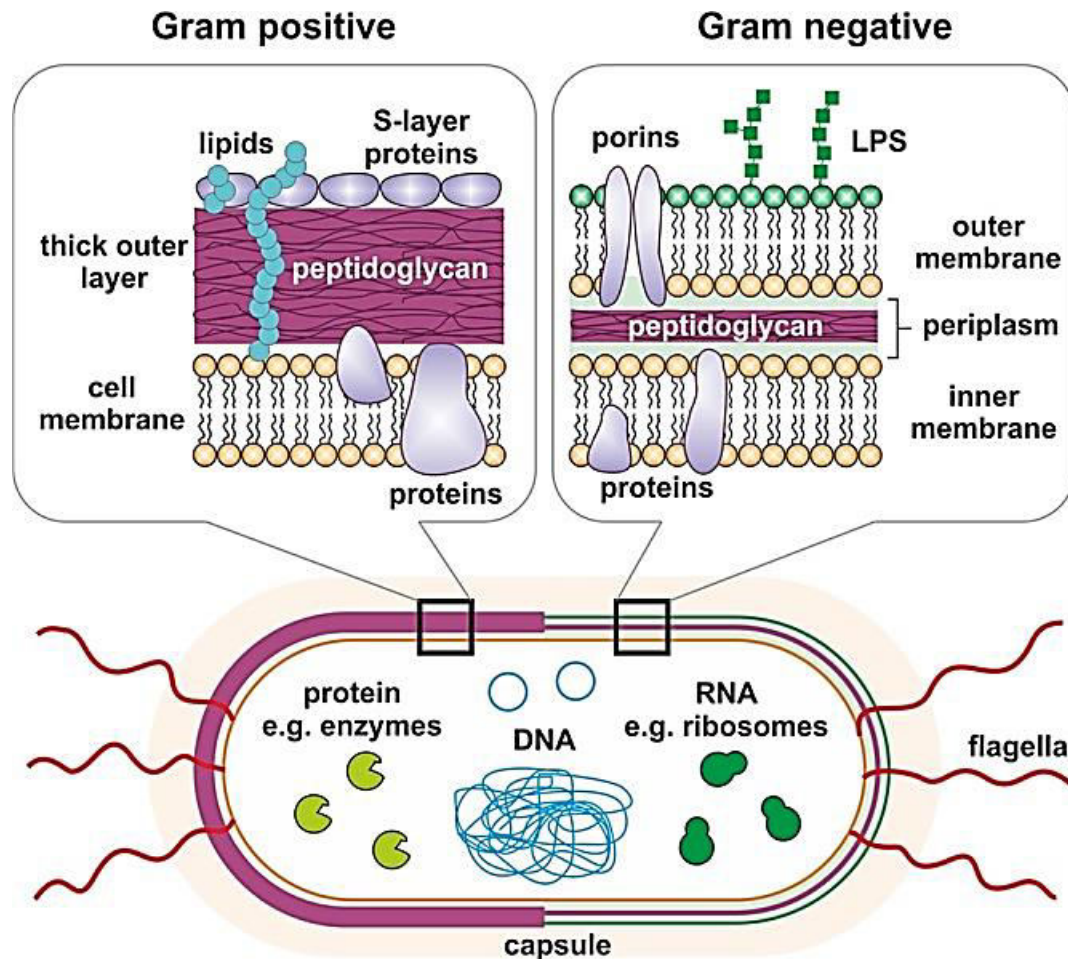
Classification of the bacteria is based on the thickness of the wall.

Bacterial Cell sensors for Pathogen Detection

Gram Negative Bacteria

1. Gram-negative bacteria do not take up the stain, as their thinner **peptidoglycan** layer is not exposed.
2. The outer lipid membrane **lipopolysaccharides** (LPS), which act as endotoxins and initiates a strong immune response in humans.
3. **Porins** are proteins that are located in the outer membrane of Gram-negative bacteria . *Escherichia coli*, *Salmonella*, *Shigella*

Bacterial Cell sensors for Pathogen Detection



<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4135896/bin/zcm9990924740002.jpg>

Bacterial Cell sensors for Pathogen Detection

Target for Biorecognition

1. A variety of **surface antigens** presented on the cell envelopes of whole bacteria, including **proteins, glycoproteins, lipopolysaccharides, and peptidoglycan**, can act as targets for bio-recognition.
2. Polyclonal antibodies raised **against specific bacterial strains** are the most commonly used for whole bacterial cell detection.

Bacterial Cell sensors for Pathogen Detection

Summary

1. A bacterial biosensor is an analytical device. It **integrates bacterium** with a physical transducer.
2. For bio sensing the **architect of the bacterial** cell is important.
3. Mostly surface antigen and polyclonal antibody are used for detection.
4. In bacterial biosensors **optical** or **impedimetric** techniques are used.

Biosensors

**Cell Biosensor
with reporter gene**



Cell Biosensor with Reporter Gene

Introduction

1. Reporter gene is a gene that is **attach to a regulatory sequence of another gene** of **interest** in bacteria, cell culture, animals or plants.
2. **The function of reporter gene** is to **help** in the **detection** of target molecule.
3. The reporter genes are **marker genes** to measure the amount of expression.
4. A variety of reporter gene is used in studies.

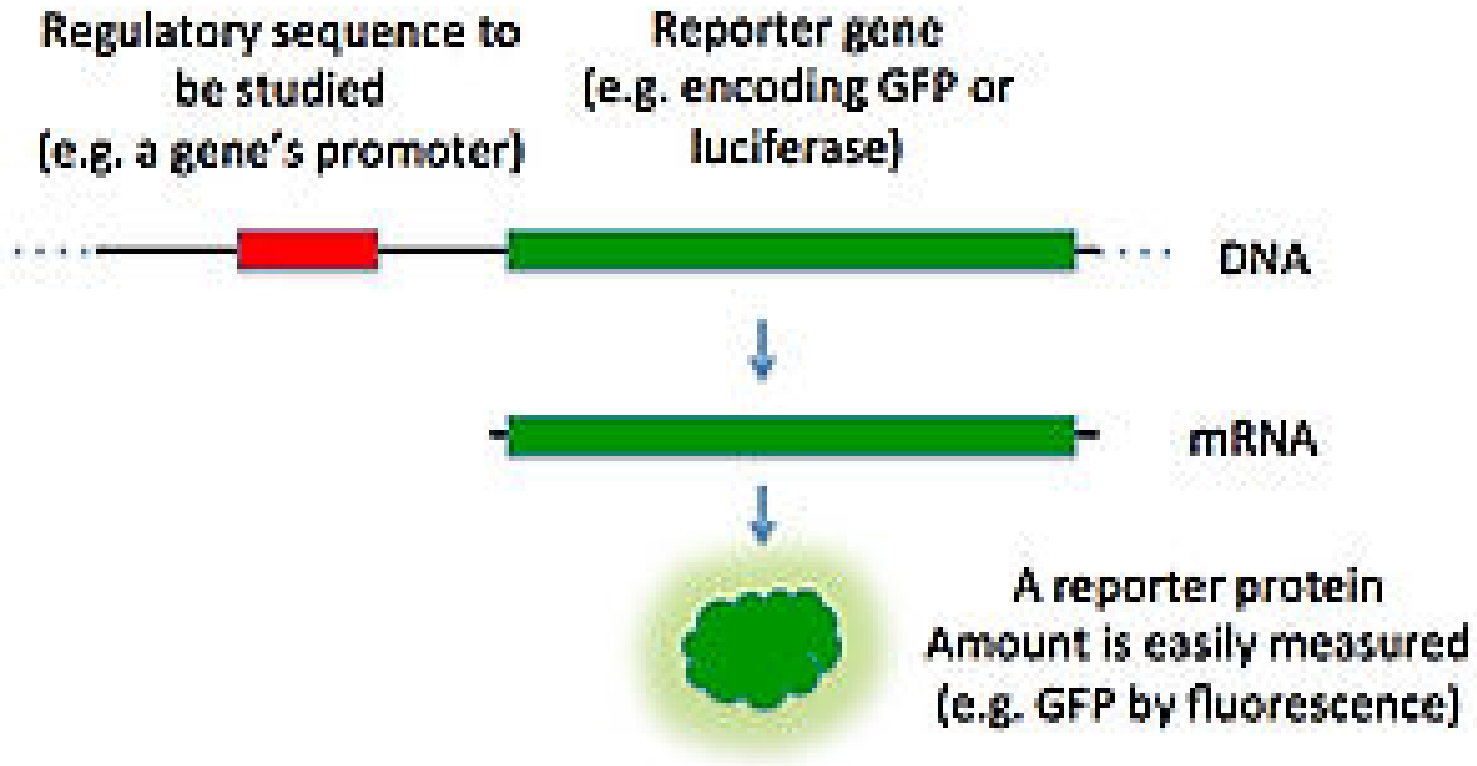
Cell Biosensor with Reporter Gene

Few common reporter Gene

The most commonly used reporter gene systems include:

1. **LacZ Reporter Gene**
2. **Luc Reporter Gene**
3. **Gfp Reporter Gene**
4. **Egfp gene** (Enhanced Green Fluorescence Protein)
5. **CAT Chloramphenicol Acetyltransferase Reporter gene**

Cell Biosensor with Reporter Gene



https://upload.wikimedia.org/wikipedia/commons/thumb/4/46/Reporter_gene.png/295px-Reporter_gene.png

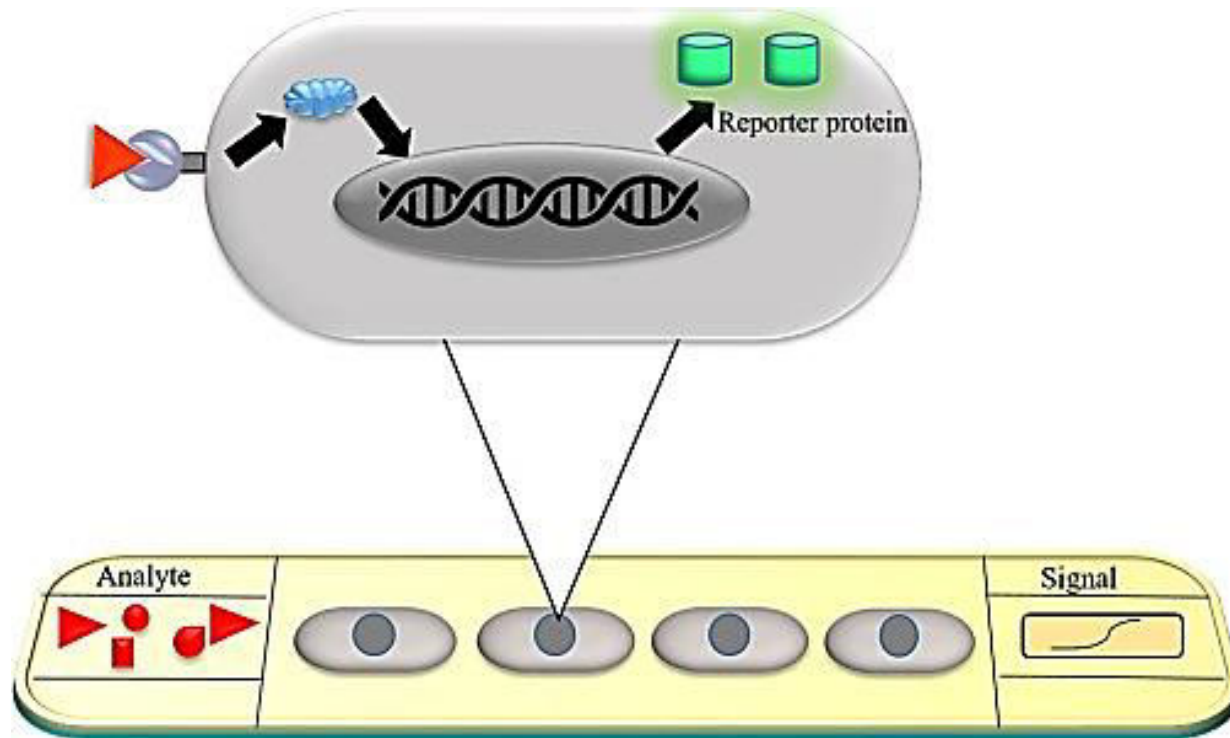
Cell Biosensor with Reporter Gene

Green fluorescent protein

1. The **protein** is coded for by a **single gene**.
2. Green fluorescent protein (GFP) is a protein composed of **238 amino acid residues**.
3. It has a **molecular weight** of 26.9 kDa.
4. The protein exhibits **bright green fluorescence** when exposed to light in the **blue to ultraviolet range**.

Cell Biosensor with Reporter Gene

Green Fluorescence Protein reporter Gene



<https://ars.els-cdn.com/content/image/3-s2.0-B9780128146798000133-f13-01-9780128146798.jpg>

Cell Biosensor with Reporter Gene

crtA reporter gene

A reporter gene crt A is responsible for **carotenoid** synthesis.

1. When applied to a sample, the crtA-based whole cell-based biosensors.
2. The change in the color of culture **media from yellow to red without** substrate is a signal for biorecognition.
3. It is considered as a good choice for rapid detection.

Cell Biosensor with Reporter Gene

luc gene and lac Z gene

Luc gene and lac Z gene requires specific substrates to generate an optical signal.

1. **The firefly luciferase (luc)** reporter has high sensitivity can detect 2fg.
2. Luc gene has a linear broad range for detection.
3. lacZ reporter gene for beta galactosidase, requires **chemiluminescent and electrochemical** substrates.

Cell Biosensor with Reporter Gene

Summary

1. A reporter gene is a regulatory sequence of an other gene that is attached to the gene of interest.
2. The product of the reporter gene can **easily identified and measured**.
3. Gfp reporter gene does not require any substrate.
4. ctr A gene changes the color of the medium due to the production of carotenes.

Biosensors

**Bacterial cell
Biosensor with
regulatory protein**

The background of the slide features a complex, layered design. It includes a grid of light gray hexagons, a faint blue circuit board pattern, and a molecular structure composed of blue and gray spheres connected by lines. The text is overlaid on this background.

Cell Biosensor with Regulatory Proteins

Introduction

Regulatory proteins regulate the level of expression of a proteins. There are two types of regulatory proteins.

1. **Repressor proteins** bind to **operators or promoters**, preventing RNA polymerase from **transcribing RNA**.
2. **Activator proteins** bind to **a site on the DNA molecule** and causes an increase in transcription of a nearby gene.

Cell Biosensor with Regulatory Proteins

Metallo-regulatory proteins

The Metallo regulators are proteins that **bind metals** .

A panel of metalloregulatory proteins **controls the expression of genes** encoding

- 1. Membrane transporters**
- 2. Metal trafficking proteins.**

These metal sensors **manage**

- **Metal homeostasis**
- **Metal resistance.**

Cell Biosensor with Regulatory Proteins

Functioning of metallo proteins.

The metal ion(s) after binding with the gene promoter can induce **different conformational changes**.

- 1. Activates** the promoter
- 2. Inhibits** the promoter

The **binding could distort the promoter structure** and can convert a **poor promoter to a strong one**.

Cell Biosensor with Regulatory Proteins

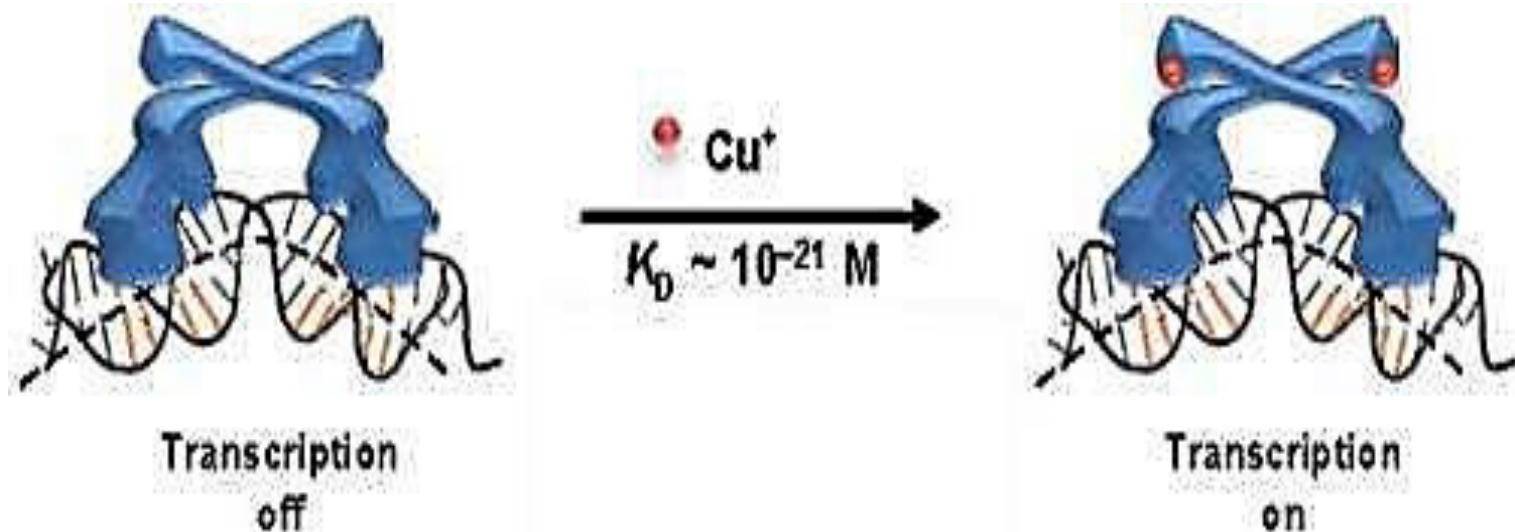
Cu ions binding proteins.

The bacteria can **detect the Cu ions, and prevent Cu toxicity by producing a protective protein.**

1. When Cu ions enter into the bacterial cells, they bind with Cu responsive gene.
2. On bonding gene changes its **configuration.**
3. Gene for the **protective protein is transcribed** and the Cu ions are shove out of the cell.

Cell Biosensor with Regulatory Proteins

Functioning of metallo Regulatory proteins.



<https://scx1.b-cdn.net/csz/news/800/2012/proteinsbarg.jpg>

Cell Biosensor with Regulatory Proteins

Detection of metal ions by using *Salmonella* **GolS** sensor

1. *Salmonella* employs a cytoplasmic **metal binding** protein, **GolB**.
2. This protein allows to detect the **presence of gold salts** in the environment.
3. **The expression** is controlled by a **regulatory protein called, GolS**, which is highly selective for Au ions.

Cell Biosensor with Regulatory Proteins

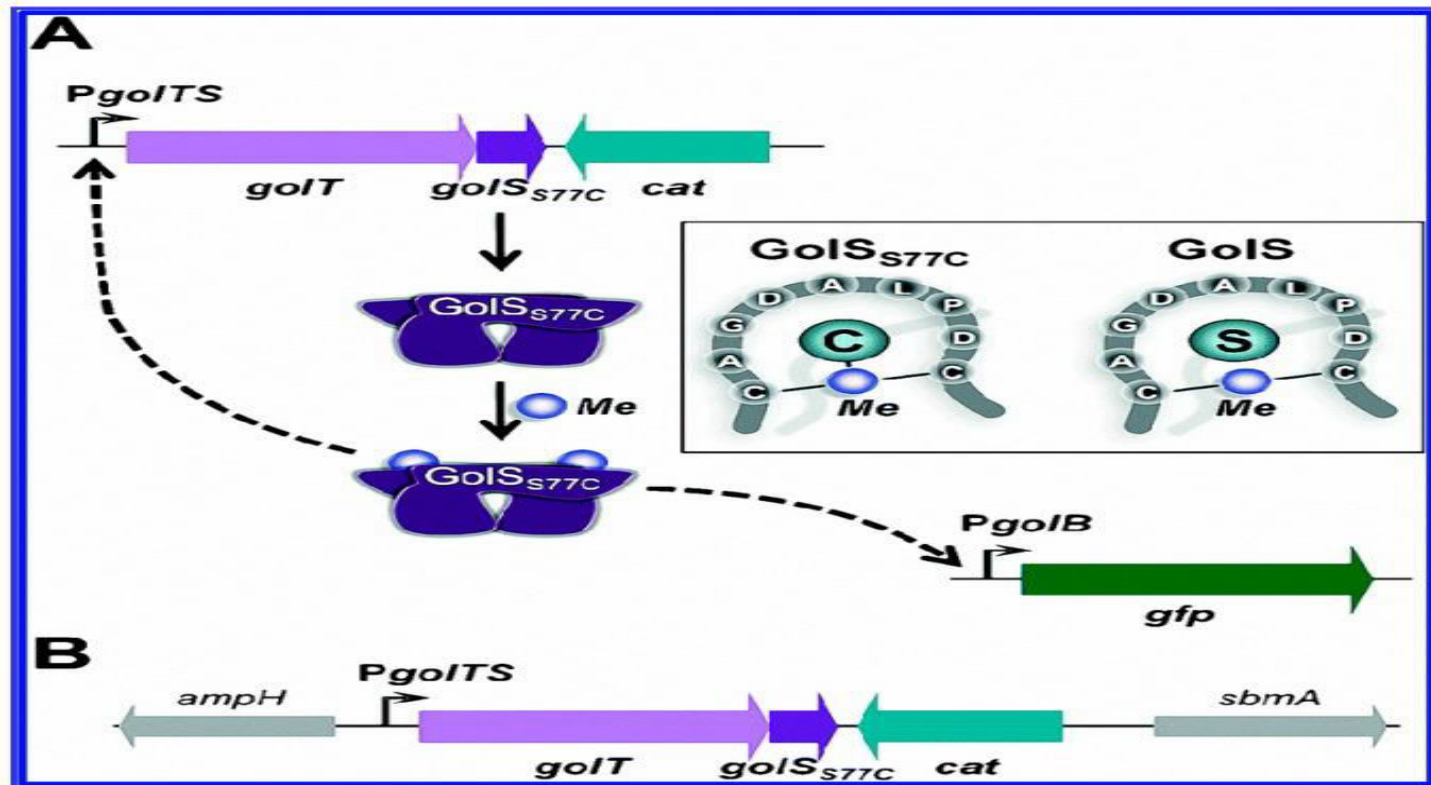
Gold sensor Protein for metal detection

The introduction of a single amino acid at the position 77 in the GolS protein.

The mutation increases the efficiency and protein was able to detection of

1. Mercury (Hg)
2. Lead (Pd)
3. Cadmium (Cd)
4. Gold (Au) ions.

Cell Biosensor with Regulatory Proteins



<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5539819/bin/sensors-17-01623-g003.jpg>

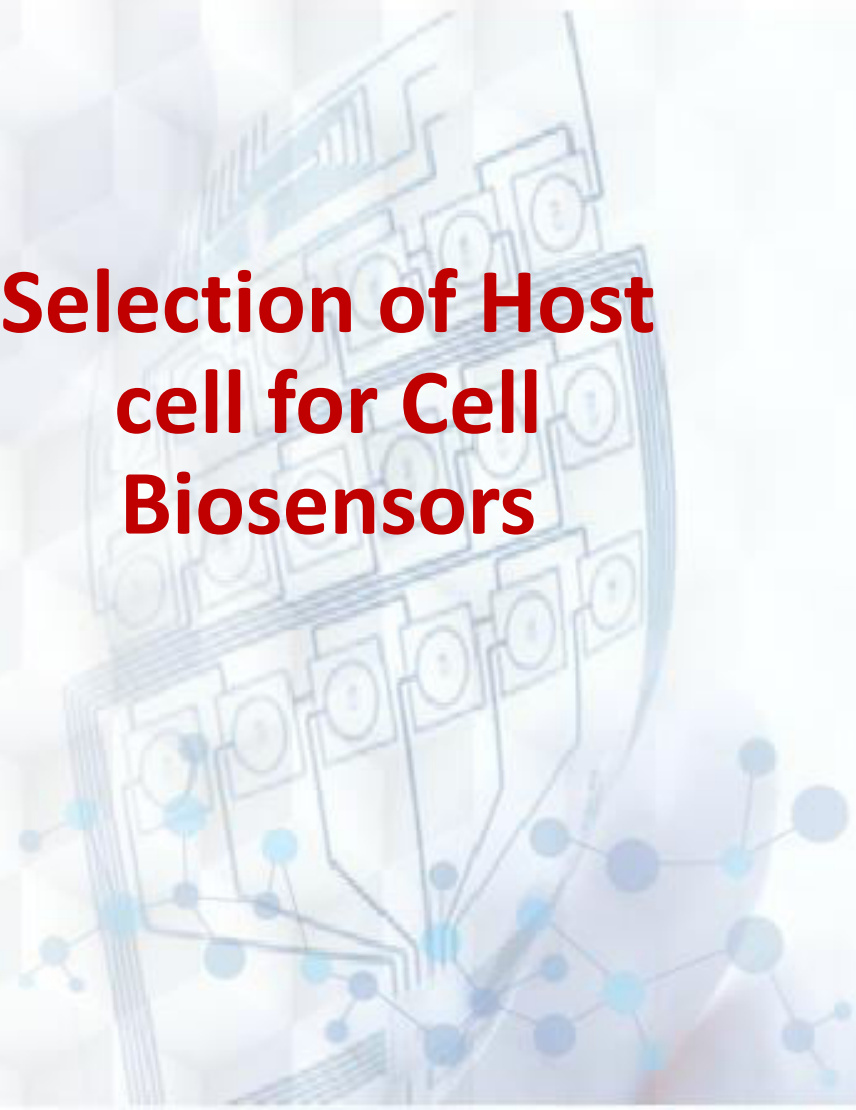
Cell Biosensor with Regulatory Proteins

Summary

1. The **expression of regulatory proteins** is used as a **signal** for the detection of **Analyte**.
2. A regulatory **gene *GolS*** present in *Salmonella* for the **detection of gold** in water is employed for the detection of metal ions.
3. The **mutated *GolS*** gene was expressed in ***E.coli*** and the whole cell was used for the detection of **Hg, Pb, Cd and Au ions** in aqueous samples.

Biosensors

Selection of Host cell for Cell Biosensors



Selection of Host Cell for Cell Biosensor

Introduction

The selection of the type of host cell is important, for the performance of whole cell biosensor. It can effect many parameters including

1. Specificity
2. Sensitivity
3. Reproducibility
4. Reliability
5. Capacity
6. Time-response,
7. Background noise

Selection of Host Cell for Cell Biosensor

Cell Biosensor to detect Pollution in sea water

Two bacterial strains could be used for pollution sensing

1. *Alcanivorax borkumensis*
2. *Pseudomonas putida*

A.borkumensis shows lower tolerance towards **pollutants**, but a **higher** tolerance towards **salinity**.

Pseudomonas putida has highest solvent tolerance and can detect the **pollution**.

Selection of Host Cell for Cell Biosensor

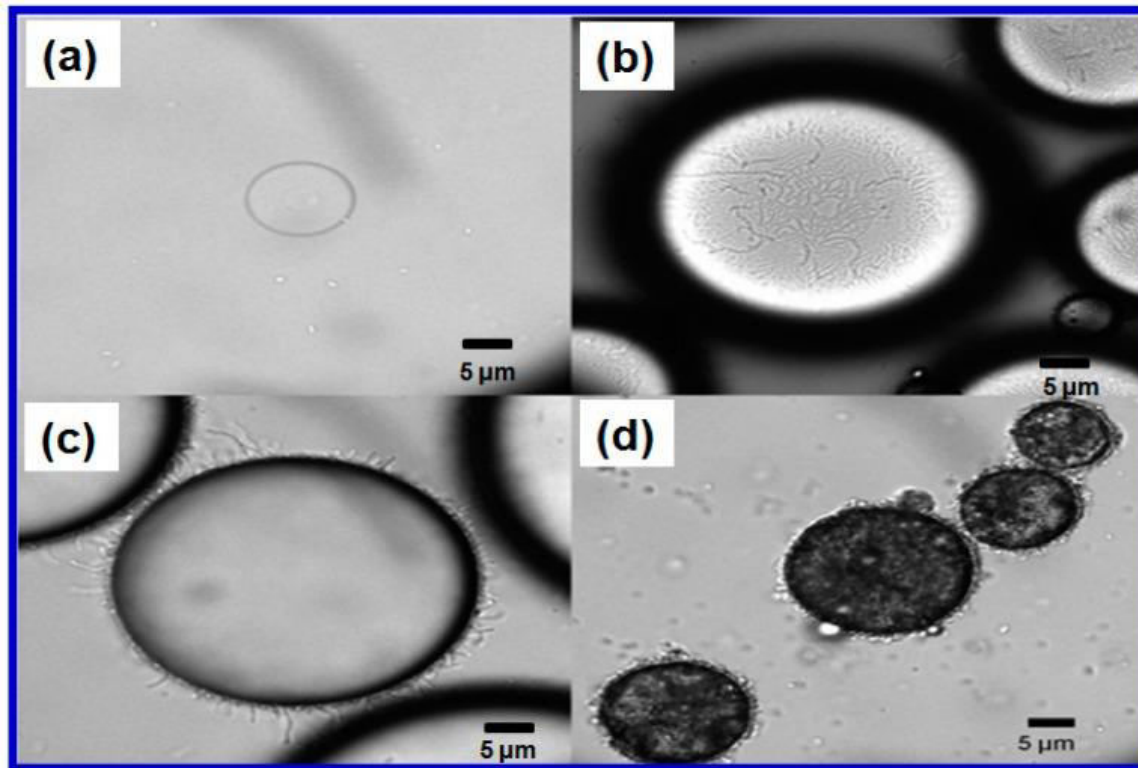
Biosensor for oil pollution.

For developing a biosensor to detect the oil pollution in sea, a **bacterial** strain should be selected that is a

1. A part of **marine microbial Ecosystem**
2. It could **resist** the composition of sea water.
3. it should have a gene for the **detection of oil**.

Acenatobacter baylyi is a **natural adhesive** to oil water interphase.

Selection of Host Cell for Cell Biosensor



https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5539819/bin/sensor_s-17-01623-g004.jpg

(a) Poor accessibility and emulsifying capability of *E. coli* (the circle) towards oil droplets. Good affinity and emulsifying capability of *Acinetobacter baylyi* (the circles) towards the surface of (b,c) the mineral oil droplets; and (d) the curde oil droplets

Selection of Host Cell for Cell Biosensor

Genetic Engineering of the Host cells

1. Selection and engineering of host cell is based on the requirements for the detection of target molecule.
2. The tailored organism give a **quantifiable signal** in the presence of an Analyte.
3. In Environment these bacterial metal-sensor are used for **specificity and sensitivity**.

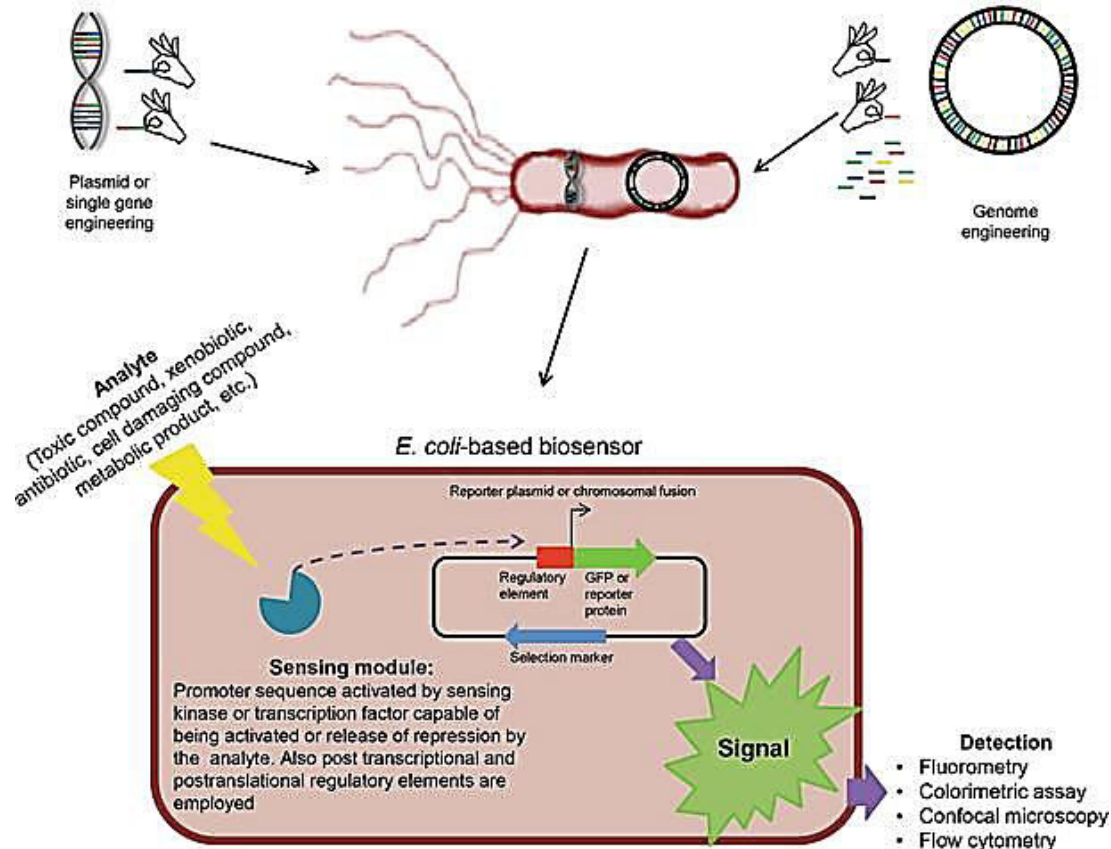
Selection of Host Cell for Cell Biosensor

E.coli for Ni ion detection

1. A biosensor was designed for the detection of metal (Ni) pollution in drinking water.
2. A series of different strains of *E.coli* was produced by recombinant DNA technology.
3. A cassette was prepared by fusing **Ni/Co metallo-regulator, promoter and luc reporter genes**.
4. *E.coli* (wild type) was proven best with ten times higher .

Selection of Host Cell for Cell Biosensor

Engineering of *E.coli* for Biosensing



<https://www.intechopen.com/media/chapter/53916/media/F1.png>

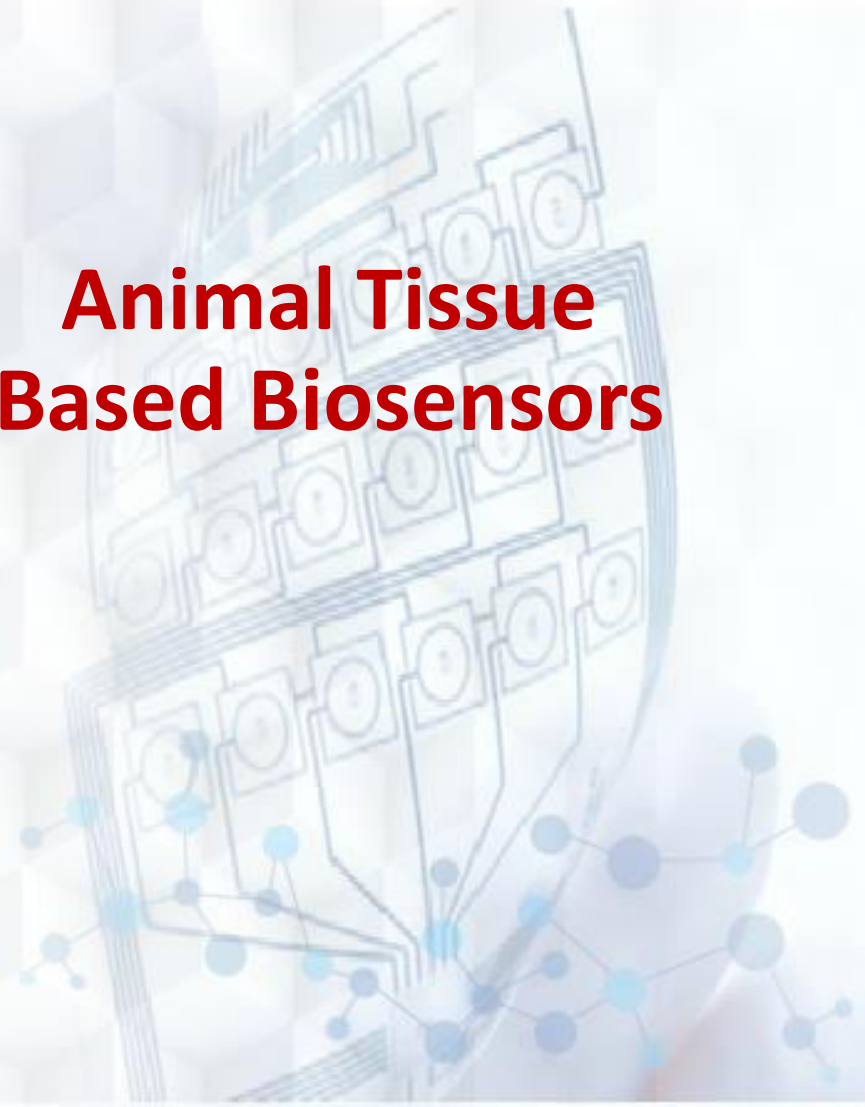
Selection of Host Cell for Cell Biosensor

Summary

1. Whole microbial cells are selected on the basis of their proteomics and genomics for use in Biosensor as a **sensing vehicle**.
2. The **choice of host bacteria** has a significant impact on the performance of a constructed whole cell-based biosensor.
3. Microbial cell are **tailored for the use as sensing vehicle** in Biosensors.

Biosensors

Animal Tissue Based Biosensors



Animal Tissue Based Biosensors

Introduction

Tissues are also used in biosensors as they contain large quantity of BRE (enzymes).

They offer a number of advantages

1. Easier immobilization
2. Higher activity
3. Stability
4. Low price
5. Presence of cofactors.

Animal Tissue Based Biosensors

Animal tissue based Biosensor

Biosensors based on tissue structures in living animals can be used to detect and measure **hormones, drugs, and toxins**.

Tissue -based biosensors can be formed from

1. Genetically modified cells or
2. Direct genetic modification

The objective is to determine the Analyte in sample..

Animal Tissue Based Biosensors

Biophotonics

Biophotonics provides the most versatile basis for tissue-based biosensors.

Light output from biosensor cells can be in the form of

1. Fluorescence
2. Bioluminescence.

Bioluminescence offers advantages of **not requiring an input source of light.**

Bioluminescence provides background signals in cell.

Animal Tissue Based Biosensors

BRET Assays

Bioluminescence resonance energy transfer (BRET), results from a **naturally occurring phenomenon**, involving

1. Resonance energy
2. Enzymatic donor and
3. Fluorescent protein acceptor.
4. Range 1-10 nm range

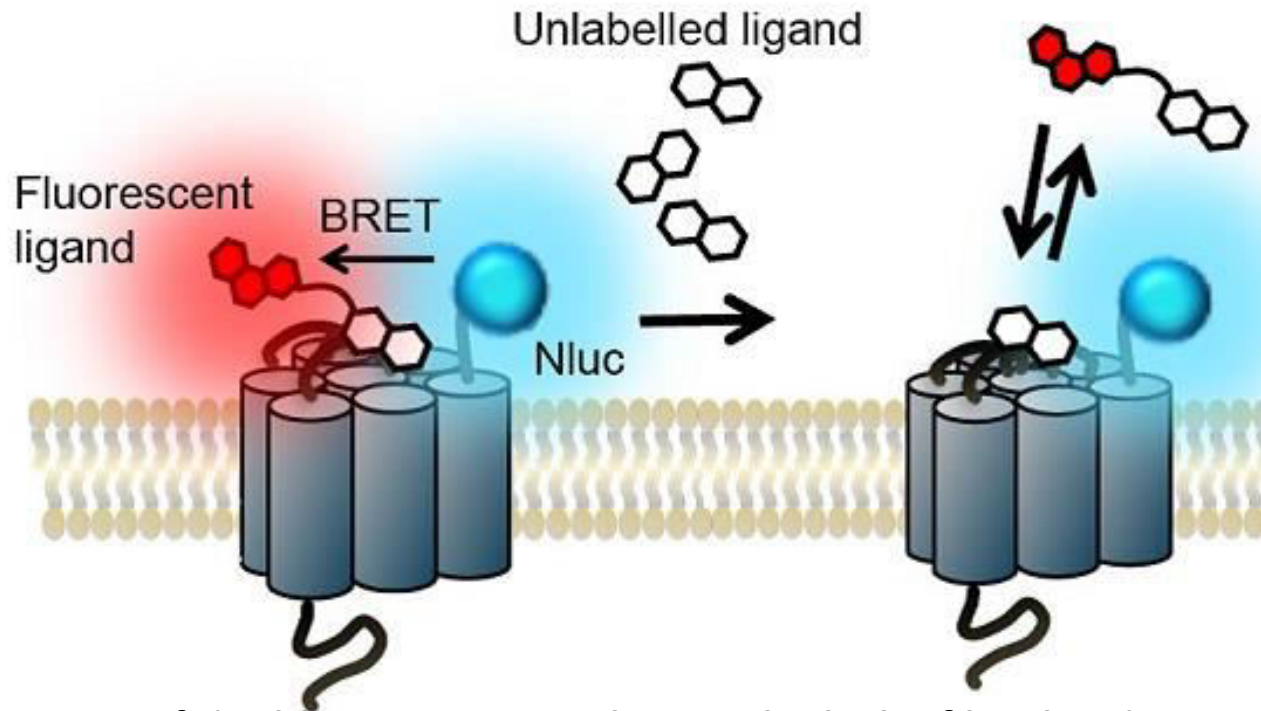
BRET an ideal platform to study **bio-molecular interaction** or physiological processes in living cells.

Animal Tissue Based Biosensors

Membrane protein - Drug interaction

1. **G protein-coupled receptors (GPCRs)** play a **key role** in **transmitting** cellular responses to **external stimuli**.
2. **Mostly (30% to 50%)** commercially available pharmaceuticals **target G protein-coupled receptors**.
3. BRET assay is used to detect the interaction of drug with membrane protein.

Animal Tissue Based Biosensors



The detection of the **blue emission** indicates the **lack of binding** between the ligand and the receptor. The detection of **red fluorescence**, which is caused by the excitation of the fluorescent indicates the **specific binding of the ligand** to the receptor.

<https://www.news-medical.net/image.axd?picture=2015%2f10%2f1-10.jpg&ts=20151027114729&ri=590>

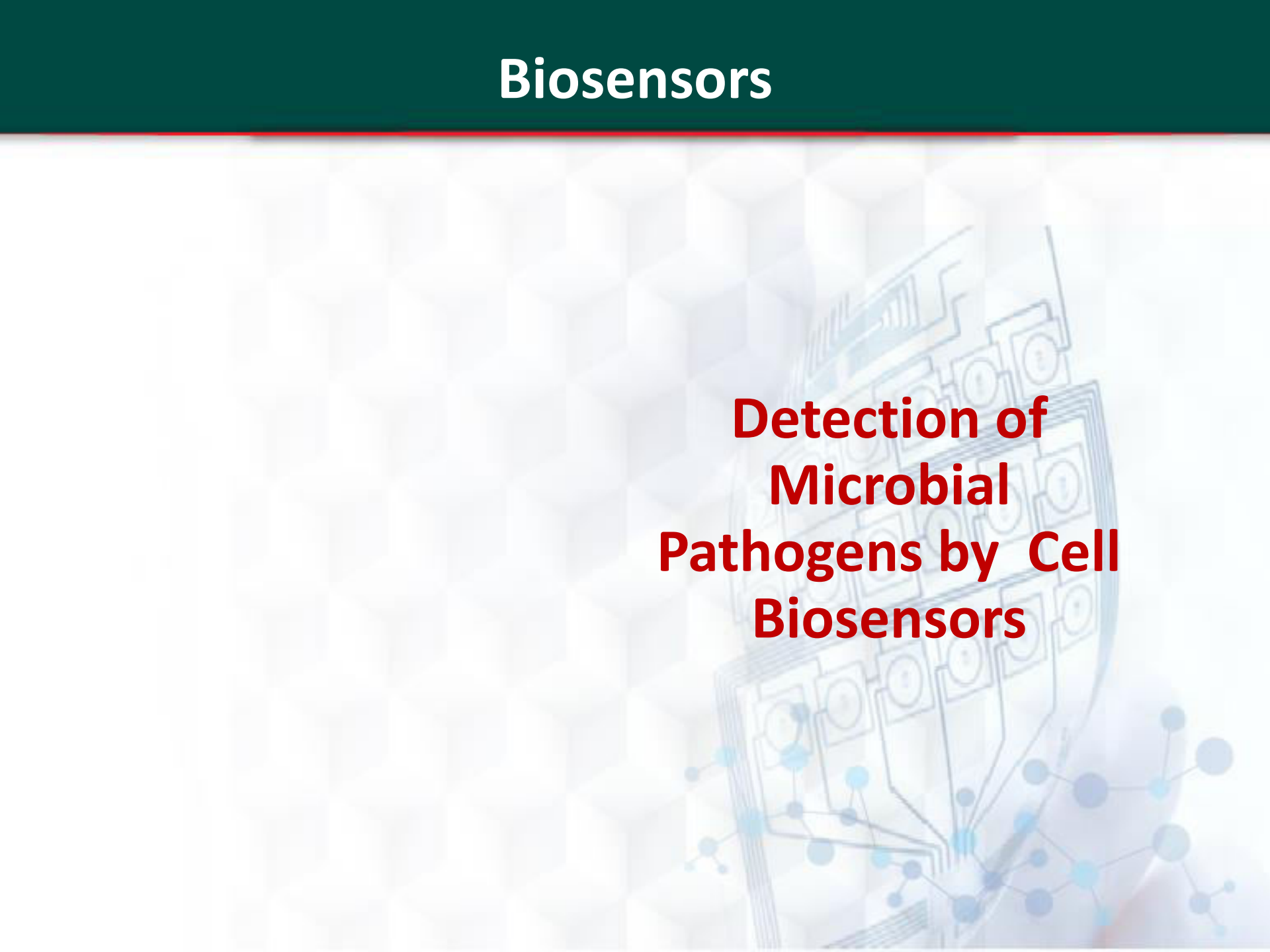
Animal Tissue Based Biosensors

Summary

1. In animal tissue based Biosensor animal tissues are used as BRE.
2. Tissues are easy to immobilize and retain the biological molecules in their intact form.
3. Biosensors are used for biomolecular interactions.
4. BRET is a technique that is used to monitor molecular interactions in tissue based biosensors.

Biosensors

**Detection of
Microbial
Pathogens by Cell
Biosensors**

The background of the slide is a light gray hexagonal grid. Overlaid on this grid is a faint, stylized illustration of a circuit board with various components and lines. In the bottom right corner, there is a molecular structure diagram consisting of blue spheres connected by lines, representing a network or a chemical structure.

Detection of Microbial Pathogens by Cell Biosensors

Introduction

1. Microbes can cause various diseases to different organisms.
2. Accurate and rapid detection of pathogens is required for timely treatment of the disease.
3. In cell based Biosensor several approaches are used for the rapid detection of pathogen.
4. Use of **virus as BRE** and **Quorum sensing** are two feasible approaches.

Detection of Microbial Pathogens by Cell Biosensors

Virus as Bioreceptor Element

Specific and direct detection of microbial pathogen could be made by using a virus as **Bio recognition element**.

1. Viruses are composed of a **protein core and nucleic acids**.
2. Virus **detects the host membrane** and **inject its DNA** inside the host.
3. By using the machinery of the host, virus replicate inside the host and causes infection.

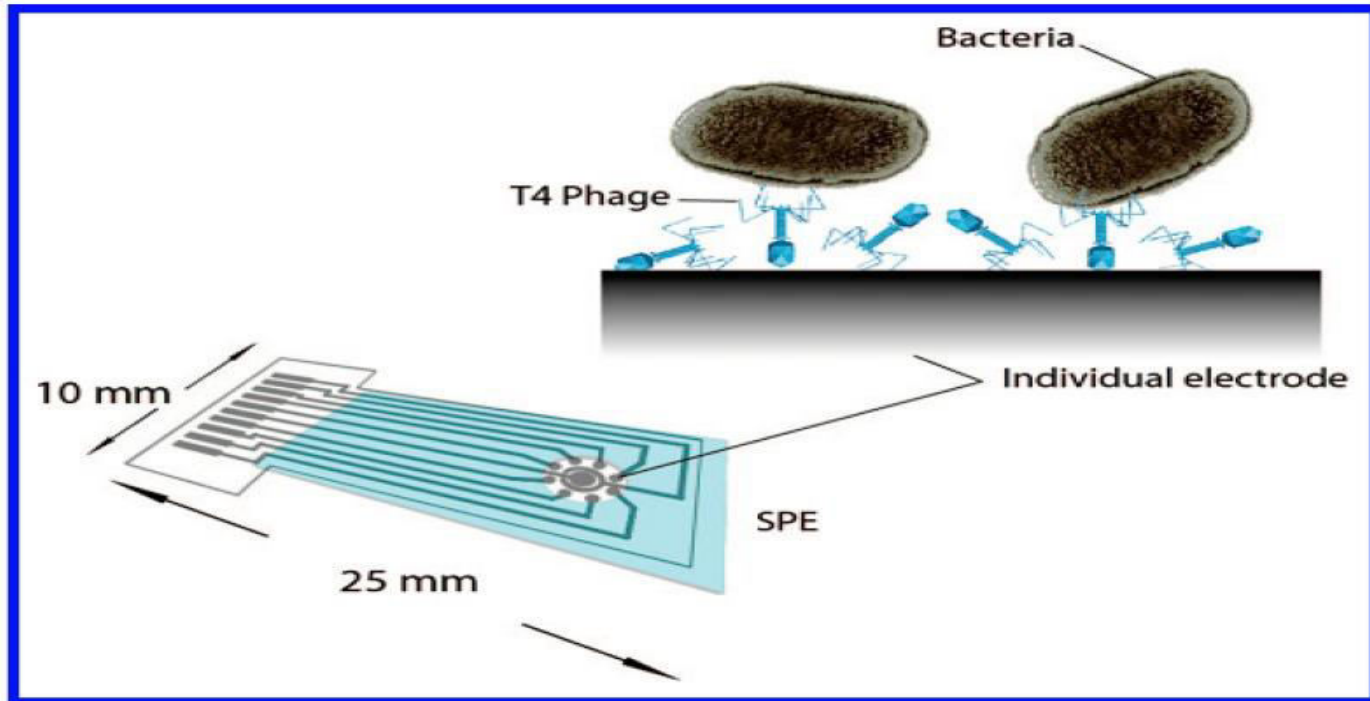
Detection of Microbial Pathogens by Cell Biosensors

Detection of *E.coli* by virus

Bacterial strain *E.coli* could be detected by using bacterial virus T4 as Bioreceptor element in a biosensor.

1. T4 is a virulent phage and use **lytic cycle**.
2. The lytic cycle leads to **destruction of the infected cells**.
3. After the infection of T4 in *E.coli* clear zones are visible on the site of bacterial colony, indicating cell lysis.

Detection of Microbial Pathogens by Cell Biosensors



Escherichia virus* T4** is a species of **bacteriophages** that infect ***Escherichia coli bacteria. Binding of T4 phage with the target host *E.coli* cells indicates the presence of pathogen in an impedance biosensor..

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5539819/bin/sensors-17-01623-g007.jpg>

Detection of Microbial Pathogens by Cell Biosensors

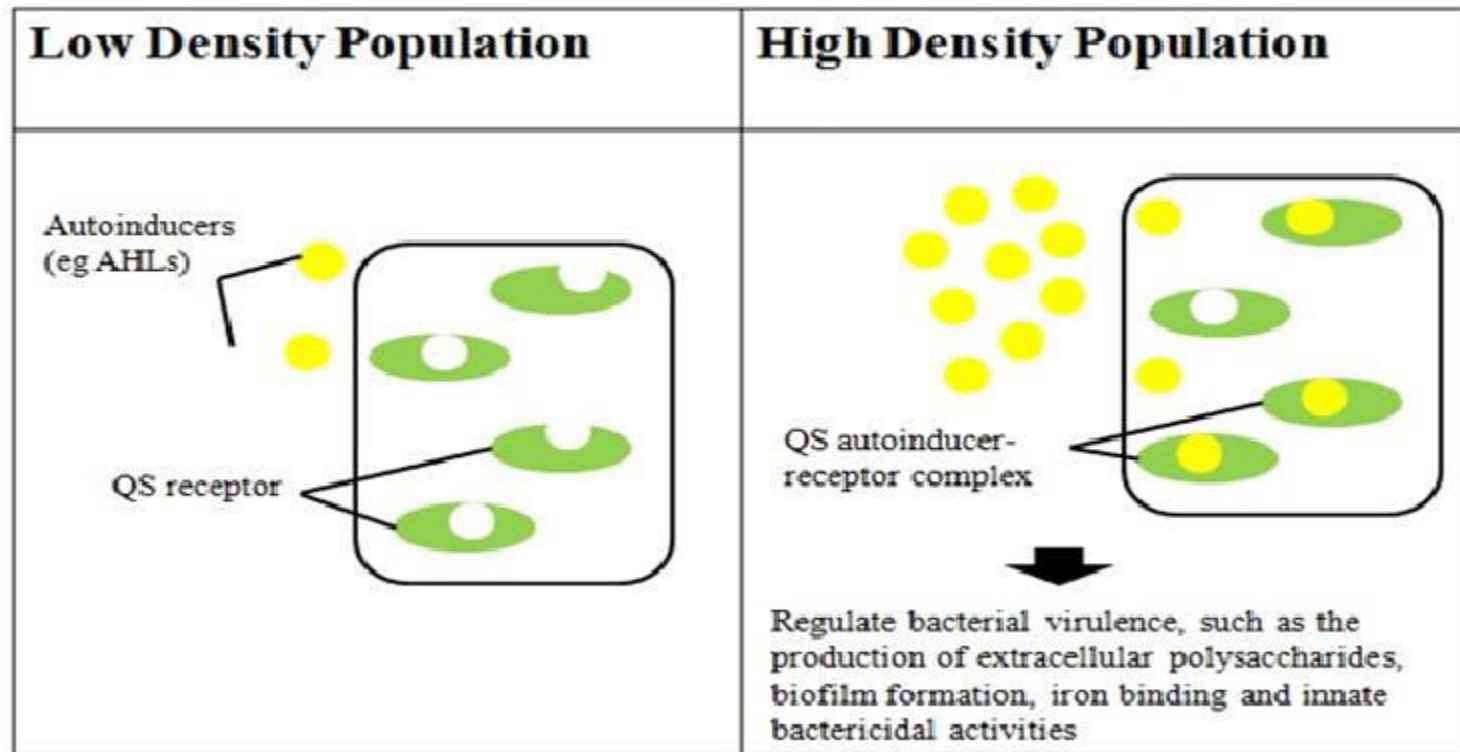
Quorum-Sensing

Quorum sensing (QS) is an intra cellular communication process, depends on the density of population. It allows the **release** of auto inducible molecules.

1. In Gram negative bacteria utilize **N-Acyl homoserine lactone** as AIM. It regulates
 - Production of polysaccharide for biofilm
 - Iron binding
 - Innate bactericidal activities.

Detection of Microbial Pathogens by Cell Biosensors

N-Acylhomoserine Lactone Production by Clinical Multidrug Resistant *Klebsiella pneumoniae*



https://www.researchgate.net/figure/Diagram-of-quorum-sensing_fig2_258957842

Detection of Microbial Pathogens by Cell Biosensors

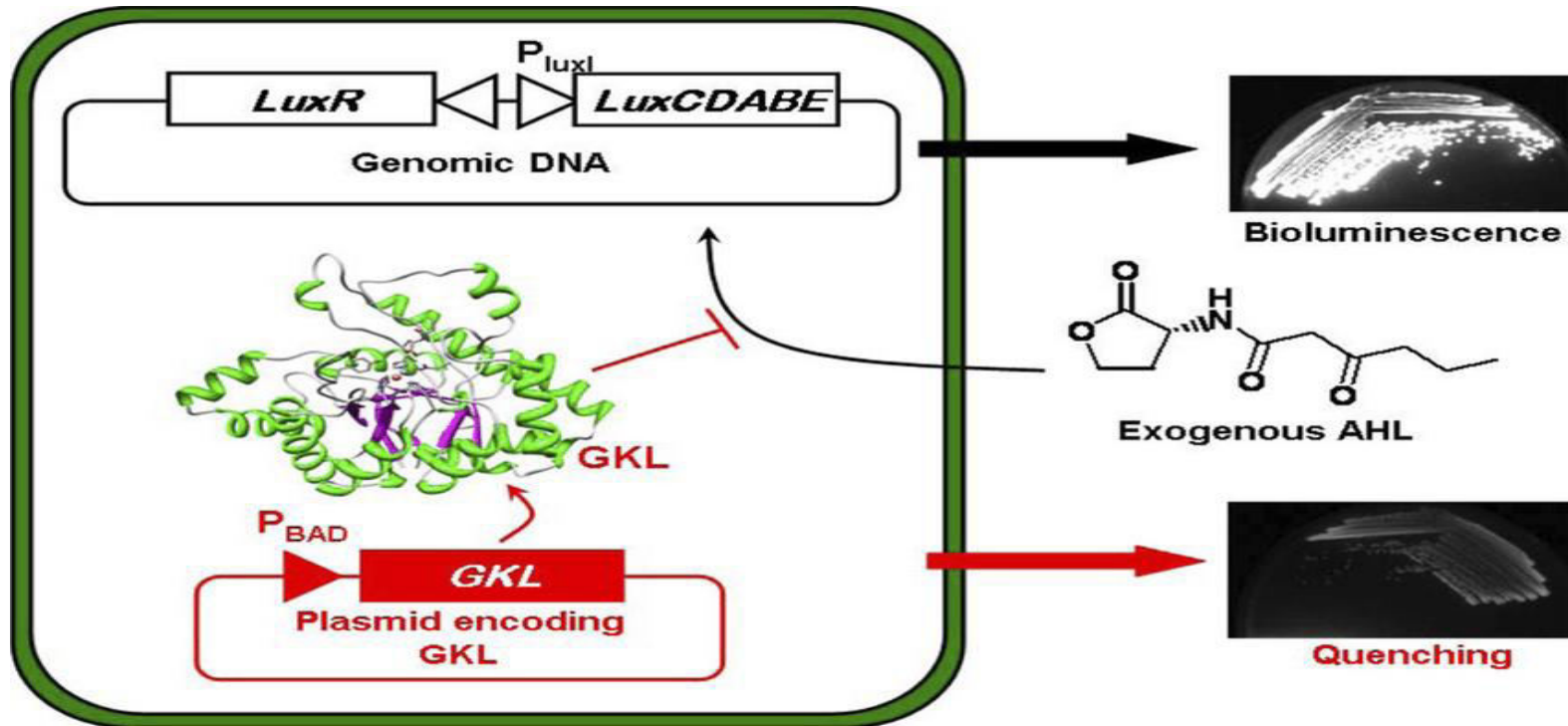
QSM and Biosensor

The *E. coli* based quorum biosensor could be designed with **two approaches**.

1. The microbial QS gene is tagged with the reporter gene. The presence of exogenous AHLs, induces the expression of reporter gene in *E.coli*.
2. AHLs is hydrolyzed to acylhomoserines by *Geobacillus Kaustophilus* lactonases, resulting in quorum-quenching of bioluminescence.

Detection of Microbial Pathogens by Cell Biosensors

Detection of Pathogen by Quorum Sensing.



https://www.researchgate.net/profile/Jeng_Chow/publication/47566252/figure/fig1/AS:448169340346368@1483863305742/Quorum-quenching-molecular-circuit-The-E-coli-based-quorum-biosensor-constitutively.png

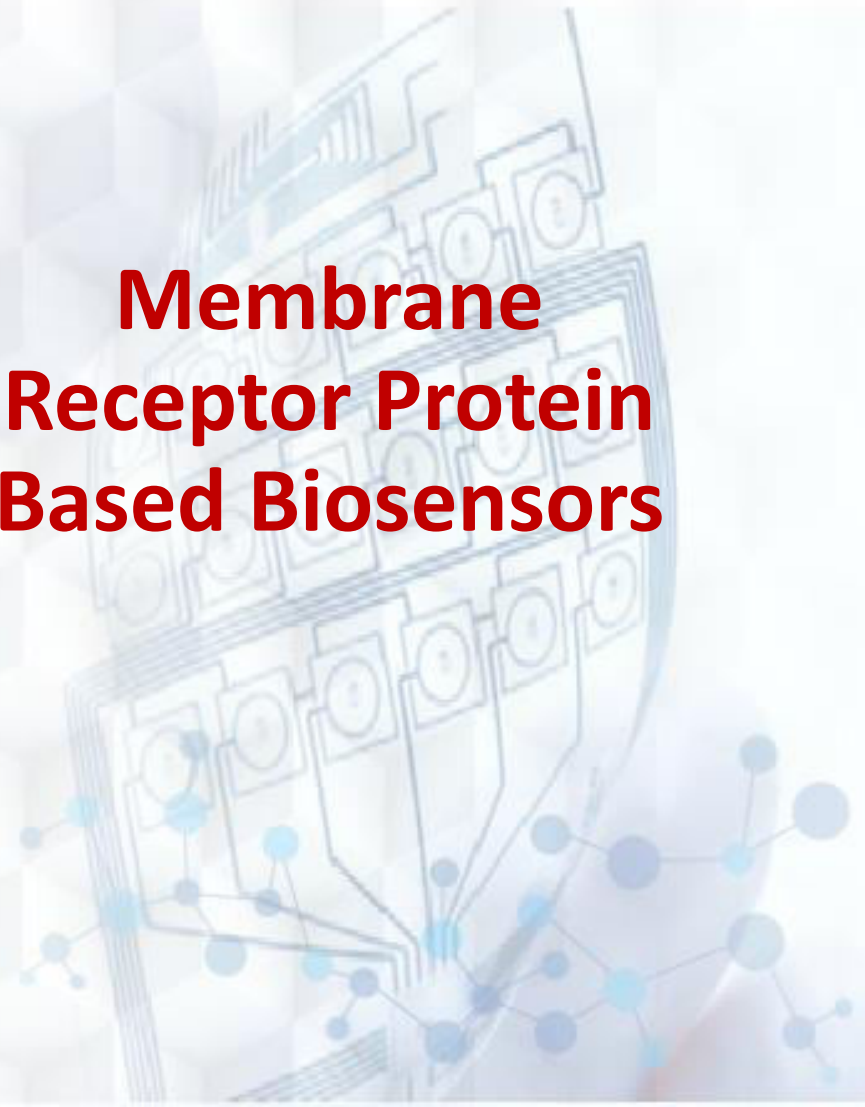
Detection of Microbial Pathogens by Cell Biosensors

Summary

1. The whole cell biosensor are successfully used for the detection of pathogens, using **virus and genetically tailored microbes**.
2. Biosensors are designed to detect the microbes on the basis of **quorum sensing** molecules.
3. The whole cell-based biosensors are **simple and can provide fast result** for detection of pathogens.

Biosensors

Membrane Receptor Protein Based Biosensors



Membrane Receptor Protein Based Biosensors

Introduction

Membrane proteins are essential components of biological membranes. There are two types of proteins.

Peripheral (Extrinsic) Protein

1. They are rich in amino acids with **hydrophilic side chains**.
2. They interact with the surrounding **water and with the polar surface** of the lipid bilayer.
3. They often contain chains of sugars (**glycoproteins**).

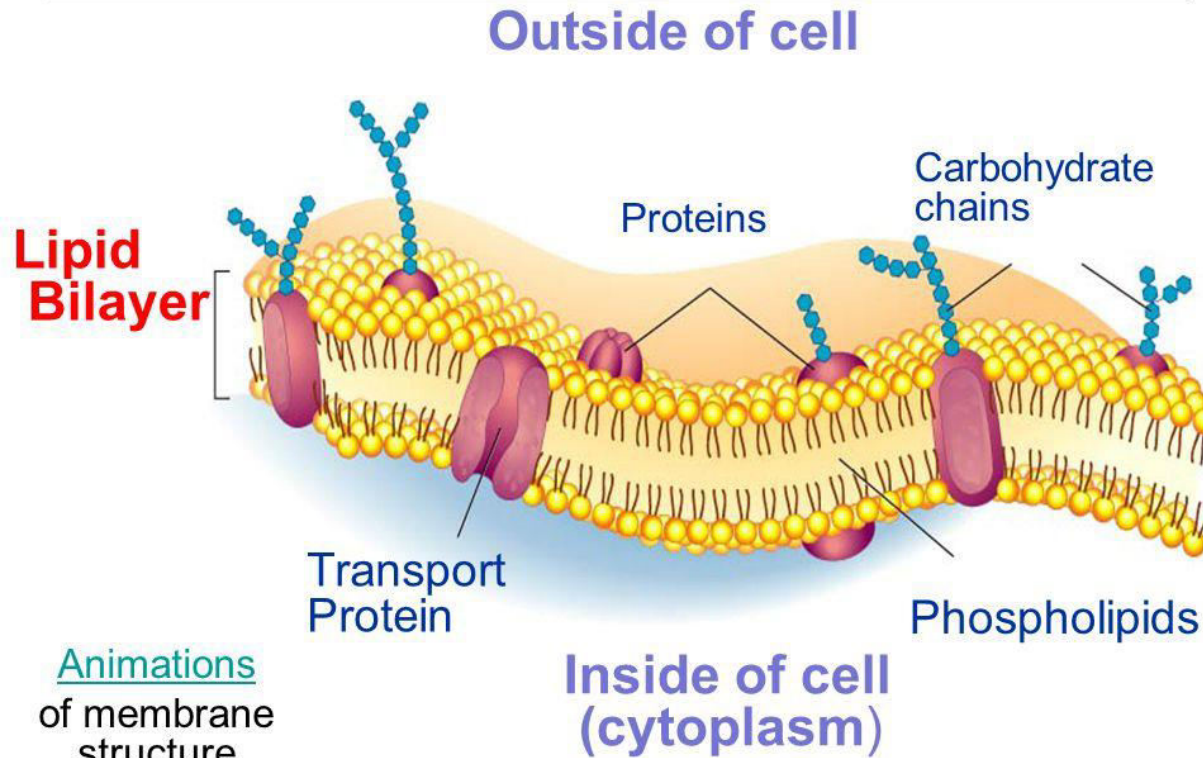
Membrane Receptor Protein Based Biosensors

Intrinsic membrane Proteins

1. Intrinsic membrane proteins contain both **hydrophilic and hydrophobic regions**.
2. The **hydrophilic portions of the protein interact with the polar heads** of the lipid molecules at each surface.
3. The **buried hydrophobic portion** of protein in the lipid bilayer are rich in **amino acids with hydrophobic side chains**.

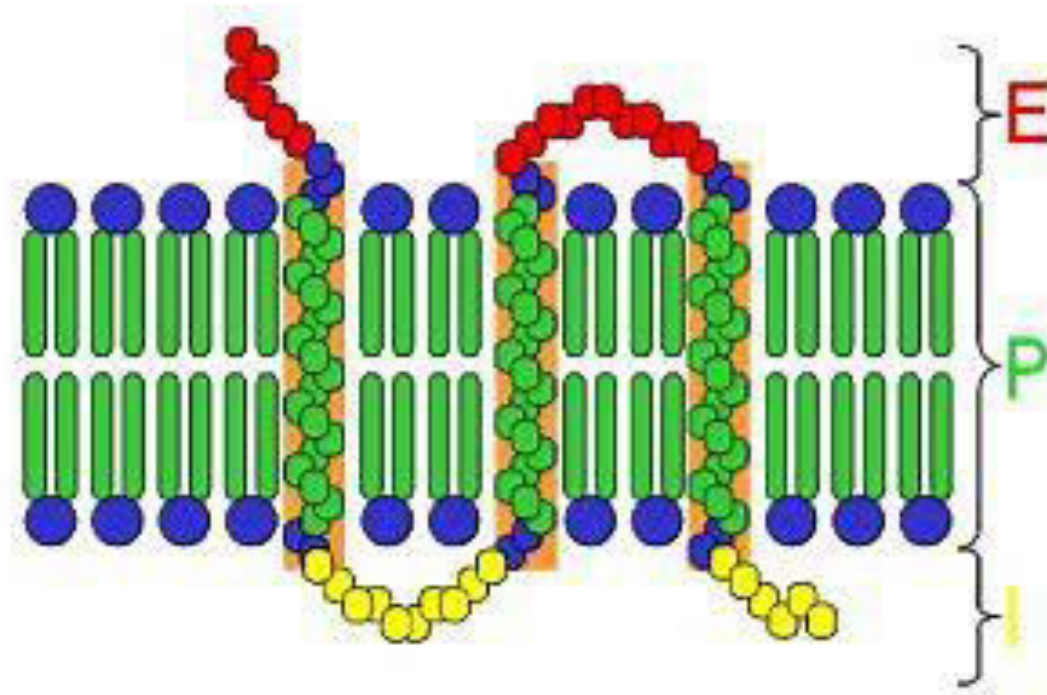
Membrane Receptor Protein Based Biosensors

Structure of the Cell Membrane



https://scienceoreilly.weebly.com/uploads/1/0/9/4/109495113/cell-membrane_orig.jpg

Membrane Receptor Protein Based Biosensors



<https://www.bing.com/th/id/OIP.KNgBw25Gako7MJoAJfssrgAAAA?w=224&h=159&c=7&o=5&dpr=2&pid=1.7>

Membrane Receptor Protein Based Biosensors

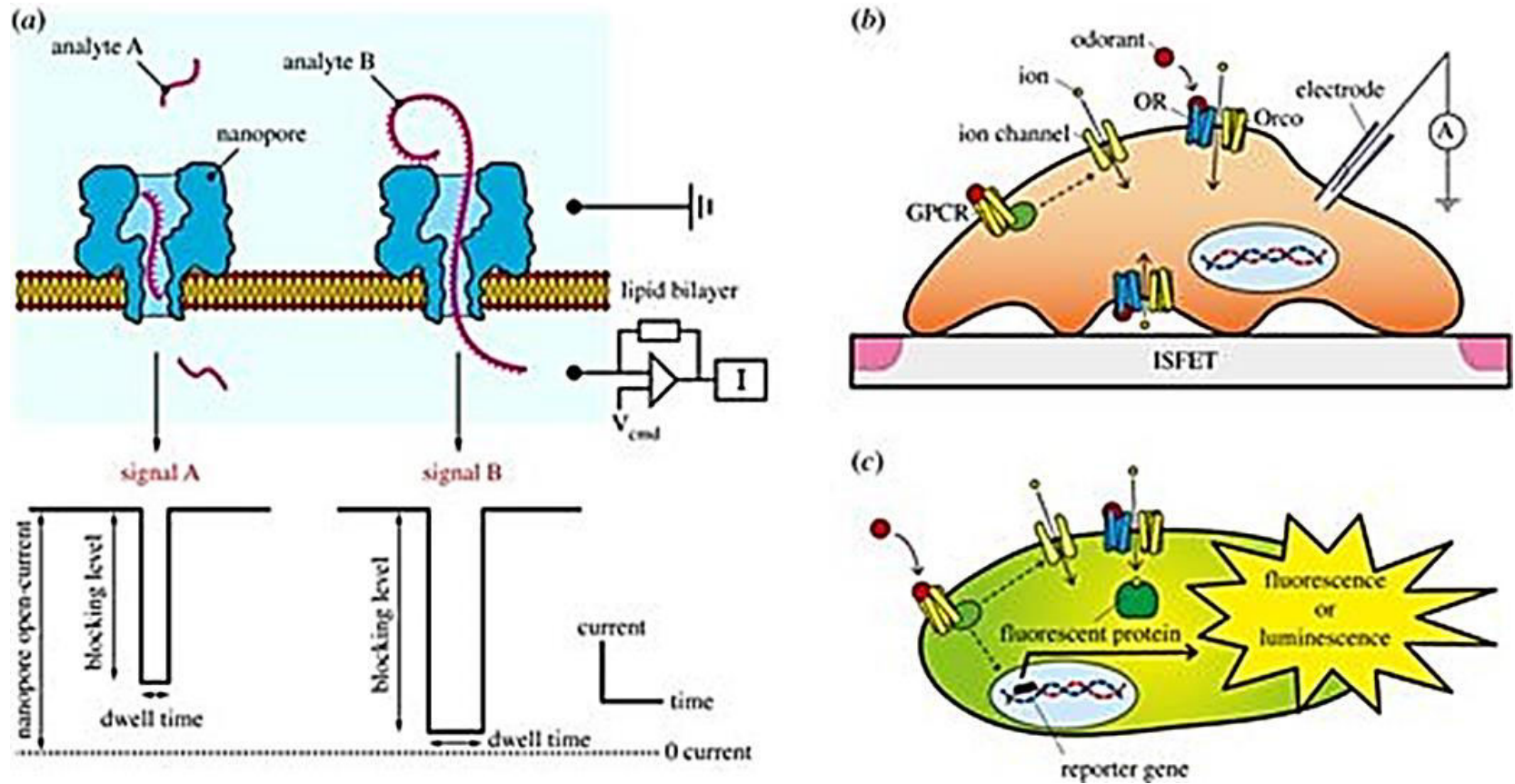
Membrane Biosensor Types

There are two types of the cell membrane protein biosensors.

1. **Lipid bilayer platforms**, the membrane proteins are **embedded in a lipid bilayer**. This layer forms a bridge between the protein and a sensor device.
2. **Cell-based platforms**, the membrane proteins are **expressed in a cultured cell**, which is **then integrated in a sensor device**.

Effort By: AARISH ALI

Membrane Receptor Protein Based Biosensors



Membrane protein-based biosensors on (a) a lipid-bilayer platform and (b,c) cell-based platforms. Membrane protein-based biosensors, Volume: 15, Issue: 141, DOI: (10.1098/rsif.2017.0952)

Membrane Receptor Protein Based Biosensors

Biosensor based on Human embryonic kidney cell

1. Human Embryo Kidney 293 cells lines are widely used in cell biology research for many years.
2. The cell line shows **reliable growth**.
3. Human embryonic kidney cells are frequently used for **foreign gene expression**.
4. They have **high transformation efficiency** for the **foreign cloned gene**.

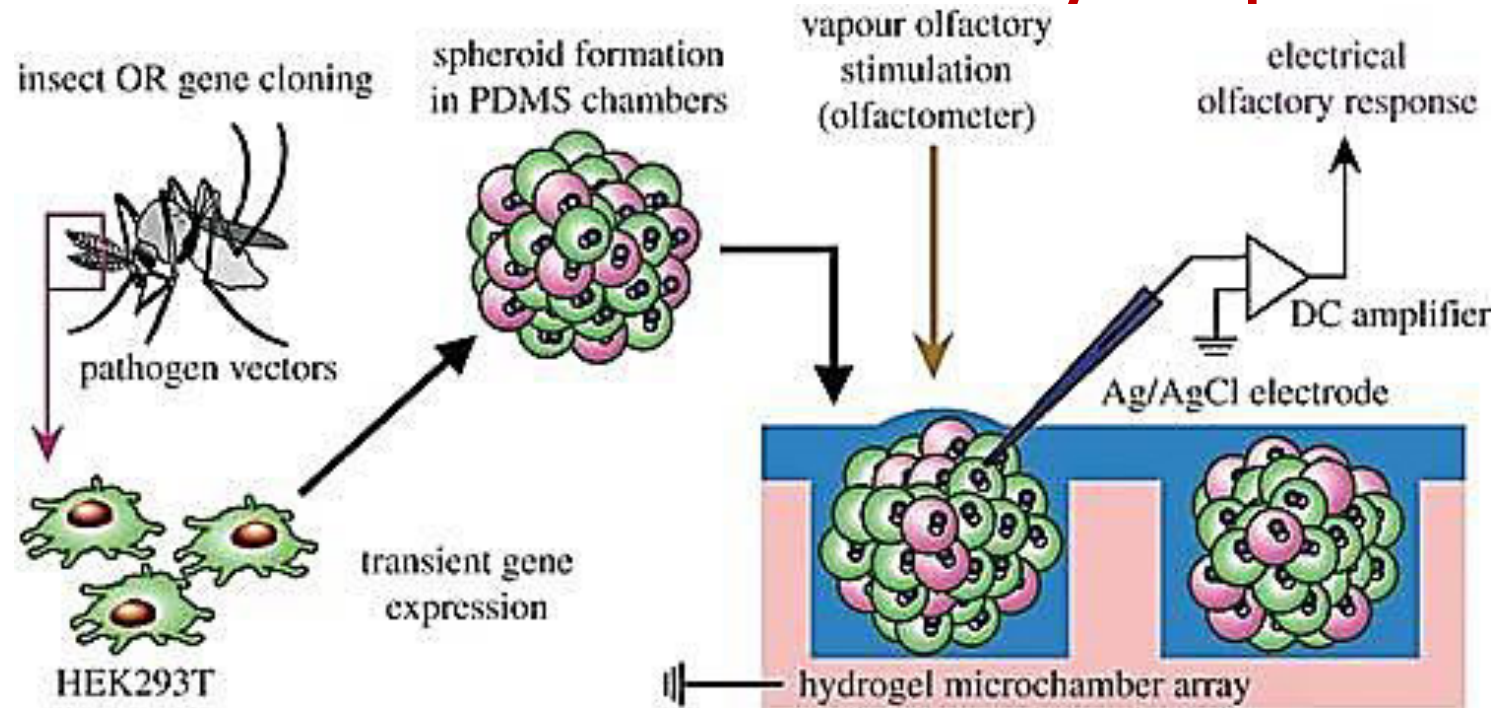
Membrane Receptor Protein Based Biosensors

Olfactory Cell Biosensor

1. Insect olfactory receptor and **OR co-receptor** were expressed in **HEK cells**.
2. The cells were placed into a **hydrogel micro-chamber** maintaining the desired **moisture conditions**.
3. Vapors of the odorant were **blown by** Olfactometer.
4. The **extracellular field potential** of the HEK spheroid was recorded using an amplifier.

Membrane Receptor Protein Based Biosensors

Cell Biosensor based on Insect Olfactory Receptor Gene



Insect olfactory receptor (OR) and OR co-receptor (Orco) were expressed in human embryonic kidney (HEK) cells that subsequently formed spheroids in polydimethylsiloxane (PDMS) chambers

<https://royalsocietypublishing.org/cms/attachment/67dd0127-3cc5-4dd5-b2b4-665ba1094621/rsif20170952f06.jpg>

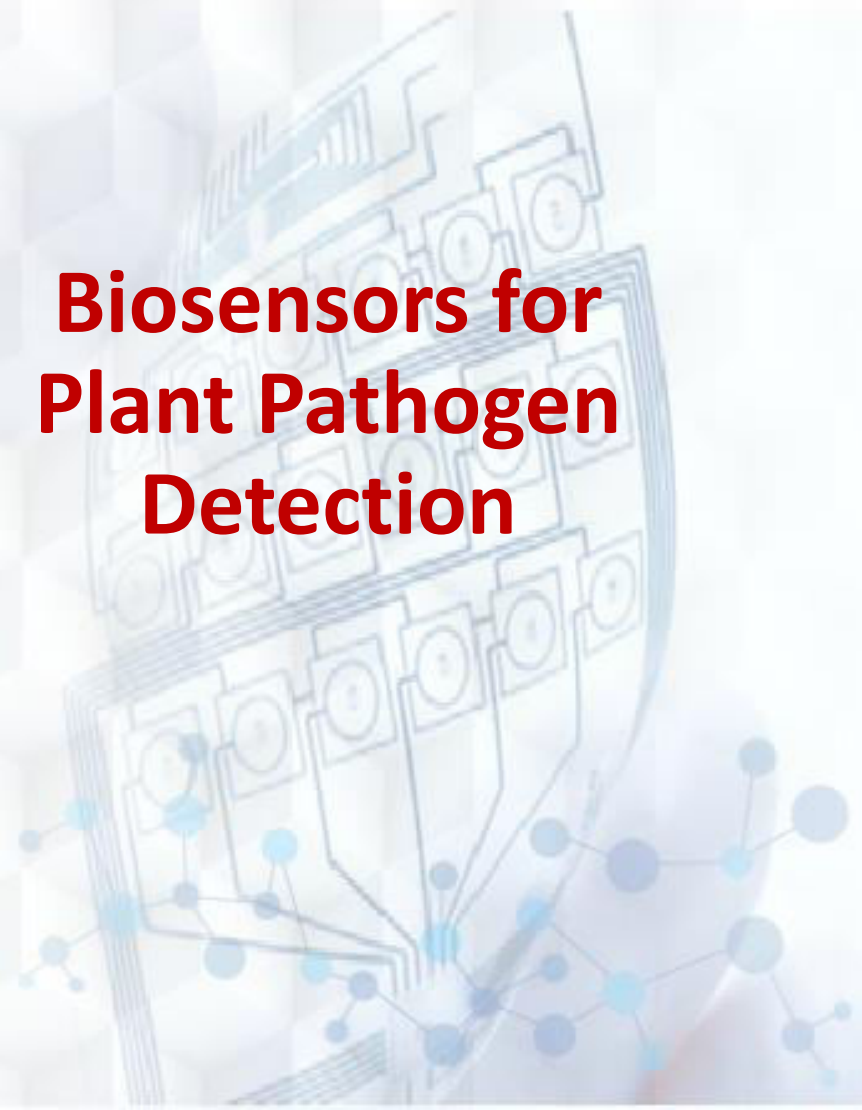
Membrane Receptor Protein Based Biosensors

Summary

1. The membrane bound protein receptor are ideal receptors for targeting the analyte.
2. There are two types of membrane biosensors.
3. Biosensor can also use only membrane as BRE.
4. The membrane bound are cloned in Human embryo kidney cell lines and are successfully used..

Biosensors

Biosensors for Plant Pathogen Detection



Biosensors for Plant Pathogen Detection

Introduction

1. A rough estimate shows that **20–30%** of the field crops are **lost annually due to microbial plant pathogens**.
2. There is an urgent need to curtail such huge loss by using latest technology.
3. Biosensor technology is a **fast growing technology** for the detection of plant pathogen, in **agriculture and food sector**.

Biosensors for Plant Pathogen Detection

Traditional Methods for detection of Microbial Pathogens

1. Traditional methods not suitable for **on-site diagnosis** of plant pathogens.
2. This approach cannot be used until the **symptoms start appearing**.
3. First it involves the **interpretation of visual symptoms** of disease.
4. It is followed by **microscopy techniques** to confirm the data.

Biosensors for Plant Pathogen Detection

Methyl salicylate Biosensor

1. A biosensor is designed to detect **methyl salicylate**
2. It is a **volatile organic** compound **released by pathogen-infected** plants via systemic response.
3. **Alcohol** oxidase and **HRP** enzymes were immobilized on to a **CNT matrix**.
4. Amperometric method was used in the development of sensor.

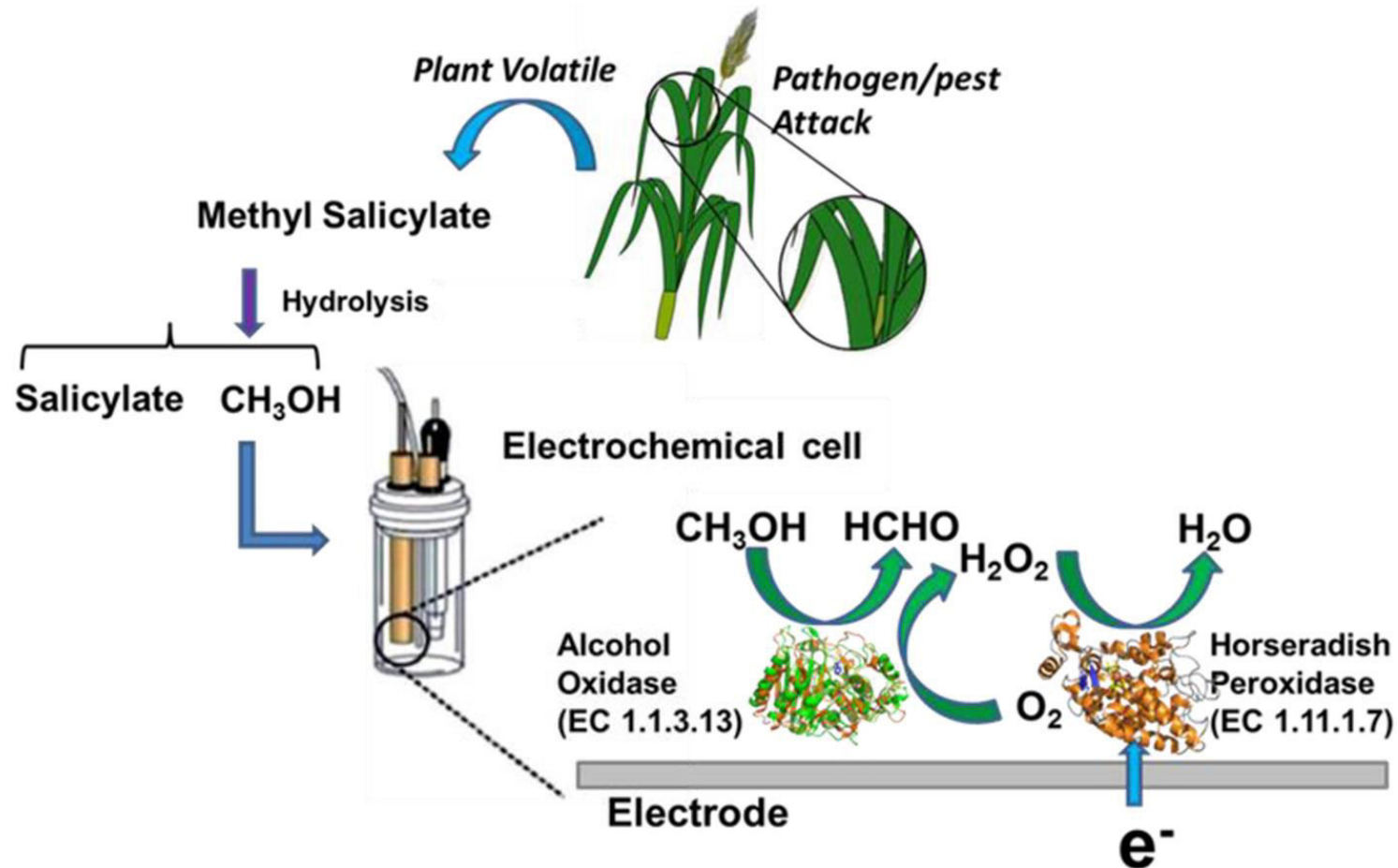
Biosensors for Plant Pathogen Detection

Working of Biosensor

1. **Hydrolysis** of methyl salicylate to form **salicylate** and **methanol**.
2. AOD converts **methanol** into **formaldehyde** and simultaneous production of **hydrogen peroxide**, in the presence of **Oxygen**.
3. Hydrogen peroxide is reduced by **HRP** to form water.
4. Generation of amperometric signals.

Biosensors for Plant Pathogen Detection

Methyl salicylate Bi-enzyme Biosensor.



Biosensors for Plant Pathogen Detection

Biosensor for pathogenic fungi

A biosensor is designed to target the **mating peptides** secreted **by pathogenic fungi**, **as an** indicator for the fungal infection.

1. Formation of **sex organs in fungi is often induced** by specific organic substances (pheromones).
2. Sex pheromones or chemicals produced by one **partner can elicit a sexual** response in the other.

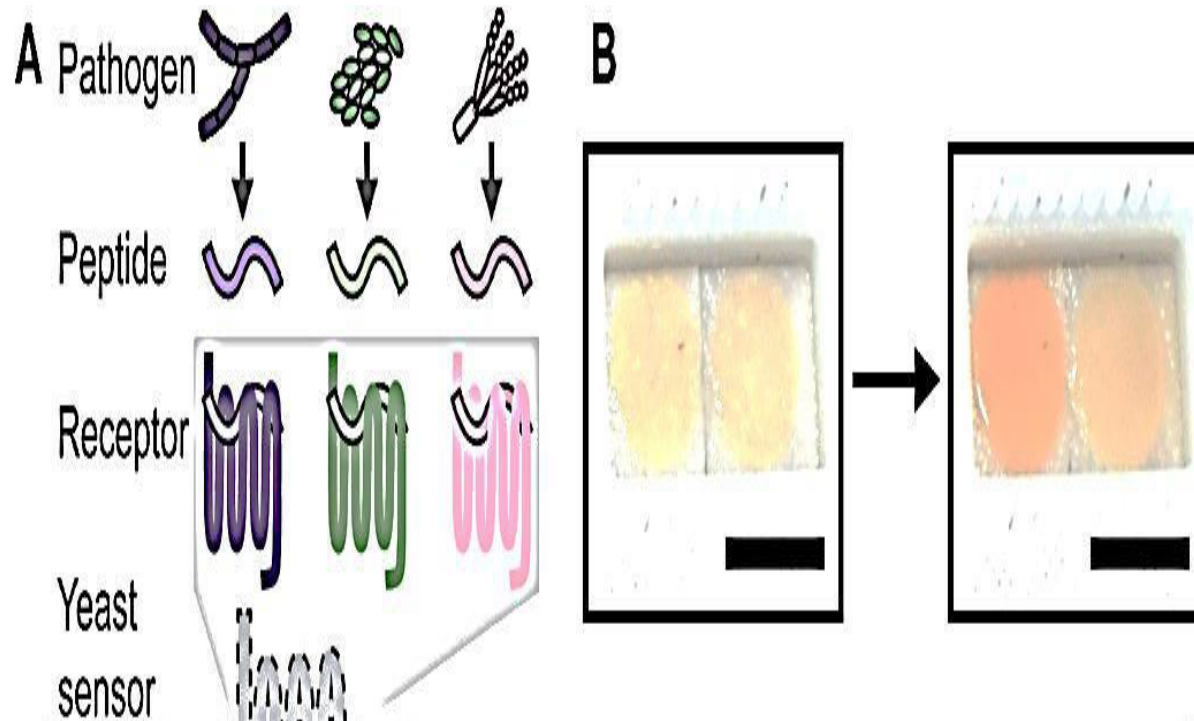
Biosensors for Plant Pathogen Detection

Working of biosensor

There are **highly specific and sensitive receptors** for the mating peptides on the fungi.

1. The detection of mating peptide leads to the **activation of mating signaling pathway**.
2. This pathway **induces the production of red lycopene pigment**.
3. The **production of pigment is a visible signal** for pathogen detection.

Biosensors for Plant Pathogen Detection



Yeast biosensor for detection of fungal pathogens. After the detection of target molecules expression of lycopene gene shows pink color signal paper-based dipstick

[assahhttps://advances.sciencemag.org/content/advances/3/6/e1603221/F1.large.jpg?width=800&height=600&carousel=1](https://advances.sciencemag.org/content/advances/3/6/e1603221/F1.large.jpg?width=800&height=600&carousel=1)

Biosensors for Plant Pathogen Detection

Summary

1. Biosensor can provide point of care testing of the plants.
2. Use of Biosensor can help to reduce the loss due to pathogen attack.
3. It is essential to select a highly specific target molecule before designing a biosensor.
4. Methyl salicylate and pheromone peptides are proven to be the excellent target molecules.

Biosensors

**Plant Biosensor to
determine Stress**



Plant Biosensors to Determine Stress

Introduction

1. In plants, **drought** conditions and **deficiency** of essential elements like **Nitrogen, phosphorus and potassium** can badly effect crop yield.
2. For a healthy crop it is required to **maintain all the nutritional requirements**.
3. **Biosensors** are designed to **determine the stress in plants** to control such deficiencies for better yield.

Plant Biosensors to Determine Stress

Water shortage in Plants

When there is a water deficiency plant shows different signs.

1. Slow down Growth
2. Reduce photosynthesis
3. Tissues Damage
4. Wilting

Some plants **show no visible signs of drought** until they have already **experienced significant harm**. It is required to detect water shortage in plants.

Plant Biosensors to Determine Stress

Biosensor to detect Water shortage

A biosensor is designed to detect drought conditions in plant by **fabricating a CNTs based electrode and stomata cells.**

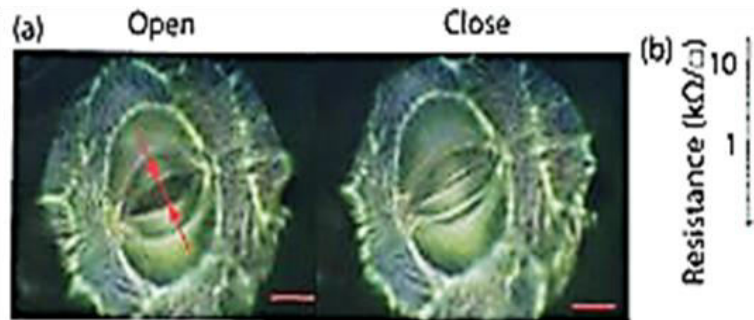
1. An ink was prepared by mixing CNTs with SDS and it was printed on the leaf surface.
2. This ink was used as an electrode, it creates an electronic circuit.

Plant Biosensors to Determine Stress

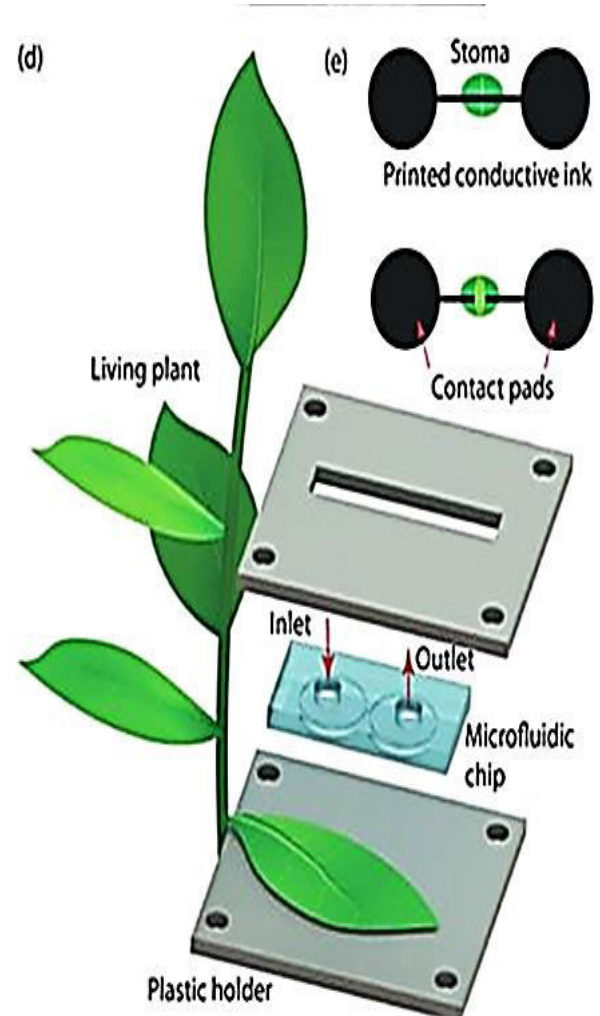
Working and Advantages

1. A plant named Peace lily was chosen for the study, because it has **large stomata**.
2. **Opening of stomata stops the current circulation** and breaks the circuit.
3. The **closing of stomata** allows the **flow of current**
4. This biosensor allows to determine the **physiological response to drought conditions**, for further studies.

Plant Biosensors to Determine Stress



https://www.researchgate.net/profile/Min_Hao_Wong/publication/320940946/figure/fig1/AS:559127677550593@1510317834288/Printing-conductive-circuits-on-Spathiphyllum-wallisii-leaf-lamina-a-Microscope_W640.jpg

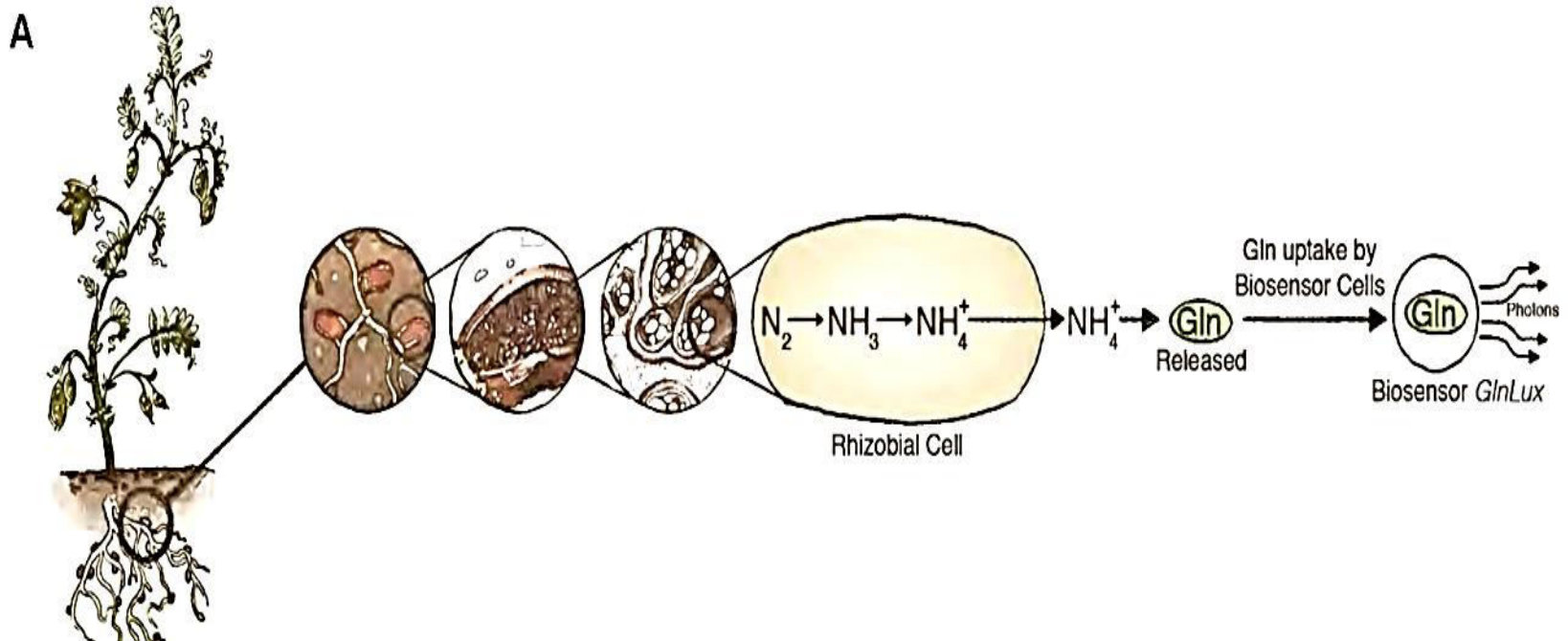


Plant Biosensors to Determine Stress

Biosensor for N Fixation

1. Plant can assimilate **atmospheric N** into **amino acids including glutamine (Gln)**, by the virtue of rhizobia.
2. **Gln** is exported to the shoots as the **major fraction**.
3. During the export Gln is **released into the atmosphere**.
4. The release of less amount of Gln could be an indicator of N deficiency.

Plant Biosensors to Determine Stress



When **Gln is released** from plant tissues to the medium, it is taken up by **GlnLux biosensor *E.coli* cells**, causing them to become metabolically active and divide, which proportionally activates a **lux operon for the generation of signals**.

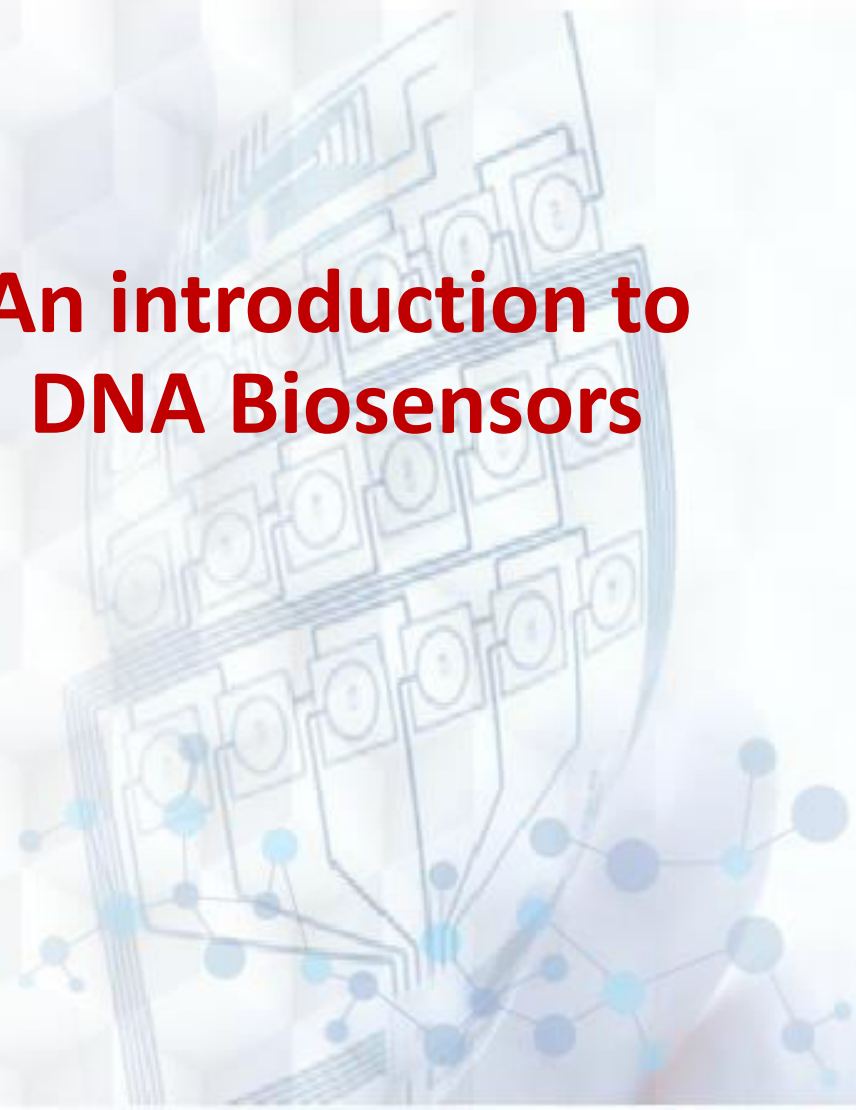
Plant Biosensors to Determine Stress

Summary

1. Biosensors are very helpful in identifying the nutritional deficiencies and drought conditions in plants.
2. A biosensor is designed to detect the drought conditions in plants. **The sensor used the opening and closing of stomata to detect the water condition in the plant.**
3. Another Biosensor is designed to detect the **N deficiency using Gln Lux promoter.**

Biosensors

An introduction to DNA Biosensors



An introduction to DNA sensors

Introduction

1. DNA biosensors are based on the **employment of DNA** molecule as a Biorecognition element.
2. There is a **selective binding between a DNA**, that is immobilized on the **electrode surface and target** molecule in the sample.
3. Generally in these Biosensors **Optical, electrochemical and Piezo electric** sensors are used.

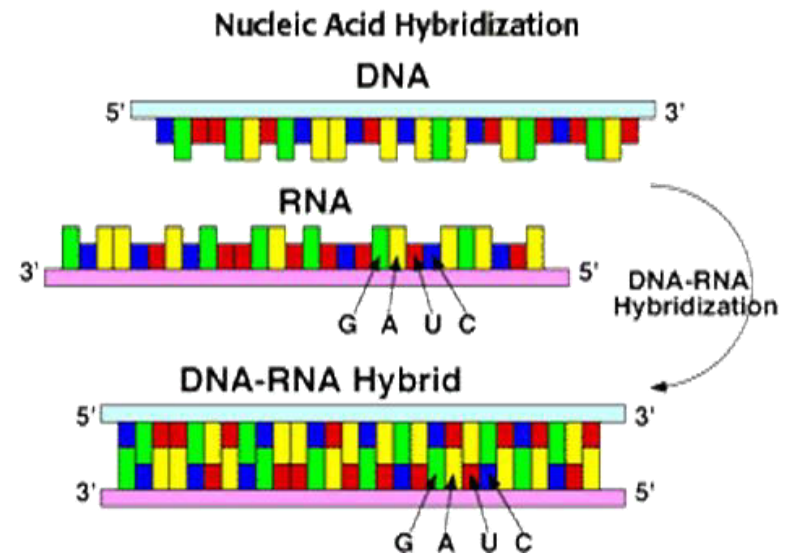
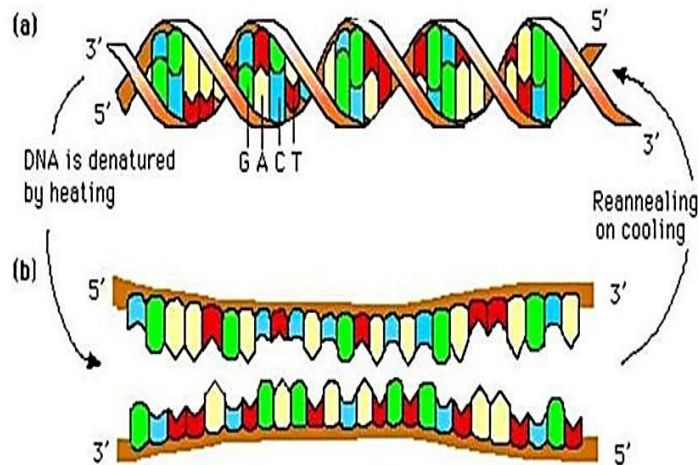
An introduction to DNA sensors

DNA as Biorecognition element

1. The binding of DNA the complementary **DNA or RNA** sequence is called **Hybridization**.
2. The binding of receptor DNA with **the proteins or other molecules** is called **inter action**.
3. DNA molecule can also inter act with other molecules such as **toxins, Vitamins**.

An introduction to DNA Biosensors

Binding of DNA with RNA and single stranded DNA molecules



<http://www3.med.unipmn.it/did/will/BiologyProject/05ta-1.gife>

An introduction to DNA Biosensors

Protein DNA interaction

Single strand of DNA **flexibly interacts with the proteins** to **adjust their position** for binding.

1. There are **two types of forces** are involved in the ssDNA and protein interaction. **Hydrogen bonding** and **Electrostatic interactions**
2. There are certain factors that can alter these forces are **ionic strength, pH and temperature**.
3. Common example Histones and DNA interaction.

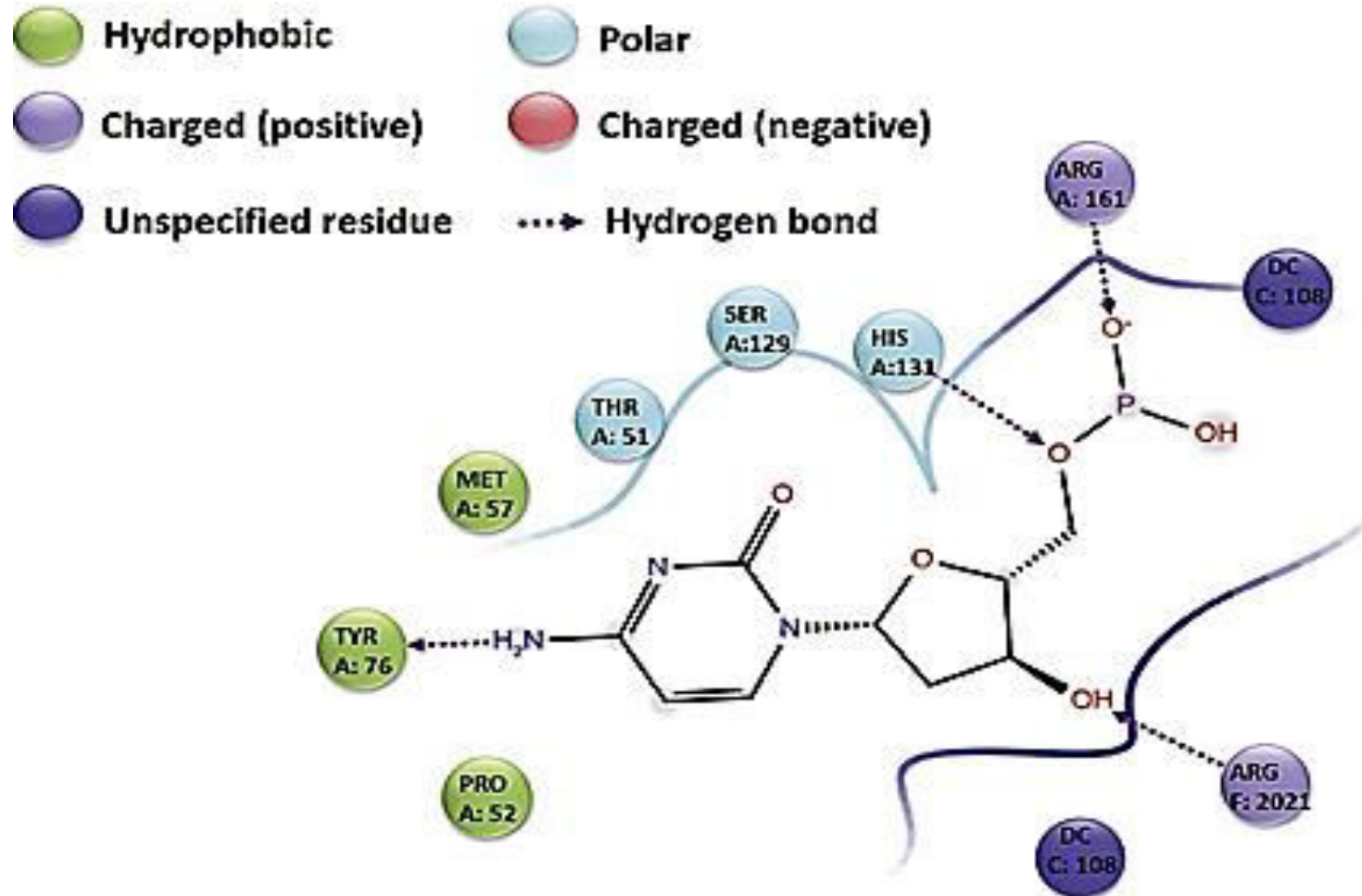
An introduction to DNA Biosensors

Hydrophobic Amino acids and Protein DNA interaction

1. **Hydrophobic** amino acid play an important role in the binding of ssDNA with the protein.
2. The amino acids having **hydrophobic character** due to their **side chain** includes.

Glycine, Alanine, Valine, Leucine, Isoleucine, Proline, Phenylalanine, Methionine and Tryptophan.

An introduction to DNA Biosensors



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An introduction to DNA Biosensors

Role of Electrostatic forces in DNA- Protein complex Formation

1. Electrostatic force or Vander wall forces are the **attractive or repulsive** interactions due to charges **between two molecules**.
2. A salt bridge (Biology) is a combination of two non-covalent interactions such as **hydrogen bonding and electrostatic interaction** within the **two biological molecules**.

An introduction to DNA Biosensors

Interaction of Histones-DNA.

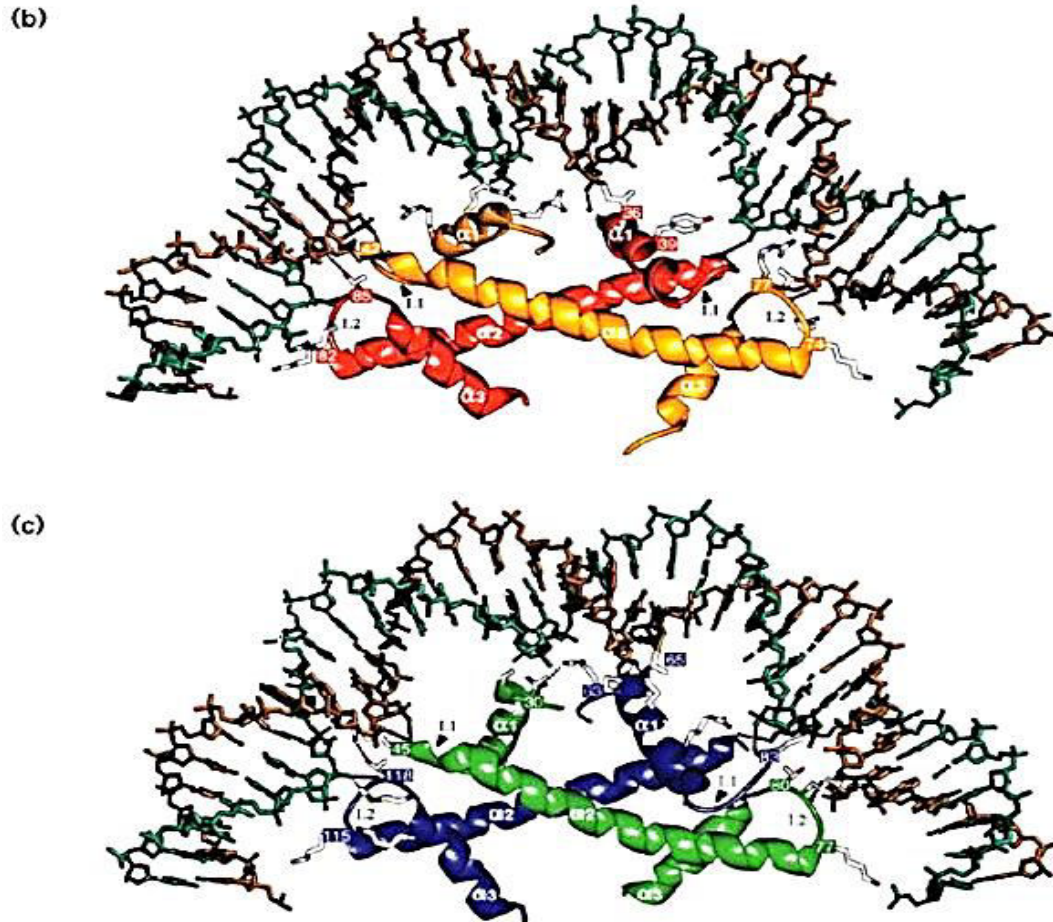
1. Polar interaction between protein helix and Phosphate groups of DNA.
2. Non polar interaction between ribose sugar and histones
3. Hydrogen bond between DNA and Amide group

An introduction to DNA Biosensors

Interaction of Histones-DNA.

4. Salt bridge between basic AA and Phosphate group.
5. Non specific interactions between the **N-terminal tail** of Histone **minor groove of DNA**.

An introduction to DNA Biosensors



Binding of Histone proteins with DNA employing weak physical forces..
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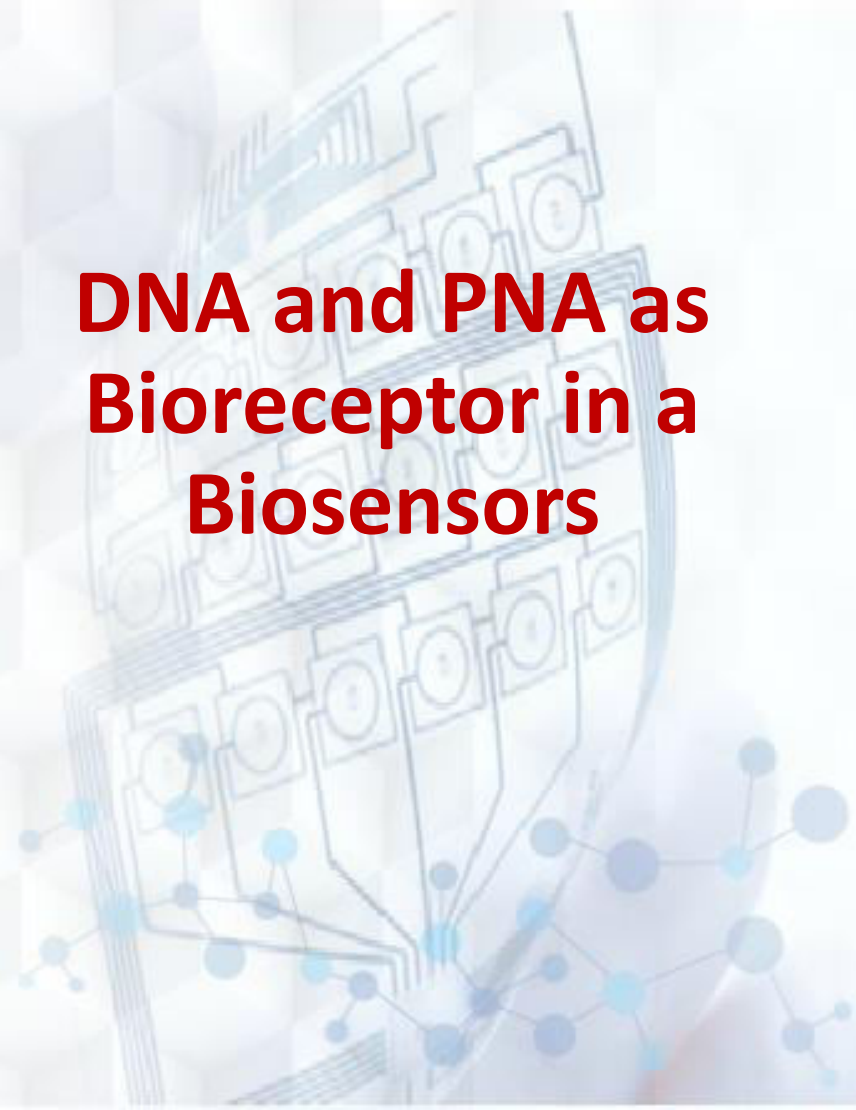
An introduction to DNA Biosensors

Summary

1. A ssDNA molecule is used as a BRE in a DNA based biosensor.
2. DNA as BRE can bind with DNA and RNA target molecules.
3. Single stranded DNA of the sensor **interacts flexibly with target protein.**
4. Different types of **weak forces play a major** role in the detection of target molecule.

Biosensors

DNA and PNA as Bioreceptor in a Biosensors



DNA and PNA as Bioreceptor in Biosensors

DNA as Bio Receptor

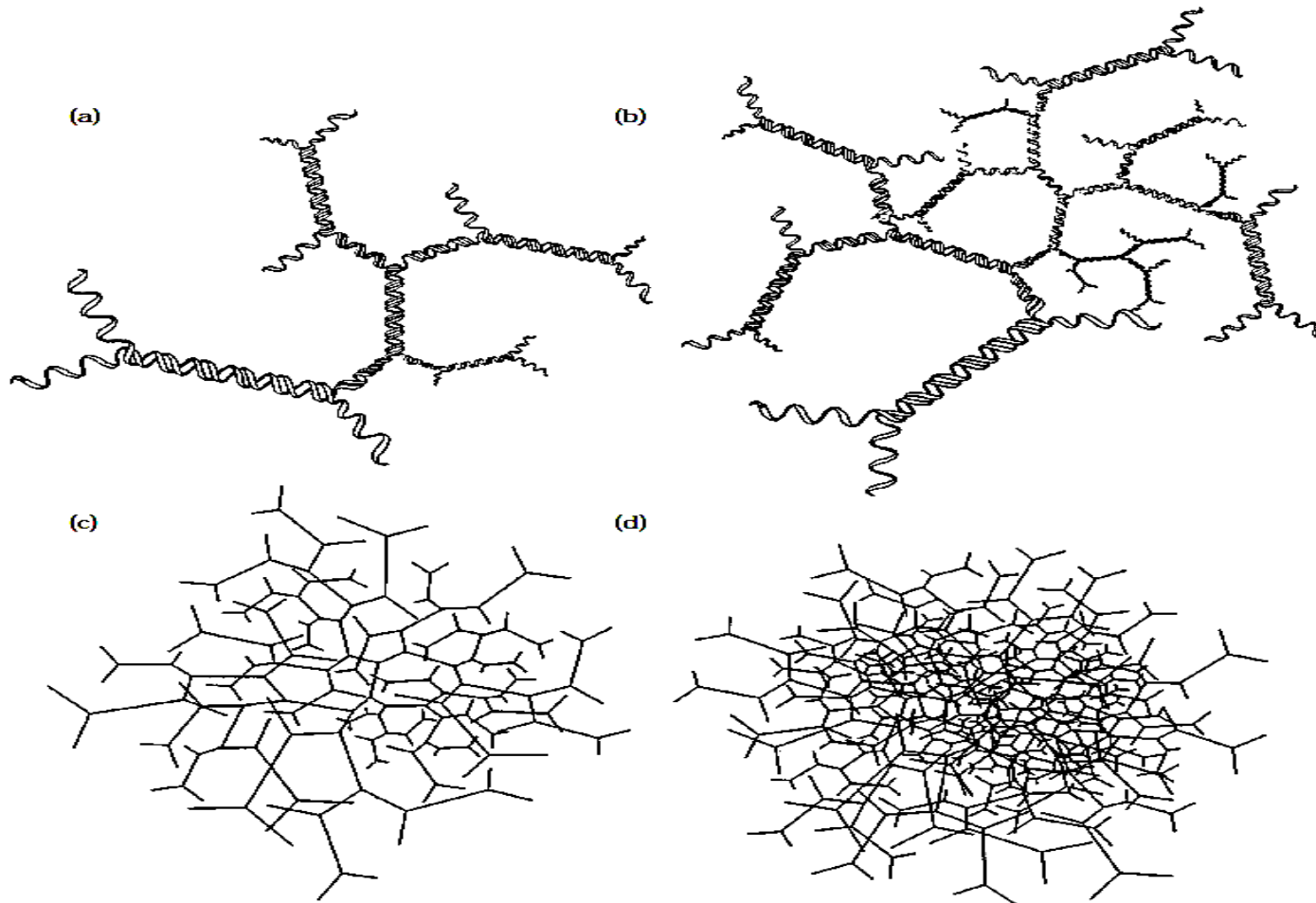
1. In DNA biosensors Small size fragments of DNA ranging from **10-100 nucleotides** are used as DNA probes.
2. ssDNA prepared by a series of **chemical reactions producing** a known sequence of nucleotides.
3. DNA biosensors are also known as **Geno sensors**.

DNA and PNA as Bioreceptor in Biosensors

DNA Dendrimers

1. Dendrimers are **tree like super structure**.
2. Each DNA Dendrimer consists of several **similar branched DNA units**.
3. DNA units connected to each other using **DNA ligase**.
4. A dendrimer is typically **symmetric around the core**, and often adopts a **spherical three - dimensional morphology**.

DNA and PNA as Bioreceptor in Biosensors



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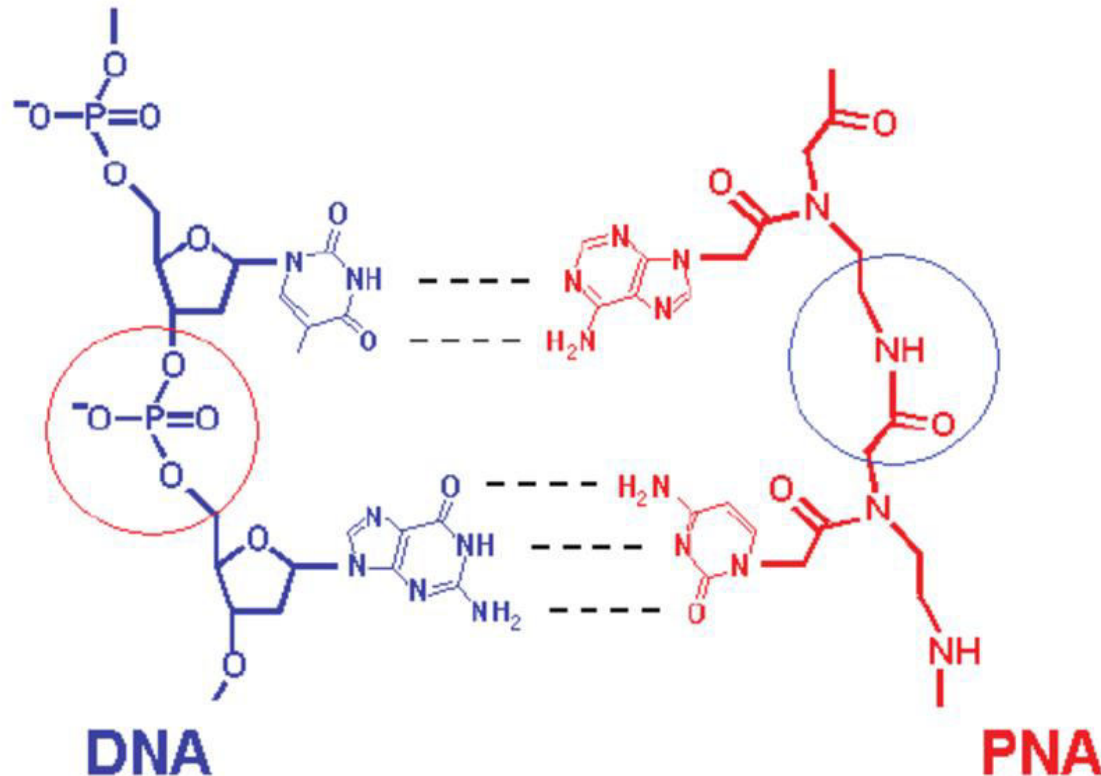
DNA and PNA as Bioreceptor in Biosensors

PNA as a receptor molecule

To increase the stability and efficiency of the DNA based biosensors a **DNA derivative PNA** is use in DNA.

1. PNA can bind with the **complementary DNA/RNA** target molecule.
2. There is no **internal repulsion in DNA-PNA dulpex** as compared to DNA-DNA duplex, that is due to the **absence of Phosphate groups**.

DNA and PNA as Bioreceptor in Biosensors



[https://www.researchgate.net/profile/Claire-](https://www.researchgate.net/profile/Claire-Marie-Pradier/publication/221906936/figure/fig1/AS:305328735440897@1449807451274/Schematic-chemical-model-of-PNA-and-DNA-molecules-showing-the-different-backbone.png)

Marie_Pradier/publication/221906936/figure/fig1/AS:305328735440897@1449807451274/Schematic-chemical-model-of-PNA-and-DNA-molecules-showing-the-different-backbone.png

DNA and PNA as Bioreceptor in Biosensors

PNA and DNA and Hybrid rate of Hybridization

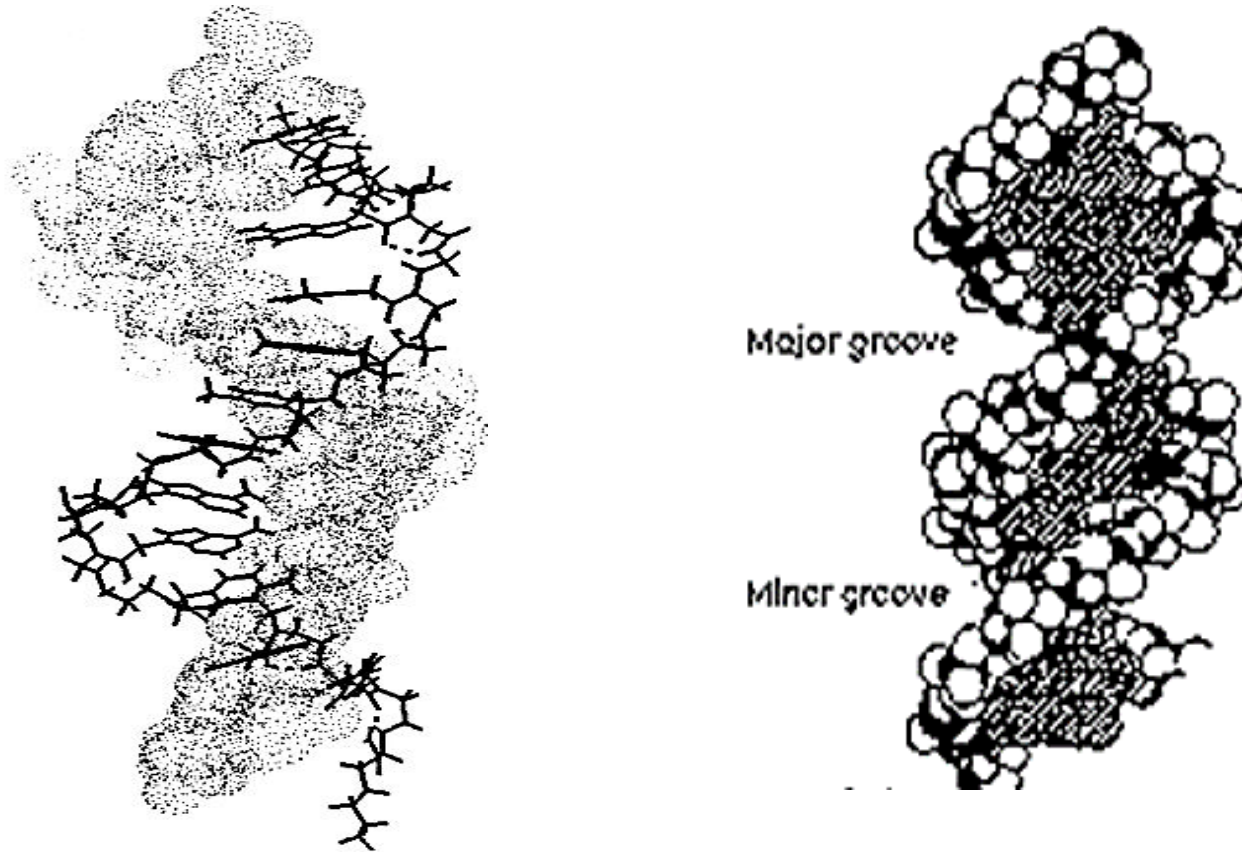
1. The **rate of hybridization** of PNA-DNA duplex increases upto **50,000 times** as compared to DNA-DNA duplex,.
2. There is no phosphate group in PNA, so there is no repulsion inside the duplex.

DNA and PNA as Bioreceptor in Biosensors

Stability of PNA and DNA Hybrid

2. The stability of DNA/PNA hybrid increased with an increase in number of bases.
3. A increase of **one base pair in DNA-PNA duplex** increases the **thermal stability** of the complex at least **upto 1°C** , as compared to DNA/DNA hybrids.

DNA and PNA as Bioreceptor in Biosensors



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https://www.researchgate.net/profile/Orn_Almarsson/publication/14783047/figure/fig4/AS:601649244545030@1520455766104/figure-fig4_W640.jpg

DNA and PNA as Bioreceptor in Biosensors

Summary

1. In DNA biosensors Short ssDNA and PNA molecules could be used to bind with their complementary sequence in RNA and DNA.
2. Different strategies are used to **increase the efficiency of sensors**, including use of **dendrimers and DNA analogue**.
3. PNA as a BRE can form a stable DNA-PNA duplex.

Final Term Syllabus

Lecture 01 to 102

Efforts By

AARISH ALI (+96569002641)

Contact us to join our esteemed group

“All About Biotech”

Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

REMEMBER IN PRAYERS
KEEP SMILING
SAY NO TO CURRUPTION
HELP TO OTHERS

Regards,
AARISH ALI & TEAM

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